Willoughby Development Control Plan

2023

Version Number	When adopted	Summary of changes
Version 1	Adopted at Council Meeting 22 May 2023, in effect after *WLEP 2012* Amendment 34 being made (30 June 2023)	Comprehensive update of WDCP, accompanying *WLEP 2012* Am 34
Version 2	Adopted at Council Meeting 23 June 2025	Housekeeping amendments across WDCP
Version 3	Adopted at Council Meeting 4 March 2026	Site specific DCP for 641-653 and 655A Pacific Highway, Chatswood (Part L, 13.1.18)

Acknowledgement of Country

We acknowledge the traditional inhabitants of the land on which we stand, the Aboriginal people, their spirits and ancestors.

We acknowledge the vital contribution that Indigenous people and cultures have made and still make to the nation that we share, Australia.

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PART A

Overview of WDCP 2023

PART A: Overview of Willoughby Development Control Plan 2023

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1 Introduction

1.1 About this plan

Willoughby Development Control Plan (Willoughby DCP) provides guidelines and controls for developments within the Willoughby local government area. This DCP repeals DCP 2006 pursuant to section 16 of the Environmental Planning and Assessment Regulation 2021. It supports the objectives and planning provisions in Willoughby Local Environmental Plan (Willoughby LEP).

All development applications submitted to Willoughby City Council are assessed in line with the performance criteria, development controls and other relevant provisions in this plan.

Each application is also assessed on the individual merits of the proposed development. A proposal may comply with the development controls in this plan but this does not necessarily ensure it will be approved.

This plan is in line with *Environmental Planning and Assessment Act 1979* (part 3, section 3.43) and *Environmental Planning and Assessment Regulation 2000* (part 3).

1.2 Aim

The aim of this part is to give an overview of the different sections of Willoughby DCP.

1.3 Objectives

This part:

a. advises when a development application is needed and the supporting information required as part of the application

b. explains how this plan relates to other state and local plans, policies, and guidelines

c. details the other parts of Willoughby DCP.

2 Framework of this plan

2.1 Land covered by Willoughby DCP

This plan applies to all land and development within the City of Willoughby local government area.

2.2 Terms and definitions

The terms ‘this plan’, ‘the plan’, ‘DCP’ and ‘Willoughby DCP’ refer to this document, Willoughby Development Control Plan.

The terms ‘Willoughby LEP’ and ‘the LEP’ refer to Willoughby Local Environmental Plan.

The terms ‘we’, ‘us’ and ‘our’ in this plan refer to the local government authority ‘Willoughby City Council’, which serves the City of Willoughby local government area.

The terms ‘local area’ and ‘our area’ refer to the City of Willoughby local government area.

The term ‘Council’ refers to the elected group of councillors for Willoughby City Council.

An ‘Authorised Officer’ is a Council employee that has been granted delegated authority to make decisions on behalf of Council.

Other definitions and meanings of the terms used in this document are the same as the terms listed in the dictionary under Willoughby LEP and any other definitions under relevant state plans, policies or codes.

2.3 Parts of this plan

This plan is divided into 12 parts, A to L.

2.3.1 Part A: Overview of Willoughby DCP

This part outlines the DCP and includes a link to the development application process. It explains the relationship of this plan to other state and local plans, policies and guidelines.

2.3.2 Part B: Residential Development

This part includes these sections and attachments:

a. Section 1 is an introduction with the aims and objectives for this part.

b. Section 2 provides the performance criteria for all types of residential accommodation.

c. Section 3 includes controls for ‘minor’ developments such as dwelling houses and dual occupancies.

d. Section 4 provides guidelines and controls for major residential development, such as residential flat buildings, manor houses, and apartments associated with shop top housing and mixed use developments.

e. Section 5 provides specific controls for additional permitted land uses such as secondary dwellings, bed and breakfast accommodation, boarding houses and ‘built-to-rent’ housing.

f. Section 6 provides specific controls for child care centres in residential areas.

g. Section 7 provides further controls for any future development of specific sites and areas.

h. Attachment 1 includes a compliance checklist table for controls in the R2 zone.

i. Attachment 2 includes a compliance checklist table for controls relating to manor houses, attached dwellings and multi dwelling housing in the R3 zone.

Attachment 3 includes a compliance checklist of controls for residential development of three or more storeys and four or more dwellings and the residential components of shop top and mixed-use developments in the R3 and R4 zones, Further controls apply to developments in R2 zones that are within a heritage conservation area. Part H details these controls.

2.3.3 Part C: Development in C4 Environmental Living Zone

This part includes specific performance criteria and controls to provide for low impact development on land close to foreshore areas and environmentally sensitive locations.

Further controls apply to developments in C4 zones that are within a heritage conservation area. Part H details these controls.

2.3.4 Part D: Commercial Development

This part includes performance criteria and controls for all types of commercial and non-residential development. This includes the commercial components of shop top housing and mixed use developments and placed based plans.

Further controls may apply to commercial and retail premises in a heritage conservation area. Part H details these controls.

2.3.5 Part E: Industrial Development

This part includes guidelines and controls for industrial development and other land uses permitted in the industrial zones.

2.3.6 Part F: Transport and Parking Management

This part applies to all developments. It provides advice and controls for traffic management, off-street car parking, bicycle parking and end-of-trip facilities, loading and unloading facilities, and other traffic and transport related matters.

It is informed by Willoughby Integrated Transport Strategy 2036. See willoughby.nsw.gov.au.

2.3.7 Part G: Vegetation Management

This part applies to any development that involves the removal of trees or vegetation.

It is made under Part 2.2 of *State Environmental Planning Policy (Biodiversity and Conservation) 2021*. See State Environmental Planning Policy (Biodiversity and Conservation) 2021 - NSW Legislation

It is supported by our Vegetation Management Guidelines. See willoughby.nsw.gov.au.

2.3.8 Part H: Heritage Items and Heritage Conservation Areas

This part is subject to Clause 5.10 (Heritage conservation) of Willoughby LEP. The clause deals with the environmental heritage of Willoughby, including heritage items, heritage conservation areas, conservation of archaeological sites, and conservation of Aboriginal objects and Aboriginal places of heritage significance.

Heritage items and an archaeological site are identified in Schedule 5 of Willoughby LEP. Heritage conservation areas are shown in the heritage maps accompanying the LEP.

Part H includes controls and measures to conserve and protect the heritage values of existing dwellings and commercial/retail buildings. It also has controls to ensure any new development is in keeping and sympathetic to the character of the particular heritage conservation area.

Any residential or commercial/retail development in a heritage conservation area must also consider the performance criteria and controls in parts B, C and D of this plan. If there is any inconsistency with the controls under parts B, C and D of the Willoughby DCP, the controls under Part H prevail.

The LEP does not identify Aboriginal objects or Aboriginal places of heritage significance. If a development is likely to disturb any natural bushland or geological features, applicants should contact Willoughby City Council to determine if a site comprises or is in proximity to an Aboriginal object or Aboriginal place of heritage significance.

2.3.9 Part I: Stormwater Management

This part provides advice and controls for stormwater management. It applies to proposals for developments that will increase the amount of impervious surface areas and that need to provide an onsite detention (OSD) system and/or rainwater reuse tank.

2.3.10 Part J: Building Sustainability

This part provides advice and controls to achieve energy efficient and environmentally sustainable buildings. It applies to any application for new development or significant alterations and additions. These controls ensure best practice sustainability measures are included in the design and construction of buildings.

2.3.11 Part K: Development near Lane Cove Tunnel Ventilation Stacks

This part applies to any new development within an 800m radius of the ventilation stacks in Sirius Road, Lane Cove West and Marsden Street, Artarmon.

2.3.12 Part L: Place Based Plans

This part provides specific guidelines and controls for development in the Chatswood Central Business District (CBD) and the following local centres:

- Artarmon
- Castlecrag
- North Willoughby
- High Street
- Naremburn
- Northbridge
- Penshurst Street
- Willoughby South.

This part also provides guidelines and controls for specific sites including in the Chatswood CBD and sites in or near local centres. The performance criteria and controls for residential and commercial development in parts B and D of this plan also apply to these place based plans.

If there is inconsistency with the controls in Parts B and D, the controls under Part L prevail.

3 Existing use rights

The existing use provisions under Division 4.11 of the *Environmental Planning and Assessment Act 1979* and Part 5 of the *Environmental Planning Regulations 2000* apply to any development that is not a permissible land use within a particular zone. This includes dwellings that are not permissible in the R3 Medium Density Residential or R4 High Density Residential Zones.

An application for change of use, alterations or extensions, or rebuilding of a non-conforming existing use, must address the standards for that type of development that would otherwise be permissible in a zone. The application will be assessed against the provisions in this plan for that type of development.

The onus of proof is on the owner of the subject property to establish that 'existing use rights' apply in line with Clause 4.65 of the Act and Section 39 of the Regulations.

4 The development application process

The forms, checklists and other details needed as part of a development application and modification application are at willoughby.nsw.gov.au.

Willoughby City Council offers applicants a pre-lodgement meeting for development applications. After the meeting we provide a written response.

Applicants can contact our Customer Services division for an appointment and the current fee for the pre-lodgement meeting.

The legal framework to lodge, assess and determine applications for development approval is in *Environmental Planning and Assessment Act 1979* (part 4, development assessment and consent). The Act is available at legislation.nsw.gov.au.

5 Notifications of development

Depending on the type and scale of development, a proposal may be notified and advertised before Willoughby City Council assesses and determines the application.

Our community participation process and practices in the planning process are set out in Willoughby Community Participation Plan. Part D of the plan details the development application notification process.

Willoughby Community Participation Plan is at willoughby.nsw.gov.au.

6 Related plans, policies and guidelines

6.1 Willoughby Local Environmental Plan

Willoughby LEP is the statutory planning instrument for land and development in the Willoughby local government area.

It provides the objectives and land use tables for residential, business, industrial, special purpose, and environmental protection zones. Each category identifies land uses that are permitted without consent, permitted with consent, and prohibited. It includes additional permitted uses under Schedule 1 for certain sites.

It also provides the principal development standards, including height of buildings, landscaped area and floor space ratio (FSR) controls.

This plan, Willoughby DCP, provides guidelines and controls that support Willoughby LEP.

6.2 Willoughby City Local Strategic Planning Statement

This is a 20-year vision for land use in the Willoughby local government area. It gives effect to the Greater Sydney Commission's North District Plan and informs this plan and Willoughby LEP.

It provides information on:

- the existing and future character of Willoughby

- future housing, jobs, and services
- future infrastructure requirements such as community facilities and transport initiatives
- protection of the environmental and heritage values, areas, and items.

The Willoughby City Local Strategic Planning Statement is at willoughby.nsw.gov.au.

6.3 State environmental planning policies

The NSW Government is responsible for many environmental planning policies (SEPPs). Some of these policies have inconsistent and overlapping controls.

The main polices and codes relevant to our local government area are listed below.

6.3.1 State Environmental Planning Policy (Exempt and Complying Development Codes) 2008

The provisions in this policy allow some structures and land uses to be carried out without approval, subject to certain development standards and other provisions. They also allow some business and building identification signs to be erected without approval.

The policy includes several codes that allow certain works, change of use and subdivision to be done as complying development.

These codes are relevant to our local government area:

- Housing Code (Part 3)
- Low Rise Housing Diversity Code (Part 3B)
- Housing Alterations Code (Part 4)
- General Development Code (Part 4A)
- Commercial and Industrial Alterations Code (Part 5)
- Industrial and Business Buildings Code (Part 5A)
- Container Recycling Facilities Code (Part 5B)
- Subdivisions Code (Part 6)
- Demolition Code (Part 7)
- Fire Safety Code (Part 8)

Willoughby DCP includes many of the same controls so that it is generally in line with these codes.
A development application (DA) is required if a proposed development does not satisfy all the criteria under the codes.

The Codes SEPP is at State Environmental Planning Policy (Exempt and Complying Development Codes) 2008.

6.3.2 State Environmental Planning Policy (Sustainable Buildings) 2022

Under this policy an application for residential accommodation must include a BASIX certificate, which has a list of commitments to achieve a prescribed level of sustainable residential development.

Willoughby City Council cannot approve a development application for residential development that exceeds $50,000, or a swimming pool or spa with a volume exceeding 40,000 litres, if it does not achieve the minimum BASIX score.

Detailed information on BASIX is available from the Department of Planning and Environment at planningportal.nsw.gov.au.

The BASIX SEPP is at State Environmental Planning Policy (Sustainable Buildings) 2022.

6.3.3 State Environmental Planning Policy (Planning Systems) 2021

The Planning Systems SEPP:

- identifies state or regionally significant development, state-significant infrastructure, and critical state-significant infrastructure
- provides for consideration of development delivery plans by local Aboriginal land councils in planning assessment
- allows the planning secretary to elect to be the concurrence authority for certain development that requires concurrence under nominated state environmental planning policies.

The Planning Policy SEPP is at State Environmental Planning Policy (Planning Systems) 2021 - NSW Legislation

6.3.4 State Environmental Planning Policy (Biodiversity and Conservation) 2021

The Biodiversity and Conservation SEPP includes:

- planning rules and controls for the clearing of native vegetation in NSW on land zoned for urban and environmental purposes that is not linked to a development application
- provisions seeking to protect and preserve bushland within public open space zones and reservations
- provisions to manage and improve environmental outcomes for Sydney Harbour and its tributaries

The Biodiversity and Conservation SEPP is at State Environmental Planning Policy (Biodiversity and Conservation) 2021 - NSW Legislation

6.3.5 State Environmental Planning Policy (Resilience and Hazards) 2021

The Resilience and Hazards SEPP includes provisions:

- for land use planning within the coastal zone, consistent with the *Coastal Management Act 2016*
- to manage hazardous and offensive development
- to provide a state-wide planning framework for the remediation of contaminated land and to minimise the risk of harm.

The Resilience and Hazards SEPP is at State Environmental Planning Policy (Resilience and Hazards) 2021 - NSW Legislation

6.3.6 State Environmental Planning Policy (Transport and Infrastructure) 2021

The Transport and Infrastructure SEPP includes provisions:

- for infrastructure in NSW, such as hospitals, roads, railways, emergency services, water supply and electricity delivery
- for child-care centres, schools, TAFEs and universities

The Transport and Infrastructure SEPP is at State Environmental Planning Policy (Transport and Infrastructure) 2021 - NSW Legislation

6.3.7 State Environmental Planning Policy (Industry and Employment) 2021

The Industry and Employment SEPP includes provisions:

- for advertising and signage in NSW

The Industry and Employment SEPP is at State Environmental Planning Policy (Industry and Employment) 2021 - NSW Legislation

6.3.8 State Environmental Planning Policy (Housing) 2021

The Housing SEPP includes provisions relating to:

- boarding houses
- build-to-rent housing
- seniors housing
- group homes
- retention of existing affordable rental housing
- secondary dwellings (granny flats)
- social and affordable housing
- short-term rental accommodation
- residential apartment development.

The Housing SEPP is at State Environmental Planning Policy (Housing) 2021 - NSW Legislation

6.4 Other state and local plans, policies and guidelines

These polices and guides also apply to residential development:

- Apartment Design Guide, NSW Department of Planning and Environment, July 2015

More details on these are in Part B (Residential Development) of this plan.

Other plans, policies and guidelines may apply to specific or different types of developments. These can be found at legislation.nsw.gov.au or on the NSW planning portal.

PART B

Residential Development

Part B: Residential Development

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1 Introduction

This part of Willoughby Development Control Plan includes guidelines and controls for low, medium and high density residential development. It also includes controls for the residential components of shop top housing and mixed use developments in the business zones.

Further controls apply to residential development in heritage conservation areas (Part H) and dwellings in the C4 Environmental Living Zone (Part C) in this plan.

1.1 Aim

This part aims to provide a consistent and equitable approach to residential development outcomes. It incorporates the provisions of relevant state planning polices and guidelines and current best practice provisions into local planning controls for the City of Willoughby area.

1.2 Objectives

This part:

a. provides numerical controls and performance criteria to ensure the assessment of development applications is consistent, including considerations of any variation to the adopted controls

b. encourages development that is compatible with the urban scale and character of the locality

c. ensures new development does not unduly impact on the residential amenity of adjoining and nearby properties

d. ensures a high standard of development that provides for good residential amenity in respect to solar access, privacy, views, tree retention and open space

e. ensures residential development maximises thermal comfort and minimises urban heat impacts in the interests of health and wellbeing, and local ecology

f. provides guidelines to enhance walkability through greening and improvements to the streetscape characters of individual localities

1.3 Checklist for development applications

There are three compliance checklists for different types of residential development in Attachments 1, 2 and 3 of this part. These cover:

- minor residential development, including single dwellings and dual occupancies in the R2 zone (Attachment 1)
- manor houses, attached dwellings and multi dwelling housing in the R3 zone (Attachment 2)
- residential flat buildings, the residential component of shop top housing, and mixed use developments comprising three or more storeys and four or more dwellings (Attachment 3).

A development application for residential development must include the relevant compliance checklist.

Notes:

- Additional controls for residential development in the Heritage Conservation Areas are provided in Section 3 under Part H (Heritage Items and Heritage Conservation Areas) of this DCP. These controls must be addressed in addition to the checklist (Attachment 1) for minor residential developments.
- Controls and a compliance checklist for residential development in the C4 zone are in Part C (Development in C4 Environmental Living Zone) of this plan.
- Shop top housing is permissible in the R4 zone and certain business zones.
- Mixed use developments are permissible in the MU1 Mixed Use zone.

2 Performance criteria

2.1 General performance criteria

These performance criteria apply to all developments for residential accommodation.

A proposed development will generally be seen as meeting the performance criteria for design and amenity issues if it complies with:

- this part and other parts of this control plan
- the relevant provisions of the state planning policies, guidelines
- Willoughby Local Environmental Plan (Willoughby LEP).

Any variation of the controls must be justified. It must consider the general performance criteria below and the provisions in other relevant plans, policies and guidelines.

2.1.1 Subdivision

The objective is to ensure new allotments have access to services and facilities and that subdivided land has the capacity for the proposed development on individual lots.

The subdivision of land should:

a. provide adequate services, including water supply, sewerage, electricity, gas, and telecommunication services

b. demonstrate stormwater disposal by gravity to Willoughby City Council's street drainage system or, if this is not possible, that an inter-allotment drainage easement can be established over downstream property or properties

c. ensure each new lot created for a dwelling or dual occupancy has a frontage to a public road

d. ensure each new dwelling within a 'multi dwelling housing' or 'attached dwelling' development has a frontage to a public road or a properly constructed internal private road

e. ensure all allotments are provided with a constructed driveway, including along the entire access handle of battle-axe allotments

f. demonstrate that each allotment has the capacity to be developed for the purpose of a dwelling house, dual occupancy, multi dwelling housing or attached dwelling (this may require a concept plan showing building envelopes on the proposed new lots)

2.1.2 Site area and lot dimensions

The objective is to ensure allotments have sufficient area to allow the effective siting of developments with a good relationship to adjoining development.

The site area and lot dimensions should:

a. ensure adequate provision is made for usable open space and sufficient area for landscaping, including deep soil zones that can support tree planting

b. allow convenient vehicle access and parking

c. enable erection of buildings that do not unduly overshadow adjoining properties

d. have regard to topographical constraints and retention of trees

e. enable effective onsite control of stormwater

f. reduce the instances of isolated properties being left with reduced development potential on land that permits medium and high-density residential development

2.1.3 Setbacks

The objective is to ensure the siting of buildings provide adequate separation for the amenity of residents, provide adequate space for landscaping (including deep soil zones) and solar access, and minimise overshadowing.

Setbacks should:

a. progressively increase as the height of the external wall increases to reduce bulk and overshadowing

b. reinforce the streetscape character of the locality

c. ensure new buildings and alterations and additions on corner lots provide a transition along the secondary street frontage between the proposed development and existing adjoining development

d. ensure that garages and carports, or access to underground parking, do not dominate the streetscape

2.1.4 Design

The objective is to encourage good environmental outcomes and a high standard of architectural design.

The design of buildings should:

a. provide durability, resilience and environmental sustainability over the long term

b. minimise overshadowing, overlooking and visual impacts on the streetscape and adjoining and adjacent properties

c. minimise carbon emissions with the sustainable choice of materials

d. use materials with a low Solar Reflectance Index (SRI) and incorporate shade structures to reduce urban heat island effects

e. ensure that residential buildings address the street and incorporate a visible and readily identifiable entry point

f. address both streets on corner lots using windows, landscaping, awnings, and other architectural elements to create visual interest

g. avoid open under-croft spaces, particularly when viewed from the street and other public places

h. ensure the orientation promotes, as far as possible, passive heating and cooling for thermal comfort and reduced carbon emissions due to mechanical heating and cooling

i. ensure the orientation, siting and height provides for reasonable sharing of views from surrounding properties and the public domain

j. provide articulation to break up the length of walls to reduce the bulk and visual impacts

k. respect the visual and aural privacy of adjoining properties by effective siting, layout and location of windows and balconies to avoid direct overlooking

l. maintain a reasonable level of solar access to adjoining properties by careful siting, height, and orientation of buildings

m. locate noise sensitive rooms and private open spaces away from noise sources such as busy roads and railway lines

2.1.5 Landscaping

The objective is to achieve good environmental outcomes and enhance the visual quality of the locality.

Landscaped areas should:

a. retain significant trees and be sensitive to site attributes such as land capability (soil type and slope), microclimate (especially access to sunlight), views and natural features

b. include deep soil zones located primarily along the street frontage, and side and rear boundaries of individual lots - these are areas of soil unimpeded by buildings or structures above or below ground

c. retain and plant trees with wide canopies within the deep soil zones to reduce the impacts of urban heat island effects and support local ecology

d. provide greening and enhancement to the street frontage to encourage the benefits of local walkability

e. consider the location and scale of buildings in the selection of species

f. be designed to minimise the impact of overlooking, maintain privacy between dwellings and minimise the dominance of buildings from adjoining properties

g. predominantly use species which are native/endemic to the locality, and ensure the vegetation types decrease surface runoff, reduce maintenance, and minimise water use

h. provide irrigation using a non-potable water supply and maximise absorption for onsite infiltration of stormwater

i. relate well to the indoor living areas and contribute to useable outdoor recreation space

j. contribute to the solar efficiency of buildings by selecting and positioning trees for shade in summer and solar access in winter

k. consider maintaining significant views from adjoining properties and the public domain

l. consider potential bushfire hazard in the selection of species and reduce any potential for soil erosion or weed establishment

m. be located to protect solar access to roof mounted solar energy systems on adjoining buildings

n. be provided on walls and roofs of larger developments at various levels of the building

2.1.6 Private open space

The objective is to provide usable and accessible private open space.

Private open space for dwellings, including secondary dwellings, should:

a. be relatively flat and have sufficient area to provide for the reasonable recreational needs of residents

b. be located to integrate with living areas, achieve privacy from the public domain and receive adequate sunlight

c. be located or screened to ensure visual and aural privacy

Notes:

- If it is necessary to locate private open space at the front of the property due to site constraints, the relationship to the streetscape should be considered in terms of planting, fencing, privacy, noise, and security.
- Private open space must not include driveways, turning areas or car spaces.

2.1.7 Private recreation facilities

The objective is to integrate private recreational facilities with the natural environment, topography of the land, and limit the impact on adjoining properties.

Private recreation facilities such as tennis courts and swimming pools should:

a. be constructed to limit the amount of cut and fill

b. be located to maintain the amenity of neighbouring properties in terms of privacy, glare or light spill from external lighting, noise and visual impacts

c. ensure adjoining properties, including reserves, are not impacted by stormwater drainage

2.1.8 Privacy

The objective is to protect the visual and acoustic privacy of residents. Developments should maintain a reasonable level of aural and visual privacy for both residents of the development and neighbours by:

a. providing effective siting, layout and location of windows, balconies, and private open space
b. avoiding elevated terraces or decks that result in direct overlooking
c. constructing privacy screens, high windowsills or translucent glazing
d. increasing building setbacks

Note:

- Greater emphasis should be placed on maintaining privacy to the living areas and private open spaces of dwellings.

2.1.9 Solar access

The objective is to protect residential amenity and maximise energy efficiency of buildings.

Developments should maintain and provide a reasonable level of solar access to both residents of the development and adjoining properties by:

a. avoiding overshadowing to living areas and private open spaces
b. planting deciduous trees in appropriate locations to maximise winter sun
c. exploring alternative design options

Note:

- Shadow diagrams may need to be submitted to show that an acceptable level of solar access can be achieved for both residents of the development and adjoining properties.

2.1.10 Service facilities and structures

The objective is to protect the residential amenity and integrate the provision of services and facilities with the design of the development.

Service facilities should be located and designed to:

a. ensure safe and convenient access to garbage and clothes drying areas by residents
b. ensure garbage areas do not have an adverse effect on the amenity of adjoining neighbours
c. contain garbage bins in an enclosed area and in the basement area for larger developments where possible
d. ensure clothes drying areas do not have an adverse visual impact on the amenity of adjoining properties
e. visually integrate within the development and not impact on the streetscape
f. ensure satellite dishes and similar structures are out of sight from the public domain and not visually intrusive on adjoining neighbours
g. ensure air conditioning units and any other noise generating plant and equipment are designed to minimise noise impacts and meet relevant environmental standards

2.1.11 Urban heat

The objective is to reduce temperatures and create a resilient framework to mitigate the extreme impacts of urban heat by including appropriate measures to improve the health, comfort and wellbeing of residents. To reduce the impacts of urban heat island effects the development should:

a. use external materials, façade elements and glazing to limit solar reflectivity (this may include retractable shade structures, 'green' walls, and roofs, and/or rooftop solar panels)

b. apply best practice water sensitive urban design (WSUD) principals

c. reduce the extent of hard surfaces

d. irrigate landscaped areas using non-potable water

e. ensure landscaped areas include large trees with wide canopies

2.1.12 View sharing

The objective is to limit the extent and impact on existing views. Developments should, as far as reasonably possible:

a. maintain existing views from adjoining and neighbouring properties

b. have particular regard to water views which are more highly regarded than land views

c. prioritise consideration of potential view loss from living areas

d. consider alternative design options to maintain significant and iconic views

2.1.13 Natural heritage

The objective is to preserve and protect significant trees/vegetation and other natural features such as rock outcrops.

Willoughby City Council has a natural heritage register which identifies items of natural heritage, including items on private properties.

A proponent should access the register to determine if the subject property has significant trees or other natural features that should be preserved and protected from any proposed development.

The register is at willoughby.nsw.gov.au/Environment/Bushland-and-Wildlife/Natural-Heritage-Register.

2.1.14 Storage areas and plant rooms

The objective is to reduce excessive areas nominated as storage area and/or plant rooms, which could be subsequently used for habitable purposes. The additional floor area often contributes to excessive excavation and additional bulk.

a. To avoid excessive excavation and additional bulk, the combined below ground floor storage areas and plant/mechanical service rooms are to have a maximum gross floor area of 5% of the total allowable floor area for a dwelling house.

3 Minor developments

Minor residential developments include construction of and alterations and additions to:

a. single dwellings

b. secondary dwellings

c. boarding houses not exceeding 300m^2 and 12 persons

d. attached and detached dual occupancies

Notes:

- A boarding house is required to be maintained and operated in a single entity; strata subdivision is not permitted.
- Depending on the size and scale, and any likely impacts, an authorised officer may determine that a proposal is seen as 'major' development.

3.1 Controls for development in the R2 zone

This plan aims to provide a consistent approach in assessing development applications in the R2 zone. Unless otherwise stated, this plan adopts the relevant controls from State Environmental Planning Policy (Exempt and Complying Development Codes) 2008 (Codes SEPP). These SEPP controls include:

- Housing Code (Part 3)
- Low Rise Housing Diversity Code on certain dual occupancies (Part 3B)
- Housing Alterations Code (Part 4).

Some of the notable exceptions from the Code SEPP controls for development in the R2 zone include the floor space ratio (FSR), landscaped area, number of car parking spaces and certain additional controls for Heritage Conservation Areas.

Notes:

- The required number of car parking spaces are provided in Part F (Transport and Car Parking Management) of this plan.
- The additional Heritage controls are provided in Part H (Heritage Items and Heritage Conservation areas) of this plan.
- Site constraints may preclude a development achieving full compliance and/or the most desirable amenity outcomes. Any variation of the numerical controls must be justified, considering the performance criteria in Section 2.1 of this part and other relevant plans, policies and guidelines.
- Any development for attached or detached dual occupancies must satisfy the objectives and design criteria in the Low Rise Housing Diversity Design Guide, Department of Planning Industry and Environment, July 2020. This is at planning.nsw.gov.au

3.2 Controls for landscaping

The minimum landscaped area controls are:

Site area (m^2)	Minimum landscaped area (m^2)
Under 200	0.25 x site area
200 – 400	(0.35 x site area) less 20
401 – 600	(0.6 x site area) less 120
601 – 1000	(0.525 x site area) less75
1001 - 1500	(0.6 x site area)less 150
Over 1500	0.5 x site area

Note:

- The minimum landscaped area includes all deep soil zones and soft landscaped areas (the minimum dimension to be included in the deep soil zone and soft landscaped area calculation is 1m).
- Deep soil zones are areas that are free of sub-ground structures such as basement car parking areas, detention tanks or the like.
- Soft landscaped areas are areas that comprise a minimum depth of 600mm to enable the establishment of vegetation, including shrubs and small trees.

Additional landscaping controls are:

a. To ensure the site of a dual occupancy is not overdeveloped, Council requires a minimum landscaped area of 35% distributed proportionately for each dwelling and potential future subdivision boundaries.

b. The landscaping at the street frontage should not be less than:
   - 70% of the area forward of the building line to the primary road where a site has a width, measured at the building line, of at least 18m.
   - 50% of the area forward of the building line to the primary road where a site has a width, measured at the building line, of less than 18m.

c. A minimum clearance of 3m should be maintained between proposed structures and trees, measured from the outside base of the tree trunk.

d. The planting of trees must be provided in the deep soil zones within the front setback area of the property, unless it can be demonstrated that the width of the block or the dimensions of the existing front setback area are inadequate or inappropriate for tree planting.

e. New trees along the streetscape should be semi-mature when planted and be compatible with existing street trees. Additional street trees may be required, at the applicant's expense, with new dwellings or major additions.

f. The siting and selection of new trees must take into account the location of solar energy systems roof mounted on adjoining properties as well as existing overhead cabling. New trees must maintain the existing level of solar access to solar energy systems, including their height at maturity. Similarly, the location and species of trees, including at maturity, must not affect existing electricity or telecommunication overhead cabling.

g. Vehicular Crossings must not result in the detrimental damage or removal of street trees.

3.3 Controls for ancillary structures

This plan adopts the development standards in the Housing Code (Codes SEPP, Part 3) for attached and detached balconies, decks, patios, terraces and verandahs located at the side and rear of a dwelling.

Any variation to these controls must meet these performance criteria:

a. the location and size of balconies, decks, patios, terraces, and verandahs must have regard to the visual and acoustic privacy needs of both residents and neighbours

b. screening devices must be integrated into the building design and have minimal negative effect on the amenity of adjoining residents

Note:

- Greater emphasis is placed on maintaining visual privacy to living areas and private open spaces than for non-living rooms including bedrooms, stairways, and bathrooms.

3.4 Controls for ancillary works

3.4.1 Car parking and vehicular access

Except for the number of car parking spaces, this plan adopts the other requirements for car parking and vehicle access in the Housing Code (Codes SEPP, Part 3).

Any variation to these controls must meet these performance criteria. Garages and carports should:

a. be located behind the building line if alternative locations (side or rear) are available to the site

b. not be visually dominant or have an adverse impact on the streetscape if they are required to be in front of the building

c. integrate with the overall design of the existing or proposed building in terms of height, form, materials, detailing and colour

Under Part F (Transport and Parking Management) of this plan, a new driveway crossing may be required if the proposal involves a change to the vehicle access arrangements and/or car parking spaces.

Also see Part F for more information on car parking and vehicular access.

3.4.2 Earthworks, retaining walls, drainage, protection of walls, and protection of trees

This plan adopts the requirements of the Housing Code (Codes SEPP, Part 3) for earthworks, retaining walls, drainage, protection of walls, and protection of trees.

Any variation of these controls must show that any excavation, fill or construction of retaining walls or any works in proximity to trees, due to a variation of the development controls, will not have any significant impacts on adjoining and nearby residential properties or the health of the trees retained on the site.

All earthworks and/or construction of retaining walls, associated with a development or not, must consider the provisions in Clause 6.2 of Willoughby LEP.

All drainage works, associated with a development or not, must consider the provisions in Part I (Stormwater Management) of this plan.

The removal or replacement planting of trees and vegetation associated with a development or not, must consider the provisions in Part G (Vegetation Management) of this plan.

3.4.3 Fences

This plan adopts the requirements of the Housing Code (Codes SEPP, Part 3) for the construction of fences. Any variation of these controls must demonstrate that the proposed fence:

a. is compatible with fencing in the streetscape

b. enables surveillance of the street for security from the dwelling

c. maintains any significant views from the public domain

d. integrates with the landscaped area of the site

e. is in keeping with the design and style of the dwelling

f. In areas of overland flow, construction of fences must allow for the natural flow of stormwater. This may require fencing to be an open type construction or raised above the finished ground level to allow the free flow of stormwater. Construction of any such fencing must remain safe during floods and not obstruct moving debris.

3.4.4 Swimming pools

This plan adopts the requirements of the Housing Code (Codes SEPP, Part 3) for the construction or installation of a swimming pool and/or spa.

Any variation of these controls must show that the swimming pool, due to the variation of the development control, will not have any significant impacts on adjoining and nearby residential properties.

4 Major developments

Major development is where the development consists of:

a. the erection of a new building

b. substantial redevelopment, refurbishment, or conversion of an existing building

This includes for boarding houses that exceed 300m^2 or 12 persons, manor houses, multi dwelling housing, attached dwellings, residential flat buildings, seniors housing, ‘build-to-rent’ housing, and the residential component of shop top housing and mixed use developments.

‘Multi dwelling housing’ is often referred to as townhouses, and ‘attached dwellings’ are often described as terraces or semi-detached dwellings.

Apartments in a ‘shop top housing’ development must be above the ground floor level comprising retail or business premises. ‘Mixed use’ developments comprise two or more different land uses, one of which can include residential apartments.

Notes:

- Redevelopment is generally seen as ‘substantial’ if the proposal will result in more than a 10% increase in ‘gross floor area’.
- Shop top housing is permissible in the R4 zone. Further controls for the commercial component for this type of development are in Part D (Commercial Development) of this plan.

4.1 Strategies and plans

Willoughby City Council has prepared several studies and strategies to set the direction for commercial and residential development in key locations in the local area.

Part L of this plan sets out specific performance criteria and controls for local centres. These include performance criteria and controls for residential flat buildings and the residential components of shop top housing and mixed used developments.

A brief description of the strategies adopted to inform these place-based plans is below.

4.1.1 Willoughby Local Centres Strategy 2036

The strategy aims to revitalise the local economy and provides the framework for future planning controls to achieve thriving, attractive and distinctive local centres.

Those centres are:

a. Artarmon

The Artarmon local centre is on Hampden Road and Wilkes Plaza, adjacent to the Artarmon railway station. The place based plan includes development opportunities for residential accommodation in shop top housing developments. Performance criteria and controls are included to maintain the fine grain shopfronts and retain the heritage values of the Artarmon Heritage Conservation Area.

b. Castlecrag

The Castlecrag local centre is on Edinburgh Road between Eastern Valley Way and Rutland Avenue/The Postern. The place based plan includes development opportunities for residential accommodation in shop top housing developments, and residential flat buildings at 95–103 Edinburgh Road. Any future development must respect the heritage values of the Griffin Heritage Conservation Area.

c. North Willoughby

The North Willoughby local centre is around the intersection of Penshurst Street and Victoria Avenue, extending south to Patrick Street. The place based plan includes development opportunities for residential accommodation in shop top housing developments.

d. High Street

The High Street local centre is on the eastern side of High Street, bounded by McClelland Street and Glover Street. The place based plan includes development opportunities for residential accommodation in shop top housing developments. Performance criteria and controls are included for upper level setbacks at the rear to protect the residential amenity of adjoining dwellings.

e. Naremburn

The Naremburn local centre is on the western side of Willoughby Road, bounded by Quiamong Street, Bongalong Street and Glenmore Street. The centre includes two heritage items. The place based plan includes development opportunities for residential accommodation in shop top housing developments. However, any future development must maintain the fine grain shopfronts, and respect the heritage values of the heritage items at 284/284A and 272–276 Willoughby Road.

f. Northbridge

The commercial precinct of Northbridge is along the major arterial roads of Sailors Bay Road, Eastern Valley Way and Strathallen Avenue. Northbridge Plaza is on the north-eastern corner of Eastern Valley Way and Sailors Bay Road. The place based plan includes development opportunities for residential accommodation within shop top housing developments, and residential flat buildings on the southern side of Sailors Bay Road and at the rear of the Northbridge Plaza.

g. Penshurst Street

The Penshurst Street commercial centre is located along, and north of Mowbray Road. There is an eclectic mix of business premises, shop top housing, and a small retail shopping strip on the eastern side of Penshurst Street near Oakville Road. The place based plan includes development opportunities for residential accommodation within shop top housing developments. Performance criteria and controls are included for upper level setbacks at the rear to protect the residential amenity of adjoining dwellings.

h. Willoughby South

The commercial area of the Willoughby South local centre predominantly fronts Willoughby Road with additional commercial areas branching off Frenchs Road. The place based plan includes development opportunities for residential accommodation within shop top housing developments, and residential flat buildings.

See the full strategy for more information and the vision for each local centre at willoughby.nsw.gov.au.

4.1.2 Chatswood CBD Planning and Urban Design Strategy to 2036

The strategy guides future private and public development over the next 20 years.

Chatswood CBD includes a large mix of land uses. These include commercial and office buildings, civic buildings, library and performance facilities, major retail complexes, banking and other services, cafes, restaurants and private health facilities and schools.

The strategy includes guidelines to achieve the vision and objectives to create employment opportunities in the CBD and residential growth on the periphery of the core. The strategy is at willoughby.nsw.gov.au.

Part L of this plan includes performance criteria and controls that allow some additional residential growth as part of mixed use development in Chatswood CBD.

4.2 Controls for medium and high density residential development

The relevant controls of the plans, policies and guidelines below apply to medium and high density residential development. This ensures a consistent approach when assessing development applications.

4.2.1 State Environmental Planning Policy (Exempt and Complying Development Codes) 2008

The statutory controls for height of building, floor space ratio and minimum lot size in Willoughby LEP apply to manor houses, attached dwellings and multi dwelling housing.

This plan also adopts many of the controls under Part 3B of the Low Rise Housing Diversity Code (Codes SEPP) for these types of development in the R3 zone. A notable exception is the landscaped area controls.

Any development for manor houses, attached dwellings and multi dwelling housing should satisfy the objectives and design criteria under Low Rise Housing Diversity Design Guide, Department of Planning Industry and Environment, July 2020. The guide is at State Environmental Planning Policy (Exempt and Complying Development Codes) 2008.

Notes:

- The term ‘manor houses’ is defined in Section 1.5 of the Codes SEPP.
- The terms ‘attached dwellings’ and ‘multi dwelling housing’ are defined in Willoughby LEP.
- If there is any inconsistency between the Low Rise Housing Diversity Design Guide and the performance criteria and controls in this part or any other parts of this plan, the provisions in this plan prevail.

4.2.2 State Environmental Planning Policy (Housing) 2021

The Housing SEPP provides design principles and planning controls that set aside any local planning controls that would prevent the development of housing for seniors or people with a disability that meets the development criteria and standards of this policy.

Note:

- The provisions and controls of the Housing SEPP apply to any proposed seniors housing development in the R2 and C4 zones.

The Housing SEPP is at State Environmental Planning Policy (Housing) 2021 - NSW Legislation.

4.2.3 State Environmental Planning Policy No 65 – Design Quality of Residential Apartment Development

The Apartment Design SEPP applies to residential apartments if the building is 3 or more storeys and contains 4 or more dwellings.

The Apartment Design SEPP is at legislation.nsw.gov.au.

4.2.4 Apartment Design Guide, NSW Department of Planning and Environment, July 2015

The Apartment Design Guide provides design criteria and general guidance on how development proposals can achieve the design quality principles in the Apartment Design SEPP.

If there is any inconsistency between this plan and the Apartment Design Guide, the requirements, standards and controls in the guide prevail.

The guide is at planning.nsw.gov.au/apartmentdesignguide.

Notes:

- Clause 50 of *Environmental Planning and Assessment Regulations 2000* requires an application that relates to a development for residential apartments to include a 'design verification statement' prepared by a qualified designer.
- The statement must show how the proposal achieves the design quality principles under SEPP 65 and how it addresses the objectives in Parts 3 and 4 of the Apartment Design Guide.
- The Apartment Design Guide applies to the residential component of a shop top housing or mixed use development in all zones where such land uses are permissible.

4.3 Specific controls for manor houses, attached dwellings, multi dwelling housing and residential flat buildings

Note:

- Where relevant, the controls below apply to other major residential developments such as boarding houses that exceed 300m^2 or 12 persons, seniors housing, the residential component of shop top housing and mixed use developments, and build-to-rent housing.

4.3.1 Site area and lot dimensions

a. Lot size:

Under Clause 6.10 of Willoughby LEP the minimum lot size for manor houses, attached dwellings, multi dwelling housing and residential flat buildings in the R3 and R4 zones is 1100m^2.

Note:

- Clause 6.10 of Willoughby LEP also includes minimum lots areas for site specific locations.

b. Amalgamation

Allotments should be amalgamated to achieve the minimum lot size. Willoughby City Council may consider a variation of the minimum site area if:

- the development is on an existing allotment that cannot be consolidated with another lot
- remaining allotments in the adjoining area are not left isolated by the proposal
- the performance criteria and other development controls under this part of Willoughby DCP are satisfied
- the neighbouring landowners are unwilling to be party to the consolidation

Notes:

- The proponent must provide written evidence that adjoining property owners were consulted and all reasonable attempts made to achieve consolidation of lots, including all reasonable offers made. This must be supported by an independent valuation of the property.
- Any variation to the minimum lot size will require a written submission under Clause 4.6 of Willoughby LEP to justify contravention of the development standard.

4.3.2 Street frontage

The minimum street frontage is 27m. This is to ensure vehicles can enter and leave a site in a forward direction, and adequate landscaped areas are provided along the streetscape.

4.3.3 Adaptable housing, access, and mobility

Adaptable housing is governed by *Australian Standard AS 4299–1995 Adaptable Housing.* Adaptable housing must be provided to accommodate people with mobility impairment, including access and facilities for people in a wheelchair.

A minimum number of adaptable dwellings must be provided for all manor houses, attached dwellings, multi dwelling housing, residential flat buildings, and the residential components of shop top housing and mixed use developments. The number of dwellings must be rounded up if there is 0.5 or more of a dwelling. The minimum numbers are:

a. 10% of single storey dwellings
b. 25% of two storey buildings
c. 33% of three storey buildings
d. 50% of residential flat buildings greater than 3 storeys
e. 50% for mixed use and shop top housing developments if lift access is provided

Under AS 4299, a dwelling is classified as Class A, B or C, depending on the number of features specified. AS 4299 designates features as ‘essential’, ‘first priority desirable’ or ‘desirable’.

Class A includes all essential and desirable features. Class B has all essential and at least 50% of desirable features, including those designated as ‘first priority’. Class C has all essential features incorporated in the design.

Willoughby City Council requires adaptable dwellings to be designed and constructed to a Class C standard; however, Class A and B are encouraged.

The architectural plans must identify which apartments/dwellings are designed as adaptable.

The communal areas of a proposed development must provide a 'continuous accessible path of travel' to assist people with disability to access facilities independently, equitably and with dignity. The development should include ramps and toilets for people who use a wheelchair; appropriate lighting and colour contrast for people with a visual impairment; tactile surface indicators for people who are blind; and any other appropriate measures to assist people with a disability.

Details are required to be submitted with the development application to demonstrate the proposal will comply with the Disability (Access to Premises – Building) Standards 2010 under the *Disability Discrimination Act 1992*.

4.3.4 Energy efficiency

Willoughby City Council encourages applying best practice principles in the design and construction of buildings to create energy efficient and environmentally sustainable buildings.

All major developments are required to comply with the relevant provisions of Part J (Building Sustainability) of this plan.

4.3.5 Bicycle and car parking

Willoughby City Council encourages alternative and sustainable transport strategies to reduce the reliance on motor vehicles as the principal mode of transport. All developments with 20 dwellings/apartments or more must submit a travel demand management plan.

More information on traffic and parking matters is in Part F (Transport and Parking Management) of this plan.

4.3.6 Water management and conservation

All major developments must provide onsite detention systems to capture and detain stormwater runoff for recycling (including irrigation systems for landscaped areas), and to mitigate the impacts of flooding.

The development application must include details of stormwater management. See Part I (Stormwater Management) of this plan for more information.

4.3.7 Urban heat

Willoughby City Council encourages applying 'best practice' principles in the design and construction of buildings and other measures to reduce the impacts of urban heat island effects.

This means increasing landscaped areas and the number of trees with large canopies, reducing hard surfaces, encouraging alternative and active transport, and applying water management, including water sensitive urban design principles.

Buildings should also use external materials, façade elements and glazing to limit solar reflectivity to minimise urban heat impacts. This may include retractable shade structures, 'green' walls and roofs, and/or rooftop solar panels.

Roofs should achieve these Solar Reflectance Index (SRI) values:

a. for roof pitch <15°, 3-year SRI minimum of 64

b. for roof pitch >15°, 3-year SRI minimum of 34

c. for terrace areas, 3-year SRI minimum of 38

Notes:

- Plans must illustrate the SRI values of all roof surfaces.
- Plans do not need to include an SRI value for areas where photovoltaic (PV) panels are mounted flat on the roof.

4.3.8 Waste management

Willoughby City Council has adopted the Waste Management Technical Guide and Development Controls by North Sydney Regional Organisation of Councils for multi dwelling housing, residential flat buildings and mixed-use developments.

The technical guide provides comprehensive information to achieve best practice design and construction of waste management and recycling systems.

The development controls provide specific requirements for internal waste storage facilities, individual bin storage areas, communal bin storage areas, bin carting routes, and access for collection vehicles.

All major residential developments must comply with the technical guide and the specific controls for multi dwelling housing, residential flat buildings, and mixed-use buildings.

A copy of Waste Management Technical Guide and Development Controls is at willoughby.nsw.gov.au.

4.3.9 Safety by design

All major developments are required to include safety and security measures to prevent criminal activity. Crime prevention measures include high visibility to front entries, careful siting of shrubs and landscape elements, and lighting of pathways or hidden spaces.

The development application must include details of the design and safety measures incorporated into the design.

4.3.10 Utility structures

These controls apply to ensure the design of the development has adequate provision for utility structures such as substations and water main boosters:

a. the application must include written advice from the energy provider and Sydney Water if these utility services are required to be provided for the development

b. all structures that are visible from the street or public domain must be suitably screened by landscaping

c. substations should preferably be located below ground level or at the rear of the property if rear lane access is available

4.3.11 Undergrounding of services

All services, including overhead electricity wires, are to be located underground for major development. This includes publicly owned land immediately outside the development site.

Notes:

- In some circumstances it may not be practical or possible to provide undergrounding of services. For example, there may not be sufficient capacity within the road reserve to accommodate more services or it may be cost prohibitive to do these works as part of the proposed development.
- Any proposed variation to this control must include a written request to justify why this requirement cannot be satisfied.
- In the case of a request to vary this requirement because it is cost prohibitive, the development application must include a detailed cost report prepared by a registered quantity surveyor.

4.4 Further controls for residential flat buildings, and the residential components of shop top housing and mixed use developments

4.4.1 Site coverage

To ensure adequate open space and reasonable privacy levels, the site coverage for residential flat buildings should progressively decrease as the height of the building increases.

The site coverage for residential flat buildings should not exceed:

a. 30% of the site area for three storey building
b. 28% of the site area for four storey building
c. 26% of the site area for five storey building
d. 24% of the site area for six storey building
e. 22% of the site area for seven storey building
f. 20% of the site area for a building with eight or more storeys

4.4.2 Building height

Section 2C of the Apartment Design Guide details matters to consider when setting the height of buildings. It includes guidelines to ensure floor to ceiling heights provide adequate daylight and solar access. It also includes guidelines to allow for articulated roof planes and building services, and/or architectural roof features within the overall height of the proposed building.

Notes:

- Clause 4.3 and 4.3A of Willoughby LEP provide the maximum height of buildings and exceptions to this.
- Generally, a 9m building height equates to 3 storeys; 12m equates to 4 storeys; 15m equates to 5 storeys; and 18m equates to 6 storeys.

4.4.3 Floor space ratio

Section 2D of the Apartment Design Guide details matters to consider when setting the floor space ratio (FSR) for buildings. The floor space ratio of a building is the theoretical maximum capacity that can be achieved. It may not always be possible to achieve the maximum floor area due to other controls or site specific constraints.

Note:

- Clause 4.4 and 4.4A of Willoughby LEP provide the maximum floor space ratio and exceptions to this.

4.4.4 Setbacks

Section 2G (Street setbacks) and 2H (Side and rear setbacks) of the Apartment Design Guide detail matters to consider for the setback for buildings. To achieve the aims in the guide, these controls apply to residential flat buildings:

a. Front setbacks: 7.5m for residential flat buildings in the R3 zone and 9m in the R4 zone

b. Side and rear setbacks: 3m plus 1.2m per storey above the ground floor level

c. Corner lots: 3.5m for residential flat buildings in the R3 zone, and 3m plus 1.2m per storey above the ground floor storey in the R4 zone.

Note:

- Car parking structures are not permitted within the front setback area.
- The total required setback applies to all floors above the third floor for side and rear setbacks.
- Balconies and verandahs not more than 1m above the finished ground level are permitted within the side and rear setback areas.
- Unroofed terraces, landings, steps, or ramps not more than 1m in height are permitted within the side and rear setback areas.
- Other intrusions into the side and rear setback areas may include: sunhoods/awnings, pergolas, gutters and downpipes, hot water heaters, rainwater tanks, and the like.

4.4.5 Open space

Section 3D (Communal and public open space), 3E (Deep soil zones) and 4O (Landscape design) of the Apartment Design Guide provide guidelines for the provision of open space that can support healthy plant and tree growth and that is environmentally sustainable. These sections also have guidelines to ensure adequate communal and private open space is provided.

To achieve these outcomes, the controls below apply to residential accommodation of three or more storeys and four or more dwellings:

a. Soft landscaped area/deep soil zones and tree planting

The site should:

1. comprise at least 35% of soft landscaped areas and deep soil zones at ground level (excluding planter boxes)
2. provide deep soil zones primarily in the front setback areas and around the perimeter of the site
3. include species with a wide tree canopy which can cover hard stand areas to reduce the impacts of heat island effects
4. include new trees within deep soil zones that are semi mature when planted to be compatible with any existing predominant street trees

Notes:

- Except for deep soil zones, all other landscaped areas, which are considered to be soft landscaping, must have a minimum soil depth of 600mm.
- All communal landscaped areas must provide an irrigation system using non-potable water.
- Advanced trees are generally a species that has been grown in a plant nursery for 2 to 5 years.
- Semi mature trees are generally a species that is transplanted with an overall height greater than 4m.
- It may not always be possible to satisfy all the controls where there is limited space for deep soil zones. Examples include shop top housing and mixed use developments in non-residential areas.

b. Private open space

The minimum area and depth of balconies and ground level private open spaces apply as detailed in Section 4E of the Apartment Design Guide.

c. For developments with 15 or more apartments, communal open space should:

1. provide 25m^2 of communal open space per each dwelling where balconies are the only form of open space
2. have a minimum dimension of 5m and minimum area of 50m^2
3. include a children's play area that has a minimum area of 30m^2, a minimum dimension of 5m and be no steeper than 1 in 20

5 Controls for other residential land uses

5.1 Secondary dwellings

A secondary dwelling may be built as complying development subject to certain standards under the Housing SEPP.

A development application is required if any part of the proposal does not satisfy the criteria under the SEPP. The application will be assessed in line with the performance criteria and relevant controls in this part of Willoughby DCP.

5.2 Bed and breakfast accommodation

Bed and breakfast accommodation is complying development under Part 4A (General Development Code) of Codes SEPP, except on bush fire prone land, and subject to certain development standards.

A development application is required if a proposed bed and breakfast accommodation development does not fully satisfy all the criteria under the code. The proposal will be assessed on its individual merits, with particular regard to any likely impacts on the residential amenity of adjoining and nearby residents.

Note:

- Bed and breakfast accommodation is only permitted in the R2 zone.

5.3 Boarding houses

Under the Housing SEPP, boarding houses that provide 'affordable housing' in an 'accessible area' may be carried out with consent, subject to compliance with certain development standards.

If a proposal for this type of development is not lodged under the Housing SEPP, applications will be assessed against the standards in the Housing SEPP, and the relevant performance criteria and controls in this plan.

A plan of management must be submitted with an application for a boarding house. It must include details on the topics below to ensure the premises is maintained and operates effectively without impacting on adjoining and nearby residents:

a. Maintenance and management of common rooms such as communal kitchen, lounge rooms, and open space areas
b. Maintenance and management of waste and recycling facilities and collection of storage bins
c. Repair, maintenance, and cleaning of the building
d. Maintenance and replacement of fire alarms and any other fire safety equipment
e. Procedures to deal with any complaints
f. Safety and security measures for access arrangements by visitors to the premises
g. Monitoring and review of management actions, and any other administration procedures

Notes:

- A copy of the plan of management must be provided to all residents and displayed in a prominent location in the premises at all times.
- The plan of management must contain provisions to ensure compliance with the *Boarding Houses Act 2012* and the *Boarding Houses Regulations 2013*.
- A boarding house is required to be maintained and operated in a single entity. Subdivision is not permitted.

5.4 Build-to-rent housing

Build-to-rent housing is permissible with consent under the Housing SEPP. Any application for this type of development will be assessed in line with the performance criteria and any relevant controls under this part of Willoughby DCP and the Apartment Design Guide.

Note:

- If the identified development standards under the Housing SEPP are complied with, Willoughby City Council cannot apply more onerous standards.

5.5 Affordable Housing

Affordable housing is to be provided in accordance with the Willoughby Local Environmental Plan 2012 and Willoughby City Council *Affordable Housing Policy.*

6 Specific controls for child care centres in residential areas

Child care centres are a commercial use; however, they are permissible in residential zones. Section 7.3 of Part D (Commercial Development) includes the provisions for Centre-based child care facilities. The following additional controls apply to child care facilities in residential areas:

a. a minimum 20m frontage is required to ensure the site is compatible with surrounding residential properties, and adequate access is available for vehicles to enter and leave the site in a forward direction

b. to ensure good access is available for emergency vehicles and evacuation options/routes, child care centres should not be located in a cul-de-sacs or no through roads, except where more than one street access and egress is available to the site, and the centre has adequate parking and pickup/drop off spaces

c. unless otherwise approved, the hours of operation must not extend beyond 7am to 7pm Monday to Friday. Any application for extended hours of operation must consider likely impacts on adjoining and nearby residents

d. a landscape plan prepared by a suitably qualified person must be submitted with the application; landscaping of the site must consider to the relevant performance criteria under Section 2.1.5 (Landscaping) in Part B of this plan.

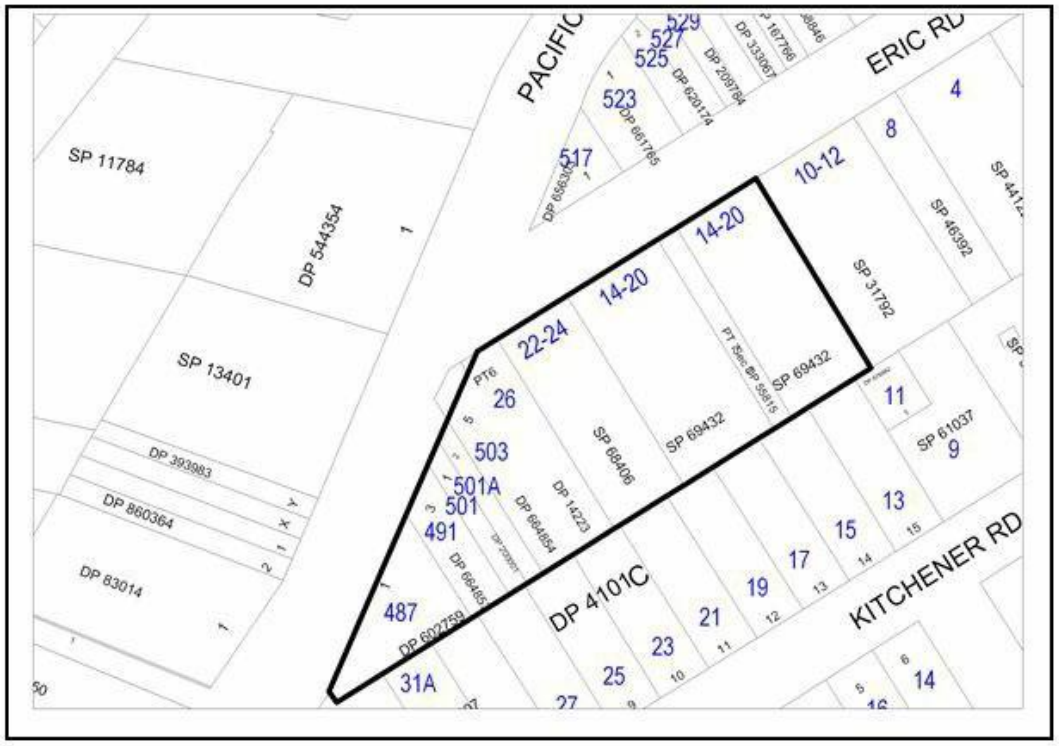
e. a site used as both a dwelling and a child care centre must provide the minimum required outdoor children's play area, and it must be separate to any outdoor areas associated with the use of the dwelling

7 Controls for specific areas

This section provides controls for any future development of specific sites and areas. These controls are further to other provisions in this part of Willoughby DCP. If there are any inconsistencies, the provisions in this section prevail.

7.1 18–26 Eric Road and 487–503 Pacific Highway, Artarmon

Map 1 18-26 Eric Road and 487-503 Pacific Highway, Artarmon



7.1.1 Rear setback

All new developments must provide a minimum rear setback of 12m. This is to provide adequate separation and landscaped area and minimise the impact on privacy and solar access to adjacent properties.

7.2 33–35 Garland Road, Naremburn

7.2.1 Pedestrian access

To achieve a 3m wide pedestrian pathway to Hallstrom Park, any new development must dedicate a 1.5m wide strip of land on the western side of 33 Garland Road and the eastern side of 35 Garland Road.

Note:

- For the purposes of calculating the floor space ratio, the total floor area can include the area of land dedicated as public open space.

7.3 Crowley Lane, Naremburn

7.3.1 Deed of Agreement

New developments bordering Crowley Lane between Nolan Crescent and Park Road must enter into a deed of agreement to purchase and consolidate half the width of thelaneway for the full width of the development site.

7.4 1–7 Hotham Street and 2 Wyvern Avenue, Chatswood

Map 2 1-7 Hotham Street and 2 Wyvern Avenue, Chatswood

7.4.1 Vehicular access

All vehicle access is to be provided from Hotham Street. This is to ensure safe egress by diverting vehicle access away from the intersection of Wyvern Avenue and the Pacific Highway.

7.4.2 Drainage/flooding

Any new development must submit a flood study to identify the existing flood extent and show the development will not adversely affect any adjoining properties. No structures other than suspended/elevated vehicular crossings are permitted over the identified flood extent.

Notes:

- Under Clause 6.10 of Willoughby LEP, the minimum site area for any future development is 3,500m^2.
- Any future development of the site must dedicate a drainage easement in favour of, and at no cost to, Willoughby City Council.

7.5 10a–14 Boundary Street, 57–63 William Street and 118–138 Archer Street, Roseville

Map 3 10a-14 Boundary Street, 57-63 William Street and 118-138 Archer Street, Roseville

7.5.1 Vehicle access

No vehicle access is allowed from Boundary Street. All vehicle access from Archer Street must be from the William Street end of the site. This is to minimise conflict with vehicles queuing at the intersections of Boundary Street and Archer Street and at Boundary Street and Pacific Highway.

7.5.2 Road widening and bicycle/pedestrian pathway

To assist with traffic movement and allow adequate space for bicycle/pedestrian pathways, all new developments must provide:

a. a 4m-wide shared public cycle/pedestrian path by dedication or public right-of-way adjacent to the railway land along the western boundary between William Street and Boundary Street

b. a 3.5m-wide strip of land along the Archer Street frontage of properties from 118–138 Archer Street, with a 6-metre radius at 138 Archer Street on the corner with Boundary Street. This is required under the Land Reservation Acquisition Map, Willoughby LEP.

7.5.3 Drainage/flooding

Any new development must submit a flood study to identify the existing flood extent and show the development will not adversely affect any adjoining properties (a clear overland flood path must be maintained).

7.5.4 Heritage item at 59 William Street

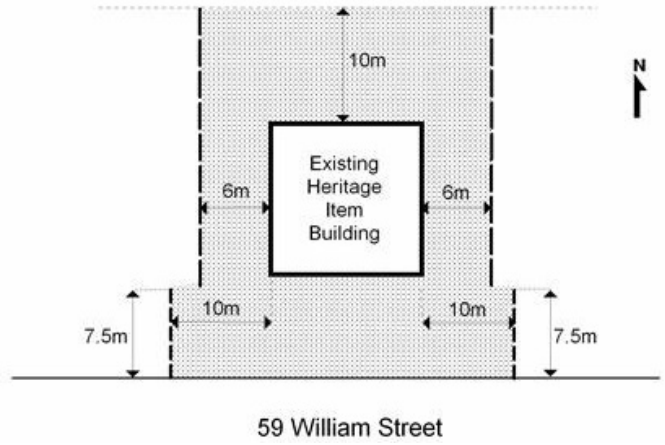
To protect the impact of any new development on the significance of the heritage item at 59 William Street, these setbacks apply:

a. 6m from the eastern and western sides of the heritage item (measured from the building wall of the item)

b. 10m from the most northern building wall of the heritage item

c. any new development to the east and west of the heritage item is to be setback from the front (William Street) boundary by at least 7.5 metres within 10 metres of the eastern and western facades of the building

Figure 1 Plan view of 59 William Street



Note:

- Under Clause 6.10 of Willoughby LEP, the minimum site area for any future development is 4,000m^2.

7.6 1–31 Walter Street and 452–460 Willoughby Road (and adjoining vacant land), Willoughby

Map 4 1–31 Walter Street and 452-460 Willoughby Road

7.6.1 General provisions

The aims and objectives of these site specific provisions are to:

a. ensure that the site is amalgamated so as to enable the substantial redevelopment of the site for high density residential development

b. ensure that the development is complementary to the scale and character of development in the area

c. encourage energy efficient design and development

d. encourage high quality, built form outcomes and achieve design excellence

e. indicate the location of vehicle access points to the site and the general location of parking areas

f. mitigate impact of traffic generation from the development on the surrounding road networks

g. control adverse environmental impacts such as wind and shadows from any development of the site

7.6.2 Specific provisions

Figure 2 Building envelope plan

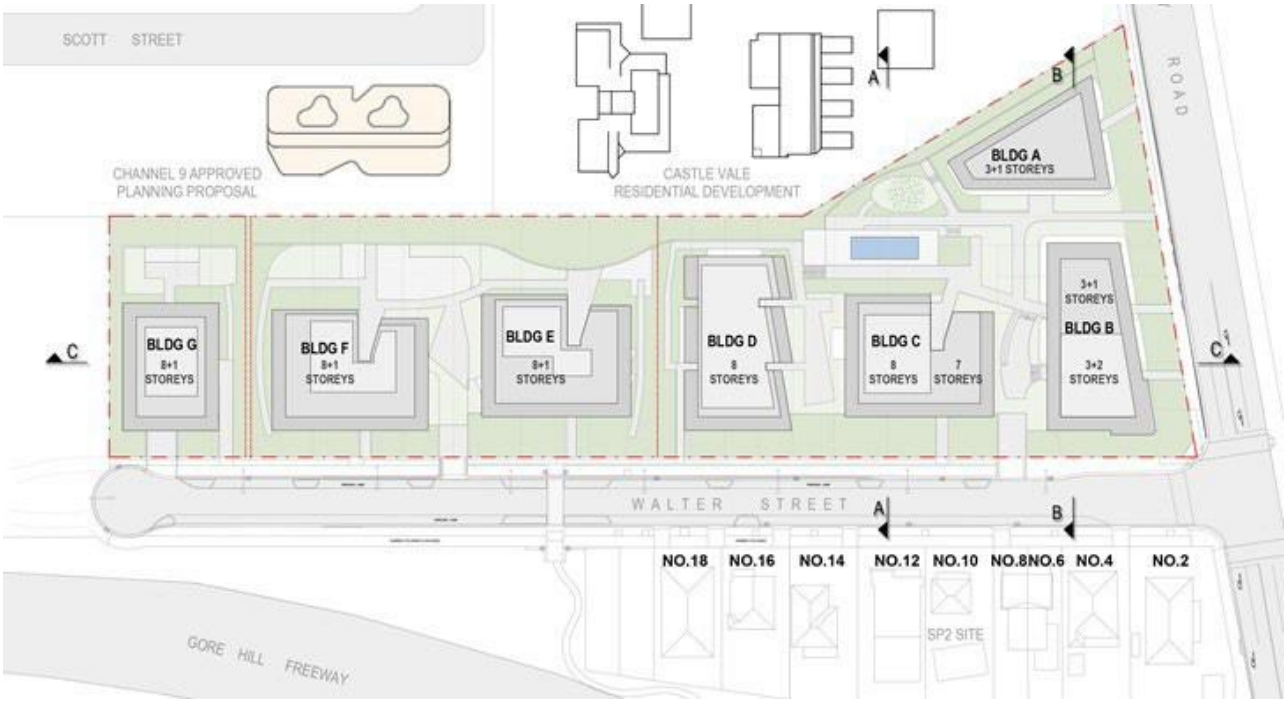


Figure 3 Indicative building envelope section A

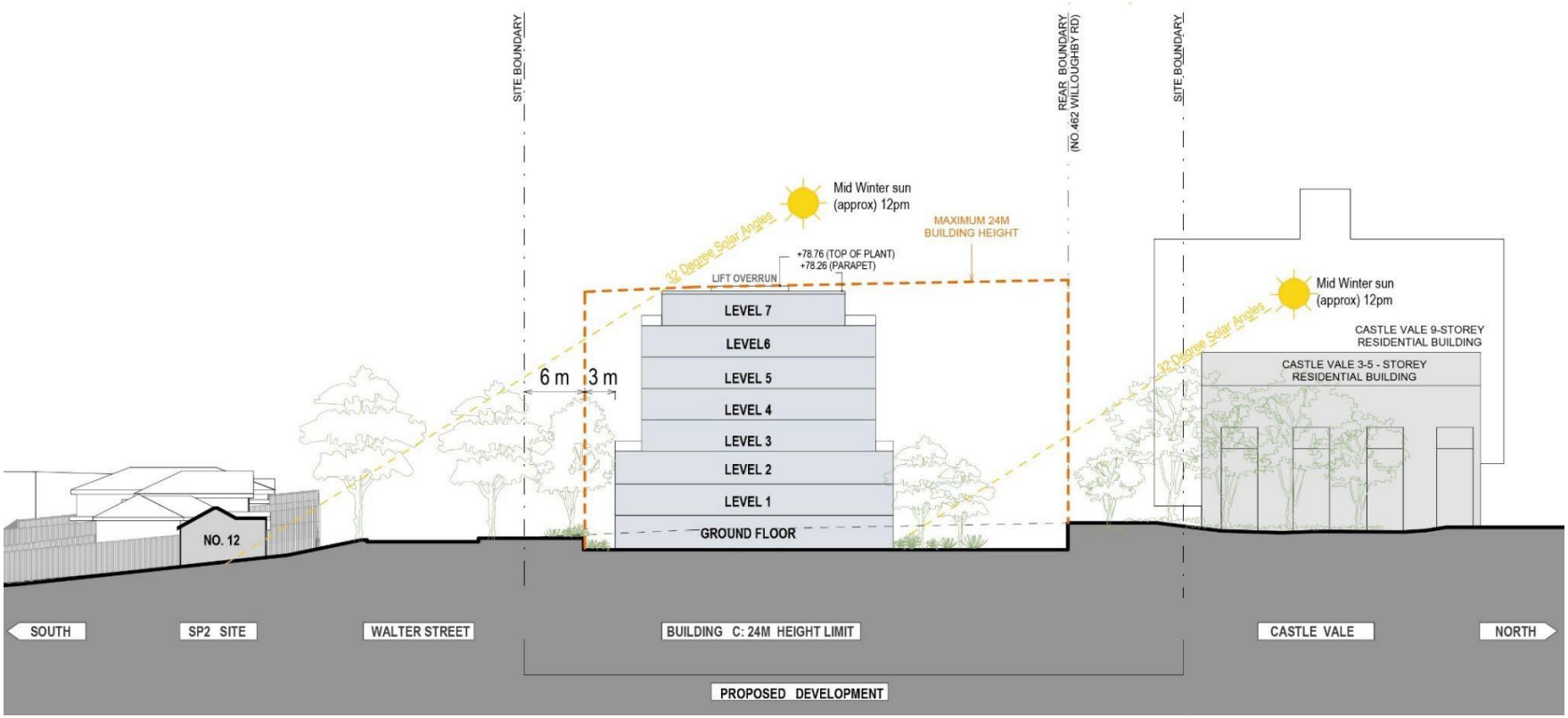


Figure 4 Indicative building envelope section B

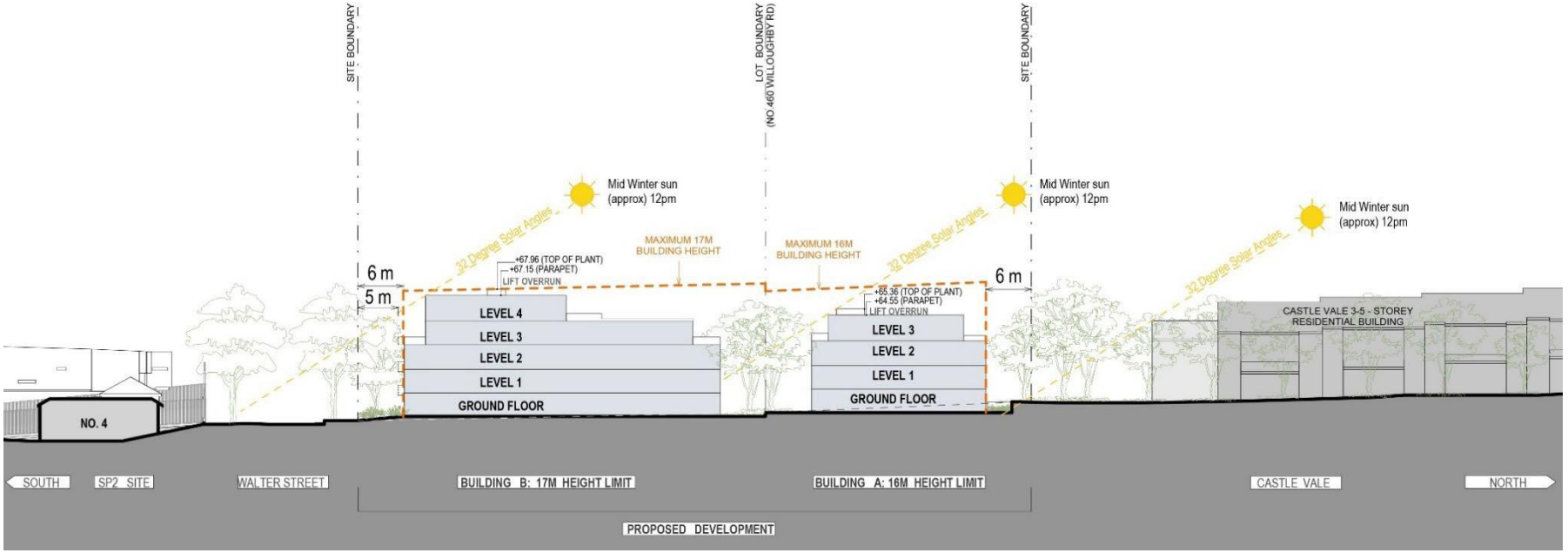
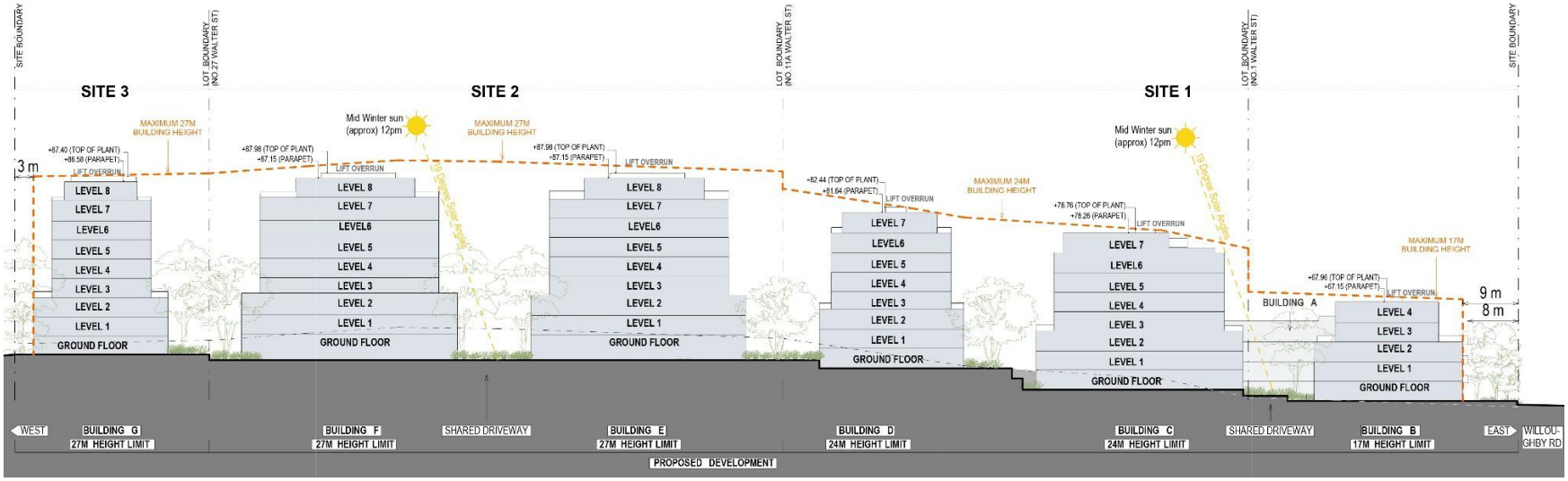


Figure 5 Indicative building envelope section C



a. Height

Objectives

The height objectives are to:

- achieve an appropriate height transition across the site responding to the existing and future desired context
- provide various height controls that maintain reasonable solar access to the private open space of nearby residential development and enhance solar access through the site

Controls

1. The height of buildings in storey are to be in line with the building envelope plan in this section.
2. The development is to provide a 3-storey podium to Walter Street with a 3m setback above to reduce impacts on the public domain and surrounding residential buildings and provide a transition in scale.

b. Design excellence

Objectives

The design excellence objectives are to:

- encourage innovative, high quality architectural design
- achieve long term durability of design and finish

Controls

1. Before granting development consent for the erection of a new building development, the Willoughby City Council must consider the quality of the architectural design and the achievement of design excellence.
2. In considering the quality of the architectural design, Council will consider feedback received from a design review panel for developments of up to 35m high.
3. Design excellence includes achieving higher building sustainability standards.
4. Any amelioration measures for environmental impacts such as architectural elements and landscaping should be integrated into the design to achieve an overall high standard of design quality.

Figure 6 Site amalgamation plan

c. Allotment size and consolidation

Objectives

The allotment size and consolidation objectives are to ensure:

- the allotments are amalgamated to ensure the site area can achieve the stated development potential, provide adequate space for landscaping, and avoid isolating any allotments subject to this section of Willoughby DCP from future development
- the site is amalgamated to enable the substantial redevelopment of the site for high density residential development
- the development is complementary to the scale and character of development in the area
- high quality, built form outcomes and design excellence is encouraged

Controls

1. New development on the site must consolidate all affected allotments.
2. The sites should be amalgamated in line with Figure 6 Site amalgamation plan.
3. A development application for the substantial redevelopment of fewer than all of the allotments subject to this section must include evidence that consolidation of all allotments has been reasonably attempted in line with NSW Land and Environment Court's planning principles.
4. Any basement car parking proposed on the site is to make provision for shared access where appropriate, with necessary easements on title in line with Section 88B of *NSW Conveyancing Act 1919*, appropriate circulation paths, and siting of breakthrough walls to allow shared/integrated basements with access from Walter Street only.

d. Setbacks and streetscape

Objectives

The setback and streetscape objectives are to ensure:

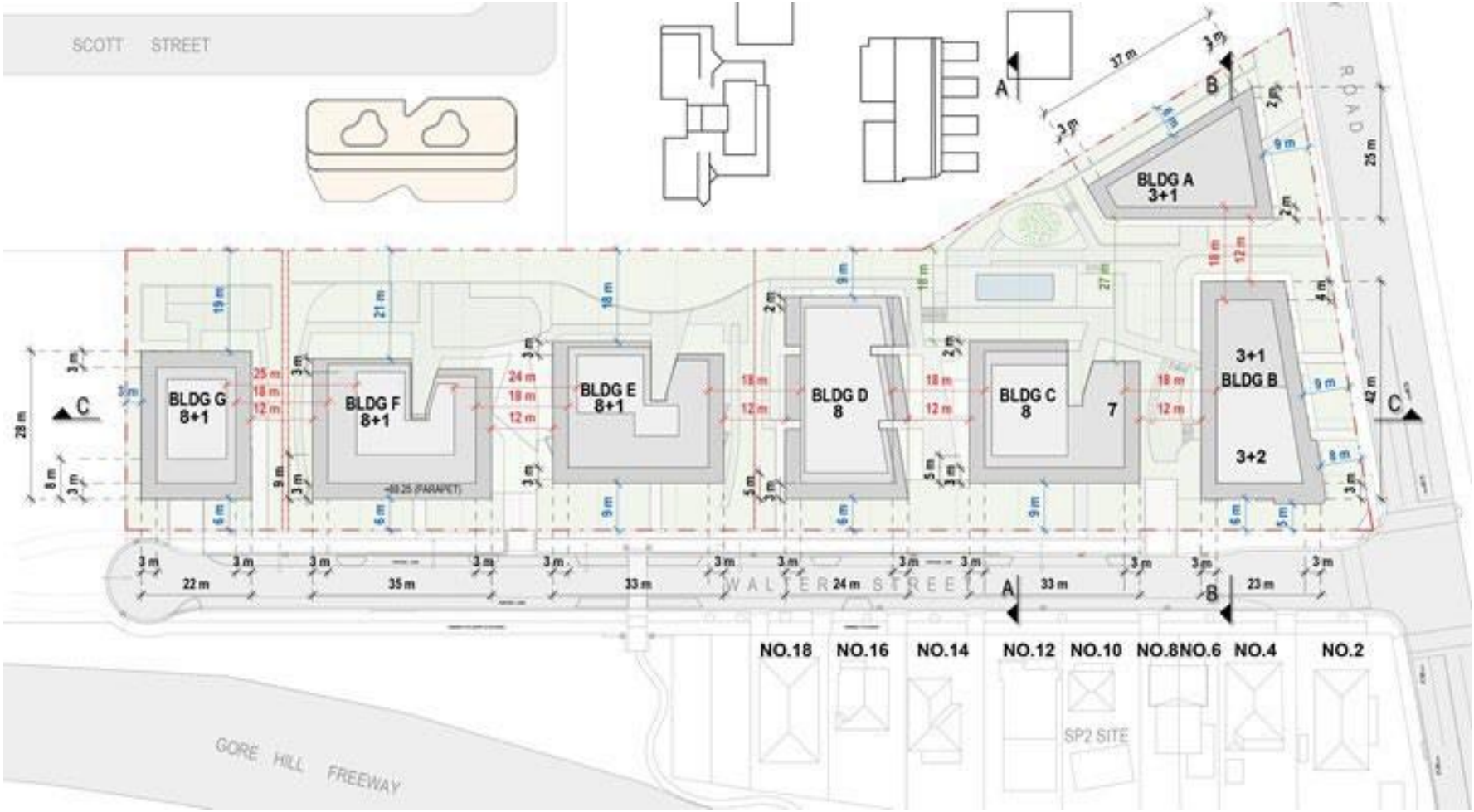
- future development appropriately recognises the streetscape and contributes to the public domain
- residential development proposed on the site is consistent with the design criteria in the Apartment Design Guide

Controls

1. There is to be a minimum 9m building setback to Willoughby Road but allowing an articulation zone of up to 2m within 10m from the corner of Walter Street and Willoughby Road.
2. There is to be a varied building setback alternating from 6m to 9m to Walter Street.
3. The below ground basement parking to Willoughby Road is to be setback a minimum of 7.5m.
4. The below ground basement parking to the northern boundary of 462 Willoughby Road is to be setback a minimum of 6m.
5. The below ground, basement parking to Walter Street setback must follow that of the building above.

6. The upper level of the Willoughby Road buildings should be setback a minimum of 3m from the level below on all sides.
7. A further 3m setback to both street frontages is required above residential level 3 to create a podium.
8. The design of the development is not to be read as a single mass from Willoughby Road or Walter Street but should be broken into a rhythm of buildings of varying width.
9. The design of the development facades is to create a visually interesting form incorporating recessed and projecting elements.
10. Separation between building forms is to be consistent with Apartment Design Guide criteria.
11. The rear setback is to be consistent with Apartment Design Guide criteria at a minimum but must on average provide for greater rear setbacks to maximise northern exposure for ground level communal open space.

Figure 7 Building envelope plan and setbacks



e. Open space and landscaping

Objectives

The open space and landscaping objectives are to ensure:

- landscaping softens and complements the development
- landscaping retains and complements the existing mature tree plantings along the northern boundary of the site
- landscaping at street level improves the amenity and appearance of the pedestrian environment and public domain

Controls

1. The trees nominated in Figure 8 Significant tree diagram and Table 1 Significant tree inventory in this part are to be retained.
2. The planting to the setback areas is to maintain clear sight lines between the entrances and the street and within planted areas.
3. Open spaces within the front setback area are to present useable and accessible open space that improves the public domain.
4. New development is to include further street tree planting. Willoughby City Council will nominate the location and species of the new planting as part of any development consent.
5. Landscaped areas are to be adequately irrigated and drained.
6. The redevelopment of the site is to include soft landscaping.
7. The maximum site coverage for each of the amalgamated sites is 30%.
8. The minimum communal open space is 25% of the site area and a minimum deep soil of 15% of the site area is to be maintained across each of the amalgamated sites.
9. Private open space is to be consistent with Apartment Design Guide criteria.
10. The minimum recreational open space is 55% of the site area across each of the amalgamated sites.
11. The minimum soft landscaping is 35% of the site area across each of the amalgamated sites.

Figure 8 Significant tree diagram

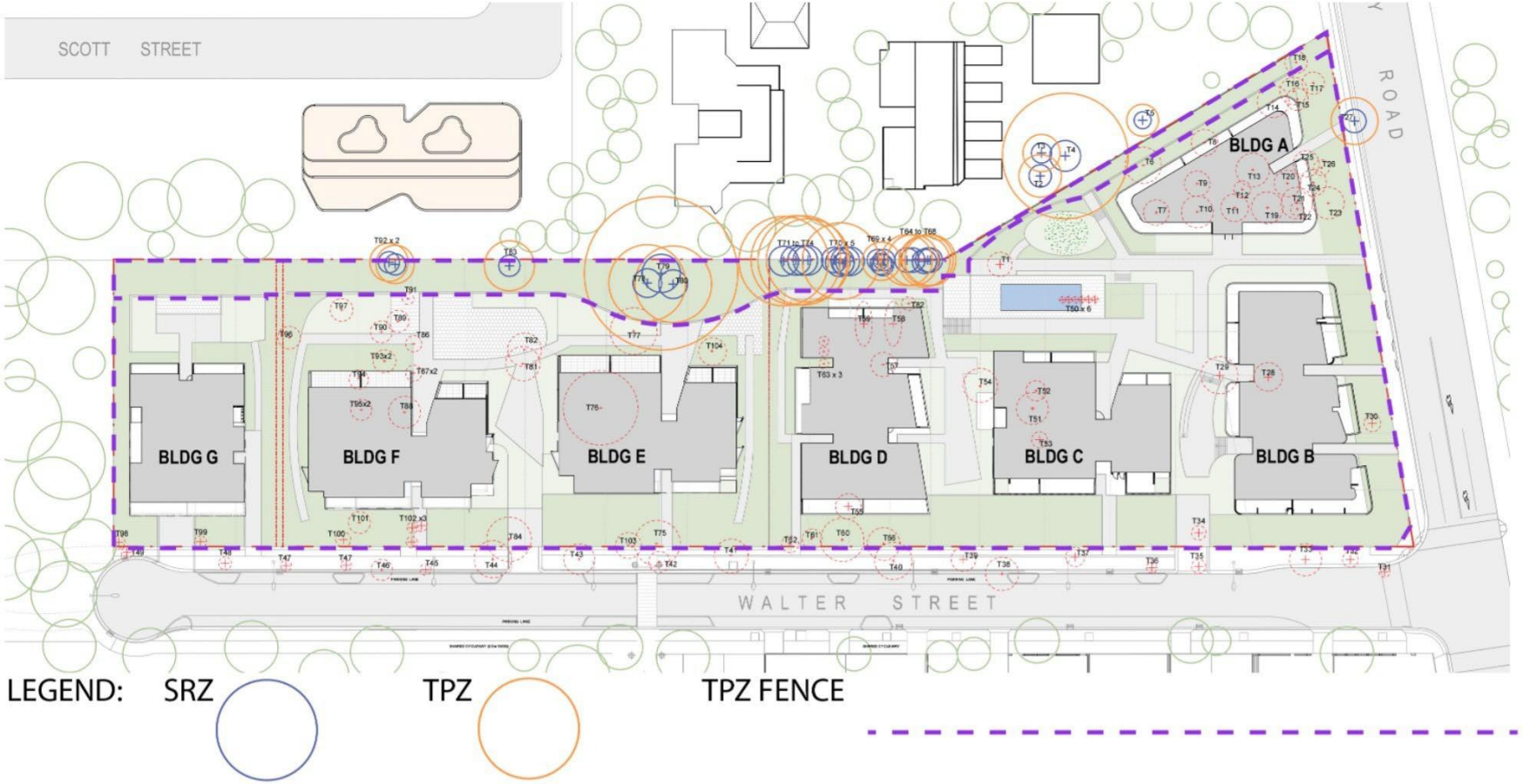


Table 1 Significant tree inventory[1,2]

Redgum tree / stand number	Structural root zone SRZ (m)	Trunk diameter DBH (mm)	Tree protection zone TPZ (m)	Proposed distance of tree protection fence/works
2	2.1	330 est	4.0	4.0
3	1.9	290	3.5	>8.0
4	2.9 (720)	980	11.8	>7.5 (basement) 6.0 (Children's play area)
5	1.6	190	3.0x2.5E/W	3.0
27	2.2	370	4.4	4.4
64	2.3	400 average	4.8	4.8
65	2.3	400 average	4.8	4.8
66	2.3	400 average	4.8	4.8
67	2.3	400 average	4.8	4.8
68	2.3	400 average	4.8	4.8
69/6	2.0	300 average	3.6	3.6
70/7	2.7	600 average	7.2	5.0 (9.7% retaining wall)
71	2.8	650	7.8	5.0 (12.2% retaining wall)
72	2.8	700	8.4	5.0 (14.5% retaining wall)
73	2.8	700	8.4	5.0 (14.5% retaining wall)
74	2.8	700	8.4	5.0 (14.5% retaining wall)
78	2.7	600	7.2	7.0
79	3.6	1200	14.4	7.0 (18.8% retaining wall)
80	2.7	600	7.2	7.0
83	2.2	390	4.7	4.5
92/9	2.0	300 average	3.6	3.6

Definition of terms from AS 4970:2009 Protection of trees on development sites:

[1] AS 4970:2009 Protection of trees on development sites

[2] TREE INVENTORY (Table 2.0, P 57, Report: Arboriculture Impact Assessment; 1-31 Walter Street & 452-462 Willoughby Road, Willoughby NSW, Redgum Horticultural 2019)

- Structural root zone SRZ defines the area required for tree stability and is required when encroachment into the TPZ is proposed. It is nominally circular with the tree trunk at its centre, expressed by the radius measured from the centre of the trunk. See AS 4970: 2009 section 3.3.5.
- Trunk diameter at breast height DBH is the nominal trunk diameter at 1.4m above ground level
- Tree protection zone TPZ is the area above and below ground to protect the tree crown and roots. The tree protection zone radius is calculated by TPZ=12×DBH, minimum 2m.

f. Access, parking and transport

Objectives

The objectives for access, parking and transport are to:

- ensure the development meets Willoughby City Council's car parking requirements for sites on major public transport corridors
- minimise the number of vehicle access points
- mitigate traffic generated from the proposed development
- ensure the design of vehicle access points minimises the impact on pedestrians and the flow of traffic
- ensure vehicle access points are unobtrusive in the streetscape but visible for motorists and approaching pedestrians

Controls

1. Proposed vehicular access is to be limited to the Walter Street frontage. No vehicle access is permitted from Willoughby Road.
2. The number of vehicle access points is to be no more than one for each amalgamated site.
3. Future development of the site must ensure measures are in place so that development will have no significant impact on the efficiency and operation of the existing surrounding road network.
4. Active transport facilities including resident and visitor bicycle parking are to be provided.
5. Electric car charge points are to be provided in any basement car park.

g. Environmental considerations

Objectives

The objectives for environmental considerations are to ensure:

- the design of the new development minimises the overshadowing impact on adjoining development
- the shape, location and height of buildings are designed to satisfy wind criteria for public safety and comfort at ground level

Controls

1. New development is to minimise further overshadowing of main private open space or living rooms of residential properties during the mid-winter period where the shadows would reduce sunlight access to less than 3 hours a day from 9am to 3pm on 21 June.
2. Any wind amelioration measures are to be integrated into the overall architectural landscape design of the site.

h. Geotechnical requirements

Objectives

The objectives for geotechnical requirements are to:

- ensure the suitability of the site for high density housing
- retain the existing topography where possible.

Controls

1. Future development is to have provisions that examine all aspects of the geotechnical environment and suitability of the site for high density housing. These provisions must be in line with the recommendations of the geotechnical assessment report prepared by JK Geotechnics dated 23 August 2019. Contact Willoughby City Council for a copy of this report.

i. Drainage easement

Objectives

The objective for draining easement is to:

- ensure appropriate drainage between adjoining properties

Controls

1. An easement is to be provided on the site in favour of the former Channel 9 site at 14 Artarmon Road to ensure satisfactory disposal of stormwater. The exact location is to be determined in consultation with and approved by Willoughby Council's development control engineer.

7.7 1A-29 Bowen Street & 6-18 Moriarty Road, Chatswood

To further guide the design of new development within the site, and to provide greater certainty regarding the future design outcome for the site, a site-specific development control plan (DCP) has been prepared to support the draft LEP amendment. Proposed DCP Controls prepared in the generally accepted format of the Willoughby DCP 2006 are outlined in the following sections.

7.7.1 General provisions

The aims and objectives of the DCP site specific provisions are to:

1. Ensure that the site is amalgamated to enable the substantial redevelopment of the site for medium to high density residential development;
2. Ensure that the development is complementary to the scale and character of development in the area;

3. Encourage energy efficient design and development;
4. Encourage high-quality built form outcomes and achieve design excellence;
5. Indicate the location of vehicle access points to the site and the general location of parking areas;
6. Mitigate impact of traffic generation from the development on surrounding road network;
7. Control adverse environmental impacts such as wind and shadows from any development of the site; and
8. Ensure the heritage listed terrace pair located at 8-10 Moriarty Road is sympathetically incorporated in the redevelopment.

7.7.2 Specific Provisions

Figure 9 Building Envelope Plan

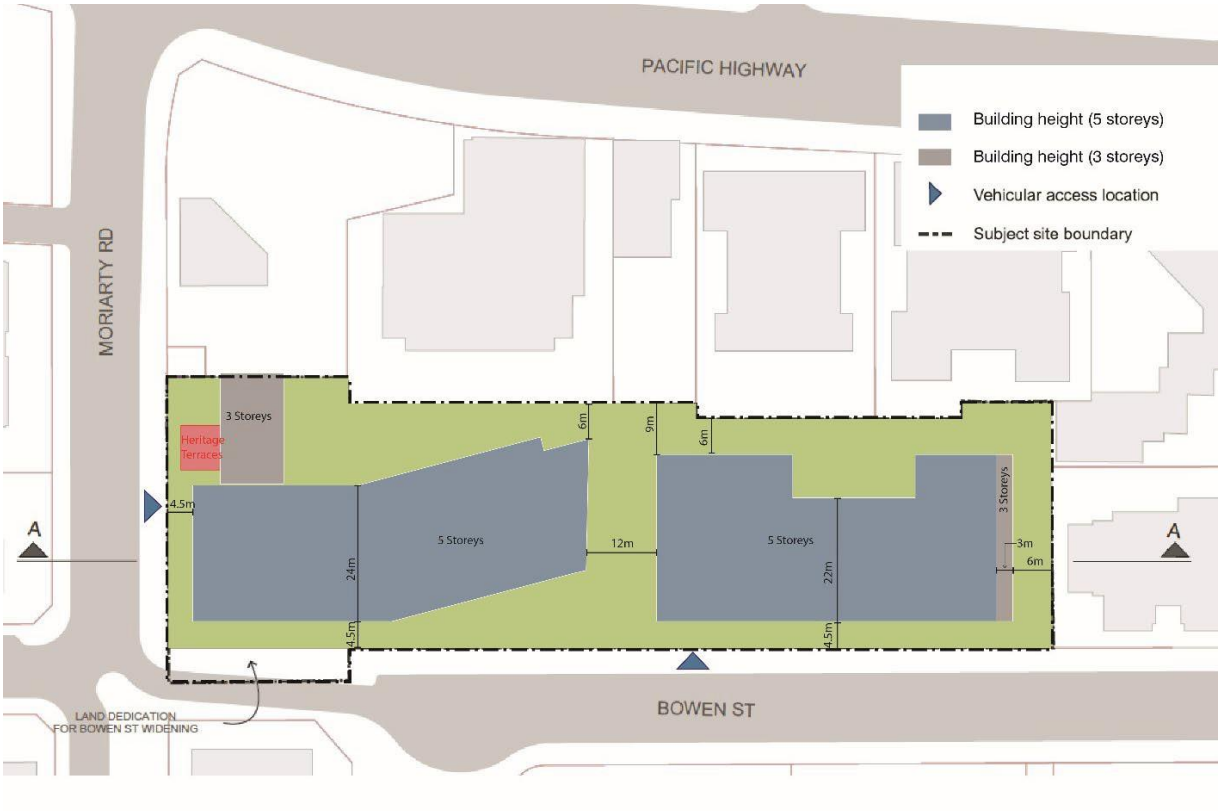
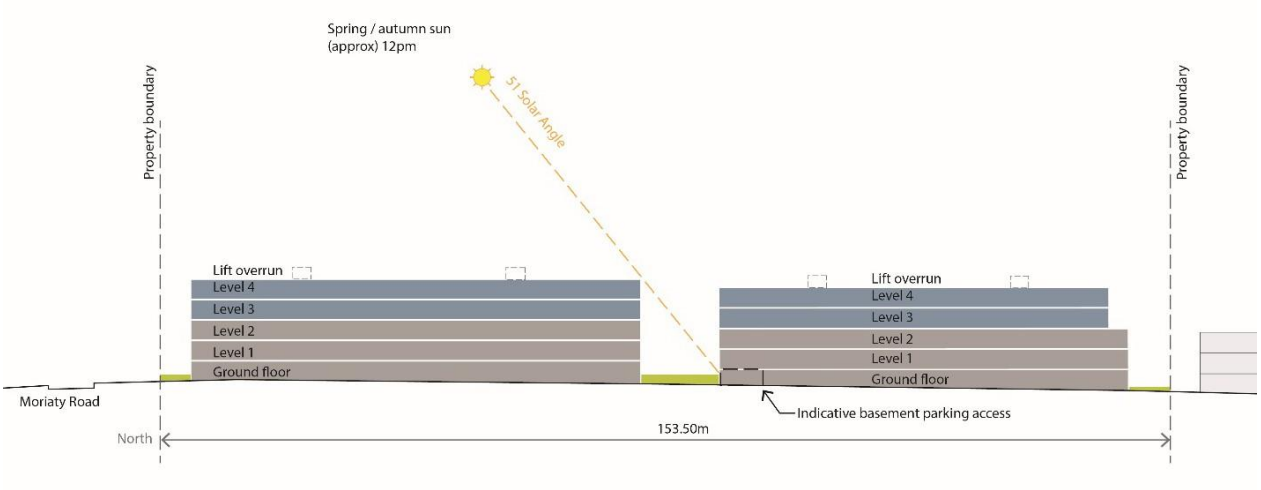


Figure 10 Indicative Building Envelope Section



a. Height

Objectives

a. To achieve an appropriate height transition across the site responding to the existing and future desired context; and

b. To provide various height controls that maintain sunlight access into nearby residential development and enhance solar access through the site.

Controls

1. The maximum height of buildings is 18 metres in accordance with the Willoughby Local Environmental Plan 2012. The height of buildings in storeys are to be in accordance with the *Figure 1 - Building Envelope Plan*.
2. The form of the development on the site is to be positioned to reduce solar impacts on the public domain and neighbouring residential properties and provide a transition in scale to The Great Northern Hotel heritage item.

b. Design Excellence

Objectives

a. To encourage innovative, high quality architectural design; and

b. To achieve long term durability of design and finish.

Controls

1. Before granting development consent for the erection of a new building development, the Council must consider the quality of the architectural design and the achievement of design excellence.
2. In considering the quality of the architectural design, the Council must consider feedback received from a Design Review Panel for developments up to 35m high.
3. Achievement of design excellence will include achievement of higher building sustainability standards.

4. Any amelioration measurers for environmental impacts such as architectural elements and landscaping are integrated into the design to achieve an overall high standard of design quality.
5. Demonstration of adaptive reuse of the Heritage Items 8-10 Moriarty Road is to be shown in accordance with Council requirements as part of any development application.

c. Setbacks and Streetscape

Objectives

a. To ensure future development recognises the existing streetscape and contributes to the public domain; and
b. To ensure residential development proposed on the site is consistent with the design criteria contained within the Apartment Design Guide.

Controls

1. Minimum building setbacks from boundaries are to be in accordance with Figure 9.
2. Design of the development is not to read as a single mass when viewed from Bowen Street.
3. The design of the development is to incorporate a visually interesting façade through the use of articulation, modulation and materials.
4. New building envelopes proposed behind the existing heritage items at 8-10 Moriarty Road must not exceed 3 storeys, and must not be positioned less than 1m from the principal building form of the heritage terrace, or not less than a 9m setback from Moriarty Road.

d. Open Space and Landscaping

Objectives

a. Retain and protect existing significant trees in the public domain;
b. Landscaping is to soften and complement the development; and
c. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment and public domain.

Controls

1. The planting to the setback areas is to maintain clear sight lines between the entrances and the street and within the planted areas.
2. Open space within front setback areas are to present useable and accessible open space that improves the public domain.
3. Additional street tree planting will be required for new development. The location and species of new planting will be nominated by Council as part of any development consent.
4. Landscaped areas are to be adequately irrigated and drained.
5. The redevelopment of the site is to provide the inclusion of soft landscaping.
6. Variation to the maximum site coverage and recreational open space requirements within Part D2 of the Willoughby DCP 2006 can be considered provided communal open space proposed

exceeds 25% of the site area, and a minimum 15% of the site is to be deep soil landscaping across the site.

7. Significant trees are to be retained and protected with appropriate building and basement setbacks in accordance with Figure 11 and Figure 12.

Figure 11 Indicative Building Envelope Section

Figure 12 Indicative Basement Plan. Significant trees in the Moriarty Road and Bowen Street streetscapes are to be retained and protected.

e. Access, Parking, and Transport

Objectives

a. The development shall meet Council's car parking requirements;

b. The number of vehicle access points to the development are minimised;

c. Traffic generated from the proposed development should be mitigated;

d. Vehicular access points are designed to minimise their impact on pedestrians and the flow of traffic; and

e. Vehicular access points should be unobtrusive in the streetscape but ensure visibility for motorists and approaching pedestrians.

Controls

1. Proposed vehicular access points are to be limited to no more than one ingress/egress per street frontage.
2. Future development of the site must ensure that measures are put in place so that development will have no significant impact on the efficiency and operation of the existing surrounding road network.
3. Active transport facilities including resident and visitor bicycle parking are to be provided.
4. Bicycle access/facilities and circulation along Bowen Street and Moriarty Road shall be encouraged, particularly in relation to providing future connectivity to the Pacific Highway, including ‘filling the gaps’ in the existing bicycle network across intersections.
5. Electric car charge points are to be provided within any basement car park.
6. Car parking rates are to be in accordance with Council requirements, or requirements for a site within a railway precinct and major public transport corridor, or to standards in the RMS Guide to Traffic Generating Development, whichever is the lesser.
7. Provide a cul-de-sac at the end of Bowen Street to facilitate vehicle turning movements through Council reservation of a strip of land at 23-29 Bowen Street as shown below in Figure 13.

Figure 13 Land Reservation Acquisition Map

f. Environmental Considerations

Objectives

a. The design of new development shall minimise the overshadowing impact on adjoining development;

b. The shape, location and height of buildings should be designed to satisfy wind criteria for public safety and comfort at ground level;

Controls

1. Any wind amelioration measures shall be integrated into the overall architectural or landscape design of a site.
2. Development is encouraged to exceed the minimum requirements of *State Environmental Planning Policy (Building Sustainability Index: BASIX) 2004*.
3. A minimum GBCA 5-star green star rating for residential flat buildings is required for any proposed development for residential flat buildings.

g. Allotment Size and Consolidation

Objectives

a. Allotments should be amalgamated to ensure that sufficient site area is available to achieve the stated development potential, adequate provision for landscaping, and to avoid isolating any allotments subject to this section of the DCP from future redevelopment; and

b. Development on any residual allotment can be satisfactorily achieved by having regard to efficient solar access design.

Controls

1. New development on the site will be required to consolidate all affected allotments.
2. A development application for the substantial redevelopment of fewer than all the allotments the subject of this section of the DCP, must be accompanied by evidence that consolidation of all allotments has been reasonably attempted.

h. Heritage

Objectives

a. To ensure the heritage listed terrace pair at 8-10 Moriarty Road is sympathetically incorporated in the redevelopment, and the heritage significance of the terrace pair is retained.

b. To mitigate impacts of the proposed development to proximate heritage items including the Great Northern Hotel at 522 Pacific Highway.

Controls

1. New building additions to the south of the existing heritage listed terrace pair at 8-10 Moriarty Road are to be generally consistent with the built form illustrated within the *Building Envelope Plan*.
2. A development application for the redevelopment of the site is to include consideration to further modulation/ articulation of new development in response to the siting of the heritage item.
3. Detailed design of proposal to specifically respond to the bulk, materiality and street wall height of the heritage item.
4. The redevelopment of the site is to retain the heritage values of the Local Heritage Items at 8-10 Moriarty Road.

Attachment 1: Controls for development in the R2 zone

Except for the floor space ratio (FSR) development standard and the landscaping controls, this component of Willoughby DCP largely aligns with Codes SEPP Part 3 (Housing Code), 3B (Low Rise Housing Diversity Code) as it applies to dual occupancies, and Part 4 (Housing Alterations Code).

If a proposed development has an estimated cost of $50,000 or more, a suitably qualified person must certify it complies with all relevant provisions in Willoughby LEP and the numerical controls in this part of Willoughby DCP.

The table comprises:

a. Development standards under Willoughby LEP.

b. General controls for dwellings, dual occupancies and attached development.

c. Amenity development standards.

d. Further development controls for detached development.

e. Further development controls for detached studios.

f. Development controls for swimming pools.

g. Development controls for fences.

h. Development controls for associated works (earthworks, retaining walls, drainage, protection of walls, and protection of trees).

i. Further controls for dual occupancies.

j. Further controls for attached dual occupancies where part of one dwelling is located above part of another dwelling.

Note*:*

- The definition of a ‘suitably qualified person’ includes a practising registered architect, a practising qualified and accredited building designer, a registered building surveyor and a certified practicing planner (CPP).

a. Development standards under Willoughby LEP

Description	Proposed	Control	Yes	No	N/A
1. Height of building (Clause 4.3)	-	8.5m	-	-	-
2. Maximum floor space ratio (FSR) (Area 1 - Clause 4.4A)		-	-	-	-
Lot area	-	-	-	-	-
<200m^2		0.65:1			
≥200m^2 and <300m^2		0.55:1			
≥300m^2 and <400m^2		0.50:1			

Description	Proposed	Control	Yes	No	N/A
≥400m^2 and <500m^2	-	0.47:1	-	-	-
≥500m^2 and <600m^2	-	0.45:1	-	-	-
≥600m^2 and <700m^2	-	0.43:1	-	-	-
≥700m^2 and <800m^2	-	0.41:1	-	-	-
≥800m^2	-	0.40:1	-	-	-
3. Minimum lot size (Clause 6.10)	-	-	-	-	-
Attached dual occupancy	-	700m^2	-	-	-
Detached dual occupancy	-	900m^2	-	-	-

b. Development controls under the WDCP for dwellings, dual occupancies and attached development

Notes:

- An application for dual occupancy development must include a concept subdivision layout.
- The site area in the compliance checklist refers to the individual lot (excluding the access handle of a battle–axe allotment) that may be created by subdivision for each dual occupancy dwelling.

Description	Proposed	Control	Yes	No	N/A
1. Landscaped area	-	-	-	-	-
Lot area:	-	-	-	-	-
Up to 200m^2	-	0.25 × site area	-	-	-
201m^2 – 400m^2	-	(0.35 × site area) - 20m^2	-	-	-
401m^2 – 600m^2	-	(0.6 × site area) - 120m^2	-	-	-
601m^2 – 1000m^2	-	(0.525 × site area) - 75m^2	-	-	-
1001m^2 – 1500m^2	-	(0.6 × site area) - 150m^2	-	-	-
>1501m^2	-	0.5 × site area	-	-	-
Notes: To be included in the calculation for landscaped area the minimum dimension must not be less than 1m The landscaped area should include a minimum 20% deep soil zone. The remainder of the landscaped area may be 'soft' landscaping having a minimum depth of 600mm.	-	-	-	-	-
Minimum landscaped area behind the building line	-	50% of the total landscaped area	-	-	-

Description		Proposed	Control	Yes	No	N/A
Minimum deep soil zone forward of the building line on a lot 18m or more wide		-	70% of the area forward of the building line	-	-	-
Minimum deep soil zone forward of the building line on a lot less than 18m wide		-	50% of the area forward of the building line	-	-	-
2. Construction of basement		-	-	-	-	-
Lot width:		-	-	-	-	-
6m – 10m		-	Not exceed 25m^2	-	-	-
>10m		-	Not exceed 45m^2	-	-	-
Note: The basement may be used for car parking and include storage areas and/or plant/mechanical service rooms. However, the combined below ground storage areas and/or plant/mechanical service rooms must not exceed 5% of the total allowable area for a dwelling house (see section 2.1.14 of this Part).		-	-	-	-	-
3. Primary road setback is to be the average of 2 adjoining dwellings, otherwise		-	-	-	-	-
Lot area:		-	-	-	-	-
Up to 300m^2		-	3m	-	-	-
>300m^2 – 900m^2		-	4.5m	-	-	-
>900m^2 – 1500m^2		-	6.5m	-	-	-
>1500m^2		-	10m	-	-	-
4. Side setbacks		-	-	-	-	-
Lot width:	Building Height:	-	-	-	-	-
Up to 10m	Up to 5.5m	-	900mm	-	-	-
Up to 10m	>5.5m – 8.5m	-	(height - 5.5m) / 4 + 0.9m	-	-	-
>10m – 18m	Up to 4.5m	-	900mm	-	-	-
>10m – 18m	>4.5m – 8.5m	-	(height - 4.5m) / 4 + 0.9m	-	-	-
>18m – 24m	Up to 4.5m	-	1.5m	-	-	-
>18m – 24m	>4.5m – 8.5m	-	(height - 4.5m) / 4 + 1.5m	-	-	-
>24m	Up to 8.5m	-	2.5m	-	-	-

Description		Proposed	Control	Yes	No	N/A
Note: excludes common side boundary walls with a nil setback or within 900mm if the adjoining building is within 900mm		-	-	-	-	-
5. Maximum height of walls abutting or within 900mm of side boundary		-	3.3m or height of adjoining building abutting or within 900mm (maximum height for detached development is 4.5m)	-	-	-
6. Maximum length of wall abutting or within 900mm of side boundary (including any detached development)		-	-	-	-	-
Lot width at the building line:		-	-	-	-	-
6m – 10m		-	20m or 50% of lot depth (whichever is lesser)	-	-	-
>10m – 12.5m		-	10m	-	-	-
Note: the length of the wall may be same as the adjoining dwelling if the length of the wall on the adjoining property is greater than these controls		-	-	-	-	-
7. Rear setbacks		-	-	-	-	-
Lot area:	Building Height:	-	-	-	-	-
Up to 300m^2	Up to 4.5m	-	3m	-	-	-
Up to 300m^2	>4.5m –8.5m	-	10m or average of 2 adjoining houses, (whichever is lesser)	-	-	-
>300m^2 – 900m^2	Up to 4.5m	-	3m	-	-	-
>300m^2 – 900m^2	>4.5m – 8.5m	-	8m	-	-	-
>900m^2 – 1500m^2	Up to 4.5m	-	5m	-	-	-
>900m^2 – 1500m^2	>4.5m – 8.5m	-	12m	-	-	-
>1500m^2	Up to 4.5m	-	10m	-	-	-
>1500m^2	>4.5m – 8.5m	-	15m	-	-	-
8. Secondary road setbacks for corner lots		-	-	-	-	-
Lot size:		-	-	-	-	-
Up to 600m^2		-	2m	-	-	-
>600m^2 – 1500m^2		-	3m	-	-	-

Description	Proposed	Control	Yes	No	N/A
>1500m^2	-	8m	-	-	-
9. Minimum front setback to a classified road (including any detached development)	-	9m	-	-	-
10. Minimum setback to a public reserve (including any detached development)	-	3m	-	-	-
11. Front setback for battle axe lot	-	3m	-	-	-
12. Minimum side and rear setbacks for aerials, awnings, air conditioners, and the like (minimum setbacks do not apply to downpipes, driveways, electricity or gas meters, gutters, and the like)	-	450mm	-	-	-
13. Setback of structures abutting or within 900mm of a rear lane	-	50% of the boundary	-	-	-
14. Maximum height of balconies, decks, patios, terraces and verandahs attached to the side and rear of dwelling	-	Floor level above existing ground level	-	-	-
Setback:	-	-	-	-	-
Up to 3m	-	2m	-	-	-
>3m – 6m	-	3m	-	-	-
>6m	-	4m	-	-	-
Note: the lot must have a minimum width of 10m and area of 300m^2	-	-	-	-	-
15. Total floor area of all attached side and rear balconies, decks, patios, terraces, and verandahs within 6m of the boundary and more than 2m above the existing ground level	-	12m^2	-	-	-
16. Minimum principal private open space	-	-	-	-	-
Lot width:	-	-	-	-	-
Up to 10m	-	16m^2	-	-	-
>10m	-	24m^2	-	-	-
4. Minimum length and width of principal private open space	-	3m	-	-	-

Description	Proposed	Control	Yes	No	N/A
5. Maximum grade of principal private open space	-	1:50	-	-	-

c. Amenity development standards

Note:

- Building elements include: an entry feature or portico; a balcony, deck, pergola, terrace or verandah; a window box treatment; a bay window or similar feature; an awning or other feature over a window; a sun shading feature; and an eave.

Description	Yes	No	N/A
c.1 Building design (does not apply to a dwelling on a battle-axe lot)	-	-	-
1. A dwelling must contain at least 1 door and 1 window to a habitable room at ground floor level facing the primary road.	-	-	-
2. A dwelling with a setback from a primary road of at least 3m may have an articulation zone that extends up to 1.5m forward of the minimum required setback from the primary road.	-	-	-
3. Building elements may be located within the articulation zone providing the maximum total area of all building elements in the articulation zone, (other than an awning or other feature over a window, a sun shading feature or an eave) must not comprise more than 25% of the area of the articulation zone.	-	-	-
4. A building element on a dwelling (other than an entry feature or portico that has the same pitch as the roof on the dwelling) must not extend more than 1m above the gutter line of the eaves of a single storey dwelling house, or above the gutter line of the eaves of a 2 storey dwelling house.	-	-	-
5. A dwelling on a corner lot must have a window to a habitable room with an area of at least $1m^2$ that faces and is visible from the secondary road.	-	-	-
6. A dwelling with a setback from a secondary road of not more than 4.5m must have at least one of the following building elements for a minimum length of 20% of the elevation of the walls that face the secondary road and that are within 4.5m of the secondary road:	-	-	-
▪ an entry feature or portico	-	-	-
▪ a balcony, deck, pergola, terrace or verandah	-	-	-
▪ a bay window	-	-	-
▪ a step of at least 600mm in depth	-	-	-

Description	Yes	No	N/A
7. Building elements listed above may be located in a secondary road articulation zone if the zone extends no more than 1m into the minimum required setback area and spans the length of the walls that face the secondary road, and if the building element comprises no more than 20% of the zone area.	-	-	-
8. Any part of a gable or hipped roof that overhangs walls that are within 4.5m of the secondary road boundary must include eaves that extend for the length of those walls and project at least 450mm, but not more than 1m from those walls.	-	-	-
c.2 Privacy screens	-	-	-
1. A privacy screen must be provided for any part of a window to a habitable room that is less than 1.5m above the finished floor level of that room if:	-	-	-
▪ the window faces and is less than 3m from a side or rear boundary and the room has a finished floor level of more than 1m above existing ground level	-	-	-
▪ the window faces and is at least 3m, but not more than 6m, from a side or rear boundary and the room has a finished floor level of more than 3m above existing ground level	-	-	-
This does not apply to a bedroom window that has an area of not more than $2m^2$.	-	-	-
2. A privacy screen of at least 1.7m, but not more than 2.2m, above the finished floor level of a balcony, deck, patio, terrace or verandah must be installed at the edge of that part of the balcony, deck, patio, terrace or verandah that is parallel to or faces towards the relevant side or rear boundary if the area of the balcony, deck, patio, terrace or verandah is at least $3m^2$ and:	-	-	-
▪ that edge is less than 3m from a side or rear boundary and the balcony, deck, patio, terrace or verandah has a finished floor level of more than 1m above existing ground level	-	-	-
▪ that edge is at least 3m, but not more than 6m from a side or rear boundary and the balcony, deck, patio, terrace or verandah has a finished floor level of more than 2m above existing ground level	-	-	-
3. A privacy screen must be provided for any part of a window in a detached studio that is less than 1.5m above the finished floor level of that room if the window faces and is less than 3m from a side or rear boundary and:	-	-	-
▪ the room has a finished floor level more than 1m above existing ground level	-	-	-
▪ the window faces and is at least 3m, but not more than 6m from a side or rear boundary and the room has a finished floor level more than 3m above existing ground level	-	-	-

Description		Yes	No	N/A
c.3 Solar access		-	-	-
1. A minimum of 3 hours of solar access must be provided for residents of the development between 9am and 3pm on 21 June to living areas, such as family rooms, rumpus, lounge/dining, kitchen, and the principal open space and recreational areas.		-	-	-
2. A minimum of 3 hours of solar access must be maintained to adjoining properties between 9am and 3pm on 21 June to living areas, such as family rooms, rumpus, lounge/dining, kitchen, and the principal open space and recreational areas.		-	-	-
c.4 Car parking and access All car parking and vehicular access must comply with the relevant provisions of Part F (Transport and Parking Management) of Willoughby DCP. The specific controls below apply to dwelling houses.		-	-	-
1. Off-street car parking space, being an open hard stand space or a carport or garage, must be provided on a lot unless:		-	-	-
▪ the lot has a width of less than 8m measured at the building line		-	-	-
▪ the alteration of, or an addition to, a dwelling and the lot does not contain an off-street car parking space		-	-	-
▪ the development is the erection or alteration of, or an addition to, attached development and the lot does not contain an off-street car parking space		-	-	-
2. The off-street car parking space for a battle-axe lot must be constructed so that vehicles can leave the lot in a forward direction.		-	-	-
3. An attached garage may only be erected on a lot that has a width of less than 8m measured at the building line if the garage is accessed only from a secondary road or lane.		-	-	-
4. An attached garage, carport or car parking space accessed from a primary road must have a minimum setback 5.5m if the dwelling is setback less than 4.5m or 1m behind the building line if the dwelling is setback 4.5m or more.		-	-	-
5. The maximum width of all garage door openings facing a primary or secondary road is 3.2m for a lot between 8m – 12m and 6m for a lot greater than 12m.		-	-	-
6. Secondary road setback for garage or carport		-	-	-
Up to 600m^2	2m	-	-	-
600m^2 – 1500m^2	3m	-	-	-
>1500m^2	5m	-	-	-
7. Rear setback for garage or carport		-	-	-

Description		Yes	No	N/A
Up to 900m^2	900mm	-	-	-
>900m^2 – 1500m^2	1.5m	-	-	-
>1500m^2	2.5m	-	-	-
A detached garage or carport of masonry construction may be built to the rear boundary if the lot area is at least 200m^2, but not more than 300m^2, and the wall of a building on the adjoining lot within 900mm of that boundary (if any) is of masonry construction and does not have a window facing that boundary.		-	-	-

d. Further development controls for detached development

Note:

- Detached development includes: a deck, patio, pergola, terrace or verandah; a cabana, cubby house, fernery, garden shed, gazebo or greenhouse; a carport or garage, and a shed.
- The setback of detached development from a primary or secondary road (other than a detached garage or carport) must be located behind the building line of the dwelling.

Description	Proposed	Control	Yes	No	N/A
1. Maximum height above existing ground level	-	4.5m	-	-	-
2. Maximum GFA for all detached development	-	-	-	-	-
Lot size:	-	-	-	-	-
Up to 300m^2	-	36m^2	-	-	-
>300m^2 – 600m^2	-	45m^2	-	-	-
>600m^2 – 900m^2	-	60m^2	-	-	-
>900m^2	-	100m^2	-	-	-
Note: the maximum overall GFA includes the dwelling, attached and detached buildings	-	-	-	-	-
3. Side setbacks for all detached development, including above ground rainwater tanks and shade structures (additional side setbacks apply to detached studios)	-	-	-	-	-
Lot width:	-	-	-	-	-
Up to 18m	-	900mm	-	-	-
>18m – 24m	-	1.5m	-	-	-
>24m	-	2.5m	-	-	-

Description	Proposed	Control	Yes	No	N/A
Note: excludes common side boundary walls with a nil setback or within 900mm if adjoining building is within 900mm	-	-	-	-	-
4. Rear setbacks for a cabana, cubby house, fernery, garden shed, gazebo, greenhouse, above ground rainwater tank, a shade structure or shed	-	-	-	-	-
Lot size:	-	-	-	-	-
Up to 900m^2	-	0.9m	-	-	-
>900 m^2 – 1500m^2	-	1.5m	-	-	-
>1500m^2	-	2.5m	-	-	-
5. Height above existing ground level and rear setbacks for detached decks, patios, pergolas, terraces and verandahs	-	-	-	-	-
6. Maximum finished floor level	-	600mm	-	-	-
7. Rear setbacks	-	-	-	-	-
Lot size:	-	-	-	-	-
Up to 900m^2	-	900mm	-	-	-
>900m^2 – 1500m^2	-	1.5m	-	-	-
>1500m^2	-	2.5m	-	-	-

e. Further development controls for detached studios

Description	Proposed	Control	Yes	No	N/A
1. Height (within 900mm of a lane and above a garage)	-	6m	-	-	-
2. Rear setbacks	-	-	-	-	-
Lot size:	-	-	-	-	-
Up to 900m^2	-	900mm	-	-	-
>900m^2 – 1500m^2	-	1.5m	-	-	-
>1500m^2	-	2.5m	-	-	-
3. Maximum GFA	-	-	-	-	-
Lot size:	-	-	-	-	-

Description	Proposed	Control	Yes	No	N/A
Up to 350m^2	-	20m^2	-	-	-
>350m^2	-	36m^2	-	-	-
(the maximum overall GFA includes detached studios)	-	-	-	-	-
4. Side and rear setbacks	-	-	-	-	-
Lot width:	-	-	-	-	-
Up to 18m	-	900mm	-	-	-
>18m	-	1.5m	-	-	-
(excludes common side boundary walls with a nil setback or within 900mm if adjoining building is within 900mm)	-	-	-	-	-
5. Maximum height of walls abutting or within 900mm of side boundary	-	3.3m or height of adjoining building but not more than 4.5m	-	-	-
6. Maximum height of walls abutting or within 900mm of side boundary if above a garage	-	height of adjoining building but not more than 6m	-	-	-

f. Development controls for swimming pools

Description	Yes	No	N/A
1. Coping around a swimming pool must not be more than 1.4m above existing ground level, and 300mm wide if the coping is more than 600mm above existing ground level.	-	-	-
2. Decking around a swimming pool must not be more than 600mm above existing ground level.	-	-	-
3. A swimming pool must be located behind the building line of the dwelling.	-	-	-
4. The swimming pool water line must have a setback of at least 1m from a side or rear boundary.	-	-	-

g. Development controls for fences

Description	Yes	No	N/A
1. A fence erected behind the building line (except on a secondary street frontage) must:	-	-	-
▪ not be higher than 1.8m above existing ground level	-	-	-
▪ if it includes an entrance gate, not have a gate that opens outward	-	-	-

Description	Yes	No	N/A
▪ if it is constructed of metal components, be of low reflective, factory pre-coloured materials	-	-	-
▪ if it is on a sloping site and stepped to accommodate the fall in the land, be no higher than 2.2m above existing ground level at each step.	-	-	-
2. A fence erected forward of the building line or on the boundary of the primary and secondary street frontage must:	-	-	-
▪ not be higher than 1.2m above existing ground level	-	-	-
▪ if it includes an entrance gate, not have a gate that opens outward.	-	-	-
▪ if it is constructed of metal components, be of low reflective, factory pre-coloured materials	-	-	-
▪ be open for at least 20% of the area of the fence that is more than 400mm above existing ground level, with any individual solid element of the fence above that height being no more than 350mm wide with a minimum aperture of 25mm	-	-	-
▪ if it is on a sloping site and stepped to accommodate the fall in the land, be no higher than 1.6m above existing ground level at each step.	-	-	-
3. Any fence must be designed so as not to restrict the flow of any floodwater.	-	-	-
4. A fence erected on bush fire prone land must be constructed of non-combustible materials.	-	-	-
5. A solid fence located at the street alignment may be constructed up to a maximum height of 1.8m, where:	-	-	-
▪ the dwelling is affected by excessive traffic volumes and noise from the street (6000 vehicles/day or 60dBA)	-	-	-
▪ the fence encloses the main private open space of the dwelling with length limited to 75% of the frontage	-	-	-
▪ fencing incorporates openings or consists of vegetation with minimum 50% transparency.	-	-	-
6. Fences along the primary and secondary road frontages that are between 1.2m and 1.8m must be setback a minimum 1m from the street alignment for the entire frontage with provision of low maintenance planting in the setback area.	-	-	-

h. Development controls for associated works

Note:

- Associated works include earthworks, retaining walls, drainage, protection of walls, and protection of trees.

Description	Yes	No	N/A
1. Excavation for the purposes of development must not exceed a maximum depth, measured from existing ground level, of:	-	-	-
▪ 1m if located not more than 1m from any boundary	-	-	-
▪ 2m if located more than 1m but not more than 1.5m from any boundary	-	-	-
▪ 3m if located more than 1.5m from any boundary.	-	-	-
2. Fill must not exceed a maximum height, measured from existing ground level, of:	-	-	-
▪ 1m if the fill is for the purposes of the erection or alteration of, or an addition to, a dwelling house	-	-	-
▪ 600mm if the fill is for any other purpose.	-	-	-
The height of fill contained wholly within the footprint of a dwelling house or any attached development or detached development is not limited.	-	-	-
3. Fill that is higher than 150mm above existing ground level and is not contained wholly within the footprint of a dwelling house or any attached development or detached development is limited to 50% of the landscaped area of the lot.	-	-	-
4. The finished ground level of the fill must not be used to measure the height of any dwelling house or any attached development or detached development.	-	-	-
5. Support for earthworks more than 600mm above or below existing ground level must take the form of a retaining wall or other structural support that:	-	-	-
▪ a professional engineer has certified as structurally sound, including in relation to (but not limited to) the ability to withstand the forces of lateral soil load, and	-	-	-
▪ it has been designed so as not to redirect the flow of any surface water or ground water, or cause sediment to be transported, onto an adjoining property, and	-	-	-
▪ it has adequate drainage lines connected to the stormwater drainage system for the site, and	-	-	-

Description	Yes	No	N/A
▪ it does not have a total height measured vertically from the base of the retaining wall or structural support to its uppermost portion that is more than the height of the associated excavation or fill, and	-	-	-
▪ it is separated from any other retaining wall or structural support on the site by at least 2m, measured horizontally, and	-	-	-
▪ if it is an embankment or batter, has a toe or top that is more than 1m from any side or rear boundary.	-	-	-
6. Development must be at least 3m from all trees identified for retention on the site (measured from the base of the trunk of the tree).	-	-	-
7. Development for the following structures may be located within 3m of a tree if works do not involve excavation or fill of more than 150mm below or above existing ground level:	-	-	-
▪ an access ramp	-	-	-
▪ a driveway, pathway or paving	-	-	-
▪ an awning, blind or canopy	-	-	-
▪ a fence, screen, or child-resistant barrier associated with a swimming pool or spa pool.	-	-	-

i. Further controls for dual occupancies

Description		Proposed	Control	Yes	No	N/A
1. Primary road setback to be the average of 2 adjoining dwellings or dual occupancies, otherwise:		-	-	-	-	-
Lot area:		-	-	-	-	-
700m^2 – 900m^2		-	4.5m	-	-	-
>900m^2 – 1500m^2		-	6.5m	-	-	-
>1500m^2		-	10m	-	-	-
2. Side setbacks*		-	-	-	-	-
Lot width:	Building height:	-	-	-	-	
Up to 24m	Up to 4.5m	-	900mm	-	-	-
Up to 24m	>4.5 – 8.5m	-	(height - 4.5m) / 4 + 0.9m	-	-	-
>24m – 36m	Up to 4.5m	-	1.5m	-	-	-

Description		Proposed	Control	Yes	No	N/A
>24m – 36m	>4.5 – 8.5m	-	(height - 4.5m) / 4 + 1.5m	-	-	-
>36m	Up to 8.5m	-	2.5m	-	-	-
Note: excludes common side boundary walls for attached dual occupancies. *see item j.1 for side setbacks relating to dual occupancies where part of a dwelling is located above part of another dwelling		-	-	-	-	-
3. Rear setbacks*		-	-	-	-	-
Lot area:	Building height:	-	-	-	-	-
700m^2 – 900m^2	Up to 4.5m	-	3m	-	-	-
700m^2 – 900m^2	>4.5m – 8.5m	-	8m	-	-	-
>900m^2 – 1500m^2	Up to 4.5m	-	5m	-	-	-
>900m^2 – 1500m^2	>4.5 – 8.5m	-	12m	-	-	-
>1500m^2	Up to 4.5m	-	10m	-	-	-
>1500m^2	>4.5 – 8.5m	-	15m	-	-	-
*see item j.2 for rear setback controls relating to dual occupancies where part of a dwelling is located above part of another dwelling		-	-		-	-
4. Secondary road setbacks for corner lots		-	-		-	-
Lot size:		-	-		-	-
700m^2 – 900m^2		-	2m		-	-
>900m^2 – 1500m^2		-	3m		-	-
>1500m^2		-	5m		-	-
*see item j.3 for secondary road setbacks for corner lots relating to dual occupancies where part of a dwelling is located above part of another dwelling		-	-		-	-
5. Dwelling configuration				-	-	-
▪ Each dwelling must face a public road				-	-	-
▪ Detached dual occupancies on a corner lot must be 3m from each other				-	-	-

Description	Proposed	Control	Yes	No	N/A
▪ Each dwelling must have a minimum width (measured at the building line) of:			-	-	-
a. 6.5m where a car space is accessed from the primary road, or			-	-	-
b. 5m in any other case			-	-	-
6. Attached balconies, decks, patios terraces and verandahs to the side or rear of the dual occupancy			-	-	-
▪ The maximum height of the floor level is 4m			-	-	-
▪ If more than 2m above existing ground level, must be setback a minimum 3m from the boundary			-	-	-
▪ If above 2m above the existing ground level, must not have a total area of more than 12m^2			-	-	-
7. The maximum width of all garage door openings facing a primary is 6m for a lot between 15m – 20m, 9.2m for a lot greater than 20m – 25m and 12m for a lot greater than 25m.			-	-	-
8. The maximum width of all garage door openings facing a secondary road is 3.2m for a lot between 12m – 15m, 6m for a lot greater than 15m – 20m, 9m for a lot greater than 20m – 25m and 12m for a lot greater than 25m.			-	-	-
9. The width of the (parent) lot must not be less than the following when measured at the building line: *(a) 15m or* *(b) 12m where the car parking is provided at the rear of the lot or in a basement and accessed only from a secondary road, parallel road or laneway.*			-	-	-

j. Further controls for attached dual occupancies where part of a dwelling is located above part of another dwelling

Description		Proposed	Control	Yes	No	N/A
1. Side setbacks The minimum side boundary setback is 1.5m. However, if the development is more than 10m behind the building line and is more than 4.5m above the existing ground level the minimum side boundary setback is: s = h-3m where (s) is the minimum setback and (h) is the height of any part of the building.		-	1.5m	-	-	-
2. Rear setbacks		-	-	-	-	-
Lot area:	Building height	-	-	-	-	-
700m^2 – 1500m^2	Up to 4.5m	-	6m	-	-	-
700m^2 – 1500m^2	>4.5m – 8.5m	-	10m	-	-	-
>1500m^2	Up to 4.5m	-	10m	-	-	-
>1500m^2	>4.5m – 8.5m	-	15m	-	-	-
3. Secondary road setbacks for corner lots		-	-	-	-	-
Lot size:		-	-	-	-	-
700m^2 – 1500m^2		-	3m	-	-	-
>1500m^2		-	5m	-	-	-

Attachment 2: Controls for manor houses, attached dwellings and multi dwelling housing in the R3 zone

The manor house controls for this component of Willoughby DCP generally aligns to Part 3B (Low Rise Housing Diversity Code) of Codes SEPP.

Many of the other controls under the Low Rise Housing Diversity Code are adopted for attached dwellings and multi dwelling housing. Notable exceptions are the landscaped area and setback controls.

A suitably qualified person must certify the proposed development complies with all relevant provisions of Willoughby LEP and the numerical controls under this part of Willoughby DCP. The table comprises:

a. Development standards under Willoughby LEP.
b. Amenity development standards.
c. Controls for manor houses.
d. Controls for attached dwellings and multi dwelling housing.
e. Controls for detached development.
f. Controls for detached studios.

Notes:

- A suitably qualified person includes a practicing registered architect, a practicing qualified and accredited building designer, a registered building surveyor and a certified practicing planner (CPP).
- If the proposed development includes construction of a swimming pools, fence, or associated works (earthworks, retaining walls, drainage, protection of walls, and protection of trees), please complete the relevant Sections f, g and h in Attachment 1.
- A concept subdivision layout must be submitted with the application.
- Unless otherwise stated the site area refers to the ‘parent lot’ which is the entire site area prior to subdivision.
- Further controls apply to individual lots that may be created by subdivision for each dwelling.
- A manor house, attached dwellings or multi dwelling housing must be consistent with the relevant design criteria in the Low Rise Housing Diversity Design Guide, Department of Planning Industry and Environment, July 2020. The requirements of Part B and any other parts of Willoughby DCP and this compliance checklist prevail to the extent of any inconsistency with the design guide.

a. Development standards under Willoughby LEP

Description	Proposed	Control	Yes	No	N/A
1. Height of building (Clause 4.3) (see 4.3A of Willoughby LEP for any exclusions)	-	As per the Height of Buildings Map	-	-	-

Description	Proposed	Control	Yes	No	N/A
2. Maximum floor space ratio (Clause 4.4 of Willoughby LEP) 3. (see Clause 4.4A of Willoughby LEP for any exclusions)	-	As per the Floor Space Ratio Map	-	-	-
4. Minimum lot size (Clause 6.10 of Willoughby LEP)	-	1100m^2	-	-	-

b. Amenity development standards for manor houses, attached dwellings and multi dwelling housing

b.1 Building design	Yes	No	N/A
1. Articulation zone	-	-	-
▪ A manor house may have a primary road articulation zone that extends up to 1.5m forward of the minimum required setback from the primary road	-	-	-
▪ Attached dwellings and multi dwelling housing may have a primary road and secondary road articulation zone that each extend up to 1.5m forward of the minimum required setback from the primary road or secondary road	-	-	-
▪ The maximum total area of all building elements in the articulation zone (other than an awning or other feature over a window, a sun shading feature or an eave) must not comprise more than 25% of the area of the articulation zone.	-	-	-
Note: Building elements include: an entry feature or portico; a balcony, deck, pergola, terrace or verandah; a window box treatment; a bay window or similar feature; an awning or other feature over a window; a sun shading feature; and an eave.	-	-	-
2. Privacy screen A privacy screen must be provided for any part of a window to a habitable room that is less than 1.5m above the finished floor level of that room if:	-	-	-
▪ the floor level of the habitable room is 1m or more, but not more than 3m, above existing ground level and the window faces a side or rear boundary and is less than 3m from that boundary	-	-	-
▪ the floor level of the habitable room is more than 3m above existing ground level and the window faces a side or rear boundary and is less than 6m from that boundary	-	-	-
▪ the floor level of the habitable room is 1m or more, but not more than 3m, above existing ground level and the window faces a dwelling on the same lot is less than 6m from that dwelling	-	-	-

b.1 Building design	**Yes**	**No**	**N/A**
▪ the floor level of the habitable room is more than 3m above existing ground level and the window faces a dwelling on the same lot is less than 12m from that dwelling.	-	-	-
These controls do not apply to a habitable room with a floor level not more than 1m above existing ground level or a window that faces a road or public space or a bedroom window that has an area of not more than $2m^2$.	-	-	-
3. Balconies, decks, patios, terraces or verandahs The edge of a balcony, deck, patio, terrace or verandah must have a privacy screen with a height of a least 1.5 above its floor level if:	-	-	-
▪ the floor level of the balcony, deck, patio, terrace or verandah is 1m or more, but not more than 3m, above existing ground level and the edge faces a side or rear boundary and is less than 3m from that dwelling	-	-	-
▪ the floor level of the balcony, deck, patio, terrace or verandah is more than 3m, above existing ground level and the edge faces a side or rear boundary and is less than 6m from that dwelling	-	-	-
▪ the floor level of the balcony, deck, patio, terrace or verandah is 1m or more, but not more than 2m, above existing ground level and the edge faces a dwelling on the same lot and is less than 6m from that dwelling	-	-	-
▪ the floor level of the balcony, deck, patio, terrace or verandah is more than 2m, above existing ground level and the edge faces a dwelling on the same lot and is less than 12m from that dwelling.	-	-	-
These controls do not apply to a balcony, deck, patio, terrace or verandah with a floor level not more than 1m above existing ground level or that faces a road or public space or that has an area of not more than $2m^2$.	-	-	-
4. Solar access	-	-	-
▪ a minimum of 3 hours of solar access must be provided for residents of the development between 9am and 3pm on 21 June to living areas, such as family rooms, rumpus, lounge/dining, kitchen, and the principal open space and recreational areas	-	-	-
▪ a minimum of 3 hours of solar access should be maintained to adjoining properties between 9am and 3pm on 21 June to living areas, such as family rooms, rumpus, lounge/dining, kitchen, and the principal open space and recreational areas.	-	-	-
Note: if the adjoining property already has less than 3 hours of solar access between 9am and 3pm on 21 June to living areas and the principal open space, every reasonable attempt should be made to provide a minimum of 3 hours of solar access)	-	-	-

b.1 Building design	Yes	No	N/A
5. Car parking All car parking and vehicular access must comply with relevant provisions of Part F (Transport and Parking Management) of Willoughby DCP. The following setbacks apply to manor houses, attached dwellings and multi dwelling housing:	-	-	-
▪ if the setback of the building from the road is less than 4.5m, a minimum 5.5m setback is required for the off-street parking space	-	-	-
▪ if the setback of the building from the road is 4.5m or more, the off-street parking space is required to be 1m behind the building line	-	-	-
▪ an off-street parking space may be an open hard stand space or carport or garage, whether attached to or detached from the dwelling	-	-	-
▪ the maximum width of all garage door openings facing a primary or secondary road is 3.2m for a lot between 6m – 12m and 6m for a lot greater than 12m.	-	-	-

c. Controls for manor houses

Note: A manor house must face a public road

Description	Proposed	Control	Yes	No	N/A
1. Lot width of the parent lot	-	27m	-	-	-
2. Landscaped area (includes the deep soil zone and soft landscaped area but excludes planter boxes)	-	Minimum 50% – 100m^2 of the site area	-	-	-
3. Deep soil zone	-	Minimum 20% of the site area	-	-	-
4. Soft landscaped area (soft landscaped area is landscaping that has a minimum depth of 600mm)	-	Minimum 20% of the site area	-	-	-
5. Dimension of all landscaped areas (areas with less than the minimum dimension are excluded from the minimum landscape area requirements)	-	Minimum width and depth is 1.5m	-	-	-
6. Minimum landscaped area behind the building line	-	50% of the total landscaped area	-	-	-
7. Minimum deep soil zone forward of the building line	-	70% of the area forward of the building line	-	-	-

Description		Proposed	Control	Yes	No	N/A
8. Primary road setback to be the average of 2 adjoining dwellings, otherwise:		-	-	-	-	-
Lot area:		-	-	-	-	-
1100m^2 – 1500m^2		-	6.5m	-	-	-
>1500m^2		-	10m	-	-	-
9. Side setbacks The minimum side boundary setback is 1.5m. However, if the development is more than 10m behind the building line and is more than 4.5m above the existing ground level the minimum side boundary setback is: s = h-3m where (s) is the minimum setback and (h) is the height of any part of the building.		-	1.5m	-	-	-
10. Rear setbacks		-	-	-	-	-
Lot area:	Building height:	-	-	-	-	
1100m^2 – 1500m^2	Up to 4.5m	-	6m	-	-	-
1100m^2 – 1500m^2	>4.5m – 8.5m	-	10m	-	-	-
>1500m^2	Up to 4.5m	-	10m	-	-	-
>1500m^2	>4.5m – 8.5m	-	15m	-	-	-
11. Secondary road setback for corner lots:		-	-	-	-	-
Lot area:		-	-	-	-	-
1100m^2 – 1500m^2		-	3m	-	-	-
>1500m^2		-	5m	-	-	-
12. Classified road setback		-	9m	-	-	-
13. Public reserve setback		-	3m	-	-	-
14. Minimum side and rear setbacks for aerials, awnings, air conditioners, and the like (minimum setbacks do not apply to downpipes, driveways, electricity or gas meters, gutters, and the like)		-	450mm	-	-	-
15. Setback of structures abutting or within 900mm of a rear lane		-	50% of the boundary	-	-	-

Description	Proposed	Control	Yes	No	N/A
16. Minimum principal private open space	-	-	-	-	-
1 bedroom or studio	-	8m^2	-	-	-
2 bedrooms	-	12m^2	-	-	-
3 or more bedrooms	-	16m^2	-	-	-
17. Minimum length and width of principal private open space	-	2m	-	-	-
18. Width of garage doors from a primary or secondary road	-	Maximum 6m	-	-	-

d. Controls for attached dwellings and multi dwelling housing

Description	Proposed	Control	Yes	No	N/A
1. Lot width of the parent lot	-	27m	-	-	-
2. Minimum width of each dwelling Note: The width of an attached dwelling is measured from the centre of a side wall adjoining another dwelling.	-	6m	-	-	-
3. Landscaped area (includes the deep soil zone and soft landscaped area) Note: Soft landscaped area is landscaping that has a minimum depth of 600mm)	-	Minimum 40% of the total site area	-	-	-
4. Deep soil zone	-	Minimum 20% of the total site area	-	-	-
5. Soft landscaped area	-	Minimum 20% of the total site area	-	-	-
6. Dimension of all landscaped areas (areas with less than the minimum dimension are excluded from the calculation of minimum area requirements)	-	Minimum width and depth is 1.5m	-	-	-
7. Landscaped area for individual lots	-	Minimum 30% of each potential lot	-	-	-
8. Minimum landscaped area behind the building line of individual lots	-	50% of the landscaped area	-	-	-
9. Minimum deep soil zone forward of the building line on individual lots	-	50% of the area forward of the building line	-	-	-
10. Primary road setback	-	6m	-	-	-

Description	Proposed	Control	Yes	No	N/A
11. Minimum setback fronting internal communal road	-	3m	-	-	-
12. Side setbacks (if the dwelling is not attached) Note: The total setback applies to each level above the second storey.	-	1.5m for 1 and 2 storey dwellings + 1.2m for each level above 2 storeys	-	-	-
13. Rear setbacks:	-	-	-	-	-
▪ one storey	-	6m	-	-	-
▪ two storeys	-	8m	-	-	-
▪ three storeys	-	10m	-	-	-
14. Secondary road setback for corner lots	-	3m	-	-	-
15. Classified road setback	-	9m	-	-	-
16. Public reserve setback	-	3m	-	-	-
17. Setback of structures from a rear lane	-	1m	-	-	-
18. Minimum principal private open space for each dwelling	-	$16m^2$	-	-	-
19. Minimum length and width of principal private open space	-	3m	-	-	-
20. Adaptable housing: (% of the total number of dwellings)	-	-	-	-	-
▪ single storey	-	10%	-	-	-
▪ two storey	-	25%	-	-	-
▪ three storey	-	33%	-	-	-
▪ four storey	-	50%	-	-	-
21. Floor to ceiling height of rooms:	-	-	-	-	-
▪ habitable rooms	-	Minimum 2.7m	-	-	-
▪ non-habitable rooms	-	Minimum 2.4m	-	-	-
22. Window for habitable rooms	-	Minimum glass area not less than 10% of	-	-	-

Description	Proposed	Control	Yes	No	N/A
		the floor area of the room			
23. Depth of habitable room	-	Maximum 2.5 x ceiling height	-	-	-
24. Solar access:	-	-	-	-	-
▪ to solar collectors on adjoining building	-	Minimum 4 hours	-	-	-
▪ to living rooms and private open space (from 9am to 3pm on 21 June)	-	Minimum 3 hours	-	-	-
25. Natural ventilation:	-	-	-	-	-
▪ unobstructed window openings	-	Minimum 5% of the floor area	-	-	-

e. Controls for detached development

- Detached development includes: a deck, patio, pergola, terrace or verandah; a cabana, cubby house, fernery, garden shed, gazebo or greenhouse; a carport or garage, and a shed.
- The setback of detached development from a primary or secondary road (other than a detached garage or carport) must be located behind the building line of the dwelling.

Description	Proposed	Control	Yes	No	N/A
1. Maximum height above existing ground level	-	4.5m	-	-	-
2. Maximum. GFA for all detached development: (the maximum overall GFA includes the dwelling, attached and detached buildings)	-	100m^2 of the site area	-	-	-
3. Side setbacks for all detached development, including above ground rainwater tanks and shade structures (additional side setbacks apply to detached studios): (excludes common side boundary walls with a nil setback or within 900mm if adjoining building is within 900mm)	-	900mm	-	-	-
4. Rear setbacks for a cabana, cubby house, fernery, garden shed, gazebo, greenhouse, above ground rainwater tank, a shade structure or shed:	-	-	-	-	-
Lot size:	-	-	-	-	-

Description	Proposed	Control	Yes	No	N/A
1100m^2 – 1500m^2	-	1.5m	-	-	-
>1500m^2	-	2.5m	-	-	-
5. Height above existing ground level and rear setbacks for detached decks, patios, pergolas, terraces and verandahs	-	-	-	-	-
6. Maximum finished floor level	-	600mm	-	-	-
7. Rear setbacks	-	-	-	-	-
Lot size:	-	-	-	-	-
200m^2 – 900m^2	-	900mm	-	-	-
>900m^2 – 1500m^2	-	1.5m	-	-	-
>1500m^2	-	2.5m	-	-	-

f. Controls for detached studios

Description	Proposed	Control	Yes	No	N/A
1. Height (within 900mm of a lane and above a garage)	-	6m	-	-	-
2. Maximum GFA (the maximum overall GFA includes detached studios)	-	36m^2	-	-	-
3. Rear setbacks	-	3m	-	-	-
4. Side and rear setbacks	-	-	-	-	-
Lot width:	-	-	-	-	-
12m – 18m	-	900mm	-	-	-
>18m	-	1.5m	-	-	-
(excludes common side boundary walls with a nil setback or within 900mm if adjoining building is within 900mm)	-	-	-	-	-
5. Maximum height of walls abutting or within 900mm of side boundary	-	3.3m or height of adjoining building but not more than 4.5m	-	-	-
6. Maximum height of walls abutting or within 900mm of side boundary if above a garage	-	height of adjoining building but not more than 6m	-	-	-
7. Separation from residential accommodation	-	3m	-	-	-

Description	Proposed	Control	Yes	No	N/A
8. Solar access to solar collectors on adjoining building	-	Minimum 4 hours	-	-	-
9. Solar access:	-	-	-	-	-
▪ to solar collectors on adjoining buildings	-	Minimum 4 hours	-	-	-
▪ to living room and private open space (between 9am – 3pm on 21 June)	-	Minimum 3 hours	-	-	-
10. Natural ventilation:	-	-	-	-	-
▪ unobstructed window openings	-	Minimum 5% of the floor area	-	-	-

Attachment 3: Controls for residential flat buildings and the residential components of shop top housing and mixed use developments in the R3 and R4 zones.

This component of Willoughby DCP aligns to the Apartment Design Guide, NSW Department of Planning and Environment, July 2015. It applies to residential flat buildings and the residential components of shop top housing and mixed use developments.

New single dwellings and dual occupancies in the R3 Medium Density zone and R4 High Density zone in the Willoughby LEP 2012 are prohibited. Where alterations and additions are proposed to existing single dwellings in the R3 and R4 zones, the R2 Low Density Residential zone DCP controls would apply (refer Part B Attachment 1).

A practicing registered architect must certify the proposed development complies with all relevant provisions of the Apartment Design Guide, Willoughby LEP and the numerical controls under this part.

The table comprises:

a. Development standards under Willoughby LEP.

b. Numerical controls, including the minimum requirements under the Apartment Design Guide.

Notes:

- The 'Site analysis checklist' in Appendix 1 of the Apartment Design Guide must be completed and included with the development application.
- The 'Development application – recommended documentation checklist' in Appendix 3 of the Apartment Design Guide must be completed and included with the development application.

a. Development standards under Willoughby LEP

Description	Proposed	Control	Yes	No	N/A
1. Height of building (Clause 4.3) (see 4.3A of Willoughby LEP for any exclusions)	-	As per the Height of Buildings Map	-	-	-
2. Maximum floor space ratio (Clause 4.4 of Willoughby LEP) (see Clause 4.4A of Willoughby LEP for any exclusions)	-	As per the Floor Space Ratio Map	-	-	-
3. Minimum lot size (Clause 6.10 of Willoughby LEP)	-	1100m^2	-	-	-

b. Numerical controls for residential development comprising three or more storeys and four or more dwellings

Description	Proposed	Control	Yes	No	N/A
1. Street frontage	-	Minimum 27m	-	-	-
2. Site coverage of the site area	-	-	-	-	-
▪ single storey	-	50%	-	-	-
▪ two storeys	-	35%	-	-	-
▪ three storeys	-	30%	-	-	-
▪ four storeys	-	28%	-	-	-
▪ five storeys	-	26%	-	-	-
▪ six storeys	-	24%	-	-	-
▪ seven storeys	-	22%	-	-	-
▪ eight or more storeys	-	20%	-	-	-
3. Height of building	-	-	-	-	-
The overall height should include:	-	-	-	-	-
▪ rooftop articulation	-	Minimum 1m	-	-	-
▪ allowance for topographic changes	-	2m	-	-	-
▪ allowance for services between floors	-	400mm	-	-	-
(the uppermost floor should be set back so it is not visible from the ground level at a 45^{o} plane and occupy not more than 60% of the floor area below)	-	-	-	-	-
4. Building depth (depending on floor to ceiling height and/or orientation)	-	12 – 18m	-	-	-
5. Building separation	-	-	-	-	-
Up to 4 storeys:	-	-	-	-	-

Description	Proposed	Control	Yes	No	N/A
▪ between non-habitable rooms	-	6m	-	-	-
▪ between habitable rooms and non-habitable rooms	-	9m	-	-	-
▪ between habitable rooms/balconies	-	12m	-	-	-
5 – 8 storeys:	-	-	-	-	-
▪ between non-habitable rooms	-	9m	-	-	-
▪ between habitable rooms and non-habitable rooms	-	12m	-	-	-
▪ between habitable rooms/balconies	-	18m	-	-	-
9 storeys and above:	-	-	-	-	-
▪ between non-habitable rooms	-	12m	-	-	-
▪ between habitable rooms and non-habitable rooms	-	18m	-	-	-
▪ between habitable rooms/balconies	-	24m Increase distance by 3m if adjacent to a lower density zone	-	-	-
(these are minimum requirements that may need to increase to achieve adequate sunlight and/or other amenity for residents)	-	-	-	-	-
6. Street setbacks	-	-	-	-	-
▪ R3 Zone	-	7.5m	-	-	-
▪ R4 Zone	-	9m	-	-	-
▪ Internal communal road	-	3m	-	-	-
7. Side and rear setbacks	-	-	-	-	-

Description	Proposed	Control	Yes	No	N/A
▪ single storey <5m	-	1.5m	-	-	-
▪ 2 storey <8m	-	3m	-	-	-
▪ storeys above the ground floor	-	3m plus 1.2m per storey	-	-	-
8. Setbacks for corner lots	-	-	-	-	-
▪ R3 Zone	-	3.5m	-	-	-
▪ R4 Zone (ground floor)	-	3m	-	-	-
▪ R4 Zone (above the ground level)	-	3m plus 1.2m per storey	-	-	-
9. Deep soil zone	-	-	-	-	-
▪ Up to 1100m^2	-	Minimum 7%	-	-	-
▪ 1100m^2 – 1500m^2	-	Minimum 10%	-	-	-
▪ >1500m^2	-	Minimum 15%	-	-	-
(the deep soil zone may be included in the total landscaped area requirement) Note: developments such as shop top housing in commercial zones which have significant site coverage at the ground level may not achieve the required amount deep soil zones.	-	6m	-	-	-
10. Soft landscaped area (soft landscaped area is landscaping that has a minimum depth of 600mm)	-	Minimum 35% of the site area	-	-	-
▪ Minimum dimension	-	6m for deep soil zone and 1.5m for soft landscaped areas	-	-	-
(areas with less than the minimum dimensions for the deep soil zone and/or the soft landscaped area are excluded from the total minimum landscaped area requirement)	-	-	-	-	-
11. Communal open space (for development with 15 or more apartments)	-	-	-	-	-

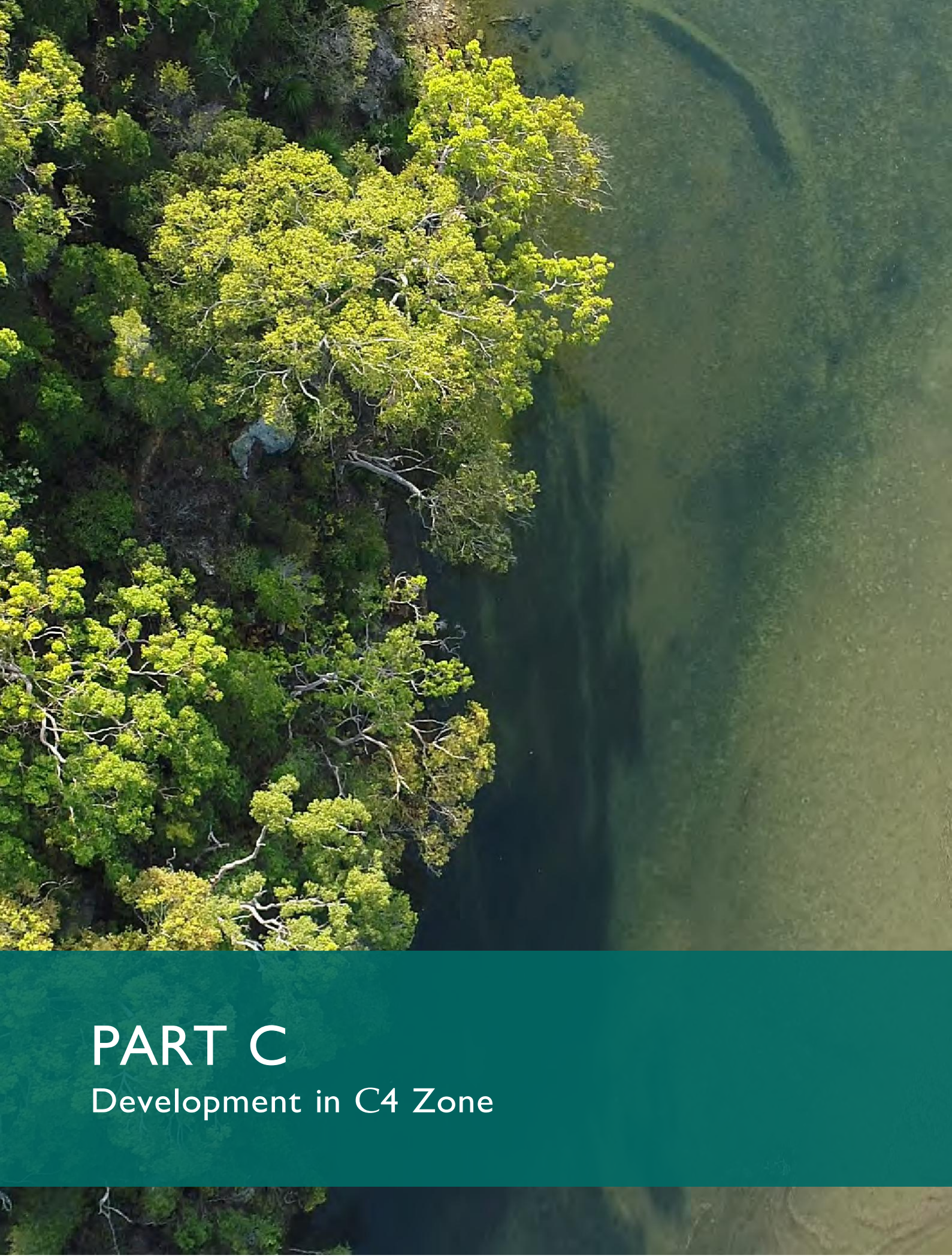
Description	Proposed	Control	Yes	No	N/A
▪ minimum area	-	25m^2	-	-	-
▪ minimum dimension	-	5m	-	-	-
▪ solar access	-	Minimum 50% over the usable space for a minimum 2 hours between 9am – 3pm on 21 June	-	-	-
12. Children's play area (for developments with 10 or more apartments in addition to any communal open space)	-	-	-	-	-
▪ minimum area	-	30m^2	-	-	-
▪ minimum dimension	-	5m	-	-	-
▪ maximum grade	-	1 in 20	-	-	-
13. Adaptable housing (% of the total number of apartments)	-	-	-	-	-
▪ 3 storeys	-	33%	-	-	-
▪ 4 or more storeys	-	50%	-	-	-
(includes apartments with a shop top or mixed use development)	-	-	-	-	-
14. Solar access to solar collectors on adjoining building	-	Minimum 4 hours	-	-	-
15. Solar access:	-	-	-	-	-
▪ to solar collectors on adjoining buildings	-	Minimum 4 hours	-	-	-
▪ to living room and private open space	-	Minimum 70% of apartments to receive a minimum 2 hours between 9am – 3pm on 21 June	-	-	-
▪ no solar access to apartments	-	Maximum 15% of apartments between 9am – 3pm on 21 June	-	-	-
16. Natural ventilation	-	-	-	-	-

Description	Proposed	Control	Yes	No	N/A
▪ cross ventilation (up to the first 9 storeys)	-	Minimum of 60% of apartments	-	-	-
▪ unobstructed window openings	-	Minimum 5% of the floor area	-	-	-
17. Apartment size and layout	-	-	-	-	-
Floor to ceiling heights:	-	-	-	-	-
▪ habitable rooms	-	Minimum 2.7m	-	-	-
▪ non-habitable rooms	-	Minimum 2.4m	-	-	-
▪ second floor level	-	2.4m for 50% of apartment area	-	-	-
Internal area of apartment:	-	-	-	-	-
▪ Studio	-	Minimum 35m^2	-	-	-
▪ 1 bedroom	-	Minimum 50m^2	-	-	-
▪ 2 bedroom	-	Minimum 70m^2	-	-	-
▪ 3 bedroom	-	Minimum 90m^2	-	-	-
▪ 4 or more bedrooms	-	Minimum 102m^2	-	-	-
(add 5m^2 for each additional bathrooms)	-	Minimum 102m^2 plus 12m^2 per additional bedroom	-	-	-
Window for habitable rooms	-	Minimum glass area not less than 10% of the floor area of the room	-	-	-
Depth of habitable room	-	Maximum 2.5 x ceiling height	-	-	-
18. Size of rooms:	-	-	-	-	-
▪ master bedrooms	-	10m^2 (excluding wardrobe space)	-	-	-
▪ other bedrooms	-	3m^2 (excluding wardrobe space)	-	-	-
▪ living/dining rooms (studio and 1 bedroom apartments)	-	3.6m	-	-	-

Description	Proposed	Control	Yes	No	N/A
▪ living/dining rooms (2 and 3 bedroom apartments)	-	4m	-	-	-
▪ width of cross-over or cross-through apartments	-	4m	-	-	-
19. Balconies	-	-	-	-	-
Size:	-	-	-	-	-
▪ studio	-	$4m^2$	-	-	-
▪ 1 bedroom apartment	-	$8m^2$	-	-	-
▪ 2 bedroom apartment	-	$10m^2$	-	-	-
▪ 3 bedroom apartment	-	$12m^2$	-	-	-
Minimum depth:	-	-	-	-	-
▪ studio	-	-	-	-	-
▪ 1 bedroom apartment	-	2m	-	-	-
▪ 2 bedroom apartment	-	2m	-	-	-
▪ 3 bedroom apartment	-	2.4m	-	-	-
▪ Private open space at ground level or adjoining a podium	-	$15m^2$ with a minimum depth of 3m	-	-	-
20. Storage	-	-	-	-	-
Size:	-	-	-	-	-
▪ studio	-	$4m^2$	-	-	-
▪ 1 bedroom apartment	-	$6m^2$	-	-	-
▪ 2 bedroom apartment	-	$8m^2$	-	-	-
▪ 3 or more bedroom apartments	-	$10m^2$	-	-	-
(50% to be located within the apartment)	-	-	-	-	-

Willoughby Development Control Plan Part B: Residential Development 2023

Description	Proposed	Control	Yes	No	N/A
21. Universal design (must satisfy Liveable Housing Australia's 'Liveable Housing Guidelines' silver level universal design features)	-	20% of the total apartments	-	-	-
22. Bicycle and car parking	-	In line with Part F of Willoughby DCP	-	-	-



PART C

Development in C4 Zone

Part C: Development in C4 Zone

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1 Introduction

The C4 Environment Living Zone contains properties with special qualities and attributes such as natural bushland, geological features, and foreshore areas. This part of Willoughby Development Control Plan (Willoughby DCP) includes information, guidelines and controls to maintain these special qualities and protect adjoining urban bushland and foreshore areas.

1.1 Aim

The aim of this part is to ensure residential and other permitted land uses result in low impact developments in environmentally sensitive locations.

1.2 Objectives

The objectives of this part are to:

a. protect environmentally sensitive foreshore and bushland areas from overdevelopment or visually intrusive development so that the scenic qualities, biodiversity, and ecological values of those areas are maintained

b. provide performance criteria and numerical controls to ensure low impact development is achieved

c. ensure development does not alter existing drainage patterns that could result in adverse sediment and erosion impacts on the foreshore

d. ensure new development includes appropriate measures to minimise the impacts of urban heat island effects

e. ensure potential impacts of sea level rise are addressed

f. ensure new development does not impact on any heritage items, heritage conservation areas, Aboriginal objects, or Aboriginal places of heritage significance

Many properties in the Castle Cove/Middle Cove, Castlecrag, Northbridge, and Chatswood West localities are zoned C4 Environmental Living. A brief description of the unique characteristics and desired future character of these localities is in Attachment 1 in this part.

1.3 Checklist for development applications

Applications for development in the C4 zone must include a completed compliance checklist from Attachment 2.

2 Performance criteria

2.1 Further performance criteria for development in the C4 zone

The performance criteria for residential development in Part B, Section 2 of this plan apply to residential development in the C4 Environmental Living Zone. However, to provide for low-impact development, prevent overdevelopment of a site, and reduce the impacts of urban heat island effects, the performance criteria below also apply.

2.1.1 Site area and lot dimensions

The site area and lot dimensions should:

a. have regard to any unique topographical features and retention of established trees and vegetation that provide a continuous wildlife corridor, and any other natural features such as rock outcrops

2.1.2 Setbacks

Setbacks should:

a. be maintained from the street frontage and bushland having regard to established building lines of dwellings in the immediate vicinity

b. reinforce the streetscape and urban bushland character of the locality

c. maintain adequate separation between any new development and the canopy of established trees

d. be sufficient to allow adequate areas for deep soil zones

2.1.3 Design

The design should:

a. integrate with the natural features of the locality and topography of the site

b. maintain view sharing to the foreshore, bushland and waterways from surrounding properties

c. not visually dominate the site due to height or bulk when viewed from waterways and public vantage points

d. ensure the type of materials for buildings and ancillary structures are sensitive to the character of the area when viewed from the street, waterways, and public open spaces

e. ensure two storey walls greater than 6m in length provide articulation to reduce the bulk of the building

f. ensure the external colour of buildings (including roofs) in areas adjacent to the foreshore and bushland use natural recessive colours with a minimum solar absorption of 0.55 under the Building Code of Australia

2.1.4 Landscaping

Landscaped areas should:

a. retain significant trees and natural features such as rock formations, watercourses, and cliff escarpments

b. provide adequate deep soil zones to accommodate large trees

c. retain and plant trees with wide tree canopies to reduce the impacts of urban heat island effects

The landscaped area controls for the C4 zone are provide in Attachment 2.

2.1.5 Private open space

Private open space should:

a. integrate with any natural features and topography of the site

2.1.6 Private recreation facilities

Private recreation facilities should:

a. be constructed to integrate with the natural environment and topography of the land

b. ensure adjoining bushland and foreshore areas are not impacted by stormwater drainage

c. be suitably screened to ensure they are not visible from the foreshore, waterways, or other public vantage points

2.1.7 Incline passenger lifts (inclinator)

a. An application for an inclinator must include the following information:

- an acoustic assessment report prepared by a suitably qualified acoustic engineer (who is a member of either the Australian Acoustical Society or the Association of Australian Acoustical Consultants) demonstrating the noise level from the inclinator will not exceed 5D(b)A above background noise when measured from the nearest property boundary.
- details of noise attenuation measures (including the design of the inclinator track and how any metal-to-metal noise will be minimised) and on-going maintenance of the inclinator
- inclinators that can be seen from a neighbouring property or a public place, including the foreshore and waterways, must:
- be situated as close as possible to the existing ground level and no closer than 2m from any side boundary
- be an open design with a maximum height of 1m above the carriage floor level
- be painted in a colour that blends in with the natural environment
- have dense planting below and adjacent to the inclinator to reduce the visual impact of the rail alignment
- not result in excessive excavation or cut through rock shelves or outcrops
- not result in the loss of significant trees
- have a privacy screen where there is a direct view within 4.5m to a window of a habitable room of another dwelling

Note:

- The SEE must include an analysis demonstrating the inclinator and any associated stairs will not result in any significant visual impacts to neighbours or from a public place.
- The installation of an inclinator must comply with the AS1735 – Lifts, escalators and moving walks, and the requirements of WorkCover NSW. Evidence of this shall be provided by the installer in the form of a signed Certificate of Compliance, which must be submitted with the development application.

2.1.8 Storage areas and plant rooms

The objective is to reduce excessive areas nominated as storage area and/or plant rooms, which could be subsequently used for habitable purposes. The additional floor area often contributes to excessive excavation and additional bulk.

a. To avoid excessive excavation and additional bulk, the combined below ground floor storage areas and plant/mechanical service rooms are to have a maximum gross floor area of 5% of the total allowable floor area for a dwelling house.

2.1.9 Solar access

The objective is to protect residential amenity and maximise energy efficiency of buildings. Developments should maintain and provide a reasonable level of solar access to both residents of the development and adjoining properties by:

a. avoiding overshadowing to living areas and private open spaces

b. planting deciduous trees in appropriate locations to maximise winter sun

c. exploring alternative design options.

Note: • Shadow diagrams may need to be submitted to show that an acceptable level of solar access can be achieved for both residents of the development and adjoining properties.

Solar access requirements to be met include:

- a minimum of 3 hours of solar access must be provided for residents of the development between 9am and 3pm on 21 June to living areas, such as family rooms, rumpus, lounge/dining, kitchen, and the principal open space and recreational areas
- a minimum of 3 hours of solar access should be maintained to adjoining properties between 9am and 3pm on 21 June to living areas, such as family rooms, rumpus, lounge/dining, kitchen, and the principal open space and recreational areas.

Note: if the adjoining property already has less than 3 hours of solar access between 9am and 3pm on 21 June to living areas and the principal open space, every reasonable attempt should be made to provide a minimum of 3 hours of solar access).

2.2 Controls

Residential developments in the C4 zone must comply with:

- the floor space ratio (FSR) under Clause 4.4A in Willoughby Local Environmental Plan (Willoughby LEP) and the landscaped area controls in Section 3.2 under Part B of the DCP
- the required number of car parking spaces under Part F (Transport and Parking Management) in this plan.

This plan also adopts many of the controls in State Environmental Planning Policy (Exempt and Complying Development Codes) 2008 (Codes SEPP) in:

- Part 3 Housing Code
- Part 3B Low Rise Housing Diversity Code
- Part 4 Housing Alterations Code.

Notable exceptions include setback and landscaping controls.

The controls for dwellings and ancillary structures are in the compliance checklist in Attachment 2 of this part.

Any variation of the numerical controls must be justified. It must consider the performance criteria in Part B Section 2 of this plan, and the further performance criteria in Section 2.1 of this part.

Notes:

- Some minor works may be permitted in the C4 zone under the Housing Alterations Code. It is up to the proponent to be satisfied that any works or demolition fully comply with the Codes SEPP.

3 Heritage items and heritage conservation areas

Several properties zoned C4 comprise a heritage item and/or are within a heritage conservation area. Willoughby LEP and Part H of this plan provide controls to ensure alterations and additions do not adversely impact on heritage items. The controls also ensure any new development is in keeping with the character of heritage conservation areas.

Any residential development in the heritage conservation areas must satisfy the provisions of this part and Part H in this plan. If there are any inconsistencies between this part and Part H, the requirements under Part H prevail.

4 Aboriginal heritage

Willoughby City Council is committed to the protection and conservation of Aboriginal objects and Aboriginal places of heritage significance. Many of these objects and sites are located on land zoned C4, and on other adjoining environmentally sensitive land, particularly around the foreshore.

This plan does not identify Aboriginal objects or Aboriginal places of heritage significance. If a development is likely to disturb any natural bushland or geological features, applicants must contact Willoughby City Council to determine if a site comprises or is in proximity to an Aboriginal object or Aboriginal place of heritage significance. We will advise applicants if an archaeological assessment report is needed as part of the development application.

5 Development adjoining bushland

The provisions of State Environmental Planning Policy (Biodiversity and Conservation) 2021 applies to development on land adjoining land that is ‘bushland zoned or reserved for public open space purposes’. The provision applies to development on land adjoining land zoned C2 Environmental Conservation and RE1 Public Recreation under Willoughby LEP.

Applications for works on land adjoining bushland should consider Willoughby City Council’s Urban Bushland Plan of Management 2014, Chapter 5 (Urban Impacts). This is at willoughby.nsw.gov.au.

6 Foreshore building line

Foreshore building lines are fixed on certain land that fronts a bay, river, creek, lake, lagoon or the harbour. They ensure development does not detract from the natural topographical features of foreshore areas, including rock outcrops, escarpments and native vegetation. Willoughby LEP contains maps showing the foreshore building lines.

Private properties with a frontage to the foreshore contain 'modified foreshore land'. The foreshore building lines on these properties coincide with the general position of buildings and significant structures along the foreshore. They encourage the protection and regeneration of land within the foreshore and prevent new structures being built forward of these lines.

Limited development is permissible in the foreshore area under Clause 6.4 in Willoughby LEP. The design and siting of any development permitted below the foreshore building line must:

a. maintain the natural landscape character and features of the foreshore, including rock outcrops, bushland, and cliff faces

b. protect the visual amenity, including views from adjoining property

c. protect the scenic quality of the foreshore when viewed from waterways and the public domain

d. protect any habitat of threatened species

e. protect water catchment quality

Notes:

- Under Clause 4.3A(5) of Willoughby LEP, the height of a building on land in the foreshore area (land between the foreshore building line and the water) must not exceed 3.5m above the existing ground level on any part of the lot.
- Land that is in or abuts the foreshore area must comply with the relevant provisions under Part 6.3 of State Environmental Planning Policy (Biodiversity and Conservation) 2021.

7 Retaining walls and sea walls

These further controls apply to retaining walls and sea walls within the foreshore building line:

a. retaining walls or sea walls should not be constructed within the foreshore building line unless it can be shown that a retaining wall or sea wall is needed to prevent landslip and/or erosion of the shoreline

b. retaining walls should generally be no more than 1m above natural ground level

c. sea walls should generally be no more than 1m above the mean high water mark

d. seawalls and retaining walls must be constructed in natural materials such as sandstone blocks or timber – unfinished concrete blocks are not permitted

Note:

- Development consent is required for development on any land below the mean high water mark of any body of water subject to tidal influence, including the bed of any such water.

8 Information to be submitted with a development application

Part A of this plan details the development application process and information that may need to accompany an application.

The information below may also be required for a development application in the C4 Environmental Living Zone:

a. a flora and fauna assessment report for properties containing and/or adjacent to significant areas of bushland, which will determine if provisions in the *Threatened Species Conservation Act 1995* apply

b. an archaeological assessment report for properties containing rock outcrops or overhanging rocks and/or located adjacent to identified or potential Aboriginal objects or Aboriginal places of heritage significance

9 Controls for additional permitted land uses

9.1 Secondary dwellings and bed and breakfast accommodation

Secondary dwellings and bed and breakfast accommodation are only permitted with consent under Willoughby LEP.

Applications for secondary dwellings and bed and breakfast accommodation in the C4 Environmental Living Zone must address the performance criteria and controls for residential development in Part B Section 2 and 3 of this plan, and the performance criteria in Section 2.1 of this part.

9.1.1 Additional controls/provisions for a secondary dwelling

The maximum height of a detached secondary dwelling is 5.7m and the maximum floor area of an attached or detached secondary dwelling is 60m^2 or 10% of the total floor area of the principal dwelling, whichever is the greater. The minimum site area for a secondary dwelling is 450m^2. A secondary dwelling cannot be subdivided from the principal dwelling.

9.1.2 Additional controls/provisions for bed and breakfast accommodation

Bed and breakfast accommodation can only be provided within the principal dwelling or an approved studio that does not contain any kitchen facilities.

An application for bed and breakfast accommodation will be assessed on the individual merits of the proposal, including any likely impacts on the residential amenity of adjoining and nearby dwellings.

Note:

- The use of a dwelling as bed and breakfast accommodation changes the building class for the dwelling under the Building Code of Australia. This means there will be new fire safety and access requirements.

9.2 Seniors housing

State Environmental Planning Policy (Housing) 2021 (Housing SEPP) provides design principles and planning controls. These set aside any local planning controls that would prevent the development of housing for seniors or people with a disability.

The Housing SEPP is at State Environmental Planning Policy (Housing) 2021 - NSW Legislation.

10 Controls for specific areas in the C4 zone

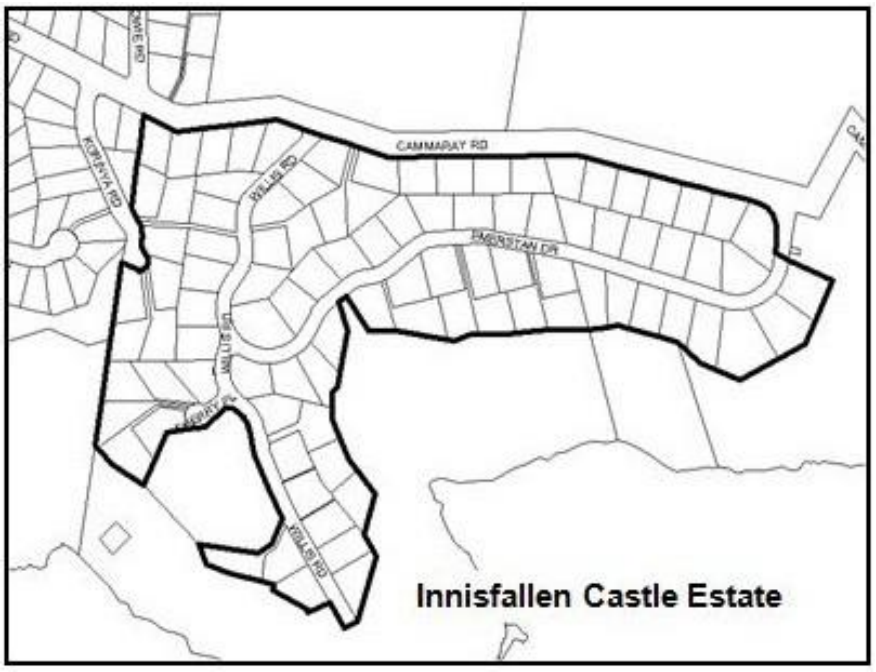
This section provides controls for any future development of specific sites and areas. These controls are further to provisions in this part and Part B of Willoughby DCP. If there are any inconsistences, the provisions in this section prevail.

10.1 Innisfallen Castle Estate

Innisfallen Castle Estate is on Castle Cove peninsula. Innisfallen Castle at 14 Cherry Place is a state significant item. The heritage values include the external fabric of the castle and curtilage, including surviving elements of the original gardens, lawns, driveway, stone edgings and wall.

Any development on adjoining or nearby properties must respect and not impact on the heritage values of the site, including views to and from Innisfallen Castle.

Map 1 Innisfallen Castle Estate



The estate is surrounded by natural bushland reserves on the foreshore of Middle Harbour. Primary and secondary building areas and height limits are identified on each property to restrict development to allow appropriate view sharing to natural bushland reserves and views over Middle Harbour.

The requirements below prevail over any other provisions in Willoughby DCP.

10.1.1 Building footprint

The erection of any building is restricted to the primary building area. Swimming pools, other minor structures such as patios and pergolas, and landscaping structures are permitted in the secondary building area. Boundary fences and walls may be permitted outside the primary and secondary building areas with approval.

10.1.2 Height of buildings

No buildings are to be erected at a height above the reduced levels specified for each individual lot. Exceptions are for chimneys, television aerials, lift towers or service installations, which may extend a further 600mm above the stated reduced levels for each allotment of land.

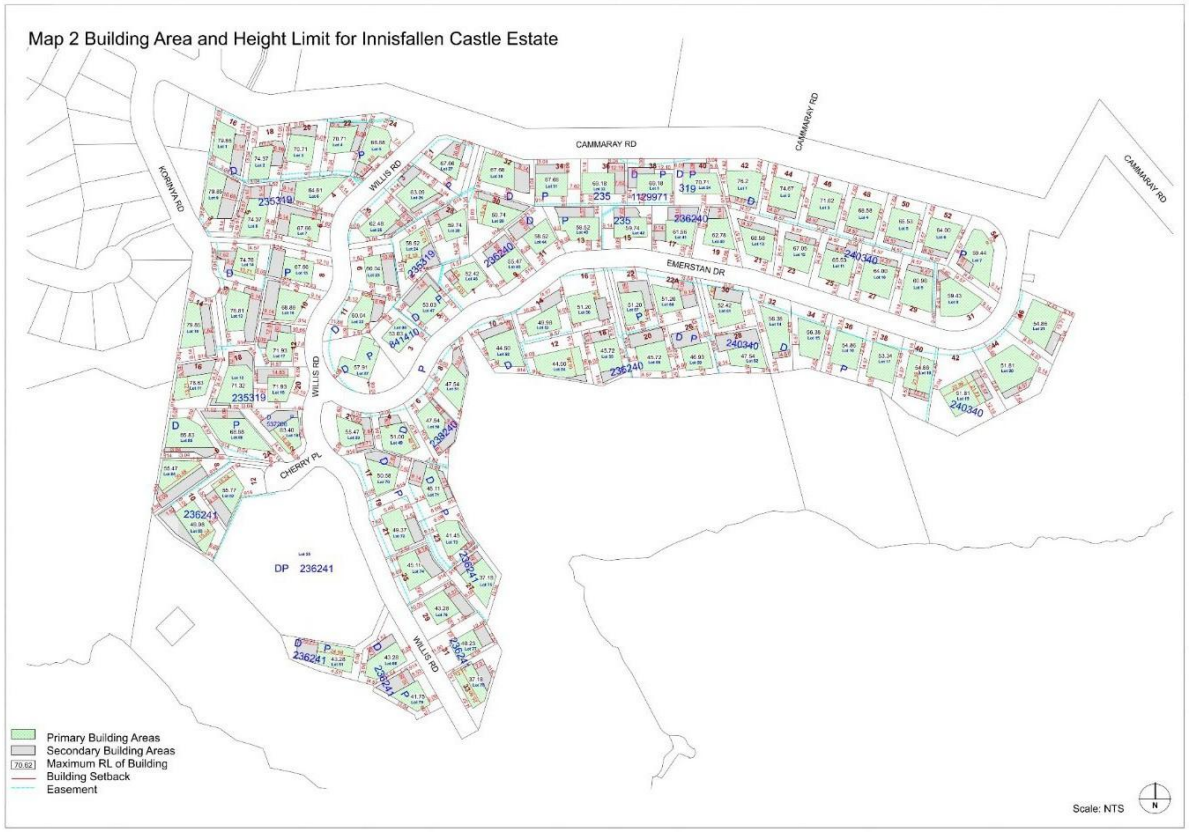
The maximum permissible height of a development is based on a conventional pitched roof design. This means:

a. the maximum height of a parapet wall is to be 1.2m less than the reduced level for each main building area

b. no roof or part of the main building is to occupy more than 50% of the air space over the area of the highest floor of the building, between the maximum permissible height of the building and at a reduced level of 2.4m below the maximum permissible height plane

The surface of a swimming pool or terraces is not to be constructed more than 1.5m above the natural finished ground level at any point.

Map 2 Building area and height limit for Innisfallen Castle Estate



10.1.3 Fencing

Fences must not be closer to the street than the building line of the dwelling or the building line for the secondary building area. Exceptions are a safety fence enclosing a swimming pool.

Fences must not consist of rough-sawn, undressed, unpainted timber or timber in its natural state.

Apart from exceptional circumstances, dividing fences must not exceed 1.8m above natural ground level.

Note:

- Willoughby City Council and adjacent property owners must agree in writing to any increase in the height of dividing fences.

10.1.4 Excavation and retaining walls

Major excavations outside the main and secondary building areas are to be kept to a minimum.

All retaining walls more than 600mm above or below the natural surface are to be within the main and secondary building areas of each site. Exceptions are walls specifically approved by Willoughby City Council.

Any works to raise or lower the surface by more than 600mm must comply with these controls:

a. retain and stabilise the face of an excavation adjacent to the external wall of the building of not more than 1.2m from the base of the embankment to the face of the building

b. construct the embankment with a slope not exceeding 4:1 and a maximum height of 3m from the base of the wall to the finished surface of the embankment

c. retain any filling adjacent to the main building to provide essential pedestrian movement to and from the building; the area must not affect the amenity of adjacent properties and should not be in the form of a terrace outside the main and secondary building area

d. the height of the surface level of the retained area is not to exceed 1.5m above natural ground level and is not to be greater than 1.5m from the face of the wall of the main building

e. where, in Willoughby City Council's opinion, the raising or lowering of the surface level does not affect the amenity of adjacent lands, a retaining wall may be up to 1.5m in height from the natural ground level to the finished surface of the wall

f. special exceptions may be granted to retaining walls for the support of excavation of fills for driveways, subject to all batters not to exceeding 1:1 in cut other than rock, and 1:1.5 in fill

Any surface adjustments should also aim to preserve existing trees.

The use of retaining walls to contain earth embankments at the street frontage is to be avoided. The only treatment permitted under this plan is the protection and beautification of the retaining wall batter.

These controls apply to the street alignment of properties in cuttings:

a. if the adjacent footpath is excavated below the surface level of the adjacent allotment, only rockeries, dwarf walls and similar treatments in natural stone are allowed along the frontage of the allotment

b. no part of the frontage is to be more than 300mm above the footpath alignment level as supplied in writing by Willoughby City Council's engineer, and no more than 450mm above the profile of the designed or specified grading for the slope of the cutting

c. the grading of the slope of the cutting is to be in solid rock at a grade of 4:1, starting from a point 300mm within the property that is at the same level as that of the adjacent footpath alignment level

d. if the adjacent footpath is not below the surface level of the adjacent allotment, the slope of the cutting is to be a grade of 1:1, starting from a point 300mm within the property that is at the same level as that of the adjacent footpath alignment level

e. the retaining wall batter may be planted with appropriate shrubs or plants providing the plants do not overhang the path; the full width of the path must be retained and free of vegetation

10.2 Tower Reserve and Linden Way Reserve

Tower Reserve and Linden Way Reserve are public open spaces that provide expansive views over Middle Harbour. To retain the outlook and views from the predominant high

points of these reserves, there are height limits for any development on the affected properties.

Map 3 Properties adjoining Tower Reserve and Linden Way Reserve

No buildings or structures are to be erected above the Australian Height Datum for the properties adjoining Tower and Linden Way Reserves. These are shown in Table 1.

Table 1 Australian Height Datum for properties adjoining Tower and Linden Way reserves

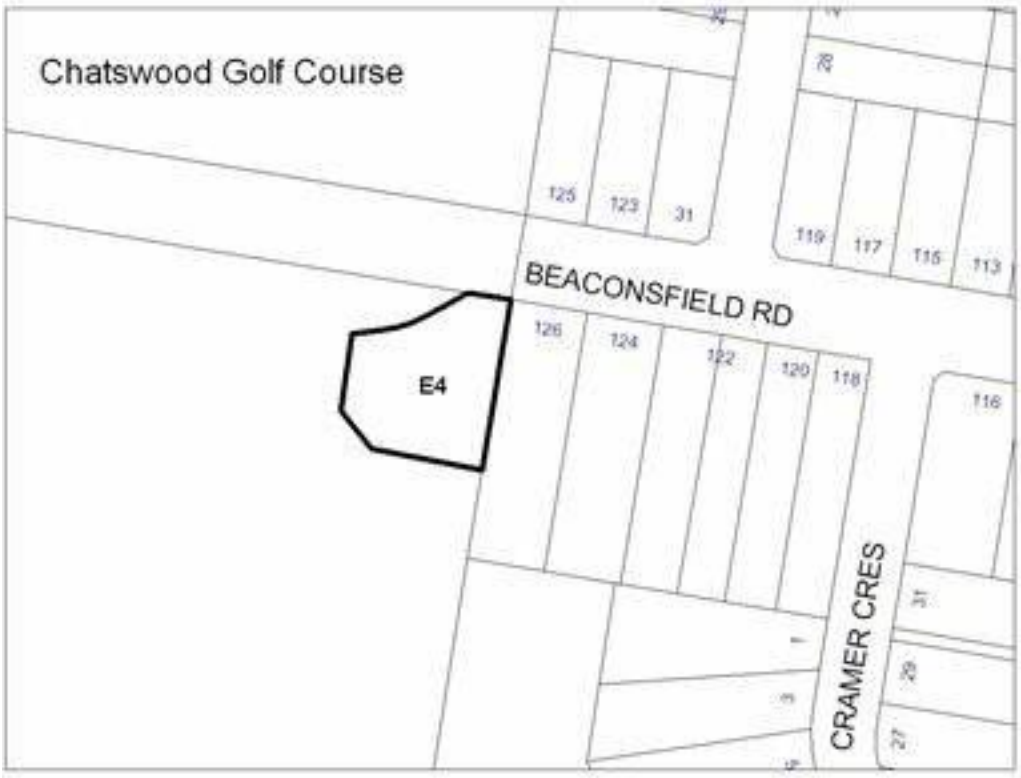
The Bulwark	**Street number**	**Australian Height Datum**
Lot 399A	No 33 (1 Tower Reserve)	85 AHD
Lot 338	No 35 (2 Tower Reserve)	90 AHD
Lot 397	No 37 (3 Tower Reserve)	90 AHD
Lot A	No 39 (4 Tower Reserve)	90 AHD
Lot B	No 41 (5 Tower Reserve)	85 AHD
The High Tor	-	-
Lot 32	No 7	88 AHD
Lot 31	No 9	84 AHD
Lot 30	No 11	84 AHD
Lot 29	No 13	86 AHD
Lot 28	No 15	84 AHD
Linden Way	-	-
Lot 27	No 25	85 AHD
Lot 26	No 27	88 AHD

Note:

- If the height is specified as a reduced level (RL) control, this refers to the RL Australian Height Datum of the uppermost point of the building. It does not include a vent, lift tower, chimney or other service installation.

11 Site specific provisions for certain land within the Chatswood Golf Club (adjacent to 126 Beaconsfield Road, Chatswood)

Map 4 Land within Chatswood Golf Club adjacent to 126 Beaconsfield Road



11.1 Objectives

The objectives of the site-specific provisions are to ensure:

a. redevelopment does not impact on bushland within the site or the immediate locality, including loss of natural vegetation and significant geological features, disruption of drainage patterns, alterations to water tables and increased bushfire hazard potential

b. the stability of the site for construction of a dwelling due to previous land fill

c. potential site contamination is considered before any development of the site due to previous landfill

d. any asset protection zones are provided within the site

e. the newly created subdivided site has access from the street and the appropriate utility services

f. a drainage easement is created over the existing reinforced concrete pipe in favour of Willoughby City Council.

11.2 Controls

These controls apply to any residential development on the subject site:

a. any development application must include a geotechnical report to confirm the stability of the site for construction of a dwelling

b. if unidentified materials are uncovered while excavating and developing the site, contamination investigations shall be done under Contaminated Land Management Act 1997 and State Environmental Planning Policy No 55

c. the development must comply with NSW Rural Fire Service requirements; Willoughby City Council will not approve any part of the Asset Protection Zone to be located outside the subject site

d. retention of trees as shown on the design concept plan for the site, which is included in this part

e. the plan of subdivision is to include:

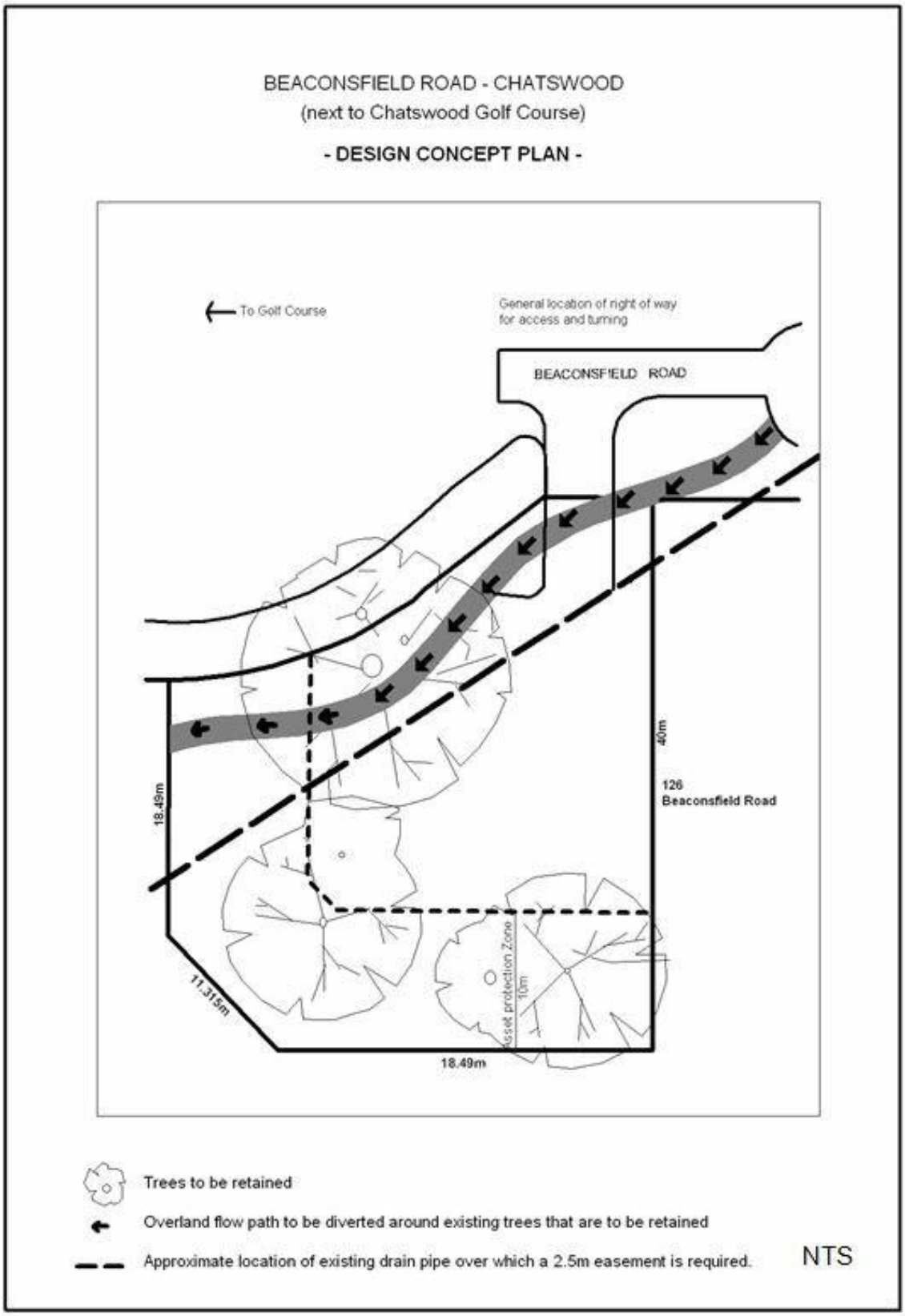
- a 5-metre-wide right of access way across Chatswood Golf Club land from the end of the Beaconsfield Road reserve to the front boundary of the proposed allotment as shown on the design concept plan; this is to benefit the proposed residential property and allow vehicles to enter and exit the site in a forward direction
- a survey of the extent of the existing 525mm diameter reinforced concrete pipe and provision of a drainage easement of 2.5m width over the existing pipe and along the allotment in favour of and dedicated at no cost to Willoughby City Council
- underground utility services from the nearest pole to the boundary of the proposed property under the relevant public utility requirements provided by the applicant

f. an unobstructed overland flow path through the property is required that directs the flow from any proposed dwelling

g. any proposed construction (including driveway and pathway access) on the site must ensure safe passage of overland flow

h. the design required as part of any application for the development of the site is to include full design details including soil erosion control, ratio of velocity and depth and the preservation of trees shown on the design concept plan. The overland flow path is to be created without extensive fill under the drip lines of the protected trees

Figure 1 Design concept plan for land within Chatswood Golf Club adjacent to 126 Beaconsfield Road



Attachment 1: Localities

This information forms the basis for managing the process of change in localities that include environmentally sensitive areas. The special qualities and attributes of these localities contribute to the pleasant living environment and amenity of the area. All proposals must consider the special qualities and attributes, and the desired future character of these localities, as part of the development application.

a. Castle Cove/Middle Cove

The localities of Castle Cove and Middle Cove extend from the gentle terrain of the inland residential areas generally east of Eastern Valley Way to the undulating terrain of the peninsula areas, which project into Middle Harbour.

Sandstone escarpments are common throughout the area, often extending some distance along the peninsula. Escarpments and sandstone tors are often visible from the harbour.

The pattern of subdivision and residential streets generally follow the contours of the land with allotments angled to obtain views of Middle Harbour and surrounding bushland. Allotments are generally large with wide frontages.

The escarpment areas are characterised by bands of dwellings with interposing layers of dense bushland vegetation contributing to the high scenic quality of the area. The character of the area is of buildings set within a natural landscape setting.

Dwellings are generally stepped to respond to the topography of the land. Traditionally, dwellings built on the low side of the street are not more than 1 storey above the roadway, and do not obstruct the views of dwellings located on the high side of the street or vistas of distant views from the roadway.

Dwellings on the high side of the street above the roadway obtain views over the dwellings below. These properties generally retain rock outcrops and natural landscaping along the street frontage.

The natural landscape comprises steeply wooded foreshore areas rising from Middle Harbour to the ridges and escarpments with species of scribbly gums, red bloodwood, black she-oak and banksia. Herbs and grasses are found on the ridgetops.

The fauna habitat of Harold Reid Reserve and the Explosives Reserve is regarded as high quality. The adjoining bushland and rear yard areas, particularly along the foreshore areas, are important corridors for fauna.

These principles should be included in the design of a proposed development to achieve the desired future character of the Castle Cove/Middle Cove area:

1. maintain the traditional pattern of setback from side boundaries to ensure good separation between dwellings and allow vistas from the roadway
2. integrate dwellings and ensure the built form is subservient to the predominant natural landscaped areas on the site
3. ensure the built form relates to the topography of the site by following the contours of the land

4. avoid dwelling heights and building elements that break the ridgeline of the peninsula areas, and ensure they are not prominent when viewed from waterways or other areas of the public domain

b. Castlecrag

Castlecrag is located partly on an undulating peninsula, which extends eastward into Middle Harbour. Large sandstone tors and overhangs are common along escarpments and are quite often visible from the harbour.

The pattern of subdivision and residential streets south of Edinburgh Road follow the contours of the land and is a legacy of Walter Burley Griffin's vision for a model residential garden suburb, designed to take full advantage of its topography and natural features. Roads are narrow and curvilinear, responding to the natural terrain with front boundaries following the curvature of the carriageway.

A central portion of the Castlecrag peninsula is the Griffin Conservation Area. Development within this area must also refer to the provisions under Part H of Willoughby DCP.

The residential area outside the Castlecrag Conservation Area comprises a variety of contemporary architectural and older style dwellings on allotments with wide frontages.

Due to the curvilinear nature of many of the streets, the front building alignment is variable, responding to the landform. The absence of kerb and gutters in many streets contributes to the almost semi-rural sense of houses sited in a bushland setting. Rocky street islands with native vegetation are a feature of the Griffin Conservation Area. The inland areas with a regular pattern of subdivision have moderately deep setback areas of informally landscaped gardens. Low masonry or palisade fencing, rockery or shrubs provide demarcation of the front boundary with the abutting grassed verges.

Traditionally, dwellings on the low side of the street are not more than 1 storey above the roadway and do not obstruct views of dwellings located on the high side or vistas of distant views from the roadway. The curvilinear residential streets following the topography of the land provide glimpses of water views and of the wooded hillsides and ridges beyond.

Dwellings are generally stepped down the site, responding to the topography of the land. The residential character of the area is generally low scale, using predominantly masonry or natural stone materials. Front gardens are informally landscaped with a predominance of native vegetation and tree cover in the area with private gardens merging into one another with minimal fences and walls fronting the street.

Remnant bushland is present along much of the Castlecrag foreshore, including the northern escarpment, providing an almost continuous stretch of bush which extends north through to Middle Cove and Castle Cove.

The vegetation in the locality is known as the Sydney Sandstone Complex. On the slopes the structure is open forest/woodland in which the dominant canopy trees are Sydney red gum, Sydney peppermint, silver top ash and red bloodwood, with a tall understorey of black she-oak and blueberry ash. Common understorey plants include Christmas bush, old man banksia, heath-leaved banksia, broadleaved and pine-leaved gum and broad-leafed hakea.

On the ridges, Sydney sandstone ridgetop woodland is characterised by scribbly gum and kunzea. In the deep gullies Sydney red gums form an emergent canopy over closed forest species, tall coachwood, lilly pilly and black wattle. Close to more exposed waterways, coastal banksia and beside sugarloaf bay, swamp oak are found.

These principles should be included in the design of a proposed development to achieve the desired future character of the Castlecrag area:

1. integrate dwellings and ensure the built form is subservient to the predominant natural landscaped areas on the site
2. ensure the built form relates to the topography of the site by following the contours of the land
3. use a variety of setbacks for site buildings to create a varied rather than regular composition within the streetscape to enhance amenity and privacy
4. avoid dwelling heights and building elements that break the ridgeline of the peninsula areas, and ensure they are not prominent when viewed from waterways or other areas of the public domain
5. use stone edging, low retaining walls and rockeries and a variety of native and exotic ground covers in landscaped gardens
6. ensure the siting and design of dwellings minimise the obstruction of views from neighbouring dwellings and vistas from roadways or public open spaces

c. Northbridge

Geologically, the peninsula area of Northbridge is Hawkesbury sandstone. The terrain slopes into Middle Harbour and is concave in form, becoming steeper as the slope recedes from the ridgeline. Large sandstone tors and overhangs are common along the escarpments, some of which are visible from Middle Harbour.

Recent development in Northbridge, particularly along foreshore properties, is characterised by more modern/contemporary dwellings. In many instances, these new dwellings have transformed the traditional character of building styles in the locality. Many of the newer dwellings are large and imposing, especially when viewed from the street and waterway.

Residential subdivisions and street patterns at the foreshore of the peninsula have been aligned to follow the contours of the land, angled to provide views of Middle Harbour and surrounding bushland. Allotments are larger with wide frontages of 15m to 25m. The escarpment areas are characterised by bands of dwellings with interposing layers of bushland vegetation. Dwellings fronting the harbour are generally contemporary in style with many presenting as 2 to 3 stepped levels when viewed from the water.

Dwellings built on the low side of the street are generally set below the road level and do not obstruct the view of dwellings located on the high side of the street or vistas of distant views from the roadway.

The vegetation of the reserves and foreshore areas of the peninsula is remnant bushland consisting of open forest/woodland structure. Dominant tree species include Sydney peppermint, red bloodwood and rusty gum.

The banding of natural vegetation between the line of buildings on the south-facing escarpment of Northbridge from Tunks Park to Ulric Lane is a distinctive quality of the area.

These principles should be included in the design of a proposed development to achieve the desired future character of the Northbridge area:

1. integrate dwellings and ensure the built form is subservient to the predominant natural landscaped areas on the site
2. maintain adequate setback from side boundaries to ensure separation between dwellings and allow vistas from the roadway
3. provide low fencing and walls (including retaining walls) on the street frontage; avoid high fences, walls, unbroken double garages, and high gates, particularly along the streetscapes which have shallow setbacks
4. avoid dwelling heights and building elements that break the ridgeline of the peninsula areas, and ensure they are not prominent when viewed from waterways or other areas if the public domain

d. Chatswood West and Lane Cove North

The Chatswood West and Lane Cove North area is characterised by prolific and large expanses of natural landscaped reserves and creeks, comprising Blue Gum Creek and Reserve to the north, Mowbray Park Reserve to the west. The reserves along Swaines Creek of OH Reid Memorial Park, Ferndale Park, Campbell Park and Coolaroo Park also contribute to the environmental quality of the locality.

The pattern of subdivision is generally regular but follows the topography of the land. Dwellings comprise a mixture of styles, predominantly bungalows from the 1920s to 1960s period.

The western edge of the locality has an important link to the Lane Cove River in terms of open space reserves and potential impacts of development on water quality, foreshore, flora and fauna, as well as the scenic quality of the river area and its tributaries.

Dwellings located on gentle terrain are sited on subdivisions of regular grid patterns, having a consistent front building alignment with moderately deep front setback areas of informally landscaped well vegetated gardens with wide grassed verges, set behind low masonry or palisade fencing of not more than 1m in height. Setbacks from side boundaries are consistent, varying from 1m to 3m in width, with garages or carports located to the side of the dwelling.

The vegetation of the reserves is Sydney sandstone gully forest. At Blue Gum Reserve, species of coachwood, water gum, blackbutt and turpentine are located to the eastern side and to the slopes north of Blue Gum Creek. Peppermint gum, angophora and red bloodwood are found to the south of Blue Gum Creek.

At O.H. Reid Reserve, the predominant species are peppermint gum and turpentine. At Ferndale Reserve, Campbell Park and Coolaroo Reserve are species of blackbutt and grey ironbark. Species in Mowbray Park are peppermint gum and angophoras with areas of blackbutt and grey ironbark. Threatened species in the area include the remnant blue gum high forest.

There are stands of significant trees on private properties such as at the top of Edgar Street, which contribute to the vegetated character of the locality. In the Jenkins/Edgar Street area there are significant stands of jacarandas which when in blossom create a blanket of mauve between the evergreen trees.

The Blue Gum Conservation Area is in the Chatswood West locality. Development within this area must refer to the provisions under Part H of Willoughby DCP.

These principles should be included in the design of a proposed development to achieve the desired future character of the Chatswood West and Lane Cove North area:

1. maintain the pattern of the existing streetscape of moderately deep setbacks to the street and setbacks from side boundaries to ensure good separation between dwellings and to allow vistas from the roadway
2. ensure the built form relates to the topography and natural features of the site, such as rocky outcrops, significant trees, and natural vegetation
3. integrate infill development and additions (including contemporary styles) to ensure they are in harmony with the scale, form, massing, and external materials of surrounding dwellings in the streetscape
4. ensure the siting and design of dwellings minimise the obstruction of views from neighbouring dwellings and vistas from roadways or public open spaces to valley views and urban bushland, particularly on the low side of a street
5. maintain a streetscape of low masonry or palisade fencing (including retaining walls) up to 1.2m in height to the street; avoid solid high fences, walls, gates, and double garages to the street, unless the dwelling has frontage to a busy road
6. avoid imposing building forms of high wall facades, particularly when close to the streetscape, and avoid high fencing to streets with shallow setbacks

Attachment 2: Controls for development in the C4 Environmental Living Zones

This attachment includes controls to ensure residential and other permitted land uses result in low impact developments in environmentally sensitive locations.

For any development with an estimated cost of $50,000 or more, a suitably qualified person must certify the proposed development complies with all relevant provisions of Willoughby LEP and the relevant numerical controls. This includes those adopted from Part 3, Part 3B (as it applies to dual occupancies) and Part 4 of Codes SEPP.

The table details:

a. Development standards under Willoughby LEP.

b. General controls for dwellings, dual occupancies and attached development.

c. Amenity development standards.

d. Additional development controls for detached development.

e. Additional development controls for detached studios.

f. Development controls for swimming pools.

g. Development controls for fences.

h. Development controls for associated works (earthworks, retaining walls, drainage, protection of walls, and protection of trees).

i. Additional controls for dual occupancies.

Note:

- A suitably qualified person includes a practising registered architect, a practising qualified and accredited building designer, a registered building surveyor and a certified practicing planner (CPP).

a. Development standards under Willoughby LEP

Description	Proposed	Control	Yes	No	N/A
1. Height of building (Clause 4.3)	-	8.5m	-	-	-
2. Maximum floor space ratio (FSR) (Area 2 - Clause 4.4A)		-	-	-	-
Lot area	-	-	-	-	-
<400m^2		0.50:1			
≥400m^2 and <500m^2		0.44:1			
≥500m^2 and <600m^2		0.40:1			
≥600m^2 and <700m^2	-	0.37:1	-	-	-
≥700m^2 and <800m^2	-	0.35:1	-	-	-
≥800m^2 and <900m^2	-	0.32:1	-	-	-
≥900m^2 and <1,000m^2	-	0.30:1	-	-	-
≥1,000m^2 and <1,100m^2		0.28:1			
≥1,100m^2 and <1,120m^2		0.27:1			

Description	Proposed	Control	Yes	No	N/A
≥1,200m^2 and <1,300m^2		0.26:1			
>1,300m^2	-	0.25:1	-	-	-
3. Maximum floor space ratio (FSR) (Area 8 - Clause 4.4A)					
Lot area					
<400m^2		0.45:1			
≥400m^2 and <500m^2		0.40:1			
≥500m^2 and <600m^2		0.36:1			
≥600m^2 and <700m^2		0.34:1			
≥700m^2 and <800m^2		0.32:1			
≥800m^2 and <900m^2		0.30:1			
≥900m^2 and <1,000m^2		0.28:1			
≥1,000m^2 and <1,100m^2		0.26:1			
≥1,100m^2 and <1,130m^2		0.25:1			
>1,300m^2		0.24:1			
4. Minimum lot size (Clause 6.10)	-	-	-	-	-
Attached dual occupancy	-	700m^2	-	-	-
Detached dual occupancy	-	900m^2	-	-	-

b. General development controls for dwellings, dual occupancies and attached development

Notes:

- A concept subdivision layout must be submitted with an application for dual occupancy development.
- The site area in the compliance checklist refers to the individual lot that may be created by subdivision for each dual occupancy dwelling.

Description	Proposed	Control	Yes	No	N/A
1. Landscaped areas	-	-	-	-	-
Lot area:	-	-	-	-	-
Under 400m^2	-	0.35 × site area	-	-	-
400m^2 – 600m^2	-	(0.5 × site area) - 60m^2	-	-	-
601m^2 – 1000m^2	-	(0.65 × site area) - 150m^2	-	-	-
1001m^2 – 1500m^2	-	(0.65 × site area) - 150m^2	-	-	-
>1501m^2	-	0.5 × site area	-	-	-

Description	Proposed	Control	Yes	No	N/A
Notes: To be included in the calculation for landscaped area the minimum dimension must not be less than 2m The landscaped area should include a minimum 30% deep soil zone. The remainder of the landscaped area may be 'soft' landscaping having a minimum depth of 600mm.	-	-	-	-	-
Minimum landscaped area behind the building line	-	50% of the total landscaped area	-	-	-
Minimum deep soil zone forward of the building line on a lot 18m or more wide	-	70% of the area forward of the building line	-	-	-
Minimum deep soil zone forward of the building line on a lot less than 18m wide	-	50% of the area forward of the building line	-	-	-
2. Construction of basement					
Lot width:	-	-	-	-	-
6m – 10m	-	Not exceed 25m^2	-	-	-
> 10m	-	Not exceed 45m^2	-	-	-
Note: The basement may be used for car parking and include storage areas and/or plant/mechanical service rooms. However, the combined below ground storage areas and/or plant/mechanical service rooms must not exceed 5% of the total allowable area for a dwelling house (see section 2.1.8 of this Part).					
3. Primary road setback	-	-	-	-	-
Up to 1500m^2	-	Average of 2 adjoining dwellings. Otherwise: 7m	-	-	-
>1500m^2	-	10m	-	-	-
4. Side setbacks	-	-	-	-	-
• One storey	-	1.5m	-	-	-
• Two or more storeys	-	2.5m	-	-	-
5. Setbacks to rear boundaries and the foreshore building line Note: the rear setback applies to attached and detached structures	-	Average of 2 adjoining dwellings. Otherwise: 15m minimum	-	-	-
6. Building envelope	-	Within 45^o from a height of 3.5m above existing ground level along the boundary of the site	-	-	-
7. Secondary road setbacks for corner lots	-	-	-	-	-
Lot size:	-	-	-	-	-

Description	Proposed	Control	Yes	No	N/A
Up to 600m^2	-	3m	-	-	-
600m^2 – 1500m^2	-	5m	-	-	-
>1500m^2	-	8m	-	-	-
8. Minimum front setback to a classified road (including any detached development)	-	9m	-	-	-
9. Minimum setback to a public reserve (including any detached development)	-	3m	-	-	-
10. Front setback for battle axe lot	-	3m	-	-	-
11. Minimum side and rear setbacks for aerials, awnings, air conditioners, and the like Note: minimum setbacks do not apply to downpipes, driveways, electricity or gas meters, gutters, and the like	-	1m	-	-	-
12. Maximum height of balconies, decks, patios, terraces and verandahs attached to the side and rear of dwelling	-	Floor level above existing ground level	-	-	-
Setback:	-	-	-	-	-
Up to 3m	-	2m	-	-	-
3m – 6m	-	3m	-	-	-
>6m	-	4m	-	-	-
13. Total floor area of all attached side and rear balconies, decks, patios, terraces, and verandahs within 6m of the boundary and more than 2m above the existing ground level	-	12m^2	-	-	-
14. Deep soil zones	-	Minimum 30% of the required landscaped area	-	-	-
15. Minimum length and width of landscaped areas (including deep soil zones) Note: if the landscaped area does not meet the minimum dimension, that area is excluded from the minimum requirement.	-	2m	-	-	-
16. Minimum landscaped area behind the building line	-	50% of the total required landscaped area	-	-	-
17. Minimum deep soil zone forward of the building line on a lot at least 18m wide	-	70% of the area forward of the building line	-	-	-
18. Minimum deep soil zone forward of the building line on a lot less than 18m wide	-	50% of the area forward of the building line	-	-	-

Description	Proposed	Control	Yes	No	N/A
19. Minimum principal private open space	-	60m^2	-	-	-
20. Minimum length and width of principal private open space	-	3m	-	-	-
21. Maximum grade of principal private open space	-	1:50	-	-	-

c. Amenity development standards

Note:

- Building elements include: an entry feature or portico; a balcony, deck, pergola, terrace or verandah; a window box treatment; a bay window or similar feature; an awning or other feature over a window; a sun shading feature; and an eave.

c.1 Building design	**Yes**	**No**	**N/A**
Note: • does not apply to a dwelling on a battle-axe lot			
1. A dwelling must contain at least 1 door and 1 window to a habitable room at ground floor level facing the primary road.	-	-	-
2. A dwelling with a setback from a primary road of at least 3m may have an articulation zone that extends up to 1.5m forward of the minimum required setback from the primary road.	-	-	-
3. Building elements may be located within the articulation zone providing the maximum total area of all building elements in the articulation zone (other than an awning or other feature over a window, a sun shading feature or an eave) must not comprise more than 25% of the area of the articulation zone.	-	-	-
4. A building element on a dwelling (other than an entry feature or portico that has the same pitch as the roof on the dwelling house) must not extend more than 1m above the gutter line of the eaves of a single storey dwelling house, or above the gutter line of the eaves of a 2 storey dwelling house.	-	-	-
5. A dwelling on a corner lot must have a window to a habitable room with an area of at least 1m^2 that faces and is visible from the secondary road.	-	-	-
6. A dwelling with a setback from a secondary road of not more than 4.5m must have at least one of the following building elements for a minimum length of 20% of the elevation of the walls that face the secondary road and that are within 4.5m of the secondary road:	-	-	-
• an entry feature or portico	-	-	-
• a balcony, deck, pergola, terrace or verandah	-	-	-
• a bay window	-	-	-
• a step of at least 600mm in depth	-	-	-

c.1 Building design	**Yes**	**No**	**N/A**
7. Building elements listed above may be located in a secondary road articulation zone if the zone extends no more than 1m into the minimum required setback area and spans the length of the walls that face the secondary road, and if the building element comprises no more than 20% of the zone area.	-	-	-
8. Any part of a gable or hipped roof that overhangs walls that are within 4.5m of the secondary road boundary must include eaves that extend for the length of those walls and project at least 450mm, but not more than 1m from those walls.	-	-	-
c.2 Privacy screens	-	-	-
1. A privacy screen must be provided for any part of a window to a habitable room that is less than 1.5m above the finished floor level of that room if:	-	-	-
• the window faces and is less than 3m from a side or rear boundary and the room has a finished floor level of more than 1m above existing ground level, or	-	-	-
• the window faces and is at least 3m, but not more than 6m, from a side or rear boundary and the room has a finished floor level of more than 3m above existing ground level.	-	-	-
Note: • does not apply to a bedroom window that has an area of not more than $2m^2$	-	-	-
2. A privacy screen of at least 1.7m, but not more than 2.2m, above the finished floor level of a balcony, deck, patio, terrace or verandah must be installed at the edge of that part of the balcony, deck, patio, terrace or verandah that is parallel to, or faces towards, the relevant side or rear boundary if the area of the balcony, deck, patio, terrace or verandah is at least 3m2 and:	-	-	-
• that edge is less than 3m from a side or rear boundary and the balcony, deck, patio, terrace or verandah has a finished floor level of more than 1m above existing ground level, or	-	-	-
• that edge is at least 3m, but not more than 6m from a side or rear boundary and the balcony, deck, patio, terrace or verandah has a finished floor level of more than 2m above existing ground level.	-	-	-
3. A privacy screen must be provided for any part of a window in a detached studio that is less than 1.5m above the finished floor level of that room if the window faces and is less than 3m from a side or rear boundary and:	-	-	-
• the room has a finished floor level more than 1m above existing ground level, or	-	-	-

c.1 Building design	Yes	No	N/A
• the window faces and is at least 3m, but not more than 6m from a side or rear boundary and the room has a finished floor level more than 3m above existing ground level .	-	-	-
c.3 Car parking and access All car parking and vehicular access must comply with relevant provisions of Part F (Transport and car parking management) of Willoughby DCP. The specific controls below apply to dwelling houses.	-	-	-
1. The off-street car parking space for a battle-axe lot must be constructed so that vehicles can leave the lot in a forward direction.	-	-	-
2. An attached garage, carport or car parking space accessed from a primary road must have a minimum setback 5.5m if the dwelling is setback less than 4.5m or 1m behind the building line if the dwelling is setback 4.5m or more.	-	-	-
3. The maximum width of all garage door openings facing a primary or secondary road is 3.2m for a lot of 8m – 12m and 6m for a lot greater than 12m.	-	-	-
4. Secondary road setback for garage or carport	-	-	-
Up to 600m^2 : 2m	-	-	-
600m^2 – 1500m^2 : 3m	-	-	-
>1500m^2 : 5m	-	-	-
5. Rear setback for garage or carport	-	-	-
Up to 900m^2 : 900mm	-	-	-
>900m^2 – 1500m^2 : 1.5m	-	-	-
>1500m^2 : 2.5m	-	-	-

d. Additional development controls for detached development

Notes:

- Detached development includes: a deck, patio, pergola, terrace or verandah; a cabana, cubby house, fernery, garden shed, gazebo or greenhouse; a carport or garage, and a shed.
- The setback of detached development from a primary or secondary road (other than a detached garage or carport) must be behind the building line of the dwelling.

Description	Proposed	Control	Yes	No	N/A
1. Maximum height above existing ground level	-	4.5m	-	-	-
2. Maximum gross floor area (GFA) for all detached development:	-	-	-	-	-
Lot size:	-	-	-	-	-
Up to 300m^2	-	36m^2	-	-	-
>300m^2 – 600m^2	-	45m^2	-	-	-
>600m^2 – 900m^2	-	60m^2	-	-	-

Description	Proposed	Control	Yes	No	N/A
>900m^2	-	100m^2	-	-	-
Note: the maximum overall gross floor area (GFA) includes the dwelling, attached and detached buildings	-	-	-	-	-
3. Side setbacks for all detached development, including above ground rainwater tanks and shade structures (additional side setbacks apply to detached studios):	-	-	-	-	-
Lot width:	-	-	-	-	-
Up to 18m	-	900mm	-	-	-
>18m – 24m	-	1.5m	-	-	-
>24m	-	2.5m	-	-	-
4. Height above existing ground level and rear setbacks for detached decks, patios, pergolas, terraces and verandahs	-	-	-	-	-
5. Maximum finished floor level	-	600mm	-	-	-
6. Rear setbacks:	-	-	-	-	-
Lot size:	-	-	-	-	-
Up to 900m^2	-	900mm	-	-	-
>900m^2 – 1500m^2	-	1.5m	-	-	-
>1500m^2	-	2.5m	-	-	-

e. Additional development controls for detached studios

Description	Proposed	Control	Yes	No	N/A
1. Height (within 900mm of a lane and above a garage)	-	6m	-	-	-
2. Rear setbacks	-	-	-	-	-
Lot size:	-	-	-	-	-
Up to 900m^2	-	900mm	-	-	-
>900m^2 – 1500m^2	-	1.5m	-	-	-
>1500m^2	-	2.5m	-	-	-
3. Maximum floor area	-	-	-	-	-
Lot size:	-	-	-	-	-
Up to 350m^2	-	20m^2	-	-	-
>350m^2	-	36m^2	-	-	-
Note: the maximum overall GFA includes detached studios	-	-	-	-	-
4. Side and rear setbacks	-	-	-	-	-
Lot width:	-	-	-	-	-
Up to 18m	-	900mm	-	-	-
>18m	-	1.5m	-	-	-

f. Development controls for swimming pools

Description	Yes	No	N/A
1. Coping around a swimming pool must not be more than 1.4m above existing ground level, and 300mm wide if the coping is more than 600mm above existing ground level.	-	-	-
2. Decking around a swimming pool must not be more than 600mm above existing ground level.	-	-	-
3. A swimming pool must be located behind the building line of the dwelling house.	-	-	-
4. The swimming pool water line must have a setback of at least 1m from a side or rear boundary.	-	-	-

g. Development controls for fences

Description	Yes	No	N/A
1. A fence erected behind the building line (except on a secondary street frontage) must:	-	-	-
• not be higher than 1.8m above existing ground level	-	-	-
• if it includes an entrance gate, not have a gate that opens outward	-	-	-
• if it is constructed of metal components, be of low reflective, factory pre-coloured materials	-	-	-
• if it is on a sloping site and stepped to accommodate the fall in the land, be no higher than 2.2m above existing ground level at each step	-	-	-
2. A fence erected forward of the building line or on the boundary of the primary and secondary street frontage must:	-	-	-
• not be higher than 1.2m above existing ground level	-	-	-
• if it includes an entrance gate, not have a gate that opens outward	-	-	-
• if it is constructed of metal components, be of low reflective, factory pre-coloured materials	-	-	-
• be open for at least 20% of the area of the fence that is more than 400mm above existing ground level, with any individual solid element of the fence above that height being no more than 350mm wide with a minimum aperture of 25mm	-	-	-
• if it is on a sloping site and stepped to accommodate the fall in the land, be no higher than 1.6m above existing ground level at each step	-	-	-

Description	Yes	No	N/A
3. Any fence must be designed so as not to restrict the flow of any floodwater.	-	-	-
4. A fence erected on bush fire prone land must be constructed of non-combustible materials.	-	-	-
5. A solid fence located at the street alignment may be constructed up to a maximum height of 1.8m, in any of these cases:	-	-	-
• the dwelling is affected by excessive traffic volumes and noise from the street (6000 vehicles/day or 60dBA)	-	-	-
• the fence encloses the main private open space of the dwelling with length limited to 75% of the frontage	-	-	-
• fencing incorporates openings or consists of vegetation with minimum 50% transparency.	-	-	-
6. Fences along the primary and secondary road frontages that are between 1.2m and 1.8m must be setback a minimum 1m from the street alignment for the entire frontage with provision of low maintenance planting in the setback area.	-	-	-

h. Development controls for associated works

Note:

- Associated works include earthworks, retaining walls, drainage, protection of walls, and protection of trees.

Description	Yes	No	N/A
1. Excavation for the purposes of development must not exceed a maximum depth, measured from existing ground level, of:	-	-	-
• 1m if located not more than 1m from any boundary	-	-	-
• 2m if located more than 1m but not more than 1.5m from any boundary	-	-	-
• 3m if located more than 1.5m from any boundary	-	-	-
2. Fill must not exceed a maximum height, measured from existing ground level, of:	-	-	-
• 1m if the fill is for the purposes of the erection or alteration of, or an addition to, a dwelling house, or	-	-	-
• 600mm if the fill is for any other purpose	-	-	-
Note: the height of fill contained wholly within the footprint of a dwelling house or any attached development or detached development is not limited	-	-	-

Description	Yes	No	N/A
3. Fill that is higher than 150mm above existing ground level and is not contained wholly within the footprint of a dwelling house or any attached development or detached development is limited to 50% of the landscaped area of the lot.	-	-	-
4. The finished ground level of the fill must not be used to measure the height of any dwelling house or any attached development or detached development.	-	-	-
5. Support for earthworks more than 600mm above or below existing ground level must take the form of a retaining wall or other structural support that must:	-	-	-
• be certified by a professional engineer as structurally sound, including in relation to (but not limited to) the ability to withstand the forces of lateral soil load	-	-	-
• be designed so as not to redirect the flow of any surface water or ground water, or cause sediment to be transported, onto an adjoining property	-	-	-
• have adequate drainage lines connected to the stormwater drainage system for the site	-	-	-
• not have a total height measured vertically from the base of the retaining wall or structural support to its uppermost portion that is more than the height of the associated excavation or fill	-	-	-
• be separated from any other retaining wall or structural support on the site by at least 2m, measured horizontally	-	-	-
• if it is an embankment or batter, have a toe or top that is more than 1m from any side or rear boundary.	-	-	-
6. Development must be at least 3m from all trees identified for retention on the site (measured from the base of the tree trunk).	-	-	-
7. Development for the structures below may be within 3m of a tree if works do not involve excavation or fill of more than 150mm below or above existing ground level:	-	-	-
• an access ramp	-	-	-
• a driveway, pathway or paving	-	-	-
• an awning, blind or canopy	-	-	-
• a fence, screen, or child-resistant barrier associated with a swimming pool or spa pool.	-	-	-

i. Additional controls for dual occupancies

Description	Yes	No	N/A
1. Dwelling configuration	-	-	-
• Each dwelling must face a public road	-	-	-
• Detached dual occupancies on a corner lot must be 3m from each other	-	-	-
2. Attached balconies, decks, patios terraces and verandahs to the side or rear of the dual occupancy	-	-	-
• The maximum height of the floor level is 4m	-	-	-
• If more than 2m above existing ground level, must be setback a minimum 3m from the boundary	-	-	-
• If above 2m above the existing ground level, must not have a total area of more than $12m^2$	-	-	-
3. The maximum width of all garage door openings facing a primary is 6m for a lot between 15m – 20m, 9.2m for a lot greater than 20m – 25m and 12m for a lot greater than 25m.	-	-	-
4. The maximum width of all garage door openings facing a secondary road is 3.2m for a lot between 12m – 15m, 6m for a lot greater than 15m – 20m, 9m for a lot greater than 20m – 25m and 12m for a lot greater than 25m.	-	-	-

PART D
Commercial Development

Part D: Commercial Development

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1 Introduction

Willoughby local government area has several commercial zones with different functions that support local commercial and retail centres and the strategic centres of St Leonards and Chatswood.

Greater Sydney Commission's Sydney Region Plan and North District Plan confirm Chatswood and St Leonards as two of Sydney's important strategic centres with significant economic and employment opportunities.

Willoughby City Council is committed to maintaining and increasing the potential of these larger centres as well as our smaller local centres. We aim to provide a strong framework to guide future commercial and employment activity across the Willoughby area.

This part of Willoughby Development Control Plan (Willoughby DCP) includes performance criteria and controls for all types of commercial development, including shop top housing and mixed use developments. *If there is any inconsistency between this part and Part L, the controls under Part L prevail.*

1.1 Aim

The aim of this part is to ensure all commercial development sustains and enhances the economic and environmental qualities of Willoughby and surrounding local government areas.

1.2 Objectives

The objectives of this part are to:

a. ensure controls do not inhibit economic development

b. ensure a high standard of urban design that makes a positive contribution to the streetscape and reinforces the importance of pedestrian areas

c. achieve energy efficient and sustainable buildings and developments that promote sustainable transport initiatives

d. ensure commercial development maximises thermal comfort and minimises urban heat impacts in the interests of health and wellbeing

e. provide a satisfactory transition and minimises any adverse effects between commercial uses and adjacent residential areas

f. ensure development is carried out in line with planning strategies for local centres, and the strategic centres of Chatswood CBD and St Leonards

g. ensure development is carried out in line with state plans, policies and guidelines

2 Strategies and plans

A set of studies, strategies and plans by Willoughby City Council set the direction for future commercial and residential development in key locations in the area.

Part L of Willoughby DCP details several place based plans. A brief description of the strategies adopted to inform these place-based plans is below.

2.1 Willoughby Local Centres Strategy 2036

This strategy aims to revitalise the local economy and provide the framework for future planning controls to achieve thriving, attractive and distinctive local centres. Those centres are:

a. Artarmon

The western side of the Artarmon local centre is on Hampden Road, next to Artarmon railway station. Wilkes Plaza, a pedestrian thoroughfare through an open space area, is on the eastern side of the station. The centre is within the Artarmon Heritage Conservation Area.

b. Castlecrag

The Castlecrag local centre is on Edinburgh Road between Eastern Valley Way and Rutland Avenue/The Postern. The centre adjoins and is partly within the Griffin Heritage Conservation Area.

c. North Willoughby

The North Willoughby local centre is around the intersection of Penshurst Street and Victoria Avenue, extending south to Patrick Street.

d. High Street

The major portion of the High Street local centre is on the eastern side of the street, bounded by McClelland Street and Glover Street. Two properties in the centre are on the north-western corner of High Street and Horsley Avenue.

e. Naremburn

The Naremburn local centre is on the western side of Willoughby Road, bounded by Quiamong Street, Bongalong Street and Glenmore Street. The centre includes two heritage items.

f. Northbridge

The commercial precinct of Northbridge is along the major arterial roads of Sailors Bay Road, Eastern Valley Way and Strathallen Avenue. Northbridge Plaza is on the north-eastern corner of Eastern Valley Way and Sailors Bay Road. There is a large council carpark at the rear of the shopping complex.

g. Penshurst Street

The Penshurst Street commercial centre is located along and north of Mowbray Road. There is an eclectic mix of business premises, shop top housing, and a small retail shopping strip on the eastern side of Penshurst Street near Oakville Road.

h. Willoughby South

The commercial area of the Willoughby South local centre predominantly fronts Willoughby Road with further commercial areas branching off Frenchs Road. The Bridge View Hotel and the facades of the shops at 549–553 Willoughby Road are local heritage items.

The performance criteria and controls in this part apply to local commercial centres. Specific performance criteria and controls for local centres are in Part L of this plan.

If there is any inconsistency between this part and Part L, the controls under Part L prevail.

Willoughby Development Control Plan Part D: Commercial Development 2023

Willoughby Local Centres Strategy 2036 is at willoughby.nsw.gov.au.

2.2 Chatswood CBD Planning and Urban Design Strategy to 2036

This strategy guides future private and public development over the next 20 years. The CBD includes a large mix of land uses. These include commercial/office buildings, civic buildings, library and performance facilities, major retail complexes, banking and other services, cafes, restaurants and private health facilities and schools.

The performance criteria and controls in this part apply to Chatswood CBD. See Part L of this plan for specific performance criteria and controls for Chatswood CBD.

If there is any inconsistency between this part and Part L, the controls under Part L prevail.

Chatswood CBD Planning and Urban Design Strategy to 2036 is at willoughby.nsw.gov.au.

2.3 State Environmental Planning Policy (Exempt and Complying Development Codes) 2008

The Codes SEPP includes provisions that affect commercial development.

The exempt development codes allow certain structures, business and building identification signs and land uses to take place without approval, subject to prescribed development standards and other relevant provisions.

The commercial and industrial alterations code (Codes SEPP, Part 5) allows certain works and change of use to be done as complying development.

A development application (DA) is required if a proposal does not satisfy all the criteria for either exempt or complying development under Codes SEPP.

The Codes SEPP is at State Environmental Planning Policy (Exempt and Complying Development Codes) 2008.

3 Major developments

Major commercial developments include:

a. the erection of a new building with a gross floor area exceeding 1,000m^2

b. substantial redevelopment, refurbishment, or conversion of commercial premises with a gross floor area exceeding 1,000m^2

c. shop top housing or mixed use developments that are three or more storeys and have four or more dwellings

d. other non-residential uses including medical and other community uses.

Notes:

- Redevelopment is generally seen as ‘substantial’ if the proposal will result in more than a 10% increase in the gross floor area.
- Depending on the size and scale, and any likely impacts, an authorised officer may determine that a proposal is major development.

- As well as the commercial zones, shop top housing is permissible in the R4 High Density Residential zone.
- Mixed use developments are only permissible in the MU1 Mixed Use zone.

4 Performance criteria and controls

The street frontage of commercial and non-residential development should generally maintain the pattern and spacing of adjoining buildings and create active street frontages to attract pedestrian traffic within the commercial precinct.

The performance criteria and controls apply to all commercial, shop-top housing and non-residential developments. Any variation of these controls must be justified and consider the performance criteria below.

4.1 Building design

4.1.1 Performance criteria

a. the type and style of building should reflect the function of the commercial precinct and create quality urban design forms that enhance the character of the existing retail/commercial area

b. materials, colours, finishes, fenestration, proportion, and scale of development should create interest, and blend with the façade of existing retail shop fronts and other commercial uses

c. the location of vehicle access and manoeuvring areas must not dominate or detrimentally affect the continuity of building façades

d. the design of buildings should create a balance between the amount of solid walls and openings of retail and commercial premise, and between the extent of horizontal and vertical elements to achieve visual interest at all levels

e. new development should reinforce existing roof forms, building height and proportions of building elements, including any distinctive parapets, gables, or ridges

f. ground level frontages should be transparent to allow interaction with the public domain by observation of activity and/or products displayed internally

g. the design should accentuate buildings on prominent corner sites and define the main points of entry to commercial/business centres; this may include a change in colours and materials, a partial additional storey, parapet extension, splayed setbacks and/or public art or landscape treatment, or other elements such as clock towers

h. the design of new buildings or alterations and additions must retain the stylistic or architectural character of the commercial precinct, particularly in heritage conservation areas

4.1.2 Controls

a. use of roller shutters in lieu of glazed facades is not permissible unless a minimum 70% of the roller shutter is transparent

b. any kind of internal security shutter, mesh gate or similar must be located a minimum of 1m behind the facade of the premises facing a public space or at the rear of any display window

c. any plant, equipment or machinery should be suitably screened from the general public

d. the above awning facades of 2-storey shop frontages should have a solid to void ratio of around 60:40

e. developments should use materials with a low solar reflectance index (SRI) and incorporate shade structures to reduce urban heat island effects

f. the entire ground floor level of a shop top housing development must be used for retail/commercial purposes; this may include loading/unloading facilities but does not include car parking

4.2 Design in heritage conservation areas

4.2.1 Performance criteria

a. ensure the heritage significance and streetscape character of the heritage conservation area is maintained

b. ensure views and vistas along streets and from public places of buildings of heritage significance are maintained

4.2.2 Controls

a. extensions and additions to buildings in conservation areas are to be designed to complement their style, form, proportions, materials, colours and overall heritage context of their setting

b. early building names and any historical painted advertising signs are to be preserved

c. the form and treatment of new additions must consider sightlines to ensure they do not dominate the heritage value of shop fronts when viewed from the street and public domain.

Note:

- For more information on heritage conservation areas, see Part H of this plan.

4.3 Awnings

4.3.1 Performance criteria

a. ensure awnings provide adequate shelter from rain and sun in locations of pedestrian activity

b. retain, reinstate or provide appropriate awnings that are designed to match traditional patterns (particularly in heritage conservation areas)

c. be designed to allow tree planting at regular intervals

4.3.2 Controls

a. if the existing awning is a traditional suspended steel box section type, it should be retained; if a new awning is needed it should match adjoining awnings and maintain the same alignment

b. new awnings should:

- be of opaque materials with glass inserts to allow light penetration to the footpath
- be continuous for the entire site frontage, including any vehicle entrance
- be set back 600mm from the footpath edge
- have a recess or opening to accommodate the growth of street trees
- have a height clearance above the footpath between 3m and 4.2m
- maintain the horizontal alignment, stepping down at regular intervals to follow the topography of the site where the footpath is sloping

c. entrances to large frontage developments can incorporate raised or arched canopy elements to highlight the entrance; these should be in scale with the building and compatible with the prevailing street awning character

Notes:

- No part of a building, including an awning, can project beyond the alignment of a road to which the building has a frontage. However, the *Roads Act 1993* enables Willoughby City Council as the road authority and owner of the road to grant approval to allow an awning to extend over a public road.
- Approval can be sought for an awning as part of a development application for a proposed development.
- If approval is only sought to erect an awning, a separate application for approval is required under the *Roads Act 1993*.

4.4 Frontages

4.4.1 Performance criteria

a. ensure no site is isolated by development on an adjoining allotment

b. provide adequate width for separation between vehicle driveways

4.4.2 Controls

a. a minimum width of 27m is required for developments that exceed 11m in height or where vehicular access is only available from the primary street frontage

b. no more than 30% of the street frontage is to be used for vehicle and pedestrian access to lower and upper levels

c. the bulk of new facades should be divided into equal units of around 6m each to reflect traditional small retail shopping frontages

d. allotments should be consolidated to reflect the general pattern of street frontages in the commercial precinct

Note:

- Development may be allowed on an existing lot that is isolated and cannot be consolidated with another lot. It may also be allowed on an existing lot if written advice is provided to Willoughby City Council that adjoining landowners are not willing to consolidate their properties.

4.5 Streetscape

4.5.1 Performance criteria

a. ensure the frontage of shops and business premises are consistent with the general pattern and spacing of buildings in the commercial precinct

b. achieve attractive streetscapes that add visual interest and amenity to pedestrian areas

c. reflect the function and character of the commercial precinct

d. integrate new development into and not in contrast with the streetscape character of established commercial areas

4.5.2 Controls

a. new developments or significant alterations and additions may be required to provide improvements to the quality of the public pedestrian domain, such as adding suitable paving, street trees and landscaping

b. if shop frontages are predominantly built along the street alignment, new developments or significant alterations and additions may be required to improve the pedestrian amenity at street level with transparent interactive frontages, including outdoor seating and/or dining areas

4.6 Car parking and vehicular access

4.6.1 Performance criteria

a. adequate provision for customer/visitor and employee car parking onsite

b. the provision of visitor bicycle spaces, and secure bicycle areas and end-of -trip facilities for employees

c. adequate turning areas provided to enable all commercial vehicles and trucks, including access for waste disposal trucks, to enter and leave the site in a forward direction

d. all vehicular access should be provided from a secondary street or laneway; if no alternative vehicular access is available, the entry/exit from the building and the footpath crossing must be designed to give priority to pedestrians

4.6.2 Controls

a. the width of the entry portal for a driveway providing access to a car parking area should not exceed 5m and the head clearance should not be more than 2.4m. If the access is also the entry to a loading dock, a head clearance of 3.6m may be allowed

b. the layout of car parking spaces must comply with AS/NZS 2890.1 and AS/NZS 2890.6 and details are to be shown on the architectural plans

c. the location, siting and grades of driveways, and driveway width must be in line with Australian Standard AS/NZS 2890.1

d. all new developments and significant alterations and additions must provide accessible car parking spaces for people with disability in line with Section D3.5 of the Building Code of Australia under the National Construction Code

e. details of swept paths may be required to demonstrate all vehicles can enter and leave the site in a forward direction

f. compliance with the provisions in Part F (Transport and Parking Management) of this plan

4.7 Loading/unloading facilities

4.7.1 Performance criteria

a. adequate provision for the safe loading and unloading of goods onsite without impacting pedestrian or vehicular traffic within the site or on adjacent streets

4.7.2 Controls

a. each commercial premises must have a separate loading facility provided off a secondary road or laneway

b. if onsite loading facilities are not available, details are to be provided of the nearest on-street loading areas and the method of conveying goods in a safe and efficient manner to and from the premises

c. all new commercial developments, and developments that involve significant demolition, alterations or additions with a floor area in excess of 1,000m^2 must make adequate provision for off-street loading and unloading facilities in line with Part F of this plan

4.8 Waste and recycling

4.8.1 Performance criteria

a. improve environmental outcomes by reducing waste and increasing source separation of materials and management of waste and recyclable materials

b. ensure adequate space is provided for the storage and access to waste and recyclable containers

c. ensure safe and hygienic processes and practices are in place for workers and contractors for the storage, handling and collection of waste and recycling materials

d. ensure adequate facilities are provided for the disposal of hazardous, medical or any liquid waste which requires special licenses and/or storage and disposal arrangements with other government agencies

4.8.2 Controls

a. all waste management facilities must comply with the Building Code of Australia and relevant Australian Standards

b. any compactors or mechanical devices must comply with occupational health and safety requirements

c. bin storages areas must:

- be suitably screened from public areas and adjoining properties
- located in areas to reduce the impacts of visual amenity, noise, and odour

d. refrigerated garbage rooms are required in either of these cases:

- the waste generated contains 20% or more by weight or volume of seafood, poultry or meat
- 50 litres or more of seafood, poultry or meat is generated in total per day, unless the waste is collected daily

e. the onsite collection point must provide adequate space for garbage vehicles to enter and leave the site in a forward direction

f. basement waste and recycling storage areas and access to these areas must have a minimum clearance height of 4.5m to accommodate waste and recycling collection vehicles

g. a resource recovery and waste management plan must be submitted with the development application (see Attachment 1 in this part)

4.9 Pollution control

4.9.1 Performance criteria

a. not cause air pollution, odour nuisance or unacceptable noise levels

b. ensure appropriate measures are in place to protect the environment and amenity of workers, and people living and other workers in the vicinity, who could be adversely affected by the operation of the development

4.9.2 Controls

a. depending on the type, scale, and location of development, the development application may need to include an acoustic report and/or other reports to address pollution control measures

b. depending on the type, scale, and location of construction works, the application may need to include a site management plan to address sediment and erosion control measures

c. the discharge of any solid, liquid, or gaseous materials must comply with the *Protection of the Environment Operations Act 1997*

d. if there is likely to be a need for the disposal of liquid waste to the sewer, Sydney Water should be contacted for its requirements on installing grease arrestors

e. new commercial buildings which include retail premises that allow cafés, restaurants or similar, must make adequate provision for the vertical discharge of exhaust from the lower floor levels

f. the hours of operation may be restricted if a particular use is likely to interfere with the residential amenity of adjoining and nearby dwellings

Note:

- The hours of operation for other permissible uses in commercial areas, such as function centres, recreation facilities and sex services premises will be assessed on the individual merits of the proposal. The assessment will consider any likely impacts on other commercial premises and/or adjoining and nearby dwellings.

4.10 Stormwater disposal and flooding

4.10.1 Performance criteria

a. ensure the installation of onsite detention systems make adequate provision for reuse and stormwater disposal

b. ensure properties identified as flood prone land provide appropriate flood mitigation measures

c. encourage water sensitive urban design measures to minimise impacts on the natural water cycle and foster ecological sustainability

4.10.2 Controls

a. depending on the type, scale, and location of development, the development application may need to include a stormwater management plan

b. if the development is on flood prone land, the development application may need to include a flood risk assessment report and/or a flood study

c. new commercial developments on land exceeding 2,000m^2 must implement appropriate water sensitive urban design measures

d. the application must show the proposal satisfies the objectives and controls in Part I (Stormwater Management) in this plan

4.11 Sustainable development

4.11.1 Performance criteria

a. the development will result in an environmentally sustainable building(s) and create energy efficient operations

b. encourage alternative and sustainable transport strategies to reduce the use and reliance on motor vehicles as the principal mode of transport

4.11.2 Controls

a. depending on the type, scale and cost of development, the development application must include a Sustainable Performance Statement, a Green Star rating report, and/or a National Australian Built Environment Rating Scheme (NABERS)

b. the application must show the proposal satisfies the objectives and controls under Part J (Building Sustainability) in this plan

c. applications for large developments exceeding 2,000m^2 must include a travel demand management plan (green travel plan); this should detail how the operation intends to modify travel decisions to and from the building so that more desirable modes of transport

are used, such as bicycles, car-pooling, mini-bus pick-up/drop off, and provision of car share spaces (see Part F in this plan for details)

Note:

- The Australian Government's commercial building disclosure program requires a NABERS Energy rating to be disclosed when office space of more than 2,000m^2 is offered for lease or sale.

4.12 Signage

4.12.1 Performance criteria

a. the number of signs is limited to avoid cluttering, distraction, and unnecessary repetition

b. corporate colours and signage on buildings do not have an adverse visual impact on adjoining and nearby properties and the public domain

4.12.2 Controls

a. the content must relate to the building and/or business on the site

b. the size of signs must be in proportion with the building

c. there must be an integrated and coordinated scheme for business identification signs for multi-tenanted buildings

d. there must be no glare or light spill from any signs onto adjoining properties

Notes:

- Signs, other than 'building identification signs' and 'business identification signs', are prohibited in the land use table for commercial zoned land.
- Many of these types of signs may be erected without approval under the relevant provisions of the Codes SEPP.
- A development application is required if any of the development standards for proposed 'building identification signs' or 'business identification signs' do not fully satisfy the exempt provisions of the Codes SEPP. Any such application will be assessed in accordance with the signage provisions under State Environmental Planning Policy (Industry and Employment) 2021.
- Third party advertising signage is prohibited in the land use table for commercial zoned land.

4.13 Safety by design

4.13.1 Performance criteria

a. ensure appropriate safety and security measures are put in place to prevent criminal activity

4.13.2 Controls

a. the design should include high visibility to front entries, lighting of pathways or hidden spaces and where applicable, careful siting of shrubs and landscape elements

Note:

- The development application must include details of the design and safety measures incorporated into the design.

4.14 Utility facilities

4.14.1 Performance criteria

a. ensure adequate provision is made and integrated into the design of the development for utility facilities such as substations and water main boosters

4.14.2 Controls

a. the application must include written advice from the energy provider and Sydney Water that states if these utility facilities must be provided for the development

b. all facilities that are visible from the street or public domain must be suitably screened by landscaping

c. substations should preferably be located below ground level or at the rear of the property if rear lane access is available

4.15 Undergrounding of services

4.15.1 Performance criteria

a. to improve the visual amenity of the urban environment

4.15.2 Undergrounding of services

a. all services, including overhead electricity wires, for major developments exceeding 2,000m^2 are to be located underground (this includes publicly owned land immediately outside the development site)

Note:

- In some circumstances it may not be practical or possible to provide undergrounding of services. For example, there may not be sufficient capacity within the road reserve to accommodate additional services or it may be cost prohibitive to do these works as part of the proposed development.
- Any proposed variation to this control must include a written request to justify why this requirement cannot be satisfied.
- If there is a request to vary this requirement because it is considered cost prohibitive, the development application must include a detailed cost report prepared by a registered quantity surveyor.

4.16 Access and mobility

4.16.1 Performance criteria

a. provide measures to assist people with a disability to access facilities independently, equitably and with dignity

b. ensure there is a ‘continuous accessible path of travel’ for people with a mobility, vision, hearing, or intellectual disability

4.16.2 Controls

a. Details are to be submitted with the development application to demonstrate the development will comply with the Disability (Access to Premises – Building) Standards 2010 under *Disability Discrimination* Act 1992

5 Further controls for St Leonards strategic centre

The St Leonards and Crows Nest 2036 Plan by the NSW Department of Planning, Industry and Environment applies to parts of the Willoughby, North Sydney, and Lane Cove local government areas.

The plan aims to strengthen employment opportunities, improve social infrastructure, and take advantage of public transport connections related to a new metro station at Crows Nest.

Willoughby LEP includes an increase in the height and floor space for commercial development on specific sites to reflect the controls in the St Leonards and Crows Nest 2036 Plan.

Willoughby City Council has also adopted these controls for specific sites in the St Leonards strategic centre:

a. 2–10 Chandos Street (the Alto Ford site): any new development on this site must provide 4 storey wall heights.

b. 12 Chandos Street (College of Law building): any new development on this site must provide 4 storey wall heights.

c. 110–120 Christie Street: any new development on these sites must provide 2 storey wall heights.

d. 207 Pacific Highway: development on this site must provide a minimum 3m wide landscaped setback along the street frontage.

The St Leonards and Crows Nest 2036 Plan 2036 is at planning.nsw.gov.au.

6 Shop top housing, mixed use and non-residential developments

The requirements under State Environmental Planning Policy (Housing) Chapter 4 – Design Quality of Residential Apartment Development and the design guidelines and controls under the Apartment Design Guide, NSW Department of Planning and Environment, July 2015 apply to the residential component of buildings that have three or more storeys and four or more dwellings.

The relevant performance criteria in Section 2.1 and controls in Section 4.2 under Part B of this plan apply to the residential components of shop top housing and mixed use developments.

The further controls below apply to shop top housing, mixed use and non-residential developments:

6.1 Setbacks

6.1.1 Front setback

a. the front (street) setback of the ground floor level may be set at the property boundary defining the street corridor with a continuous edge

b. the first floor level is required to be setback a minimum 2m from the street frontage

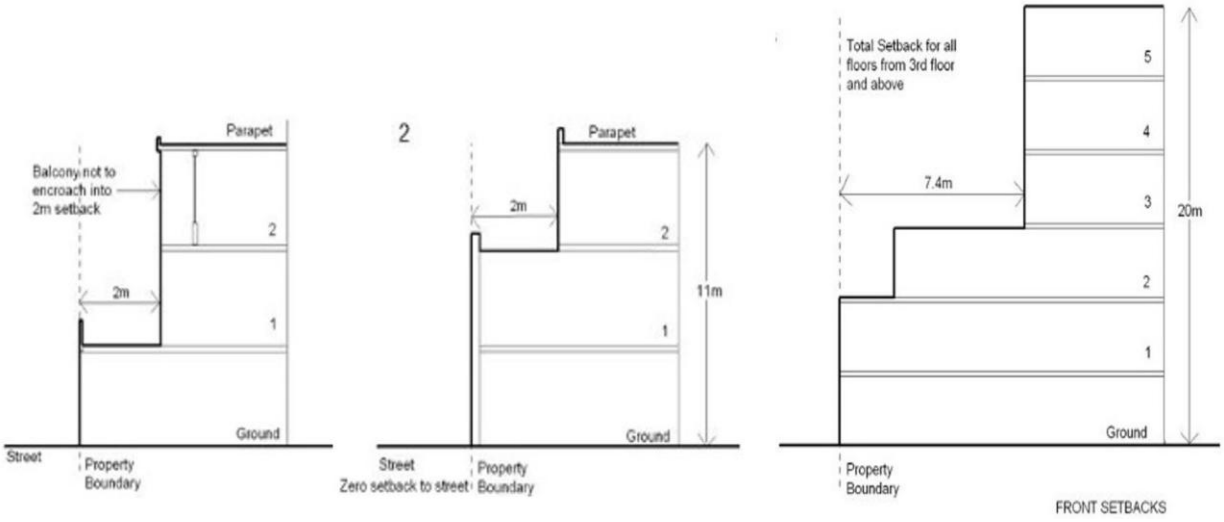
c. balconies are not to encroach into the required setback of the level below

d. the first floor level may have a zero setback if it comprises commercial uses and/or it is in keeping with established adjoining developments

e. the third floor level and above is required to be setback 5m plus 1.2m for each level above the third level

f. the total required setback applies to all floors above the third floor.

Figure 1 Front setbacks



6.1.2 Side setbacks

a. the ground floor level may have a zero setback

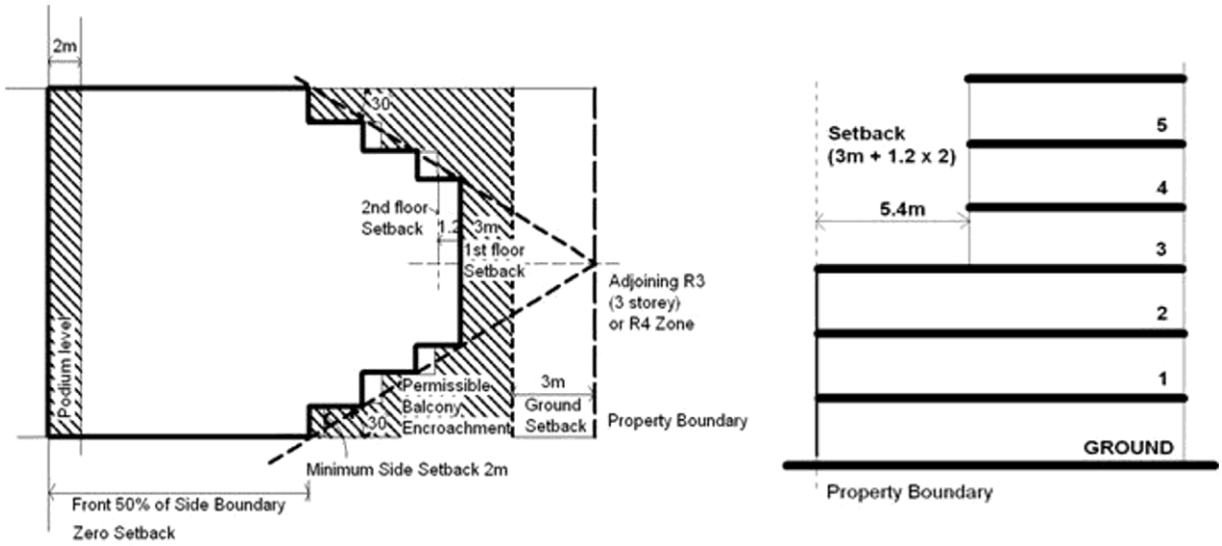
b. the first and second floor level are required to have zero setback for a maximum of 50% of the length of the side boundary located within the front half of the site

c. the side boundary setbacks for the first and second floors for the remainder of the building are to comply with the building envelop determined by a line projected horizontally at 30^{o} from the side boundary from halfway along the boundary towards the rear of the site

d. the third floor and above is required to have a setback of 3m plus 1.2m for each level above the third level

e. the total required setback applies to all floors above the third floor

Figure 2 Side setbacks



6.1.3 Rear setbacks adjoining commercial, industrial or high density residential zoned land

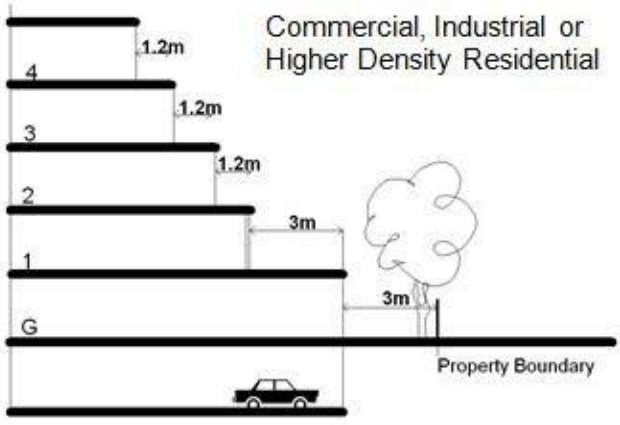
The rear setbacks below apply to developments adjoining commercial, industrial, medium density (R3 zone with a 12m height limit) and high density (R4 zone) residential land:

a. 3m from the ground floor level

b. a further 3m for the first floor level

c. and a further 1.2m for each additional floor level

Figure 3 Rear setbacks for developments adjoining commercial, industrial, R3 (12m height limit) and R4 zoned land



6.1.4 Rear setbacks adjoining low and medium density zone land

The rear setbacks below apply to developments adjoining low density (R2 zone) or medium density (R3 zone with a height limit of 9m). These setbacks apply whether or not the land is separated by a laneway:

a. 3m from the ground floor level

b. plus 6m for the first floor level

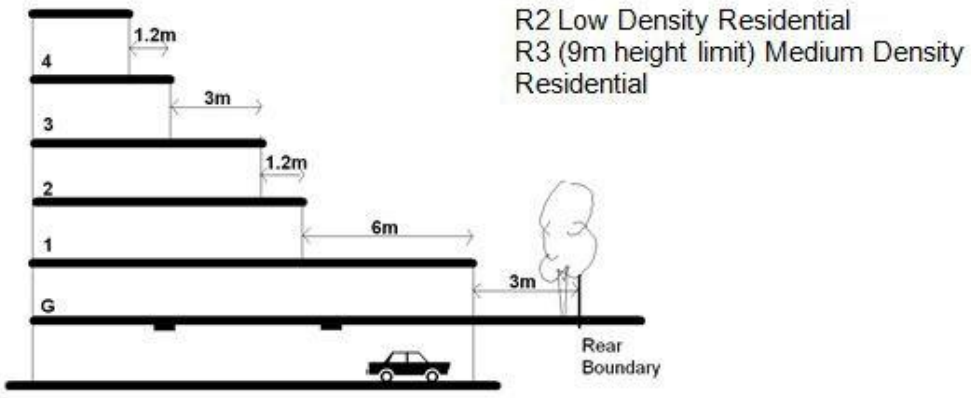
c. plus 1.2m for the second floor level

d. plus 3m for the third floor level

e. plus 1.2m for the fourth floor level

f. plus 3m for the fifth and each additional floor level

Figure 4 Rear setbacks for R2 and R3 (9m height limit) zones



Notes:

Willoughby City Council may consider a variation of the side and rear side boundary setback requirements if:

- the boundary line is on the boundary of a secondary road, laneway or reserve
- the wall height and length matches an existing building on the adjoining side of the site

6.2 Landscaping

a. a 3m wide deep soil zone must be provided along the rear boundary adjoining residential zoned land

b. the deep soil zones must be landscaped with trees that when mature, reach a minimum height of 15m and a minimum 3m wide tree canopy

c. a planter box with a minimum internal width of 1m must be provided along the edge of a balcony or terrace on the first floor level that faces low or medium density zoned land

d. a planter box with a minimum internal width of 400mm must be provided along the edge of a balcony or terrace on all other floors facing low or medium density zoned land (other than the street frontage)

e. all planter boxes must not exceed 1m in height, have a minimum soil depth of 600mm, and be landscaped with dense screen planting

f. a minimum of 20% of any podium and a minimum 20% of any rooftop open space must be landscaped

g. details of the type and height of shrubs and trees within the deep soil zones, planter boxes, podium and/or rooftop open space must be shown on the landscape plans

6.3 Car parking and access

a. access to the residential car parking area must be separated from the commercial car parking and loading areas

b. shared car spaces may be provided for residential visitors and customers if unimpeded access is available at all times

c. vehicle movements for loading and unloading must be separated from all car parking areas

6.4 Building mass and bulk

a. buildings over 11m should have a defined podium level

6.5 Signage

a. signs are restricted to shop fronts, awnings and under awning signs

b. any signs must consider the visual impact on residential occupants, particularly in terms of illumination and light spill

7 Specific land uses

7.1 Sex services premises

Sex service premises, commonly referred to as brothels, are not a prohibited land use in E1 Local Centre and E2 Commercial Core commercial zones. However, there are strict limitations on where they may be located.

A development application for a sex services premises must include this information:

a. number of sex workers and support staff, including any security personnel

b. hours of operation

c. name of the proprietor

d. floor plan showing the entry and exits from the premises, and number and description of rooms within the premises, for example reception/office, waiting or lounge area, and sex worker rooms

e. description of safety by design measures

f. site plan showing the location of places of public worship, schools, community facilities, child-care centres, parks, playgrounds, hospitals, medical centres, bus stops, dwellings or any place in the vicinity of the premises that may be regularly frequented by children within a 200m radius

These controls apply to sex services premises:

g. must not be located within public view from a place of public worship, school, community facility, child care centre, hospital, medical centre, or any place in the vicinity of the premises that may be regularly frequented by children

h. must not form clusters within areas where brothels are permissible

i. approval will not be granted for another sex services premises located within a 100m of another existing approved premises

j. the operation is limited to not more than 10 sex worker rooms

k. owners, operators, managers and staff must be aware of and understand the 'Health and safety guidelines for sex service premises in NSW' by SafeWork NSW

'Health and safety guidelines for sex service premises in NSW' is at safework.nsw.gov.au.

Note:

- Willoughby City Council's environmental health officers, and authorised officers of the NSW Health and WorkCover, have the powers to conduct regular inspections of all sex services premises without notice to determine if the premises are complying with the required guidelines and health standards.

7.2 Outdoor dining areas

Outdoor dining areas can provide vitality to the streetscape in commercial precincts. Willoughby City Council encourages the use of footpaths and public space for outdoor dining areas in conjunction with a café or restaurant, providing they are established in appropriate locations.

Any application to lease a public footway from Willoughby City Council for an outdoor dining area must be associated with a food and drinks premises, café or restaurant. These conditions apply:

a. the minimum area for an outdoor dining area is $2m^2$

b. the public footway space must have a minimum width of 3.5m to the curb-line

c. an outdoor dining area should not be next to a bus stop, taxi rank, Australian Post mailbox, or automatic teller machine (ATM)

d. the outdoor area must not interfere with or prevent access to existing street elements such as street furniture, pedestrian crossings, traffic lights, street trees, planter boxes, and street or directional signs; a clear width of 2m should be maintained from these structures

e. the layout is consistent with other outdoor dining areas in the vicinity so that pedestrian circulation and flow is not obstructed

f. a minimum 900mm must be maintained between the curb-line and the outdoor dining area

g. the outdoor dining area must be immediately in front of the 'food and drinks premises', café or restaurant and not encroach on footpath space in front of adjoining premises unless written approval is provided by the owner or tenant of the adjoining premises; this approval must be provided if there is a change of tenant/owner or annually, whichever comes first

h. if the premises is on a corner, the outdoor dining area may occupy both street frontages subject to the same restrictions for a single fronted premises

An application for an outdoor dining area must include the information below:

1. details of furniture demonstrating that it is of high standard, safe and strong, wind resistant, and able to maintain a quality presentation

2. details of the type and height of barriers (barriers must not be used to completely enclose the outdoor dining area)
3. details of any business identification signs on the furniture and barriers; no third party advertising is permitted unless it is associated with the operation of the premises, such as the brand of coffee used

Notes:

- The use of a building, including an outdoor dining area, cannot extend beyond the alignment of a road to which the building has a frontage; but as the road authority and owner of the road, Willoughby City Council can grant approval to allow part of a footpath to be used as an outdoor dining area (*Roads Act 1993*).
- Approval can be sought for an outdoor dining area as part of a development application for a proposed development, but a separate approval to lease the land must be obtained from Willoughby City Council.
- If approval is only sought to use the footpath as an outdoor dining area, a separate application for approval is required under the *Roads Act 1993*.
- Furniture includes tables, chairs, umbrellas, bins, barriers, heating devices, and the like.
- No permanent structures are permissible, and all furniture must be removed if the business ceases to operate or there is a change of use or ownership, unless separate approval is obtained from Willoughby City Council.

7.3 Centre-based child care facilities

Centre-based child care facilities are permitted with consent in all commercial and residential zones. The definition of a centre based child care facility is in the dictionary in Willoughby LEP. It includes the use of a building or place that provides long day care, occasional child care, out-of-school-hours care (including vacation care), preschool care or an approved family day care venue.

Notes:

- An approved family day care venue is a place, other than a residence where an approved family day care service (within the meaning of *the Children (Education and Care Services) National Law (NSW)*) is provided.
- A centre-based child care facility does not include a building or place used for home-based child care or school-based child care.

A centre-based child care facility requires development consent from Willoughby City Council.

Notes:

- Development consent is conditional on the appropriate licensing approval from the NSW Department of Education – Early Childhood Directorate.
- Any approval by Willoughby City Council will not necessarily result in the issue of a license from the NSW Department of Education – Early Childhood Directorate.

- Applicants are encouraged to contact Willoughby City Council's Children's Services division before lodging a development application to obtain further information about rules and regulations, and licensing and operational requirements.

7.3.1 Site suitability

The selection of a site must have regard to all potential environmental health hazards or safety risks, including:

a. Electromagnetic fields

A child care centre should not be within 300m of mobile phone towers and base stations, transmission line easements or other sources of significant electromagnetic radiation.

b. Noise

Sufficient separation should be provided from busy roads and rail corridors to avoid adverse noise impacts. A child care centre should also be designed to mitigate any unreasonable noise impacts on adjoining and nearby properties.

These controls apply:

1. the use of the premises including indoor/outdoor play areas, traffic on site and mechanical plant shall not exceed the background noise level by more than 5dB at the most affected point on or within any receiving property boundary
2. the internal noise levels within indoor play or sleeping areas of the centre, when it is operating at full capacity, shall not exceed an Leq,1hr 40 dB(A)
3. the intrusive noise onto any outdoor play or activity area of the centre, when it is operating at full capacity shall not exceed an Leq,1hr 55 dB(A)

Note:

- An acoustic report prepared by a suitably qualified person must be submitted with the application showing that compliance with these requirements can be achieved.

c. Pollution and contamination

The location of child care centres, including outdoor play areas, should consider:

- pollution from vehicle fumes or any other air pollutants
- proximity to any water cooling/warming systems that may pose a risk of airborne contamination
- proximity to land use activities that may cause contamination, such as service stations

7.3.2 Emergency evacuation

Development applications for all child care centres must include an emergency evacuation plan. This plan must include this information:

a. the mobility of children and how this is to be accommodated during an evacuation, including the use of emergency evacuation cots

b. the location of a safe congregation area, away from the evacuated building, busy roads, other hazards and the evacuation points of other residents or tenants within the building or surrounding buildings

c. if the child care centre is part of a larger building or complex, that the emergency evacuation plan is complementary and consistent with other emergency evacuation plans in place

d. the supervision of children during the evacuation and at the safe congregation area with regard to the capacity of the child care centre and the child-to-staff ratios

Notes:

- Centre-based child care facilities in commercial buildings must demonstrate that suitable access is available to designated play areas, and effective emergency evacuation procedures can be provided for centres located above the ground floor level.
- Off-street car parking must be provided in line with the requirements of Part F (Transport and Parking Management) in this plan.

7.3.3 Specific controls for child care facility in residential areas

a. a minimum 20m frontage is required to ensure the site is compatible with surrounding residential properties, and adequate access is available for vehicles to enter and leave the site in a forward direction

b. to ensure good access is available for emergency vehicles and evacuation options/routes, child care centres should not be located in a cul-de-sacs or no through roads, except where more than one street access and egress is available to the site, and the centre has adequate parking and pickup/drop off spaces

c. unless otherwise approved, the hours of operation must not extend beyond 7am to 7pm Monday to Friday. Any application for extended hours of operation must consider likely impacts on adjoining and nearby residents

d. a landscape plan prepared by a suitably qualified person must be submitted with the application; landscaping of the site must consider to the relevant performance criteria under Section 2.1.5 (Landscaping) in Part B of this plan.

e. a site used as both a dwelling and a child care centre must provide the minimum required outdoor children’s play area, and it must be separate to any outdoor areas associated with the use of the dwelling

Attachment 1: Resource recovery and waste management plan

The resource recovery and management plan must include this information:

a. The type and estimated quantity of garbage generated, and number of collections per week.

b. The type and estimated quantity of recycling materials generated, and number of collections per week.

c. An estimate of any garden or organic waste generated per week, and number of collections, if any, per week.

d. The size and number of mobile garbage bins and/or bulk bins to be used to manage garbage collection.

e. The type, size and number of recycling containers and any associated equipment to be used to manage recyclable materials.

f. The type, size and number of containers and any associated equipment, including any composting facilities, to be used to manage garden or organic materials.

g. Details of the location, size and access to garbage and recycling spaces/rooms.

h. Details of any refrigerated garbage rooms required for the disposal of seafood, poultry, meat, or other organic materials.

i. Details of the waste management arrangements, including safe and hygienic access to garbage and recycling spaces/rooms by employees and contractors. This should include:

    1. name of the contractor
    2. times, days and frequency to collect recyclable and general waste materials
    3. type and size of waste and recyclable collection vehicles to be used in the operation
    4. measures to achieve source separation of garbage and recyclable materials
    5. display of signs to ensure source separation and correct use of bins
    6. access to water and provision of high-pressure hose for washing bins and bin storage, and drainage to sewer
    7. person or persons responsible for cleaning bins and bin storage areas
    8. ventilation for bin storage areas
    9. features to prevent vermin entering bin storage areas
    10. method, distance, and grade of pathway to wheel mobile garbage bins to collection points
    11. driveway access to ensure vehicles can enter and leave in a forward direction
    12. education and any incentives to encourage correct waste management

PART E
Industrial Development

Part E: Industrial Development

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1 Introduction

The Willoughby local government area comprises three industrial areas in Artarmon, East Chatswood, and Lane Cove North. Artarmon and East Chatswood are the two main industrial areas.

Various studies and analysis have identified that Willoughby comprises a critical part of the already low level of industrial and urban services land in Greater Sydney. This makes those areas in Willoughby particularly important to retain and manage for the future.

This part of Willoughby Development Control Plan (Willoughby DCP) includes guidelines and controls for industrial development and other appropriate land uses that produce goods and services to meet the current and future needs of a growing population.

Willoughby City Council is committed to maintaining and increasing the potential of existing industrial service hubs, and consequently increasing employment opportunities.

Willoughby Industrial Lands Strategy 2036 supports this by increasing the floor space provision in certain localities. Appropriate changes are included in Willoughby Local Environmental Plan (Willoughby LEP).

1.1 Aim

The aim of this part is to ensure development takes place in a way that sustains and enhances the economic and environmental qualities of Willoughby and surrounding local government areas.

1.2 Objectives

The objectives of this part are to:

a. ensure controls do not inhibit economic development

b. improve the design of buildings and aesthetic quality of streetscapes, and ensure any land uses conform to environmental and hazard reduction guidelines

c. achieve energy efficient and sustainable buildings and developments that promote sustainable transport initiatives

d. ensure developments maximise thermal comfort and minimise urban heat impacts in the interests of health and wellbeing

e. include appropriate controls to ensure industrial development does not pollute or adversely affect adjoining land

f. protect the viability of business zones by limiting the size of ancillary offices and showrooms used in conjunction with industrial premises

g. accommodate other permissible land uses because of their specific building or site requirements or operational characteristics

h. preclude activities that are incompatible or inappropriate for industrial zones, and should otherwise be in established residential or commercial zones

i. ensure adequate provision is made for off-street car parking and loading facilities

j. optimise the efficiency and effectiveness of freight handling and logistic networks

k. minimise adverse effects of industrial activity on adjoining or nearby residential properties

2 Strategies and plans

2.1 Willoughby Industrial Lands Strategy 2036

This strategy states that any loss of existing industrial zoned land would further compromise the local and regional economies of the North Shore area. It finds that it is vitally important to protect the integrity of the industrial areas and ensure their uses are appropriate.

This part of Willoughby DCP aligns with the principles and intent of the strategy. Willoughby Industrial Lands Strategy 2036 is at willoughby.nsw.gov.au.

2.2 State Environmental Planning Policy (Exempt and Complying Development Codes) 2008

The Codes SEPP includes provisions that affect industrial development.

The exempt development codes allow certain structures, business and building identification signs and land uses to be carried out without approval, subject to prescribed development standards and other provisions.

The commercial and industrial alterations code (Codes SEPP, Part 5) also allows certain works and change of use to be carried out as complying development.

The Codes SEPP is at State Environmental Planning Policy (Exempt and Complying Development Codes) 2008.

Note:

- A development application is required if a proposal does not satisfy all the criteria for either exempt or complying development under the Codes SEPP.

3 Major developments

Major industrial developments include:

a. the erection of a new building with a gross floor area exceeding 2,000m^2

b. substantial redevelopment, refurbishment, or conversion of buildings with a gross floor area exceeding 2,000m^2

Notes:

- Redevelopment is generally seen as ‘substantial’ if the proposal will result in more than a 10% increase in the ‘gross floor area’.
- Depending on the size and scale, and any likely impacts, an authorised officer may determine that a proposal is ‘major’ development.

4 Performance criteria and controls

It is important to maintain clusters of traditional industrial activities and retain the integrity of existing industrial areas. It is equally important to allow a level of flexibility to accommodate emerging high technology and allied health related industries.

There is no minimum lot size for industrial development. However, the area and dimensions of the lot, and the internal layout of buildings, must be suitable for the intended use.

The general performance criteria and controls apply to all industrial developments. Any variation of these controls must be justified and consider the general performance criteria below.

4.1 Design

4.1.1 Performance criteria

a. provide a flexible building design such as large spans, adaptable floor plates and adequately sized and appropriately located loading facilities

b. provide adequate natural lighting and ventilation to create a safe and healthy work environment

c. avoid bulky roof forms and blank facades (the façade should be modulated and articulated to reduce the bulk)

d. use variations in materials, colours, and textures to improve the visual appearance of the development and contribute to the aesthetic quality of the streetscape

4.1.2 Controls

a. open storage areas, truck parking areas, and any activities likely to generate emissions, are to be suitably screened and located no closer than 10m from the boundary of any dwellings within a residential zone

b. exposed blank walls are to be finished to a high standard to minimise the potential for graffiti

c. glazing must avoid glare to occupants of adjoining and nearby buildings, pedestrians and motorists

d. rooftop parking and structures such as plant rooms, air conditioning, exhaust systems and the like are to be suitably screened from public view

e. any new development or alterations and additions must not decrease solar access to primary living areas and private open space of adjoining residential properties by more than 20%; this may require increased setbacks, reorientation and/or lowering the building height

f. office and staff facilities should not be near intrusive noise, vibration, fumes, or any other pollution sources such as heavy machinery, industrial equipment and loading docks

g. development on corner lots must address both street frontages in terms of façade treatment, articulation of elevations and landscaping

4.2 Safety by design

4.2.1 Performance criteria

a. ensure appropriate safety and security measures are put in place to prevent criminal activity

b. provide a clear definition of entry points when viewed from the street for staff and visitors that are linked to car parking areas and pedestrian paths

4.2.2 Controls

a. the design should include high visibility to front entries, lighting of pathways or hidden spaces and careful siting of shrubs and landscape elements

Note:

- The development application must include details of the design and safety measures included in the design.

4.3 Fences

4.3.1 Performance criteria

a. front fences contribute to an attractive setting for the development when viewed from adjoining properties and the public domain

b. any fencing complements the development and does not impact on the visual quality of the streetscape

4.3.2 Controls

a. all fencing, except for dwarf walls up to a maximum height of 1m above the footpath level must be set back and provide a minimum 2m wide landscaped area

b. no front fencing shall exceed a maximum of height of 1.8m

c. In areas of overland flow, construction of fences must allow for the natural flow of stormwater. This may require fencing to be an open type construction or raised above the finished ground level to allow the free flow of stormwater. Construction of any such fencing must remain safe during floods and not obstruct moving debris

4.4 Paving and external hard stand areas

4.4.1 Performance criteria

a. paving and hard stand areas are suitably drained to prevent stormwater runoff into adjoining properties

4.4.2 Controls

a. all hard stand areas used for outdoor storage, parking, loading and/or manoeuvring vehicles must be sealed and drained to an appropriate onsite detention system

4.5 Setbacks

4.5.1 Performance criteria

a. provide adequate area for landscaping, solar access to adjoining properties and minimise overshadowing

b. provide adequate separation between incompatible land uses such as dwellings

c. increase setbacks progressively as wall heights increase to reduce the bulk and scale of building

d. ensure car parking areas forward of the building line are setback so they do not dominate the street frontage

4.5.2 Controls

a. in East Chatswood, the front boundary setback is a minimum 4.5m

b. in Artarmon, the front setback is:

- a minimum 4m for properties facing the Pacific Highway
- a minimum 12m along the western side of Herbert Street
- a minimum 10m from the Royal North Shore Hospital, on the southern boundary of the land being part Lot 1 DP 591747, Herbert Street Artarmon
- a minimum 9m along the rear of 1, 5,7,9,10 and 11 George Place
- 3m elsewhere

c. in Lane Cove North (Epping Road), the front setback is to be a minimum of 15m

d. where located on a corner lot, boundary setbacks on side street are to be a minimum of 2m

e. a rear setback of 3 metres is to be provided on properties backing on to major roads, access streets, rear lanes, railway lines, and public places

f. development abutting residential properties or residentially zoned land must have a minimum setback of 3 metres; this distance is to be increased by 1.2m for each 3.5m by which the height of the building exceeds 3.5m

g. setbacks on corner blocks must enable sufficient sightlines for traffic under the relevant Australian Standard. (AS 2890)

h. all setback areas are to be landscaped to Willoughby City Council's satisfaction; no car parking, storage of materials, garbage facilities or the like is to occur within the setback area, and any security fencing must be screened and located behind the setback area

i. setbacks along the western side of Herbert Street, St Leonards are to provide for substantial street tree planting, footpath paving and widening to create a boulevard style gateway entry to Willoughby City through the industrial area (Council Resolution 99/538 26 July 1999)

4.6 Landscaping

4.6.1 Performance criteria

a. retain any existing significant trees and vegetation

b. ensure the type of landscaping is sensitive to the site attributes such as streetscape, land capability, microclimate, natural landform, existing vegetation, and vistas and views to and from the site

c. provide adequate open space for employees

d. maximise absorption for onsite infiltration of stormwater

e. contribute to improving the streetscape character of the locality

4.6.2 Controls

a. all front, side and rear setbacks are to comprise deep soil zones with a minimum width of 2m

b. all other landscaped areas must have a minimum width of 2m and a minimum depth of 600mm

c. all existing trees are to be shown on the landscape plans and details provided on how the trees to be retained will be protected during construction

d. a minimum setback of 4m from the outside of the trunk of existing trees is required from any building structure

e. planter boxes are to be integrated into the building structure and have a minimum internal width of 600mm

f. all landscaped areas (including planter boxes) are to be planted with native low maintenance species, provided with automatic watering systems, and suitably drained

g. for sites that exceed 1500m^2, a minimum 10m^2 of outdoor eating and sitting area is to be provided for staff in a suitably landscaped area that receives adequate sunlight from 12pm to 2pm

h. any new development or significant alterations and additions may be required to provide additional street trees along the street frontage

i. a minimum 2m wide deep soil zone is required at the rear of any site that abuts an access street, service road, railway line, public place, residential property, or residentially zoned land

j. trees with wide tree canopies are to be planted in deep soil zones to reduce the urban heat island effect

k. the deep soil zones must be landscaped with trees that when mature, reach a minimum height of 15m and a minimum 3m wide tree canopy

l. any electrical substation required for the site must be shown on the landscape plans and suitably screened from public view

4.7 Car parking and vehicular access

4.7.1 Performance criteria

a. make adequate provision for visitor and employee car and bicycle parking onsite

b. provide adequate turning areas to enable all commercial vehicles and trucks, including waste disposal trucks, to enter and leave the site in a forward direction

c. ensure access and location of car parking areas, loading bays and manoeuvring areas do not unreasonably impact on adjoining or nearby residential properties

4.7.2 Controls

a. the layout of car parking spaces must comply with AS/NZS 2890.1 and AS/NZS 2890.6 (details are to be shown on the architectural plans)

b. the location, siting and grades of driveways, and driveway width must be in line with Australian Standard AS/NZS 2890.1

c. all new developments and significant alterations and additions must provide accessible car parking spaces for people with disability in line with Section D3.5 of the Building Code of Australia under the National Construction Code

d. details of swept paths may need to show that all vehicles can enter and leave the site in a forward direction

e. any dedicated truck parking areas are to be suitably screened and located no closer than 10m from the boundary of any dwellings in a residential zone

f. all visitor car parking for industrial unit complexes is to be provided in a common car parking area

g. development must comply with the provisions in Part F (Car parking and Transport Management) of this plan

4.8 Loading and unloading facilities

4.8.1 Performance criteria

a. make adequate provision for the safe loading and unloading of goods on-site without impacting pedestrian or vehicular traffic within the site or on adjacent streets

b. ensure these facilities do not unreasonably impact on adjoining or nearby residential properties

4.8.2 Controls

a. each tenancy must have a separate loading facility

b. all loading and unloading areas must not be located closer than 10m from the boundary of any dwellings within a residential zone

c. all new industrial developments, and developments that involve significant demolition and/or alterations and additions must make adequate provision for off-street loading and unloading facilities in line with Part F of Willoughby DCP.

4.9 Access and mobility

4.9.1 Performance criteria

a. provide measures to assist people with a disability to access facilities independently, equitably and with dignity

b. ensure there is a 'continuous accessible path of travel' for people with a mobility, vision, hearing, or intellectual disability

4.9.2 Controls

a. details are to be submitted with the development application to demonstrate the development will comply with the Disability (Access to Premises – Building) Standards 2010 under *Disability Discrimination Act 1992*

b. include ramps and toilets for people who use a wheelchair; appropriate lighting and colour contrast for people with vision impairment; tactile surface indicators for people who are blind; and any other appropriate measures to assist people with disability

4.10 Waste and recycling

4.10.1 Performance criteria

a. improve environmental outcomes by reducing waste and increasing source separation of materials and management of waste and recyclable materials

b. ensure adequate space is provided for the storage and access to waste and recyclable containers

c. ensure safe and hygienic processes and practices are put in place for workers and contractors for the storage, handling and collection of waste and recycling materials

d. ensure adequate facilities are provided for the disposal of hazardous, medical or any liquid waste which requires special licences and/or storage and disposal arrangements with other government agencies

4.10.2 Controls

a. all waste management facilities must comply with the Building Code of Australia and relevant Australian Standards

b. any compactors or mechanical devices must comply with occupational health and safety requirements

c. bin storages areas must be:

- suitably screened from public areas and adjoining properties
- located in areas to reduce the impacts of visual amenity, noise, and odour

d. refrigerated garbage rooms are required in either of these cases:

- the waste generated contains 20% or more by weight or volume of seafood, poultry or meat
- 50 litres or more of seafood, poultry or meat is generated in total per day, unless the waste is collected daily

e. the onsite collection point must provide adequate space for garbage vehicles to enter and leave the site in a forward direction

f. basement waste and recycling storage areas and access to these areas must have a minimum clearance height of 4.5m to accommodate waste and recycling collection vehicles

g. the development application must include a resource recovery and waste management plan (see Attachment 1 in this part)

4.11 Pollution control

4.11.1 Performance criteria

a. not cause air pollution, water pollution, odour nuisance or unacceptable noise levels

b. appropriate measures are put in place to protect the environment and amenity of workers, and people living and other workers in the vicinity, who could be adversely affected by the operation of the development

4.11.2 Controls

a. depending on the type, scale, and location of development, the development application may need to include an acoustic report and/or other reports to address pollution control measures

b. depending on the type, scale, and location of construction works, the application may need to include a site management plan to address sediment and erosion control measures

c. the hours of operation may be restricted to 7am to 6pm Monday to Saturday (no work to take place on Sundays) for industrial premises that are likely to interfere with the residential amenity of adjoining and nearby dwellings

d. the development application may need to include details of loading and unloading times (including waste collection), and vehicle movements and their routes, if these activities are likely to interfere with any adjoining or surrounding residential properties

e. the discharge of any solid, liquid, or gaseous materials must comply with the *Protection of the Environment Operations Act 1997*

f. if there is likely to be a need to dispose of liquid waste to the sewer, Sydney Water should be contacted for its requirements on installing grease arrestors

g. appropriate pollution control measures must be installed to prevent stormwater pollution

Note**:**

- The hours of operation for other permissible uses in industrial areas, such as places of public worship, recreation facilities and sex services premises are assessed case by case.

4.12 Stormwater disposal and flooding

4.12.1 Performance criteria

a. ensure onsite detention systems make adequate provision for reuse and stormwater disposal

b. ensure properties identified as flood prone land provide appropriate flood mitigation measures

c. encourage water sensitive urban design measures to minimise impacts on the natural water cycle and foster ecological sustainability

4.12.2 Controls

a. depending on the type, scale, and location of development, the development application may need to include a stormwater management plan

b. an application for development on flood prone land may need to include a flood risk assessment report and/or a flood study

c. new industrial development on land exceeding 2,000m^2 must implement appropriate water sensitive urban design measures

d. the application must show the proposal satisfies the objectives and controls of Part I (Stormwater Management) in this plan

4.13 Sustainable development

4.13.1 Performance criteria

a. ensure the development will result in environmentally sustainable buildings and create energy efficient operations

b. encourage alternative and sustainable transport strategies to reduce the use and reliance on motor vehicles as the principal mode of transport

4.13.2 Controls

a. depending on the type, scale and cost of development, the development application must include a Sustainable Performance Statement, a Green Star rating report, a National Australian Built Environment Rating Scheme (NABERS) commitment agreement and/or details of compliance with National Construction Code

b. the application must show the proposal satisfies the objectives and controls under Part J (Building Sustainability) in this plan

c. major developments must submit a travel demand management plan (green travel plan); this should detail how the operation intends to modify travel decisions to and from the industrial premises so that more desirable modes of transport are used, such as bicycles, car pooling, minibus pick-up/drop off, and provision of car share spaces (see Part F in this plan for details).

4.14 Signage

4.14.1 Performance criteria

a. the number of signs is limited to avoid cluttering, distraction, and unnecessary repetition

b. corporate colours and signs on buildings do not have an adverse visual impact on adjoining and nearby properties and the public domain

4.14.2 Controls

a. the content must relate to the building and/or business on the site

b. the size of signs must be in proportion with the building

c. there must be an integrated and coordinated scheme for business identification signs for multi-tenanted complexes

d. there must be no glare or light spill from any signs onto adjoining properties

e. directory boards identifying tenants must be located next to the entrance of the complex and not impact on landscaped areas or access points to adjoining properties

Notes:

- Signs, other than building identification signs and business identification signs, are prohibited in the land use table for general industrial (E4) zoned land.
- Many of these types of signs may be erected without approval under the Codes SEPP.
- A development application is required if any of the development standards for proposed 'building identification signs' or 'business identification signs' do not satisfy the exempt provisions of the Codes SEPP. Any such application will be assessed in accordance with the signage provisions under State Environmental Planning Policy (Industry and Employment) 2021.
- Third party advertising signs are prohibited in the land use table for industrial zoned land.

4.15 Utility facilities

4.15.1 Performance criteria

a. ensure adequate provision is made and integrated into the design of the development for utility facilities such as substations and water main boosters

4.15.2 Controls

a. the application must include written advice from the energy provider and Sydney Water that states if these utility facilities must be provided for the development

b. all utility facilities that are visible from the street or public domain must be suitably screened by landscaping

c. substations should preferably be located below ground level or at the rear of the property if rear lane access is available

4.16 Undergrounding of services

4.16.1 Performance criteria

a. to improve the visual amenity of the urban environment

4.16.2 Undergrounding of services

a. all services, including overhead electricity wires, for major developments exceeding 2,000m^2 are to be located underground (this includes publicly owned land immediately outside the development site)

Note:

- In some circumstances it may not be practical or possible to provide undergrounding of services. For example, there may not be sufficient capacity within the road reserve to accommodate additional services or it may be cost prohibitive to do these works as part of the proposed development.
- Any proposed variation to this control must include a written request to justify why this requirement cannot be satisfied.
- If there is a request to vary this requirement because it is considered cost prohibitive, the development application must include a detailed cost report prepared by a registered quantity surveyor.

5 Specific land uses

5.1 High technology industries

Willoughby City Council encourages high technology industries, particularly if these include a significant component of research and development operations. However, these types of 'clean' industries must not occupy industrial zoned land if the predominant use is more likely to be a commercial/business type of activity that may be better suited to an office environment in a commercial zone.

A development application for a high technology industry must include:

a. an outline of the operation, including the core business activity

b. information on the end products

c. details of the internal layout showing each component of the operation, including 'research and development' areas, production, assembly, packaging, warehouse, ancillary office and/or laboratories

d. the number and type of staff

If the application shows the operation is a high technology industry best located in an industrial area, Willoughby City Council may allow more office space, providing it does not exceed 50% of the total gross floor area.

5.2 Ancillary office and showrooms

Generally, office and showrooms should only be a relatively minor component of an industrial use that relates to the service activity, manufacture, assembly or warehousing of goods.

The total combined office/showroom of an industrial premises must not exceed 20% of the total gross floor area.

However, the exceptions below apply to industrial activities on the general industrial (E4) zoned land bounded by Campbell Street, Cleg Street, Herbert Street, the Gore Hill Freeway, and the Pacific Highway:

a. if the development is on land fronting the Pacific Highway, the maximum floor area to be used for ancillary office and/or showroom space is 50% of the total gross floor area

b. if the development is not on land fronting the Pacific Highway, the maximum floor area to be used for ancillary office and/or showroom space is 30% of the total gross floor area

Notes:

- Training rooms are calculated as part of the maximum permissible office/showroom area.
- Filing compactus are not considered as warehouse space and those areas will be calculated as part of the maximum permissible office/showroom area.
- A clear delineation must be maintained between the office/showroom areas and the production, warehouse, and other industrial areas of the premises.

5.3 Vehicle repair stations and vehicle body repair workshops

A vehicle repair station or vehicle body repair workshop must comply with these controls:

a. all vehicles waiting for repair or any other vehicles associated with the business are to be stored within the building or onsite in dedicated car parking spaces

b. one separate visitor car space per two 'work bays' is required for vehicles waiting to be serviced

c. a mechanical workshop bay, spray painting booth or car oven drying bay are deemed to be work bays

d. in the absence of any defined work bays, one separate visitor car space is required for every 36m^2 of factory floor area for vehicles waiting to be serviced

e. any further visitor car parking spaces may be used for vehicles waiting to be serviced

f. adequate area must be provided to ensure vehicles (including tow trucks) can enter and leave the site in a forward direction

g. water discharged from a car wash bay must comply with any trade waste agreement with Sydney Water

Notes:

- all vehicles brought to the premises for servicing must not stand or park in adjacent or nearby streets while unattended by the owner of that vehicle or while it is under the control of the manager or staff of the premises
- generally, these types of uses will not be granted approval if they are adjoining or adjacent to dwellings or residential zoned land

5.4 Specialised retail premises

Specialised retail premises are permitted on certain industrial land under Item 2 (Use of certain land at Herbert, Cleg and Frederick Streets and Reserve Road, Artarmon) and Item 34 (Use of certain land at Chatswood and Roseville) under Schedule 1 of Willoughby LEP.

Any application for this type of development must show the proposed use satisfies the definition of 'specialised retail premises' under Willoughby LEP.

5.5 Sex services premises

Sex service premises, commonly referred to as brothels, are not a prohibited land use in industrial zones. However, there are strict limitations on where they may be located.

A development application for a sex services premises must include this information:

a. number of sex workers and support staff, including any security personnel
b. hours of operation
c. name of the proprietor
d. floor plan showing the entry and exits from the premises, and number and description of rooms within the premises, for example reception/office, waiting or lounge area, and sex workers rooms
e. description of safety by design measures
f. site plan showing the location of places of public worship, schools, community facilities, child-care centres, parks, playgrounds, hospitals, medical centres, bus stops, dwellings or any place in the vicinity of the premises that may be regularly frequented by children within a 200m radius

These controls apply to sex services premises:

a. must not be located within public view from a place of public worship, school, community facility, child-care centre, hospital, medical centre, or any place in the vicinity of the premises that may be regularly frequented by children
b. must not form clusters within areas where brothels are permissible
c. approval will not be granted for another sex services premises that is located within a 100m of another existing approved premises
d. the operation is limited to not more than 10 sex worker rooms
e. owners, operators, managers, and staff must be aware and understand the 'Health and safety guidelines for sex service premises in NSW' by SafeWork NSW

'Health and safety guidelines for sex service premises in NSW' is at safework.nsw.gov.au

Note:

- Willoughby City Council's environmental health officers, and authorised officers of NSW Health and WorkCover, have the powers to conduct regular inspections of all sex services premises without notice to determine if the premises are complying with the required guidelines and health standards.

Attachment 1: Resource recovery and waste management plan

The resource recovery and management plan must include this information:

a. The type and estimated quantity of garbage generated, and number of collections per week.

b. The type and estimated quantity of recyclable materials generated, and number of collections per week.

c. An estimate of any garden or organic waste generated per week, and number of collections, if any, per week.

d. The size and number of mobile garbage bins and/or bulk bins to be used to manage garbage collection.

e. The type, size and number of recycling containers and any associated equipment to be used to manage recyclable materials.

f. The type, size and number of containers and any associated equipment, including any composting facilities, to be used to manage garden or organic materials.

g. Details of the location, size and access to garbage and recycling spaces/rooms.

h. Details of any refrigerated garbage rooms required for the disposal of seafood, poultry, meat, or other organic materials.

i. Details of the waste management arrangements, including safe and hygienic access to garbage and recycling spaces/rooms by employees and contractors. This should include:

    1. name of the contractor
    2. times, days and frequency to collect recyclable and general waste materials
    3. type and size of waste and recyclable collection vehicles to be used in the operation
    4. measures to achieve source separation of garbage and recyclable materials
    5. the display of signs to ensure source separation and correct use of bins
    6. access to water and provision of high pressure hose for washing bins and bin storage, and drainage to sewer
    7. person or persons responsible for cleaning bins and bin storage areas
    8. ventilation for bin storage areas
    9. features to prevent vermin entering into bin storage areas
    10. the method, distance, and grade of pathway to wheel mobile garbage bins to collection points
    11. driveway access to ensure vehicles can enter and leave in a forward direction
    12. education and any incentives to encourage correct waste management.

PART F

Transport and Parking Management

Part F: Transport and Parking Management

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1 Introduction

Willoughby City Council is committed to promoting Travel Demand Management by encouraging the use of active and public transport and minimising the adverse effects of car use in a way that sustains and enhances the economic and environmental qualities of the local government area.

Increasing the supply of car parking tends to encourage a greater number of vehicle trips. This increases congestion and impacts negatively on the city environment. We carefully consider when off-street car parking is allocated for developments and the amount of car parking allocated.

This part of Willoughby Development Control Plan (Willoughby DCP) outlines the transport requirements for off-street car parking, bicycle parking and end-of-trip facilities, loading/unloading facilities, and provisions for alternative transport modes.

1.1 Aim

The aim of this part is to reduce and manage the demand for car use and associated parking. This part also encourages the use of other alternative modes of transport.

1.2 Objectives

The objectives of this part are to:

a. ensure developments make adequate provision for a reasonable number of off-street car parking spaces, including accessible (disabled) parking spaces

b. ensure developments make adequate provision for parking and end-of-trip facilities for bicycles

c. ensure developments make adequate provision for electric and autonomous vehicles, car share spaces, and other alternative modes of transport

d. provide controls for the safe, convenient, and efficient movement of pedestrians, bicycles and vehicles for developments

e. ensure the access and design of car parking areas contribute positively to the public domain

f. reduce the demand for private car use and off-street car parking by employing the principles of travel demand management.

2 Strategies/studies

2.1 Willoughby Integrated Transport Strategy 2036

Willoughby Integrated Transport Strategy 2036 is Willoughby City Council's overarching framework for transport planning and initiatives to 2036. This part of Willoughby DCP aligns with the principles and intent of this important strategy for the local government area. The strategy is at willoughby.nsw.gov.au.

2.2 Review of Parking Rates

Consultants Cardno (now Stantec) carried out studies and provided two reports that reviewed the DCP parking rates, including bicycle parking and end-of-trip facilities.

The work stated that parking polices must reflect the need for parking across different types and uses. It also recognised that an oversupply of parking is just as undesirable as an undersupply.

The report recommended that parking rates may be reduced by employing the principles of travel demand management (TDM). This would generally need to be supported by a travel demand management plan (see Section 6 of this part).

The first report was supported by a Technical Memorandum dated 20 October 2022 which was entitled *Review of Parking Rates Addendum: Regional Shopping Centre Parking Rates.* This memorandum provided recommended minimum and maximum parking rates for regional shopping centres (premises greater than 30,000m^2), located in Chatswood and St Leonards. These rates were considered by Council and supported.

A second report investigated the inclusion of minimum parking rates for land uses in the Chatswood, St Leonards and the Artarmon railway precinct. This provide a 'banded rate' with a maximum and minimum rate for these railway precincts which Council has supported. The study also investigated the parking rates in local centres such as Castlecrag.

3 Parking provisions in the railway precincts of Chatswood, St Leonards and Artarmon.

The report recommended the following controls in the railway precincts: (See **Table 1** for details of a range of development types)

- Between 0.1 and 0.25 spaces for residential development in the Chatswood/St Leonards CBDs and between 0.1 and 0.5 in the Artarmon railways precincts (depending on the number of bedrooms).
- The minimum car parking rate for primary/destination developments (e.g. office/business, supermarkets and bulky good retail premises) be approximately 2/3 of their maximum rate.
- The minimum car parking rate for regional shopping centres (greater than 30,000m^2 in Chatswood and St Leonards) be just over 1/2 of their maximum rate.
- The minimum car parking rate for certain supporting/ancillary developments (e.g. small retail and food and drink premises) be approximately 1/3 of their maximum rate.
- The minimum car parking rate for certain unique uses (e.g. hospitals, medical centres, childcare centres) be between 1/3 and 1/2 of their maximum rate.
- A development application is required if a proposal does not satisfy all the criteria for either exempt or complying development under the Codes SEPP.

These rates were also considered by Council and supported.

3.1 Parking provisions outside of the railway precincts of Chatswood, St Leonards and Artarmon

The demand for car parking will vary from one development to another and in different locations. The parking rates outside of the railway precincts are 'target' rates. These rates are neither maximum nor minimum. However, any proposal to vary the number of spaces must be justified and consider the location and parking demand generated by the development.

A reduction of the target rate for developments along and within 200m of a major public transport route may be considered if the development is close to a regular bus service.

Major public transport routes are:

- Pacific Highway south of Albert Avenue
- Victoria Avenue west of Penshurst Street
- Willoughby Road
- Strathallen Avenue
- Penshurst Street south of Victoria Avenue
- Sailors Bay Road (for land zoned E1 Local Centre)
- Eastern Valley Way
- Smith Street
- High Street
- Mowbray Road

If part of a property falls within the 500m radius of the Artarmon railway precinct or within 200m of a major public transport route, the whole of that property is considered to be within the railway precinct or major public transport route.

- See Sections 5.3 and 5.4 of this part for details on visitor and accessible car parking.
- Unless otherwise stated, if the total number of car spaces is not a whole number, the number of spaces required must be rounded down.
- Any additional car spaces that may be allowed by Willoughby City Council will be included in the calculation of the gross floor area.
- If the development is an extension/modification to an existing development, more car parking space is only required if increasing the floor area creates further parking demand.

This second report also reviewed local centre parking rates, including bicycle parking and end-of-trip facilities. With respect to car parking rates for Castlecrag, the Stantec report stated that this suburb is highly dependent on private vehicle travel due in part to its geography, relatively poor walkability and access to public transport, as well as its older demographic.

The assessment found that Castlecrag is skewed towards a more aged and less accessible community with higher dependence on private vehicles, however not to an extreme extent that significantly segregates Castlecrag from other local centres such as Northbridge, Willoughby South, Willoughby High Street or Penshurst Street.

In relation to the Quadrangle site (100 Edinburgh Road) site, specific DCP controls in *Part L Section 13 Controls for Specific Sites* includes site specific car parking rates including accessible parking rates which will prevail over parking rates that apply in Part F.

The analysis determined that Council could reasonably approve an increase in the amount of car parking above the target rate in local centres including the number of disabled parking spaces, provided adequate justification is provided.

The proposed justification in local centres would need to include provision of the following:

- The assessment of likely demand for parking spaces, and
- Whether it is appropriate to allow a variation in the supply of spaces.

These are two separate considerations, outlined further below:

1. Car Parking Demand Assessment

An application to vary the number of car parking spaces required by the target car parking rates must be accompanied by a Car Parking Demand Assessment. The Car Parking Demand Assessment must address the following matters, to the satisfaction of the responsible authority:

- The likelihood of multi-purpose trips within the locality which are likely to be combined with a trip to the land in connection with the proposed use
- The variation of car parking demand likely to be generated by the proposed use over time
- The short-stay and long-stay car parking demand likely to be generated by the proposed use
- The availability of public transport in the locality of the land
- The convenience of pedestrian and cyclist access to the land
- The provision of bicycle parking and end of trip facilities for cyclists in the locality of the land
- The anticipated car ownership rates of likely or proposed visitors to or occupants (residents or employees) of the land
- Any empirical assessment or case study.

2. Appropriateness of Varying the Parking Supply

Before granting a permit to vary the number of parking spaces, Council must consider the following, as appropriate:

- The Car Parking Demand Assessment
- Any relevant local planning policy or incorporated plan
- The availability of alternative car parking in the locality of the land, including:
  a. Public car parks intended to serve the land
  b. On street parking in non-residential zones
  c. Streets in residential zones specifically managed for non-residential parking.
- On street parking in residential zones in the locality of the land that is intended to be for residential use
- The practicality of providing car parking on the site, particularly for small lots
- Whether a Green Travel Plan has been provided
- Existing and likely future traffic volumes on the surrounding road network and the nature of this network
- Any adverse economic impact a shortfall of parking may have on the economic viability of any nearby activity centre
- Any credit that should be allowed for car parking spaces provided on common land or by a cash-in-lieu payment or similar

- Local traffic management in the locality of the land
- The impact of the variation of car parking spaces on local amenity, including pedestrian amenity and the amenity of nearby residential areas
- The environmental implications of providing parking with particular regard to vegetation and landscape impacts
- The character of the surrounding area and whether varying the car parking provision would result in a positive urban design outcome
- Any other relevant consideration.

The recommendations contained in the Stantec report were considered and supported by Council and are now included as DCP requirements. **Table 1** at the end of this part details car parking requirements for a range of development types.

Maps showing Chatswood CBD, the St Leonards precinct and the Artarmon railway precinct are in Attachments 1, 2 and 3.

3.2 Motorcycle parking

Adequate provision must be made for motorcycle parking for all new developments and, where practicable, for alterations, additions and change of uses. All motorcycle parking requirements are minimum rates.

As a minimum, one space for residents and one visitor space is required for residential development comprising 10 or more dwellings, and one space for employees and one visitor space is required for commercial, retail, industrial and other non-residential buildings exceeding 500m^2.

Notes:

- If the total number of spaces is not a whole number, the number of spaces required must be rounded down.
- Motorcycle spaces must have a minimum dimension of 1.2m x 3m.

Table 2 at the end of this part details the requirements for motorcycle spaces**.**

3.3 Bicycle parking and end-of-trip facilities

The social, economic, and environmental benefits of bicycle use have created a greater emphasis on promoting and prioritising bicycle parking facilities.

The Stantec report recommends a minimum bicycle parking rate that is greater than existing demand to support ongoing and future shifts towards more sustainable transport mode targets.

Research has found that an increase in quality end-of-trip facilities is correlated with increased bicycle use. The provision of showers, personal lockers and change rooms also has the benefit of accommodating other people who walk or exercise regularly.

All new developments and wherever practicable, bicycle parking and end-of-trip facilities should be provided for alterations and additions and change of uses.

The bicycle parking and end-of-trip facilities are minimum requirements. As a minimum, all developments (excluding dwelling houses and dual occupancies) should provide 2 bicycle parking spaces to ensure at least a small amount of parking is provided at all destinations.

Table 3 at the end of this part details the requirements for bicycle parking. Table 4 details end-of-trip facilities.

3.4 The layout, design, and security of bicycle parking facilities

The layout, design and security of bicycle parking facilities must comply with the minimum requirements under Australian Standard 2890.3 (2015). The Standard classifies bicycle parking facilities by the level of security provided. These are:

a. Class A: a space within an individual bicycle locker that provides a high security locking system

b. Class B: a space within a secure room or structure that is accessed with devices such as keys, codes or swipe cards for communal areas (chain mesh fencing is not suitable for Class B facilities)

c. Class C: a space that allows a bicycle frame and wheels to be locked to a 'bicycle parking device' using the cyclists own 'locking device'

Notes:

- Class A and B bicycle parking facilities are intended for use by residents/employees of the development and must be in a secured area that is not accessible to the general public.
- Class C bicycle parking facilities are intended for use by visitors to the building and must be located in visible publicly accessible locations.
- AS2890.3 (2015) defines a 'bicycle parking device' as 'a construction made of high security material designed to provide stability to bicycles placed in or against the device'.
- Bicycle racks and rails are examples of a bicycle parking device.
- End-of-trip facilities include personal lockers for clothes, showers and change rooms.
- Where the total number of bicycle spaces and end-of-trip facilities is not a whole number, the number of spaces and facilities required must be rounded up.

4 Car parking and driveway design

4.1 Car parking

These controls apply to all car parking and access arrangements:

a. the layout of car parking spaces and structures must comply with AS/NZS 2890.1 and AS/NZS 2890.6

b. column locations must comply with AS/NZS 2890.1 (columns shall not be located within circulation aisles)

c. the headroom for an accessible (disabled) car space must comply with the requirements of AS 2890.6

d. car parking space accessed from a rear lane must provide a minimum setback of 1m with a 1m splay to the rear boundary

e. except for assigned residential parking spaces, the length of any blind aisle shall not exceed the width of six 90° parking spaces plus 1m

f. one way aisles/access handles may only be used for areas where there is no general public access, there are less than 6 parking spaces and where appropriate measures are provided to allow vehicles to pass within the site

g. ventilation must be provided under the relevant standards for basement car parking areas

h. for all developments other than dwelling houses and all developments with vehicular access from a State road frontage (Pacific Highway, Eastern Valley Way, Strathallen Avenue, Willoughby Road, Penshurst Street, Millwood Avenue, Fullers Road between Pacific Highway and Millwood Avenue and Boundary Street), vehicles must be able to turn around within the site to ensure they can enter and leave the site in a forward direction. Where more than two dwellings share a driveway, vehicles must be able to traverse the driveway and leave the site in a forward direction.

i. small car spaces are not permitted for residential developments, including single dwellings

j. for other than residential, the number of small cars shall be a maximum of 5% of the total parking spaces provided

4.2 Vehicle crossing/driveways

A new vehicle crossing is required if the proposal involves a change to the vehicle access arrangements or the number or location of car parking spaces and the existing vehicle crossing has any of these features:

a. is in a dilapidated condition with multiple cracks

b. is a trip hazard to pedestrians along the footpath alignment

c. there is evidence of significant damage from scraping on the road/laneway

d. is incomplete or has been constructed with non-approved materials

e. the crossing is unsafe due to its alignment with the proposed driveway or car parking space(s)

f. is in an unsafe location and there is a better alternative location for a new driveway

g. is proposed to be replaced as part of the development application

This requirement for a new crossing does not apply to proposed structures such as pergolas or open carports over an existing hard stand area.

Before construction, the applicant must obtain a separate vehicle crossing permit (construction of crossover on council property) from Willoughby City Council.

A new driveway crossing must comply with these controls:

h. the location, siting and grades of driveways, and driveway width must be in line with Australian Standard AS/NZS 2890.1

i. crossings must be separated from each other at the kerb and must be a minimum of 600mm from the side boundary for single dwelling / dual occupancy residential properties and 1.2m for other developments.

j. the distance between adjacent crossings must be less than 1.5m or greater than 6m to deter vehicles from attempting to park between driveways

k. driveways must not be located closer than:

- 2m from a street tree or 1.5m from any other tree or as otherwise directed by Willoughby City Council
- 1m from a power pole
- 1m from a council stormwater pit

l. if there is parking onsite for more than 6 vehicles, the driveway width must be sufficient to allow two vehicles to pass within the site

m. sight distance requirements at the property boundary must be provided in line with AS/NZS 2890.1

n. for single dwellings, only one crossing is permitted per property to improve streetscape, maximise on-street parking and reduce the number of conflict points between vehicles and pedestrians

o. dual occupancy developments may be permitted to have two vehicle crossings, providing the overall width of the two crossings does not exceed 30% of the property frontage

p. to maximise pedestrian safety, vehicles are to be at 90^0 to the boundary when leaving the site (where parking spaces are not in-line with the vehicle crossing, swept path diagrams are to show this requirement)

q. the maximum grade for all driveways, including to single dwellings, for the first 6m within the site shall be 1:20 for downgrades away from the property boundary.

r. vehicle crossings are to be comply with the requirements of Council's standard drawing SD105

s. vehicular Crossing must not result in the detrimental damage or removal of street trees

4.3 Access handle

For any access handle for battle-axe allotments serving two or more dwellings, these requirements apply:

a. the minimum width of the access handle is 4.8m

b. reciprocal rights of way are to be provided over each separate access strip

c. a passing bay in line with AS/NZS 2890.1 is required for access handles greater than 30m

d. on an arterial or sub-arterial road, the access driveway for the first 6m from the property boundary is to be 5.5m wide

e. vehicles are to traverse the access handle in a forward direction both entering and leaving the site.

4.4 Mechanical car parking systems and turntables

Willoughby City Council does not generally support mechanical car parking systems or turntables, and under no circumstances will they be allowed for accessible or visitor car parking.

An applicant must show there is no other reasonable car parking arrangement available. They must ensure that safety, emergency and noise issues relating to the ongoing operation of a mechanical car parking system and/or turntable have been addressed to Willoughby City Council's satisfaction.

Mechanical systems and turntables must satisfy all of these requirements:

a. the site is constrained by existing conditions

b. they are only used to access spaces assigned to a particular dwelling/unit/tenant/business

c. there is adequate standing space within the site for vehicles waiting to use the system

d. turntables may be used to ease turn paths but must not be the sole means to allow vehicles to turn around within the site; vehicles must be able to turn around using multi-point turns without using the turntable

e. a passing bay is to be provided at the entry to the car lift, to ensure a vehicle leaving the site is able to pass a vehicle waiting to use the car lift.

Note:

- The purpose of item (d) is to ensure vehicles can enter and leave the site if there is a malfunction with the turntable.

4.5 Electric vehicles

More households are transitioning to owning electric vehicles. All new dwellings and new dual occupancies, and significant alterations and additions to existing dwellings and dual occupancies are required to make provision for the installation of Level 2 electric vehicle (EV) charging stations.

As a minimum, new dwelling, new dual occupancies, and any significant alterations to existing dwellings or dual occupancies must make provision for at least a 40A (single phase) dedicated circuit to each garage.

Notes:

- Alterations and additions to an existing dwelling is regarded as ‘significant’ if the proposal has an estimated cost in excess of $100,000.
- The following are excluded from this control:
  - works not requiring a licenced electrician
  - works external to the principle dwelling (e.g. pools, secondary dwellings, boatsheds, and outbuildings)
- Notwithstanding the above, provision for EV charging stations needs to be provided in garages and car ports, even if they are external to the principle dwelling.
- Provision for EV charging must be made for all new major residential and non- residential developments (see section 5.6 in this Part for guidelines on EV charging).

5 Major development

In this part of Willoughby DCP, the definition of ‘major’ developments include:

a. new residential developments for attached dwellings, multi dwelling housing, residential flat buildings, seniors housing, boarding houses and hostels exceeding 300m^2 and 12 persons, shop top housing and mixed use developments

b. the erection of new buildings exceeding 1,000m^2, including commercial, industrial, institutional and other non-residential buildings

c. substantial redevelopment, refurbishment, or conversion of an existing building

Notes:

- Redevelopment is generally seen as ‘substantial’ if the proposal will result in more than a 10% increase in the ‘gross floor area’.
- Depending on the size and scale, and any likely impacts, an authorised officer may determine that a proposal is regarded as ‘major’ development.
- Institutional and non-residential buildings include health service facilities, recreation facilities, centre-based child care facilities, community facilities, a place of public worship, and other miscellaneous developments.

5.2 Parking and traffic studies

All new major developments and developments that involve substantial alterations and additions must include a parking and traffic impact assessment report with the development application. This report may also be required for a change of use that is likely to have adverse traffic and/or on-street parking impacts.

The report must assess the parking and traffic impacts of the development on the surrounding arterial and local road network. It must also identify any external traffic-related infrastructure requirements for the proposed development.

Note:

- Any traffic or parking counts done for a parking and traffic impact assessment report should not be done during public or school holiday periods.

5.3 Off-street loading/unloading facilities

All new major commercial, retail and industrial developments, and developments that involve substantial redevelopment with a floor area in excess of 500m^2 for commercial/retail developments and 1,000m^2 for industrial developments, must make adequate provision for off-street loading and unloading facilities.

A loading/unloading bay/dock for these types of developments must comply with these controls:

a. loading bay dimensions must conform with AS 2890.2

b. onsite turning areas must be provided to ensure service and delivery vehicles can enter and leave the site in a forward direction (the swept path design templates shown in AS 2890.2 must be used to determine the layout of service areas)

c. the minimum headroom must be in line with AS 2890.2

d. adequate provision must be made for garbage compactor units

e. internal waste collection areas must have a headroom clearance of at least 6.6m

f. splays must be provided to the loading bay areas and at the driveway to ensure adequate sight distances for pedestrians

For other existing or smaller commercial, retail and industrial developments, including change of use, the size and number of loading bays/docks will depend on the type, size and scale of the proposed development. This includes the frequency of deliveries, size and bulk of goods,

size of trucks, availability of on-street loading zones; and intended use of the commercial, retail or industrial premises.

As a minimum, provision should be made for a small rigid vehicle to access the site.

Note:

- The statement of environmental effects (SEE) or the parking and traffic impact assessment report must include details of the anticipated volume and frequency of deliveries, and the size of vehicles necessary to service the proposed development.

5.4 Visitor car parking

The visitor car parking rates in Chatswood CBD, St Leonards precinct and the Artarmon railway precinct are maximum rates. Visitor car parking spaces outside these areas, including major public transport routes, are target rates.

All visitor car parking spaces must be grouped together, sign posted and provided in a convenient and readily accessible location.

5.5 Accessible car parking for people with disability

All new developments and significant alterations and additions to major developments must, as a minimum, provide accessible car parking spaces for people with disability in line with Section D3.5 of the AS2890.6 Building Code of Australia under the National Construction Code.

For Class 2 buildings that contain two or more sole occupancy units, the required number of accessible spaces is calculated on the number of adaptable units required for that development. See Part B, Section 4.3.3 of Willoughby DCP to determine the number of adaptable units required for medium and high-density housing.

The National Construction Code identifies the class of building for different types of development. A guide to the building classifications is in Attachment 4.

Table 5 at the end of this part details the required number of accessible car parking spaces.

Accessible spaces must be clearly marked and provided in a communal car parking area. These spaces are not to be allocated to any individual unit or tenant.

There must be a continuous accessible path of travel from all accessible parking spaces to the entrance of the premises. In basement car parking areas, lifts must provide access to all levels.

Notes:

- Accessible car parking spaces must comply with the design requirements of AS/NZS 2890.6.
- Where the total number of accessible spaces is not a whole number, the number of spaces required is to be rounded up.
- The required number of accessible car spaces for unspecified developments will be assessed on the individual merits of the proposal, with regard to the nature and scale of the proposed development.
- Where practicable, accessible car parking spaces should be provided for all developments, including a change of use.

5.6 Stack car parking

Stack or tandem car parking is where one or more vehicles need to be moved to allow another vehicle to enter or exit a car space. This type of car parking is only permitted if the affected vehicles are allocated to the same owner or tenant of residential, commercial, retail or industrial premises. It is also only permitted if there are no safety issues and not more than 25% of vehicles are in a stacked parking arrangement.

Notes:

- Stack or tandem car parking will generally only be considered for employee vehicles associated with low turnover commercial, retail and industrial uses having less than a total of 12 car parking spaces.
- Stack or tandem car parking is not permitted for accessible or visitor car parking.
- Adequate area to manoeuvre vehicles in a stacked parking arrangement must be available on site for all major residential, commercial, retail and industrial developments.

5.7 Electric vehicle charging

Demand for public electric vehicle (EV) charging stations is set to increase. Willoughby City Council is committed to extending this infrastructure as suitable locations are identified. It is also necessary to make provision for electric vehicle charging on private properties. All types of new major residential and non-residential developments must be designed and constructed with appropriate electrical infrastructure to allow for future EV charging points.

1. The following controls apply to new residential flat buildings, multi dwelling housing, shop top housing and mixed use developments:

    a. All garages and car spaces allocated to an individual residential apartment must make provision for:

        - Level 2: single or 3-phase electric vehicle supply equipment with a power range of 7kW-22kW, as defined by NSW Electric and Hybrid Vehicle Plan (Future Transport 2056) from Transport for NSW, which provides faster, more secure charging.

    b. All communal car parking areas within a new major residential development must make provision for:

        - A minimum 5A per phase electrical capacity must be provided per space e.g:
            - If there are 4-9 spaces per level, provide one dedicated 63A three-phase EV charging switchboard per level.
            - If there are 10-19 spaces per level, provide one dedicated 100A three-phase EV charging switchboard per level.
            - If there are 20-39 spaces per level, provide one dedicated 200A three-phase EV charging switchboard per level.
            - If there are 40-80 spaces per level, provide one dedicated 400A three-phase EV charging switchboard per level.

In addition:

- Switchboards shall be dedicated and centrally located to facilitate simple connection of EV charging.

- Common areas should provide Level 2 cabling to each space, or where this is not possible, at least the provision of cable trays sufficient to accommodate electric circuitry between each car space and the dedicated EV charging switchboard.

2. The following controls apply to all new commercial, industrial and other major developments

    a. All communal car parking areas for new commercial, industrial and all other major developments must make provision for:

        - A minimum 5A per phase electrical capacity must be provided per space e.g:
            - If there are 4-9 spaces per level, provide one dedicated 63A three-phase EV charging switchboard per level.
            - If there are 10-19 spaces per level, provide one dedicated 100A three-phase EV charging switchboard per level.
            - If there are 20-39 spaces per level, provide one dedicated 200A three-phase EV charging switchboard per level.
            - If there are 40-80 spaces per level, provide one dedicated 400A three-phase EV charging switchboard per level

Notes:

- Switchboards shall be dedicated and centrally located to facilitate simple connection of EV charging.
- Individual industrial units, commercial/retail premises and other non-specified major developments are encouraged to provide EV charging points to facilitate the growing demand for electric vehicles.
- ‘Other’ major developments include, but not limited to: boarding houses; child care centres; gymnasiums; specialised retail premises (bulky goods stores), etc.
- EV charging may be required for ‘significant’ alterations and additions to major non-residential developments. Alterations and additions are generally regarded as ‘significant’ if the proposal will result in more than a 10% increase in the ‘gross floor area’. The requirement for the provision of EV charging will have regard to the nature and extent of the proposal, and practicality of retrofitting the electrical capacity for the building.

5.8 Car share spaces

There is a growing trend for individuals and businesses to use private car share vehicles as an economical alternative to car ownership. Car share schemes are generally more viable in locations where private car ownership is discouraged, such as Chatswood CBD, St Leonards precinct and the Artarmon railway precinct.

A reduction in car parking may be justified if car share spaces are provided and a legally binding car share scheme is put in place in lieu of the nominated car parking rates.

The Provision of car share spaces must satisfy the following controls:

a. one car share space may be substituted for 10 car parking spaces, whether they are maximum or ‘target’ rates in accordance with Table 1 (Parking Requirements)

b. car share spaces must be nominated on the plans submitted with the development application

c. all car share spaces are to be:
    - publicly accessible 24 hours a day, seven days a week
    - located together in a communal area with safe pedestrian access
    - located in an accessible area with access from a public road
    - located in a mobile phone signal area to ensure connectivity to the car share scheme
    - located adjacent to appropriate electrical infrastructure to allow for EV charging points
    - integrated with the streetscape and suitably screened with appropriate landscaping where the space is external
    - clearly designated as a car share space by signage and line marked

d. on-site car share spaces are to be retained as common property by the owner or the Owners Corporation of the site and not sold or leased to an individual owner, occupier or any external entity at any time

e. submission of documentary evidence that the owner or Owners Corporation has entered into a legally binding agreement for a car share scheme with a suitable car share operator (details to be submitted to Council prior to the issue of an Occupation Certificate)

f. the use and operation of the car share spaces must be managed by the owner or the Owners Corporation or contracted to a care share operator (details to be submitted to Council prior to the issue of an Occupation Certificate)

g. the car share spaces must be included in the land title to allow public access via covenants, building or strata management statements, by-laws or any other necessary instrument (details to be submitted to Council prior to the issue of an Occupation Certificate)

h. the owner or Owners Corporation must have appropriate insurance, including public liability (details to be submitted to Council prior to the issue of an Occupation Certificate)

Note:

- ***Car share scheme*** means a scheme in which a body corporate, an unincorporated body or a public authority owns or manages and maintains vehicles for shared or communal use and hires those vehicles exclusively to members of the scheme for occasional use for short periods of time, on demand and on a pay-as-you go basis.
- Car share spaces are to be rounded down to a whole number if it is not a whole number.
- On-street parking permits will not be available for owners or occupiers of any new developments that are subject to the car parking provisions under Part F (Transport and Parking Management) of the WDCP.

5.9 Autonomous vehicles

The use of autonomous vehicles may be a reality in the foreseeable future. Production and active testing is underway in major cities around the world to ensure the safe and efficient operation of these vehicles. This means appropriate infrastructure to support the use of autonomous vehicles needs to be in place for new developments.

All car parking areas must provide clear line marking around the base of columns, kerbs and driveway paths. There must be clearly defined (white or yellow) definition to columns and poles to improve visibility to autonomous vehicles.

Note:
As technology evolves, further requirements may be required to support the safe and efficient operation of autonomous vehicles.

5.10 Further controls for medium and high-density residential accommodation

These further controls apply to attached dwellings, multi dwelling housing, residential flat buildings, seniors housing, boarding houses and hostels exceeding 300m^2 and 12 persons, shop top housing, and mixed use developments comprising residential units:

a. visitor parking must be suitably grouped, clearly marked and conveniently located

b. visitor parking is to be designed to comply with Class 2 Medium Term parking in line with AS/NZS2890.1

c. any security for residents' vehicles must be installed to ensure that it does not impede access to visitor spaces

d. visitor car parking may be located forward of the building line providing it does not detrimentally impact the streetscape

e. all above ground car parking areas must be well landscaped and include the use of absorptive surfaces such as pavers or 'grasscrete' to soften the appearance of these areas

f. resident visitor car parking spaces must be available at all times for shop top housing and mixed use developments

g. the provision of car wash bays is not encouraged; if a car wash bay is proposed, the space must be connected to the sewer and serviced by recycled or tank stored water, and the development application must include the hydraulic details

h. provision must be made for removalist vans to park, load and unload onsite for all developments of more than 12 units or where no kerbside parking is available in front of the site on a classified road

i. pedestrian access to dwellings should be separate from vehicular access to the site

j. for residential accommodation of between 6 and 18 units, bulk waste storage bins must be provided on the site and adequate provision must be made for garbage vehicles to enter and leave the site in a forward direction

k. for residential accommodation of 4 or more storeys or 18 or more units with basement car parking, bulk waste storage bins must be provided in the basement with adequate height and provision for garbage vehicles to enter and leave the site in a forward direction

5.11 Additional controls for a centre-based child care facility

A centre-based child care facility includes a building or place that provides long day care, occasional child care, out of hours school care, preschool or an approved family day care venue.

These further additional controls apply to all types of child care centres:

a. access driveways widths are to be in line with AS/NZS 2890.1

b. car parking is to be provided on site for setting down and picking up children

c. vehicles must enter and leave the site in a forward direction

d. vehicle parking must be separated from any outdoor play areas to ensure safety and vehicle emissions do not affect children

e. pedestrian access must be separated from the access and manoeuvring of vehicles in the driveway and car parking areas

These controls apply to pre-school and long day care centres in residential areas outside Chatswood CBD:

f. a minimum 20m frontage is required or a corner block location in all residential areas to allow for a one-way drive through entry and exist

g. a drive-through drop-off and pick-up area is required at the rate of 1 car space per 10 children within the driveway

h. the width of the driveway must allow for a vehicle exiting the site to pass a parked vehicle

These controls apply to child care centres in Chatswood CBD:

i. staff parking is to be clearly marked to reflect that they are for the exclusive use of staff associated with the operation of the child care centre between 7am and 7pm

j. a vehicle drop-off/pick-up area is to be provided at the rate of 1 space per 8 children

k. if the drop-off/pick-up area is in a basement car park area, it must be within 20 metres of a lift providing direct access to the child care centre

l. if the drop-off/pick-up area is not provided onsite, the matter will be referred to Willoughby City Council's traffic committee to consider

m. the drop-off/pick-up area should be clearly marked to reflect that it is for the exclusive use of the childcare users during the centre's morning and afternoon peak times, generally from 7am to 9.30am and 4pm to 6.30pm

Note:

- Depending on the size, scale, and location of the child care centre in a commercial area outside of Chatswood CBD, the number of car spaces and any other controls will be considered on the individual merits of the proposed development.

6 Alternative and sustainable transport strategies

Willoughby City Council encourages alternative and sustainable transport strategies to reduce the use and reliance on motor vehicles as the principal mode of transport.

Alternative transport strategies can be achieved by including a 'travel demand management plan' (also known as a 'green travel plan') in the business operation.

This plan has details on how to modify travel decisions to and from a business premises so that more desirable modes of transport are used and to reduce the adverse impacts of car travel. Examples include use of bicycles, carpooling, mini-bus pick-up/drop off, public transport services and provision of car share spaces.

Strategies in a travel demand management plan may include:

a. provide information to staff on public transport options available
b. provide staff with a discount or subsidy on public transport costs
c. provide a bus to pick up and drop off staff to the nearest railway station or bus interchange
d. provide staff with cycling allowances, loans and/or insurance
e. provide secure bicycle storage and shower and change room facilities
f. introduce a staff car sharing scheme for fleet vehicles
g. use private transport services or public transport for work related journeys
h. create a carpool scheme to and from work and provide priority parking for staff who carpool with 2 or more passengers

Subject to a written agreement and conditions of consent to implement a travel demand management plan, Willoughby City Council may reduce the required number of car parking spaces for a development.

7 Laneway widening

Increased urban development often requires the local road network to be upgraded to cope with additional traffic and improve access and traffic circulation. Willoughby City Council has identified a number of laneways that need to be widened to accommodate future commercial and residential developments. Affected properties must dedicate a portion of land when redevelopment occurs.

The affected properties are identified in Attachment 5**.**

Notes:

- Redevelopment of many properties has already resulted in the dedication of land for future lane widening purposes.
- Additional road works may be required to mitigate associated traffic impacts arising from a proposed development.
- The ‘gross floor area’ or ‘floor space ratio’ is calculated on the existing site area (before any dedication of land).

Table 1 Car parking rates

Land use category	Development type	Chatswood CBD and St Leonards precinct (maximum rates)	Artarmon railway precinct (maximum rates)	Chatswood CBD and St Leonards precinct (minimum rates)	Artarmon railway precinct (minimum rates)	All other areas, including major public transport route (target rates)
Residential	a. dwelling houses, attached, and detached dual occupancies	1 space/dwelling	1 space/dwelling	1 space/dwelling •	1 space/dwelling • 0	1 space/dwelling with 2 bedrooms or less; 2 spaces/dwelling with 3 bedrooms or more
Residential	b. manor houses, attached dwellings and multi dwelling housing not exceeding 2 storeys or more than 4 dwellings in the R3 zone	0.5 space per studio and 1, 2, 3 or more bedroom units; 1 visitor space per 7 dwellings	0.5 space per studio and 1 and 2 bedroom units; 1 space per 3+ units; 1 visitor space per 7 dwellings	• 0.1 spaces per studio/ 1-bedroom flat • 0.2 spaces per studio/ 2-bedroom flat • 0.25 spaces per studio/ 3+ bedroom flat	• 0.1 spaces per studio/ 1-bedroom flat • 0.2 spaces per studio/ 2-bedroom flat • 0.25 spaces per studio/ 3+ bedroom flat	0.5 space per studio and 1 bedroom unit; 1 space per 2 and 3 or more bedroom units; 1 visitor space per 7 dwellings
Residential	c. affordable housing, including boarding houses and group homes	The minimum number of car parking spaces must comply with Housing SEPP 2021	The minimum number of car parking spaces must comply with Housing SEPP 2021	The minimum number of car parking spaces must comply with Housing SEPP 2021	The minimum number of car parking spaces must comply with Housing SEPP 2021	The minimum number of car parking spaces must comply with Housing SEPP 2021

Land use category	Development type	Chatswood CBD and St Leonards precinct (maximum rates)	Artarmon railway precinct (maximum rates)	Chatswood CBD and St Leonards precinct (minimum rates)	Artarmon railway precinct (minimum rates)	All other areas, including major public transport route (target rates)
Residential	d. seniors housing, including residential care facility, hostels and self-contained dwellings	The minimum number of car parking spaces must comply with Housing SEPP 2021	The minimum number of car parking spaces must comply with Housing SEPP 2021	The minimum number of car parking spaces must comply with Housing SEPP 2021	The minimum number of car parking spaces must comply with Housing SEPP 2021	The minimum number of car parking spaces must comply with Housing SEPP 2021
Residential	e. multi dwelling housing, residential flat buildings and attached dwellings with more than 2 storeys and 4 or more dwellings	0.5 space per studio and 1, 2, 3 or more bedroom units; 1 visitor space per 7 dwellings	0.5 space per studio and 1 and 2 bedroom units; 1 space per 3+ units; 1 visitor space per 7 dwellings	• 0.1 spaces per studio/ 1-bedroom flat • 0.2 spaces per studio/ 2-bedroom flat • 0.25 spaces per studio/ 3+ bedroom flat	• 0.1 spaces per studio/ 1-bedroom flat • 0.2 spaces per studio/ 2-bedroom flat • 0.25 spaces per studio/ 3+ bedroom flat	0.5 space per studio and 1 bedroom unit; 1 space per 2 and 3 or more bedroom units; 1 visitor space per 7 dwellings

Land use category	Development type	Chatswood CBD and St Leonards precinct (maximum rates)	Artarmon railway precinct (maximum rates)	Chatswood CBD and St Leonards precinct (minimum rates)	Artarmon railway precinct (minimum rates)	All other areas, including major public transport route (target rates)
Residential	f. dwellings in shop top housing and mixed-use developments (further car parking for the commercial component is to be provided in line with the requirements for commercial and retail premises)	0.5 space per studio and 1, 2, 3 or more bedroom units; 1 visitor space per 7 dwellings	0.5 space per studio and 1 and 2 bedroom units; 1 space per 3+ units; 1 visitor space per 7 dwellings	• 0.1 spaces per studio/ 1-bedroom flat • 0.2 spaces per studio/ 2-bedroom flat • 0.25 spaces per studio/ 3+ bedroom flat	• 0.1 spaces per studio/ 1-bedroom flat • 0.2 spaces per studio/ 2-bedroom flat • 0.25 spaces per studio/ 3+ bedroom flat	0.5 space per studio and 1 bedroom unit; 1 space per 2 and 3 or more bedroom units; 1 visitor space per 7 dwellings
Hotels, motels and serviced apartments	g. hotels, motels and serviced apartments (if a restaurant or function room is included, the parking rates for that use is to be in line with the relevant parking rate)	0.25 space/room	0.75 space per room	0.1 spaces per room	0.25 spaces per room	0.75 space per room

Land use category	Development type	Chatswood CBD and St Leonards precinct (maximum rates)	Artarmon railway precinct (maximum rates)	Chatswood CBD and St Leonards precinct (minimum rates)	Artarmon railway precinct (minimum rates)	All other areas, including major public transport route (target rates)
Bed and breakfast accommodation	h. bed and breakfast accommodation	The minimum number of car parking spaces must comply with Codes SEPP 2008	The minimum number of car parking spaces must comply with Codes SEPP 2008	The minimum number of car parking spaces must comply with Codes SEPP 2008	The minimum number of car parking spaces must comply with Codes SEPP 2008	The minimum number of car parking spaces must comply with Codes SEPP 2008
Office/business/retail	i. office and business premises	1 space/400m^2	1 space/75m^2	1 space per 670 m^2	1 space per 100 m^2	1 space/60m^2
Office/business/retail	j. retail premises (excluding supermarkets)	1 space/70m^2	1 space/50m^2	1 space per 200 m^2	1 space per 145 m^2	1 space/33m^2
Office/business/retail	k. retail premises (supermarkets)	1 space per 70 m^2	1 space per 50 m^2	1 space per 100 m^2	1 space per 70 m^2	1 space/33m^2
Office/business/retail	l. Regional shopping centres (greater than 30,000 m^2) *(applies only to additional/new regional shopping centre floor space).*	1 space per 40 m^2	N/A	1 space per 70 m^2	N/A	N/A

Land use category	Development type	Chatswood CBD and St Leonards precinct (maximum rates)	Artarmon railway precinct (maximum rates)	Chatswood CBD and St Leonards precinct (minimum rates)	Artarmon railway precinct (minimum rates)	All other areas, including major public transport route (target rates)
Office/business/retail	m. specialised retail premises (these controls apply to garden centre, plant nursery, hardware and building supplies, landscaping material supplies, timber yards and the like)	1 space/150m^2 of showroom and outdoor areas accessible to the public	1 space/125m^2 of showroom and outdoor areas accessible to the public	1 space per 200 m^2	1 space per 200 m^2	1 space/100m^2 of showroom and outdoor areas accessible to the public
Office/business/retail	n. wholesale supplies (in a business zone)	1 space/70m^2	1 space/50m^2	1 space per 200 m^2	1 space per 200 m^2	1 space/33m^2
Motor vehicle services	o. service stations	5 holding bays per service/workshop bay; plus office and retail space in line with office and retail rates	5 holding bays per service/workshop bay; plus office and retail space in line with office and retail rates	N/A	N/A	5 holding bays per service/workshop bay; plus office and retail space in line with office and retail rates
Motor vehicle services	p. vehicle repair stations	5 holding bays per service/workshop bay; plus office space in line with office rates	5 holding bays per service/workshop bay; plus office space in line with office rates	N/A	N/A	5 holding bays per service/workshop bay; plus office space in line with office rates

Land use category	Development type	Chatswood CBD and St Leonards precinct (maximum rates)	Artarmon railway precinct (maximum rates)	Chatswood CBD and St Leonards precinct (minimum rates)	Artarmon railway precinct (minimum rates)	All other areas, including major public transport route (target rates)
Motor vehicle services	q. vehicle sales and hire premises	1 space per 200m^2 of display area for customer parking; plus 5 holding bays per service/workshop bay; plus office space in line with office rates	1 space per 200m^2 of display area for customer parking; plus 5 holding bays per service/workshop bay; plus office space in line with office rates	N/A	N/A	1 space per 200m^2 of display area for customer parking; plus 5 holding bays per service/workshop bay; plus office space in line with office rates
Child care centres	r. long day centres	1 space/20m^2	1 space/20m^2	1 space per 60 m^2	1 space per 60 m^2	1 space/20m^2
Child care centres	s. pre-school child care centres (within 500m of railway station)	1 space/66m^2	1 space/66m^2	1 space per 80 m^2	1 space per 80 m^2	1 space/32.5m^2
Child care centres	t. occasional care and out-of-school care centres (within 500m of railway station)	1 space/26m^2	1 space/26m^2	1 space per 80 m^2	1 space per 80 m^2	1 space/20m^2
Food and beverage	u. food and drinks premises (includes restaurant, café and take away food and drinks premises)	1 space/50m^2	1 space/50m^2	1 space per 145 m^2	1 space per 145 m^2	1 space/33m^2

Land use category	Development type	Chatswood CBD and St Leonards precinct (maximum rates)	Artarmon railway precinct (maximum rates)	Chatswood CBD and St Leonards precinct (minimum rates)	Artarmon railway precinct (minimum rates)	All other areas, including major public transport route (target rates)
Food and beverage	v. pubs	1 space/50m^2	1 space/50m^2	1 space per 145 m^2	1 space per 145 m^2	1 space/33m^2
Health and community services	w. hospitals	3 spaces per bed	3 spaces per bed	2 spaces per bed	2 spaces per bed	3 spaces per bed
Health and community services	x. medical centres (includes consulting rooms, medical centres and veterinary hospitals/clinics)	2 spaces per consulting room	2 spaces per consulting room	1 space per consulting room	1 space per consulting room	2 spaces per consulting room
Industrial	y. industrial activity, including high technology industries	1 space/100m^2 of factory floor area (including any storage areas); plus 1 space/60m^2 of office area and showroom area accessible to the public	1 space/100m^2 of factory floor area (including any storage areas); plus 1 space/60m^2 of office area and showroom area accessible to the public	N/A	N/A	1 space/100m^2 of factory floor area (including any storage areas); plus 1 space/60m^2 of office area and showroom area accessible to the public

Land use category	Development type	Chatswood CBD and St Leonards precinct (maximum rates)	Artarmon railway precinct (maximum rates)	Chatswood CBD and St Leonards precinct (minimum rates)	Artarmon railway precinct (minimum rates)	All other areas, including major public transport route (target rates)
Industrial	z. self-storage units	1 space/300m^2 of storage area; plus a minimum of two visitor parking spaces are to be provided at the front of the premises in a publicly accessible area for customers.	1 space/300m^2 of storage area; plus a minimum of two visitor parking spaces are to be provided at the front of the premises in a publicly accessible area for customers	N/A	N/A	1 space/300m^2 of storage area; plus a minimum of two visitor parking spaces are to be provided at the front of the premises in a publicly accessible area for customers
Industrial	aa. warehouse, distribution centres, and wholesale supplies (in an industrial zone)	1/300m^2 of internal and external storage areas plus 1 space per 60m^2 of office space	1/300m^2 of internal and external storage areas plus 1 space per 60m^2 of office space	N/A	N/A	1/300m^2 of internal and external storage areas plus 1 space per 60m^2 of office space
Sex services premises	bb. Sex services premises	2 spaces per room	2 spaces per room	1 space per room	1 space per room	2 spaces per room

Notes:

1. If a specific land use is not included in this table, the car parking requirements will be assessed on the merits of the application. A parking and traffic impact assessment report may be needed with the application to determine the number of car spaces required.
2. Where the amount of parking required is not a whole number, the number of spaces required will be rounded down to the nearest whole number.

Table 2 Motorcycle parking rates

Zone type and usage	Chatswood CBD, St Leonards precinct and Artarmon railway precinct	All other areas, including major public transport routes
Business zones	1 space per 20 car parking spaces; 1 visitor space per 10 motorcycle spaces	1 space per 30 car parking spaces;1 visitor space per 15 motorcycle spaces
Industrial zones	1 space per 20 car parking spaces; 1 visitor space per 10 motorcycle spaces	1 space per 30 car parking spaces;1 visitor space per 15 motorcycle spaces
Residential zones and the residential components of shop top and mixed-use developments	1 space per 20 car parking spaces; 1 visitor space per 10 motorcycle spaces	1 space per 20 car parking spaces;1 visitor space per 10 motorcycle spaces

Notes:

- Motorcycle parking rates for land uses not included in this table will be assessed on the merits of the application
- As a minimum, one space for residents and one visitor space is required for residential development comprising 10 or more dwellings
- As a minimum, one space for employees and one visitor space is required for commercial, retail, industrial and other non-residential buildings exceeding 500m^2
- Visitor spaces are included in the total number of spaces required

Table 3 Requirements for bicycle parking and end-of-trip facilities

Zone type and usage	Chatswood CBD, St Leonards precinct and the Artarmon railway precinct	All other areas, including major public transport routes
Business	1 Class A or B bicycle space per 10 car parking spaces Minimum 1 Class C bicycle space or 1 space per 10 Class A or B bicycle spaces, whichever is the greater	1 Class A or B bicycle space per 20 car parking spaces Minimum 1 Class C bicycle space or 1 space per 20 Class A or B bicycle spaces, whichever is the greater
Industrial	Minimum 1 Class A or B bicycle space or 1 space per 10 car parking spaces, whichever is the greater Minimum 1 Class C bicycle space or 1 space per 10 Class A or B bicycle spaces, whichever is the greater	Minimum 1 Class A or B bicycle space or 1 space per 20 car parking spaces, whichever is the greater Minimum 1 Class C bicycle space or 1 space per 20 Class A or B bicycle spaces, whichever is the greater
Residential (excluding dwelling houses and dual occupancies) and	1 Class A or B parking spaces per 10 units 1 Class C (rails/racks) per 10 apartments for visitors	1 Class A or B parking spaces per 20 units 1 Class C (rails/racks) per 20 apartments for visitors

Zone type and usage	Chatswood CBD, St Leonards precinct and the Artarmon railway precinct	All other areas, including major public transport routes
residential components of shop top and mixed-use developments		

Table 4: End-of-trip facilities for non-residential developments

Showers	Change rooms	Lockers
One shower per 5 bicycle parking spaces (Class A or B)	One change room per shower	Two personal lockers per bicycle space

Notes for Tables 3 and 4:

- All bicycle parking and end-of-trip facilities for land uses not included in these tables will be assessed on the merits of the application.
- If no car parking or reduced car parking is provided in the Chatswood CBD, St Leonards precinct or the Artarmon railway precinct, bicycle parking and end-of-trip facilities must be provided and calculated on the maximum number of car parking spaces that would otherwise be required.
- Where more than one shower or change room is required for end-of-trip facilities, separate male and female facilities must be provided.

Table 5 Accessible car parking requirements

Class of building	Number of accessible spaces
Class 1a	Nil
Class 1b	1 space for each accessible or adaptable unit
Class 2	Whichever is the greater: a. Minimum 1 resident and 1 visitor space for developments comprising 10 or more units b. 1 space/4 accessible or adaptable units + 1 visitor space for developments comprising 50 or more car parking spaces
Class 3	Whichever is the greater: a. Minimum 1 resident and 1 visitor space for developments exceeding 300m^2 and 12 persons b. 1 space for each accessible or adaptable unit (10% of spaces must be provided as a visitor space) c. 3% of the total car parking spaces (10% of spaces must be provided as a visitor space)
Class 4	1 accessible space
Class 5, 6, and 8	In parking areas with 5 or more spaces, whichever is the greater:

Class of building	Number of accessible spaces
	a. 1 employee and 1 visitor space b. 3% of the total car parking spaces (10% of spaces must be provided as a visitor space)
Class 7 and 9b	In parking areas with 5 or more spaces, whichever is the greater: a. 1 space b. 3% of the total car parking spaces
Class 9a	Whichever is the greater: a. 1 employee and 1 visitor space b. 4% of the total car parking spaces (10% of spaces must be provided as a visitor space)

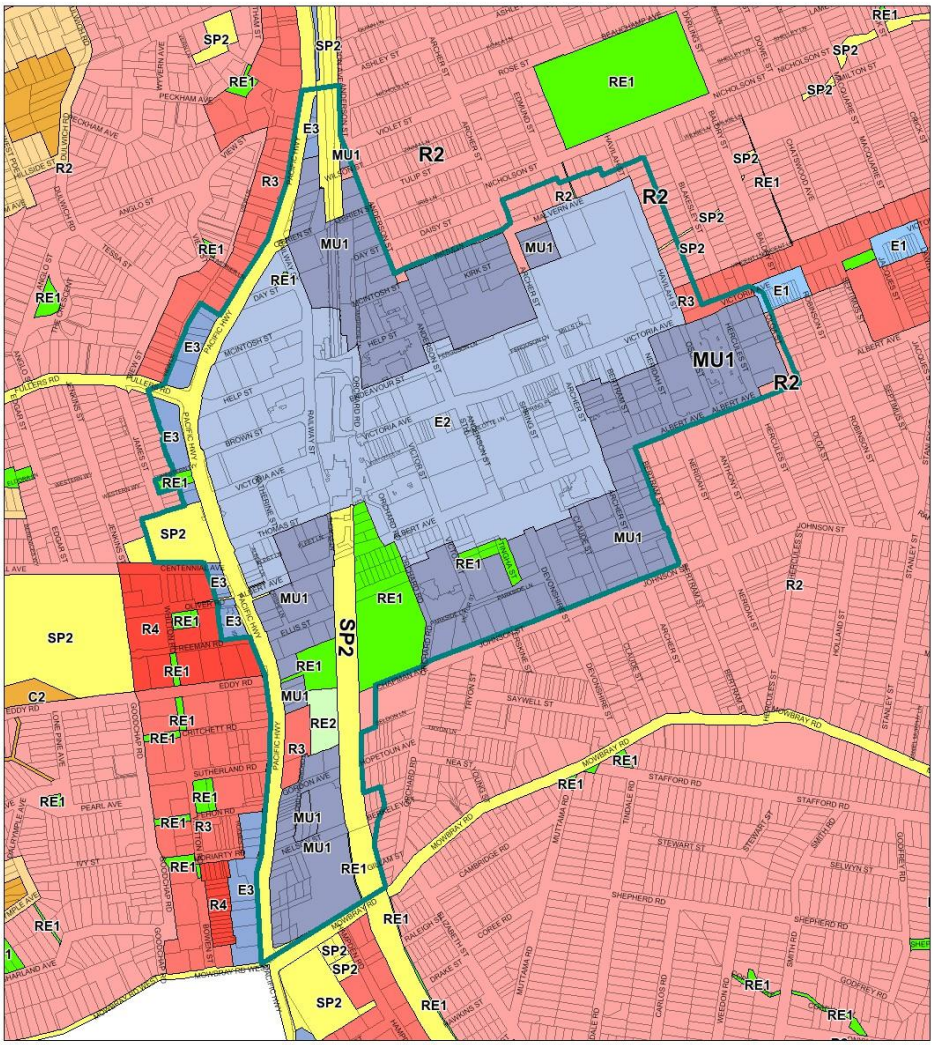
Notes:

- Accessible parking is included in the total number of car parking spaces required. However, if no parking is required, the minimum number of employee/resident and visitor spaces must be provided.
- Additional accessible car parking spaces may be required in line with Section D3.5 of the Building Code of Australia under the National Construction Code.
- If the requirements under Section D3.5 change, the accessible car parking controls under this part of Willoughby DCP change accordingly.
- Applicants may apply a ‘performance solution’ under the Building Code of Australia to satisfy the provision of accessible car parking in lieu of the ‘deemed-to-satisfy solution’ requirements.

Attachment 1: Chatswood CBD

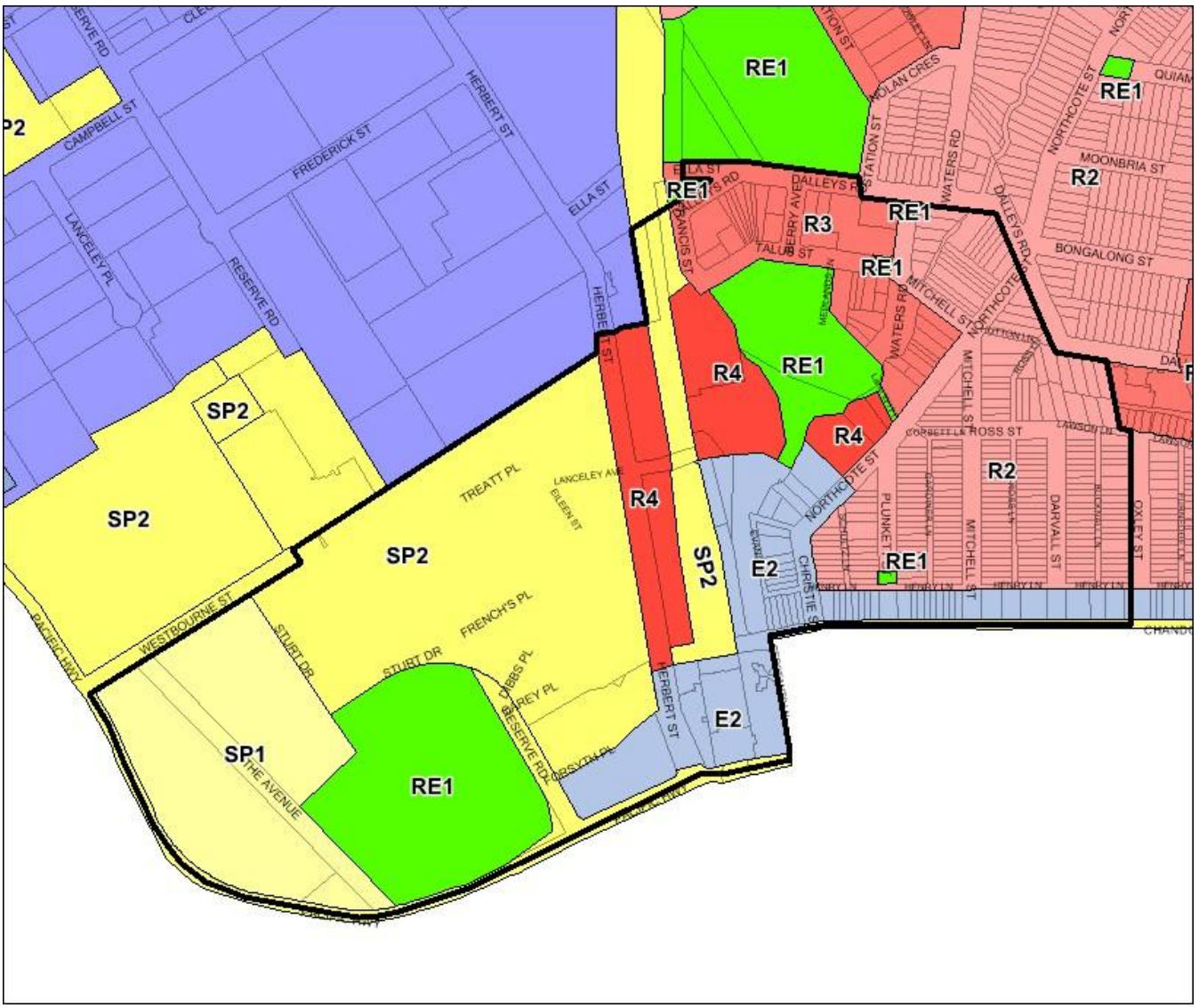
Map 1 Chatswood CBD

Chatswood CBD Willoughby 2012 Local Environmental Plan
Land Zoning Map



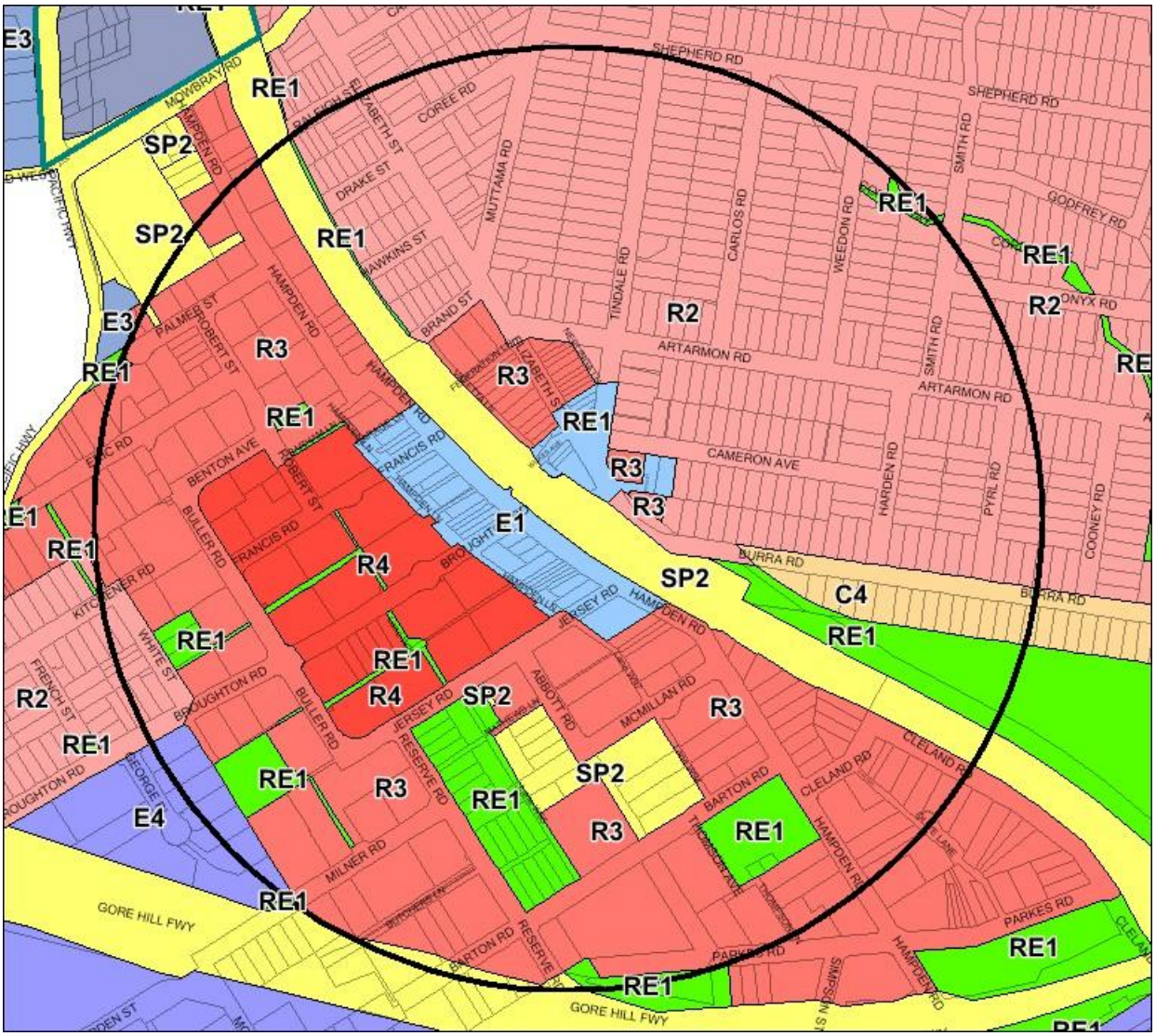
Attachment 2: St Leonards precinct

Map 2 St Leonards precinct



Attachment 3: Artarmon railway precinct

Map 3 Artarmon railway precinct



Attachment 4: Building Code of Australia building classifications

This table identifies the class of building and the different types of development:

Table 6 Building Code of Australia building classifications

Class of building	Types of development
Class 1a	Single dwellings. This includes a detached house or semi-detached dwellings, semi-detached dual occupancies, terraces, townhouses and villa units with fire resisting walls and no basement carpark.
Class 1b	Boarding house, guest house, hostel, or the like, not exceeding 300m^2 and not more than 12 residents.
Class 2	Building containing two or more sole occupancy units. This includes dual occupancies/duplexes (where one unit is above the other), serviced apartments with a kitchen, apartments in shop-top housing, and residential flat buildings.
Class 3	A residential building, other than a Class 1 or 2 building, including; boarding house, hostel, backpackers accommodation, and retirement villages.
Class 4	A single dwelling within a class 5, 6, 7, 8 or 9 building.
Class 5	An office building used for professional or commercial purposes, excluding Class 6, 7, 8 or 9 buildings.
Class 6	Shops and buildings used for supply of services directly to the public, including; café, restaurant, hairdresser's shop, public laundry, showroom, and service station.
Class 7a	A carpark.
Class 7b	Warehouses, storage/display of goods for sale by wholesale.
Class 8	A laboratory or building (factory) for the production, assembly, altering, repairing, packing, finishing or cleaning of goods for trade, sale or gain.
Class 9a	Healthcare building.
Class 9b	Assembly building including halls, libraries, schools, early childhood centres, church, theatre, nightclub/disco, cinema, and the like.
Class 9c	Aged care building.
Class 10a	Non habitable building such as private garage, carport, shed or the like.
Class 10b	Non habitable structures such as fences, retaining walls, swimming pools, or the like.

More information is at abcb.gov.au/Resources/Publications/Education-Training/Building-classifications.

Attachment 5: Laneway widenings

Table 7 Properties affected by laneway widening

Lane or road name	Properties affected	Suburb	Laneway widening required
George Brain Lane	327–341 Penshurst Street	Chatswood	1.2m from west side of west leg
George Brain Lane	72 Victoria Avenue	Chatswood	1.2m from west side of east leg
George Brain Lane	72–80 and 92–96 Victoria Avenue	Chatswood	2m from north side of southern leg
Stirling Lane	202–212, 218–224 Sydney Street; 334–348 and 352 Penshurst Street	Chatswood	2m from east side
Thomas Lane	73 Albert Avenue; 2 Thomas Street	Chatswood	3m widening on western side
View Lane, between Leplastrier Lane and Fullers Road	734, 754, 760, and 772–778 Pacific Highway; 1–7 and 25 View Street	Chatswood	2m from both sides
View Lane, north of Leplastrier Lane	786, 794, 812A–832 Pacific Highway; 49 View Street; 8–10 View Lane	Chatswood	2m from east side
Whitton Road between Moriarty and Fehon Roads	2 Fehon Road (Lot 3 DP 171741)	Chatswood	3m from east side
Albert Avenue	34, 34A and 34B Albert Avenue	Chatswood	3m from north side
Burke Lane, between Waters Road and Station Street	25 Station Street; 36A–40 Park Road; 54–56 Waters Road	Naremburn	1.2m from north side
Kershaw Lane, parallel to Willoughby Road and Wheatleigh Street	141 and 155 Willoughby Road	Naremburn	1.2m from west side
Lawson Lane west of Oxley Street	2, 4 and 6 Lawson Lane	Naremburn	1.2m from north side
Appleton Lane, between Euroka and Bellambi Streets	157–173 and 179–181 Sailors Bay Road; 10 Kiola Road and Lot 1 DP 1048708	Northbridge	6 metre continuation of existing laneway between Euroka and Bellambi Streets
Jeffery Lane, east of Woonona Road	288–294 Sailors Bay Road	Northbridge	1.2m from north side
Wagschall Lane	262–278 Sailors Bay Road	Northbridge	1.2m from north side
Dougherty Lane	40–44 Forsyth Street	North Willoughby	1.2m from north side

Lane or road name	Properties affected	Suburb	Laneway widening required
Iceworks Lane, between Alexander Ave and McClelland Street	181–195 High Street	North Willoughby	1.2m from west side
Adamson Lane, between Hudson Ave and Harris Street	569–593 Willoughby Road	Willoughby	1.2m from west side
Harris Lane, between Harris Street and Frenchs Road	12 Harris Street; 525–535, 545–547 and 553 Willoughby Road	Willoughby	2m from west side
Nathan Lane	449, 453– 465 and 485 Willoughby Road	Willoughby	1m from west side
Osborn Lane, off Harris Lane	6–8A Harris Street	Willoughby	1.2m from north side
Tulloh Lane, west of Tulloh Street	46–52 Frenchs Road; 68 Tulloh Street	Willoughby	2m from north side
Tulloh Lane, east of Tulloh Street	30–42 Frenchs Road	Willoughby	1.2m from north side
Hampden Lane between Jersey Road and Garage Lane.	64-66, 70-80, 84, 88-92, 96-118, 122-134 Hampden Road (even numbers only) and 2A Broughton Road.	Artarmon	3m from east side

PART G

Vegetation Management

Part G: Vegetation Management

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1 Introduction

This part of Willoughby Development Control Plan (Willoughby DCP) is made under Part 2.2 of State Environmental Planning Policy (Biodiversity and Conservation) 2021.

1.1 Aim

The aim of this part is to:

a. declare vegetation that requires a landholder to seek a permit for removal under Clause 2.6 of SEPP (Biodiversity and Conservation) 2021.

b. promote best practice and sustainable vegetation management on development sites

1.2 Objectives

The objectives of this part are to:

a. prevent unnecessary damage or removal of trees and other vegetation

b. maintain and enhance the urban landscape

c. recognise and protect vegetation considered significant due to heritage, cultural, social and ecological criteria

d. enhance species diversity, connectivity and recovery of fragmented habitat

e. increase tree canopy to mitigate urban heat island effects

f. facilitate the removal of undesirable exotic plants, noxious weeds, and any other inappropriate plantings, and replace these with suitable vegetation species

g. provide a consistent framework to assess applications that require a permit for clearing vegetation and pruning or removal of trees

h. provide a consistent framework to assess development applications for development sites that require clearing vegetation and pruning or removal of trees

i. provide safe and enjoyable living spaces for residents and visitors

2 Permits for clearing vegetation

2.1 When permits are required

A permit from Willoughby City Council is required to clear vegetation and prune or remove a tree if:

a. the tree has any of these dimensions:
    - a height exceeding 4 metres
    - a trunk girth (circumference) exceeding 600 millimetres measured at 1.4 metres above ground level
    - a crown spread exceeding 3 metres

b. the tree is a locally native species that is representative of the original vegetation of the area

c. the vegetation is within a defined wildlife corridor or has known wildlife habitat value

d. the tree is visually prominent from the street or surrounding properties and makes a positive contribution to the visual character of the locality

Note**:**

- A copy of the approval permit to prune or remove vegetation must be displayed on the front boundary of the property 72 hours prior to and during the course of the works being undertaken. A permit cannot be granted to clear native vegetation that exceeds the biodiversity offsets scheme threshold (see Clause 10(2) of Vegetation SEPP).

2.2 When permits are not allowed

A permit will not be allowed to clear vegetation or prune or removal of a tree if:

a. the vegetation is part of a heritage item listed in Schedule 5, Part 1 of Willoughby Local Environment Plan

b. the vegetation is within a heritage conservation area listed in Schedule 5 under Part 2 of Willoughby Local Environment Plan

c. the vegetation forms part of an Aboriginal object or is within an Aboriginal place of heritage significance

d. the vegetation is listed as an item under Willoughby Natural Heritage Register

3 Development applications for clearing vegetation

If a permit is not allowed to clear vegetation or prune or remove a tree under clause 2.2 above, a development application must be submitted to Willoughby City Council.

This is in line with Part 4 of the *Environmental Planning and Assessment Act 1979.*

Some exemptions apply and these are listed below.

4 Exemptions for clearing vegetation

4.1 Exemptions for risk and to maintain heritage significance

A permit or development application to clear vegetation or prune or remove a tree is not required if:

a. the vegetation or tree is dying or dead and is not required as habitat for native animals

b. the tree is a risk to human life or property

c. the proposed activity is of a minor nature or is to maintain a heritage item, Aboriginal object, Aboriginal place of heritage significance or heritage conservation area and the activity will not adversely affect its heritage significance

Notes:

- While a permit or development application is not required in the circumstances above, Willoughby City Council still requires evidence of these circumstances.

- In exceptional circumstances a permit is not required to remove a tree if we are satisfied there is an imminent risk to human life or property.

4.2 Other exemptions

A permit or development application to clear vegetation or prune or remove a tree is also not required if:

a. the vegetation or tree does not fall into any of the criteria listed in Section 2.2.1 of this part

b. the activity is selective pruning for up to a total of 10% of an individual tree's crown over a 5-year period

c. the vegetation is declared a weed under *Biosecurity Act 2015*

d. the activity is for pruning or removal of recognised horticultural varieties of fruit trees grown for fruit production

e. the activity is to clear vegetation or remove trees under the NSW Rural Fire Service bushfire 10/50 rules at rfs.nsw.gov.au

f. the activity is pruning or removal of the species listed in Attachment 1 (undesirable species exempt from permit approval)

g. the activity is on land under the care, control or management of Willoughby City Council

h. the activity is due to development that requires consent under Part 4.2 of the *Environmental Planning and Assessment Act 1979* (see Section 5.1 of this plan)

i. the clearing of vegetation, pruning or removal of trees is authorised under other legislation, including:

   - State Environmental Planning Policy (Exempt and Complying Development Codes) 2008
   - *Biodiversity Conservation Act 2016*
   - *Rural Fires Act 1997*
   - *Electricity Supply Act 1995* – only Ausgrid can carry out works on trees on public land affecting power lines. This cannot be carried out by residents or arborists not engaged by Ausgrid. Call 13 13 65 or see ausgrid.com.au/In-your-community/Tree-trimming
   - *State Emergency Services Act 1989*
   - *Water Management Act 2000*
   - *Roads Act 1993*

Notes**:**

- Exemptions for clearing vegetation, pruning and removal of trees only apply to private land.
- The written consent of the property owner is required if the applicant is not the owner.
- It is recommended that any removal or pruning of trees is carried out by a suitably qualified arborist. There are significant penalties for wrongful removal or excessive pruning of trees.

4.3 Controls for pruning

Pruning works allowed in line with this part must also comply with Australian Standard AS 4373. They can only consist of these pruning classes:

a. crown lifting: removal of the lower branches

b. crown thinning: selective removal of branches that does not alter the overall size of the tree

c. deadwooding: removal of dead branches

d. formative pruning: pruning of young and established trees with the general aim of directing plant growth and/or developing a sound structure

e. pollarding: a specialised pruning technique that establishes branches ending in a pollard head of buds and vigorous shoots

f. reduction pruning: the removal of the ends of branches to lower internal lateral branches or stems to reduce the height and/or spread of the tree

g. remedial (restorative) pruning: the removal of damaged, diseased or lopped branches back to undamaged tissue in order to induce the production of shoots from latent or adventitious buds, from which a new crown will be established.

5 Development that requires consent and development sites

5.1 Applications as part of development that requires consent

The clearing of vegetation, pruning or removal of trees due to development that requires a consent is assessed as part of the development application process.

Notes:

'Development' includes:

- use of land
- subdivision of land
- the erection of a building
- the carrying out of a work
- the demolition of a building or work
- any other act, matter or thing that may be controlled by Willoughby Local Environmental Plan

5.2 Controls for development sites

These controls apply to clearing vegetation, pruning or removal of trees:

a. unless the clearing of vegetation or the pruning and removal of trees is exempt under this part, vegetation or trees must not be cleared, pruned or removed until consent is obtained for the proposed development

b. the proposed development is to be sited and designed to retain and minimise the impact on any trees or any other significant vegetation

c. suitable replacement planting may be required if an application shows the removal of trees and vegetation is required and there is no other reasonable alternative design

d. the landscape and/or architectural plans must identify the type and species of trees and vegetation to be retained or proposed to be removed, including those on adjoining sites within 5m of the boundary

e. an arborist report prepared by a suitably qualified arborist may be required to justify clearing vegetation, pruning or removal of trees; the report must include the condition of all trees and vegetation on the site

f. where trees and vegetation proposed to be retained may be affected by construction works, including any trees or vegetation on public land, a tree protection plan must be submitted with the development application

Note**:**

- See the relevant landscaping provisions in this plan for controls relating to heritage conservation areas and specific land uses such as residential accommodation, commercial, industrial, recreational and institutional developments, including child care centres and place of public worship.

6 Replacement trees

If Willoughby City Council consents to the removal of an existing tree it will require the replanting of trees at a rate of 3:1.

Replacement trees are to be cared for by the land owner until established to a size that is covered by the vegetation controls.

If tree replacement is not possible, the applicant may pay a fee for each tree not replanted on the subject site. The fee payable is listed in the fees and charges section at willoughby.nsw.gov.au.

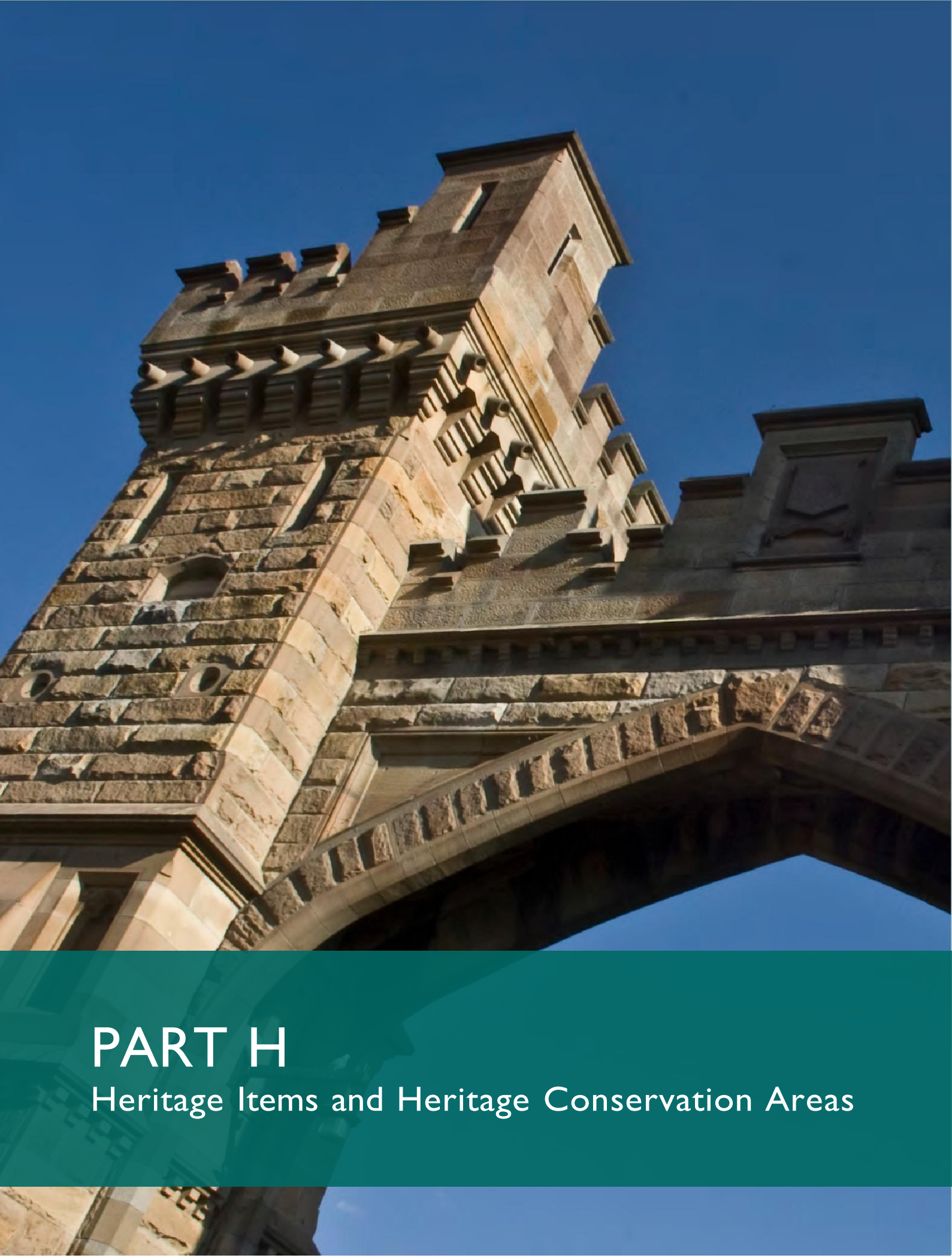
7 Vegetation management guidelines

This part is supported by our vegetation management guidelines which provide further information for landowners and consultants. These are available at willoughby.nsw.gov.au.

Attachment 1: Undesirable species exempt from permit approval

Table 1 Undesirable species exempt from permit approval

Common name	Botanical name	Restrictions
Cootamundra wattle	*Acacia baileyana*	No restrictions
Golden wattle	*Acacia siligna*	No restrictions
Box elder maple	*Acer negundo*	No restrictions
Tree of heaven	*Ailanthus altissima*	No restrictions
Evergreen alder	*Alnus jorullensis*	No restrictions
Nettle berry	*Celtis spp.*	No restrictions
Camphor laurel	*Cinnamomum camphora*	Only if less than 10m in height
Cotoneaster (all species)	*Cotoneaster spp.*	No restrictions
Leighton cypress	*Cupressocyparis, leylandii* "Leighton Green"	No restrictions
Loquat tree	*Eriobotrya japonica*	No restrictions
Coral tree	*Erythrina spp.*	No restrictions
Narrow-leaved black peppermint/willow peppermint	*Eucalyptus nicholii*	No restrictions
Rubber tree	*Ficus elastia and cvs.*	No restrictions
Honey locust	*Gleditsia triacanthos*	No restrictions
Silky oak	*Grevillea robusta*	No restrictions
Pyramid tree/sally wood/itchy bomb tree/cow itch tree	*Lagunaria petersonia*	No restrictions
Privet (all species)	*Ligustrum spp.*	No restrictions
Liquidambar	*Liquidambar styraciflua*	No restrictions
Oleander	*Nerium oleander*	No restrictions
African olive	*Olea europaea var. africana*	No restrictions
Canary Island date palm	*Phoenix canariensis*	Only if trunk is less than 7m in height
Frangipani	*Plumeria spp.*	No restrictions
Poplar (all species)	*Populus spp.*	No restrictions
Black locust	*Robina pseudoacacia*	No restrictions
Willow (all species)	*Salix spp.*	No restrictions
Cocos palm	*Syagrus romanzoffianum*	No restrictions
Chinese tallow	*Triadica sebifera*	No restrictions
Umbrella tree	*Schefflera spp.*	No restrictions
Evergreen ash	*Fraxinus griffithii*	No restrictions



PART H

Heritage Items and Heritage Conservation Areas

Part H - Heritage Items and Heritage Conservation Areas

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1 Introduction

This part applies to properties, places and heritage conservation areas subject to Clause 5.10 Heritage Conservation of Willoughby Local Environmental Plan (Willoughby LEP).

A list of heritage items and heritage conservation areas is in Schedule 5 of Willoughby LEP. A heritage map in the LEP identifies each heritage item and heritage conservation area.

These properties and places, because of their associations, their history, or their intrinsic qualities – their heritage significance – are required to be appropriately managed and respected when planning for new development.

Part H of this plan seeks to control development to ensure heritage significance is retained.

Conservation is the process of retaining these significant qualities. It involves the processes of maintenance, preservation, restoration, reconstruction and adaptation. It is aimed at preserving the significant fabric of a heritage item or heritage conservation area.

The performance criteria and relevant controls for residential development in Part B of this Plan apply to residential development in the Heritage Conservation Areas. However, to maintain the heritage values of these areas, the provisions and controls under this Part prevail over any other Parts of the WDCP.

This part should be read in conjunction with Attachment 1: Design Principles for the Conservation of Environmental Heritage.

1.1 Aim

The aim of this part is to protect and conserve heritage items, archaeological sites, Aboriginal objects and Aboriginal places of heritage significance. This part also aims to ensure any new development is sympathetic to the heritage item and is in keeping with the heritage values and character of the heritage conservation areas.

1.2 Objectives

The objectives of this part are to:

a. guide future development within a framework of conservation, including the restoration and/or remedial works to retain heritage items and contributory buildings within a conservation area

b. ensure that the significance of heritage items is identified and retained

c. ensure that the heritage significance, special streetscape and landscape character of heritage conservation areas is maintained

d. ensure alterations and extensions to existing buildings respect those buildings and do not compromise the significance and character of individual heritage items and heritage conservation areas

e. ensure new sustainable development respects the context and is sympathetic in terms of form, scale, character, bulk, orientation and setback, fabric, colours and textures and does not mimic or adversely affect the significance of heritage items and heritage conservation areas and their settings

f. encourage a sustainable high quality of design for any new development in achieving compatibility with the heritage significance of individual heritage items and heritage conservation areas

g. provide controls for the development of land within the vicinity of heritage items and heritage conservation areas

Note:

- This part is not intended to inhibit genuinely innovative design solutions that meet the above aims and objectives and comply with the design principles in this part. If there is any inconsistency between these heritage controls (Part H) and other guidelines in this plan, these heritage controls prevail.
- Applications to demolish a heritage item or contributory buildings within a heritage conservation area are not supported by Council, and approval will only be granted in exceptional circumstances where it can be demonstrated that the building is not capable of retention and/or restoration.

1.3 Aboriginal heritage

Willoughby City Council is committed to the protection and conservation of Aboriginal objects and Aboriginal places of heritage significance. Many of these objects and sites are located on environmentally sensitive land, particularly around the foreshore.

Willoughby LEP does not identify Aboriginal objects or Aboriginal places of heritage significance. Willoughby City Council should be contacted to determine if the site comprises or is near an Aboriginal object or Aboriginal place of heritage significance.

1.4 Demolition

Objectives

1. conserve individually listed heritage items and the general building stock that contributes to the significance of a heritage conservation area
2. ensure that, if approval is granted, any replacement development enhances the significance of the heritage conservation area

Requirements

Council does not support demolition of a heritage item or a contributory building in a heritage conservation area, and any application will only be considered where the building is not capable of retention and/or restoration.

Applications for the total or partial demolition of buildings, or works that are listed heritage items or on sites in heritage conservation areas, Willoughby City Council will assess:

a. the heritage significance of the building or work, including its contribution to the streetscape in heritage conservation areas

b. the opportunities for adaptation and whether the building or work would be incapable of reasonable or economic use

c. if the building or work constitutes a danger to its users or occupiers or to the public

d. whether the proposed redevelopment is compatible with the heritage conservation area.

In its determination of an application to demolish a heritage item or contributory building, Council will give consideration to any relevant Planning Principles established by the Lane and Environment Court.

If demolition of a heritage item is proposed, we may first refer the application to the National Trust of Australia (NSW) or any other relevant bodies.

If demolition of a heritage item of state significance is proposed, in line with Willoughby LEP, we will notify the Heritage Council of NSW.

Council will require reconstruction if there is any unauthorised removal of detail or significant decorative elements.

Willoughby City Council will not grant consent for demolition in a heritage conservation area unless it has considered the future development of the site. All applications for total demolition or demolition of significant parts of a building should include:

- A report from a structural engineer specialising in work on heritage buildings or structures. This should detail the structural condition if the proposal claims it is beyond repair, and evidence that stabilisation and/or the retention of the building or structure is impossible
- A heritage impact statement, heritage conservation management plan or heritage conservation management strategy as applicable, detailing the heritage significance of the building or structure or contribution to the heritage conservation area
- Other professional reports where relevant, such as from an archaeologist or historian.

If consent is granted for demolition of a heritage item or a component of a building in a heritage conservation area, an archival record of the existing building and grounds will need to be submitted. This record must be in line with Heritage NSW guidelines.

Any infill or replacement development must respect the heritage values and significance of the area and comply with Clause 5.10 of Willoughby LEP.

Notes*:*

If an Authorised Officer is not entirely satisfied that the dilapidation report demonstrates retention or restoration of the building is impossible, Council will engage an independent and suitably qualified expert to peer review the report at the applicant's expense.

1.5 Information requirements

Schedule 5 of Willoughby LEP identifies heritage items and heritage conservation areas. Any proposed development to these items or places, or near to these, is subject to Clause 5.10 of Willoughby LEP.

As well as the requirements in Part B of this plan, development applications for properties subject to Clause 5.10 of Willoughby LEP and Part H of this plan must include:

a. a heritage impact statement (HIS) and/or heritage conservation management plan or heritage conservation management strategy

b. a site plan to scale, showing the location of the heritage item or component of a heritage conservation area, contours and heights being relative levels to the Australian Height

Datum, and any other features affected by the proposal (streetscape, neighbouring structures, outbuildings, significant landscape features, views and vistas)

c. in the case of development in heritage conservation areas, an elevation of the principal façade to the primary streetscape(s) showing adjoining buildings

d. proposed new works should be clearly distinguished by means of diagrams, drawings and photographs from the existing original building or work and other improvements, including landscaping, where these are affected

e. schedules and samples of external materials, finishes and colour schemes.

1.5.1 Heritage impact statement

Development applications must include a heritage impact statement (HIS) for heritage items, or for land in the vicinity of a heritage item or for a building, work, relic, site or place within a heritage conservation area.

The statement must identify the heritage significance of an item, place or area, the impacts of any proposed changes, and how these impacts will be mitigated.

The length of the statement will vary depending on the scale and complexity of the proposal. A brief account included in the statement of environmental effects (SEE) may be sufficient for minor work that will have little impact on the significance of a heritage conservation area. A more extensive report would be required for more complex proposals or those that will have a major impact on the item.

Notes:

Applications should show that consideration has been given to the conservation and heritage of the item or component of a heritage conservation area under Part H of this plan.

Applicants should refer to:

- Local Government Heritage Guidelines, Heritage NSW at heritage.nsw.gov.au.
- Development application checklist – supplementary information, Willoughby City Council, available at willoughby.nsw.gov.au.

1.5.2 Heritage conservation management plan

Proposals for change to individually listed heritage items may also need to include a heritage conservation management plan. Applicants should follow the guidelines by Heritage NSW at heritage.nsw.gov.au.

A conservation management plan can be an important tool in caring for a heritage item. It provides a guide to future care and use of the item, including any new development as it 'sets out what is significant in a place and, consequently, what policies are appropriate to enable that significance to be retained in its future use and development' (J S Kerr, *The Conservation Plan*, National Trust NSW, 2000).

The conservation management plan must be prepared by a qualified heritage professional. It may be needed in these cases:

- to accompany an application for approval under *Heritage Act 1977* (see Local Government Heritage Guidelines Chapter 7: Determining applications for work to heritage items, Office of Environment & Heritage, Department of Premier & Cabinet, at heritage.nsw.gov.au)
- to support an application for site-specific exemptions from *Heritage Act 1977* approvals (see Heritage Council of NSW Heritage Information Series: Standard exemptions for works requiring Heritage Council approval, at heritage.nsw.gov.au). Note: as of 1 December 2020, procedures for standard exemptions for heritage items listed on the State Heritage Register or subject to an Interim Heritage Order, have changed. See heritage.nsw.gov.au for more information.
- as a framework for an agreed management approach to a heritage item.

1.5.3 Heritage conservation management strategy

A heritage conservation management strategy may be an acceptable alternative to a heritage conservation management plan. It is a briefer document that gives an overview of conservation approaches and management guidance.

The strategy must be prepared by a suitably qualified heritage professional. It may be needed in these cases:

- for use with items of local significance
- for use with items of state significance for which no major changes or interventions are planned in the short to medium term that have the potential to materially affect the item
- as an interim planning document for NSW State Heritage Register items pending the preparation of a standard heritage conservation management plan.

Note:

- See Heritage NSW at heritage.nsw.gov.au for guidelines on preparing a conservation management strategy.

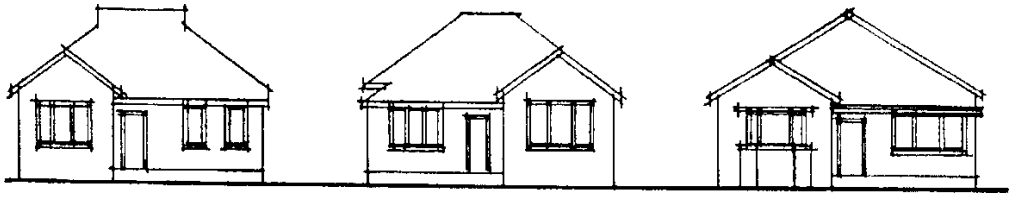
2 General conservation controls

To ensure new development respects and enhances the heritage significance of its context, the requirements below apply to all development covered by this part.

Further requirements also apply to the 12 individual heritage conservation areas. These are set out in Section 3 of this part.

2.1 Planning and design principles

Figure 1 Typical built form of dwellings in heritage conservation areas



Willoughby City Council will not consent to the alteration, extension or erection of a building or other works that alter the existing improvements on land that is listed as a heritage item or located in a heritage conservation area without considering:

- setting
- scale
- massing and form
- proportion
- detail

2.1.1 Setting

Objectives

1. provide an appropriate visual setting for heritage items and buildings in heritage conservation areas, including landscaping, fencing and carparking
2. maintain and enhance the existing heritage significance of the streetscape and the vicinity
3. ensure new development respects the established patterns in the streetscape, including setbacks, siting, landscaped settings, carparking and fencing

Requirements

a. The side and front setbacks are to be typical of the spacing of buildings both from each other and from the street in that locality so that the rhythm of buildings in the streetscape is retained.

b. Except for car parking and fences, no new structures should be built forward of the established street building line.

c. An adequate curtilage including landscaping, fencing and any significant trees, are to be retained.

d. The established landscape character of the locality including height of canopy and density of boundary landscape plantings should be retained in any new development.

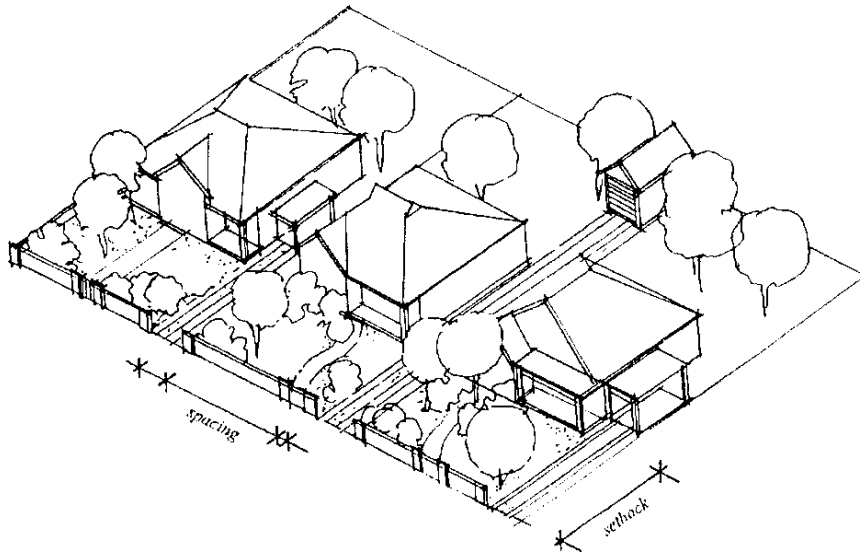
e. Development in the vicinity of a heritage item or a heritage conservation area should respect the visual curtilage of that item.

f. New developments must respect the existing significance of the streetscape and the vicinity.

g. The amenity and privacy of back gardens should be retained.

h. Views are to be protected.

Figure 2 Development pattern along a street showing setbacks and spacing between buildings



2.1.2 Scale

Objective

1. ensure the scale of new development is in harmony with the streetscape and does not dominate or compete with existing heritage items, reduce their contribution and importance to their context, or destroy an existing pattern of development

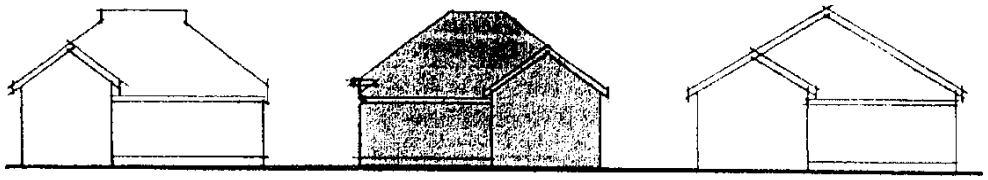
Requirements

a. The scale of the new work must relate visually to the scale of adjacent buildings that are heritage items or are located in a heritage conservation area. This includes the height, bulk, density and number of storeys. Unless the application can clearly show that greater scale would be appropriate in the individual circumstances, new buildings and additions are to be of the same scale as the surrounding development.

b. Extensions must not visually dominate or compete with the original scale of the existing buildings to which they are added or altered.

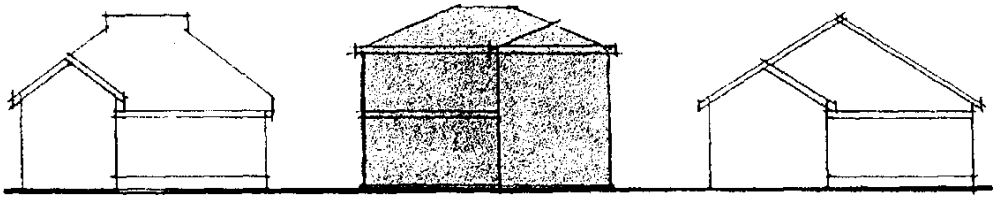
c. New buildings must not visually dominate, compete with or be incompatible with the scale of existing buildings of heritage significance or contributory value either on the site or in the vicinity of the proposal.

Figure 3 Example of a pattern of harmonious scale



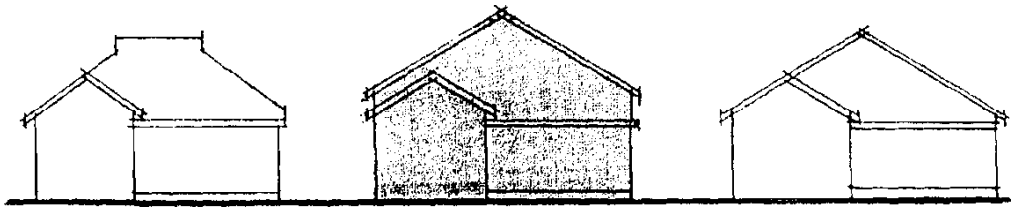
In Figure 3, the height, bulk and number of storeys in the new development (centre image) are consistent with surrounding development.

Figure 4 Example of a new development that ignores the established pattern of scale



In Figure 4, the new 2-storey development (centre) ignores the established pattern and dominates the surrounding buildings. While the height is consistent, the bulk of the new development and the number of storeys do not respect surrounding development.

Figure 5 Example of a new development that is harmonious with the scale of surrounding buildings



In Figure 5, the height of the new 1.5 storey development (centre) is similar to the 2-storey example in Figure 4. However, in this example the second storey is reduced in bulk, and the house retains a single storey character.

2.1.3 Massing and form

Objectives

1. ensure new development acknowledges dominant massing and form of the heritage item or heritage conservation area, and is in harmony with existing significant fabric and form, and with the surrounding streetscape
2. ensure that the form of new development is compatible with or complements the heritage significance of its context

Requirements

a. Extensions should not visually dominate or compete with the original form of the existing buildings they alter.

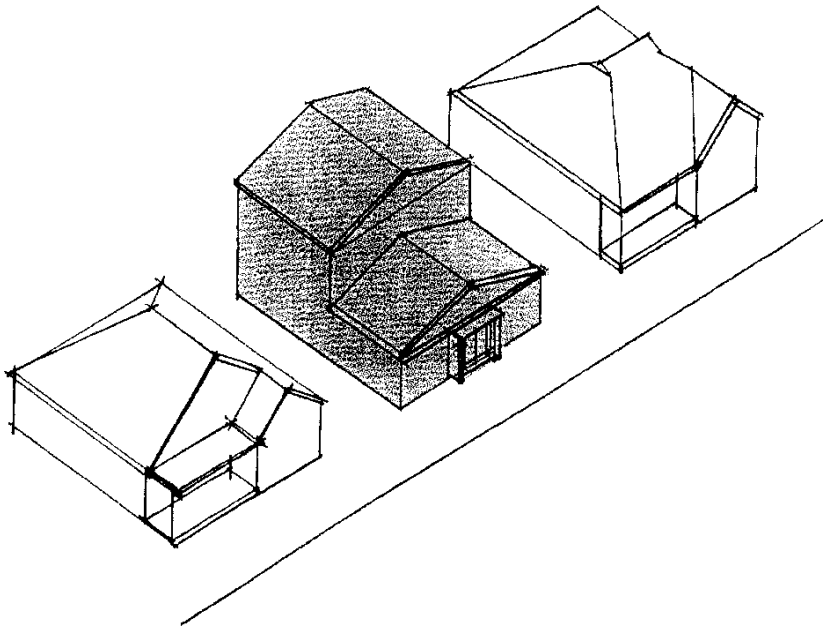
b. New buildings should not visually dominate, compete with or be incompatible with the form of existing buildings of heritage significance or contributory value on the site or in the vicinity of the proposal.

c. New buildings and extensions should have a similar massing, form and arrangement of parts to existing buildings of heritage significance in any heritage conservation area.

d. Maximum wall heights of any extension should be the same as the existing house.

e. Alterations, additions and new buildings must comply with the building envelope controls to achieve compliance with roof pitch, eaves and ridge heights, and maximum overall height limit.

Figure 6 Example of a new development (centre image) that does not respect the massing and form of surrounding buildings



In Figure 6, the new development (centre) does not respect the massing and form of the surrounding buildings which are characterised by:

- a dominant hipped and gabled primary roof form
- an offset projecting gabled bay
- an offset verandah with recessed entry.

The wall height of the extension is greater than the existing house.

2.1.4 Proportion

Objectives

1. ensure new development respects the proportions of elements of existing heritage fabric
2. ensure new development has regard to the architectural character and style of the heritage item or heritage conservation area setting

Requirements

a. New work and extensions should respect the proportions of major elements of significant existing fabric including doors, windows, openings and verandahs.

Figure 7 Example shows existing patterns of proportion

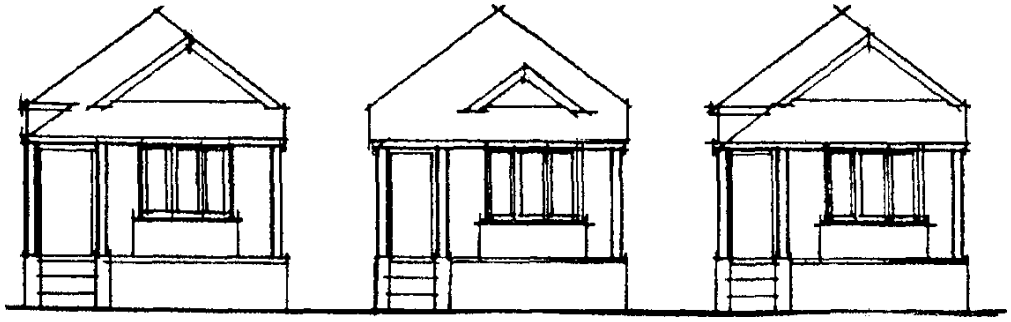
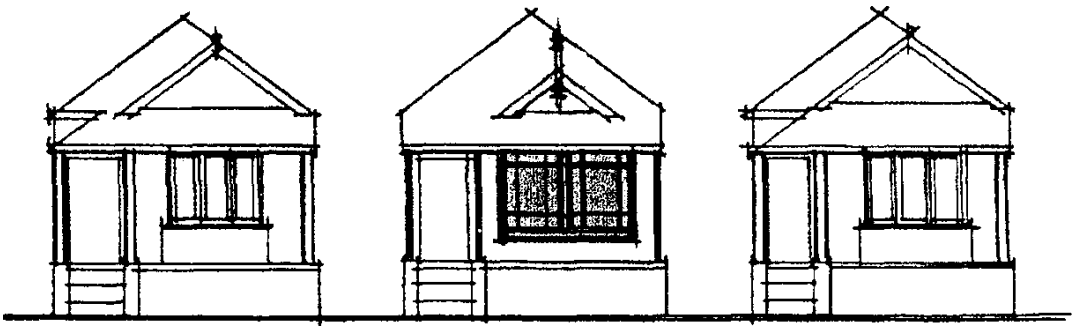


Figure 7 shows existing patterns of:

- simple vertically proportioned windows in groups of three
- simple bargeboard detail to gable end
- robust timber verandah posts.

Figure 8 Example of a new development that does not respect existing patterns of proportion



In Figure 8, the new development (centre) does not respect existing patterns by selecting:

- multipaned square proportioned windows in a group of two
- an over scaled finial to gable end
- slender steel verandah posts.

2.1.5 Details

Objectives

1. ensure new development has a level of detail appropriate to its context
2. ensure that new development has regard to the architectural character and style of the heritage item or heritage conservation area setting but does not incorporate elaborate new detailing in a period style that would prevent interpretation of what is original and what is new

Requirements

a. New work and extensions in heritage conservation areas should have a level of detail that is similar to and complements that of surrounding heritage fabric.

b. New work should adopt a simple character which uses external finishes, colours and textures that complement the heritage fabric, rather than mimic inappropriate heritage decoration and/or detailing.

Figure 9 Example shows existing patterns of detailing

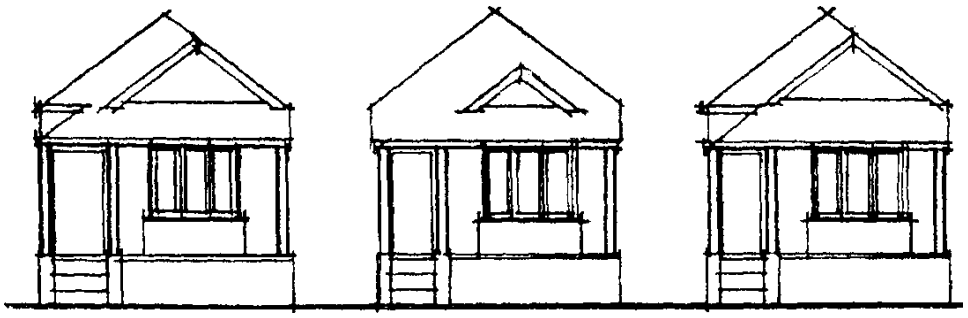
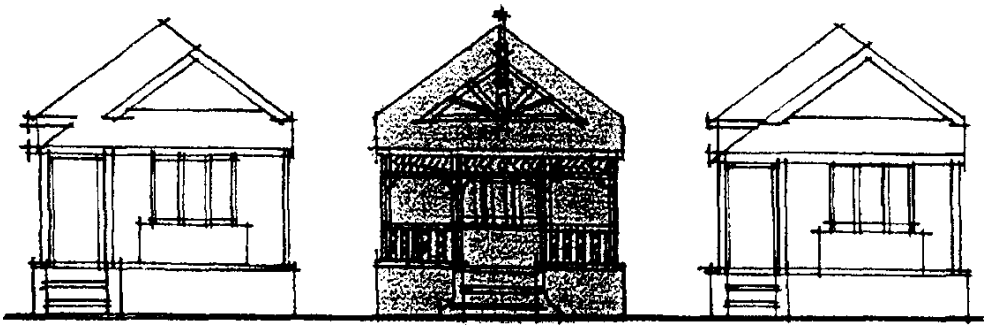


Figure 9 shows existing patterns of:

- simple verandah detailing
- simple timber bargeboards with roughcast gable infill
- simple timber casement windows.

Figure 10 Example that does not respect existing patterns of detailing



In Figure 10, the new development does not respect existing patterns by using:

- elaborate timber fretwork balustrade and valance to verandah
- elaborate gable end detail
- multipaned aluminium windows.

2.2 Design elements

Alterations and extensions to heritage items or components of heritage conservation areas and new infill development in heritage conservation areas (except for Griffin Heritage Conservation Area) must conform with the provisions below for individual elements of the buildings and settings.

2.2.1 Roofs

Objective

1. retain the characteristic scale and massing of roof forms within the heritage conservation area
2. retain the characteristic scale and massing of significant roof forms of heritage items

Requirements

a. Roofs of extensions are to match the existing roof in form, pitch and eaves, and be in proportion with the existing building.

b. Attic rooms are to use existing roof forms that retain the streetscape appearance of the existing building.

c. New buildings must have roofs that reflect the size, shape, pitch, eaves and ridge heights, and bulk of existing roofs in the locality, and be in proportion with the proposed building.

d. Roof elements such as dormers and skylights should not be located where visually dominant and must be kept below the ridge line.

e. Skylights must not be used on the front plane of roofs.

f. Chimneys must be retained.

g. The location of attic room extensions must not adversely affect significant views or vistas.

h. Where possible, structures attached to the exterior roof should avoid being located where they are visible from the street on the principal elevations of buildings.

2.2.2 Façades

Objectives

1. retain the existing façade, fabric, scale and massing and character of original development, in terms of the proportions of façades
2. ensure that new development does not disturb or reduce the importance of original verandahs
3. ensure new verandahs do not conflict with the heritage significance or significance of the place or building

Requirements

a. Two storey façades must only be used if surrounding development is of a predominantly two storey scale.

b. Bay widths are to match those of surrounding significant development.

c. The form and materials of principal elevations are not to be altered. This includes the removal of the external skin or rendering of exterior walls, unless it is associated with acceptable reconstruction works.

d. If altering existing houses, the original sunhoods, blinds, awnings and skirts to principal elevations should be retained and repaired. Authentic construction or reconstruction is supported.

e. If altering existing buildings, original verandahs are to be retained and restored. Infilling of verandahs is not supported. Additional verandahs must not compete with the importance of the original and must be simple in design, and based on existing detail or an understanding of appropriate designs for each period or style.

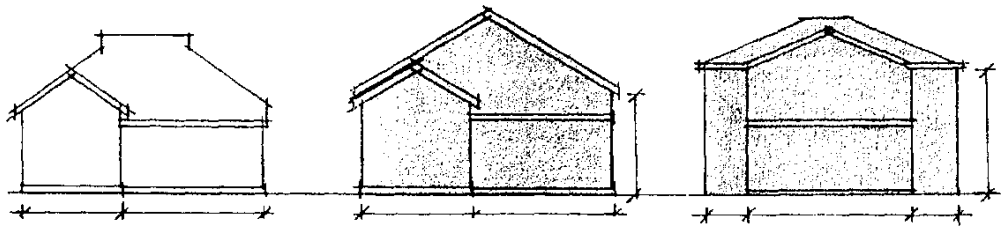
f. New buildings must consider the significance and design of verandahs in the locality, the methods of their incorporation in building designs and their harmonising role in streetscapes.

g. Alterations to original façades of heritage significance are not supported.

h. The proposed works are to be sympathetic to and/or not detract from the style, character and interest of the building and place. Designs are encouraged where the massing, details, materials and colours reflect the type of façade historically used in each locality, without relying on replicating these features.

i. Balconies on front façades are not appropriate unless consistent with the original architectural style.

Figure 11 Examples that respect and ignore existing patterns of wall height and bay width



In Figure 11, the original building (left) shows an existing pattern of single storey scale and stepped massing. The new development (centre) respects existing patterns through the control of wall heights and bay widths. The new development (right) ignores existing patterns using uncharacteristic wall heights and bay widths.

2.2.3 Doors and windows

Objective

1. original elements are retained
2. the character and patterns of any new door and window openings and their construction clearly relate to the proportions, placement and scale of fenestration patterns of the existing heritage fabric

Requirements

a. Retain and repair/restore original doors and windows to principal elevations. Authentic reconstruction is encouraged. Original leadlight and coloured glass panes should be kept.

b. New doors and windows in additions are to be compatible with the proportions, position, size and detailing of existing doors and windows.

c. Doors and windows in new buildings are to be compatible with the proportions, position and size of those typical of the locality.

Figure 12 Example of the existing pattern of windows and doors

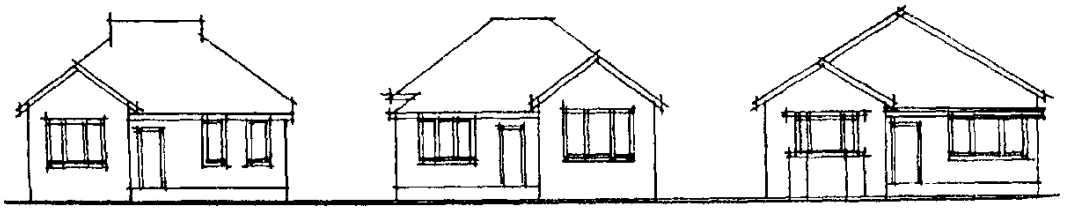
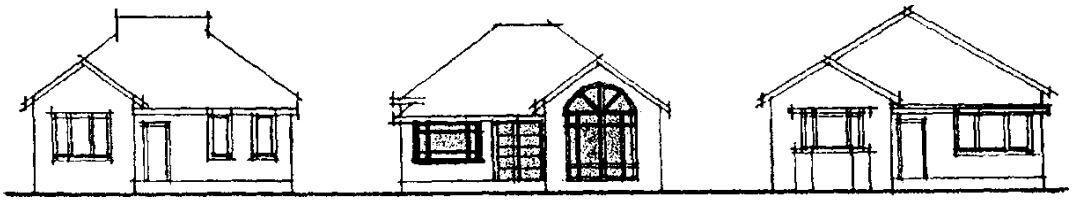


Figure 12 shows an existing pattern of simple, vertically proportioned timber casement windows in bays of three, and single leaf doors.

Figure 13 Example of a new development (centre) that ignores existing patterns of windows and doors



In Figure 13, the new development (centre) ignores existing patterns. It uses an elaborate horizontally proportioned aluminium window and an uncharacteristically large arched window and a double leaf door.

2.2.4 Materials and colours

Objective

1. the selection of materials and colours is based on an understanding of the original finishes
2. finishes in new development should have regard to the significance and character of the heritage item or of development in the street or heritage conservation area, and the likely impact of that proposed work

Requirements

a. Roofs – New or replacement roof materials are to match existing materials or use approved alternative materials appropriate to the style and location.

b. Façades – Matching materials should be used when repairing the fabric of external surfaces. New development should use materials similar to or compatible with that of original buildings in the locality. Any new face brickwork should carefully match the colour and texture of the brick, the type of jointing, and mortar colour. Original unpainted brickwork, sandstone and blockwork must not be rendered or painted. New buildings and additions in heritage conservation areas should employ colour schemes that do not detract from traditional colour schemes in the locality.

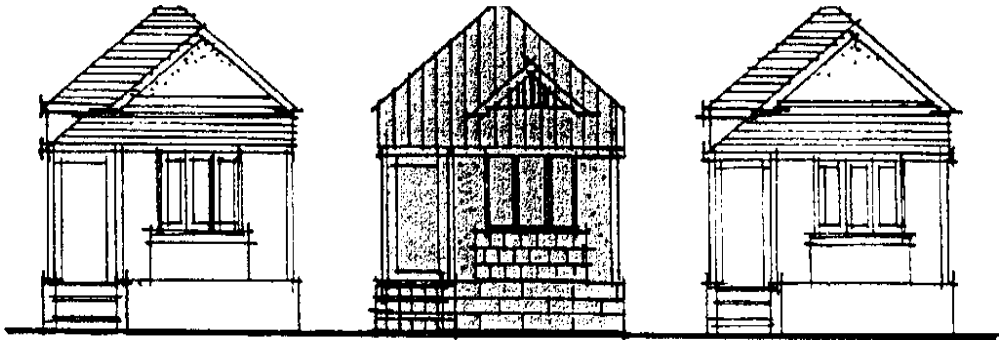
   Doors and windows – Original doors and windows are to be retained in existing buildings. New doors and windows must be of design materials and finishes to match the original or an approved alternative that fits with the characteristics of the locality. Wide section

aluminium windows may be considered as an alternative for timber windows in new development.

c. Fences – Front fencing must be of materials characteristic to the locality and particular to the street.

d. Paving and driveways – materials for paving of pathways may include tessellated tiles for Federation and Victorian styles, or suitably textured and coloured finishes: plain or stencilled concrete is not acceptable. Preferred materials for driveways include wheel strips, brick paving or gravel.

Figure 14 Example of a new development (centre) that does not reflect the external materials and finishes of surrounding buildings



In Figure 14, the new development (centre) does not reflect surrounding buildings in the selection of external materials and finishes. Roof materials and wall surfaces are particularly critical.

2.2.5 Carparking

Objectives

1. allow for reasonable onsite car parking while retaining the character and significance of the heritage conservation area or heritage item
2. ensure car parking facilities do not have any adverse visual impact on heritage streetscapes
3. ensure garages, carports and driveways are visually discreet

Requirements

a. Access

- Existing rear lane access is to be used in preference to front access.
- Existing side vehicular access is to be used.
- Driveways are to be to side boundaries and not central.
- Development that removes existing access must not preclude future carports or garages behind the building line.

b. Location

- Open stand car spaces may be provided forward of the building line

- Garages and carports are to be located behind the building alignment where physically possible
- If no alternative exists and the frontage of the property is of a sufficient width, a single carport located forward of the building line, set back from the street and offset to side boundary may be considered.

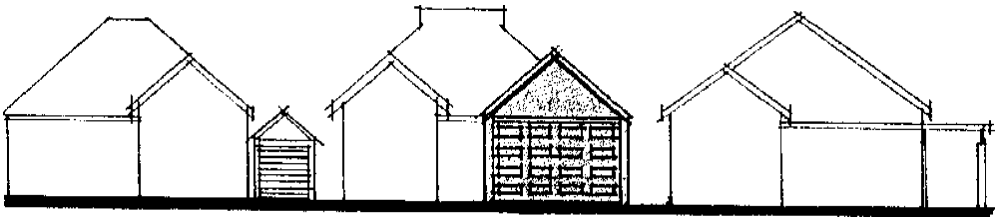
c. Scale

- Maximum width of a driveway at street frontage is to be 3.5m.
- Garages and carports are to occupy no more than 20% of street frontages.
- Car parking structures should be smaller in scale compared to the residence.
- Structures forward of the building line must be designed to minimise their bulk with a maximum eaves height of 2400mm.
- Flat roof structures of sympathetic materials and detail are acceptable.

d. Appearance

- Materials, form and details of carparking structures are to harmonise with and be subservient to the residence.
- A similarity in colour of garage doors and wall surfaces may reduce impact to street and is favoured.
- Structures forward of the building line must be screened with vegetation.
- Garage doors and structures are to be recessed behind the primary façade to create a shadow line.

Figure 15 Example of carport and double garage that are forward of the building alignment



In Figure 15, the double garage and carport (centre image) are forward of the building alignment. This dominates and destroys the significance of a heritage streetscape.

2.2.6 Fences

Objective

1. provide fencing that reinstates the original form of fencing
2. ensure any new fencing is consistent with and does not detract from the established patterns of the street

Requirements

a. Front fencing, and side fencing forward of building line, is to have a scale consistent with existing streetscape, generally:

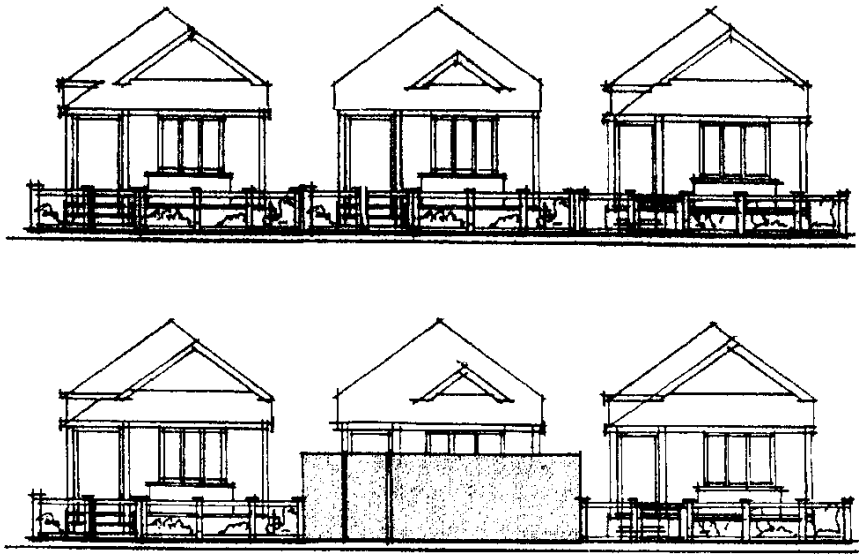
- masonry fencing is be to 700mm maximum
- open fencing (such as pickets or palisade) is to be 1200mm maximum
- hedging is to be 1200mm maximum
- lapped and capped timber fencing on side boundary is to be 1200mm maximum
- retain the original fences where practical
- modifications should match the original where possible
- fencing must be simple with a level of detail compatible with the house.

b. Acoustic fencing – If properties front state or regional roads (such as Mowbray Road, Penshurst Street), a solid wall to 1.6m high set 1m behind a low front fence is acceptable if screen hedge planting is installed and maintained.

Note:

- In Griffin Heritage Conservation Area there should be no fencing or only low fencing of up to 300mm. See the specific controls for Griffin Heritage Conservation Area in this part.

Figure 16 Example of high solid fencing



In Figure 16, the high solid fencing in the new development (centre) destroys the harmony of the streetscape and prevents views of the dwellings and gardens.

2.2.7 Garden elements, including paving and driveways

Objective

1. retain or reinstate landscaped settings for heritage items and components of heritage conservation areas

Requirements

a. Hard surfaces are to be kept to a minimum. As a guide, 70% of the area forward of the building line is to be a deep soil zone.

b. Screening of hard surfaced areas with wide tree canopies is encouraged.

c. Garden structures are to be appropriate to primary buildings in terms of scale, style and materials.

2.2.8 Outbuildings

Objective

1. ensure outbuildings do not detract from the heritage significance of the heritage item or heritage conservation area through inappropriate siting, excessive scale, bulk or visibility

Requirements

a. When assessing any application to erect a garden shed or store, Willoughby City Council will consider:
    - the location of the proposed structure in relation to the principal building, boundaries and other details of the site
    - the form, scale, materials and colours; colours and materials should be recessive, and height should not exceed 1.2m
    - its relative prominence and visibility from the street frontage or frontages of the site and neighbouring properties
    - the need for landscaping such as screening or planting to ensure it is well integrated with its intended site
    - the retention of any significant outbuildings that form part of a historical curtilage.

2.3 Development of corner allotments within heritage conservation areas

Objective

1. acknowledge corner blocks are more significant in defining the character of the area because they are visible from two streets and an important component in distant vistas
2. provide specific guidelines to ensure the characteristics of the heritage conservation area are considered from both streets

Requirements

a. The significant parts of the original house must be retained which include its main frontage and side frontage. Non sympathetic rear additions generally do not require retention.

b. The scale of additions and alterations are to respect the existing ridge or eaves heights.

c. If additions are attached, detailing (including finishes and materials) is to be appropriate to the original.

d. If additions are detached or infill development is proposed, contemporary solutions that respect the scale, bulk and detailing of the original without mimicry are preferred.

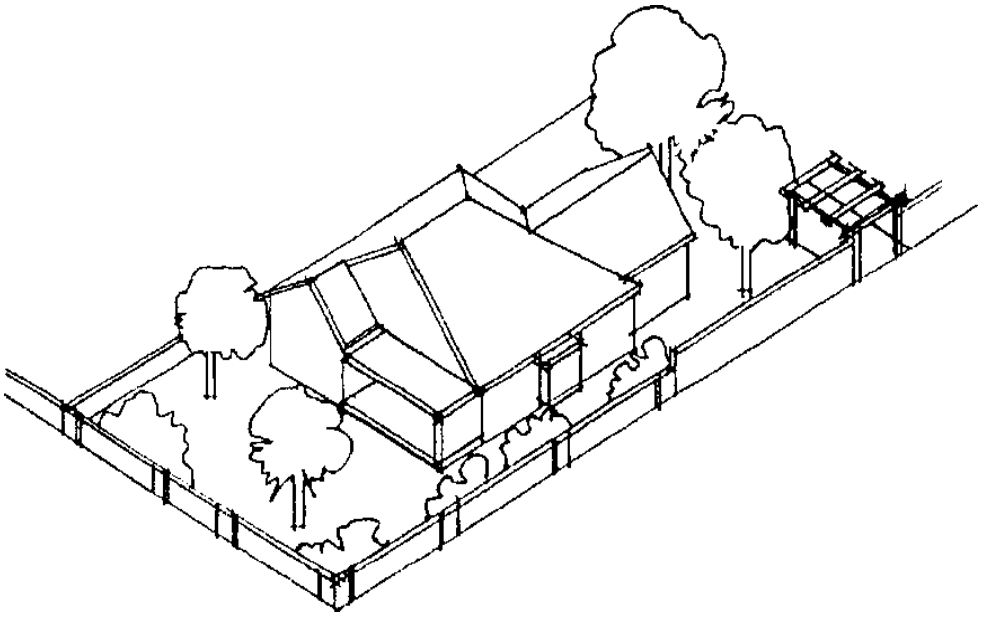
e. Carparking must be located to the rear of the side frontage. Double garages forward of the building line are not acceptable.

f. Fencing to the side frontage must not exceed 1800mm in height.

g. Landscaping is required to both street boundaries, and a landscaping concept plan is required with the submission of a development application.

h. New development or additions must be located to minimise any impact on existing prominent trees.

Figure 17 Example of a corner block development where both street fronting elevations are considered



2.4 Subdivision

Objectives

1. retain the development and subdivision pattern of heritage conservation areas including their characteristic rhythm and spacing of built form
2. retain significant curtilages, views and vistas and landscape elements associated with individual heritage items
3. retain the original subdivision pattern in Griffin Heritage Conservation Area

Requirements

Subdivision of land must comply with the minimum allotment size requirements of Willoughby LEP and with the design guidelines for infill development.

Subdivision applications must include adequate plans, showing the building envelopes, siting and setbacks of the proposed buildings. This applies to application for development in the vicinity of a heritage item, on land where heritage items are located, or in heritage conservation areas.

These applications must show that:

a. the allotment and building spacing, such as frontage widths, side and front boundary setbacks, are typical so that:

- the rhythm of buildings in the streetscape of heritage conservation areas is retained
- vistas and views to and of heritage items and significant buildings, especially the principal elevations of buildings, are not interrupted or obscured
- the landscape quality of the streetscape in heritage conservation areas is retained
- the setting of the heritage item and a satisfactory curtilage, including important landscape and garden elements, are retained

b. the scale and form of proposed new construction or buildings is compatible with, and does not detract from, the significant dominant heritage elements of the streetscape in heritage conservation areas or of individual heritage items

c. the new development preserves the essential qualities of the streetscape and building styles that form part of the locality's heritage significance

d. the original qualities and styles have been used a guide to the desired character, even if more recent development in the vicinity has varied the older development standards

e. the subdivision does not require demolition of existing building stock or re-arranged vehicular access and carparking (on or off the site), that would adversely affect the streetscape in heritage conservation areas or the principal elevations of heritage items

f. the contours and any natural features of the site have been retained and respected.

2.5 Infill development

Objectives

1. ensure infill development achieves a sympathetic relationship with nearby heritage items or the heritage conservation area in terms of its scale, massing, character, setback, orientation, materials and detailing
2. ensure infill development respects the established streetscape, and the patterns of development, including setbacks, siting, landscape settings, carparking, height, dominant ridge line and building envelope
3. ensure infill development displays architectural 'good manners' by respecting the significant characteristics of nearby and adjoining development

Requirements

a. Infill can be contemporary in design. But the scale, form and detail of the infill must not detract from the scale, form, unity, cohesion and predominant character of the building and streetscape and landscape around it.

b. Infill development in the vicinity of a heritage item must respect the visual curtilage of that item.

c. Infill development must not visually dominate, compete with or be incompatible with the scale (size, height and bulk) of existing buildings either on the site or in the vicinity of the proposal.

d. Infill development must be sited to correspond with the existing pattern of relationships between buildings and their sites. Front boundary setbacks are to be equivalent to those of neighbouring buildings. Side setbacks must be consistent with existing patterns.

e. Infill design is to be integrated into the established character of the area and heritage buildings. Incorporating basic design elements such as the characteristic roof form and massing of the original development, proportions of windows, doors and verandahs.

f. Infill design must not visually dominate, compete with or be incompatible with the form of existing buildings of significance, either on the site or in the vicinity of heritage items.

g. New development must use materials and colours similar to or compatible with that of original buildings in the locality.

h. Contemporary kit/project designs which purport to be ‘heritage homes’ are generally poorly integrated mixtures of design elements from different eras and do not fulfil the objectives for infill development in heritage conservation areas or adjacent to heritage items.

i. For infill development in Griffin Heritage Conservation Area the management policies and controls for that area are to apply, in particular the Griffin design principle of subordinating buildings to the natural landscape.

2.6 Secondary dwellings

Controls for secondary dwellings in heritage conservation areas or on sites occupied by heritage items must comply with the controls set out in this part including, compliance with:

- floor space ratio (FSR)
- setbacks
- proportion
- massing and form
- setting
- details
- materials and finishes
- specific requirements for individual conservation areas.

2.7 Semi-detached dwellings

There are a number of semi-detached dwellings (semis) within the Willoughby local government area. These incorporate a number of architectural styles developed between the 1880s and the 1930s. Several significant examples of these styles are included as individual heritage items or are within the conservation areas listed in Willoughby LEP. Applicants should be conscious of the significance of these styles before considering any alterations and additions to semi-detached dwellings.

Objectives

1. retain the concept of the semi-detached dwelling as one of a pair or group of dwellings, maintaining traditional scale, character and established streetscape values

2. ensure alterations and additions to a semi-detached dwelling are appropriate to the established scale and building envelope of the original building and to the predominant development in the vicinity
3. see alterations and additions to a semi-detached dwelling as an extension of the general form of the existing building envelope, related in form and detail to the adjoining semi-detached dwelling
4. allow the potential for complementary development of an adjoining semi-detached dwelling

Requirements

a. Any alteration to an individual semi-detached dwelling should recognise it as one of a pair or group of similar, identical or complementary buildings. Any extension should be carefully integrated with the building it is attached to, both in its present form and on the assumption that the adjoining owner may wish to add extensions in the future.
b. Applicants should show how the adjoining semi-detached dwelling could be treated to maintain character.
c. Any first floor addition should be set back from the principal street frontage of the building to avoid altering a substantial portion of the existing roof over the front of the building and to locate the bulk of new development towards the rear. Additions should be located behind the main gable or hipped feature of the street frontage.
d. The style, pitch, material, profile and colour of the proposed roof should match, complement and extend the existing roof form of the building.
e. Characteristic features of the existing roofscape should be identified and where appropriate incorporated into the proposed extension. The positioning and proportion of gables, the use of parapets and gambrel and dormer roof forms should be considered in the context of the surrounding original development.
f. Flat roofed areas should only be contemplated where they are not seen from the street or other important viewing positions in the vicinity of the building. Uncharacteristic roof forms and details will not be considered appropriate if they impinge on the street character of the adjoining or nearby semis.
g. Roof forms, which contribute excessively to the visual bulk of the building, such as Mansard roof extensions will not be permitted. Contemporary roof form additions to the rear of traditional semis may be acceptable if the visual impact to the street is minimised.

   Where extensions are proposed to only one half of a pair of semi-detached dwellings, consideration should be given to the roof design as an extension of the established roof lines of the undeveloped semi. This may involve the agreement of the adjoining owner but will avoid the appearance of a blank dividing wall and will conceal changes to the roof line.
h. The roof design is to prevent stormwater spillover onto the adjoining semi.
i. Dormer roofs shall be used in a manner that is characteristic of the particular style of the subject semi.
j. New roofing should maintain or replace the original roofing of the building in material, profile and colour. Where the roofs of adjoining semis are currently different from each

other, the new roof finish should match the adjoining dwelling as closely as possible, unless it can be demonstrated that such a finish is of undesirable architectural merit.

k. Willoughby City Council will only allow dormer windows, balconies, skylights and solar panels on the front façade where they will not have an adverse streetscape impact.

l. Unpainted surfaces should remain unpainted.

m. Where possible it is preferable that both semis forming a pair propose comprehensive and matching additions to be carried out at the same time. However, it is possible to design alterations and additions to one semi of a pair or terrace group which does not detract unreasonably from the architectural integrity of the pair or group. The design approach will vary depending on the form of the semi design. If symmetry is the dominant characteristic it should be respected. If asymmetry gives the appearance of a single building this should be respectfully acknowledged in the design to maintain that character.

2.8 Contributory Buildings

This section is designed to further clarify Council's intent in guiding development of buildings considered to make a positive contribution to the heritage character and significance within heritage conservation areas.

Where an existing building contributes to the heritage significance of the conservation area, alterations and additions should be designed and sited to ensure the retention of any contributory features or characteristics of the building and the streetscape of the heritage conservation area in which they are located. The loss of this significant fabric weakens the integrity and authenticity of heritage conservation areas, so a cautious approach is required. Development to contributory items should not result in a situation where only the front façade of a building is kept, without retaining any of the significant structural fabric behind the front exterior wall.

Objectives

1. Ensure that sufficient structural fabric of contributory buildings is retained and conserved to ensure the heritage contribution of the building to the conservation area is maintained.
2. Ensure that additions or changes to the appearance of contributory buildings within heritage conservation areas respect the original built form, architectural style and character.
3. Encourage proposed development to be contained to the less significant parts of contributory buildings.
4. Avoid a situation where only the front façade of a building is kept.

Requirements

a. Street elevations and visible side elevations must not be significantly changed. Additions must be located to the rear or to one side of the building to minimise impact on the streetscape.

b. Significant structural features for the front principal portion of the building are to be retained. This can include the first four rooms of the dwelling, staircases and fireplaces.

c. Principal roof forms visible from the street, including roof pitch, roof plans, eaves height and chimneys are to be retained, with the exception of additions such as rear dormers and skylights.

d. Changes to materials (including roofs and walls) on elevations visible from a public place are not supported. Original face brickwork must not be rendered, bagged or painted. The removal of external brickwork skin is not supported.

3 Heritage conservation areas

3.1 Introduction

For each of the City of Willoughby's heritage conservation areas, this section provides:

- **History**, which outlines the historical context of the area, and significant characteristics of the individual heritage conservation areas, to enable an understanding of the heritage significance of the area.
- **Description**, which outlines the significant characteristics of the individual area to enable an understanding of the heritage significance of the area.
- **Statement of significance**, which states why the area is significant.
- **Management policies**, which set out controls to ensure that this heritage significance is retained in any new development.
- **Controls for future development**, which are tables that outline specific requirements so the design of new development is compatible with the heritage significance to be conserved and assists in achieving the desired future character.
- **Map** of the heritage conservation areas under Willoughby LEP.

Heritage conservation areas are representative of particular phases in the development of the City of Willoughby area. They have distinctive historic, stylistic and streetscape values that are essential to retaining the heritage of Willoughby. Components that contribute to this significance are to be retained, and new development must reflect and reinforce this character.

The special characteristics required for retention are summarised below for each area, together with specific policies and development controls for achieving this.

Note:

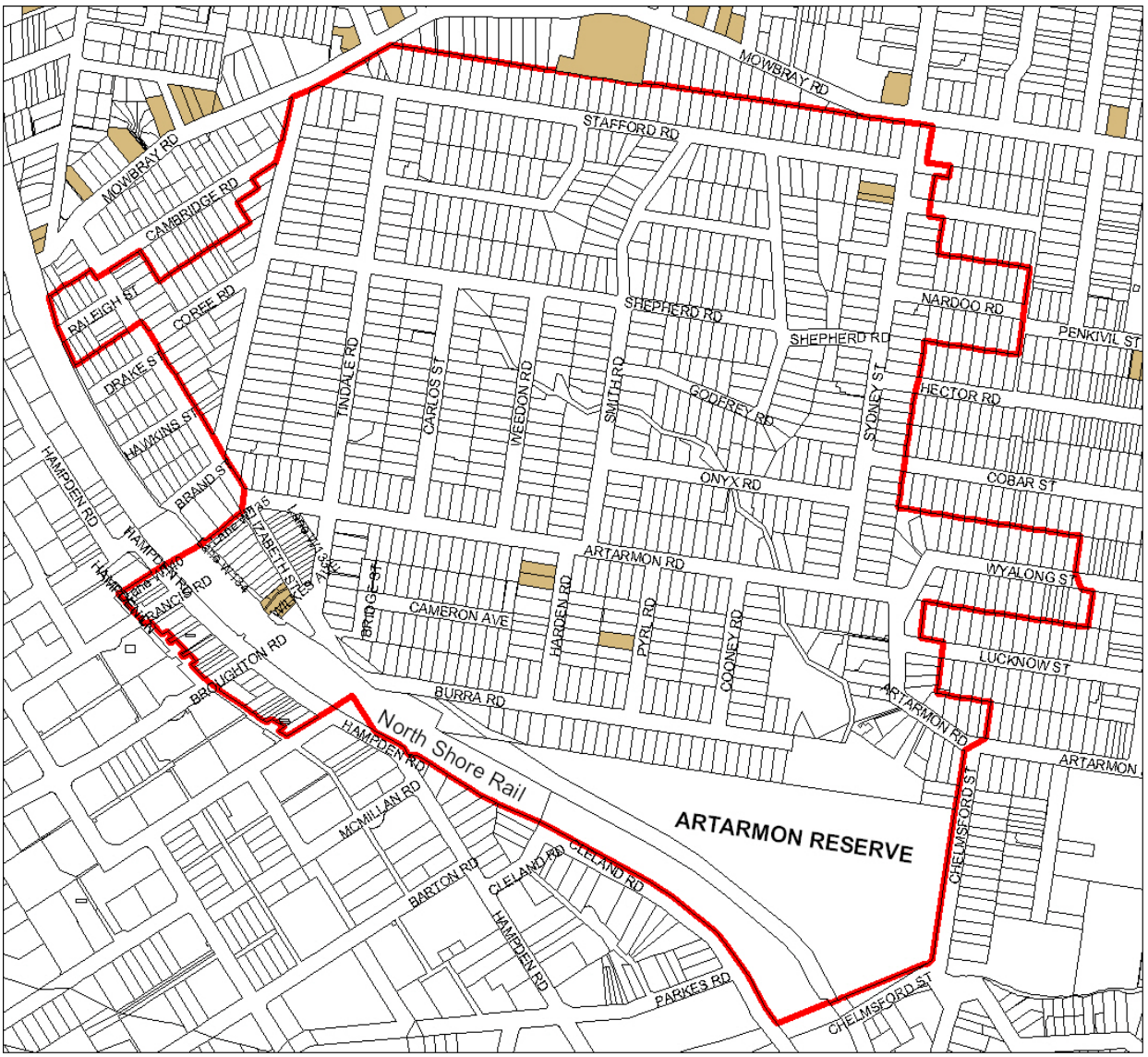
- The intent of these controls is not to inhibit genuinely innovative design solutions but to ensure that sustainable design and new development respects and responds to the heritage significance of the area.
- The Heritage Impact Statement (HIS) and/or the Statement of Environmental Effects (SEE) required under Section 1.5 of this Part must address the 'Management policies' and 'Controls for future development' for the relevant Heritage Conservation Area.

Figure 18 Example of heritage properties in the City of Willoughby area

3.2 Artarmon Heritage Conservation Area

3.2.1 Location

Figure 19 Map of Artarmon Heritage Conservation Area



3.2.2 History

The earliest land grants in the Artarmon area were made in 1793–94, by Major Francis Grose of the NSW Corps. These grants were intended for farming but were used only for grazing. In 1810, Governor Macquarie granted 150 acres to the General–Provost Arthur Gore. Gore then bought out his neighbours and by 1815, he owned most of the land as far west as the Pacific Highway. Gore named his farm ‘Ardthelmon’ after his home in Ireland. This period of prosperity was short lived. By 1818, Gore had lost all but a small portion of land on which he built Artarmon House (the site of the current TAFE and Institute of Technology).

The Artarmon Heritage Conservation Area was developed in two stages. The major streets, such as Artarmon Road and the western precinct closest to the station, such as Muttama Road, were partially developed before the First World War. Consequently these are

characterised by development from the Federation era to a greater degree than other streets. There are also some grand villas and houses of this period on the eastern edge along early transport routes, such as Sydney Road. The remainder and bulk of the area was developed in the 1920s and 1930s and is predominantly bungalow development with a few semi-detached cottages on the eastern edge. Some two-storey flat buildings, dating from the 1930s, are located closer to the station.

The National Trust of Australia (NSW) recognised the importance of the area by classifying it as an urban conservation area in 1989.

3.2.3 Description

A rectilinear subdivision pattern has been superimposed on a hilly area of moderate slopes. The regularity of the lot layout is interrupted by several well vegetated drainage reserves flowing across to the remnant bushland of Artarmon Reserve.

The original subdivision pattern of narrow carriageways and wide grassy verges (usually planted with mature street trees, such as brush box and jacaranda) prevails, except for Artarmon and Sydney Roads, which have wider carriageways taking the main traffic volumes to and from Artarmon Station and shopping centre.

The combination of undulating landform, well established street trees and other vegetation (including mature trees behind houses, remnant bushland in the drainage reserves and the pattern of usually generous lot sizes) creates vistas of leafy ridges and valleys and a wide range of interesting streetscapes with houses stepped down the terrain.

Houses are either unpainted red-brown brick and unglazed terracotta tile Federation styles or a rich variety of 1920s to 1930s bungalows. The latter range from typical California Bungalow styles to the transitional 1930s Interwar bungalows of unpainted brown to liver brick with red or brown tiled roofs. The character is predominantly single storey other than the small number of Interwar flats and relatively few second storey additions. Good individual examples including some of the older flat buildings, with Art Deco or Spanish Mission characteristics, also occur.

The denser pattern of settlement near Artarmon Station is a good example of Interwar flat development providing comfortable housing that is well integrated with nearby public transport and commercial opportunities. The well proportioned flat buildings form a cohesive residential precinct of generally consistent two storey scale, of similar materials and with shared modest embellishments, including decorative brickwork. Such characteristics enhance their unity with the earlier group of Edwardian commercial buildings at Wilkes Plaza.

3.2.4 Statement of significance

The heritage conservation area is outstanding for its intactness, with few unsympathetic intrusions occurring. The wide range of largely intact California and Interwar bungalows as well as Federation housing in generally good condition, occur in either groupings of consistent styles or subtle blends of successive periods to produce a mix of interesting and varied streetscapes. The area is significant as a harmonious and unified 1910s – 1920s lower north shore residential area whose development relates to the development of the railway.

3.2.5 Key period of significance

The key period of significance is 1900 to 1940.

3.2.6 Management policies

a. Retain and enhance the original form, scale and detail of existing buildings which contribute to the character of streetscapes and the heritage significance of the area.

b. Retain single storey character of the area.

c. Retain characteristic palette of materials of the area, particularly the use of unpainted dark monotone face brick, sandstone bases, painted timber trim and unglazed terracotta roof tiles.

d. Retain intactness of area by controlling alterations and additions, which should be set at the rear and not visible from the street.

e. Additions must not result in excessive changes in scale, or bulk, or the introduction of visually intrusive and dissimilar materials particularly on prominent sites, including corner sites.

f. Two storey development and second storey additions are inappropriate unless the original streetscape appearance of the building is retained, or in areas adjacent to original two storey flats. Sloping sites may enable two storey additions to the rear, which must preserve the predominantly single storey scale of the detached housing and must not offend the individual building's roof shape or interrupt the streetscape, unified by similarly patterned roofs.

g. New front fences must reconstruct original details or repeat the pattern of low unpainted masonry fences and dense boundary plantings.

h. Retain existing side driveways and rear car parking facilities.

i. The site cover and siting of new development must not lead to the loss of significant landscaping either onsite or in verges. In this regard driveways are to be minimal with wheel strips as the preferred solution.

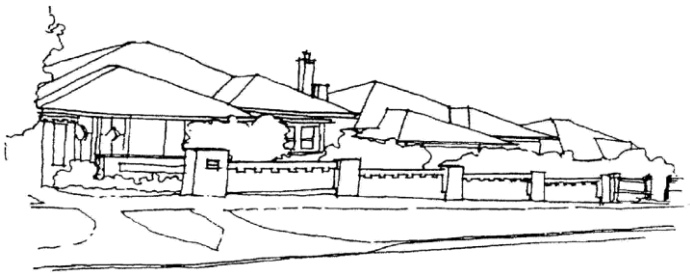
j. Protect mature trees in rear gardens which form a backdrop to the streetscapes.

k. Street plantings and reserve vegetation must be maintained and encouraged, particularly in lesser vegetated streets. Where a site is bushfire prone land landscaping should conform with the NSW Rural Fire Service 'Planning for Bushfire Protection' guidelines.

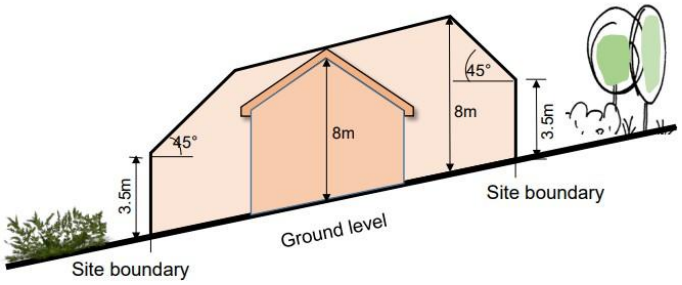
l. Existing vistas resulting from the undulating topography must be maintained.

m. The character and significance of the railway station and the Hampden Road shops as historic elements and as a buffer to the high rise development of West Artarmon must be respected in any new development.

Figure 20 Example of properties in the Artarmon Heritage Conservation Area



3.2.7 Controls for future development – Artarmon: R2 Residential

a. Scale	Generally:	Uniform low scale, detached bungalows, few semi-detached cottages and flat buildings at station.
	1. Storeys	1 (2nd storey within roof forms or basement is acceptable)
	2. Floor space ratio	0.4:1: This figure represents an upper limit to floor space ratio (FSR). It accounts for the existing pattern of development and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Rectilinear subdivision superimposed on moderate slopes. Average 675m^2 15m (12.5m minimum) frontages. Uniform setbacks within streets or within groups of houses. Generous well landscaped front and rear yards, mature native and exotic plantings, trees to rear gardens.
	1. Setbacks	Side: 0.9m – 2.5m (driveway).
	2. Carparking	Single garage to side or rear of dwelling, or part of basement.
	3. Front fencing	Low unpainted face brick.
c. Form/ massing	1. Roof	Simple, mainly hipped with some gabled forms.
	2. Façade	Symmetrical or asymmetrical, generally stepped massing, incorporating porch or verandah.
	3. Building envelope	Roof pitch: 25 – 30° Eaves height: 3.5m from floor level Ridge height: 6 – 6.5m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and to the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP. **(a) Building Envelope on Sloping Site** 

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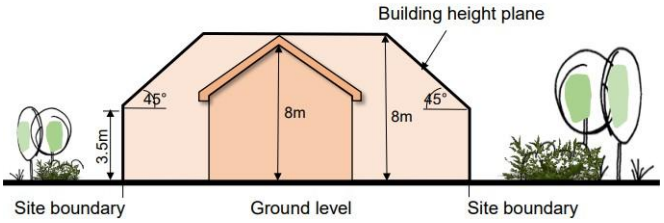
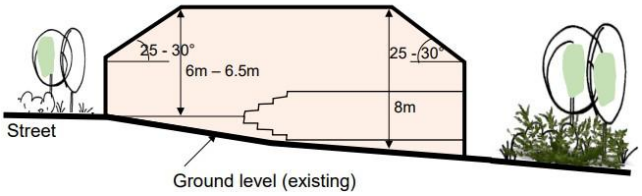
<td>
(b) Building Envelope on Flat Site


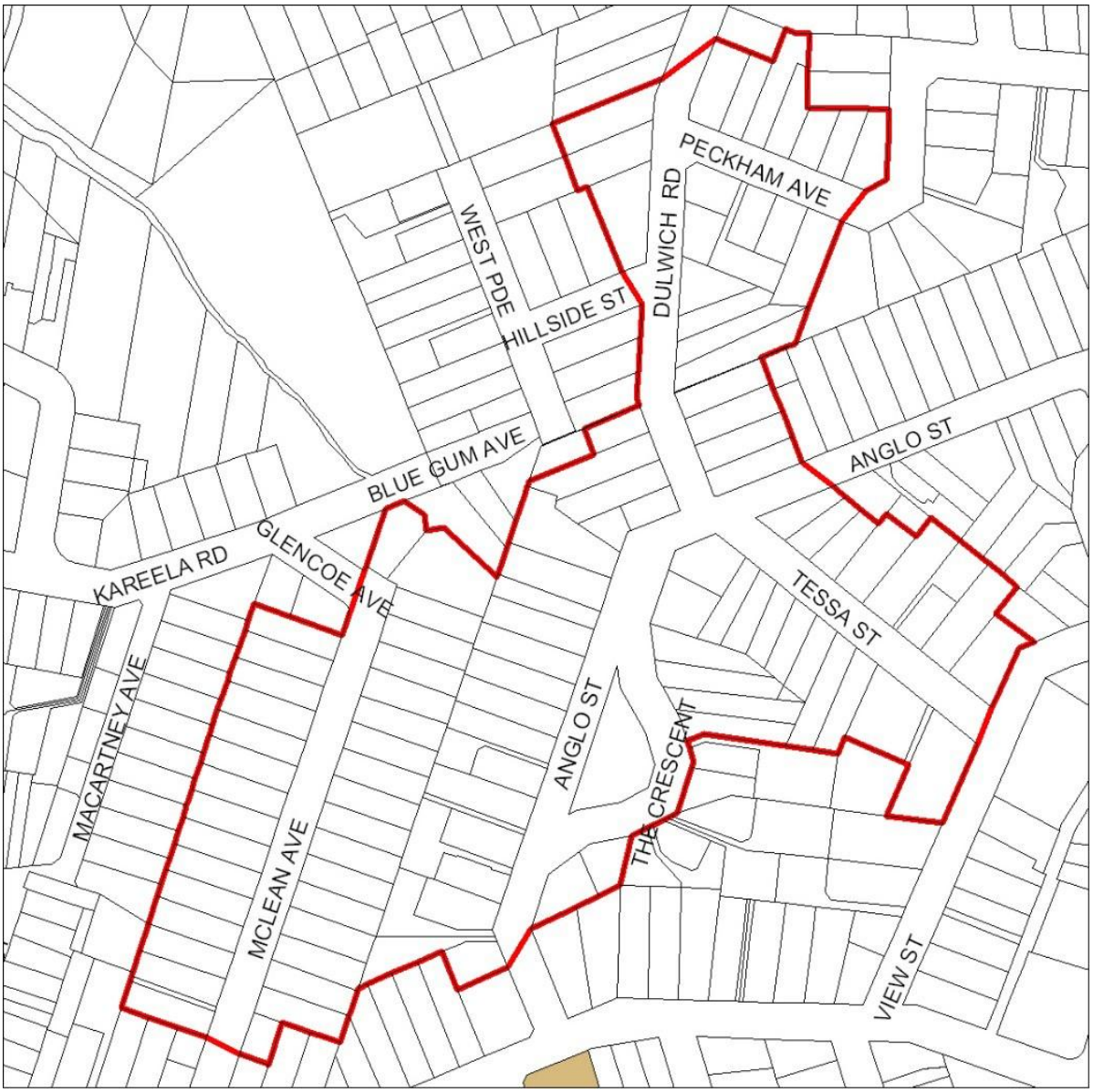
(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height
Roof pitch: 25 – 30°
Eaves height: 3.5m from floor level
Ridge height: 6 – 6.5m from floor level


</td>
</tr>
<tr>
<td colspan="3">Notes:
• The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
• The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
• There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).</td>
</tr>
<tr>
<td rowspan="4">d. Materials/ details</td>
<td>1. Roof</td>
<td>Unglazed terracotta Marseilles tile
Strapwork or shingle gable end details.</td>
</tr>
<tr>
<td>2. Walls</td>
<td>Red brown, brown or liver smooth face brick: all unpainted face brick to be retained, roughcast to upper wall surfaces.</td>
</tr>
<tr>
<td>3. Windows and doors</td>
<td>Timber casement or sash windows, some leadlight, timber glazed and panelled entry doors.</td>
</tr>
<tr>
<td>4. Joinery and decoration</td>
<td>Decorative brickwork, use of sandstone/ render trim, robust verandah detailing.</td>
</tr>
</table>

3.3 Blue Gum Heritage Conservation Area

3.3.1 Location

Figure 21 Map of Blue Gum Heritage Conservation Area



3.3.2 History

John McMillan was granted 60 acres by the Crown in 1840. This grant extended south from Corona Avenue to Isaac Nicholls' 1805 Crown Grant known as Kings Plains. Originally heavily timbered, the proposed conservation area has a history of timber felling followed by orchards and small farms. A dairy was known to have been operated in nearby Kareela Road. McMillan's Farm Subdivision was up for sale in 1897, with properties offered along Gordon Road (now Pacific Highway) between Wyvern and Corona Avenues. A number of substantial residences were already built in Wyvern and Findlay Avenues at that time.

The Anglo Park Estate which included residential allotments fronting Anglo, View, Tessa, Fullers and Gordons Roads and larger allotments fronting The Crescent and Tessa St (south side) were offered for sale in 1906. The Anglo Park Estate was developed by the Anglo Australian Assets Company. This subdivision represents the second wave of residential development in the area.

The last major wave of residential development occurred in the period 1913 – 1920, and is represented in the following streets West Parade, Hillside Street, Dulwich Road (1913), Peckham Avenue (1914), and McLean Avenue (1918) and Macartney Avenue (1922). The McLean Estate was formerly a seven and a half acre orchard worked by Duncan McLean, and leased to Chinese market gardeners. McLean Avenue was formed in 1917. Allotments in McLean Avenue were up for sale as the 'Ellerslie Estate' in 1919. Kareela Road, Blue Gum Road, Kooba Avenue and Glencoe Avenue are later developments.

3.3.3 Description

The area is dominated by Interwar housing, predominantly 1920s and 1930s bungalows. The streets close to the Pacific Highway (such as Tessa Street) include several fine Federation houses. Both Federation and Interwar cottages harmonise in their form, single storey scale, palette of materials (red/brown face brick, terracotta roof tiles) and garden settings. The undulating topography, irregular street layout, wide grass verges, mature street trees, and well – vegetated reserves with remnant blue gum high forest (an identified threatened species) complement the well established private gardens to create a verdant bushland feel.

3.3.4 Statement of significance

The area is representative of residential development of the late Federation and Interwar periods. Many streetscapes within the area have a uniformity of housing style including form materials and detailing that gives the area a harmonious appearance. The predominance of the 1920s bungalow type illustrates the important influence of American housing ideals and styles on Australia. The precinct has a strong physical and historical connections with the blue gum high forest and the topography.

3.3.5 Key period of significance

The key period of significance is 1906 to 1920.

3.3.6 Management policies

a. Retain and enhance the original form, scale and detail of existing buildings and the individual significance of the streetscapes.

b. Retain single storey character of the area. Sloping sites may enable two storey additions to the rear, which must preserve the predominantly single storey scale of the detached housing and must not interrupt the streetscape.

c. Retain characteristic palette of materials of the area, particularly the use of dark monotone face brick, sandstone bases, painted timber trim and terracotta roof tiles.

d. Retain intactness of area by carefully controlling alterations and additions, which must be set at the rear or where not visible from the street.

e. Additions should not result in excessive changes in scale, or bulk, or the introduction of visually intrusive and dissimilar materials particularly on prominent sites, including corner sites.

f. Retain the pattern of low masonry fences and landscaped front gardens.

g. Retain existing side driveways and rear car parking facilities.

h. The siting of new development must not result in the loss of significant landscaping either on – site or in verges. In this regard driveways should must be minimal with wheel strips being the preferred solution.

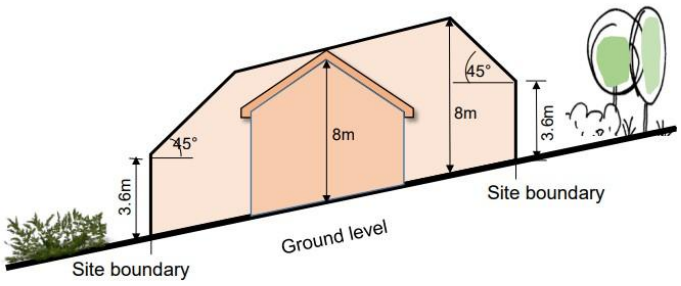
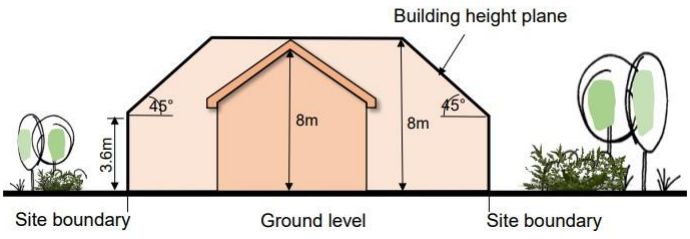
i. Protect mature trees in rear gardens which form a backdrop to the streetscapes. Protect remnants of the blue gum high forest.

j. Where a site is bushfire prone land, landscaping should conform with the NSW Rural Fire Service ‘Planning for Bushfire Protection’ guidelines.

k. Street plantings and reserve vegetation must be maintained and encouraged. Retain and enhance remnants of blue gum high forest.

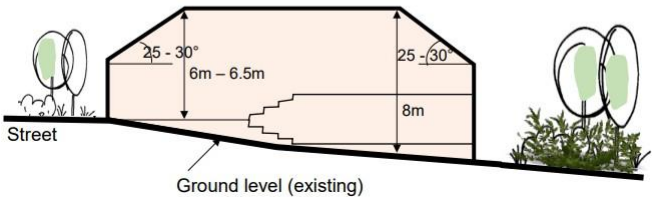
Figure 22 Example of properties in the Blue Gum Heritage Conservation Area

3.3.7 Controls for future development – Blue Gum Heritage Conservation Area

a. Scale	Generally:	Medium domestic scale, detached residences.
	1. Storeys	Generally 1 (2nd storey within roof forms and basements where topography dictates).
	2. Floor space ratio	0.4:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the character of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivisions	Generally:	Irregular, many large allotments.
	1. Setbacks	Variable front and side setbacks consistent within groups of houses.
	2. Landscaping	Well established informal gardens with fairly dense shrubbery and mature trees, including some remnant blue gum high forest. See management policies in this part.
	3. Carparking	Side driveways to rear/side parking.
	4. Front fencing	Low unpainted face brick or timber picket.
c. Form/ massing	1. Roof	Generally dominant hipped and gabled roof forms.
	2. Façade	Stepped massing with protruding verandahs/ entry porches and bays.
	3. Typical building envelope	Roof pitch: 25 – 35° Eaves height 3.6m from floor level Ridge height: 6 – 6.5m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and to the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP. **(a) Building Envelope on Sloping Site**  **(b) Building Envelope on Flat Site** 

<table>
<tr><td></td><td></td><td>(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height

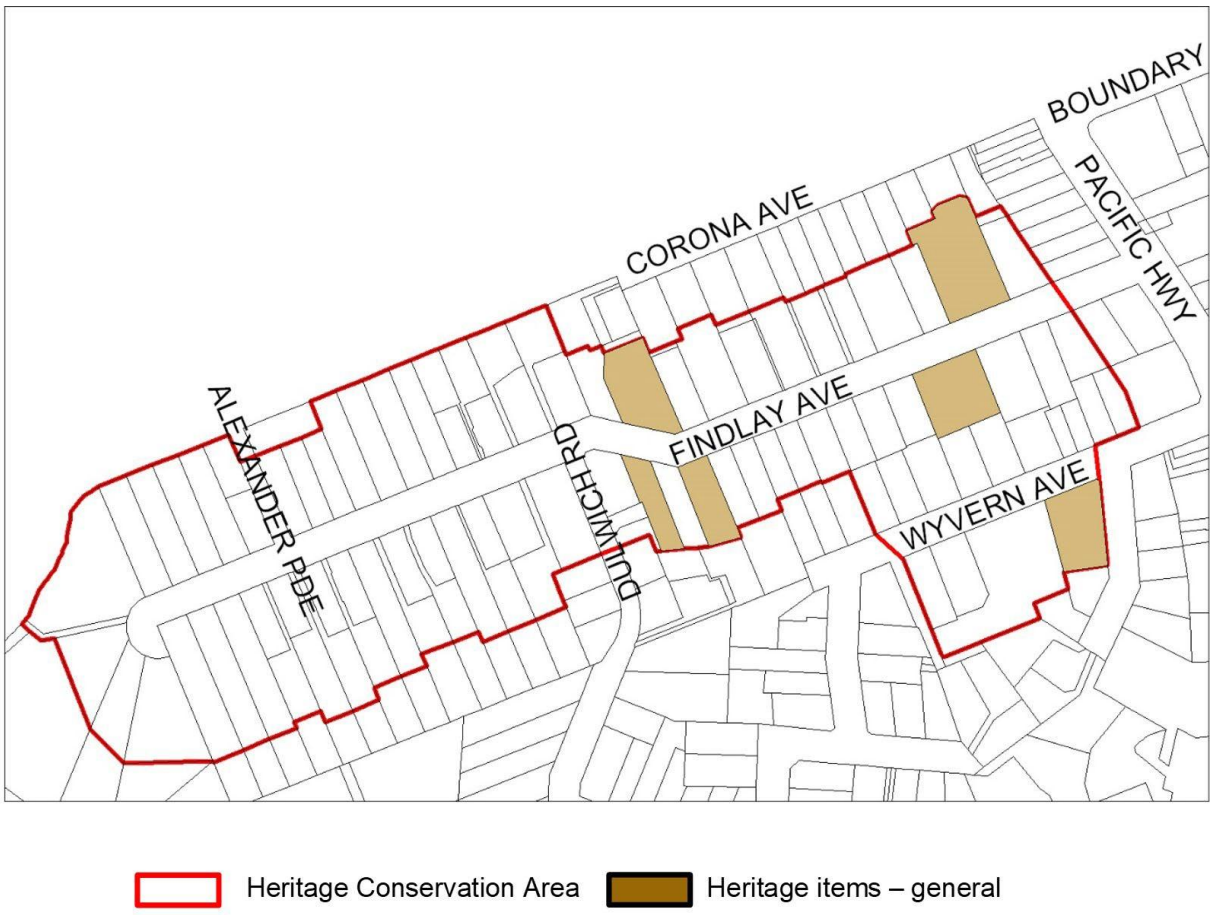
Roof pitch: 25 – 30°
Eaves height: 3.6m from floor level
Ridge height: 6m – 6.5m from floor level

 
</td></tr>
<tr><td colspan="3">Notes:
• The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
• The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
• There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).</td></tr>
<tr><td rowspan="4">d. Materials/ details</td><td>1. Roof</td><td>Unglazed terracotta Marseilles pattern tile.</td></tr>
<tr><td>2. Walls</td><td>Generally red or brown smooth face brick: all unpainted face brick to be retained.</td></tr>
<tr><td>3. Windows and doors</td><td>Timber casement sash windows often in groups of 3 in bays to front façade. Single or double leaf panelled and glazed entry doors.</td></tr>
<tr><td>4. Joinery and decoration</td><td>Heavy masonry verandah columns with stone or render capping, stone base coursing, simple gable ends.</td></tr>
</table>

3.4 Findlay and Wyvern Avenues Heritage Conservation Area

3.4.1 Location

Figure 23 Map of Findlay and Wyvern Avenues Heritage Conservation Area



3.4.2 History

Findlay and Wyvern Avenues formed a part of a Crown land grant to John McMillan in 1840, and was later bought up by Richard Hayes Harnett. Findlay Avenue dates from around 1898, and was named after Miss Findlay, who married Frederick White, an early landholder who lived in this street. Findlay Avenue was home to the first Chatswood Public School. Wyvern Avenue developed at around the same time and is thought to have been named by Harnett.

3.4.3 Description

This area can be broadly divided into two localities which reflect the pattern and timing of development, through the dominant Federation and Interwar Bungalow styles.

The flatter sections near the Pacific Highway have a variety of Federation era housing, including gracious Queen Anne style grand villas, imposing two storey mansions in spacious gardens, plus a pair of unusual two storey semi-detached houses in the Arts and Crafts style (Wyvern Avenue). These occupy larger sites, though some original lots have been subdivided to intersperse a few California and 1930s Bungalows. Wide verges with mature, usually brush

box, street trees complement the well established original gardens to create an attractive avenue effect. Most houses are intact with a wealth of original decoration and detailing and notable architectural features, including graceful slate roofs. The significance of this locality lies in its largely coherent Federation character, which is richly expressed in an array of individually interesting and surprisingly varied buildings.

The sloping sections are generally subdivided into smaller, narrower fronted lots. The street frontages are occupied by a mix of California and 1930s bungalows, interspersed with the occasional Federation bungalow and, furthest from the Highway, a few more recent unsympathetic developments (Findlay Avenue). About a third of the lots have been re-subdivided enabling battle-axe development which generally does not disrupt the streetscape. Most houses are single storey, with the few two storey buildings in Findlay Avenue being set well below the roadway or on very large allotments. The streetscape presents a generally consistent single storey scale. However, it is more remarkable for the mix of largely intact original buildings (some of which occur in harmonious groups of similar or related styles) than for its uniformity.

3.4.4 Statement of significance

The area possesses a streetscape integrity due to the development during one period and the excellent state of preservation of the houses and their garden settings. The area has a uniformity of housing style including form materials and detailing that gives the area a harmonious appearance. The predominance of the 1920s bungalow type illustrates the important influence of American housing ideals and styles on Australia.

3.4.5 Key period of significance

The key period of significance is 1898 to 1940.

3.4.6 Management policies

a. Retain and enhance the original form, scale and detail of existing buildings.

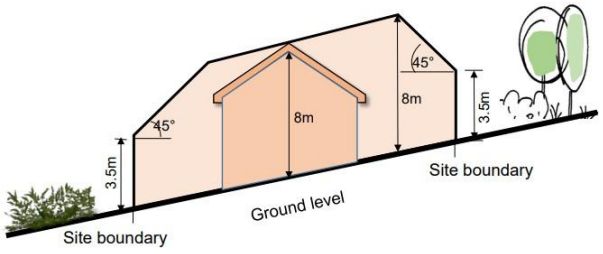
b. Maintain the individual character of streetscapes which are of heritage significance, particularly as seen in harmonious or consistent groupings of buildings.

c. Ensure that any re-subdivision and infill development on larger sites is well integrated with the individual character of existing original buildings, the local streetscape and the setting of important heritage items.

d. Retain the balance between built and landscaped elements by maintaining street trees and encouraging the retention of private gardens.

e. Where a site is bushfire prone land landscaping should comply with the NSW Rural Fire Service ‘Planning for Bushfire Protection’ guidelines.

3.4.7 Controls for future development – Findlay and Wyvern Avenue

a. Scale	Generally:	Medium / large domestic scale.
	1. Storeys	Generally 1 (2nd storey within roof forms acceptable), 2 storey on large allotments.
	2. Floor space ratio	0.4:1 for land zoned R2 Residential. This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP. For land at 52 – 84 Findlay Avenue zoned C4 Environmental Living refer to Willoughby LEP, Clause 4.4 Floor Space Ratio, including the table for Maximum FSR in Area 2.
b. Setting/ subdivision	Generally:	Irregular, many large allotments, several have been subdivided to create battle axe blocks.
	1. Setbacks	Variable front and side setbacks consistent within groups of houses.
	2. Landscaping	Well established gardens with fairly dense shrubbery and mature trees. (See Management Policies).
	3. Carparking	Side driveways to rear/ side parking.
	4. Front fencing	Low unpainted face brick or timber picket.
c. Form/ massing	1. Roof	Generally dominant hipped and gabled roof forms.
	2. Façade	Generally stepped massing with protruding verandahs/ entry porches.
	3. Typical building envelope	Roof pitch: 25 – 30° Eaves height 3.5m from floor level Ridge height: 6 – 6.5m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP. **(a) Building Envelope on Sloping Site** 

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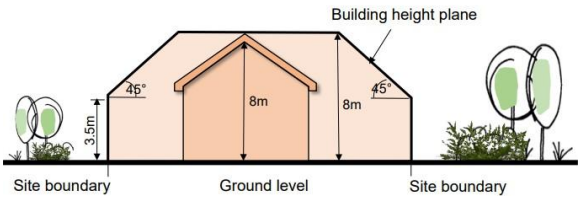
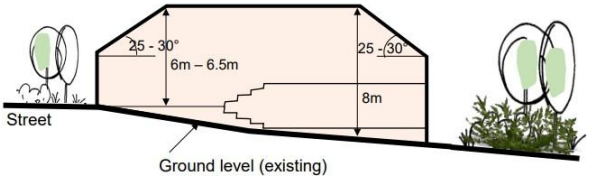
(b) Building Envelope on Flat Site


(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height
Roof pitch: 25 – 30°
Eaves height: 3.5m from floor level
Ridge height: 6 – 6.5m from floor level


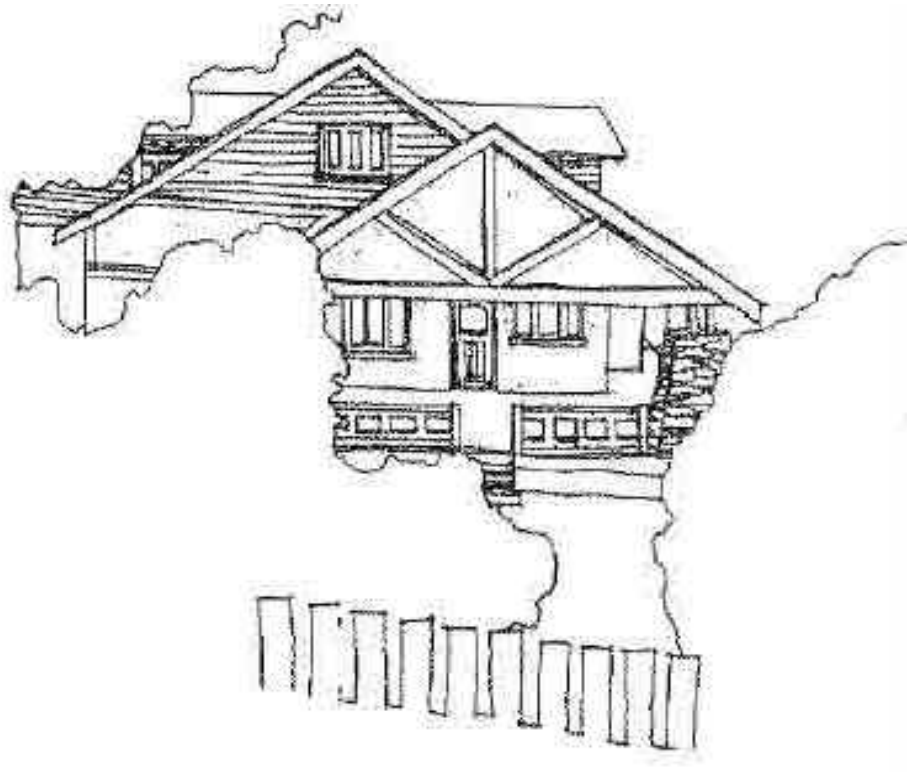
</td>
</tr>
<tr>
<td colspan="3">
Notes:

The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).

</td>
</tr>
<tr>
<td rowspan="4">d. Materials/ details</td>
<td>1. Roof</td>
<td>Unglazed terracotta Marseilles tile and slate.</td>
</tr>
<tr>
<td>2. Walls</td>
<td>Generally red or liver smooth face brick: all unpainted face brick to be retained.</td>
</tr>
<tr>
<td>3. Windows and doors</td>
<td>Timber casement sash windows often in groups of 3 in bays to front façade, single or double leaf panelled and glazed entry doors.</td>
</tr>
<tr>
<td>4. Joinery and decoration</td>
<td>Heavy masonry verandah columns with stone or render capping, stone base coursing, simple gable ends.</td>
</tr>
</table>

Figure 24 Examples of two properties in the Findlay and Wyvern Avenues Heritage Conservation Area

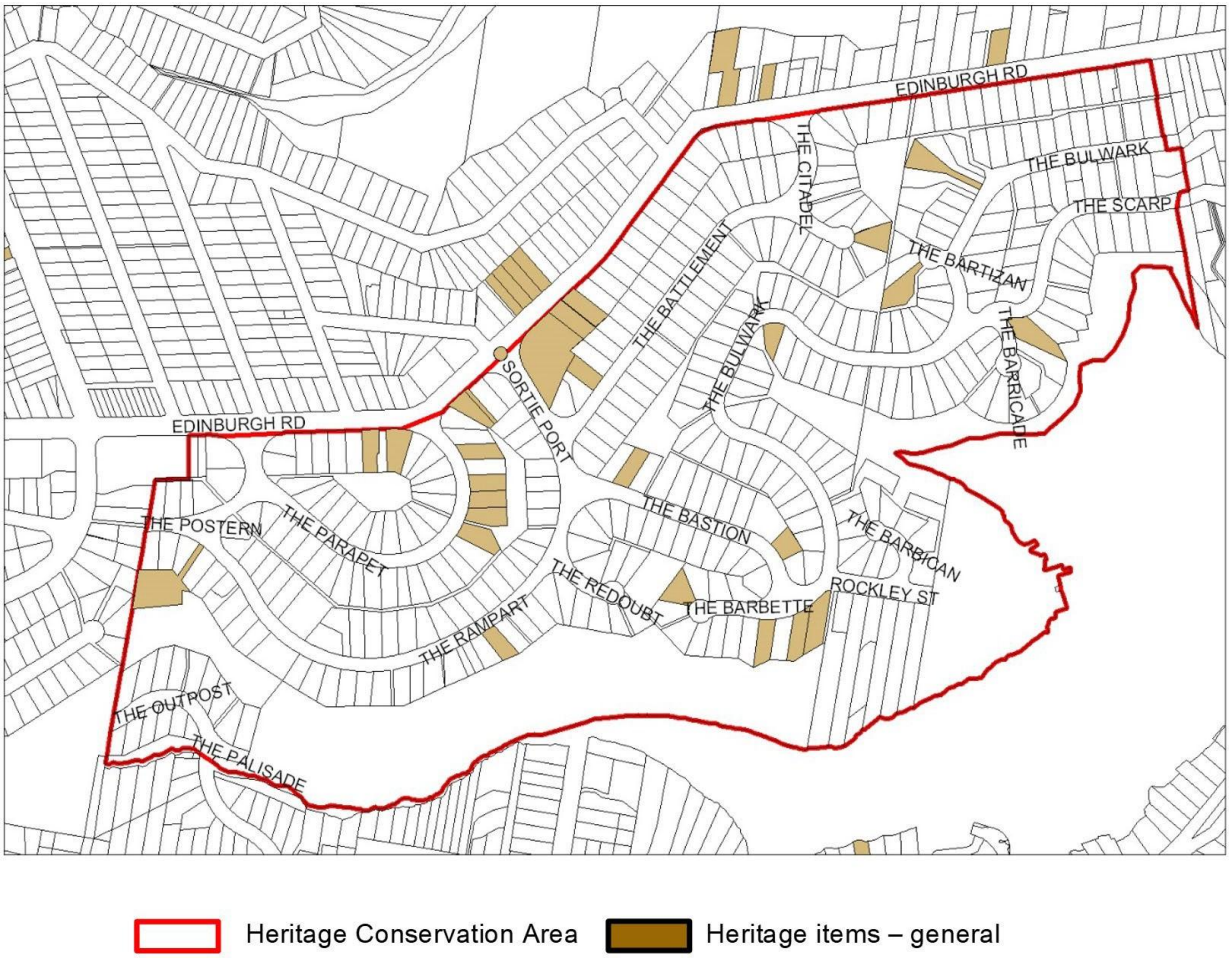
Figure 25 Further example of a property in Findlay and Wyvern Avenues Heritage Conservation Area



3.5 Griffin Heritage Conservation Area: C4

3.5.1 Location

Figure 26 Map of Griffin Heritage Conservation Area



3.5.2 History

Walter Burley Griffin and his architect wife Marion Mahony Griffin founded the Greater Sydney Development Association (GSDA) in 1920. They set about establishing a residential development that was sympathetic with the natural environment and in sharp contrast to the red roofs and grid streets that characterised Sydney at the time.

In 1921, the GSDA purchased 650 acres at Middle Harbour, including the southwest part of Castlecrag. Griffin designed the Castlecrag Estate, as it became known, in sympathy with the natural environment, creating bushland reserves that preserved the major landforms and rock outcrops; foreshore reserves; a delightful network of walkways; and roads that followed the contours and respected the landforms.

In 1926 Walter and Marion Griffin bought, in their own names, two portions that adjoined the eastern boundary of the Castlecrag Estate. This they called the Haven Estate. Griffin extended the same principle of contoured road design and reserves to the subdivision of this land.

The houses were designed to harmonise with the landscape and were built of sandstone quarried from the site and/or Knitlock, a concrete building block patented by Griffin that had a crushed sandstone finish. Many sites had a house plan attached and the houses were carefully located on the sites with stepped setbacks from the road to respect the views and privacy of other houses. Most houses had flat roofs to maximise the retention of views and covenants controlled all development.

Fourteen houses designed by Griffin were built at Castlecrag but evidence of over forty house designs exists. GSDA exercised control of design and siting of buildings through covenants affecting the construction materials used, number of buildings on the site, setbacks, design of fences, advertising signs, and required contribution to upkeep of public reserves, to safeguard the general high class residential character of the area. Eric Nicholls varied the covenants to allow pitched roofs and brick walls. Willoughby City Council is not bound to enforce these covenants but they may be enforced by owners of properties within the estate that comply with the covenants.

The National Trust of Australia (NSW) recognised the importance of the area by classifying it as an urban conservation area in 1981.

3.5.3 Description

Within residential properties, the lack of front fences and separate garages retain the transition from street to reserve that Griffin intended by minimal building within the first 10m zone of the property. The intent was that the house was to exist in the bush landscape without the artificiality of property boundaries.

Griffin's main aims in designing the estate were to get away from the traditional suburb with its rectangular grid street pattern, imposed upon the landscape without thought for topography, and to demonstrate that architecture could be subordinate to and harmonious with the landscape. To Griffin, architecture, the site planning, town planning and landscape design were inseparable. Any structure had to fit into the overall landscape of the area – being harmonious rather than obtrusive. He achieved this integration through his design philosophy – narrow winding roads following contours, linked open space networks, lack of fences, unobtrusive houses, retention of bushland in the reserves, walkways, nature strips and road islands. Roads are unobtrusive because of the proliferation of natural vegetation, their winding nature, the grass and bushland verges, the sandstone kerbs and narrow widths. All these aspects, the natural landforms of rock outcrops, cliffs and gullies and the rock cuttings for the road and also the culverts are extremely important and of great significance in the Griffin Heritage Conservation Area.

The form of houses proposed by Griffin was single storey, or in rare cases two storey buildings based on a series of projecting wings radiating from the central area and fireplace. The elevational treatment tended towards strong horizontals because of a design intention to integrate with the landscape. This is reinforced by the use of masonry blocks with strong horizontal window lintels and flat roofs. The finishes tended toward natural materials such as stone and the use of colours which reflect the bushland setting.

3.5.4 Statement of significance

The Castlecrag and Haven Estates are outstanding early examples of subdivision which respected the landscape character of an area, created community environments and provided

shared views. This process was initiated by Walter Burley Griffin and Marion Mahony Griffin who were influential architects in the United States and Australia, and were known for their appreciation of the opportunities provided by landscape which was embodied in their plan for Canberra. The significance of the estates is heightened by the extant works of the Griffins which are nationally and internationally recognised.

The estates are a larger more complete demonstration of principles Griffins had developed in the United States and their first seven years in Australia. These innovative principles involved subdivision, contoured roads embedded in the sandstone topography and engineering processes that were aimed at conserving the sandstone terrain, stream systems, indigenous bushland and harbour foreshore, and provided extensive reserves and walkways through the estates that created an integrated open space network. The distinctive nomenclature the Griffins gave to the reserves and roads expresses the castle-like quality of the Castlecrag peninsula.

The estates embody the sense of community and social connection from the Griffins, the first investors and the Griffins' friends to a continuing community with a strong sense of connection to the place. Long established community facilities include the Haven Amphitheatre, Community Centre, Griffin shops, tennis courts, reserves and walkways.

The estates are unique in their application of small lots in a spectacular harbour setting with public vistas and filtered views providing the suburban ideal subordinated to the landscape.

3.5.5 Key period of significance

The key period of significance is 1921 to 1935.

3.5.6 Management policies

a. Retain and where possible reinstate the original subdivision pattern and linked system of public reserves and pedestrian pathways. Retain road islands, rock cuttings, sandstone retaining walls, sandstone kerb and guttering, grass road verges and bushland nature strips.

b. Maintain and where possible reinstate natural topography, landforms, the natural ecology/drainage ways and ponding areas, and vegetation types.

c. Buildings should be placed to respect natural features which historically has resulted in a variety of setbacks. This principle should be adhered to in new development.

d. Permit new development which fits in with the original subdivision pattern, and which blends with and preserves as much as possible of the natural landscape, its remnant bushland and rocky terrain.

e. New development must carefully follow the contours of the land to minimise bulk, and cutting and infilling.

f. The height, scale, bulk, massing and proportion, site cover, location and visibility of new development must be such that:

    - built forms are subordinate to the natural landscape
    - the spacing between buildings and rhythms of the streetscapes are retained, or opened up to create vistas of the natural landscape

- primary views from nearby and adjoining dwellings and public reserves, roadways, pathways and drainage reserves are not obstructed
- undercroft areas are not visually intrusive in the landscape when viewed from the water or the land
- buildings are highly articulated in plan and elevation
- amenity of public reserves is maintained as public open space

g. Materials and colours must blend inconspicuously with the predominant colours of the local bushland.

h. Significant heritage items including landscape must not be encroached upon.

i. Carparking, including garages are to be designed and sited to retain the unique character of the usually well vegetated narrow winding roadways and public open spaces.

j. No fences, screens or gates other than very low fencing (300mm in height) forward of the building line.

k. No fences or screens adjoining public reserves and pathways:

- indigenous vegetation to be used to achieve privacy; and

- where special security requirements can be demonstrated, fencing to a maximum of 1200mm may be permitted to be constructed of timber posts or metal frame posts with lightweight open mesh infill. Acceptable styles include arris rail or piped rail and diagonal mesh, horse wire or childproof weldmesh in unpainted galvanised finish. Where bushfire prone land is identified non-combustible materials must be used.

l. No child restraint barriers surrounding a swimming pool are to be located on the property boundary adjoining public reserves and pathways. Instead:

- the barrier must be setback 3m from the property boundary
- the barrier should not exceed 1500mm
- a minimum 3m wide landscaped area shall be provided between the barrier and a side or rear property boundary

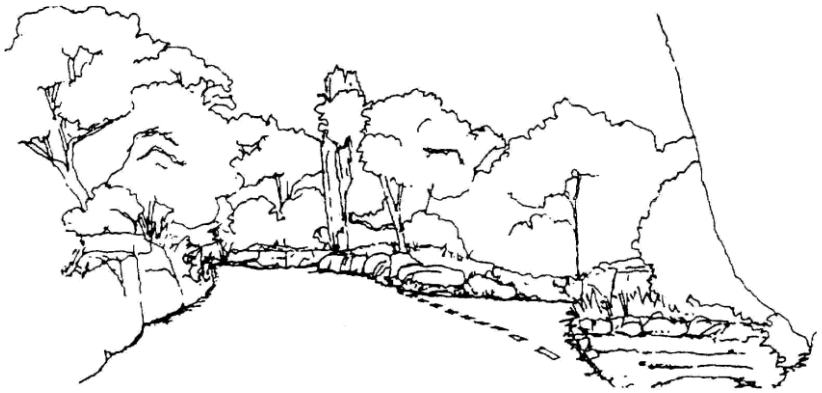
Notes:

1. The provisions of the Swimming Pool Act 1992 must be satisfied
2. These requirements will be included as a condition of consent for any approval granted for a swimming pool.

m. Locally indigenous vegetation is to be used for landscaped areas, including private open space adjoining Griffin walkways, reserves and public open space.

n. Where a site is bushfire prone land, landscaping should conform with the NSW Rural Fire Service ‘Planning for Bushfire Protection’ guidelines.

Figure 27 Illustration of the landscape in the Griffin Heritage Conservation Area



3.5.7 Controls for future development – Griffin: C4 Environmental Living

a. Scale	Generally:	Low scale, maintain and reinstate predominance of native landscape over built form. Ensure no encroachment by new buildings into public vistas.
	1. Storeys	Generally 1 – 1.5, maximum 2.
	2. Floor space ratio	The maximum gross floor area (GFA) of a building on land that is in the Griffin Heritage Conservation Area is the GFA specified in column 2 of the table. This figure represents an upper limit to GFA. It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. The floor space ratio calculated for the site area (m^2) is: under 400m^2: 0.45:1 400m^2–500m^2: 0.40:1 501m^2–600m^2: 0.36:1 601m^2–700m^2: 0.34:1 701m^2–800m^2: 0.32:1 801m^2–900m^2: 0.30:1 901m^2–1000m^2: 0.28:1 1001m^2–1100m^2: 0.26:1 1101m^2–1,300m^2: 0.25:1 >1300m^2: 0.24:1 See Clause 4.4A (Exceptions to Floor space ratio) of Willoughby LEP.
	3. Massing	The scale and massing of new development is to respect the Griffins' objective to have all built forms subservient to the landscape. Development in the vicinity of heritage items shall respect the scale and massing of those items and ensure public view corridors to them.

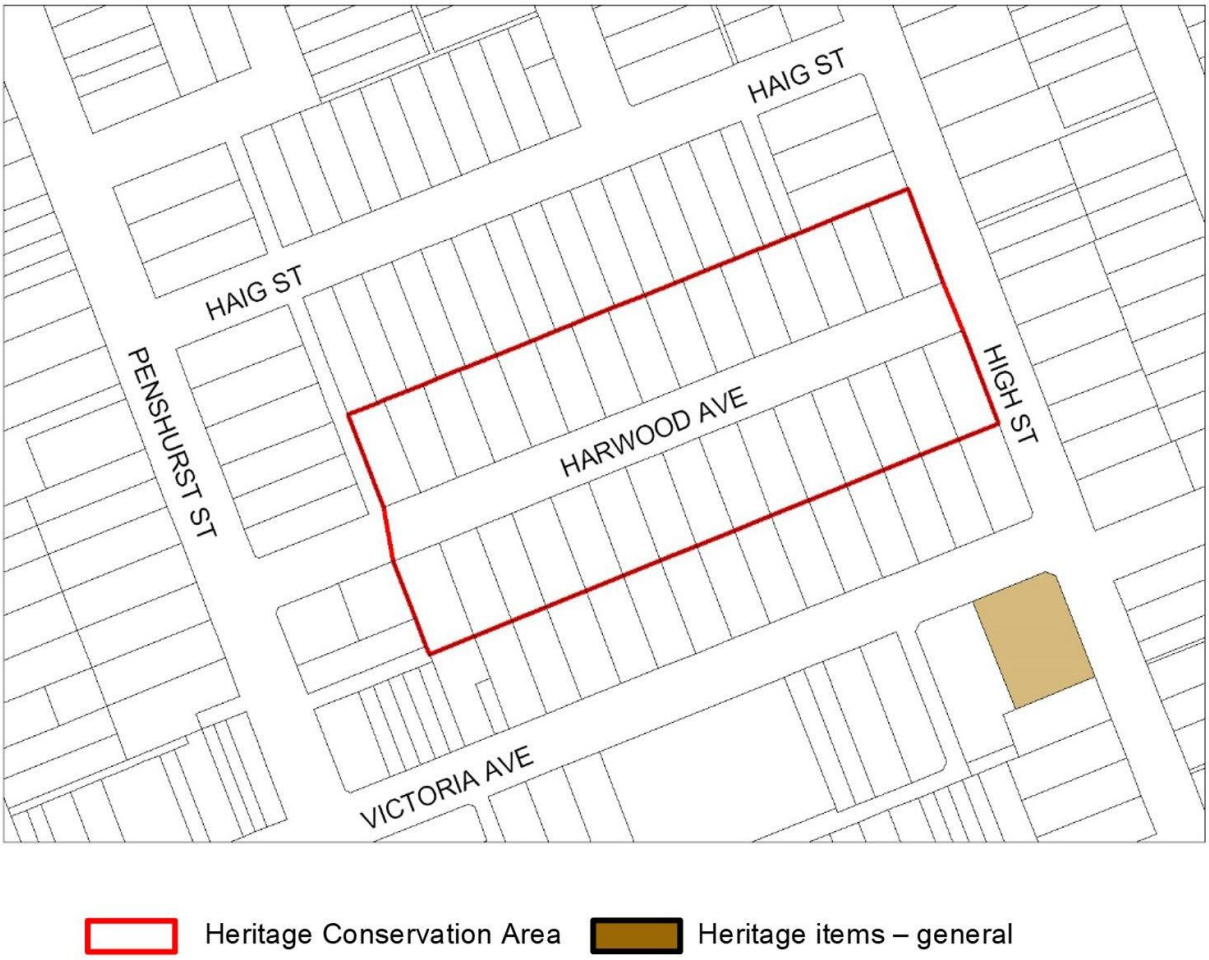
b. Setting/ subdivision	Generally:	Retain and enhance original Griffin subdivision and its intention with emphasis on the natural qualities, views and topography of the site.
	1. Setbacks	Natural features, public walkways, or existing heritage buildings: 3m Street: variable, typically 6 – 10m Public reserves: min 10m from the rear boundary with reserve. One side: 3m Façades visible to pathways or reserves are to incorporate setbacks stepped in plan to create articulation and reduce visual bulk.
b. Setting/ subdivision	2. Landscaping	Informal, heavily landscaped front gardens dominated by indigenous shrubs and trees. Maintain, enhance and protect indigenous landscape, natural features, landforms and vegetation (For bushfire prone land refer to Management Policies).
	3. Carparking	Small scale, carparking behind building line where possible, else 5m setback for carports. Reduce impact of cars through the use of lightweight open carports and absorbent surface materials (not interfering with public view from street). Open hard stand parking spaces or lightweight open sided carports, with absorbent ground surface materials preferred rather than enclosed garages to reduce the impact of cars.
	4. Preservation of views from public places	Retain public views of waterways and other significant outlooks.
	5. Fencing	(See management policies).
	6. Gates behind building alignment	Behind building alignment, no gate or 'see through'/open mesh form gates to maximum 1200mm is preferred. Gates permitted on boundaries adjoining public pathways and reserves. (See management policies).
c. Form/ massing	1. Roof	Simple flat roofs or low pitched hipped roof forms in recessive neutral colours.
	2. Façade	Stepped, well articulated elevations, projecting bays asymmetrical building forms, horizontal emphasis. Avoid use of strong vertical elements such as fenestration or columns.
	3. Building envelope	Roof pitch max. 22.5° Height: 8m max in line with Clause 4.3 (Height of buildings) of Willoughby LEP. Ground floor plane: follow contours of landforms. Heights relative to ground level (existing)

	4. Substantial alterations	Development involving substantial alterations is to achieve side boundary setbacks that open up vistas or views between houses.
d. Materials/ details	1. Roof	Bitumen, corrugated metal sheet, terracotta tile or concrete tile in recessive neutral colours.
	2. Walls	Stone, render or timber in recessive neutral colours which blend inconspicuously with the predominant colours of the local bushland. Materials used to enhance horizontal emphasis.
	3. Windows and doors	Horizontal emphasis to groups of windows.
	4. Joinery and decoration	Horizontal emphasis.

3.6 Harwood Avenue Heritage Conservation Area

3.6.1 Location

Figure 28 Map of Harwood Avenue Heritage Conservation Area



3.6.2 History

Harwood Avenue is a short street of identical, average sized lots on gently sloping land that was developed in the 1920s. Originally part of John Street, the Avenue was renamed in 1921 after George Harwood who owned the nearby Harwood Estate. The first rate assessment in Harwood Avenue was in 1921.

3.6.3 Description

The housing is almost entirely fairly modest California or transitional bungalow styles of the 1920s that are typical of much of Willoughby. The street is unified by the consistent use of face brick and Marseilles pattern terracotta tile as well as a consistency in building form, scale and setbacks. Façades feature some leadlight windows generally in groupings of three, rough face sandstone base coursing, and limited use of timber shingles, vertical battens and roughcast to gable ends and window skirts. Carparking is located behind the building alignment and front gardens are open and well landscaped with low fencing, providing an

appropriate setting for the bungalows. The street is enhanced by its closure at Penshurst Street and by the mature, consistent brush box street tree planting.

3.6.4 Statement of significance

This is an outstanding example of a 1920s residential development in near original condition. Very few additions or irreversible alterations have occurred to the original housing stock, and its coherent character is retained. The area possesses a streetscape integrity due to the development during one period and the excellent state of preservation of the houses and their garden settings. The area has a uniformity of housing style and setting including form materials, detailing, fencing and setbacks that gives the area an harmonious appearance. The predominance of the 1920s bungalow type illustrates the important influence of American housing ideals and styles on Australia.

3.6.5 Key period of significance

The key period of significance is 1921 to 1930.

3.6.6 Management policies

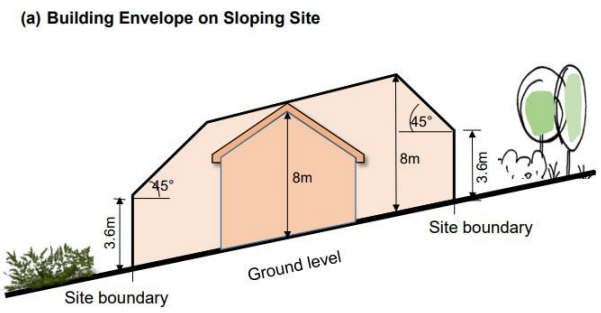
a. Retain the dominant bungalow character of the street by maintaining all original 1920s – 1930s buildings and their consistency in scale, style, detailing, spacing, setbacks and subdivision pattern.

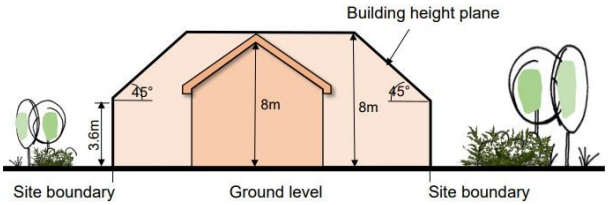
b. Any new development (including carparking structures) must retain the principal elevations of the bungalows and be set back behind the main body of the original building envelope. Additional floor space may be achieved within the roof space at the rear of houses.

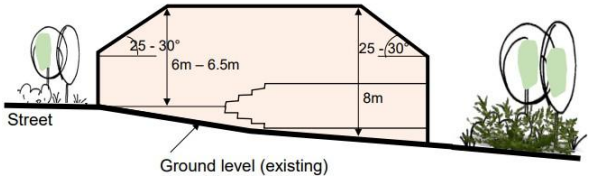
c. Maintain the avenue character by retaining the original brushbox street trees and largely original style front gardens. Any street tree renewal should be replaced with brushbox trees.

Figure 29 Illustration of a local streetscape in Harwood Avenue Heritage Conservation Area

3.6.7 Controls for future development – Harwood Avenue

a. Scale	Generally:	Uniform low domestic scale, detached bungalows.
	1. Storeys	1 (2nd storey within roof forms acceptable).
	2. Floor space ratio	0.4:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Consistent grid layout, lots average 565sqm, 15.24m frontages.
	1. Setbacks	Uniform front setbacks: 6m Variable side setbacks: 0.9 – 3m
	2. Landscaping	Open front gardens with lawns and low exotic shrubbery.
	3. Carparking	Single low scale rear or side garages accessed by side driveways.
	4. Front fencing	Low unpainted brick, timber and wire fencing.
c. Form/ massing	1. Roof	Dominant broad simple low roof, 2 – 3 gables facing street, hipped to rear, some single gables; with cross gable; flat roofed porches and verandahs to street.
	2. Façade	Typically asymmetrical stepped façades, with protruding bays, recessed porches and verandahs.
	3. Building envelope	Roof pitch: 25 – 30° Eaves height: 3.6m from floor level Ridge height: 6 – 6.5m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP. 

<table>
<tr><td></td><td></td><td>
(b) Building Envelope on Flat Site


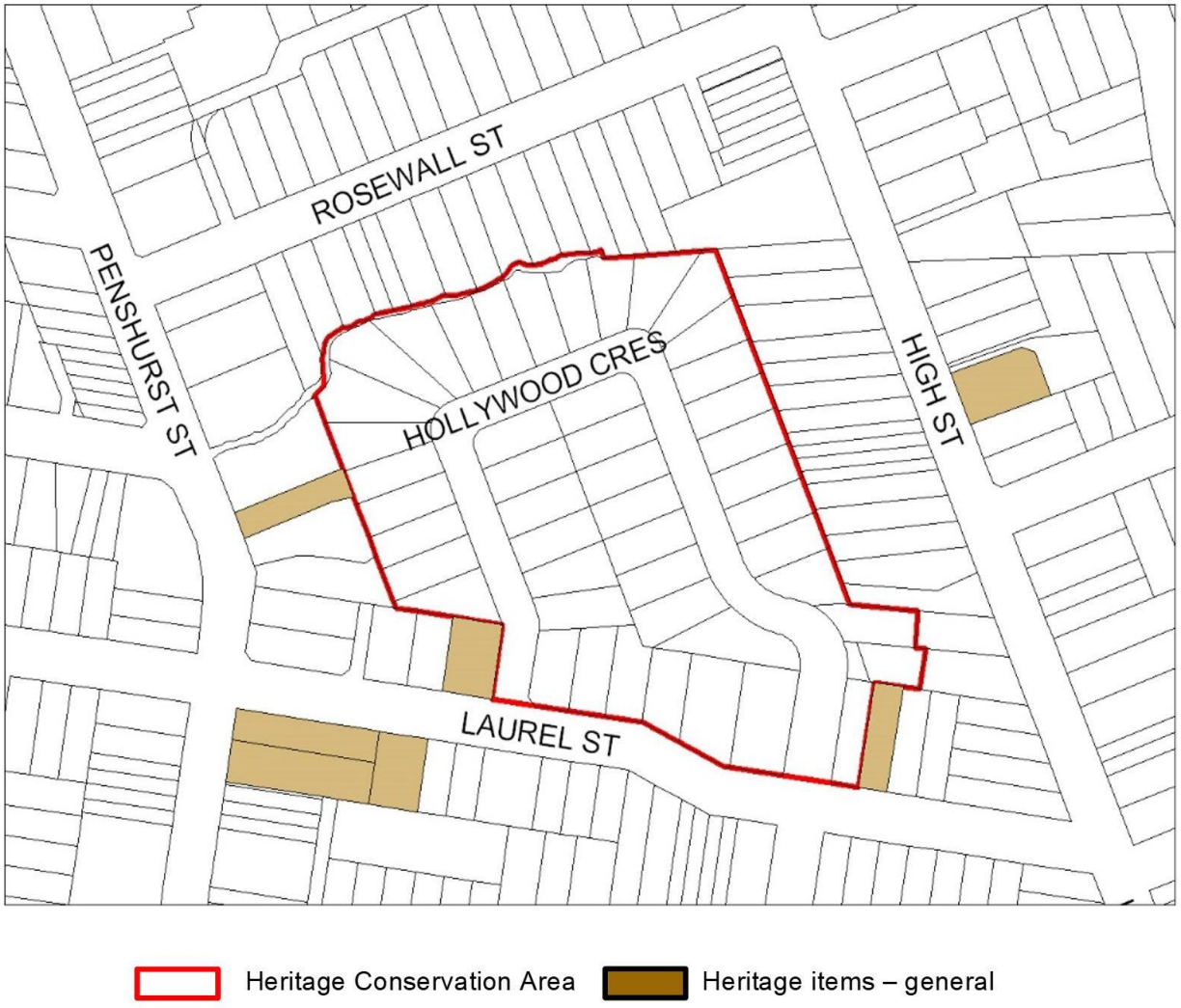
(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height
Roof pitch: 25 – 30°
Eaves height: 3.6m from floor level
Ridge height: 6 – 6.5m from floor level


</td></tr>
<tr><td colspan="3">Notes:
• The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
• The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
• There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).</td></tr>
<tr><td rowspan="4">d. Materials /details</td><td>1. Roof:</td><td>Unglazed terracotta Marseilles tiles, exposed eaves.</td></tr>
<tr><td>2. Walls</td><td>Liver or red smooth face brick: all unpainted face brick to be retained, render or sandstone base coursing, rough cast to upper wall surfaces.</td></tr>
<tr><td>3. Windows and doors</td><td>Large bays/boxed timber casement windows in groups of 3 to street with skirts and awnings, single leaf timber panelled glazed entry doors.</td></tr>
<tr><td>4. Joinery and decoration:</td><td>Simple lattice or shingles and /or vertical battering to gable ends, simple ventilators, massive battered verandah columns in brick, stone or roughcast.</td></tr>
</table>

3.7 Hollywood Crescent Heritage Conservation Area

3.7.1 Location

Figure 30 Map of Hollywood Crescent Heritage Conservation Area



3.7.2 History

The area of Hollywood Crescent was originally a part of the Higgins Estate, and was at one time cultivated by Chinese market gardeners. It is believed that Higgins received an inheritance from a relative in USA with connections in Hollywood, prompting the development of the estate in 1926–28 by Cramer Bros, Real Estate Agents of Crows Nest, prominent local business men and politicians, who assumed trusteeship for Higgins Buildings Limited. Cramer Bros named the Crescent and decided to build Californian bungalows.

3.7.3 Description

Hollywood Crescent's winding horse-shoe shape is a rare and marked contrast to the grid pattern of the bulk of Willoughby's streets of this era and is reminiscent of the 'prime garden' suburb layouts found in the Appian Way, Haberfield or the early South Sydney suburb of

Daceyville. The grassy verges and regular plantings of brush box enhance the garden suburb vision.

The subdivision pattern is most unusual, including radial divisions with all lots facing the 'U' shaped road. Consequently, the lots are extremely varied in shape and frontages. The north boundary of the conservation area is partially formed by the alignment of Sugarloaf Creek.

Housing in the area consists predominantly of 1920s bungalows on medium sized lots. The street is unified by the consistent use of face brick and terracotta tile as well as uniformity in building form, scale and setbacks. The mature regular street tree plantings, changing orientation of bungalows, landscaped front gardens and curving street layout, creates a picturesque sequence of vistas along the Crescent.

3.7.4 Statement of significance

This is a rare local example of a curving tree-lined residential development of the 1920s. The area possesses a streetscape integrity due to the development during one period and the excellent state of preservation of the houses and their garden settings, and the consistent and mature street plantings. The area has a uniformity of housing style and setting including form, materials, detailing, setbacks and fencing that gives the area a harmonious appearance. The predominance of the 1920s bungalow type illustrates the important influence of American housing ideals and styles on Australia.

3.7.5 Key period of significance

The key period of significance is 1926 to 1935.

3.7.6 Management policies

a. Retain and enhance the open garden suburb character and dominant Bungalow style of the streetscape.

b. Retain the style, scale, setbacks and spacing of the original buildings.

c. Second storey additions to be designed to integrate into roof forms, and maintain the scale and character of the original bungalows.

d. Encourage low fencing and open front gardens.

e. Maintain the existing brush box street trees and wide grassy verges.

f. Reinstate brush box street trees where other species have been planted.

g. Retain existing access to rear/side carparking facilities.

h. Ensure that the scale of development in the vicinity of Hollywood Crescent does not impact on the significance of the streetscape, or on vistas into the Crescent.

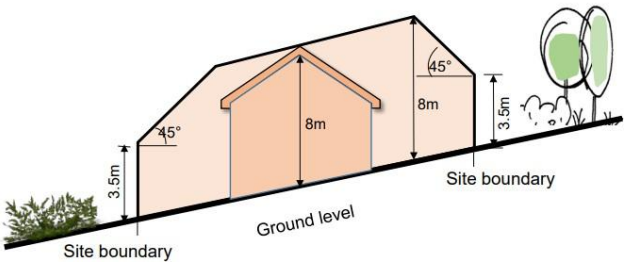
Figure 31 Illustration of the Hollywood Crescent streetscape

3.7.7 Controls for future development – Hollywood Crescent

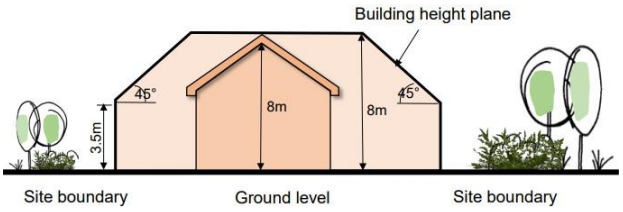
a. Scale	Generally:	Uniform low domestic scale, detached bungalows.
	1. Storeys	1 (2nd storey additional roof forms in sympathetic style acceptable – see management policies).
	2. Floor space ratio	0.4:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Rare patterns of vegetation and radical lots fronting 'U shaped' road, variable lot sizes and frontages.
	1. Setbacks:	Front setbacks follow curve of street front: 6 – 7m, variable side: 0.9 – 3m.
	2. Landscaping	Established open garden to street, lawn with mixed exotic shrubs.
	3. Carparking	Simple, low scale single garages to rear or side accessed by side driveways.
	4. Front fencing	Low unpainted face brick, timber and unpainted brick, or sandstone.
c. Form/ massing	1. Roof	Broad dominant gables (2 – 3) fronting street, hipped to rear, low pitch or flat roofed verandahs.
	2. Façade	Bold stepped massing with projecting bays and porches, typically asymmetrical, recessed verandahs to street.
	3. Building envelope	Roof pitch: 25 – 30° Eaves height: 3.5 m from floor level Maximum ridge height: 6 – 6.5 m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and to the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP.

<table>
<tr><td></td><td></td><td>

(a) Building Envelope on Sloping Site



(b) Building Envelope on Flat Site

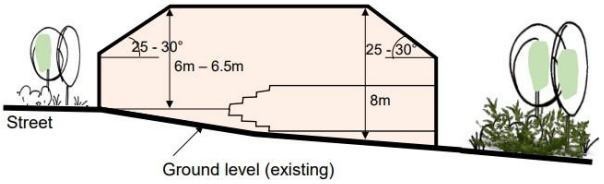


(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height

Roof pitch: 25 – 30°

Eaves height: 3.5m from floor level

Ridge height: 6 – 6.5m from floor level



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<tr><td colspan="3">

Notes:

- The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
- The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
- There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).

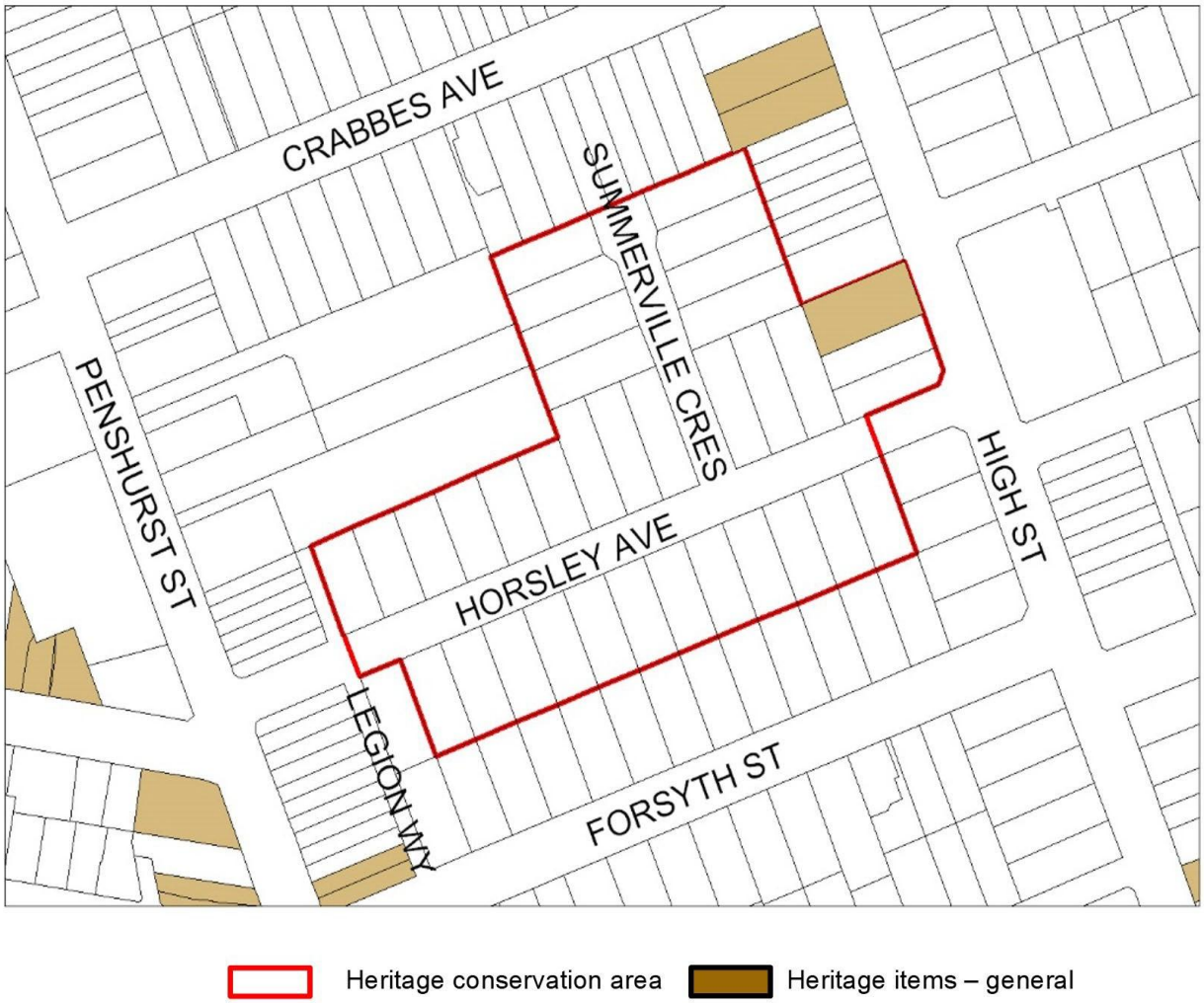
</td></tr>
<tr><td rowspan="2">d. Materials/ details</td><td>1. Roof</td><td>Unglazed terracotta Marseilles tiles
Exposed eaves, fibre cement gable infill.</td></tr>
<tr><td>2. Walls</td><td>Red/liver smooth face brick: all unpainted face brick to be retained, stucco to upper wall surfaces and sandstone base courses.</td></tr>
</table>

	3. Windows and doors	Bay/boxed timber casement windows in groups of 2, 3 or 4 with shingled skirts/awnings to street, single leaf semi glazed entry doors.
	4. Joinery and decoration	Battened or shingled gable ends, squat masonry columns and timber posts to verandahs.

3.8 Horsley Avenue Heritage Conservation Area

3.8.1 Location

Figure 32 Map of Horsley Avenue Heritage Conservation Area



3.8.2 History

Part of Richard Archbold's original grant of 1825, later granted to John Stirling (Chairman of the Bank of Australia) in 1850. These grants comprised 600 acres bounded by Victoria Ave, High St, Mowbray Rd to almost Stanley St and continuing parallel with that street to Victoria Avenue. Stirling sold the entire 600 acres to William Lithgow four years later. Lithgow subdivided immediately into lots of varying sizes ranging from small farms of 25 acres to housing lots, and put the whole area up for public auction with little success. In 1858, Richard Horsley bought an 8-acre portion of Lithgow's Estate and settled in Willoughby. Horsley, a pioneer orchardist and gardener, had arrived in the colony with his wife in 1857 on board the 'Plantagenet'.

Horsley's land was bounded by Penshurst St (east side), High Street (west side) and Forsyth Street (north side), and extended half way to Crabbes Avenue. Horsley established a

successful orchard on this site growing vegetables, flowers and operating a small farm. The family home ‘Seaview’ was on the east side of Penshurst Street (now demolished). Richard’s oldest son William bought ‘Kianga’ at 190 High Street in 1911 for a family home. It later became a convent and then the Catholic Education Office. Richard died at ‘Seaview’ in 1915, and the farm passed to his youngest son Francis. Francis built a house and boat shed at Sugarloaf Bay and leased the farm to Chinese market gardeners. After Francis’ death, the property was subdivided as the ‘Heart of Willoughby Estate’, and sold in 1926 and 1928. This subdivision included allotments fronting Horsley Avenue, the north side of Forsyth Street, and sections of High Street and Penshurst Street.

Summerville Crescent was developed by Cramer Bros. estate agents of Crows Nest in 1935. Charles Irving Cramer and John Oscar Cramer were very successful businessmen; John later became Mayor of North Sydney, and a formative member of the Liberal Party.

3.8.3 Description

Horsley Avenue is a narrow street with consistent street plantings of Crepe Myrtle. The eastern end is dominated by St Thomas’ School and Church, and the western corner features Interwar cottages and flats. The houses are Interwar bungalows built of dark red/brown and liver brick with rough face sandstone base courses and detailing. The roof forms are predominantly hipped with vertically battened gables. Many houses feature leadlight windows in the street facing façades. The low face brick fencing, consistent setbacks, side or rear carparking arrangements and well vegetated front gardens create a streetscape impression that is consistent with the heritage values of the housing.

Summerville Crescent is a short cul-de-sac lined with small-scale bungalows, which appear to have been speculatively developed. The bungalows have glazed terracotta tiled hipped roofs, some with parapeted gables, and are distinguished by idiosyncratic contrasting brick detailing. This detailing is reflected in front fencing. Carports are located to the side and rear of the houses.

3.8.4 Statement of significance

Summerville Crescent is a rare example of a street speculatively developed in the 1930s that has a strong sense of unity and aesthetic harmony due to the close stylistic relationships of the cottages. Collectively the houses are important as a group of modest bungalows reflecting popular taste of the late Interwar period with a high degree of consistency in form and fabric, and displaying fine brick detailing. The Crescent is an example of a cul-de-sac subdivision, a popular form of subdivision for capitalising on larger remnant allotments during the later Interwar period.

Horsley Avenue is an intact example of a late 1920s residential development in near original condition. Few additions or irreversible alterations have occurred to the original housing stock, and its coherent character is retained. The area has a uniformity of housing style and setting including form, materials, detailing, fencing and setbacks that gives the area a harmonious appearance. The predominance of the 1920s bungalow type illustrates the important influence of American housing ideals and styles on Australia.

3.8.5 Key period of significance

The key period of significance is 1926 to1940.

3.8.6 Management policies

a. Retain the dominant Bungalow character of Horsley Avenue by maintaining all original 1920s –1930s buildings and their consistency in scale, style, detailing, spacing, setbacks and subdivision pattern.

b. Retain the individual character of Summerville Crescent by retaining original face brick bungalows with their individual brick detailing. Any future development must take the form of single storey additions to the rear or side which do not dominate the original building stock.

c. Any new development (including carparking structures) must retain the principal elevations of the bungalows and be set back behind the main body of the original building envelope.

d. Maintain the avenue character by retaining the street trees and largely original style front gardens.

3.8.7 Controls for future development – Horsley Avenue

a. Scale	Generally:	Uniform low domestic scale, detached bungalows.
	1. Storeys	1 (attic within roof forms acceptable).
	2. Floor space ratio	0.4:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Consistent grid layout, lots average 550sqm, 15m frontages.
	1. Setbacks	Uniform front setbacks: 6m Variable side setbacks: 0.9 –3m
	2. Landscaping	Open well vegetated informal front gardens with lawns and low exotic shrubbery.
	3. Carparking	Single low scale rear or side garages accessed by side driveways.
	4. Front fencing	Low unpainted face brick fencing.
c. Form/ massing	1. Roof	Dominant broad simple low roof, predominantly hipped, some street facing gables and parapeted gables.
	2. Façade	Typically asymmetrical stepped façades, with protruding bays, recessed porches and verandahs.
	3. Building envelope	Roof pitch: 25 –30° Eaves height: 3.5m from floor level Ridge height: 6 –6.5m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and to the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP.

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<tr>
<td></td>
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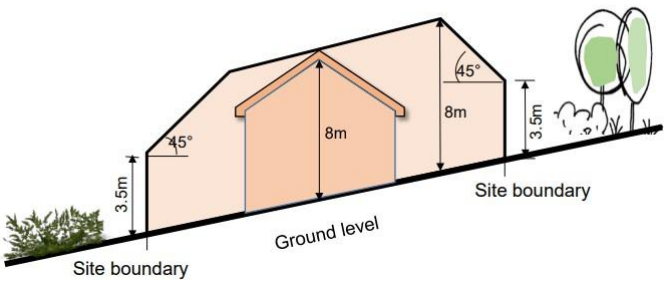
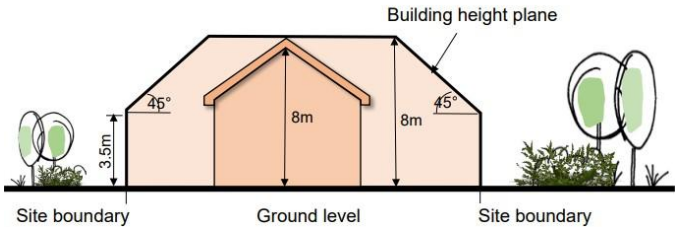
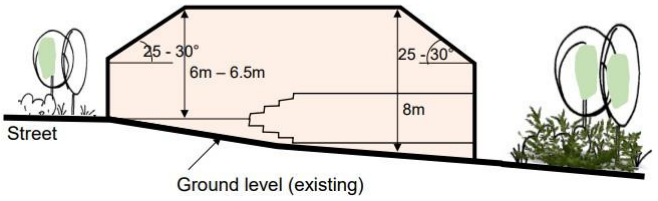
<td>
(a) Building Envelope on Sloping Site


(b) Building Envelope on Flat Site


(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height
Roof pitch: 25 – 30°
Eaves height: 3.5m from floor level
Ridge height: 6 – 6.5m from floor level


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<tr>
<td colspan="3">
Notes:

The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).

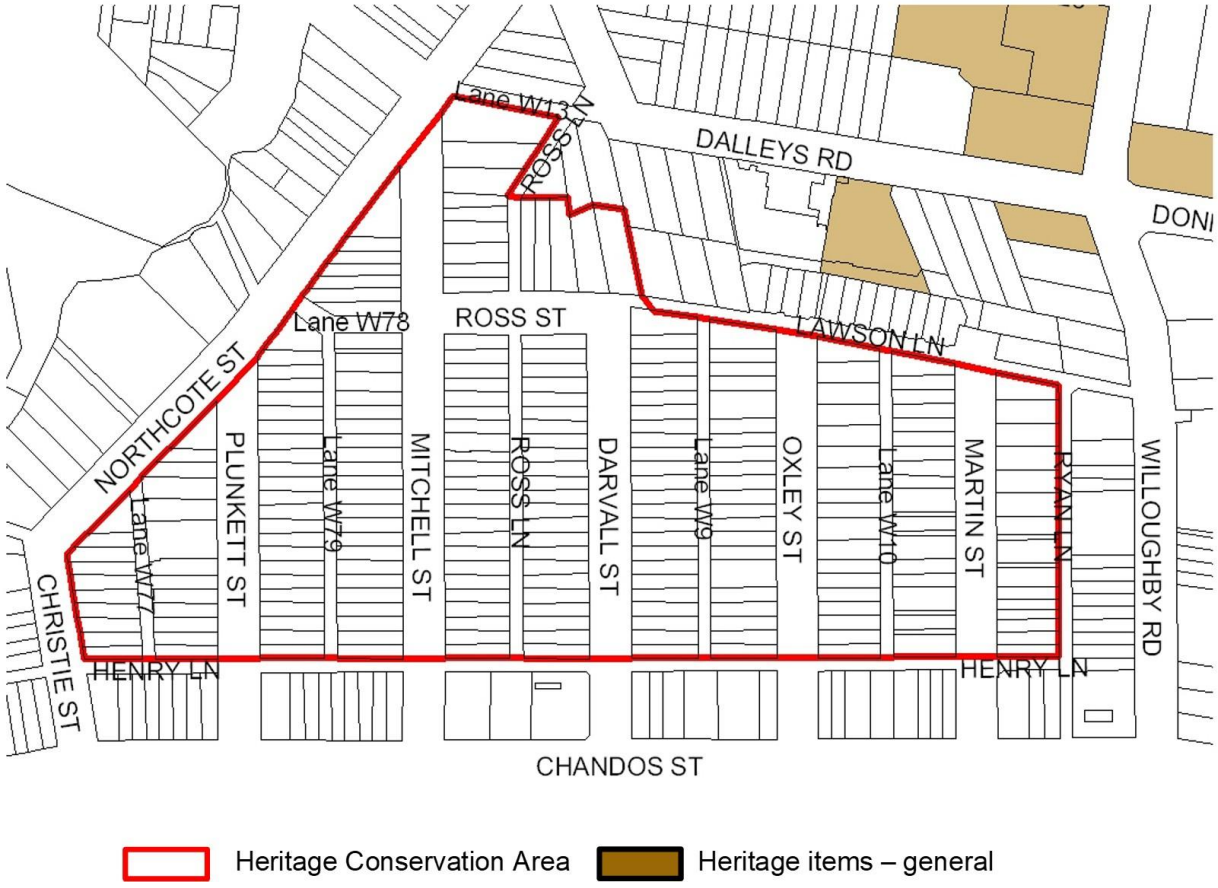
</td>
</tr>
<tr>
<td rowspan="2">d. Materials/ details</td>
<td>1. Roof</td>
<td>Terracotta Marseilles pattern tiles, exposed eaves.</td>
</tr>
<tr>
<td>2. Walls</td>
<td>Liver or dark red/brown smooth face brick: all unpainted face brick to be retained, render or sandstone base coursing.</td>
</tr>
</table>

	3. Windows and doors	Large bays/boxed timber casement windows in groups of 3 to street with skirts and awnings, single leaf timber panelled glazed entry doors, some leadlight to the street façades.
	4. Joinery and decoration	Simple vertical battening to gable ends, decorative brick detailing to street façades, some sandstone trim, massive battered verandah columns in brick, stone or roughcast.

3.9 Naremburn Heritage Conservation Area

3.9.1 Location

Figure 33 Map of Naremburn Heritage Conservation Area



3.9.2 History

The Naremburn Heritage Conservation Area lies on land previously belonging to Thomas Broughton. Broughton's Estate was developed after his death in 1901. Many of the streets in the locality are named after friends of Thomas Broughton including John Bayley Darvall and Sir James Martin. The extension of the tramline along Willoughby Road to Willoughby in 1898 prompted the development of the area.

The area was subdivided in 1903 and was known as the 'Crows Nest Subdivision'. The subdivision set out a uniform grid pattern of very small, narrow frontage allotments of about 250 square metres in area. The area developed quickly, largely because of its proximity to the city of Sydney, which at that time was reached by tram and ferry.

The suburb boomed until the 1930s from which point a decline set in. By the 1950 the entire heritage conservation area and land south of Chandos Street was zoned light industrial for redevelopment in the County of Cumberland Planning Scheme, reflecting low community esteem for the area. Redevelopment did not occur and the subsequent 30 years saw a complete turnaround in the status and public perception of the area with the National Trust of

Australia (NSW) recognising the importance of the area by classifying it as an urban conservation area in 1983.

3.9.3 Description

The nature of this development is reflected in the consistency of the building stock which primarily comprises rare, single fronted, detached Federation style brick cottages. Wider fronted cottages (though still small) on amalgamated lots, and some timber construction can be found scattered throughout the otherwise homogeneous area.

The cottages directly address the street from narrow front setbacks behind small front garden plots. Individual streetscapes vary in the blending of built and landscaped elements. Martin Street's wider level carriageway and the general lack of street tree planting creates an urban landscape, dominated by buildings. In Plunkett, Darvall and Oxley Streets, established street trees in generally wider verges are often complemented by more densely planted garden plots and more sloping landforms, attractively blend buildings and vegetation. Darvall Street's rows of towering mature *Phoenix canariensis* palms provide a dramatic and decorative setting for the tiny cottages. Rear service lanes originally used for sanitary collection now provide vehicular access.

3.9.4 Statement of significance

This is a unique, cohesive and generally intact Federation era suburb which has remained so largely because of the small size of the cottages and the restrictive subdivision pattern. The closely spaced cottages create a set of similarly patterned and cohesive streetscapes – uninterrupted vistas of almost uniform roof shapes and similarly massed façades in original materials. These small buildings have aesthetic significance for their shared Federation style characteristics and the individual expression of these in a wealth of original detailing, generally in good repair. Their turned and fretted woodwork, strapwork, shingled and pebbledash gable infills and coloured glazing distinguishes them individually and further integrates the group. The area has historic and aesthetic significance as a highly intact precinct of working class cottages of Federation period developed in response to the tramline extension to Willoughby.

3.9.5 Key period of significance

The key period of significance is 1903 to 1915.

3.9.6 Management policies

a. Retain the original building stock.

b. Retain the homogeneous Federation streetscape characterised by uniform small, low domestic scale single fronted cottages, with regular setbacks and spacing.

c. Alterations, additions and infill development must not result in excessive site cover that eliminates all useable landscaped area and private open space between the main dwelling and the rear lane or rear boundary.

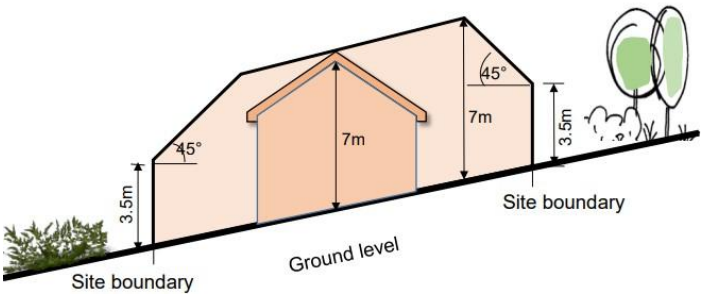
d. Second storey additions are inappropriate, and attic rooms will only be permitted at the rear of existing cottages where the original principal elevations can be maintained.

e. Rear lane vehicular access must be retained by any alterations and additions.

f. Infill development must retain the predominant cottage forms on the street elevations.

g. Any new development which abuts the heritage conservation area must provide an appropriate interface between the commercial and residential precincts through its scale, form, materials and landscaping.

3.9.7 Controls for future development – Naremburn

a. Scale	Generally:	Uniform low scale detached cottages.
	1. Storeys	1 (attic only where streetscape appearance unaltered).
	2. Floor space ratio	0.65:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Homogenous narrow fronted lots Average: 250m^2.
	1. Setbacks	Uniform front: 2.5 –4m side: 0.9 –3m
	2. Landscaping	Small front garden plots.
	3. Carparking	Rear lane access only.
	4. Front fencing	Low unpainted face brick, timber picket.
c. Form/ massing	1. Roof	Simple hipped roof forms with single offset gable or gablet to street.
	2. Façade	Simple, single fronted, narrow verandah Offset entry, some offset projecting gabled bays.
	3. Building envelope	Roof pitch: 30 –35° Eaves height: 3.5m from floor level Ridge height: 5.5m from floor level Building height: 7m from natural ground level to the vertical distance between ground level (existing) and to the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP. **(a) Building Envelope on Sloping Site** 

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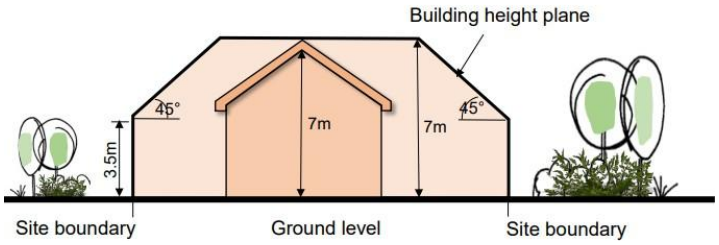
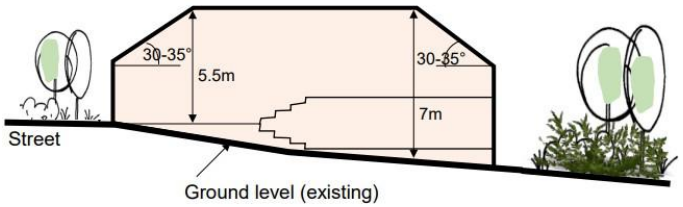


(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height
Roof pitch: 30-35°
Eaves height: 3.5m from floor level
Ridge height: 5.5m from floor level


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<tr><td colspan="3">Notes:
• The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
• The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
• There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).</td></tr>
<tr><td rowspan="4">d. Materials/ details</td><td>1. Windows and doors</td><td>Simple narrow timber casements or double hung, may include coloured glazing.
Half glazed panelled timber doors with coloured glass high lights.</td></tr>
<tr><td>2. Roof</td><td>Unglazed terracotta Marseilles tiles, corrugated metal sheet, or slate; exposed eaves.</td></tr>
<tr><td>3. Walls</td><td>Red smooth face brick: all unpainted face brick to be retained, painted weatherboard, roughcast or shingle gable infill.</td></tr>
<tr><td>4. Joinery and decoration</td><td>Simply patterned timber bargeboards, finials, verandah posts, gable brackets, decorative gable infills may include ‘sunburst’ motif.</td></tr>
</table>

Figure 34 Illustration of local streetscape in Naremburn

3.10 Naremburn Central Township Heritage Conservation Area

3.10.1 Location

Figure 35 Map of Naremburn Central Township Heritage Conservation Area

3.10.2 History

Naremburn Central Township comprised two Crown Grants of about six acres each, made in 1853 and 1854 to a Dugald MacPherson. The original Central Township was bounded by Brook Street, Garland Road, Central Street and Adolphus Street East, although properties on the northern side of Slade Street were also included in the early development. Residential development on Adolphus Street occurred concurrently with this commercial development. As

the first settlement in the Naremburn area, the township was expected to become the commercial centre of the north side. The development of this area can be attributed to its close proximity to the city, which at that time was reached by ferry. In the early 1880s, Market Street had a number of shops, including a cabinet factory, post office, newsagent, butcher and a market site, several of which survive. Many small businesses and home industries thrived until 1898 when the tramline from Crows Nest to Willoughby took trade elsewhere. Since the construction of the Warringah Freeway in the 1970s, Naremburn Central Township has been isolated from the rest of the suburb.

3.10.3 Description

None of the earliest housing of wattle and daub or bark remains, but some stone construction and many simple Georgian and Victorian weatherboard cottages from the 1880s –1900s survive. The streetscapes are now dominated by modest Federation cottages, California Bungalows and semis, interspersed with a few grand late Victorian 2 storey residences. There are some Interwar flat buildings on the corner of Central and Slade Streets.

Situated on a gently sloping plateau that falls away steeply to the north and east, the subdivision pattern is fairly irregular with many narrow allotments to Slade, Adolphus, Market and Probate Streets and larger allotments to Garland Road and Central Streets.

3.10.4 Statement of significance

The subdivision pattern, together with the wide range of architectural styles represented, reflects the continuing incremental growth of the township from the mid-1800s through to the 1930s and has created a mixed yet harmonious streetscape character. The area has retained its historical and social significance as the oldest commercial development on the North Shore and as an early working class residential area. The wide range of buildings, including some shops and the modest streetscape character are evidence of this.

3.10.5 Key period of significance

The key period of significance is 1880 to 1930.

3.10.6 Management policies

a. Retain the historical and social significance of the area as an early commercial centre and working class residential area by retaining the existing streetscape character and contributory individual buildings.

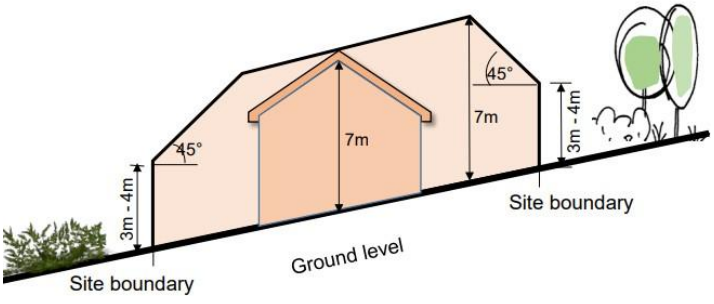
b. Examples of early residences and places of business are to be retained.

c. Additions must respect the scale and form of existing buildings, and the modest character of the general development.

d. Infill development must respect the existing development patterns and modest character of the heritage conservation area, in terms of typical lot size, dominant building setbacks, scale, character and form.

e. Carparking structures must not be dominant in the streetscape.

3.10.7 Controls for future development – Naremburn Central Township: R2 Residential

a. Scale	Generally:	Small domestic scale cottages, semi-detached and terraces, (some flat buildings).
	1. Storeys	1 (2nd storey located within roof) 2 storey flat buildings in established locations.
	2. Floor space ratio	0.5:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Fairly irregular, generally narrow allotments, with large lots to Garland Road and Central Street.
	1. Setbacks	Variable, consistent within groups of houses.
	2. Landscaping	Small front gardens with low shrubbery.
	3. Carparking	Generally on street.
	4. Front fencing	Low unpainted face brick, timber picket, & timber and wire.
c. Form/ massing	1. Roof	Simple hipped and gabled roof forms.
	2. Façade	Mostly simple asymmetrical stepped façades.
	3. Typical building envelope	Roof pitch: 25 –35 Eaves height: 3 –4m from floor level Ridge height: 5 –6m from floor level Building height: 7m from natural ground level to the vertical distance between ground level (existing) and to the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP. **(a) Building Envelope on Sloping Site** 

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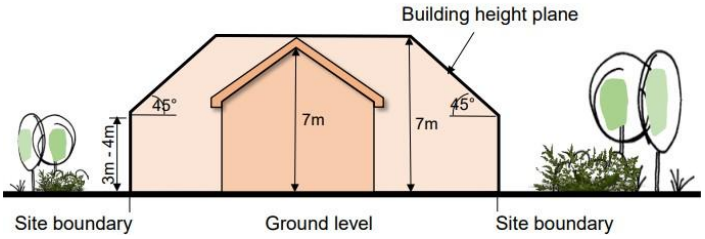
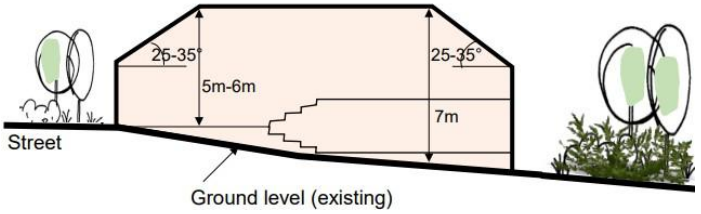


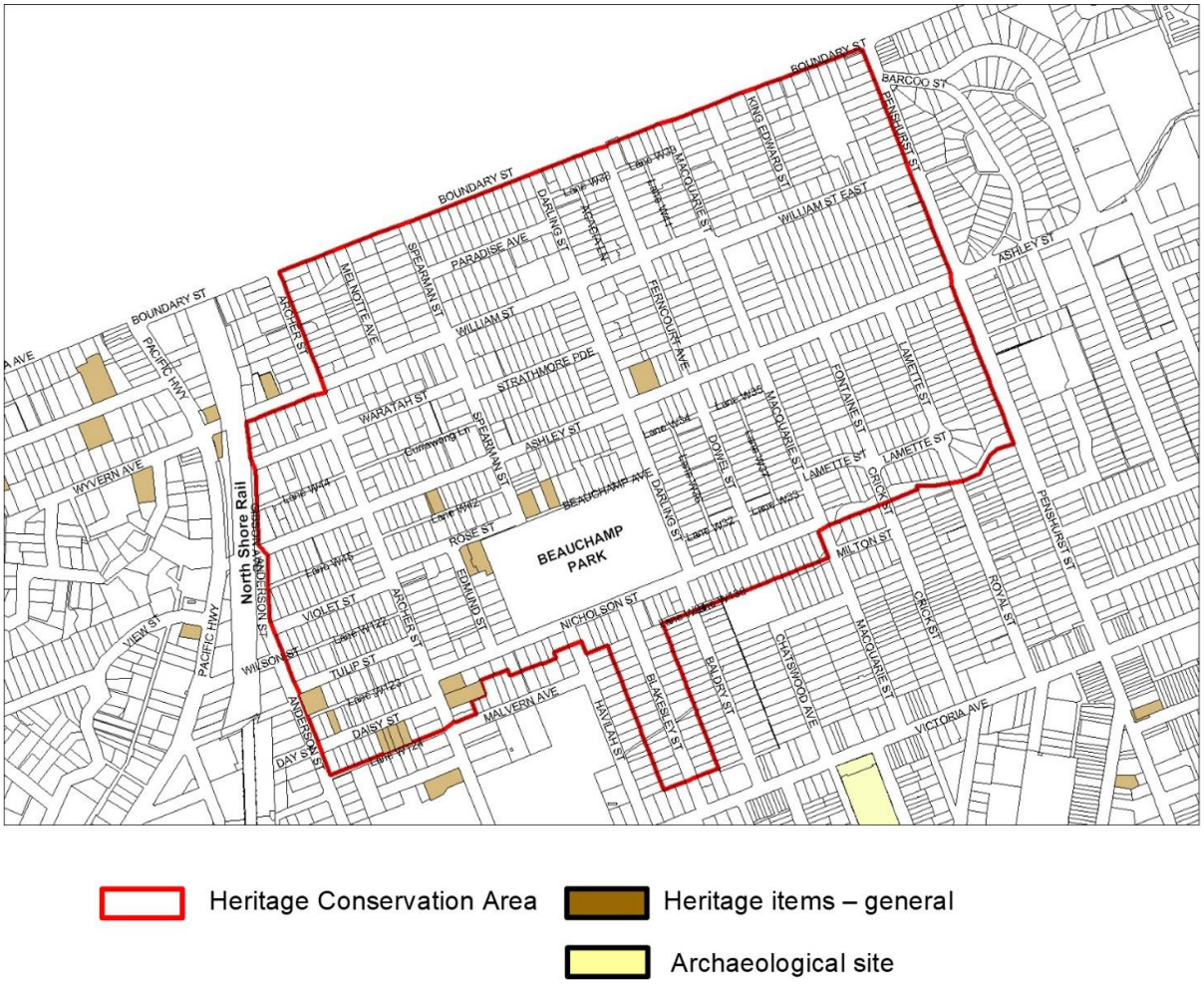
(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height
Roof pitch: 25-35°
Eaves height: 3m-4m from floor level
Ridge height: 5m-6m from floor level


</td></tr>
<tr><td colspan="3">Notes:
• The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
• The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
• There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).</td></tr>
<tr><td rowspan="4">d. Materials/ details</td><td>1. Roof</td><td>Unglazed terracotta Marseilles tile, corrugated metal sheet.</td></tr>
<tr><td>2. Walls</td><td>Rendered masonry, weatherboard, or face brick: (all unpainted face brick to be retained), often with stone or render base courses.</td></tr>
<tr><td>3. Windows and doors</td><td>Timber casement or sash,
Single leaf timber panelled entry door.</td></tr>
<tr><td>4. Joinery and decoration</td><td>Simple and modest.</td></tr>
</table>

3.11 North Chatswood Heritage Conservation Area

3.11.1 Location

Figure 36 Map of North Chatswood Heritage Conservation Area



3.11.2 History

In 1825, Richard Archbold was granted 600 acres of Crown land approximately bounded by Victoria Avenue, Boundary Street, Anderson Street and High Street. The land reverted to the Crown and lay dormant until 1850 when it was sold to John Stirling of the Bank of Australasia. William Lithgow, retired Auditor General of Colonial Accounts and a shareholder of the Bank, purchased and subdivided the land, and put 400 acres up for sale in 1854 as part of the 'Township of North Sydney' plan. Sales were slow and it was not until the later decades of the 19th century, with news of the railway line, that development took off.

The 1880s was a decade of vigorous land subdivision in Chatswood, and with the arrival of the railway in 1890, high class building activity flourished in the area. Beauchamp Park was proclaimed in 1899, and was named after the Governor of NSW, William Lygon, the 7th Earl of Beauchamp. After the First World War, a further wave of subdivisions and building activity added to this stock of grand Federation homes set on large allotments.

In the 1950s Boundary Road was realigned and properties fronting the road were reduced. At this time the Chatswood CBD was proposed to extend east to Neridah Street and north almost to the alignment of Kirk Street. The National Trust of Australia (NSW) recognised the importance of the area by classifying it as an urban conservation area in 1982.

3.11.3 Description

North Chatswood's original dwelling stock is predominantly single storey detached housing in Federation or Interwar bungalow styles. However, there are a few two storey Federation mansions, the occasional late Victorian house and some semi-detached 'bungalows'.

Lot sizes vary, although most streets have similar width frontages. Buildings are usually consistently spaced and of similar scale and bulk within each streetscape even where this is composed of varied styles. Mature landscaped gardens are an important quality of this area.

The grid pattern subdivision is laid over gentle to moderate slopes. Most streets have well-established street trees, notably brush box and canary island palms, which enhance the high residential amenity of the orderly streetscapes. The 'square' formed by Beauchamp Park is a focus in the townscape.

The form of original development in streetscapes near Chatswood CBD is generally maintained where rear lanes accommodate separate access to off-street car parking. Otherwise, onsite parking facilities are usually unobtrusive, since the spacing of houses typically allows side access to rear parking.

3.11.4 Statement of significance

North Chatswood Heritage Conservation Area is a good example of early North Shore residential development. The housing stock describes the progressive overlays of development taking place over half a century. The essential scale, form and spacing of the original dwellings is predominant, even where original architectural detailing has been lost, though much of this is still intact in fine residential buildings and as a general townscape impression.

3.11.5 Key period of significance

The key period of significance is 1880 to 1930.

3.11.6 Management policies

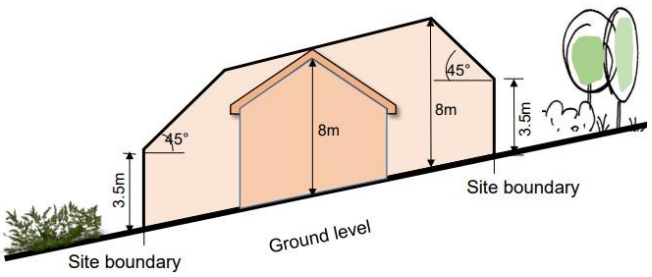
a. Retain the scale, form, massing and detailing of original buildings and the typical scale, rhythm and character of individual streetscapes.

b. Retain the predominant single storey scale of the heritage conservation area by preventing visible two storey additions or intrusive two storey infill development.

c. Retain landscaped settings by minimising hard surfaces forward of the building line, locating carparking structures behind the building alignment, and maintaining the existing patterns of low/semi-transparent fencing to the street.

d. Infill development must respect existing patterns of development including the spacing, rhythm and siting of built forms, the landscaped settings, and the scale massing, form and materials of the general development.

3.11.7 Controls for future development – North Chatswood: R2 Residential

a. Scale	Generally:	Low/medium residential scale, detached.
	1. Storeys	Generally 1 (2nd storey within roof forms acceptable).
	2. Floor space ratio	0.4:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Variable lot sizes, but mostly similar frontages.
	1. Setbacks	Consistent with patterns of individual streetscape.
	2. Landscaping	Mature well vegetated front garden with exotic shrubs and trees.
	3. Carparking	Rear lane accessed garaging, or carports to side of dwellings behind the building alignment, else single open stand area forward of building line.
	4. Front fencing	Low unpainted masonry or timber picket, palisade or timber and wire.
c. Form/ massing	1. Roof	Dominant hipped and gabled roof forms with flat low/ pitch or integral verandas.
	2. Façade	Predominantly stepped massing, usually asymmetrical with projecting bays & porches.
	3. Building envelope	Roof pitch: 25 –35 Eaves height: 3. 6m from floor level Ridge height: 6 –6.5m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and to the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP. **(a) Building Envelope on Sloping Site** 

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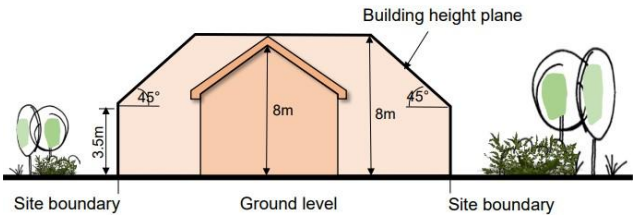
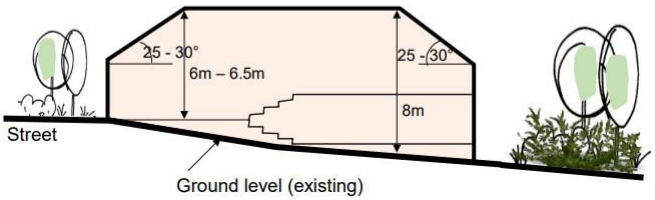


(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height
Roof pitch: 25 – 30°
Eaves height: 3.5m from floor level
Ridge height: 6m – 6.5m from floor level


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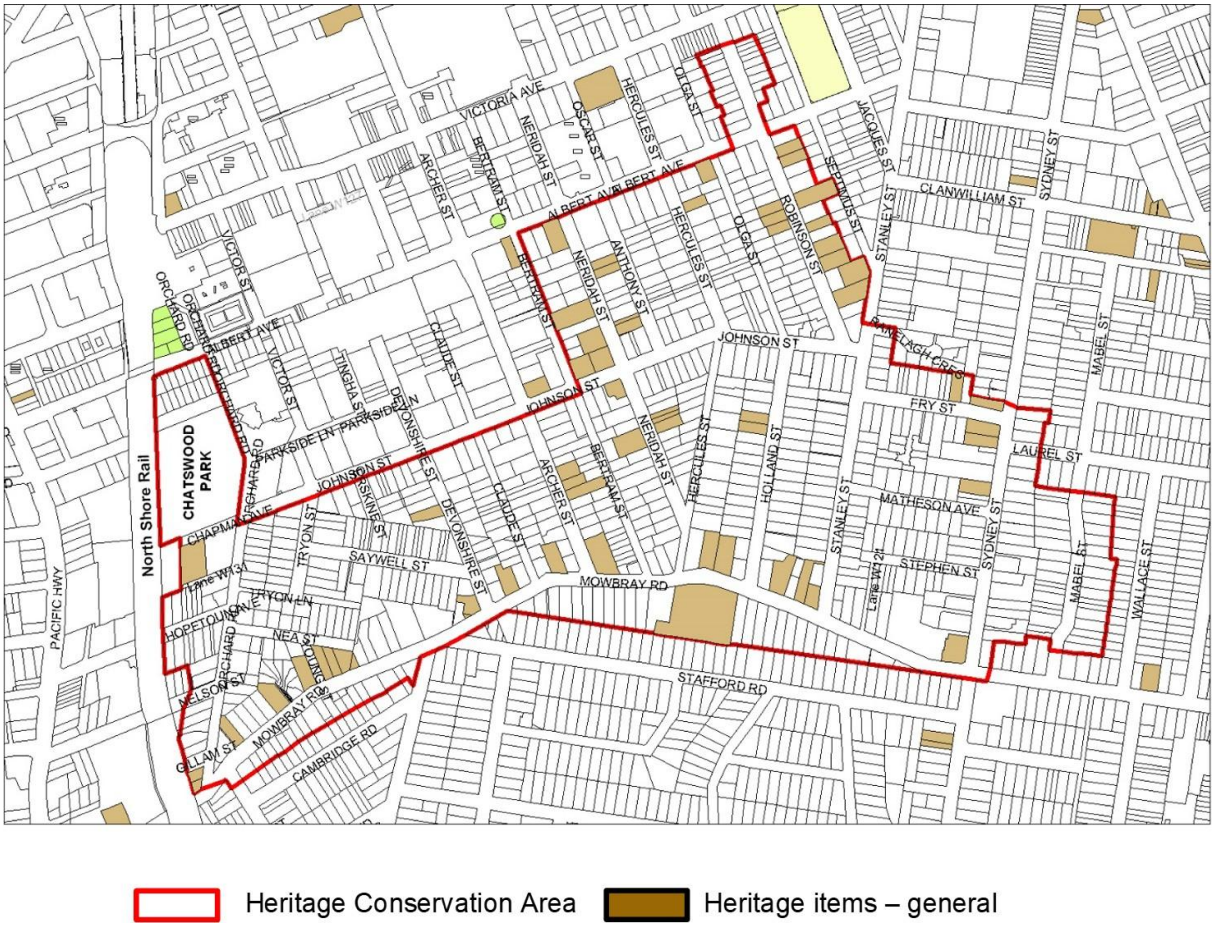
<tr><td rowspan="4">d. Materials/ details</td><td>1. Roof</td><td>Unglazed terracotta Marseilles tile, slate.</td></tr>
<tr><td>2. Walls</td><td>Red/ brown monotone smooth face brick: all unpainted face brick to be retained, render/ stucco to upper surfaces, render/stone to base course.</td></tr>
<tr><td>3. Windows and doors</td><td>Simple painted timber casement or sash windows often in bay groupings to street façade.</td></tr>
<tr><td>4. Joinery and decoration</td><td>Varied; simple timber verandah and gable end decoration.</td></tr>
</table>

Figure 37 Illustration of local streetscape in North Chatswood

3.12 South Chatswood Heritage Conservation Area

3.12.1 Location

Figure 38 Map of South Chatswood Heritage Conservation Area



Heritage Conservation Area Heritage items – general

3.12.2 History

In 1840, Governor Sir George Gipps proclaimed a grant of 160 acres to James Blackett and Richard Read, in trust for Maria Brown. This land was bounded roughly by the Artarmon Estate to the south, Victoria Avenue to the north, Stanley Street to the east and Devonshire Street to the west. The area east of Stanley Street was part of an earlier 1825 Crown Grant to Richard Archbold. This land was bought and subdivided by William Lithgow and put up for sale in 1854 with limited success.

Urban development commenced in earnest in the 1870. With the news of the impending arrival of the railway, the 1880s was a decade of vigorous land subdivision in Chatswood. High class building activity flourished in the area. Richard Hayes Harnett's 'Willoughby Park' Estate which extended from Mowbray Road to Victoria Avenue and from Devonshire Street to Stanley Street was offered for sale in 1884. Development occurred on a succession of major streets crossing Johnson, Street: Devonshire, Archer, Neridah, Hercules, and Robinson. The allotments were divided into generous blocks, and were developed with residences in the late Victorian, Federation and California Bungalow styles. The lesser streets in between (Claude,

Bertram, Anthony, Oscar, Olga, and Septimus) provided service and rear lane access. The area west of Devonshire Street included ‘Mackenzie's Subdivision’ (Nea and Orchard Road), ‘Orchard Road Estate’ (Orchard Road, Tryon, and Johnson Streets) and Saywell ‘Willoughby Heights Estate’ which were all subdivided in the 1880s.

After the First World War, a further wave of building activity added to this stock of grand Federation homes set on large allotments. Bales Park, formerly Central Park, is an important focus for the South Chatswood precinct. Planned by the Council in 1928 on the site of a Chinese market garden, land was resumed for the park in 1930 and in the 1950s. In the 1940s and 1950s many rear gardens were subdivided and developed. In the 1950s, the town centre was proposed to extend east to Neridah Street and south to Albert Avenue.

The National Trust of Australia (NSW) recognised the importance of the area by classifying it as an urban conservation area in 1982.

3.12.3 Description

As a whole, South Chatswood has a mixed character: styles range up to California and Interwar bungalows, which are interwoven among the dominant Federation and late Victorian era buildings through the progressive resubdivision of what were often originally generous and gracious early estates.

As well as a few modest Victorian cottages, South Chatswood is remarkable for its concentration of imposing late Victorian mansions. Along Mowbray Road, their location on a prominent ridgeline and their well established gardens (including tall, highly visible exotic species), add to the landmark qualities of the more palatial buildings.

These typically two storey mansions and many later Federation grand villas, are usually listed as individual heritage items. They also have considerable significance in their own right in addition to the distinctive qualities they impart to the streetscapes.

There is great variety in the predominant Federation style from single storey semidetached cottages to rambling grand villas and rare two storey architect-designed mansions. Some streetscapes, as in Robinson and Neridah Streets, display a highly consistent Federation character, even where the scale and siting of buildings varies.

The streetscapes of South Chatswood are unified by the generous landscaped settings of both the modest cottages and mansions. These gardens are characterised by mature exotic shrubbery and lawns, and low fencing allowing leafy vistas to and from the houses.

3.12.4 Statement of significance

South Chatswood Heritage Conservation Area is a good example of early north shore residential development. Much of the dwelling stock retains its original detailing and distinctive architectural features. Most retain the original scale and basic form, which, apart from the two storey mansions, is predominantly single storey. The Mowbray Road streetscape is significant as one of Willoughby’s earliest streets and most important routes, and because of this significance it displays a range of substantial and high quality residences from the early phase of development of the area, complemented by later development of the Interwar period. The heritage conservation area displays a high level of amenity and originality in its development as an early residential suburb up to the Second World War.

3.12.5 Key period of significance

The key period of significance is 1880 to 1930.

Figure 39 Illustration of a local streetscape in South Chatswood

3.12.6 Management policies

a. Retain the scale, form, massing and detailing of original buildings and the typical scale, rhythm and character of individual streetscapes.

b. Retain the predominant single storey scale of the heritage conservation area by preventing visible two storey additions or intrusive two storey infill development.

c. Retain landscaped settings by minimising hard surfaces forward of the building line, locating carparking structures behind the building alignment, and maintaining the existing patterns of low/semi transparent fencing to the street.

d. Infill development must respect existing patterns of development including the spacing, rhythm and siting of built forms, the landscaped settings, and the scale massing, form and materials of the general development.

e. Protect the special significance of the Mowbray Road streetscape by retaining the fabric and enhancing the landscaped settings of the original development, and by ensuring that new development does not dominate or detract from this significance.

Development in the minor streets (Bertram, Claude, Anthony and Olga) are to be designed to be compatible with the South Chatswood Heritage Conservation Area in relation to:

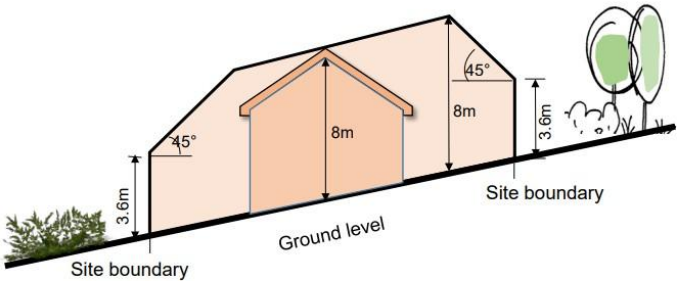
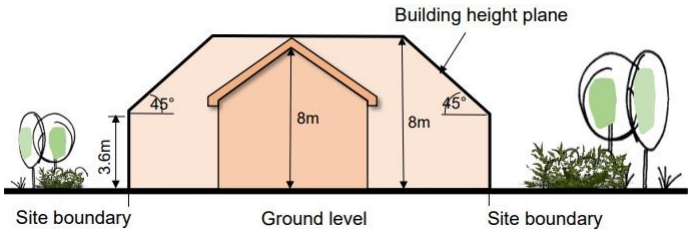
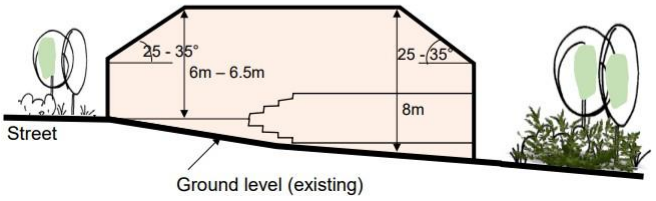
- The spacing of buildings and driveways
- Proportion of primary façade elements
- Form and pitch of roof elements
- The scale of the building form
- Massing of new building forms (particularly asymmetrical forms and room in roof forms)

- Single carport structures
- Use of materials in roofing front and side façades
- Height and materials of front fencing
- Predominantly soft landscaped front yard areas.

In these streets, the intent of this plan is that over time development will more closely reflect the historic character of the South Chatswood Heritage Conservation Area. The existing controls for future development guidelines apply to all streets in the heritage conservation area.

3.12.7 Controls for future development – South Chatswood

a. Scale	Generally:	Low/medium residential scale
	1. Storeys	1 (2nd storey within roof forms acceptable)
	2. Floor space ratio	0.4:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Variable lot sizes
	1. Setbacks	Consistent with patterns of individual streetscapes.
	2. Landscaping	Mature well vegetated front gardens with exotic trees and shrubs.
	3. Carparking	Use rear lane access or side access to garages and carports behind building line, open stand forward of building line.
	4. Front fencing	Low unpainted masonry, timber picket, palisade or timber and wire.
c. Form/ massing	1. Roof	Dominant hipped and gabled roof forms, verandahs to front elevations.
	2. Façade	Generally stepped massing with projecting bays, some simple symmetrical forms.
	3. Building envelope	Roof pitch: 25 –35° Eaves height: 3.6m from floor level Ridge height: 6 –6.5m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and to the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP.

		(a) Building Envelope on Sloping Site  **(b) Building Envelope on Flat Site**  **(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height** Roof pitch: 25 – 35° Eaves height: 3.6m from floor level Ridge height: 6m – 6.5m from floor level 
Notes: • The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site. • The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level. • There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the 'Height of Buildings Map' in accordance with Clause 4.3 under the WLEP).		
d. Materials/ details	1. Roof	Unglazed terracotta Marseilles tile, slate, corrugated metal sheet.
	2. Walls	Red/brown monotone smooth face brick: all unpainted face brick to be retained, render or stucco to upper surfaces, stone/render to base.

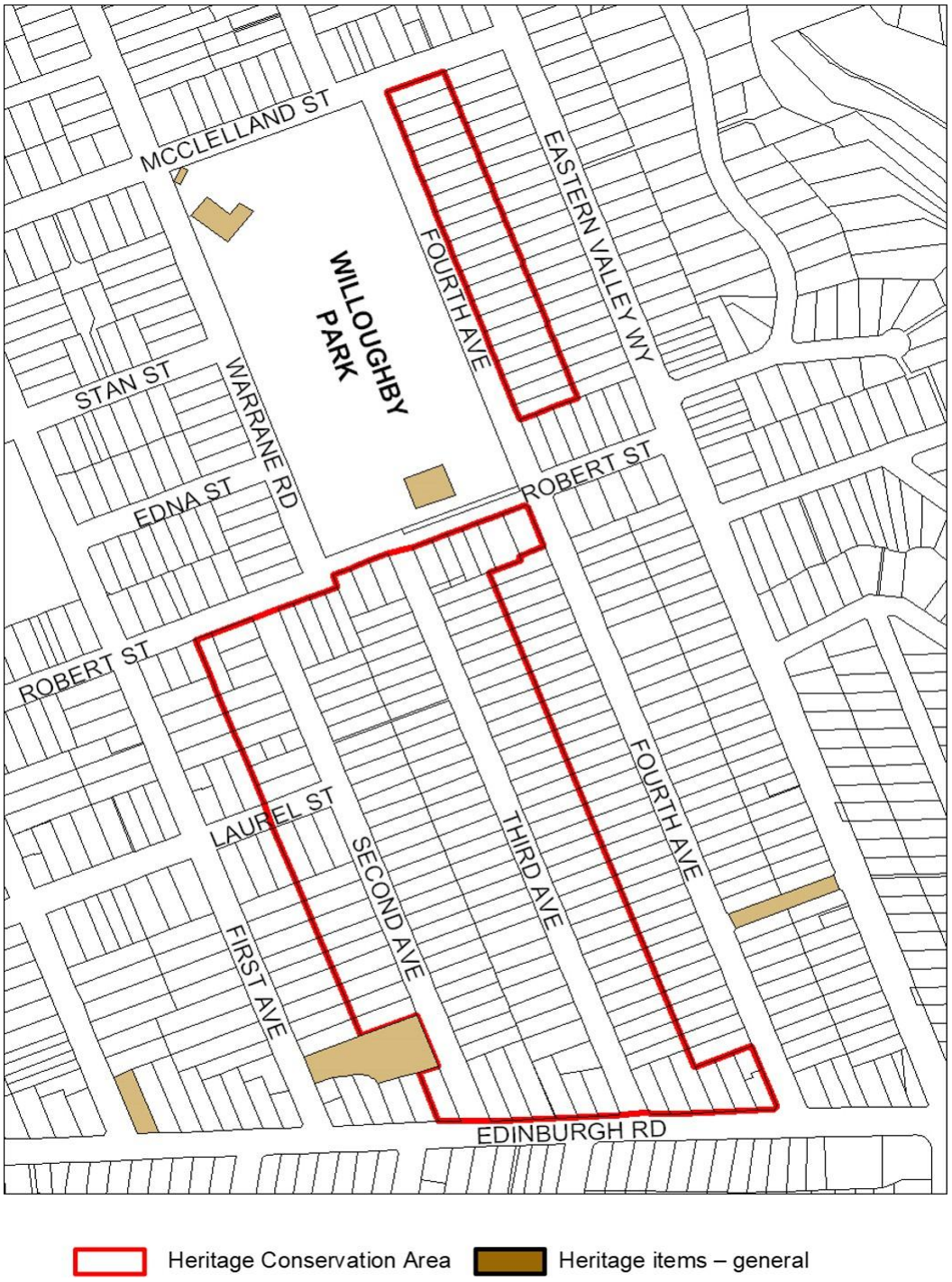
	3. Windows and doors	Timber casement and sash windows often grouped in bays to street elevation.

Figure 40 Further illustration of a local streetscape in South Chatswood

3.13 Willoughby Park Heritage Conservation Area

3.13.1 Location

Figure 41 Map of Willoughby Park Heritage Conservation Area



3.13.2 History

The Willoughby Park Heritage Conservation Area is situated on crown grants to Edward Henry Herring (1858), James Harris French (1857), and James William Bligh (1857). James Forsyth

set up ‘Rosewall Tannery’, Willoughby’s first tannery in Stan Street Willoughby in 1869. The site was suitable as there was a wattle grove where tanners where able to source wattlebark for their industry and creek providing water. James left the business to his sons Robert and Thomas in 1882. Robert and Thomas bought up the land in the vicinity of their tannery in the 1880s, and leased much of it to Chinese market gardeners, who cultivated the land along Sugarloaf Creek. The creek now runs in concrete pipes. The Forsyth family sold the tannery to Broomham Brothers in 1907. The NSW Government resumed the site in 1957 for use as a bus depot.

In 1907 the site of Willoughby Park was purchased for a park, following representations by ratepayers regarding the scarcity of parkland in the Ward. The park is a focus for the local community. The entrance gates came from ‘Penshurst’ the home of William Muston in Penshurst Street. ‘Greenacre’ now in Second Avenue but originally accessed from Edinburgh Road, was built in 1904 as a home for Professor Pitt Cobbett, first Professor of Law at the University of Sydney. Cobbett left Willoughby in 1910 –11, and Tresillian was established in ‘Greenacre’ in 1927. The land was subdivided and sold as the Greenacre, Thompson’s, Sunnybank, Forsyth’s and the Tyneside Estates.

The construction of the Eastern Valley Way as a relief project for the unemployed in the 1930s coinciding with the rebuilding of the Northbridge Suspension Bridge and the completion of the Harbour Bridge opened up the area for further residential development.

3.13.3 Description

The Willoughby Park precinct displays a consistency in housing stock and setting creating a community focus of great visual harmony. The housing comprises single storey bungalows most with one or two gables fronting the street, constructed in unpainted red brown face brickwork modulated by the use of areas of roughcast or shingles, and terracotta Marseilles pattern roof tiles. The houses are set in landscaped gardens with low fencing, allowing pleasant leafy vistas up and down the streetscapes and views to houses.

3.13.4 Statement of significance

The core of the Willoughby Park Precinct is an area of high amenity and visual harmony as well as historic and social value that provides an important community focus. Established in 1907, the Park has been the centre of many important community activities since the 1920s. The Interwar housing stock that surrounds the Park provides an appropriate setting. The rows of consistently pitched gabled roof forms set along these gently sloping streets create pleasant, rhythmic vistas, which, together with their consistent setbacks, the wide grassy verges and mature street trees and gardens creates a sense of spaciousness and design harmony. Parts of the area possess a streetscape integrity due to the development during one period and the excellent state of preservation of the houses and their garden settings. A uniformity of housing style including form, materials and detailing gives these streetscapes a harmonious appearance. The predominance of the 1920s bungalow type illustrates the important influence of American housing ideals and styles on Australia.

3.13.5 Key period of significance

The key period of significance is 1907 to 1930.

3.13.6 Management policies

a. Retain the coherent low scale California Bungalow style streetscape.

b. Preserve the outlook to Willoughby Park.

c. Retain Interwar buildings: original detailing should be retained or reinstated.

d. New development must be limited to extensions to the rear, and development predominantly contained within the original building envelope as seen on principal elevations.

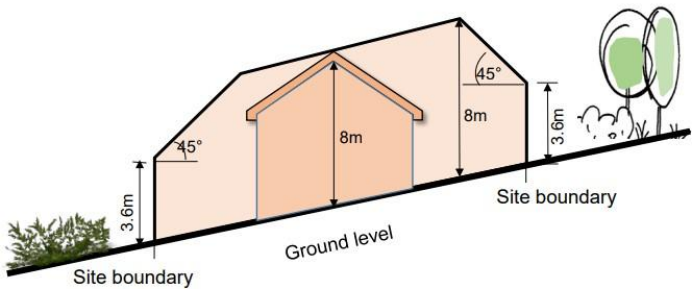
e. Retain existing side and rear parking spaces.

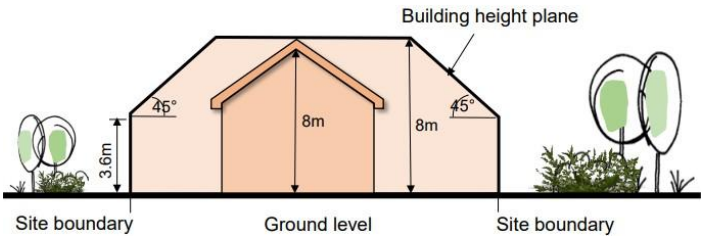
f. Where parking cannot be provided to the rear or side of the house, facilities are to be limited to single open stand areas, or lightly framed carports, designed and sited to be subordinate to the main building.

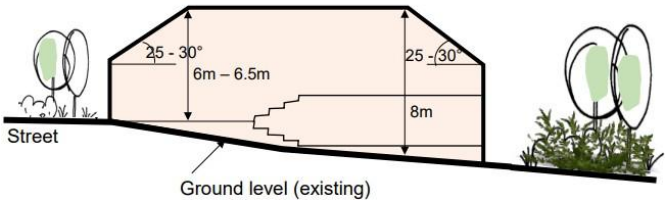
g. Retain the garden settings of the housing by retaining low front fencing and landscaped front gardens.

Figure 42 Illustration of a street front

3.13.7 Controls for future development – Willoughby Park

a. Scale	Generally:	Medium/low domestic scale detached bungalows.
	1. Storeys	1 (attic within roof forms acceptable).
	2. Floor space ratio	0.4:1: This figure represents an upper limit to floor space ratio (FSR). It takes the existing pattern of development into account and provides for some additional floor area without compromising the significance of the heritage conservation area. See Clause 4.4 (Floor space ratio) of Willoughby LEP.
b. Setting/ subdivision	Generally:	Consistent, generally small frontages of 12.5m, lots average 560m^2.
	1. Setbacks	Uniform front setback to 6.5m, side setbacks allow driveway or 900m.
	2. Landscaping	Small open front gardens, generally lawn with shrubs.
	3. Carparking	Side access where possible, single open spaces forward of building line.
	4. Front fencing	Low unpainted brick, sandstone, or timber and wire.
c. Form/ massing	1. Roof	Dominant broad simple gables (1 – 3) fronting street, hipped to rear.
	2. Façade	Large flat unadorned planes, simple asymmetrical massing with verandah and protruding gables to street façade.
	3. Building envelope	Roof pitch: 23 – 30° Eaves height: 3.6m from floor level Ridge height: 6 – 6.5m from floor level Building height 8m from natural ground level to the vertical distance between ground level (existing) and the highest point of the building in line with Clause 4.3 (Height of buildings) of Willoughby LEP. **(a) Building Envelope on Sloping Site** 

<table>
<tr><td></td><td></td><td>
(b) Building Envelope on Flat Site


(c) Building Height with Heritage controls relating to roof pitch, eaves height and ridge height
Roof pitch: 25 – 30°
Eaves height: 3.6m from floor level
Ridge height: 6m – 6.5m from floor level


</td></tr>
<tr><td colspan="3">Notes:
• The building envelopes for (a) and (b) are determined by planes projected at 45^{O} from a height of 3.5m at the side boundaries of the site.
• The building envelope for (c) is determined by the eaves and ridge heights above the ground floor level.
• There are variable maximum height limits for dwellings in the Heritage Conservation Areas (the maximum height for a specific site can be identified on the ‘Height of Buildings Map’ in accordance with Clause 4.3 under the WLEP).</td></tr>
<tr><td rowspan="4">d. Materials/ details</td><td>1. Roof</td><td>Unglazed terracotta Marseilles tile, metal or bitumen on flat roofed areas, exposed eaves.</td></tr>
<tr><td>2. Walls</td><td>Dark red or liver face brick: all unpainted face brick to be retained, rough cast to upper wall surfaces.</td></tr>
<tr><td>3. Windows and doors</td><td>Casement windows typically in groups of 3, some boxed frames with skirts or awnings, single panelled or glazed entry doors.</td></tr>
<tr><td>4. Joinery and decoration</td><td>Simple strapwork and timber shingle gable infill, heavy brick/timber verandah detailing, some roughcast/ contrasting brick trim.</td></tr>
</table>

Attachment 1: Design principles for the conservation of environmental heritage

This attachment outlines the design principles for the conservation of environmental heritage. It supplements Willoughby LEP and Willoughby DCP, Part H: Controls for Heritage Items and Heritage Conservation Areas.

1 Conservation principles

The fundamental objectives applying to all heritage items and heritage conservation areas are to ensure that:

- the fabric which makes a place significant is conserved
- new development is sympathetic.

In simple terms, conservation is all about extending the existence or usefulness of resources and thoughtful maintenance is a key to successful conservation.

The approach to conservation is based on principles set out in a document known as the ‘Burra Charter’. This Charter establishes the nationally accepted standard for the Conservation of places of cultural significance. These principles assist in deciding what is significant, and what works are appropriate.

Article 1 of the Burra Charter defines conservation as 'all the processes of looking after a place so as to retain its cultural significance. It includes maintenance and may according to circumstance include reconstruction and adaptation and will be commonly a combination of more than one of these'.

1.1 The importance of fabric

The ‘fabric’ of a building or heritage conservation area refers to the physical material of which it is comprised. To varying degrees the fabric represents and expresses the history and significance of the place. Thus when original fabric is removed from a heritage building or a heritage conservation area, the significance of that place is diminished.

Careful judgement is also needed to avoid damaging external wall surfaces by trying to reverse old changes; some of these may also prove to be valuable evidence of a building's history. For example, while it may be appropriate and relatively easy to remove an unsympathetic 1960s verandah enclosure comprising aluminium windows and fibro construction from a standard 1920s California Bungalow. But an earlier verandah enclosure with 1930s detailing, such as leadlight windows and more solid construction, may be significant and could be worth retaining.

Beware of irreversible changes to the fabric of a building such as painting brickwork. These changes diminish the value of the property. People can become more aware of what is authentic and what is not and are often willing to pay for the more intact original property. It can also lead to long term increased maintenance. More appropriate ways of reviving the appearance of a building are generally possible.

The three preliminary steps to be taken before beginning any conservation work are:

- research
- professional advice
- attention to detail.

1.1.1 Step 1: Research

In the case of older buildings, knowing what not to do is as important as knowing about what has to be done.

Looking at similar buildings in your locality can help in working out why things were done the way they were or seeing what decayed or missing parts of a building may have looked like.

Careful inspection of the building itself can reveal evidence of original detailing. For example, a verandah may have been removed, holes in walls may indicate a former handrail, and paint lines may reveal the profile of a moulding or the shape of a former corrugated iron roof and nail holes may show the sites of cast iron brackets or timber friezes. Carefully scraping protected or difficult-to-paint areas, (such as under window sills and in tight corners,) can reveal original colour schemes.

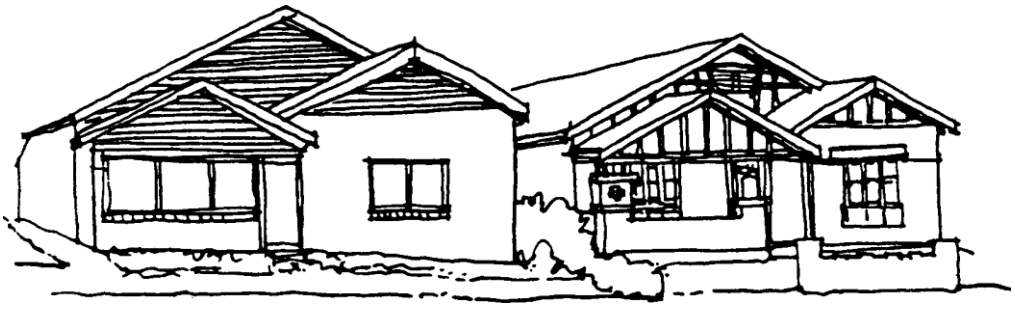
Attachment 1 Part 3 Useful publications and references – provides a sample of publications and websites which can help owners of buildings to learn about their design and significance, construction materials and methods, and conservation techniques.

Documentary research through archives, such as historical societies or Council records, may locate an old photograph, title deeds may indicate the building's layout over time, or approaches to previous owners can also help in seeing what is original and identifying the significant building fabric.

Maintenance, repair and restoration work can then be carried out with the objective of conserving the maximum possible amount of this fabric.

Remember – retaining the original is preferable to replacement with reproduction.

Figure 43 Illustration of streetscape



1.1.2 Step 2: Professional advice

Obtain advice from suitably qualified professionals, who are skilled in heritage work. Do not rely on guesswork – seek professional advice whenever you have any real doubt. A project which involves considerable expenditure and effort should start with sound research and advice and conclude with confident and competent execution.

Heritage NSW has a ‘Heritage Consultants Directory’ at heritage.nsw.gov.au

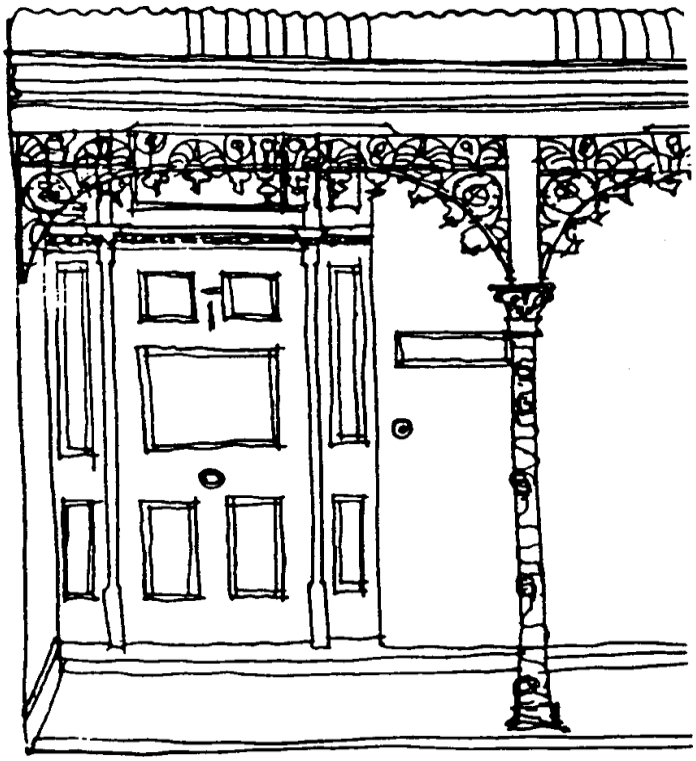
1.1.3 Step 3: Detail

A building is designed with a consistent approach to detail, and it is important to first understand this approach. New work should be relevant to the design of the building or place; it is not appropriate to imitate something from another era, place or site.

Maintain existing features. Adding decorative elements such as false shutters and aluminium lace detracts from the significance and value of the property. Remember the original absence of decorative elements can be distinguishing. Example include on Georgian buildings or austere interwar bungalows.

Wherever possible, reinstate missing components, such as joinery, verandahs, chimneys and the like but do not add what was never there. A 1910s wooden picket fence is not appropriate to a 1930s house: this sort of detail diminishes the significance and value of both the individual item and of the streetscape.

Figure 44 Illustration of heritage frontage



1.2 Maintenance and restoration

The first principle of conservation is **maintenance** – looking after buildings and places as and when they need it. Maintenance and repairs assist in ensuring the ongoing significance of a building and enhances its longevity. Appropriate maintenance is perhaps the most influential aspect of conservation.

Regular maintenance should be an important part of every building owner's approach to their property management. It means that problems such as water penetration or termite infestation, which could lead to severe deterioration in buildings, do not get out of hand. Maintenance should have priority over restoration or reconstruction.

Restoration and **reconstruction** mean returning the existing fabric of a building or work to a known earlier state. Restoration may involve removing insignificant accretions or reassembling existing components without the introduction of new materials, while reconstruction introduces materials – new or old – into the fabric to replace missing elements.

Maintenance, restoration and reconstruction should aim at compatibility with the original design and materials, this often means using original materials for structural as well as aesthetic reasons. Original materials should be matched with suitable second-hand materials if possible.

Mixing modern and traditional materials without careful consideration can be problematic. The long life of traditional materials and methods testifies to their practicality. Substitute materials such as aluminium windows instead of timber framed or concrete tiles for original unglazed terracotta will not have the same aesthetic qualities as the original. Bear in mind also that older building components of 'imperial' dimensions are larger than their metric equivalents when replacement is unavoidable.

The principles in this section on traditional methods and materials are a general guide. The information provided needs to suit the subtleties of the particular maintenance, restoration or reconstruction case.

1.2.1 Roofs

Roofs of older style buildings are usually a very dominant element of their design. Certain roofing materials relate to particular styles of houses. Because the roof is so dominant, its shape and cladding material are often major unifying factors in the heritage conservation areas.

The repair of slate roofs of some Victorian and Federation buildings is a job for a skilled tradesperson. Problems usually result from delamination of the slates, requiring replacement, or loose fixings, perhaps requiring new fixing battens and re-fixing with copper nails.

Patching with recycled slate is practical but, without sufficient care, a ‘patch work’ roof can result. If necessary, concentrate on the front elevation, using slates from rearward facing slopes. Use second-hand slate to patch the less important faces, or use corrugated iron on an inner slope that is not visible to the street.

Some of the many types of modern roof slate now available are unsuitable being too thick, with too rough a texture and may also be too heavy for the original roof structure. Asbestos or fibre cement imitation slates are not recommended as a substitute for real slate.

The use of bituminous paper or plastic sarking may enable the TV antenna to be placed inside the roof space, although many slaters prefer not to use sarking so that leaks can be readily located. TV aerials should not be attached to chimneys.

If unglazed terracotta ‘Marseilles’ tiles on Federation houses need replacing, sound second-hand tiles or new similar tiles of the correct Imperial size should be used. Avoid concrete and glazed tiles, which do not have the subtle colour and texture of the older-style Federation tiles, though these may be suitable on some later Bungalow styles.

Where the original roof fabric has been replaced with a modern material, re-roofing in the appropriate style will enhance the appearance of the house. If you cannot afford to replace the

authentic slate roof on Federation or Victorian cottages, then corrugated iron (now galvanised steel or zincalume) also often used at this time, could be a sympathetic alternative.

A natural finish or a dull prefinished grey colour which most resembles natural weathered iron is preferred, unless the roof has always been painted, when a colourbond finish in period colours could be considered in lieu of repainting. Corrugated iron is suitable for subsidiary roofs on Victorian and Federation houses, such as rear skillions or back verandahs, which were often treated with less substantial materials than the main roof structure.

Using the correct gutter shape such as ogee, half round or quad profiles and the round downpipes common until the early 20th century is important for a thorough finish: where original gutter profiles are evident, consider their reinstatement. Regular cleaning to remove leaves, which produce corrosive acids on decomposing, and painting their insides with bituminous or tax-epoxy paints, can prolong the life of gutters.

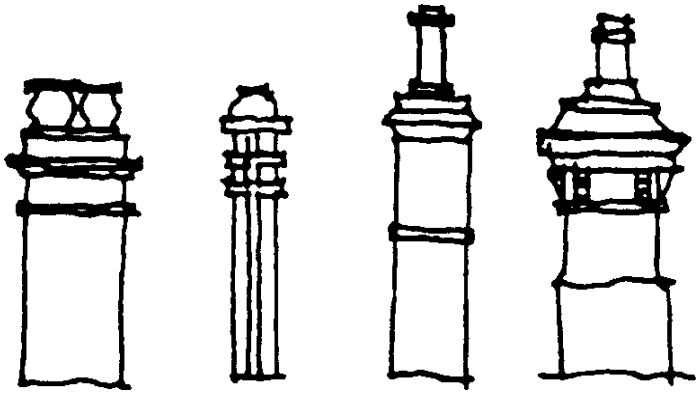
Figure 45 Half-round and Ogee guttering

Damaged or missing decorative tile trimmings and chimney pots found on Federation houses can be replaced either from demolition saleyards or by new products. When fitting ridge crests and finials, the bedding mortar should be stained with red oxide pigment to match the tile colour.

Do not block roof or rafter ventilation holes as good ventilation reduces the likelihood of rot in roof timbers and soffits.

Chimneys add scale and character to buildings of all periods. Chimneys should not be removed when roofs are reclad, the flashings need replacing, or the chimneys themselves need repair. While acids, derived from soot and smoke, attack the mortar pointing of chimneys and loose pots or bricks and flaunching (the mortar weathering on top) are dangerous and allow moisture penetration, these can be repaired by an experienced tradesperson. Reconstruction of chimneys, a distinctive feature of Victorian and Federation houses, is also possible and desirable.

Figure 46 Illustration of various chimney forms



Chimneys are a distinctive feature of Victorian and Federation houses

For more advice see the maintenance series information sheets by Heritage NSW at heritage.nsw.gov.au. In particular see advice on:

- slating, tiling and roof plumbing
- corrugated roofing
- metalwork.

1.2.2 Walls

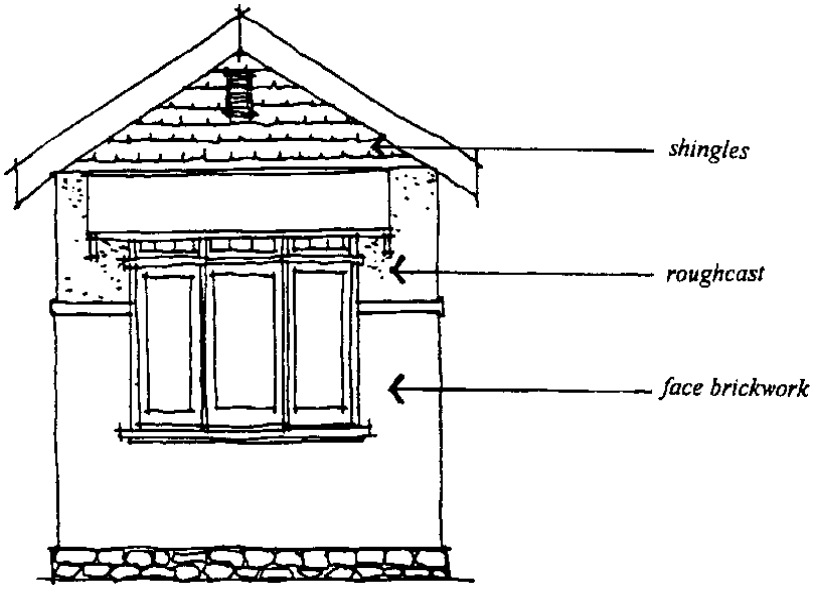
The coating of render, or stucco, often applied to Victorian houses, protected their exterior walls: it should not be removed to expose the original sandstock brickwork. This destroys the character and significance of the building and allows the destructive entry of moisture: soft, porous sandstock bricks decay and crumble on removal of their protective render. A solid sandstone wall without stucco protection also soaks up moisture which passes through to destroy internal finishes. Cracked or chipped render should be replaced with a similar compatible render (not cement) and usually painted over.

The old face bricks of Federation and Bungalow styles, however, should not be painted. Many Federation houses have a pattern of pressed red brick or red-ochre dressed front elevations, with tuck pointing and contrasting blues or buffs for arched lintels and string courses. Cheaper, variable common bricks were used for the lesser elevations. Any repair work should maintain the distinction between front and sides, which also occurs in standard California Bungalows, though these tend more towards liver brick shades. The textured clinker brick, sometimes in variegated red brown colours with herringbone patterning of 1930s bungalows, should also be retained as a distinctive feature of this period.

Paint can be chemically removed from brickwork by skilled tradespeople. Sandblasting is an extremely dangerous process for old buildings, as it removes the outer burnt skin of the brickwork, leaving it exposed to weathering with a pockmarked ragged appearance.

Mortars for repointing and repair should match the colour, texture and mix of the original. Old mortars were usually a mixture of lime, putty and sand. Professional skills are often needed in selection of mortar mixes for fragile older buildings. Lime mortar can accommodate some movement, while harder inflexible cement mortars cannot, thus causing cracking in the brickwork. Badly rusted and swollen lintel bars should be replaced to prevent walls cracking.

Figure 47 Shingles, roughcast and face brickwork



Rising or falling damp may be present: its cause may be simple or very complex. Leaking gutters and drains may saturate adjoining ground; cement floors inside or outside may force ground water to rise up walls. It is not always necessary to introduce or replace a damp course to stop rising damp. Improved ventilation and the elimination of water sources may solve the problem but specialist advice, sometimes from a number of sources, should be obtained and compared before deciding on suitable action.

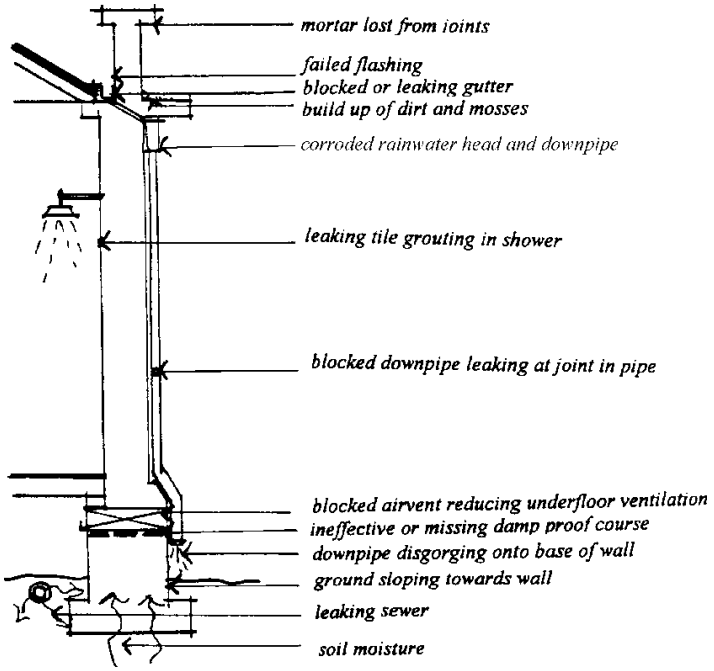
In replacing deteriorated timber shingles, for example on gable ends or bay windows, any pattern in the arrangement or variations in size and depth of courses should be matched to retain the original texture. Compatible soaker flashings and sarking should be used in fixing.

Gable ends and bay windows in Federation and early bungalow houses may be clad with pebble-dash or rough cast on timber or metal lathes or fibre cement sheet with vertical battens concealing joints. Replacement of lathing when it fails through rot or rust is essential to prevent moisture penetration: if correctly applied, the vertical surfaces should bell out towards the bottom to shed rainwater. Modern ‘metal’ weatherboards are not suitable for replacing timber weatherboards in rear extensions or early houses.

For more advice see the maintenance series information sheets by Heritage NSW at heritage.nsw.gov.au. In particular see advice on:

- rising damp
- salt attack and rising damp
- sodium bicarbonate to remove paint from historic buildings
- treating biological growths on historic masonry
- cracking of buildings due to shrink/swell in clay soils
- drought – related cracking to buildings
- repointing lime mortar joints – some important points.

Figure 48 Diagram from 'Rising damp', an information sheet by NSW Heritage Office



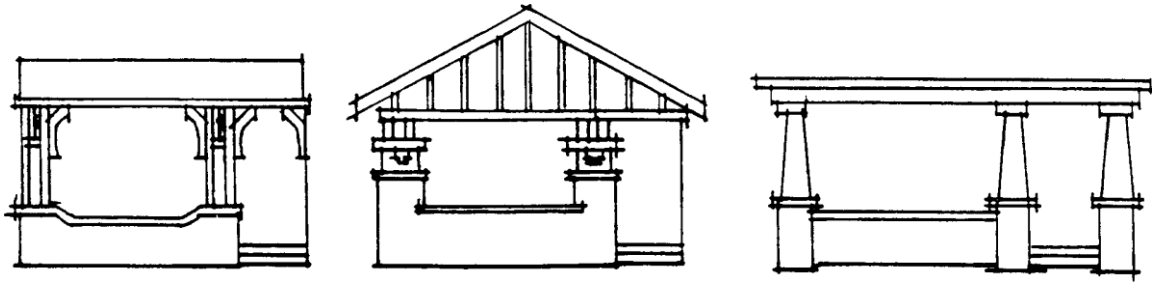
1.2.3 Verandahs

Verandahs, balconies and porches are often distinctive architectural features, whose detailing, materials and proportions are important stylistic elements to be carefully maintained and repaired.

The removal of windows and other enclosures to verandahs and balconies, where these detract from the heritage significance of an individual building or break the rhythm of a whole street of similar buildings, is encouraged. However, the value of retaining such enclosures needs to be carefully considered in terms of both their architectural contribution and whether the added amenity they provide would assist the long-term conservation of the building.

When verandahs are restored, the original walls and structures should be carefully inspected for evidence of earlier detailing, which can usually be reproduced.

Figure 49 Federation, California and Interwar bungalows



1.2.4 Windows, doors and decorative joinery

Inadequate maintenance often shows first in deterioration of timber work. Experienced joinery contractors can copy subtle details where replacement is needed because of the condition of

the original fabric. Inappropriate replacement of doors and windows amounts to defacing a building. It is also important to recognise that repair of joinery is a job for experts.

Where it has been damaged, the wide vocabulary of detail in the turned and lattice woodwork of Federation and other significant houses should be copied carefully. Wherever possible, handrails, balustrades and verandah posts should be reinstated in the original style. Any recycled materials must be consistent with the building's design. Similar interior joinery including skirting boards, picture rails and architraves, should be reinstated in the original style where possible.

Timber windows, doors and shopfronts were generally painted externally, not stripped back and varnished. Apart from looking wrong, externally applied varnish usually breaks down quickly, leaving timber unprotected.

Modern acrylic paints are easier to apply and recoat and generally more durable than the older oil-based paints. However, acrylic paints are more thermoplastic and difficult to remove, so are often less suited to door and window frames than for other external paintwork. Be careful that thick paint coatings built up over time do not distort door and window frames, making closure tight and difficult: if necessary, strip paint back. Modern mastic sealants (silicon) can help seal joints between joinery and brickwork, particularly where movement has widened gaps between masonry and timber framework.

The wide variety of leadlights and figured or coloured glass panes found up to the Second World War may be virtually irreplaceable today. Beware of amateur leadlight repairers: appropriately trained professionals should be entrusted with repairs so that there is minimum disruption of valuable original pieces.

Door and window hardware items, such as casement fasteners, handles and stays, were often decorative items in older houses: generally these can be cleaned and rendered operable.

Steel hinges and fastenings (screws) are generally incompatible with the American redwood used for joinery such as door and window frames in Federation houses. As rust can accelerate timber decay, these are often better replaced by brass fittings.

For more advice see the maintenance series information sheets from Heritage NSW at heritage.nsw.gov.au. In particular see advice on:

- paint finishes
- timber repairs.

1.2.5 Shopfronts

It is a common misconception that the maintenance of old shopfronts is too costly and often shop owners do not appreciate the contribution that older shopfronts can make to their business. With simple maintenance and appropriate signs, old shopfronts can be conserved to create effective marketing images of great value to the individual business and the broader business community, especially where groups of shops of consistent or harmonious design give a distinctive identity to the commercial centre. This identity contributes to shopping ambience that encourages and promotes business.

Signs should be carefully designed so they are integrated with and do not overwhelm the building's form. The building's façade massing, proportion and fenestration can suggest suitable opportunities for signs.

1.2.6 Colour schemes and finishes

The reintroduction of traditional colour schemes has become popular, with several commercial ranges of 'heritage' colours and well researched reference texts available to assist building owners in choosing authentic exterior and interior period colour schemes. The tones of old photographs (including black and white) can indicate varying colours and the relative intensity of shades on different elements of a building to assist in selecting an appropriate colour scheme. Paint scrapings may enable building owners to approximate early or sequential colour schemes.

While reinstating original or authentic colour schemes helps reinforce the character of a house, 'over-restoring' buildings to an elaborate appearance that they never possessed is not appropriate. The patina of age can give an appealing visual quality to a street or area.

An understanding of traditional colour schemes appropriate to a particular building or area will promote an approach which can then be employed to produce a sympathetic contemporary colour scheme which reinforces the character of the area.

Though these guidelines are mostly concerned with the exterior of buildings, interior details are important and can influence the resale value of buildings.

Lime plaster which is a common lining material in houses of Federation and earlier eras should be patched or replaced with matching material (not cement which is too hard and unyielding). Damaged cornices and ceiling roses can be replicated in fibrous plaster.

Patient work on interiors may occasionally reveal intact decorative paint features such as dado stripes, stencilling or wallpapers. While sometimes old decoration can be seen on the surface under oblique light, often it is not, so razor blades or scalpels should be used carefully where owners are keen to expose earlier finishes and recreate these.

For more advice see the maintenance series information sheets by Heritage NSW at heritage.nsw.gov.au. In particular see advice on:

- paint finishes
- plaster finishes

1.2.7 Landscaping, fences and gates

Fences, gates and landscaping have a large impact on the streetscape. Planting consistent with the established garden character can enhance the architecture and may even hide previous unsympathetic alterations. The maintenance or restoration of a garden suited to the taste and style of the era adds to the authentic restoration of a house.

In some cases, garden beds, trees and other landscaping elements, such as paths of paving bricks or tessellated ceramic tiles, have their own heritage value. For example, the well established gardens with their landmark trees provide remarkable settings for some Victorian mansions along the Mowbray Road ridgeline. So do the bushland settings of the Griffin houses in Castlecrag. These settings are integral elements of these particular styles.

Be aware of changing taste and fashion in landscaping as much as buildings. In general, early Victorian and Federation era gardens were based on traditional English designs, using introduced plantings in ordered and symmetrical patterns. Federation gardens introduced curving garden paths and beds and combined exotic and native species but a strong trend toward indigenous species developed with the California Bungalow.

If you are reinstating or adding to an authentic garden design for your house, consider not only the historical examples, but try to consider your garden from all angles: from the street, as an entrance to your home and from the inside looking out to maximise all your efforts.

Because of their proximity to the street, fences have a strong visual impact: fences and gates sympathetic to the style of the house can maintain and unify the character of both the individual item and the street just as inappropriate fencing can destroy it. Keep surviving original fences in good condition, where deteriorated, save and repair as much as possible before using new, closely matching elements to replace what has been lost. Be cautious about using reproductions when the originals can be maintained.

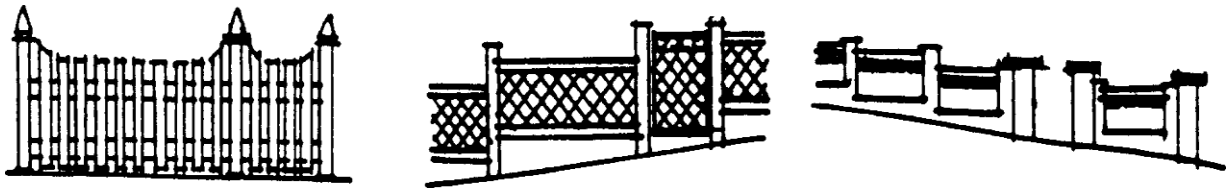
Traditionally, the finest and most elaborate fences were placed towards the street, with rough sawn paling fences on the rear and sides of properties: sometimes a tall, latticed fence and gate separated front and rear gardens.

In Willoughby, many streetscapes are distinguished by their low fences (for example, the Artarmon Heritage Conservation Area), or the absence of fencing (for example, the Griffin Heritage Conservation Area) and these qualities should be retained where this is characteristic of the streetscape and does not compromise the amenity of individual properties.

For more advice see the maintenance series information sheets by Heritage NSW at heritage.nsw.gov.au. In particular see advice on:

- Heritage gardens and grounds

Figure 50 Illustration showing the fence styles of Federation, California and Interwar bungalows



1.2.8 Installing modern services

Install new services (such as ductwork, pipewort, wiring conduits, air conditioners and TV antennae) inconspicuously to cause least damage to the fabric. Consider using sub-floor or roof spaces or bury them underground. Pull cord switches are a viable alternative to chasing electrical wiring. Floor mounted power points are an alternative to damaging skirtings.

Fittings should be unobtrusive. Historical recreations should be avoided unless there is evidence for such designs being used in the original building.

Avoid powerful heating and cooling systems, which may cause dryness and cracking or internal condensation. Supplementary humidity control may be appropriate.

1.2.9 Solar panels and hot water systems

Solar panels should be sensitively located and designed to avoid intruding on the heritage item or cohesiveness of the heritage conservation area. Placement for properties with north facing main elevations will be more difficult as this may impact on a building's heritage values. Placement on the side or rear elevations reduces the impact. Placement to the rear, particularly for solar hot water systems or mounted solar systems reduces the impact further.

If an alternative exists, solar panels are not to be located on the primary frontage and are to be located to limit visibility from the street.

Where no alternative is possible solar panels may be located on the primary frontage if justification has been provided that there is no suitable alternative location.

The quantity of solar panels on the primary street frontage or in a highly visible location are to be limited to the amount that services the house.

The impact on heritage significance may be greater where the collector is mounted on a stand at angle to the roof plane, or roof top tanks are installed.

Where possible, the mounting of collectors on stands should be avoided unless the installation is very discreetly located. The tank should be installed inside the roof space or on ground within near proximity of the solar collector.

Ensure that the weight of the new equipment can be borne by the heritage building supporting structure (such as roof rafters). The supplier or installer will have information on the weight of the equipment.

To minimise damage over the long term, use the minimum number of fixing holes. As solar collectors have a finite lifespan, minimising fixture points will assist with future removal and replacement. Also limit the number of holes made into the loft space for pipework and cabling.

Meters should be positioned so they are not immediately obvious from the street.

When maintenance is carried out there is the potential for foot traffic to damage any fragile roof cladding. Discuss with the installer how often equipment will need routine maintenance and how it will be accessed.

1.3 Alterations and additions

Alterations and additions to buildings should be made to their less prominent areas – to the rear or set back from their principal elevations – and carefully scaled and designed to enable the heritage character of the building to be maintained. This approach accepts that many buildings are changed during their lives – to better satisfy their original purpose, adaptation for some new use, or to make them more attractive to their occupants. Poorly designed and built alterations and additions might provide functional solutions but can ruin any building, marring its appearance and detracting from its value.

Any project to alter or add to a building should start by carefully assessing the heritage significance of the building involved and its context, to suggest what changes can be made. While additions should reflect their time, it requires considerable skill to make changes appear as a subtle and natural amendments to the building rather than changes that overpower and diminish the significance of the original building.

1.3.1 Sympathetic development

All new development affecting heritage items and properties in a heritage conservation area – whether alterations, additions or infill – should acknowledge the heritage values of the existing place or building. It should respond to the established heritage significance of the building, place and neighbouring area by seeking to avoid any reduction in its value or any undesirable intrusion into the streetscape.

The degree to which additions or changes to heritage items and properties in a heritage conservation area should reflect the style of the significant fabric depends on the characteristics of the individual building and the area. This influences whether the new work should blend in, be imperceptible, or be clearly distinguishable as new work.

An alteration or extension must consider:

- the architectural character and style of the building concerned
- the characteristics of development in the locality.

The proposed size, siting and function of the addition or alteration and the significance, intactness and features of the building and its setting or locality also help determine the most suitable solution.

Alterations or additions to a building should be done in a similar manner to the original structure, avoiding the destruction of important elements such as chimneys, windows and gables. Unsympathetic alterations, which may include the use of modern roofing and other materials, the removal of chimneys and verandah joinery, and the enclosure of verandahs, detract not only from the house itself but may damage the significance of neighbouring houses and the streetscape.

A sympathetic alteration or addition will blend and harmonise with the building. Even where the new work is suitably intended to be distinct from the original, it will fulfil the essential criterion of ‘belonging’ to the building, in terms of scale, proportion and massing.

1.3.2 Principles

a. Scale

Scale refers to size. Do not overwhelm the original. An alteration or extension should not be of a size or scale that dominates the original building, challenging or destroying its identity, or changing its contribution and importance in a significant context.

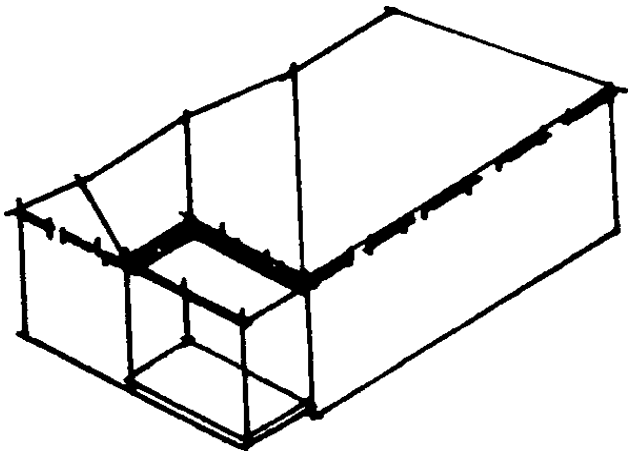
For example, cottages cannot be turned into large houses or mansions without major changes of character that are inconsistent with the conservation of their significance as small-scale cottages. Major changes in scale to individual buildings, which are components of a heritage conservation area, can also destroy the consistency, unity and cohesiveness of a group or a streetscape.

Height and bulk are critical in determining scale. In this respect, the building envelopes identified for specific heritage areas and generic building types are a guide to the desirable height and bulk.

Maintain consistent levels. In houses, the main wall plate height, which determines the external wall height and the point from which the roof structure and form springs, should be

consistent throughout a building. Otherwise this may not conform with the pattern of a consistent plate height throughout the main rooms, with a lower roof area and plate height to the rear for less important rooms or service areas.

Figure 51 Main wall plate (eaves height)



b. Form and massing

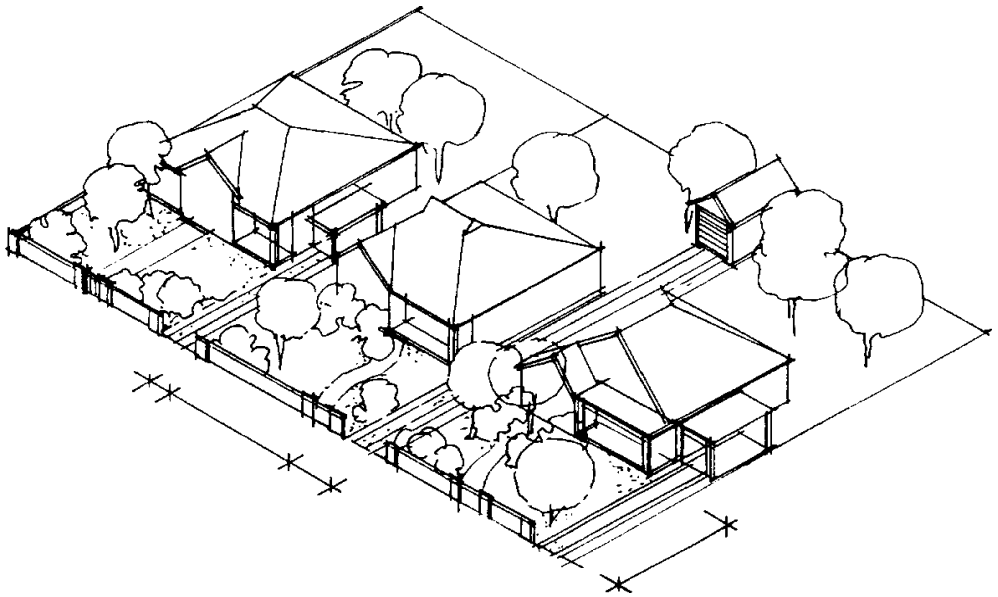
Form and massing refers to the shape of building elements and the arrangement of these elements. If an addition is proposed, the characteristic form and massing of the existing building or of the locality should be observed and reflected in the new work.

The massing of a building may be symmetrical or asymmetrical. The distinctive qualities of many buildings depend on the balance between the symmetry or asymmetry of their various elements. For example, small Victorian cottages usually have windows symmetrically positioned on either side of the central entry door on their front façade. Small Federation cottages typically have asymmetrical massing with a projecting gabled bay almost always placed to one side of the street façade.

c. Siting

Additions are usually best suited to the rear or side, to allow the significance of the original building to be maintained. Avoid alterations to the street façade to minimise the impact of change.

Figure 52 Illustration showing the traditional development pattern and siting of houses



d. Design and proportion

Proportion refers to the relative size of distinct parts. Respect the existing built form; roof type and pitch; verandah support spacing; the proportion and groupings of doors and windows; the ratio of solid to voids. Roof form and pitch should closely match the existing, whilst the repetition of eaves and gable projections and details can help tie in new to old. The size, proportion and position of existing openings should be respected.

e. Materials

Use materials that reflect and harmonise with original materials to maintain the significance of the building. Avoid materials where they do not respond to the original palette of materials in the locality. For example, aluminium windows, flush faced doors, non decorative posts or columns, quarry tiled or slate floors, painted face brickwork, arched brick colonnades and modern cladding materials.

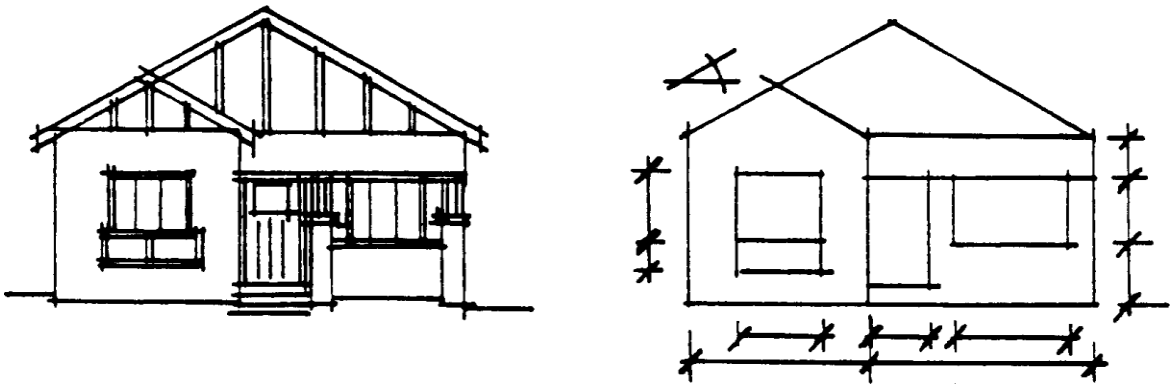
f. Consistency

Be consistent with the original. Observe the massing, scale, proportion, character and details of the existing fabric when blending new with old. Consider traditional solutions – it is easier to successfully add a traditional form than a contemporary form. But modern designs can be an acceptable solution if well designed and executed.

g. Intactness

When adding to an existing building keep the original fabric intact and distinct, whilst subtly yet clearly distinguishing new work, to avoid confusion in reading the history of the building.

Figure 53 Illustration showing traditional proportions and roof pitch on older homes



1.3.3 Types of additions to houses

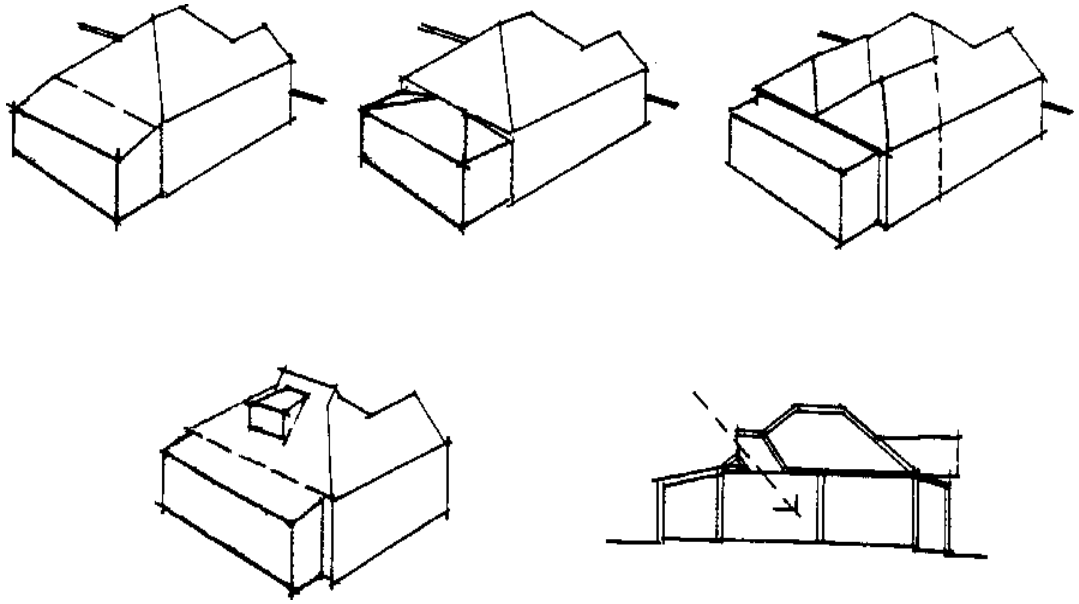
Certain design approaches are more suitable to the building's significance, architectural style, siting, the accommodation needs and the proposed use. Flat-roofed extensions and extensions which rely on 'Cape Cod', popups, A-frame and mansard roofs, are not suitable for most traditional houses.

a. Simple ‘lean-to’

This is usually the cheapest form of addition and can be used extensively with a variety of styles. Preferably sited at the rear of the house, this type of addition can also be used to form a new verandah or in combination with a wing. The use of a verandah addition, either at the rear or the side, may be a means of sympathetically relating a new extension with the existing house.

If the addition links into an existing roof the materials should be matched. Where the lean-to addition is constructed at the rear and not visible from the street, materials need not match the original. Depending on the land's slope, the size of a lean-to addition will be limited by the need to provide adequate head height (A wing addition overcomes this difficulty). Light access can be improved with a lean-to addition by adding gablets or skylights to the rear face of the roof.

Figure 54 Examples of 'lean-to' additions



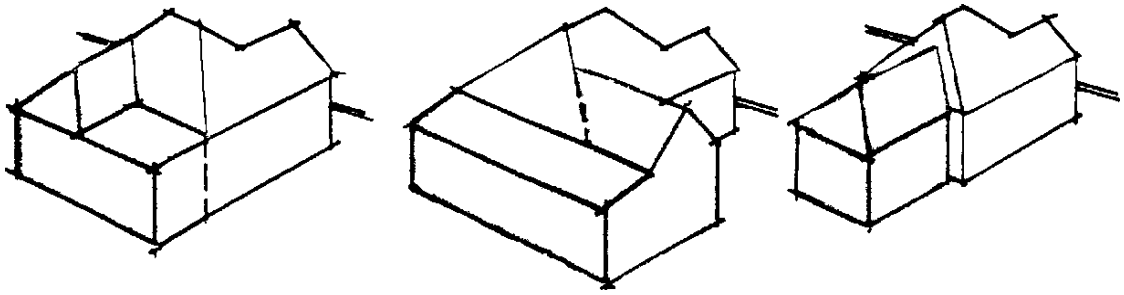
b. Wing additions

Wing additions should be made at the side or rear of the building to minimise disruption to the main elevation and the streetscape. The size and slope of the land and the form of the house will dictate the location of the wing, which is generally better suited to asymmetrical buildings.

Where the original design has a picturesquely irregular plan and roof shape, an additional wing, gablet or dormer to one side may reinforce this theme. However, it should not dominate and should not be located in the same elevation as the original work. This can be achieved by stepping the extension back. Side additions should not remove the possibility of car access to the rear of the site unless there is alternative rear lane access.

The existing roof form and pitch can be repeated and new detailing should reflect the gables, hips, eaves, ventilation, window hoods and other projections of the main roof.

Figure 55 Examples of wing additions



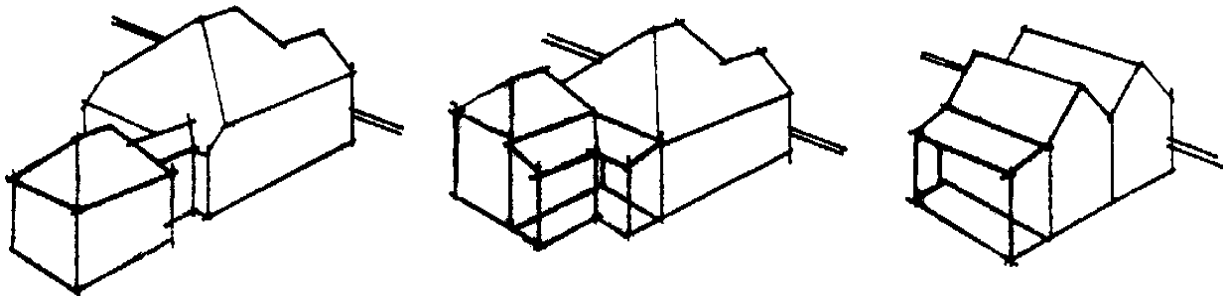
c. Pavilion

The pavilion solution involves constructing a separate block element which may repeat the general form of the original house. It is particularly appropriate where the architectural proportions, scale, symmetry or detailing of the main house would be adversely affected by another type of addition. For example, a pavilion addition could be used where the original

design would be upset by a lean-to or wing addition, or where the use of either of these would result in the loss of natural light or views.

Pavilions may be connected to the main building by a breezeway or small utility area, such as laundry or bathroom. Pavilions may be the best solution for long deep allotments and can also be functionally integrated with landscaping to provide internal courtyards.

Figure 56 Examples of pavilion type additions

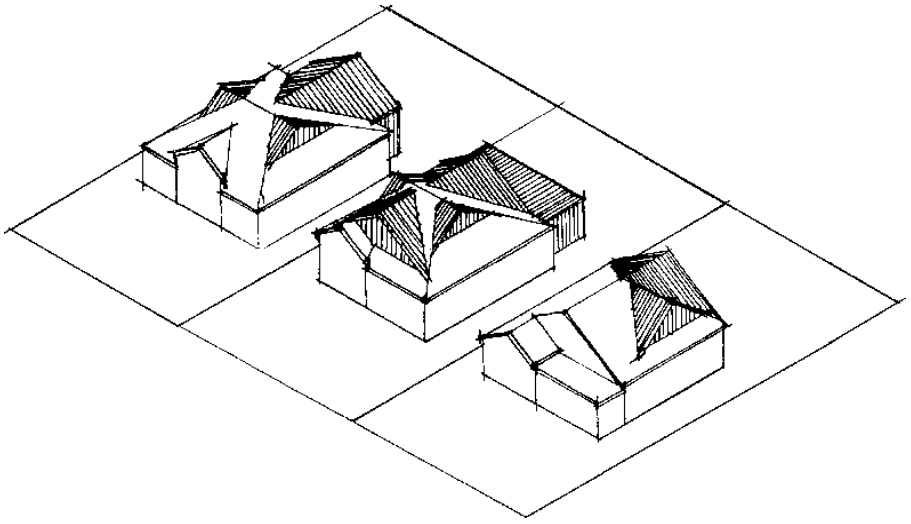


d. Roof

Providing the existing roof volume is sufficiently large and the roof profile is generally maintained, one or more habitable attic rooms may be created. Dormer windows should not dominate the original composition of roof elements through excessive scale and should be located to the rear.

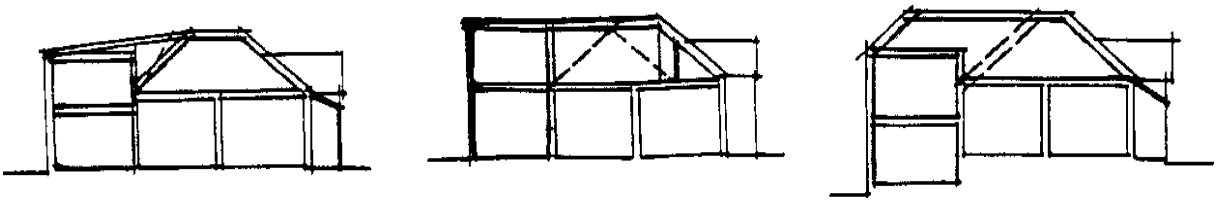
The roofs of Federation-style houses are characterised by multi-hipped and gabled roofs. Additions to the roofs of these houses should be in a traditional gabled, hipped or lean-to form, matching the materials and roof pitch of the original structure.

Figure 57 Examples of sympathetic roof additions



Roof additions should be confined to the rear of the existing building and should not alter the essential form and character of single storey buildings, for example, simple, modest Federation cottages should not be transformed into elaborate two storey ‘Queen Anne’ style mansions.

Figure 58 Types of roof additions



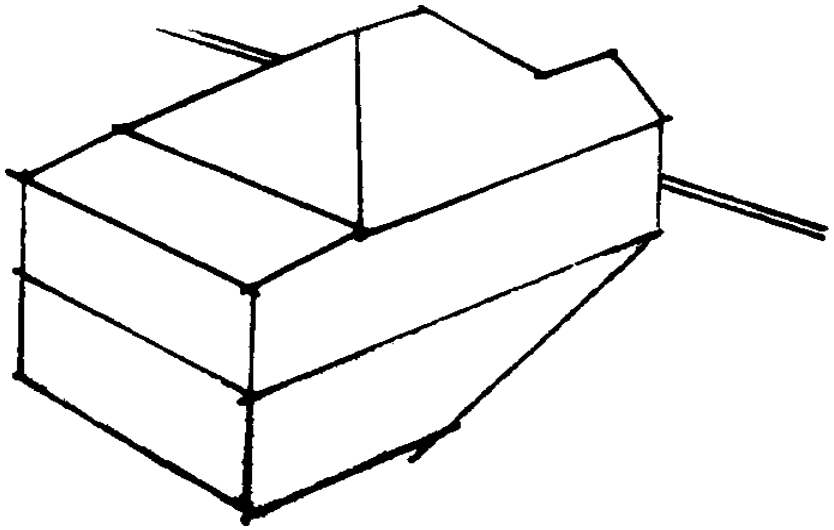
e. Basement

A basement addition is created by constructing rooms within the foundation area of an existing house. Houses built on sloping sites may possess a large underfloor area which can often be excavated to provide additional living space.

Before initiating this form of construction, carefully consider the type of materials to be removed, access for machinery and equipment, the removal of piers and installation of alternative support structures, drainage and the possible need for retaining walls. Care must be exercised to ensure adequate underfloor cross ventilation and to avoid water seepage or rising damp. Engineering advice is recommended and will probably be required.

Additional considerations include the placement of stairs and their effect on other areas of the house, the appearance of new windows and exterior doors which should be designed in terms of the overall appearance of the building and retention of most of the original foundation materials, such as sandstone. Garages can often be successfully accommodated in undercroft areas.

Figure 59 Example of a basement addition



1.3.4 Car parking and access

Garages and carports should preferably be located to the rear of the house and not dominate the streetscape. Priorities for accommodating the car are:

- locate at rear, with access from rear lane
- locate at rear, with access from the front

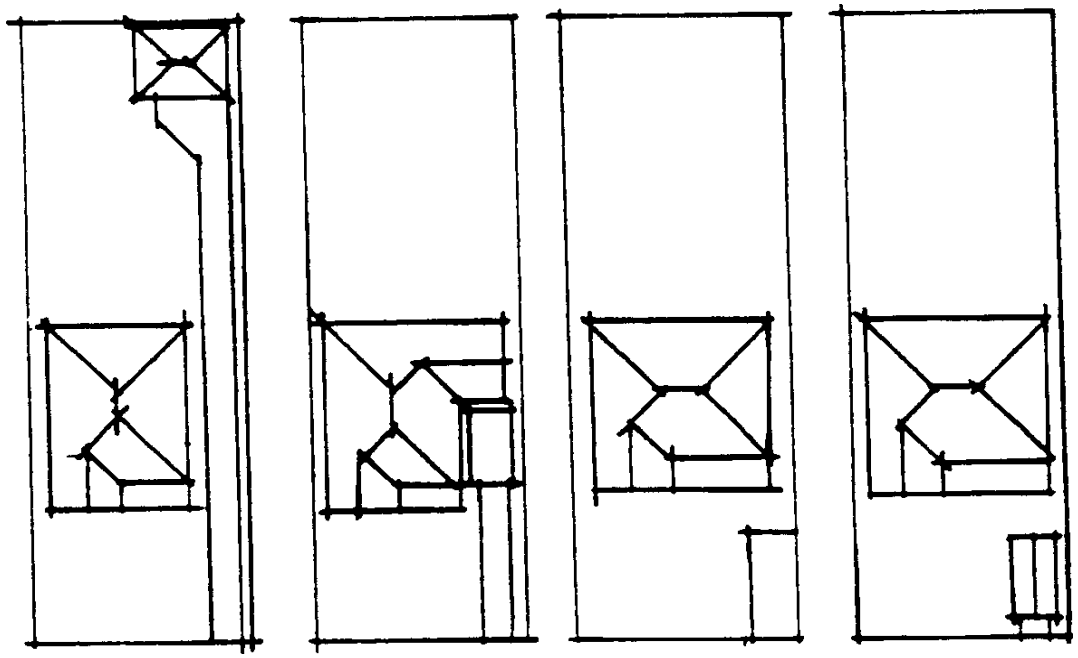
- locate at side of house, setback behind house alignment
- provide an uncovered, paved parking space at the front.

Rear lane access should be used wherever available in preference to interrupting the street façade and setting with additional driveways and parking facilities.

Extensions and additions should not sever the possibility for car access alongside the dwelling to the rear of the site. Where new garages at the rear of the site behind the house would be seen from the street, their design should not detract from the qualities of the house. The horizontal proportions of many roller shutter doors are generally unsuitable, and tiltadoors clad with vertical panels may be more appropriate.

Single car openings are preferred to double garages and stacked parking of cars may be encouraged to minimise the width of the structure as seen on the street frontage. Garages should have simple hipped, gabled or skillion roof forms of the smallest practicable scale in a given situation: the use of lean-to or skillion roofs over any attached to the main part of the structure will help minimise scale.

Figure 60 Examples of car parking and access options



Where car parking is to be provided beside the building, it should preferably be uncovered. If one covered space is needed, a simple, unobtrusive lean-to carport would generally be preferred to a more elaborate structure. Great care is needed to ensure that any attached structure fits in with the main horizontal lines of the building such as the eaves line or top plate and that it does not obscure major features such as windows, doorways or verandahs. Side carports should be set behind the building alignment.

Garages may be incorporated unobtrusively in the undercroft of a building or provided by excavation of the front alignment on sloping sites but only in cases where these options have been ‘traditional’ solutions in that locality.

Car parking facilities forward of the main building should be open stand areas.

Simple, open, lightly framed, pergola-like structures are preferred to elaborate carports, even where these emulate elements of the main building design. The open pergola type can complement Federation or California Bungalows without competing with, obscuring or overwhelming the main building in the way that elaborate structures, particularly with heavy tiled, visually dominant, gabled or hipped roofs, tend to do.

Where a carport is designed to emulate elements of the main building, great care is needed to ensure that it's siting, scale, materials and visual complexity are appropriate, and that it complements rather than competing with or obscuring the main building: out-of-scale details can visually detract from the more sophisticated contours of early buildings.

1.3.5 Pools and outbuildings

Major structures such as sheds, swimming pools and barbecue areas should be designed and sited carefully. These are best sited to the rear, or otherwise where lot width allows, beside the house. As with other extensions and additions, these will not be acceptable where access to existing or likely future car parking facilities at the rear of the site is cut off, and would necessitate the introduction of new, obtrusive car parking facilities.

Inground pools are less conspicuous than above-ground types. The pool and its surrounds should be designed with the aesthetics of the period in mind to harmonise with the house and its landscaped setting.

The use of close-jointed brick paving or stone flagging, with internal finishes in subdued tones such as deep greens, blues or browns, is highly compatible with Victorian and Federation sites. Pebble aggregate, quarry tile or mosaic tile aprons are equally suitable for some later bungalow and functionalist styles.

Changing rooms, equipment stores, plant rooms or garden sheds will be less obtrusive if built as simple sheds. These can be similar to modest early garage designs or designed as pavilions repeating some of the design elements of the house, provided these are of modest scale and not over elaborate.

1.4 Infill development

1.4.1 'Filling in the gaps'

Infill development must achieve a sympathetic relationship with either nearby heritage items or the heritage conservation area. Infill needs to display architectural 'good manners' by respecting the significant characteristics of nearby and adjoining development.

In this document, infill development includes new development on the site of a heritage item, or on a vacant or newly subdivided allotment in a heritage conservation area, or on sites which are in the vicinity of heritage items or heritage conservation areas. It is not an alteration or addition, but a new structure not attached to the heritage item. The principles that apply to alterations and additions also apply to infill development.

1.4.2 Imitation compared with sympathetic design

New buildings in heritage conservation areas should understand the characteristics of the past as these resulted from specific historic parameters. Reinterpreting these characteristics in a way appropriate to the present allows one to appreciate the original work in its context, and not confuse the old with the new.

The preferred approach is to design in a sympathetic but openly contemporary fashion to recognise existing patterns and to interpret them in a manner appropriate to the current social situation. Infill presents design constraints that often need special skill and ingenuity to be both contemporary and compatible.

Infill can be openly contemporary in design when it is well integrated with and relates harmoniously to its older original neighbours. It is essential that the scale, form and detail of the infill does not detract from the scale, form, unity, cohesion and predominant character of the building and development (streetscape/landscape elements) around it. The particular sensitivity of the whole setting must be respected.

Many contemporary kit/project home designs which purport to be ‘heritage homes’ are in fact poorly integrated mixtures of design elements from different eras- a pastiche of older styles and materials, often of a completely different scale to the original older buildings and badly proportioned. These infill developments are intrusive elements which detract from the cohesiveness and amenity of the heritage conservation areas, and clearly offend the essential underlying principle of sympathetic development.

The preferred approach with infill development is:

- Keep it simple. Do not try to copy a mixture of design elements from different eras.
- Incorporate basic design elements such as the characteristic roof forms and massing of the original development to help integrate the new work.
- Do not assume that contemporary project home ‘heritage’ designs will automatically suit any site or setting. Carefully observe and respect local nuances of design to ascertain if they are suitable.

1.4.3 Principles

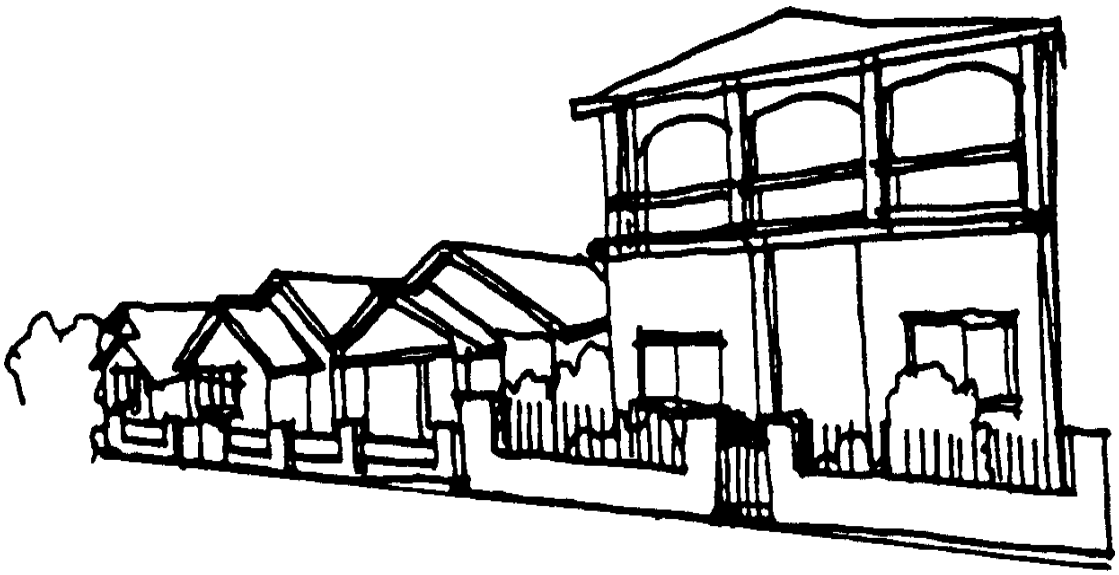
The following principles apply:

a. Character

The first principle of infill is to be guided by the established character of the area, so it is essential that the elements that contribute to the special qualities of the place are clearly identified. The aim is to harmonise with and complement the existing streetscape fabric or fabric of individual heritage buildings, not compete with it. Infill should not try to dominate its surrounds but should relate sympathetically to existing scale, mass and proportion. As the town character varies throughout Willoughby, infill should be particularly influenced by its individual street setting or context. Good infill relates well to its context, whether this is an individual heritage site, a group of significant original buildings or a streetscape of heritage value.

Willoughby City Council requires applicants to present a streetscape assessment with an application, as an important part of the design process and as a useful tool for assessment.

Figure 61 Out of character development



b. Scale

The scale (size, height and bulk) of a new building is of paramount importance. It should not dominate or compete with its neighbours or destroy an existing pattern of single or two-storey development.

Most of the City of Willoughby's heritage conservation areas have a consistent single storey scale: this is frequently an essential unifying principle. On the other hand, Naremburn Township is remarkable for its diverse mix of built scales, as are some parts of South Chatswood and Findlay/Wyvern Avenues, where there is a scatter of two storey buildings and larger mansions among the predominant single storey original housing. The existing diversity of scale in some places might allow a greater freedom of approach, but if uniformity of scale is part of the heritage value of the area, it must be respected accordingly.

Buildings are said to be 'in scale' when the elements fit – when they appear to belong in their setting, when there is a respectful relationship that is often recognisably 'right'. Often this means similar height, size and proportion to adjacent buildings.

'Landmark' buildings in heritage conservation areas are special cases: these may be individual heritage items, such as mansions, public or commercial buildings. In such cases, any adjacent or nearby infill should relate to the scale of the existing development around the landmark and respect its prominence. The presence of a landmark building, which may be a two storey building in a 'sea' of cottages, should not be used as a precedent for increasing the scale of infill development.

The relative scale of new buildings should consider the profile of original buildings – that is, the heights of the main ridgelines, or perhaps parapets in the case of commercial buildings, top plates/eaves level (or awnings of commercial buildings) and ground floor levels street or natural ground levels.

- Ensure the building relates in scale to its site and setting.
- Use heights, scale and bulk of original existing buildings as reference points.

- Make sure the parts are in scale with the whole.

Figure 62 Illustration showing development out of scale with the surrounding houses

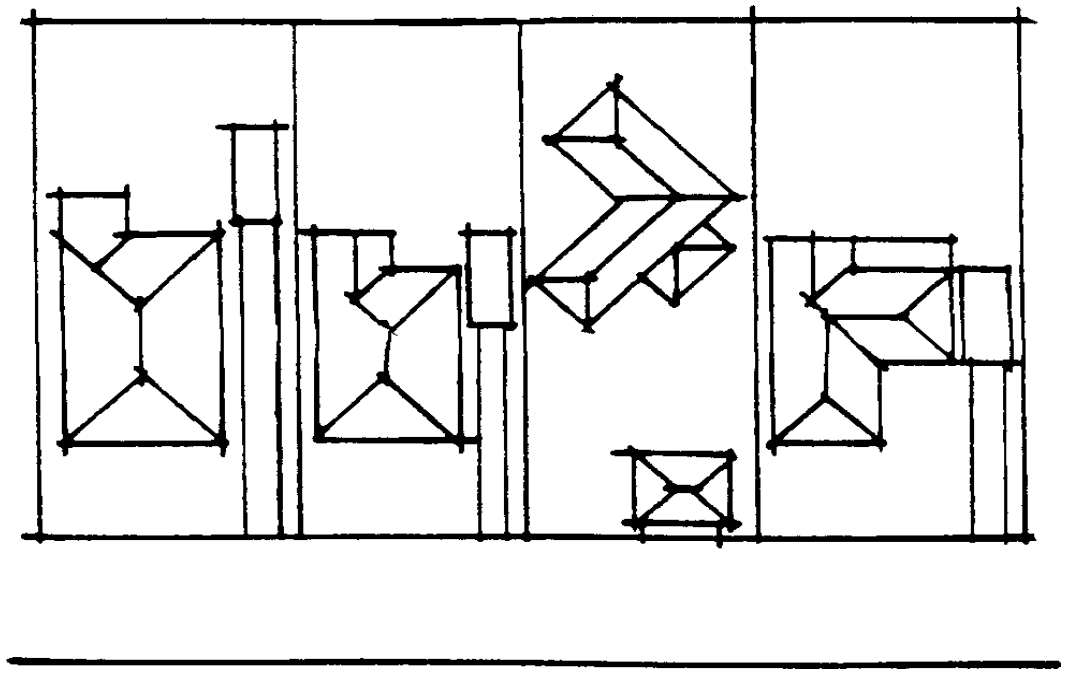
c. Siting and setbacks

The relative siting and setback of buildings is also important in forming the character of the streetscape and the relationship between adjoining buildings.

New buildings should be sited to correspond with the existing pattern of relationships between buildings and their sites. Front boundary setbacks should be equivalent to those of neighbouring buildings. Side distance setbacks should also be similar to the rhythm of development in the immediate vicinity.

Where existing buildings observe formal setbacks, or have historically been placed in a certain pattern relative to adjoining streets, that pattern must be considered in the location of any new building. If subdivisions challenge those patterns, the particular approach adopted must be justifiable in terms of its effects on heritage significance.

Figure 63 Sitings and setbacks should respect existing patterns



d. Form and massing

The form and massing of a building is its overall shape and the arrangements of its parts. Roofs, façades and verandahs are the primary elements of mass in heritage buildings. The distinctiveness of many older styles may be found in the way the massing of larger elements was broken down into smaller.

Infill design should identify the predominant form and massing and then design in sympathy with these forms. For example, the apparent bulk of a building may be reduced by breaking the primary façades into smaller components that reflect the character of their neighbours.

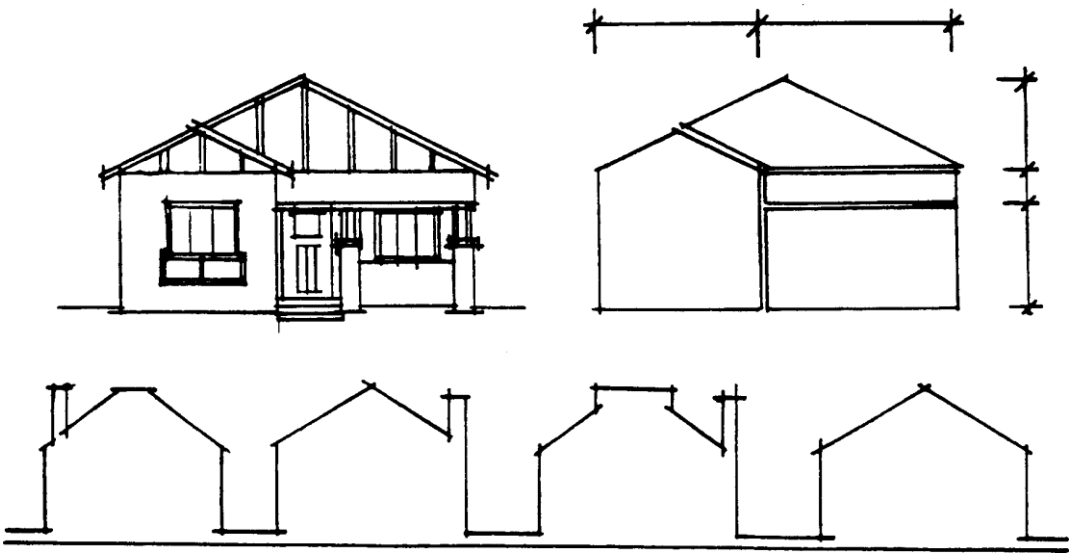
- Follow established practice of recessing doorways and emphasising display windows in retail buildings.
- Maintain the proportion and skyline interest of old building parapets when inserting new infill buildings amongst old commercial buildings.

The shape of the roof and the pattern it makes against the sky is often distinctive in heritage conservation areas. The form and character of the roof of new buildings can be the most decisive influence upon its successful integration within an established setting. Although the roof reflects the scale, setback, siting and plan form of the building it shelters, its pitch, form and shape will greatly affect the skyline appearance of a street, and may enable an otherwise very different building to blend in with the forms of buildings around it.

Hipped or gabled, simple or complex, steeply-pitched or flat – these aspects of a roof's design will determine its harmony or otherwise with neighbouring properties.

- Respect adjacent scale, heights and massing.
- Locate the bulk of larger new buildings to the rear.
- Consider the importance of maintaining roof form and rhythm.

Figure 64 Building examples that respect form and massing



e. Proportion

Traditionally older buildings up to the 1930s use vertical proportions reflecting the construction technology of the day. There was a regular ratio of solid wall to openings which reflected the economy of smaller spans. It produced a common pattern and proportion in the doors and windows. Where wider openings were required, this was achieved by multiples of the standard (vertically proportioned) doors, windows and bays.

Modern technology and construction permits much greater spans and often features a horizontal emphasis. New infill buildings in heritage conservation areas should preserve the proportions of the surrounding development, even when using modern materials. The proportions and details of openings displayed by a new building will be read as part of the overall design of the building which contains them, and should only depart from established norms in the street if they are balanced by other factors, such as by being placed under shading verandahs or overhangs. Large windows can also be divided into smaller vertically proportioned units, often with structural mullions (supports). Avoid contemporary ‘pseudo heritage’ mullion patterns that do not resemble to any typical traditional styles.

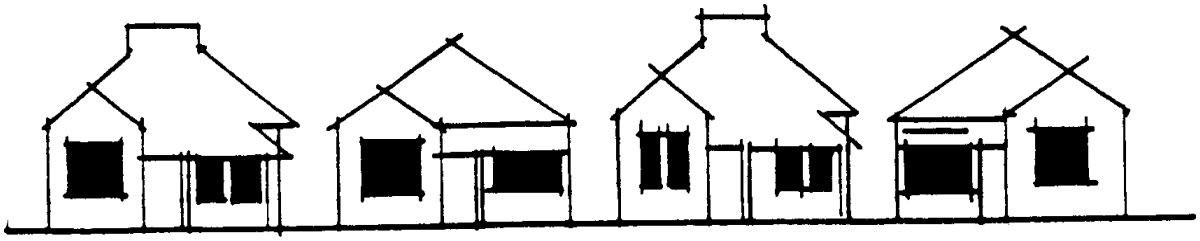
Similarly, modern garage doors can be divided into smaller vertical bays to diminish their visual impact.

Figure 65 A traditional garage door, which reflects the proportions of an older house

The proportions of any architectural decorative features should respect those of the original features from which they derive.

- Maintain characteristic proportions of design elements.
- Modulate building façades and maintain rhythm of fenestration.

Figure 66 Buildings that maintain characteristic proportions



f. Materials and colours

Materials and their colours will also influence the degree to which any new building will blend with or intrude on the general street or group character established by its neighbours. The materials used in a new building might be completely different from those around its site but can be brought into an overall picture of harmony by careful colour selection.

Materials and details of surrounding buildings need not be copied but can be used as a reference point for infill development. Observe how traditional forms were typically broken into smaller massing elements by using contrasting materials and colours.

Many materials that reproduce original features may be too expensive to use widely even if seen as appropriate. It is usually preferable to use harmonious contemporary materials well. A clean contemporary design solution that observes all the principles of infill is likely to be more successful.

- Avoid fake or synthetic detailing: Do not ‘dress up’ a contemporary design with superficial historic detail.

- There is no need to slavishly follow past styles except in heritage restoration: use simple, sympathetic but contemporary detailing.

g. Characteristic design elements

Architectural devices and design elements may be important contributions to the character of particular heritage conservation areas. For example, verandahs may be a common feature of street elevations. Chimneys can also be an important part of the skyline, visually grouping buildings together. The use of materials in certain feature elements may also be a signature of a place and part of its subsequent identity.

Some of these elements can be used in a way that makes no immediate or direct reference to architectural style or fashion, but establishes the family resemblance or neighbourly connection with the other buildings and spirit of the heritage conservation area.

2 Typical building styles and periods

The bulk of Willoughby's heritage is important because it demonstrates the area’s residential development over time. Understanding the features of the major building styles is essential to evaluate the significance of individual buildings and to decide how a building or place can be maintained, restored, reinstated, adapted for another use, extended or added to.

If infill development is proposed, it is essential to understand the locality's dominant styles and how these combine with the landform, subdivision pattern and landscaping to form distinctive streetscapes. This helps design new development that fits in with the heritage significance of the locality or nearby heritage items.

The distinguishing features of the major building styles found in housing in the City of Willoughby area are summarised here to help you identify the period of your house. Important commercial and public buildings may show these or other styles. Further references are listed at the end of this section.

Buildings are rarely ‘pure’ examples of these ‘typical’ styles. Some buildings may combine elements of several styles, as these naturally evolved. For example, the Federation Bungalow has aspects of a California Bungalow. But generally the massing and proportion of the building together with other architectural detailing enables one style to be distinguished from another. Some of Willoughby's streetscapes are significant because they display a consistent style while others are remarkable and attractive because they blend successive building styles and periods of development in a distinctive way.

Figure 67 Illustration of traditional house styles

2.1 Victorian – c1840 – c1890

Houses reminiscent of early Colonial buildings persisted in Victorian times. The earliest remaining buildings in Willoughby are usually modest single storey cottages in simple, symmetrical Georgian and Regency styles.

2.1.1 Victorian Georgian/Regency

These styles are similar enough given their small number in Willoughby to be treated as one. They share an ordered approach in their symmetrical façades and simple rectangularity of overall shape and individual elements.

As in other colonies in warm and hot climates, the early Australian house protected its principal rooms from the sun by introducing a verandah. The development of new materials meant that roofing slate or corrugated iron were generally used on the main roof with a separate verandah roof of iron. Corrugated iron bullnosed verandahs were often painted in wide stripes of alternating colours, a ‘motif’ derived from canvas awnings used at the turn of the 18th century.

Machine tools for shaping timber joinery led to turned posts, balusters and other decorative features. With glass available in larger panes, the 12 pane window of the old Colonial period gradually gave way to the window in which each sash was either divided into two panes of glass or was glazed with a single sheet.

Regency buildings had more sophisticated decoration including simple projecting mouldings. Examples include architraves around doors and windows, louvered timber shutters for sun protection, open-work iron columns, or columns of classical shapes, and an elegantly drooping concave or ogee corrugated iron roof to verandahs. Another feature was a cantilevered balcony with a balustrade of iron or timber.

Figure 68 Example of a typical Victorian period cottage

a. Scale

- Small and low domestic scale – mainly single storey with occasional two storey houses in Willoughby.

b. Massing

- General roof shape: Simple hipped roof, or occasional gabled form of moderate pitch.
- Building envelope: Simple formal rectangular shapes with symmetrical plain, undecorated or subtly decorated façades; sometimes cantilevered balcony projection on occasional two storey Regency type.
- Façade openings/proportions: Regular symmetrical openings, usually from a verandah stretching across the entire front façade.

c. Materials and details

- Roof: Mainly corrugated iron, some slate in Willoughby; close eaves; low simple often rendered chimney sometimes with moulded capping.
- Walls: Sometimes exposed brick work, sometimes rendered some sandstone and weatherboard in Willoughby.
- Joinery and decoration: Simple geometric or turned wooden verandah posts; simple mouldings including stucco architraves; cast iron verandah posts and valances.
- Windows and doors: Sash window with small (often 12) panes, some with large panes; casement or French windows; panelled door; fanlight or transom light.
- Verandahs: Usually separate corrugated iron bullnose or drooping convex profile roofs supported by slender, simple verandah posts of classical or rectangular design or open-work or slender cast iron columns.

Figure 69 Typical two-storey Victorian period house

2.1.2 Victorian Gothic/Italian/Filigree

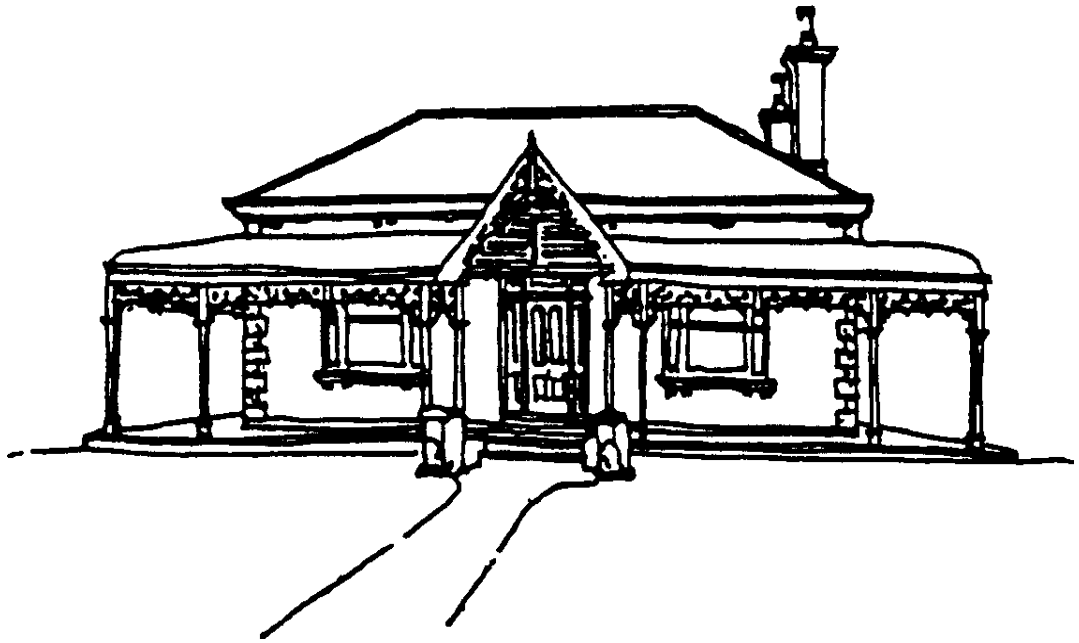
Willoughby has a small number of these buildings mainly in Gothic and Italianate styles. Their asymmetrical massing and elaborate decoration are usually seen as typically Victorian. These styles grew out of the British Victorian fascination with ‘the picturesque: many Britons, finding their nation in the vanguard of 19th-century industrialisation, felt the need for regular doses of nostalgia to counteract the harsh and ugly aspects of modern life’.

The Italianate style drew on images of country villas. A feature common to house design since, ‘The Great Australian Asymmetrical Front (where the main bedroom pokes out a metre or two towards the street beyond the rest of the house)’ began with Victorian Italianate.

‘Rustic Gothic drew on a romantic image of a rural, vaguely medieval past: it was an anti-monumental style applied to houses in the suburbs or country’ and was more modest in scale and decoration than the Italianate.

Some of Willoughby's two storey mansions have a Victorian Filigree character, the principal feature being an intricately textured verandah screen shading the main building mass. This style is an extension and embellishment of the Australian verandah tradition, generally expressed by a two storied verandah on an often relatively simple masonry form with classical medieval or Italianate detailing. A growing demand for more ornate architecture after the gold rush was largely met by the extensive use of decorative cast-iron components, by the 1870s designed and made in Australia, sometimes using local flora themes.

Figure 70 Example of a Victorian house



a. Victorian Gothic scale

- Intimate and modest scale – usually single storey in Willoughby – despite steeply pitched roof and prominent gables.

b. Victorian Gothic massing

- General roof shape: Steep pitched (45º-60º) roof with prominent gables over projecting room bays.
- Building envelope: Irregular, asymmetrical massing, though some of the single storey houses in Willoughby have more formal principal elevations with a verandah to one side of a projecting gabled room bay on the front façade.
- Façade openings/proportions: Decorative façade with asymmetrical arrangement of openings.

c. Victorian Gothic materials and details

- Roof: Usually slate, some iron roofs tall mediaeval looking sculptured chimneys.
- Walls: Grey stucco walls sometimes imitating stone, stone walls, brick walls with decorative quoining.
- Joinery and decoration: Elaborately carved decorative timber ‘traceried’ bargeboards and fretwork; turned finials and pendants.
- Windows and doors: Tall narrow casement windows with leadlight panels; bay windows; groups of windows; masonry mullion.
- Verandahs: Intricate cast iron fringes and brackets to cast iron verandah posts.

Figure 71 Example of a Victorian Gothic style

a. Victorian Italianate scale

- Imposing, even for single storey houses, with ‘landmark’ elements of towers and faceted bays.

b. Victorian Italianate massing

- General roof shape: Generally low to medium pitched hipped roof with octagonal or pyramidal tower elements, such as over the projecting front room bay, typically found in more modest single storey houses.
- Building envelope: Asymmetrical but still formal, principal elevation.
- Façade openings/proportions: Grouped openings of windows and colonnades.

c. Victorian Italianate materials and details

- Roof: Usually slate decorative moulded chimney pots.
- Walls: Usually grey stucco with moulded ornamental finishes.
- Joinery and decoration: Moulded architraves; bracketed eaves.
- Windows and doors: Curved, arched or stilted segmental window heads; faceted bay windows; coloured and etched glass of reds, blues, greens, purples and yellows; elaborate four panel doors with raised mouldings.
- Verandahs: Deep verandahs (loggias); encaustic or tessellated tile or marble hall and porch floors some iron filigree work on valances, posts, balustrades.

Figure 72 Example of a Victorian style

2.2 Federation – c1890 – c1915

The main strands of Federation housing found in Willoughby are the Queen Anne style and to a lesser extent the Arts and Crafts and Federation Bungalow styles, though some architect-designed domestic or public buildings exemplify other styles, for example, the Federation Tudor/Gothic of Innisfallen Castle, the Federation and Romanesque of The Willoughby Hotel and the Federation Gothic of Northbridge Suspension Bridge and St Leonards Catholic Church, Naremburn.

2.2.1 Federation Queen Anne

In the Federation period, wood replaced cast iron for structural and ornamental components of verandahs: ‘The use of steam and, later, electricity to operate tools such as the bandsaw, jigsaw and the lathe made it possible for pieces of wood of many shapes and sizes to be made quickly, easily, relatively cheaply and in large quantities’.

Timber posts, balustrades, brackets and valances had a chunkier quality than wrought iron which by the early 20th century was seen as rather old fashioned. Some buildings used both: the few buildings in Willoughby combining timber and iron decoration do not have the intricately textured screen that was the epitome of the Federation Filigree style.

The other major material changes were the use of unglazed terracotta Marseilles pattern tile roofing – though slate roofing with terracotta decoration (ridge cappings and finials) also occurred in the early period – and reddish face brick walls, usually tuck pointed on the street façade, which replaced Victorian grey stucco finishes.

The roof became a dominant feature with its steeper pitch and, on larger houses, an ensemble of varied roof shapes evolving from a usually hipped, pyramidal main roof. A prominent gable usually faced the street and the verandah tended to be incorporated in the main roof.

The Federation Queen Anne style embodied influences of the English and the more extravagant American styles of the same name, along with some distinctly Australian features.

This was the dominant style of Australian domestic architecture during the decades immediately before and after 1900, a period of accelerated suburban expansion based on improved public transport. The first growth spurt in Willoughby occurred at this time with more dense urban settlement near the new railway stations and along tram routes. Willoughby has a wide range of Federation housing types in a variety of locations as a result.

a. Scale

- Moderate domestic scale of usually single storey in Willoughby, varying from picturesque but imposing larger two storey mansions to elegant grand villas, modest single dwellings and narrow fronted workers' cottages.

b. Massing

- General roof shape: Roof dominant with an ensemble of varied roof shapes of moderate to steep pitch; prominently featured subsidiary gables of different sizes; conical tower-like or pyramidal ‘candle snuffer’ roof elements; multiple chimneys, tall with decorative cappings such as lantern shaped with terracotta pot or pots.
- Building envelope: Asymmetrical, informal, picturesque massing, including projecting room bays and diagonally projecting corners; deliberate avoidance of three dimensional simplicities. Smaller houses often have simple plan-shapes with corridor from front door past living rooms and bedrooms to the kitchen, bathroom and laundry at the rear: typically one room at the front projects forward towards the street and a veranda extends across the remainder of the frontage in the more modest houses and worker's cottages.
- Façade openings/proportions: Varied informal ensemble of openings, usually tall narrow windows, often grouped.

c. Materials and details

- Roof: Slate gradually supplemented by the more common unglazed French Marseilles tile; terracotta ridge and apex ornaments.
- Walls: Warm red brick, often dichromatic, with tuck pointing on street façades; some contrasting rough cast walling; half timbered effects in gable; occasional pressed metal wall and verandah ceiling sheeting.
- Joinery and decoration: Painted timber detailing; timber gable screen; bracketed projecting or ‘flying’ gable; ornamental timber frieze or valance often with turned or fretwork elements; sometimes curvilinear Art Nouveau timber ornaments.
- Windows and doors: Casements with Art Nouveau inspired leadlights; sometimes casements on front façade but double-hung on side walls with multi-paned, (sometimes coloured glass) upper sashes; accent windows, including round windows, bay windows and oriels.
- Verandahs: Wide verandah, often wrapped around more than one side on larger houses; timber posts with ornamental brackets, balustrades and valances.

2.2.2 Cottage forms

Most of Willoughby's Federation housing was fairly modest. Naremburn's unique and cohesive settlement of small single fronted workers' cottages developed within walking distance of the

station, brick pits and other industry. And, on the other hand, two storey mansions and commodious single storey grand villas were scattered or found in enclaves in select new suburbs and on more isolated, larger land holdings.

While sharing similar characteristic Federation materials and design elements, there are considerable differences in scale, complexity, massing and richness of decoration between the grand villas, houses and single fronted workers' cottages of this period as illustrated below. The roof profile of the smaller forms, while still prominent visually, is much plainer: cottages tend to have a single gabled room bay to one side of the main hipped (sometimes gambrel) form, while the single fronted cottages in Naremburn usually have either a reduced version of this roof shape or a single gable facing the street with a hipped roof behind.

Figure 73 Examples of Federation style buildings

2.2.3 Arts and Crafts

Compared with the Queen Anne, the Arts and Crafts style is less common in Willoughby: it tends to be more informal in planning, massing, fenestration and landscaping and have a more unassuming, homely feeling. The roof is a dominant element, featuring gables (with barges or parapets) and/or hips of medium to steep pitch with prominent eaves, sometimes wide enough to require brackets.

In contrast with Queen Anne red brick, painted pebble-dash stucco (rough cast) is commonly used on exterior walls and chimneys, together with other materials having earthy, 'natural' colours and textures, such as wooden shingles, often used in gable ends, and stone, often used for basework and trim. In Willoughby, sandstone is much used for these purposes and also for fences of the period. Touches of sinuous Art Nouveau detail are also common, for example in woodwork and stained glass.

Figure 74 Examples of Arts and Crafts style

2.2.4 Federation Bungalow

This is a transitional style, which is a forerunner of the ground-hugging interwar California Bungalow, with which it shares qualities of homely simplicity and robust honesty. It differs from the earlier Federation styles in its broad simple roof planes, often featuring an awning-like gabled roof with the ridge parallel to the street and with the main roof extending over the verandah.

It is generally single storey but sometimes has attic rooms in the roof space, which have small 'eyelid' dormer windows, though this is very rare in Willoughby. Eaves are wide with exposed rafters. A prominent gable facing the street is often decorated with wall hung stained shingles and roof ventilators.

Verandahs have simple masonry pier supports and/or simple, sturdy timber posts. Walls are rough cast and face brick, sometimes in contrasting sections.

Stone is also used in Willoughby for basework, verandah columns and pier cappings. Leadlights are used sparingly, usually in simple rectangular or diamond grid patterns.

Figure 75 Examples of Federation style bungalows

2.3 Interwar Modern – c1915 – c1940

2.3.1 California Bungalow

This earthy and unpretentious style was based on a distinctive type of rustic, Japanese-influenced single storey detached house. This developed in the Los Angeles suburbs to become the standard unit of US west coast suburbia by the outbreak of the First World War. The Australian version also became the most common housing style of its time, being embraced by speculative builders from the early 1920s until the Depression. Willoughby has a rich variety of California Bungalows.

The hallmarks of the standard bungalow are a visually dominant, low-pitched spreading roof, with wide overhanging eaves, exposed rafters and barge boarded gables facing the street. This is on a simple rectangular house plan, with walls of dark red or purplish, liver-coloured face brick. A distinctive feature is the deep, shady verandah under a low pitched or flat roof that is supported by substantial masonry piers, sometimes with squat colonettes or grouped timber posts.

In Willoughby, orange-red Marseilles pattern tiles most commonly cover the main roof, while unobtrusive corrugated iron, or usually bituminous felt, is used on flat veranda roofs.

Bargeboards are visually prominent but plain, often with raked edges to accentuate the sweep of the roof. Simple designs of strap work, shingling and ventilators are common end gable infills. External timber joinery is generally plain, compared with Federation decoration, and painted. Window frames, often mounted on the outside face of the wall, sometimes have skirts of shingles or boards.

Occasionally walls are pebbledash though pebbledash accents are more common, for example, on heavy tapered masonry verandah pylons or as infills on low veranda walls.

Smooth river stone detailing, such as verandah supports or as an external chimney wall, is very rare in Willoughby. But rusticated sandstone is frequently used in foundations, decorative masonry finishes and low walls.

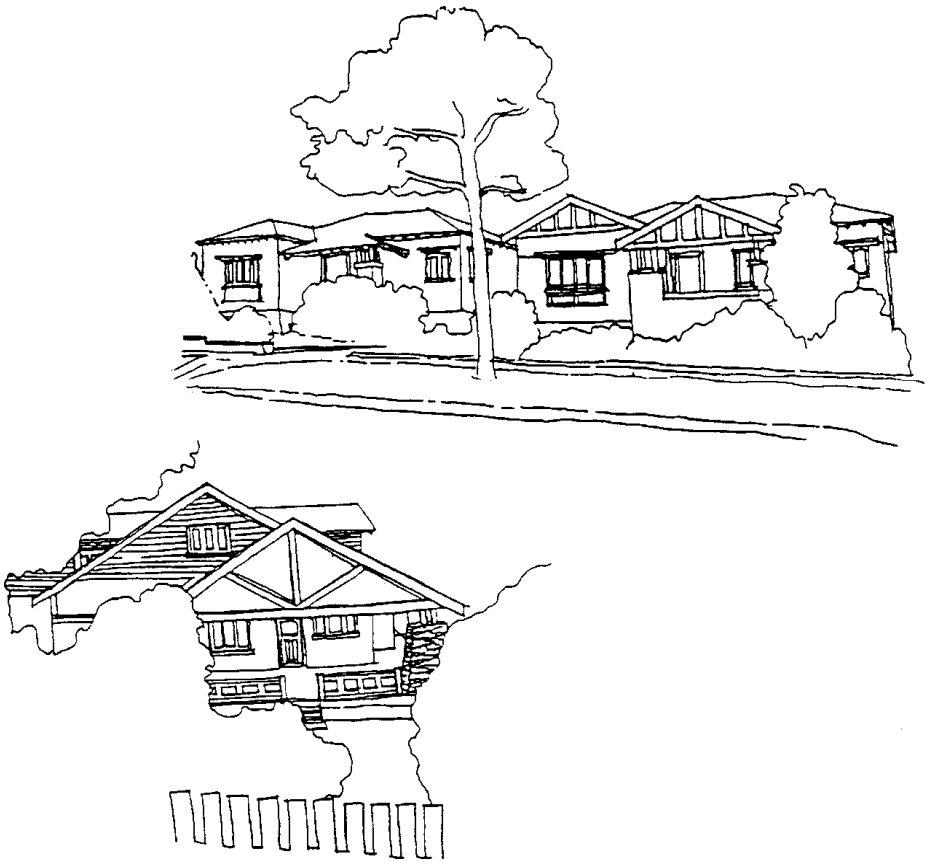
The bungalow usually has only one prominent main fireplace to the living room/main bedroom, with a second chimney to the kitchen/laundry, compared with the multiplicity for earlier styles, and the dominant chimney is sometimes placed externally on or near the street elevation. Window openings are usually grouped, especially on street façades and to main rooms, with casement sashes favoured, sometimes with stylised geometric leadlight glazing having coloured glass, often blue, accents. Half-glazed, usually single, entry doors are typical.

Figure 76 Examples of California bungalow style

A separate garage, typically for a single car, began to appear to the side and behind the house in the backyard. This was usually a small unobtrusive shed with double hinged timber doors, either lightly framed in timber or steel with a corrugated iron roof or in the same brick and tile as the main house.

Post and wire fences replaced the timber picket fences associated with the former Victorian and Federation styles. Some Willoughby houses still display the timber pergola entryways typical of the era.

Figure 77 Further examples of California bungalow style



2.3.2 1930s Bungalow

The Depression ushered in a period of more austere buildings: the bulk of Willoughby's small houses from this period are usually fairly plain dark red to liver coloured brick bungalows with hipped (concrete or Marseilles pattern) tile roofs.

Massing is still informal but the roof is lower and visually less dominant than the standard California Bungalow. The distinctive gables facing the street are replaced by a simple hipped roof, which sometimes has a smaller secondary hip over a small centrally placed verandah or porch. In others, the main roof is gabled with the ridgeline parallel to the street. Eaves decreased in size: exposed rafters were replaced by boxed eaves. Chimneys became lower, narrower and less decorative. Sometimes the colour of the roof became an important decorative element with variegated multi-coloured tiles.

The deep verandah of the California Bungalow also decreased in size during the 1930s until it disappeared to be replaced by a recessed entry – sometimes a porch with decorative brickwork or slender Georgian style flanking columns. In transitional styles, the formerly open verandah of the California Bungalow was often closed-in with glazing similar to the rest on the front façade.

Sometimes the unpainted face brick walls feature clinker or textured bricks, with occasional herringbone patterns or chequered brick nogging. Rusticated sandstone under crofts, which occasionally house single car garages, are common on sloping sites and some sandstone detailing, often on columns, occurs.

Windows and doors became visually important decorative elements in what was otherwise an often plain façade in the later houses. Boxed casements, sometimes bayed and supported on corbelled brickwork, gave way to recessed double hung timber framed windows. Leadlight glazing, often in geometric patterns with light blue, opaque white or textured glass accents, was one of the few decorative features.

Earlier houses generally have half-glazed single entry doors, some with textured glass panels. Double glazed entry doors and French doors also occur while the later plainer bungalows tended to have more solid doors with smaller glazed areas.

A single car garage became standard: this was generally provided as a separate building behind the house – usually a simple shed with double opening, sometimes half-glazed, timber doors – or sometimes as a low wing to the side of the house. Occasionally, where lot width or topographic restrictions discouraged these more typical options, the garage was built in to a front boundary wall or incorporated as a front projecting wing of the house itself.

Figure 78 Example of a 1930s bungalow

The 1930s Depression also saw the period revival styles introduced in the 1920s in the houses still being built for the middle and upper classes. Willoughby has a few simple, symmetrical rectangular Interwar Georgian Revival houses with their restrained face brick or plain stucco walls and moderately pitched, tiled hipped roofs.

Generally, however, this and other revival styles are found in Willoughby less as clear examples in themselves than as influences modulating the more common ‘spec built’ 1930s brick bungalow. For example, Georgian columns may decorate the portals of the entry porch of an otherwise plain liver brick bungalow or the Spanish Mission style may be seen in the three half-arched windows on a red brick façade.

Many bungalows show one or more aspects of the popular Interwar Old English (Tudor revival) style. This style has asymmetrical massing, medium pitch gabled roofs usually of tiles (sometimes in variegated colours) with timber bargeboards and imitation half-timbered gable infills, tall brick chimneys, or textured clinker face brick walls with herringbone or chequered brick nogging, bay windows, casement sashes with diamond pattern leadlight glazing, and arched entry porches.

The final expression of the speculatively built popular 1930s bungalow incorporated a mix of revival styles usually in simplified form.

Figure 79 Further examples of 1930s bungalows

2.3.3 Spanish Mission

The Spanish Mission style, developed in California after the 1890s, created an image of a romantic past, which was popularised in the Interwar period by Hollywood stars who used it for their luxurious, well-publicised homes. The style imported to Australia was seen as an attractive option during the Depression for a ‘middle class’ house, a cinema or even a service station.

Spanish Mission houses have generally asymmetrical massing, round-headed arched openings (usually in groups of three) of, for example, windows onto loggias, which are another distinctive feature. The arches are supported by either twisted ‘barley sugar’ or chevron decorated baroque columns or plain, heavy piers.

Walls are stucco with an exaggerated texture ‘to simulate peon-built adobe masonry’, usually cream painted. The medium pitch hipped or gabled roofs often have half-round orange terracotta tiles of Roman or Spanish flavour: the chimney and gables sometimes have a tile capping, with stucco finished up to the underside of the tiles edging decorative parapets. The building is often remarkable for the amount of ornament, including black wrought iron work, for example, as lanterns, balustrades or open work screens in arched window or loggia openings. There are sometimes ledged and boarded window shutters and heavy timber doors with decorative wrought iron screens over their small glazed lights.

Figure 80 Examples of Spanish Mission style

2.3.4 Functionalist

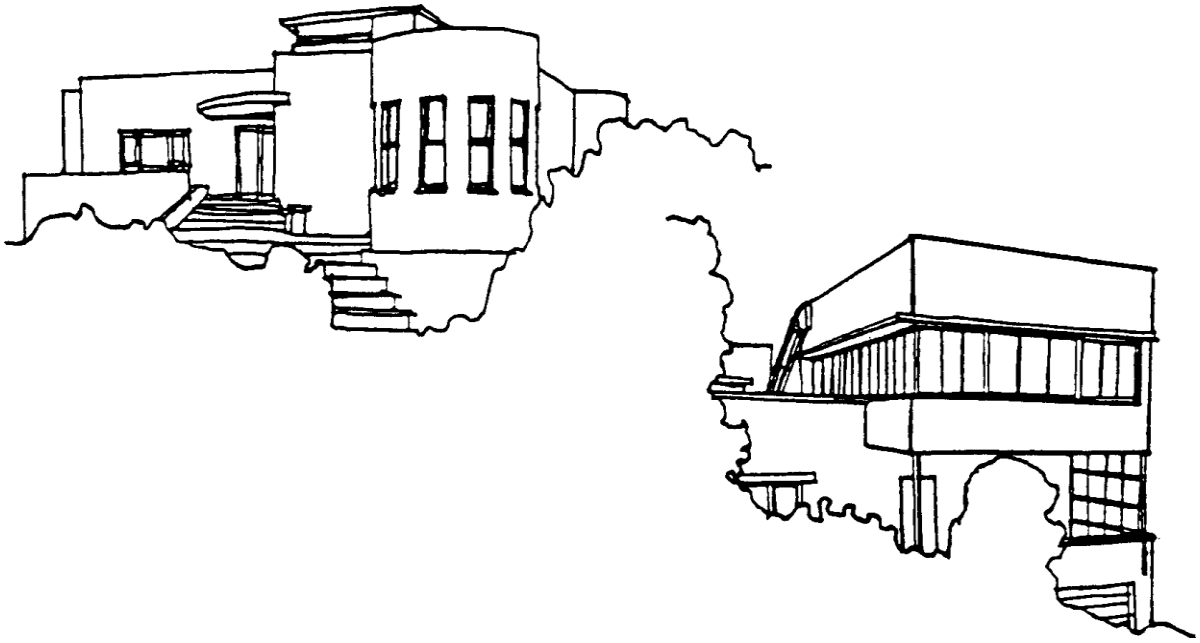
The Interwar Functionalist style was a major departure which had its roots in Europe and in the idea that the designed object should be 'functional' above all else. It was the forerunner of the international styles which developed more vigorously after the Second World War. While not a widely popular movement, the distinctive ‘ocean liner’ or ‘waterfall’ house was its mainstream expression in domestic architecture for the middle and upper classes. Willoughby has a scattering of these as well as a few more radical, unadorned Bauhaus influenced ‘modern’ houses though most of these are from the post war period.

The Functionalist style is characterised by an asymmetrical massing of simple geometric shapes and contrasting horizontal and vertical motifs. Major changes in materials were the use of steel and reinforced concrete to achieve wide spans, continuous windows and cantilevered balconies, hoods, roofs (even in domestic buildings) and metal framed, usually steel windows, also in large horizontal expanses. The horizontal ‘ribbon’ window was a major stylistic change from the vertical openings of earlier styles. Glass bricks are also used.

The roof is often concealed by a parapet. Walls usually have plain surfaces of light toned cement or face brick. While one strand of this style emphasises sharply rectangular geometric lines, the more usual ‘ocean liner’ or ‘waterfall house’ has curving shapes, rounded corners, including curved glass windows, wrapping around corners.

The most common expression of the ‘waterfall house’ was a brick cottage with horizontal rounded corners on its front façade, alternating bands of brick and cream stucco, horizontal steel casement windows on its corners, porthole accent windows (including a porthole on the front door) and boxed flush eaves to its low pitched hipped roof of cement tiles.

Figure 81 Examples of Functionalist homes



2.3.5 Modernism

Modernism followed the dictum of ‘less is more’. The modern house was designed to be functional and streamlined with a minimum of decoration, often painted white. Typically it had a lightweight structure, large expanses of glass and a flat roof. Internal planning was based on a geometric grid. It observed the landscape, typically from an elevated terrace or balcony, rather than engaged directly with it.

Modernist architecture has these features:

- little or no ornamentation
- factory made parts
- man-made materials such as metal and concrete
- emphasis on function
- rebellion against traditional styles.

2.3.6 The Sydney School

This group of architects emerged during the late 1950s.

The members of the Sydney School had a desire to work with the native Sydney landscape, designing buildings that responded closely to the sites and maintained strong links between internal spaces and the natural world. The sites were usually secluded bushland blocks, often steeply sloping. Their favoured palettes were economical and simple materials; they often used exposed timber beams, floorboards and clinker bricks which blended with the colours of the bush.

Influenced by Frank Lloyd Wright, Japanese architecture, Brutalism and the Arts and Crafts movement, the Sydney School tended to leave the materials and structure of a building exposed.

The Sydney School house often could not be seen from the street. It was married to the landscape, sometimes planned around a feature, such as a rock or tree. It was oriented to light and prevailing breezes. Sydney School architects loved natural materials – typically, brick or timber left in near-original state.

3 Useful publications and references

Heritage consultants

Heritage NSW has a directory of heritage professionals and consultants at heritage.nsw.gov.au

How the Australian House Developed

Architecture in Australia: A History by J.M. Freeland (Cheshire, Melbourne, 1968)

Australia's Home: Its Origins, Builders and Occupiers by Robin Boyd (Melbourne University Press, Melbourne, 1961)

Identifying Types of Buildings

A pictorial guide to identifying Australian Architecture by Richard Apperly, Robert Irving and Peter Reynolds (Angus & Robertson, Sydney, 1989)

Restoring Houses

Getting the details right: restoring Australian houses 1890s – 1920s, Department of Planning (The Flannel Flower Press, Queensland, 1989)

The Federation house: Australia's own style by Hugh Fraser (Landsdowne, Sydney, 1986)

‘The Federation house: a restoration guide by Ian Evans (Flannel Flower Press, Sydney, 1983)

Conservation of Federation houses – Seminar, 16 May 1981, Heritage Council of New South Wales (ISBN 7240 4501 5)

Australian houses of the 20s and 30s by Peter Cuffley (The Five Mile Press, Fitzroy, Victoria, 1989)

The California Bungalow in Australia by Graeme Butler (Lothian Books)

The Australian Home by Ian Evans (Flannel Flower Press, Sydney, 1983)

Colour Schemes

Colour Schemes for Old Australian Houses by Ian Evans, Clive Lucas and Ian Stapleton (The Flannel Flower Press, Sydney, 1984)

Gardens

Traditional Gardens in Australia by Peter Cuffley (The Five Mile Press, Balwyn, Victoria, 1991)

Interwar Gardens: a guide to the history, conservation and management of gardens of 1915 – 1940, The National Trust of Australia (NSW) Parks and Conservation Committee

Useful websites

- Willoughby City Council: willoughby.nsw.gov.au
- Heritage NSW: https://www.environment.nsw.gov.au/research-and-publications/publications-search
- World of Old Homes: oldhouses.com.au

PART I

Stormwater Management

Part I: Stormwater Management

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1 Introduction

As impervious areas increase due to continuing urban development there is greater pressure on both the natural waterways and the built drainage systems. Our council stormwater drainage systems (including the natural streams and creeks) are operating at full capacity. Apart from environmental and pollution consequences, the increase in impervious surface areas leads to more urban flooding and further costs to maintain and replace council stormwater drainage infrastructure.

To mitigate these impacts, Willoughby City Council requires most new developments to install onsite detention (OSD) systems to reduce and control the extent of stormwater runoff from a site during storm events.

OSD systems control stormwater peak flows from a site by using tanks or other types of storage systems to temporarily store stormwater. This discharges into the council drainage system at a controlled rate.

These systems are required for all developments. However, rainwater reuse tanks may be installed to offset the requirement for OSD in minor developments such as dwellings, dual occupancies, boarding houses not exceeding 300m^2 and 12 persons, and secondary dwellings. The stored rainwater can be used for washing cars, watering lawns and gardens, topping up chemically treated swimming pools, toilet flushing and washing clothes.

There are three technical standards relevant to water management in Attachments 1, 2 and 3 of this part. A summary of the technical standards is in Section 6 of this part.

1.1 Aim

The aim of this part is to increase the number of relatively small low-cost storage systems on individual properties. This will help limit the peak discharge to predevelopment flow rates for all storm intensities and durations, including a 1% annual exceedance probability (AEP) storm event.

1.2 Objectives

The objectives of this part are to:

a. provide a safe and effective framework for the control, reuse and disposal of stormwater

b. reduce flooding risk in urban areas and protect council stormwater drainage infrastructure

c. maintain public health and safety

d. use water resources efficiently

e. encourage water sensitive urban design (WSUD) measures to minimise impacts on the natural water cycle and foster ecological sustainability

f. improve water quality and protect the scenic landscape and recreational values of bushland, natural watercourses and receiving waters

g. prevent, mitigate and control land degradation

2 Minor developments

2.1 Rainwater reuse tanks for minor developments

Minor developments include construction of, and alterations and additions to:

- single dwellings
- attached and detached dual occupancies
- boarding houses not exceeding 300m^2 and 12 persons
- secondary dwellings.

Alterations and additions include ancillary structures such as garages, carports, sheds, studios, swimming pools, studios, gazebos and the like.

Rainwater reuse tanks are an effective method of capturing and disposing of stormwater for minor developments that discharge directly into the council stormwater drainage system. The tanks help reduce the extent of overland flow and the risk of flooding in urban areas. They also provide cost savings for households by reducing the consumption of water from the reticulated system.

To promote the installation of rainwater reuse tanks for minor developments, Willoughby City Council will discount all or part of the OSD storage capacity requirement. This is in cases where the overflow from the rainwater tank can discharge by gravity into the council stormwater drainage system.

Further OSD systems may be required if the at-grade impervious surface areas cannot drain by gravity into the council stormwater drainage system and there is likely to be inundation to downstream properties.

Using rainwater solely for outdoor purposes such as watering the garden is a poor use of the stored rainwater. This is due to mismatches between the seasonal rainfall and outdoor water usage patterns. For example, during extended wet periods the stored rainwater is unlikely to be used for outdoor purposes such as watering gardens or washing cars.

To maximise the reuse of rainwater, the rainwater tank should be plumbed to indoor facilities such as toilets and laundries. This will ensure a steady rate of use regardless of seasonal rainfall. A combination of indoor and outdoor use of stored rainwater optimises the conservation of the potable water supply and provides some capacity for OSD.

This means that all rainwater reuse tanks must be plumbed to bathrooms/WC for toilet flushing and laundries for washing clothes, unless stated otherwise.

2.2 Size of rainwater reuse tanks for minor developments.

Table 1 Size of rainwater reuse tanks for new dwellings

Item	New dwellings	Requirement for sites that fall directly to a street or via an easement	Requirement for sites with a charged system or onsite disposal system
1	New dwellings on lots greater than 400m^2.	10kL	10kL
2	New dwellings on lots less than 400m^2.	5kL	5kL

Table 2 Size of rainwater reuse tanks for new dual occupancies and boarding houses

Item	New dual occupancies and boarding houses	Requirement for sites that fall directly to a street or via an easement	Requirement for sites with a charged system or onsite disposal system
3	New attached and detached dual occupancy on potential lots greater than 400m^2/dwelling.	10kL per dwelling	10kL per dwelling
4	New attached and detached dual occupancy on potential lots less than 400m^2/dwelling.	5kL per dwelling	5kL per dwelling
5	New attached dual occupancy on potential strata subdivision lots.	5kL per dwelling	5kL per dwelling
6	New boarding houses (not exceeding 300m^2 and 12 persons).	10kL	10kL

Table 3 Size of rainwater reuse tanks for new attached dwellings

Item	New attached dwellings	Requirement for sites that fall directly to a street or via an easement	Requirement for sites with a charged system or onsite disposal system
7	New attached dwellings on potential lots greater than 400m^2/dwelling.	10kL per dwelling	10kL per dwelling
8	New attached dwellings on potential lots less than 400m^2/dwelling.	5kL per dwelling	5kL per dwelling

Table 4 Size of rainwater reuse tanks for alterations and additions

Item	Alterations and additions (including attached secondary dwellings)	Requirement for sites that fall directly to a street or via an easement	Requirement for sites with a charged system or onsite disposal system
9	Alterations and additions to a minor development that involves significant demolition on lots greater than 400m^2.(i)	10kL	10kL
10	Alterations and additions to a minor development that involves significant demolition on lots less than 400m^2.(i)	5kL	5kL

Item	Alterations and additions (including attached secondary dwellings)	Requirement for sites that fall directly to a street or via an easement	Requirement for sites with a charged system or onsite disposal system
11	Alterations and additions to a minor development that will result in more than 60% impervious areas on lots greater than 400m^2.	10kL	10kL
12	Alterations and additions to a minor development that will result in more than 60% impervious areas on lots less than 400m^2.	5kL	5kL
13	Alterations and additions to minor developments with an estimated cost of more than $50,000 and impervious area between 50 – 60%.	5kL	5kL
14	Alterations and additions to a minor development with an estimated cost that is less than $50,000 and impervious area between 50 – 60%.	3kL(ii)	5kL
15	Alterations and additions to a minor development that do not result in impervious areas more than 50%.(iii), (iv)	none	5kL for lots >400m^2 3kL for lots <400m^2
16	Despite items 11, 12, 13 & 14 above, alterations and additions to minor developments that do not result in more than 5% over the existing impervious areas, and the area of works does not exceed 25m^2.(iii), (iv)	none	5kL for lots >400m^2 3kL for lots <400m^2

Table 5 Size of rainwater reuse tanks for detached secondary dwellings

Item	Detached secondary dwelling	Requirement for sites that fall directly to a street or via an easement	Requirement for sites with a charged system or onsite disposal system
17	New detached secondary dwellings and alterations and/or additions to convert an existing out buildings/studio to a detached secondary dwelling.	3kL	5kL

Table 6 Size of rainwater reuse tanks for other controls

Item	Other controls	Requirement for sites that fall directly to a street or via an easement	Requirement for sites with a charged system or onsite disposal system
18	Irrespective of the development type, a rainwater reuse tank is not required for sites that drain directly into Middle Harbour or Lane Cove River.(v)	none	none
19	Irrespective of the development type, a 5,000L rainwater reuse tank is required for sites that drain into a natural watercourse, creek or bushland.(v)	5kL	5kL

Notes from tables:

i. Demolition is ‘significant’ if the extent of demolition is more than 50% of the external fabric of the existing building.

ii. The rainwater reuse tank is only required to be plumbed to outdoor facilities.

iii. A 3,000-litre rainwater tank may be required under the State Environmental Planning Policy (Building Sustainability Index: BASIX) 2004 for any development with an estimated cost of $50,000 or more or a swimming pool that has a volume of more than 40,000 litres.

iv. Where a rainwater reuse tank is not required, if the site cannot drain by gravity into the council stormwater drainage system, measures such as infiltration and/or onsite detention systems may be needed. These are to ensure downstream properties are not affected by stormwater runoff from the subject property.

v. For more information on approvals and guidelines to drain directly into Middle Harbour, Lane Cove River, natural waterways, creeks or bushland, see technical standard 1 – stormwater management in Attachment 1.

Additional notes:

- Any variation to the size of rainwater tanks must be justified. This will be assessed case by case.
- A written request to vary the size of the rainwater reuse tank and/or the plumbing requirement to indoor facilities must be prepared by a suitably qualified person.
- For more information OSD systems and the installation and discharge of overflow for rainwater reuse tanks, see technical standard 1 – stormwater management in Attachment 1.
- The table in this section provides the minimum size of rainwater tanks for minor developments. Households are encouraged to provide larger rainwater reuse tanks if adequate space is available.

2.3 General submission requirements

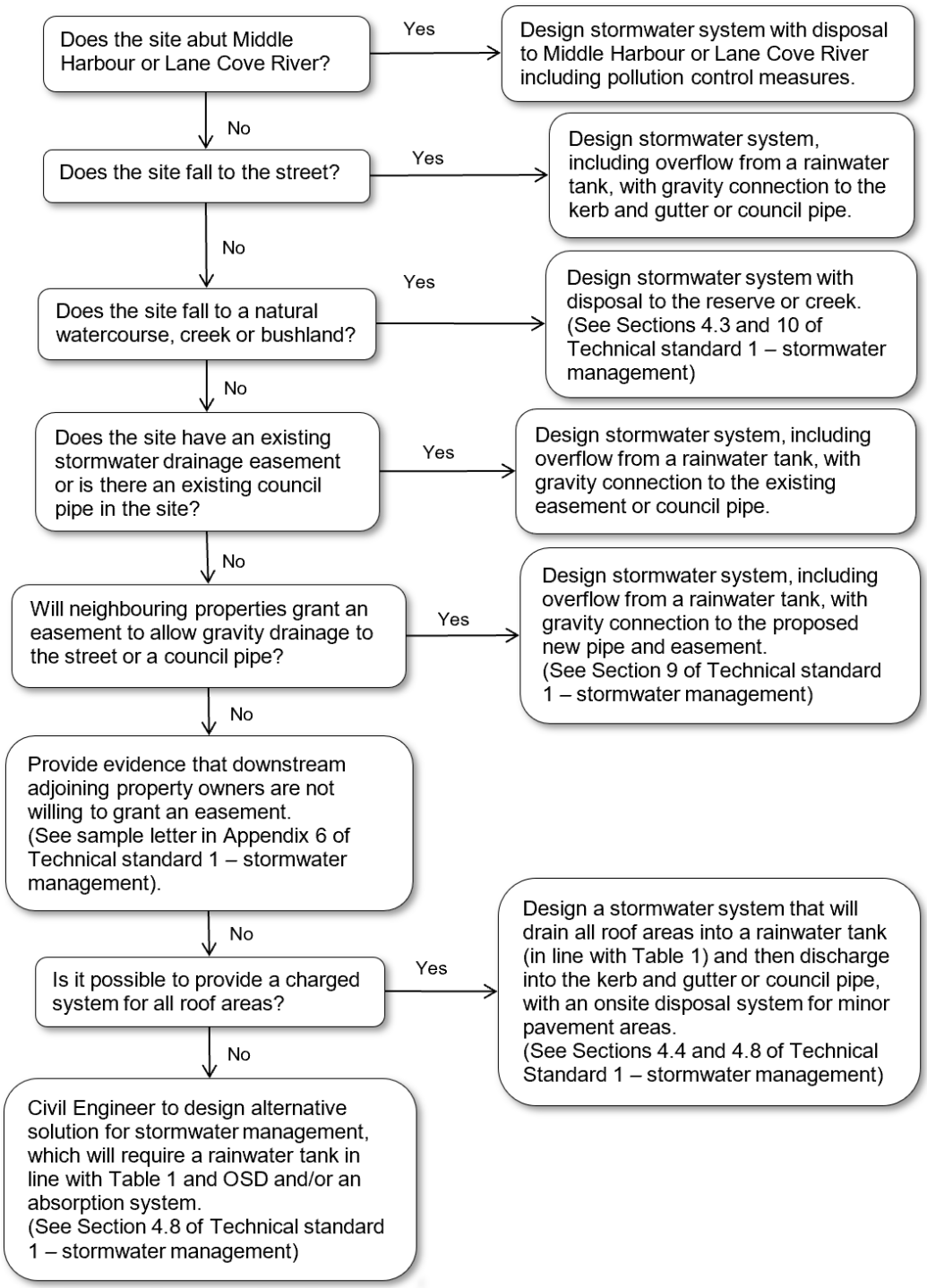
All applications for minor developments must include stormwater drainage details and an impervious area calculation plan prepared by a suitably qualified person. ‘Impervious areas’ include all existing and proposed new built structures such as roofs (measured to the eaves, including any gutters), driveways and paths, and paved areas. Swimming pools are also impervious areas.

Open slatted timber decks may be excluded if the area below the deck is unpaved.

Other partially permeable surfaces such as gravel driveways are included in the impervious area calculation. This is because these areas may be concreted or paved at a later date under State Environmental Planning Policy (Exempt and Complying Development Codes) 2008 without Willoughby City Council approval.

2.4 Flow chart for the design and discharge of stormwater systems for minor developments

Figure 1 Flow chart for the design and discharge of stormwater systems for minor developments



2.5 Controls for rainwater reuse tanks

All development application for minor development or an application for a complying development certificate (CDC) must comply with these controls:

a. above ground rainwater tanks should not exceed 2.4m in height above the existing ground level; this height restriction includes any stand for the tank

b. rainwater tanks must be located at least 450mm from any property boundary

c. rainwater tanks must be positioned or screened so as not to be visible from the street frontage

d. rainwater tanks must be in a neutral, non-reflective colour to ensure tanks are compatible with the character of the locality and do not impact on adjoining buildings

e. above ground tanks must be built to ensure the area directly beneath any outlet tap and/or drainage plug is contained (or bunded) and drained to the nearest stormwater drain

f. rainwater tanks must not be installed over or immediately adjacent to a water pipeline, sewer pipeline or council stormwater pipeline or easement, unless it is installed with approval and in line with any requirements of the public authority that has responsibility for the pipeline

g. rainwater tanks are not to be located within overland flow paths or flood zones

h. minor developments may include OSD systems instead of or as well as the requirement for rainwater reuse tanks; however, the OSD system must be designed by a suitably qualified engineer

i. rainwater reuse tanks may be located below the building footprint, such as voids and below decks, providing they are accessible for maintenance and replacement; any tank located below the building footprint must be designed to ensure that any leakage can drain to a point outside of the building to avoid any flooding of habitable rooms

j. a sign measuring no less than 400mm x 200mm is to be permanently attached and displayed within the immediate vicinity of the rainwater reuse tank/s. The wording should state:

   ‘This rainwater retention and reuse system is required by Willoughby City Council. It is an offence to alter any part of the system without written consent from the council. The registered proprietor should keep the system in good working order by regular maintenance including removal of debris.’

k. a ‘charged system’ for rainwater reuse tanks for minor developments is permissible if the overflow can drain directly into the street or via an easement (see Section 4.4 of technical standard 1 – stormwater management)

l. if the overflow from a rainwater reuse tanks cannot drain directly under gravity into a street, or discharge into a watercourse or bushland, an applicant may need to obtain a downstream easement or connect into an existing inter-allotment drainage easement (see Section 9 of technical standard 1 – stormwater management)

m. if an easement cannot be obtained or the property cannot connect into an existing inter-allotment drainage easement, alternative measures may be needed to mitigate impacts

on downstream properties; this may include installing OSD tanks and rainwater tanks together with other measures such as onsite disposal systems (see Section 4.8 of technical standard 1 – stormwater management)

n. any variation to the controls for the installation of rainwater reuse tanks will need to be justified and assessed on its individual merits; a written request to vary the control must be prepared by a suitably qualified person

o. if multiple rainwater tanks are provided, and are not interconnected, the roof area draining to each tank must be proportionate to the tank size

3 Major developments

3.1 Onsite detention (OSD) for major developments

All developments that are not seen as 'minor' are deemed to be 'major' development, including:

a. residential developments for: multi dwelling housing; attached dwellings; boarding houses exceeding 300m^2 and 12 people; hostels; residential flat buildings; and seniors housing

b. commercial, industrial, institutional, shop top housing and mixed-use developments, and recreation facilities

c. all development that generate a water demand in excess of 5,000 litres per day

Note**:**

- Institutional development includes: community facilities, educational establishments, child care centres, places of public worship, and public administration buildings.

All major developments must provide OSD systems designed to capture and detain stormwater runoff for all storm events up to and including the 1% AEP storm event. The OSD system must be in line with Willoughby City Council's technical standard 1 – stormwater management and AS/NZS3500.3.

The statement of environmental effects (SEE) and stormwater management plan included with a development application must address water sensitive urban design (WSUD) and recycling and pollution control measures (see technical standard 1 – stormwater management).

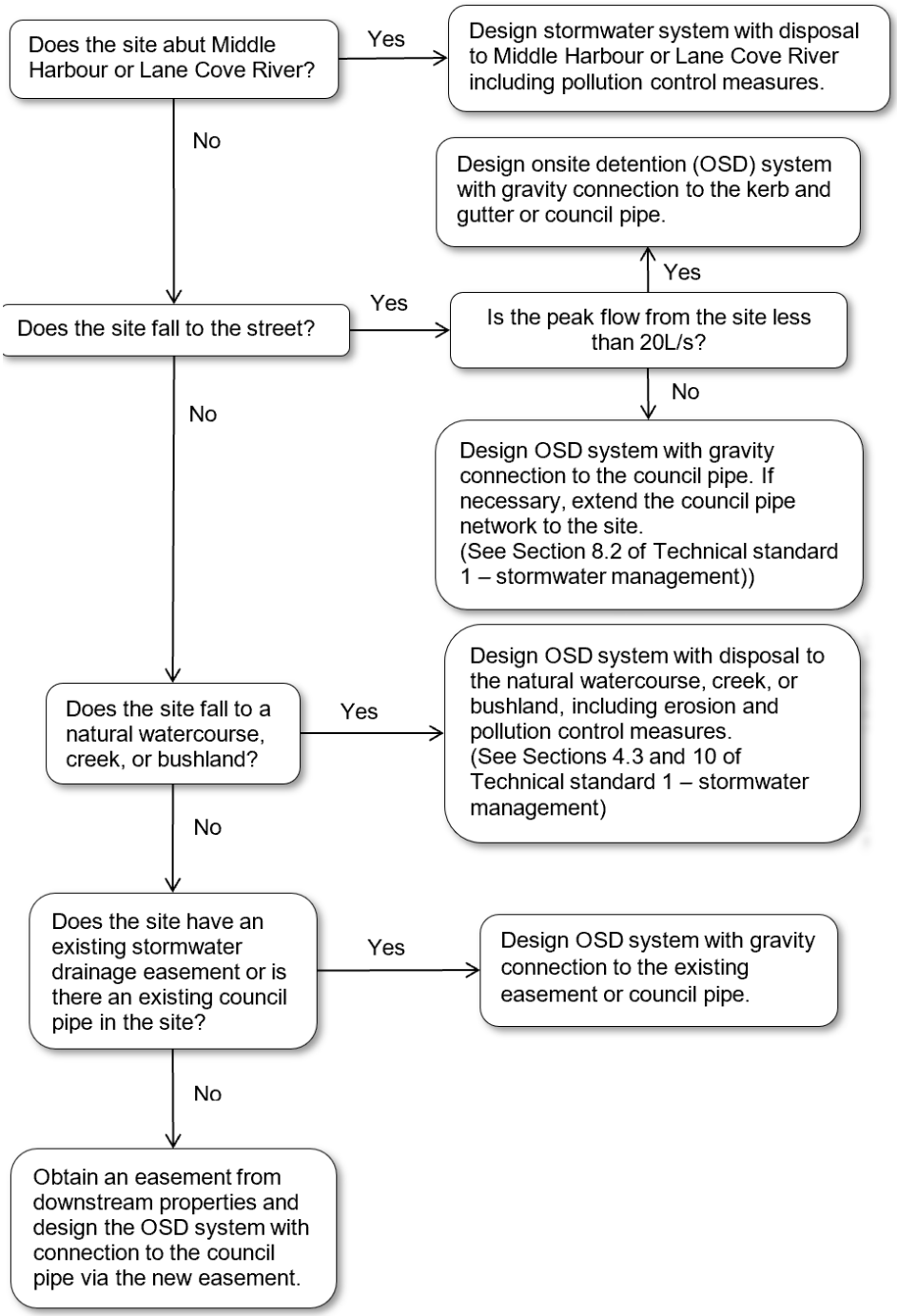
The stormwater management plan must be prepared by a suitably qualified engineer.

3.2 Exceptions

Onsite detention is not required if Willoughby City Council is satisfied that the discharge of stormwater from a property does not pass through any council owned drainage infrastructure before reaching the receiving waters of Middle Harbour or Lane Cove River. Drainage infrastructure includes any pipe, culvert, lined channel or other restrictive structure. However, certain requirements apply for the discharge of stormwater into natural watercourses or bushland (see Section 4.3 and Section 10 of technical standard 1 – stormwater management).

3.3 Flow chart for the design of stormwater systems for major developments

Figure 2 Flow chart for the design of stormwater systems for major developments



3.4 Discharge of stormwater

Stormwater is required to discharge via gravity, with pipes or channels having a minimum grade of 1% fall. If the overflow from an OSD system cannot drain directly under gravity into a street or an existing easement or discharge into a watercourse or bushland, an applicant must obtain a downstream easement or connect into an existing inter-allotment drainage easement (see Section 9 of technical standard 1 – stormwater management). For flows greater than 20L/s, it may be necessary to extend the council pipe system to the site at the developer's expense.

3.5 Water quality

For all major developments, water quality improvement measures are to be provided in line with technical standard 1 – stormwater management.

3.6 Water Sensitive Urban Design (WSUD)

An integrated approach to the management of stormwater in urban areas is the use of WSUD measures. It is one way we can protect council stormwater drainage infrastructure and make a significant contribution toward sustainability.

WSUD integrates the management of the natural soil and water resources on a development site by retaining or restoring natural site features. All major developments must consider and, where practical, implement appropriate WSUD measures.

A detailed explanation of WSUD by Sydney Water is at sydneywater.com.au.

Landscaping can play an important role in achieving WSUD to help control stormwater pollution and reduce stormwater runoff.

This includes:

- planting native grasses, groundcovers or mulched garden beds
- the use of plants with a high water demand to filter nutrients and reduce runoff
- the construction of depressions, swales, contour banks, rock channels, pebble paths or similar measures to capture and retain runoff

A civil engineer and landscape architect can assist in providing appropriate WSUD measures to include in the landscape design of a proposed development.

3.7 Further information

For some larger or more complex developments, the stormwater management plan should include a comprehensive water cycle management strategy.

This includes developments that:

a. incorporate 15 or more dwellings

b. accommodate 50 or more residents, occupants or employees

c. generate a water demand in excess of 5,000 litres per day

d. involves the creation of 2,500 square metres or more of impervious surface areas

e. involves the subdivision of 2,500 square metres or more of land for commercial or industrial purposes.

The plan should include:

f. the site constraints and the existing environment within its catchment context

g. design principles, objectives and performance standards for water cycle outcomes (post-development stormwater volume discharged from the site during a typical rainfall year should not exceed 90% of the volume that would be expected if no measures were applied to reduce the stormwater volume)

h. water management measures to meet the objectives of this plan and associated guidelines and technical standards or policies adopted by other relevant government agencies

i. an infrastructure program that integrates all aspects of water cycle management

j. strategies to ensure effective ongoing maintenance of onsite water management measures, maintenance and regulatory requirements

k. arrangements to monitor and maintain the water management strategy to satisfy the objectives of this plan

For major and/or more complex developments, applicants are encouraged to have a pre-development application meeting to confirm the level of information required as part of the application.

4 Subdivision

Applications for subdivisions that create a new lot must include a concept stormwater plan. The plan is to show the proposed method of stormwater drainage from each lot in the proposed subdivision. The maximum flow to the kerb and gutter from the un-subdivided lot shall be 20L/s. All lots must have a gravity drainage system, with pipes and channels having a minimum grade of 1%. If easements are required over downstream properties, they are to be shown on the plans, and written advice provided to confirm that the downstream property owner(s) will grant an easement.

For subdivisions that cannot drain directly into the street or via an easement, an OSD system must be provided on the site before the plans are registered. Details of the OSD system must be shown on the concept plans.

Note:

- For subdivision of an approved dual occupancy or attached dwellings where no building or ancillary works are proposed, no further stormwater measures are required.

5 Fences

In areas of overland flow, construction of fences must allow for the natural flow of stormwater. This may require fencing to be an open type construction (eg, palisade, mesh) or raised above the finished ground level to allow the free flow of stormwater. The open style fencing or open area to allow passage of stormwater, must extend from the ground level to a minimum of the 1% AEP flood level. Construction of any such fencing must remain safe

during floods and not obstruct moving debris. Details are to be submitted with the development application.

6 Applications for development

6.1 Site specific requirements

In preparing a development application or an application for a complying development certificate (CDC), please check the relevant technical standards for a stormwater management system. These vary depending on the development and discharge conditions.

The information included with the application should consider these questions:

a. is the development at risk from slope instability, reactive soils, erosion hazards, acid sulphate soils, land contamination or archaeological relics?

b. is the land affected, or potentially affected by flooding from stormwater drains, overland flow paths, drainage easements, watercourses or open channels?

c. does the site drain to adjacent land which is a watercourse, riparian community, bushland or reserve?

d. Does the development propose to cut into the natural surface or change the natural surface levels?

e. does the site fall to the rear and/or into neighbouring properties?

If the answer to any of these questions is ‘yes’, a suitably qualified engineer must be engaged to prepare a stormwater management plan. The design of this plan must:

f. consider ground levels and the location of existing stormwater drains when designing new drainage facilities

g. consider potential site areas which may maximise rainwater and stormwater collection for reuse

h. consider potential recycling facilities and reuse options

i. consider the potential use of existing landscape features as part of the proposed stormwater source controls

j. identify how the design responds to the site constraints and opportunities

Applications for minor developments that drain into a natural waterway or bushland must include all relevant information and a stormwater management plan prepared by a suitably qualified person.

6.2 Soil and water management plan

All applications must address sediment and erosion control measures. A soil and water management plan specifying the proposed measures to control erosion and pollution sources of water both during and after the construction phases will be required where:

a. a proposed development will expose a soil surface area greater than 250 square metres

b. the site adjoins public open space or a watercourse

c. the site is on a slope exceeding 18 degrees

d. involves cut or filling of the land which will alter the rate, volume or direction of overland flow

e. the development contains 10 or more car parking spaces

For more information see technical standard 3 – sediment and erosion control.

7 Technical standards

7.1 General

Willoughby City Council has three technical standards. These include details relating to stormwater management, floodplain management, sediment and erosion control, and water quality. These technical standards deal with the provision of onsite detention (OSD) systems, rainwater reuse tanks, flood mitigation, and erosion and pollution control measures.

The technical standards will assist proponents to provide the appropriate level of information to satisfy the aims and objectives of this plan. These documents are provided in Attachments 1, 2 and 3. A summary of each technical standard is provided below:

7.2 Technical standard 1 – stormwater management (Attachment 1)

This document provides guidelines and requirements for stormwater management systems, including OSD, rainwater tanks and discharge requirements. It also includes requirements for water quality improvement measures.

All developments must provide OSD systems that temporarily store stormwater before this is discharged into the council drainage system.

For minor developments, Willoughby City Council encourages the installation of rainwater reuse tanks to offset the requirement for OSD. Rainwater reuse tanks are required to capture roof water to be used for non-potable purposes such as washing cars, irrigating gardens, topping up chemically treated swimming pools, flushing toilets and washing clothes.

This document includes requirements for the installation and discharge of overflow from a rainwater reuse tank.

Note:

- A Positive Covenant and Restriction on Use of Land is not generally required for rainwater reuse tanks for minor developments. However, some minor developments may require this covenant to ensure any overflow from a rainwater reuse tank or OSD system is retained and maintained to prevent inundation to downstream properties.
- To ensure OSD systems are maintained, the covenant must be put on the title of the subject property for major developments.

Major developments must incorporate pollution control measures to ensure an acceptable level of water quality during and after construction. This is to mitigate the amount of pollutants entering council stormwater drainage systems, which ultimately flow into Middle Harbour or the Lane Cove River.

7.3 Technical standard 2 – floodplain management (Attachment 2)

Applications must address the risk of inundation and flooding if the site is adjacent to a creek, drainage reserve, stormwater pipeline or within a low point. The assessment is based on the 1% annual exceedance probability (AEP) event, also referred to as the 1 in 100 years average recurrence interval (ARI) storm event.

A flood study, prepared by a suitably qualified person, must be submitted for developments deemed to be at risk so that the overland flow volume, depth, velocity and extent of inundation can be ascertained. Applicants of properties identified as floodprone land are encouraged to have a pre-development application meeting to confirm the level of information required with the application.

The flood study should address all requirements and the information under Section 12 of technical standard 2 – floodplain management.

Note:

- An applicant can obtain a Section 10.7 Certificate (previously a 149 Certificate) or contact Willoughby City Council's engineering division to confirm if a property is identified as floodprone land.
- If the property is identified as floodprone land, a flood information certificate is available from Willoughby City Council. The application form and fees are at www.willoughby.nsw.gov.au.

7.4 Technical standard 3 – sediment and erosion control (Attachment 3)

All applications involving earthworks or the generation of additional stormwater runoff must include a site management plan detailing the proposed method for sediment and erosion control measures. This document provides guidelines on simple, practical steps that can be taken to reduce the risk of polluting runoff from construction sites during storm events.

Attachment 1 – Technical standard 1 – stormwater management

1 Introduction

This technical standard provides detailed stormwater management requirements and controls to comply with Part I (Stormwater Management) under Willoughby Development Control Plan (Willoughby DCP).

Willoughby City Council encourages sustainable development to help preserve the natural environment and protect private and public land from the increased probability of being flooded due to upstream development.

The increase in impervious surfaces due to urban development also puts greater pressure and further costs on Willoughby City Council to maintain and replace its drainage infrastructure. To mitigate these impacts, we require most new developments to install onsite detention (OSD) systems to reduce and control the extent of stormwater runoff from a site during storm events.

The requirement for OSD systems for any new development on a site is addressed as part of the development assessment process. The appropriate method of onsite detention depends on the type, size and scale of the proposed development.

This technical standard provides information and controls to help limit the peak discharge to predevelopment flow rates for all storm intensities and durations up to and including a 1% annual exceedance probability (AEP) storm event.

2 Objectives

The objectives of this technical standard are to:

a. regulate development to ensure it does not contribute to increased flood risk

b. manage the change in discharges from development sites by source control

c. minimise adverse environmental impacts caused by increased stormwater runoff by reducing the total volume of runoff being discharged from individual properties

d. minimise the use of mains supplied water by encouraging water conservation through the reuse of rainwater

e. provide an effective framework for the installation and ongoing maintenance of OSD systems and rainwater reuse tanks

f. ensure compliance with all codes, guidelines and legislation relating to OSD and rainwater reuse tanks

g. minimise water pollution from new developments

h. protect the water quality of natural watercourses and receiving waters by preventing further deterioration of the water quality, riparian zones and aquatic ecosystems due to development

i. maintain public health and safety

To achieve these objectives and ensure OSD systems operate correctly, the maximum flow rate allowed to discharge from a particular site (permissible site discharge or PSD) must be controlled by a restriction such as an orifice plate. In this way, the collective use of OSD systems can reduce the peak discharge to try and match the capacity of the downstream drainage system. This will save the community money by not requiring larger pipe systems. It will also reduce damage to the environment and improve water quality by preventing increased levels of scour, erosion and sedimentation being transported downstream towards the estuaries.

3 Application of this technical standard

This technical standard applies to all development/building works that require approval under *Environmental Planning and Assessment Act 1979* or *Local Government Act 1993*.

In line with Part I of Willoughby DCP, development falls into two categories; ‘minor’ development and ‘major’ development. Most minor developments will only be required to install a rainwater reuse tank to offset the requirement for OSD.

4 General water management requirements

4.1 Stormwater design elements

4.1.1 Standards

Stormwater drainage systems are to be designed and constructed in line with AS/NZS 3500.3, Australian Rainfall and Runoff by Geoscience Australia, Willoughby City Council’s Design Standards (AUS-SPEC) and the National Construction Code.

4.1.2 Surface inlet pits and grated trench drains

Surface inlet pits and grated trench drains with flush fitting grates are to be installed where necessary within the site to collect stormwater runoff and direct it to the disposal location. They must be:

a. sized to be capable of accepting the design flows

b. located to prevent runoff from entering buildings and garages

c. located to prevent long term ponding of stormwater

d. located so that runoff does not affect pedestrian access to buildings and so that concentrated stormwater does not flow over any public footpath or adjacent properties

All required grated trench drains are to be a minimum width of 200mm. For driveways falling to the street, a grated trench drain is to be provided across the driveway adjacent to the property boundary.

A stormwater pit is to be provided adjacent to the property boundary, before discharge to the council drainage system or kerb and gutter, to ensure the system can be maintained. For single dwelling properties the pit is to be a minimum of 450 x 450mm internal dimensions. For all other developments the pit is to be 600 x 600mm.

All stormwater junction pits and kerb inlet pits installed in council road reserve and public open space must be designed and installed in line with Willoughby City Council's Design Standards and AS/NZS 3500.3. Pits are to be provided at:

- low points
- every change in pipe size, level, direction or grade
- junction points of multiple pipes

4.1.3 Stormwater pipes

The minimum allowable pipe size across the footway to the street is a 100mm diameter sewer grade UPVC pipe. This should have a converter through the kerb consisting of a 75mm high by 125mm wide by 4mm thick galvanised RHS (rectangular hollow section) steel, with a minimum grade of 1%.

When discharging directly to the kerb and gutter, the maximum velocity is 2.5m/s and the maximum flow is 20L/s. Any site with a discharge of greater than 20L/s is required to connect directly into the council's piped stormwater system via a pit and/or pipe.

All pipes within the council road reserve, other than direct connections to the street kerb and gutter, or public open space must be a minimum of 375mm diameter reinforced concrete pipe.

For stormwater pipes in the road reserve or public open space, the minimum pipe grade is to be 1%. Grades of 0.5% are only acceptable for pipes larger than 225mm diameter, where a grade of 1% is not achievable and the pipe can be shown to be self-cleaning. For pipes within private property, minimum grades shall comply with AS/NZS 3500.3.

Pipes with a gradient greater than 20% must have anchor blocks at the top and bottom of the inclined section and at intervals not exceeding 20m.

The depth of cover on all pipelines is to be in line with AS/NZS 3500.3 and manufacturer's guidelines.

4.2 Surface runoff retention tanks

Water stored in underground tanks that are designed to capture runoff from paved or other ground surfaces may be used for outdoor irrigation and other non-potable uses.

These systems must:

a. incorporate suitable treatment measures before storage in the tank, such as a first flush and filter system

b. be connected to a sub-surface or drip irrigation system rather than a hose tap, unless treatment systems are provided that meet requirements for contact with the water

c. not be connected to indoor water features without suitable treatment in line with the national Australian Guidelines for Water Recycling: Managing Health and Environmental Risks

d. have all fixtures connected to the system marked: 'Not suitable for drinking'

e. have the tank enclosed and all inlets screened to prevent the entry of foreign matter and to prevent mosquito breeding

f. have the tank sited in a location where it will not affect the structural integrity of any nearby buildings

g. have the overflow connected to the onsite detention system (if required) and to the site drainage system

h. be designed by a suitably qualified engineer

4.3 Discharge into natural watercourses or bushland

Approval is required to discharge stormwater, including the overflow from a rainwater reuse tank or OSD system, directly into a natural watercourse, creek or bushland reserve.

Uncontrolled water discharge into natural watercourses and creeks increases embankment erosion and streambed scour. Applicants must ensure that any watercourse or creek bank and bed is protected against erosion and scour at the point of discharge.

Uncontrolled water discharge into bushland increases erosion, weed growth and causes long-term degradation of the bushland. If an applicant has approval to discharge stormwater through bushland, the discharge must be conveyed to the nearest substantial drainage line or watercourse via a natural-looking rock-lined channel or underground pipeline. The construction of the channel can be incorporated into the existing landscape and/or drainage lines and rock outcrops while protecting existing trees and vegetation.

The system for conveying the stormwater or overflow must incorporate:

a. energy dissipation structures that reduce the velocity of the stormwater discharge (see technical standard 3 – sediment and erosion control)

b. facilities for retention of gross pollutants and sediment; these structures and facilities are to be constructed on private property (see Section 11)

Applicants are responsible for the rehabilitation of any disturbed bushland area as a result of the drainage works being undertaken.

A plan detailing any proposed drainage and remedial works within a natural watercourse, creek or bushland must be prepared by a suitably qualified person and submitted to Willoughby City Council for approval.

4.4 Controls for a charged system

Charged systems are permitted to connect downpipes to rainwater tank, where it is not possible to provide a gravity drainage system. They are also permitted for minor development where it is not possible to drain the site via gravity and where it is not possible to obtain an easement.

For a charged system draining to a rainwater tank, these requirements are to be met:

a. hydraulic grade line calculations are to be done by a suitably qualified person demonstrating the proposed system will have sufficient operating head; a freeboard of at least 500mm is to be allowed between the lowest roof gutter level and the hydraulic grade line at the top of the respective downpipe; alternatively, the eaves gutter level is to be a minimum of 1.5m above the top water level in the rainwater tank and a maximum of 2.0m

b. the pipe system is to be fully sealed to a minimum of 500mm above the top water level in the pipe

c. a grated cleanout pit must be established adjacent to all system low-points and provided with a screw-capped sealed extension of the respective main charged drainage line that connects to an onsite dispersal trench system or stormwater pit

d. leaf guards, mosquito mesh and/or an appropriate flap valve must be established over the inlet pipes to the stilling pit to minimise mosquito nuisance

e. exposed aerial drainage will not be approved by Willoughby City Council, except for guttering and vertical downpipes and diagonal lines where they are feeding directly into a rainwater tank

For a charged system within the site that drains to the kerb and gutter, these requirements are to be met:

f. there is a gravity flow (min 1% grade) across the road reserve from the property boundary to the street gutter to preclude the possibility of street water backflow

g. a minimum of 1.5 metres head height must be available from the roof gutter or rainwater tank to the invert of the inlet in the stilling pit and a maximum of 1.5 metres head height between the invert level of the inlet in the stilling pit and the base of the downpipe

h. hydraulic grade line calculations are to be done by a suitably qualified person demonstrating that the proposed system will have sufficient operating head; a freeboard of at least 500mm is to be allowed between the lowest roof gutter level and the hydraulic grade line at the top of the respective downpipe or overflow from the rainwater tank

i. the discharge pipe must be fully sealed to a minimum level of 1 metre above the invert level of the council street gutter or 500mm above the top water level in the system, whichever is greater

j. a grated cleanout pit must be established adjacent to all system low-points and provided with a screw-capped sealed extension of the respective main charged drainage line that connects to an onsite dispersal trench system

k. leaf guards, mosquito mesh and/or an appropriate flap valve must be established over the inlet pipes to the stilling pit to minimise mosquito nuisance

l. exposed aerial drainage will not be approved by Willoughby City Council, except for guttering and vertical downpipes and diagonal lines where they feed directly into a rainwater tank

4.5 Pumps

Generally stormwater drainage from all properties should be by gravity. The use of pumps is generally only permitted for distribution of water from rainwater reuse tanks.

Pumps may be used to drain seepage from underground basements and a minor amount of direct runoff from basement driveway ramps. The pumps are required to be dual submersible pumps and sized and constructed in line with Section 9.4 of AS 3500.3.

Wet wells must have a minimum storage capacity of $3m^3$ and be designed and constructed in line with AS/NZ 3500 Part 3.

Direct connection of a pump's rising main to the kerb will not be permitted. The rising main must be directed to an OSD system, at a level above the top water level, or a stilling pit, before discharge.

4.6 Sub-soil drainage

Sub-soil drainage systems may be necessary in certain types of development. These are to be designed and constructed in line with AS/NZ 3500.3. Sub-soil drains should not be directly connected to the street kerb. Instead they should be connected under gravity to an internal stormwater drainage system via a pit.

Water in the system should be disposed of in a way that will not adversely affect adjacent properties. Applicants are advised to consult a hydraulic engineer.

4.7 Overland flow paths

Existing overland flow paths, including flows from adjacent allotments, are to be preserved and retained. Naturally diffuse surface waters (sheet flows) are not to be concentrated.

Catchment flood studies or drainage analysis must be carried out for sites where there is a risk or record of flooding from overland flow.

Approval to straighten, widen, line or pipe open channels may be granted in some instances, subject to environmental and hydrological considerations.

No structure or fill is to be placed within the flow path where it could cause a rise in the flow depth, increase the velocity beyond the allowable safety limit as defined by Australian Rainfall & Runoff (current edition) by Geoscience Australia, or have an adverse effect on adjacent properties.

Fences within overland flow paths are to be open style that allows the passage of floodwaters.

4.8 Onsite disposal systems

Onsite disposal, including dispersion and infiltration, as a primary method of stormwater disposal is generally not permitted as it can cause inundation of downstream properties in built environments. In addition, dispersion at the rear of properties backing onto bushland could change the moisture content in the bushland area and have an adverse impact on the local flora. Onsite disposal is not acceptable for major development.

For minor development only, where it is not possible to obtain a downstream easement, onsite disposal may be used if these controls are met:

a. it is demonstrated that no drainage easement exists either over adjoining properties nor are readily available through negotiation

b. it is demonstrated that all other alternatives, including charged systems, have been comprehensively examined and demonstrated to be inappropriate and ineffective

c. rainwater tank(s) are installed to capture runoff from all roof areas for reuse with an effective capacity in line with Section 2.2 of Part I of Willoughby DCP. The tanks are to be connected to supply non-potable use including toilet flushing, laundry systems, car

washing, and landscape irrigation. Overflow from rainwater tank(s) is to be piped to the onsite dispersion system

d. surface runoff from hard paved areas shall be collected via a filtration system before discharging into the system to prevent blockage by silt and/or debris

e. site conditions are suitable for onsite disposal

The design of onsite absorption systems are to be prepared by a suitably qualified civil or geotechnical engineer. The design must show that the soil and terrain conditions can accommodate an infiltration or soil absorption system and consider the soil types, slope of the land, level of the water table, and contamination and hydraulic conductivity of the soils.

The design must comply with these criteria:

f. the base of the trench is to be at least 1 metre above the underlying water table or rock stratum

g. a minimum of 95% of the total site impervious area is to drain to the rainwater tank; a maximum of 5% of the impervious area, generally driveways and paved areas, may bypass the rainwater tank and be drained directly to the disposal system

h. the system is at least 5 metres from downstream property boundaries and 2 metres from side boundaries

i. the system is at least 3 metres from any buildings

j. the system is not located where it could have an adverse impact on any existing or proposed tree

k. if the system is being constructed in conjunction with any new structure (including residential buildings) the foundations of the structure are to consist of pier and beam. The piers are to be to a solid stratum

l. the system must be at least 1 metre from pavements that are subjected to vehicle traffic

m. a debris/silt collection pit is to be constructed immediately upstream of the absorption system

n. the system is to consist of an Evertrench Jumbo 410 or similar product with 20mm river gravel wrapped with geofabric, with 1 lineal metre of trench for every 10m^2 of impervious area draining to the trench, including areas draining to the rainwater tank

For dual occupancies and other minor developments where there is variation to these criteria, a detailed soil assessment and test report from a qualified geotechnical engineer confirming that the site is suitably permeable is required.

The design details and hydraulic calculations are to be prepared by a suitably qualified civil/hydraulic engineer. These are to show such a proposal would have no adverse effect on any building, downstream property, soil stability, local vegetation, drainage system or watercourse.

The design must include these details:

- the system will enable infiltration of up to a 5% AEP storm event for all storm durations without surcharging onto neighbouring properties
- the system can completely drain within 72 hours

- the infiltration area shall be the area of the base(s) of the trench(s) only
- a 50% clogging factor is to be added to the trench area
- the trench aggregate fill to have a minimum of 35% void
- installation to allow access pit(s) for cleaning

If test results detail that absorption is not suitable or the minimum trench sizes detailed above cannot be achieved, an onsite stormwater detention (OSD) system is required as well as a rainwater tank. The OSD system is to limit peak flows from the site to the 1% AEP flow from the fully pervious site, before disposal on site. All impervious areas must drain to the OSD system.

4.9 Permeable paving

Permeable paving may be used as part of a water sensitive urban design (WSUD) system provided for a development, and to meet water quality objectives, particularly on major developments or developments draining to reserves, watercourses, bushland or the harbour.

Permeable paving should not be used in areas that:

a. comprise impermeable soils with a hydraulic conductivity of less than 0.36mm/hr

b. have a surface of rock or shale

c. have a slope greater than 5%

d. have a high water table

e. receive high vehicular traffic or regular use by heavy vehicles

f. are downstream of areas likely to contribute significant amounts of silt, sediment, debris or windblown material (due to the potential for such material to result in clogging of permeable paving)

Soil assessment and permeability testing must be done as part of the design process for permeable paving. In the case of shallow soil cover over rock, testing is required to ensure that seepage will not cause negative impacts on downstream properties. Assessment and test results are to be included with the application for any development that includes permeable paving as part of its stormwater management system.

5 Rainwater reuse tanks

5.1 General

Willoughby City Council encourages the installation of rainwater reuse tanks instead of OSD systems for minor developments. Applicants should see the tables in section 2.2 and the flow charts in section 2.4 of Part I of Willoughby DCP to determine the required size of the rainwater reuse tank and discharge method of overflow for particular types of minor development.

Sydney's water supply is treated to drinking water standard. However, approximately 54% of domestic water usage in an average Sydney household could be supplemented by water stored in rainwater tanks.

NSW Health Department does not recommend the use of rainwater tanks for drinking purposes where reticulated potable water supply is available. For more information, contact NSW Health Department or see health.nsw.gov.au/environment/water.

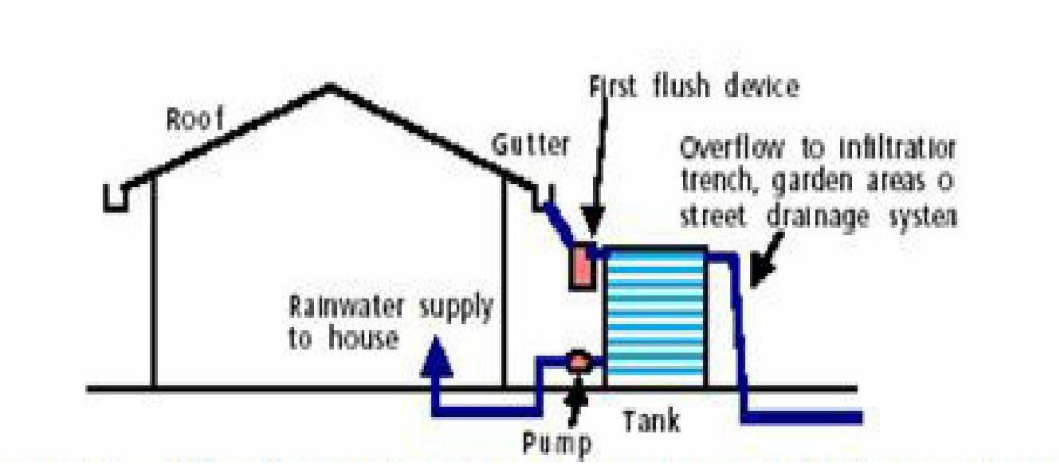
The figure in this section shows the percentage of potable water used in an average Sydney household for domestic purposes.

Figure 3 Typical household water usage

Note:

- Sourced from Sydney Water (2021)

Figure 4 Key elements of a domestic rainwater system



5.2 Exempt development

Subdivision 32 of *State Environmental Planning Policy 2008* (Exempt and Complying Development Codes) allows some above ground rainwater reuse tanks to be installed without a development application. See the policy for detailed requirements.

If an above ground rainwater reuse tank is constructed or installed on or in a heritage item in line with the policy, the tank must be located in the rear yard.

Installation of rainwater ranks that fall under the policy should also consider this technical standard and the provisions in Part I of this plan.

5.3 Bush fire prone land

Rainwater tanks installed on bush fire prone land must comply with these requirements:

a. have a gate valve fitted to the water tank (a 38mm Storz coupling will assist NSW Fire and Rescue)

b. have an additional gate valve fitted to enable the use of a portable pump

c. place a SWS (static water supply) symbol in a conspicuous position at the front of the property

Note:

- Bush fire prone land is identified on the Willoughby City Council Bush Fire Prone Land Map at willoughby.nsw.gov.au.

5.4 Tank construction

a. a rainwater tank must only collect rainwater from roof gutters and downpipes and from a water supply service main

b. underground rainwater tanks are to be fully sealed to prevent runoff from the ground surface and groundwater entering the rainwater tank

c. the rainwater tank must be fitted with a first-flush bypass system; the system is to be designed to cause a minimum of 1mm of the initial run-off from the roof area to bypass the tank thereby acting as a separator to reduce pollutants entering the tank (note that filters are not permitted)

d. the tank must be rigid walled and structurally sound and be prefabricated or constructed from prefabricated materials that are designed and manufactured for use as a rainwater tank; bladder type tanks are not acceptable

e. the tank must be assembled and installed in line with the instructions of the manufacturer or supplier

f. the tank must be placed on a structurally adequate base in line with the manufacturer's or engineer's details

g. the tank, and any stand, must be installed and maintained in line with any requirements of Sydney Water which is responsible for the supply of potable mains water to the premises on which the tank is installed

h. the tank is not to be fixed to any wall of a building unless certified by a practicing structural engineer

i. the tank must be enclosed and any inlet to the tank must be screened to prevent the entry of foreign matter, animals or insects

j. tanks must be designed for ease of flushing, desludging and general maintenance and, where applicable, comply with Australian/New Zealand Standard 2179.1 "Metal shape or sheet rainwater goods, and metal accessories and fasteners"

k. tanks are to be accessible for maintenance and replacement and structurally separate from habitable parts of a building; to cater for leakage or bursting of tanks, an overflow path is to be provided to direct water away from habitable areas

l. tanks must be in a neutral, non-reflective colour that is compatible with the character of the locality and adjoining buildings

m. tanks must be clear of the 1% AEP flood extent. They may be elevated on stands above the flood level, in accordance with Technical Standard No2.

5.5 Interconnected water supply systems and mains top up

To ensure a continuous water supply from the rainwater tank and ensure adequate storage for rainfall events, an interconnection to the potable water supply or a maximum storage volume (mains top-up zone) is to be set when installing the system.

The interconnection to the potable water supply may be via a three-way flow switching system provided that suitable backflow protection is in place. A three-way flow switching system selects water from the rainwater tank for its intended purpose. When the rainwater tank is low or the pump fails, the system automatically reverts to the potable water supply.

For a mains top up, the maximum level is to be based on the daily household usage expected from the tank and set for one day's usage. Reference should be made to the Sydney Water website when calculating the average daily usage patterns for the household. See sydneywater.com.au.

More information on plumbing requirements on backflow prevention for rainwater tanks and interconnected systems is also on the Sydney Water website. See sydneywater.com.au.

5.6 Overflow

Any overflow from the rainwater tank must be drained by gravity directly to an existing council stormwater system, OSD system, bushland or watercourse.

For underground rainwater tanks, the invert level of the discharge pipe must be above the 1% AEP flood level. There must be no other connections to the overflow pipes, such as surface water inlet pits. This is to prevent foreign matter entering the pipelines and flowing back into the rainwater tank.

5.7 Pumps

Any motorised or electric pump used to draw water from the tank or used to transfer water between tanks must:

a. comply with NSW Office of Environment and Heritage guidelines

b. be installed by a licensed electrician if it is a permanent electrical pump

5.8 Signage

A sign must be affixed to the tank clearly stating that the water in the tank is rainwater.

Taps supplied by water from the tank must be clearly labelled indicating the source of water.

Pipes from the tank are to be labelled rainwater/non-potable water in line with current Australian and Sydney Water standards.

6 Onsite stormwater detention

In line with Part I of Willoughby DCP, all developments that are not seen as ‘minor’ are deemed to be ‘major’ development. All major developments must provide onsite stormwater detention (OSD) systems and/or other water management strategies, such as water sensitive urban design (WSUD). A stormwater management plan, with hydraulic and hydrological concept plans, must be prepared by a suitably qualified engineer and included with the development application.

When full or partial redevelopment of a site is proposed, OSD must be provided to cover all the impervious area. No credit will be given for existing impervious areas.

OSD is also required for minor development on sites that fall to the rear, where geotechnical test results detail that absorption is not suitable and an onsite disposal system is required (see Section 4.8 for details).

6.1 Exemptions

OSD is not required where discharge of stormwater from a property does not pass through any council owned drainage infrastructure before reaching the receiving waters of Middle Harbour or Lane Cove River. However, other stormwater management systems are required to maintain water quality and prevent erosion (see Sections 10 and 11 of this technical standard for further details).

6.2 Design requirements

OSD systems must be designed in line with Willoughby City Council's policy as follows:

a. the volume of storage or the site storage requirement (SSR) for a site must be in line with Table 7

Table 7 Site storage required

Zone	Volume of storage required m^3/ha	Volume of storage required $m^3/100m^2$
1	327	3.27
2	360	3.6
3	380	3.8
4	315	3.15

Note:

- See the onsite detention drainage zone map (Appendix 1) for the required zone.

b. the Permissible Site Discharge (PSD) for a site must be in line with Table 8

Table 8 Permissible site discharge (PSD)

Zone	Permissible site discharge L/s/ha	Permissible site discharge L/s/100m2
1	225	2.25
2	170	1.7

Zone	Permissible site discharge L/s/ha	Permissible site discharge L/s/100m2
3	180	1.8
4	136	1.36

Note:

- See the onsite detention drainage zone map (Appendix 1) for the required zone.

c. where flow from impervious area bypasses the detention system, the PSD from the system is to be reduced by a rate equal to the flow from the bypass area in the 1% AEP storm event. Where the area bypassing the system is over 5% of the total impervious area, then a DRAINS model is to be prepared, to confirm that for the 1% AEP storm event the PSD is achieved and confirm the required storage volume for the reduced outflow

d. if the PSD from the entire site is to be reduced to 20L/s to allow discharge to the kerb and gutter, a DRAINS model is to be prepared to confirm the required storage volume

e. High Early Discharge (HED) chambers are not permitted.

f. a spillway with an overland flow route is to be provided in the event that a storm higher than the design storm occurs, or the OSD system malfunctions. The flow route must be capable of carrying the flows for a 1% AEP storm, assuming that the outlet to the OSD system is fully blocked. Finished ground levels of the route must be shown on the plan. Piped overflow and overflow via an internal weir to an overflow pit / chamber with a piped outlet is not acceptable. The overflow path must be in a visible location at ground level, so any blockage of the system can be rectified

g. all roof and impervious areas shall drain through the OSD system for all storms up to an including the 1% AEP storm event. If the internal major flow path does not drain to the system, then the pipe network, including gutters and downpipes, is to be designed for the 1% AEP storm event

h. all floor levels adjacent to the OSD storage, or the overland flow path/spillway from the OSD storage, are to be a minimum of 300 mm above the maximum design storage water surface level. The freeboard may be reduced to 150mm for garages only

i. maximum above ground storage ponding depths are to be:

   - 200mm deep in areas such as driveways and car parking areas
   - 300mm deep in other storage areas such as landscaping or gardens

   Floatable landscaping material, such as tree bark chips, is not permitted in above ground storage ponding areas. The depth may be increased to 1000mm, if the storage area is fully fenced with pool fencing, it is possible to “walk out” of the ponding area (maximum 1:8 batters) and warning signs provided to indicate sudden rise in water levels

j. volumes of storage in landscaped areas to be increased by 20%, to allow for vegetation growth

k. a minimum of 0.36m^3 of the storage must be provided in the form of below ground storage such as a pit, which is to form the outlet of the OSD system. For storage on car parks, a minimum 25% of the site storage requirement is to be stored underground

l. crate style cell units, such as Atlantis Drainage Cells, are not acceptable to Willoughby City Council for OSD systems, as they are prone to siltation and blockage and are difficult to maintain

m. all below ground OSD tanks must be accessible for maintenance, with the access pit as near to the outlet as possible. Note the provisions of AS 2865 – Safe Working in Confined Spaces. For below ground tanks:

- step irons are required at the outlet access grate of below ground storage tanks with a depth of 900mm or greater
- OSD storage outlet discharge flows of 20 l/s or more shall be connected directly to the council underground drainage system or an extension of the system
- below ground tanks shall be a minimum of 500mm deep
- all below ground OSD tanks are to have an extra access grate diagonally opposite the outlet for ventilation and cleaning purposes. The minimum size of any access grate is to be 600mm x 900mm
- the below ground tanks and pits are required to drain completely dry at the cessation of any storm and do not require sediment traps at the outlet point
- tanks are not to be located beneath habitable floors
- In multi-unit developments, all access point to OSD tanks and associated water quality improvement tanks are to be located in common areas.

n. orifice plates used to restrict the outflow must be machined to the exact dimension as calculated from minimum 3mm thick stainless steel, or 3mm thick steel galvanised after machining. The minimum diameter of any orifice shall be 65mm. They must be bolted to the pit walls or permanently fixed in the pit by some approved method so that they cannot be easily removed. A pipe outlet is not recommended for discharge control

o. a stainless steel or galvanised mesh screen with a minimum area of 75 times the orifice area shall be latched over the orifice plate. To this end, Maxi Mesh (rh3030) is desirable. A handle attached to the screen is desirable

p. all storage outlets must have inverts above the 1% AEP floodway level of any nearby creeks or overland flow routes, or be designed with hydraulic grade line analysis to be above the downstream 1% AEP water level in the case of connection to council's underground system

q. for all OSD systems, a Positive Covenant and Restriction on Use of Land will be required to be placed on the Title in favour of Willoughby City Council. These instruments shall be created under Section 88B for newly created lots or under Section 88E(3) of the Conveyancing Act 1919 where no new lot is created. The purpose of this is to ensure that the registered proprietor has care, control and maintenance obligations of the OSD system

r. a plaque measuring no less than 400mm x 200mm and of a material acceptable to Willoughby City Council is to be permanently attached and prominently displayed close to the OSD system. This plaque will advise occupiers of the property of the existence of the OSD system and that the system must not be altered without written consent from Willoughby City Council.

The required wording for the plaque is:

> "This is an onsite detention system. It must not be altered in any way without written consent from Willoughby City Council. The owner shall regularly clean the system."

Where needed for work health and safety requirements, a 'Danger' sign no less than 450 mm x 300 mm is to be displayed at the entrance of the tank advising that only persons with confined space training should enter that tank.

The suggested wording for the sign is:

> "Danger. Confined space. No entry without confined space training."

s. the constructed OSD installation must be certified by a suitably qualified and competent professional engineer, (generally CP Eng qualification) and state that it complies with Willoughby City Council's Design Standards (AUS-SPEC), all relevant codes and technical standards, and is in line with the approved plans.

6.3 Submission requirements

All plans, calculations and details submitted must include this information:

a. total impervious area in m^2. Separately show the total roof area and total paved area including all driveways, carports and pathways. Swimming pools will be considered impervious where the high-level overflow is not connected to the sewer. Details of the proposed high-level overflow system are to be included in the submission when seeking exemption from the impervious area calculation

b. floor levels of all existing and proposed buildings and structures on the site

c. existing and proposed surface levels and contours. These contours shall extend 5 metres outside the property

d. location of all existing and proposed trees

e. the volume of storage being provided, with dimensions and level shown so that the storage volume can be easily calculated. This volume must be in line with the figures in Table 7

f. show that stormwater flows up to the 1% AEP storm event from all impervious areas (including roofs) are conveyed to the OSD system, either by a major flow path or an appropriately sized pipe network

g. should the permissible site discharge be reduced or throttled calculations showing the outflow versus storage relationship will be required for the critical storm calculations of the size of the orifice plate or other approved outlet control system will be required

h. show adequate details of an overflow spillway and overland flow path. Willoughby City Council may require calculations to show that the flow path can cope with the 1% AEP storm event

i. all levels are to be given to Australian Height Datum (AHD). Benchmark information is available from the Survey Control Branch of the Department of Lands (Land & Property Information Section) and Sydney Water

j. show the invert level of the proposed discharge outlet point at the street kerb or council trunk drainage system, and that all connections from OSD systems are above the 1% AEP water surface level

k. the OSD checklist (Appendix 5) is to be completed, signed and included in the application.

The standard of presentation required for ease of checking the plan should indicate a north point, reduction ratios, existing and finished floor levels, construction dimensions, details and cross-sections.

6.4 Post-development documentation

a. the construction of the OSD installation must be supervised and certified by a suitably qualified engineer, which must state that it complies with Willoughby City Council's OSD policy, all relevant codes, standards and also that it is in line with the approved plans. Appendix 2 contains a draft certificate of hydraulic / hydrological compliance

b. on completion of the OSD system, a registered surveyor must verify critical levels and storage volumes. A suitably qualified engineer must prepare the works-as-executed plan with any changes shown in red, and complete the attached OSD record of installation.

7 Maintenance of stormwater management systems

7.1 Maintenance

Regular maintenance of stormwater management systems, including rainwater tanks and OSD systems, is to be done to ensure they are working effectively.

Rainwater tanks are to be checked for sludge every two to three years. Gutters, leaf screens or guards are to be inspected and cleaned on a regular basis. Discharge control pits and the associated trash rack of OSD systems are to be inspected and cleaned on a regular basis, not exceeding six-monthly intervals.

7.2 Legal requirements

To ensure the continued effectiveness of stormwater water management systems, the applicant must place a restriction on the property title in favour of Willoughby City Council. This must be in the form of a Positive Covenant and/or Restriction on Use of Land under the *Conveyancing Act 1919* for onsite detention systems. Examples of Section 88B and 88E Instruments for the legal protection of stormwater management systems are in Appendix 3 of this standard.

8 Controls for discharge from a property into the council drainage system

8.1 General

All surface water, roof water not connected to a rainwater tank, and the overflow from a rainwater tank must be conveyed by gravity to the council drainage system, a watercourse or the harbour. All methods of discharge into or through public open space require the approval

of Willoughby City Council. Public open space includes road reserves, drainage reserves, public car parks, public parks, public reserves and bushland).

Work, including new connections, may not be done on the council drainage system without approval from Willoughby City Council, and it may only be done in line with current Australian Standards and AUS-SPEC specifications. Willoughby City Council must be given a minimum of two working days' notice before any works are done on public roadways or involving council owned or operated structures, including new pipe systems.

Development sites that discharge less than 20 litres per second are permitted to discharge directly to the street kerb using 125x75x4 galvanised rectangular hollow section (RHS). Only one discharge line is permitted across the nature strip from each property.

The pipe discharging to the street kerb should be within the frontage of the subject property at an angle no less than 60^{o} to the kerb line. If drainage is beyond the subject property frontage, council's street drainage system is to be extended using a minimum 375mm diameter reinforced concrete pipe (RCP) with a kerb inlet pit to a point near the frontage of the property.

In exceptional circumstances and only for minor developments, Willoughby City Council may consider the extension of property drainage line for a maximum of 20m or the frontage of one property. The drainage line is to cross the nature strip perpendicular to the kerb with two 45^{o} bends to turn the pipe running directly behind, and parallel to the street kerb.

For any stormwater pipeline proposed to discharge directly into a council-owned stormwater drainage system such as a pipeline, channel or kerb inlet pit, the application may need to include hydrological and hydraulic analyses. These must confirm that there will be no surcharges due to backwater effects from the system.

For any connection to a council pit, pipe, channel etc., the pipe connection must be flush with the pit or pipe wall, and not extend into the pipe or pit.

When connecting to a council pipe, pit or drainage channel, a minimum of two working days' notice must be given to Willoughby City Council before starting the work. This is to allow for an inspection of the works when the connection is made.

8.2 Construction of a public pipeline

The proposed stormwater management system may include a public pipeline or other stormwater infrastructure built by the developer in the council road reserve or public open space as a condition of approval. Usually such infrastructure should be designed for all storm events up to and including a 5% AEP storm event, with a minimum pipe size of 375mm and a minimum grade of 1%. However, where the existing road may be unable to carry excess flows in larger storm events and where there may be danger to persons or risk of property damage, Willoughby City Council may require a larger storm event as a basis for the design. More information on public stormwater drainage infrastructure design requirements is in Willoughby City Council's AUS-SPEC.

Plans for any proposed public pipeline are to be submitted to Willoughby City Council for approval under the *Roads Act 1993*. Specific conditions for construction, including inspection requirements and work-as-executed drawings are to be provided as part of the approval of any plans.

8.3 Discharge to roads or lands controlled by other statutory authorities (integrated development)

An applicant who proposes connecting the site discharge to a stormwater drainage system under the control of another statutory authority must have the authority's written approval for the work. The approval must be submitted to Willoughby City Council before applying for a construction certificate.

8.4 Controls for discharge into a new inter-allotment drainage easement and into an existing drainage easement or drainage reserve

8.5 New inter-allotment drainage easements

The acquisition of downstream easement(s) may be necessary if the site is unable to discharge into the council stormwater drainage system by gravity. Documentary evidence of the registration of the drainage easement(s) must be provided to Willoughby City Council before issue of a construction certificate. Any easements required for a development must be shown on the linen plan for any subdivision.

The proposed pipeline in the inter-allotment drainage easement must have adequate capacity to convey the 1% AEP uncontrolled runoff from the development site. This pipeline must be constructed before starting any other works. At completion, the applicant must provide certification to Willoughby City Council from a suitably qualified engineer and works-as-executed drawings from a registered surveyor that the installation of the pipeline has been completed and complies with the approved drawings, the current Australian Standards and Willoughby City Council's AUS-SPEC. The registered surveyor must also certify that all drainage structures are constructed wholly within the drainage easement(s).

If the downstream property owner does not grant the necessary easement, the developer may be required to apply to the Supreme Court under Section 88K of the *Conveyancing Act 1919* for the matter to be adjudicated.

8.6 Existing inter-allotment drainage easement

An applicant may propose discharging runoff through an existing pipeline across an adjoining property. In this case the applicant must provide evidence that the property being developed has the right to use the inter-allotment drainage easement.

The applicant may also need to provide a hydrologic and hydraulic assessment indicating that the existing pipeline has adequate capacity to carry the 1% AEP uncontrolled runoff from the development site.

If the pipeline is unable to convey the additional discharge from the development, the applicant will have to upgrade the pipeline. In this case, the applicant will have to supply these documents to Willoughby City Council for approval:

a. evidence in the form of a legal agreement showing that the property has the right to drain via the existing easement

b. design details of the proposed upgraded stormwater pipeline

The applicant must construct the approved upgraded pipeline in the easement before the starting of any works. At completion, the applicant must provide certification to Willoughby City Council from a suitably qualified engineer and works-as-executed drawings from a registered surveyor that the installation of the pipeline has been completed in line with the approved drawings, current Australian Standards and Willoughby City Council's AUS-SPEC. The registered surveyor is to certify that all drainage structures are constructed wholly within the drainage easements.

8.7 Connection to a council drainage easement or drainage reserve

For any proposed stormwater pipeline discharging directly into a council-owned stormwater drainage system such as a pipeline, channel, inlet pit or discharge onto a vegetated drainage reserve, the applicant may need to submit hydrological and hydraulic analyses. These are to confirm that there will be no surcharges due to backwater effects within the system or adverse impacts to neighbouring property.

8.8 Drainage easement widths

These easement widths are required for inter-allotment (or private) drainage pipelines:

Table 9 Easement width for private drainage pipelines

Pipe (mm diameter)	Drainage easement width (m)
<300	1.25
300	1.50
375, 475	2.00
525, 600,675	2.50
750, 825, 900, 1050	3.50
1200, 1350, 1500	4.00
1650, 1800	4.50

An inter-allotment drainage pipeline is to be contained within an easement of not less than a 900mm width.

The following easement widths are required for council pipelines:

Table 10 Easement width for council drainage pipelines

Pipe (mm diameter)	Drainage easement width (m)
<300	2.50
300	2.50
375, 475	2.50
525, 600,675	2.50
750, 825, 900, 1050	3.50
1200, 1350, 1500	4.00
1650, 1800	4.50

When installing pits in a drainage easement, the easement boundaries must be a minimum of 300mm clear of the pit. This also applies to any other drainage structure.

In exceptional circumstances and subject to Willoughby City Council approval, for example where the existing width between a house and fence is limited, drainage easements may be of variable width.

8.9 Structures within easements

Willoughby City Council will not approve the construction of any permanent structure or the placing of fill over a drainage easement if the structure or fill will prevent or hinder overland flows, construction, reconstruction, maintenance, cleaning or access to the pipeline or easement. Permanent structures include habitable dwellings, eaves, balconies, garages, impervious fences, swimming pools and retaining walls.

Paving and demountable carports over a drainage line or easement may be approved subject to Willoughby City Council assessment. Structures adjacent to pipelines must be sufficiently clear of easement boundaries or independently supported to avoid placing any load within the zone of influence of pipeline elements. Footings must be extended to a minimum of 100mm below the invert of the pipe or to solid rock.

9 Controls for discharge from a property into natural watercourses, bushland or the harbour

Stormwater systems on properties draining to natural watercourses, bushland or the harbour must include:

a. water quality improvement systems (see Section 11 Water Quality for further details)

b. energy dissipation systems

For properties draining to a watercourse over natural bushland, a rock lined channel is to be provided from the site to the watercourse, or other measures as agreed with Willoughby City Council.

For major developments draining to bushland or natural watercourses, a rainwater reuse tank or appropriate WSUD measures are to be incorporated in the stormwater management system, to reduce water velocities and minimise erosion.

For minor developments draining to bushland or natural watercourses, a 5kL rainwater tank is to be provided.

Permission is required from Willoughby City Council to discharge stormwater directly into a natural watercourse, creek or bushland reserve. A plan detailing any proposed drainage and remedial works on council land must be submitted to Willoughby City Council for approval.

10 Water quality

All major developments must address water quality during and after construction. Under *the Protection of the Environment Operations (POEO) Act 1997*, allowing pollutants to enter any watercourse is an offence.

10.1 Water quality during construction

Construction sites have the potential to significantly affect water quality and contribute to the degradation of the natural environment through poor management of soil, water and materials on the site.

Good management practices on a construction site for the control of water quality includes erosion and sediment control as well as other construction materials such as chemicals and waste.

See Willoughby City Council's technical standard 3 – sediment and erosion control and, for larger sites to Managing Urban Stormwater, Soils and Construction – Landcom 2004, for information on managing sediment and erosion during the construction phase.

10.2 Permanent post development water quality controls

For all developments draining to bushland, creeks or the harbour and for all major development, stormwater runoff is to be treated before leaving the site.

10.2.1 General controls

a. all stormwater flows from regular rainfall events up to and including the 1:2 year annual recurrence interval (ARI) are to be captured and treated before discharge to the council stormwater drainage system

b. the treatment measure may include one or more of these features:
   - retention – pond, wetland or basin
   - retention and filtration – bioretention system, sand filter, rain garden
   - retention and volume loss –rainwater tank, surface runoff retention tank, infiltration system
   - filtering and conveyance – grassed swale, filtration system
   - Gross Pollutant Trap (GPT) – offline diversion system, inline diversion system, trash rack, centrifugal displacement systems, pit basket
   - any system that can capture and retain the specified pollutant load
   - any other appropriate technique suitable to the needs of the site

c. the treatment measure is to be placed as close as possible to the pollution source within the development site

d. the pollution retention efficiency must be maintained up to the design discharge and should not decrease with the build-up of materials

e. in storm events greater than that of the design discharge or if the storage capacity of captured material is exceeded, the storage system must not allow any release of the previously captured material

f. the system must be designed to be able to bypass flows greater than the design discharge without blocking or overtopping

g. the system must be designed to meet appropriate public health and safety standards

h. the design should ensure that there is minimal risk of mosquito breeding within the system

10.2.2 Performance objectives and criteria

Objectives

Table 11 Water quality performance objectives

Issue	Objectives
Runoff volumes and flow rates / stormwater quality	1. minimise impervious areas connected to the stormwater drainage system without causing uncontrolled property runoff 2. maximise reuse of roofwater for non-potable uses 3. maximise use of vegetated flow paths 4. use stormwater pollutant traps 'at source' where appropriate
Riparian vegetation and aquatic habitat	5. protect and maintain natural wetlands, watercourses and riparian corridors 6. protect and maintain all natural (or modified) drainage channels within the site that possess either baseflow, defined bed and/or banks, or locally occurring native riparian vegetation
Flow	7. ensure alterations to natural flow paths, discharge points and runoff volumes from the site are negligible – generally no increase in the 50% and 1% AEP storm event peak flows 8. the frequency of bank full flows should not increase as a result of development
Natural bushland	9. minimise the impact of stormwater discharge on bushland areas

Requirements

Table 12 Water quality requirements

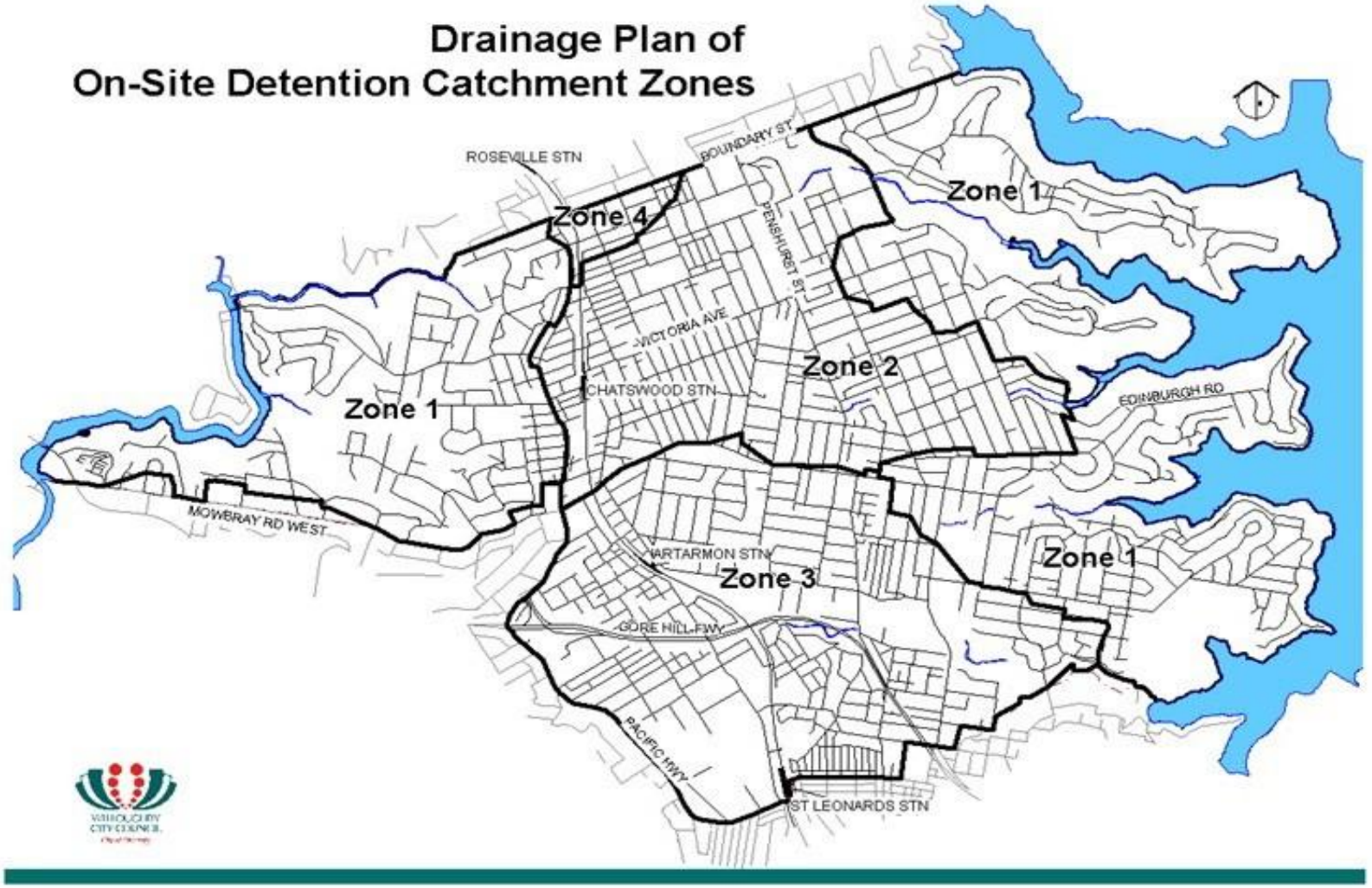
Pollutant	Average annual pollutant load reduction (%)
Gross pollutants	90%
Total suspended solids (TSS)	85%
Total phosphorus (TP)	60%
Total nitrogen (TN)	45%
Hydrocarbons	See below

Determination of water quality targets are to be achieved in line with Sydney Water guidelines or using a MUSIC model. If the MUSIC model is used, parameters are to be in line with the parameters in Sydney Catchment Authority's document 'Using MUSIC in Sydney's Drinking Water Catchment'.

Developments with more than 5 parking spaces and commercial and industrial development are to provide a treatment system that specifically targets hydrocarbons or incorporates a raingarden / bio-retention basin. Runoff from all parking areas, driveways or access roads must drain to the treatment system or raingarden / bio-retention basin.

Appendix 1 – Onsite detention catchment zones

Figure 5 Drainage plan of on-site detention catchment zones


Drainage Plan of
On-Site Detention Catchment Zones
ROSEVILLE STN
BOUNDARY ST
Zone 4
Zone 1
PENSHURST ST
VICTORIA AVE
CHATSWOOD STN
Zone 2
Zone 1
EDINBURGH RD
MOWBRAY RD WEST
ARTARMON STN
Zone 3
Zone 1
GORE HILL FWY
PACIFIC HWY
ST LEONARDS STN

Appendix 2 – Onsite stormwater detention record of installation and certificate of hydraulic/hydrological compliance

Record of installation of onsite stormwater detention

Job number:

Development application number:

Drawing number:

Project at:

Table 13 On-site stormwater detention installation parameters

Attributes	Designed parameters	Constructed parameters
Tank dimensions (LxWxH) (m)		
Orifice size (mm)		
Orifice plate type		
Hydraulic head (m)		
Storage volume (m^3)		
Permissible site discharge (l/s)		
Debris screen type		
Debris screen area (m^2)		
Overland flow provided		
Number of access grates		
Plaque(s) attached to wall		
Step irons provided (over 0.9m grate-invert)		
Plumber's certificate number for rainwater tank installation		

Comments:

Certificate of hydraulic/hydrological compliance

I _____of _____(a professional engineer competent to practise in the field of stormwater drainage design) have inspected the above completed onsite stormwater detention system. This includes verification of the generally + or -10% acceptable limits and tolerances of the system in this policy.

I certify the works have been constructed in line with the approved design details for the project, subject to satisfactory maintenance. This is except for any variations that do not affect the performance of the system, which are highlighted in the works-as-executed drawings.

Signature:

Qualifications:

Position:

Membership number:

Telephone:

Appendix 3 – Samples of draft terms for positive covenant and restriction on use of land

Detention system

To be used when a deposited plan is being registered

Terms of positive covenant _____referred to in the above-mentioned plan

The registered proprietor of the land (the 'proprietor') will ensure the onsite stormwater detention system (the 'system') on the land is:

- cleaned, maintained and repaired for all pits, basins, tanks, pipelines, orifice plates, trench barriers, walls, earth banks and other structure, gutters, leaf gutter guards, down pipes, pipe connections and any associated systems to the system
- regularly kept clean and free from grass clippings, silt and other debris to ensure efficient operation at all times

The proprietor agrees to provide Willoughby City Council access to the system with necessary materials and equipment at all reasonable times and on reasonable notice, and without notice in the case of an emergency.

The proprietor agrees access will be provided for Willoughby City Council to:

a. view the state of repair of the system

b. ascertain if there has been any breach of the terms of this covenant

c. do any work required to remedy a breach of the terms of this covenant if the proprietor has not, within 14 days of receiving written notice from Willoughby City Council of a breach, taken steps to remedy this

In the case of point c) above, Willoughby City Council may recover the cost of this remedial work from the proprietor on demand as a liquidated debt.

Name of authority empowered to release, vary or modify the positive covenants referred to: **Willoughby City Council**.

Terms of section 88b instrument for the legal protection of onsite stormwater

To be used when a deposited plan is being registered

Terms of restriction on the use of land:_____ **referred to in the above mentioned plan**

The registered proprietor of the land (the 'proprietor') agrees, with respect to the onsite stormwater detention system (the 'system') on the land, to:

a. not erect or permit any building, structure or erection on the whole or erection on this part of the land except:
    - dividing fences
    - buildings, structures and erections that are required to maintain the system
    - other structures approved by Willoughby City Council
b. not carry out any alterations to the system including, detention levels, controlled outflows, grates, pipes, orifice plate, mesh screen, gutters, leaf gutter guards, downpipes, pipe connections and any associated systems of the system without first gaining consent in writing Willoughby City Council

Name of authority empowered to release, vary or modify the restriction on the use of land referred to: **Willoughby City Council**.

Terms of section 88e(3) instrument for the legal protection of onsite stormwater detention system

To be used with Form 13PC from the Department of Lands

Terms of positive covenant

The registered proprietor of the land (the 'proprietor') will ensure the onsite stormwater detention system (the 'system') on the land is:

- cleaned, maintained and repaired for all pits, basins, tanks, pipelines, orifice plates, trench barriers, walls, earth banks and other structure, gutters, leaf gutter guards, down pipes, pipe connections and any associated systems to the system
- regularly kept clean and free from grass clippings, silt and other debris to ensure efficient operation at all times

The proprietor agrees to provide Willoughby City Council access to the system with necessary materials and equipment at all reasonable times and on reasonable notice, and without notice in the case of an emergency.

The proprietor agrees access will be provided for Willoughby City Council to:

a. view the state of repair of the system

b. ascertain if there has been any breach of the terms of this covenant

c. do any work required to remedy a breach of the terms of this covenant if the proprietor has not, within 14 days of receiving written notice from Willoughby City Council of a breach, taken steps to remedy this

In the case of point c) above, Willoughby City Council may recover the cost of this remedial work from the proprietor on demand as a liquidated debt.

Name of authority empowered to release, vary or modify the restriction on the use of land referred to: **Willoughby City Council**.

Terms of section 88b instrument for the legal protection of onsite stormwater detention system

To be used with Form 13RPA from Department of Lands

Terms of restriction on the use of land

The registered proprietor of the land (the 'proprietor') agrees, with respect to the onsite stormwater detention system (the 'system') on the land, to:

a. not erect or permit any building, structure or erection on the whole or erection on this part of the land except:
    - dividing fences
    - buildings, structures and erections that are required to maintain the system
    - other structures approved by Willoughby City Council
b. not carry out any alterations to the system including, detention levels, controlled outflows, grates, pipes, orifice plate, mesh screen, gutters, leaf gutter guards, downpipes, pipe connections and any associated systems of the system without first gaining consent in writing Willoughby City Council

Name of authority empowered to release, vary or modify the restriction on the use of land referred to: **Willoughby City Council.**

Appendix 4 – Rainfall intensity

Table 14 Rainfall intensity in mm/hr for various durations and return periods

Duration (hours)	1 year	2 years	5 years	10 years	20 years	50 years	100 years
0.083	97.9	125.	159.	178	204.	237.	262.
0.100	91.8	118.	149.	167.	192.	223.	247.
0.167	75.2	96.7	124.	139.	159.	186.	206.
0.333	54.9	70.9	91.7	104.	120.	141.	156.
0.500	44.7	57.9	75.3	85.6	99.0	117.	130.
1.000	30.5	39.6	52.0	59.4	69.1	81.8	91.5
2.000	20.2	26.2	34.8	39.9	46.5	55.3	62.0
3.000	15.7	20.5	27.2	31.3	36.5	43.5	48.8
6.000	10.2	13.4	17.9	20.6	24.1	28.7	32.3
12.000	6.72	8.79	11.8	13.6	16.0	19.1	21.5
24.000	4.45	5.82	7.86	9.10	10.7	12.8	14.5
48.000	2.90	3.80	5.16	6.00	7.07	8.48	9.59
72.000	2.19	2.88	3.92	4.56	5.38	6.47	7.32

List of coefficients to equations for calculation of intensity frequency duration (IFD)

$\ln(I)=a+b(\ln(T))+c(\ln(T))^2+d(\ln(T)^3+e(\ln(T))^4+f(\ln(T))^5+g(\ln(T))^6$

I = Intensity in millimetres per hour

T = Time in hours

Table 15 Values of coefficients for IFD calculation

Years	a	b	c	d	e	f	g
1	3.4171	-0.5782	-0.0314	0.00714	0.000801	-0.0001313	-0.0000369
2	3.6781	-0.5738	-0.0320	0.00726	0.000784	-0.0001541	-0.0000311
5	3.9521	-0.5605	-0.0350	0.00702	0.001072	-0.0001620	-0.0000370
10	4.0849	-0.5537	-0.0365	0.00708	0.001194	-0.0001887	-0.0000353
20	4.2350	-0.5485	-0.0378	0.00720	0.001314	-0.0002139	-0.0000347
50	4.4041	-0.5427	-0.0392	0.00746	0.001396	-0.0002625	-0.0000286
100	4.5167	-0.5382	-0.0400	0.00733	0.001460	-0.0002589	-0.0000302

Values of $(t.I^{0.4})$

Where I = Intensity in millimetres per hour

Table 16 Rainfall duration intensity in minimum mm/hr for various return periods

Dur (hr)	Dur (min)	1 year	2 years	5 years	10 years	20 years	50 years	100 years
0.083	5	31.16	34.36	37.83	39.57	41.79	44.37	46.19
0.100	6	36.58	40.45	44.40	46.48	49.14	52.18	54.35
0.167	10	56.41	62.38	68.90	72.12	76.11	81.04	84.41
0.333	20	99.18	109.86	121.77	128.06	135.60	144.64	150.61
0.500	30	137.17	152.12	168.98	177.87	188.53	201.56	210.23
1.000	60	235.44	261.35	291.44	307.37	326.54	349.34	365.36
2.000	120	399.32	443.10	496.38	524.29	557.39	597.41	625.37
3.000	180	541.54	602.52	674.68	713.66	758.91	814.08	852.39
6.000	360	911.47	1016.59	1141.42	1207.40	1285.61	1378.65	1445.38
12.000	720	1542.70	1717.62	1932.35	2045.26	2182.63	2342.86	2456.45
24.000	1440	2616.41	2912.94	3284.97	3483.21	3716.34	3992.51	4196.72
48.000	2880	4409.12	4912.54	5552.04	5897.30	6297.39	6772.53	7114.10
72.000	4320	5911.01	6595.39	7461.02	7926.28	8468.31	9116.86	9578.29

Note:

- For use with kinematic wave equation (equation 14.2 in Australian Rainfall and Runoff, 1987 – see technical note 3).

Appendix 5 – Onsite stormwater detention system design checklist

Table 17 Onsite stormwater detention system design checklist

Item	Description	Response
1	Has a spillway with an overland flow route been provided?	yes / no
2	Has the minimum freeboard been provided between habitable floor levels and the 1:100yr flood level? See Section 10 of Willoughby City Council's Floodplain Management – technical standard 3 for freeboard requirements.	yes / no
3	Has a minimum of 300mm freeboard been provided between the habitable floor levels and the OSD design storage topwater level?	yes / no
4	Above ground storage maximum depth • over driveways and carparking (mm) • maximum depth over landscaping or garden areas (mm) • minimum below ground storage (m^3)	 ____mm ____mm ____m^3
5	Below ground OSD tanks must be accessible for maintenance in line with AS 286,5 – Safe Working in Confined Spaces. Have step irons been provided for tanks deeper than 900mm?	yes / no
6	Total impervious area including driveways, carports, pathways and swimming pools not incorporating high-level overflow system (m^2)	____m^2
7	Existing, and proposed surface levels and contours have been provided (contours extend 5m beyond property boundary)	yes / no
8	Location of all trees has been provided	yes / no
9	Calculated volume of storage according to Part I Section 2.2 (m^3)	____m^3
10	Calculated permissible site discharge according to Table 8 (l/s)	____l/s
11	Calculated orifice diameter (65mm minimum) (mm)	____mm
12	Type of mesh screen provided	____
13	Does the major overland flow path within the site for runoff from all impervious areas drain to the OSD system or has the pipe network been designed for the 1% AEP storm event?	yes / no
14	Direct connections to the council drainage system have been checked against backwater effect and drowned orifice?	yes / no
15	Is the base of the OSD tank graded flush with the orifice invert level?	yes / no
16	Has a minimum 1125mm × 75mm × 4 RHS been provide for a discharge through the street kerb?	yes / no
17	All levels have been given to Australian Height Datum (AHD)	yes / no

Name of design engineer:

Qualifications of the design engineer:

Appendix 6 – Standard easement request letter

Date:

Street address:

Postcode: NSW

Dear:_____

I/we are proposing to redevelop our property at:_____

Before we can proceed with this proposal Willoughby City Council has advised us that we have two options for the drainage of stormwater.

Option 1 – drainage easement

This option is the preferred method and is to obtain a drainage easement to convey the stormwater runoff from our property. This would convey our stormwater under your property via a pipe to :_____

This option requires you to grant us a drainage easement through your property. We would be responsible for all costs for creating the easement, together with any consideration for the use of your property determined by an independent valuation or later agreement. Alternatively we are prepared to offer you $_____* as compensation for the right to drain our stormwater under your property.

Option 2 – onsite disposal system

An alternative option is to install an onsite disposal system on our property. This discharges stormwater into an underground trench and allows absorption of the stormwater flow into the ground, with overflow over the ground surface and into downstream properties.

However, the runoff and seepage from this system may flow towards your property and could cause some dampness and overland flow issues because of the slope of the land.

Could you please indicate your position on this matter so we can advise Willoughby City Council to enable our application to progress.

Yes: I/we are willing to grant you a drainage easement:

Name:

Address:

Signature:

No: I/we are not willing to grant you a drainage easement. We understand that our property will be subject to some overland flow and that we do not want to accept any compensation for a drainage easement as

Name:

Address:

Signature:

* compensation amount offered to be in line with market value for easements, and not a nominal amount.

Attachment 2 – Technical standard 2 – floodplain management

1 Background

Like many other major cities of the world, Sydney and in particular Willoughby have been subjected to intense development and redevelopment pressures, including development on areas that are affected by flooding. These areas are mainly near the creek and drainage systems in the City of Willoughby local government area.

There are economic and other pressures for potential developers to build on these flood affected lands. However, Willoughby City Council has to ensure that the community and its assets are protected from future flooding. This technical standard is designed to manage these competing demands.

This standard provides consistent guidelines and criteria for property development of flood affected land. We will assess development applications in line with these guidelines. This will help maximise the value of these flood affected properties and minimise the potential environment impacts from the development.

Stormwater catchments in the local government area are generally dissected by the Pacific Highway, which forms a ridge of the area. Catchments to the west of the Pacific Highway drain into pipes, open drains and creeks which eventually flow into the Lane Cove River. Catchments to the east of the highway drain into several complex manmade channels, pipes and creeks before discharging into the Middle Harbour.

Due to the complex creek system and the limited capacities of an aging drainage network, some land adjoining to these systems may experience stormwater inundation during times of heavy rainfall. In most cases, these drainage systems cannot fully convey the runoff when there is heavy rainfall. Land adjoining these systems may experience overland flow when the capacity of the drainage network is exceeded. Similarly, most of the lands adjoining the natural creek systems also experience overflow during these events.

1.1 Historical flood records

According to our historical flood record, major rainfall events have previously caused serious flooding in parts of the area.

These storms are:

- 1955: the event recorded in Chatswood was a 1 in 20-year average recurrence interval (ARI) or (5% AEP) event
- 1986: the event was between the 1 in 20 and 1 in 50 (2% and 5% AEP) year event
- 10 April 1998: the event exceeded the 1 in 100 year ARI (1% AEP) in a critical storm duration

Because extreme storms do not occur regularly, computer modelling is used to ascertain flooding from these events. We have done detailed flood studies on most of the local government area to ascertain the risk and extent of flooding. The results have been used to

determine the extent of the flood affected properties in the area that should be subject to development controls.

1.2 Floodplain risk management studies

Willoughby City Council has completed these flood studies and floodplain risk management plans:

Floodplain risk management studies and plan

- Flat Rock Creek
- Sugarloaf Creek

Full catchment flood study

- Blue Gum Creek
- Sailors Bay Creek
- Swaines Creek

Mainstream flood study only

- Scotts Creek

Lyall & Associates also assessed local overland flooding in the City of Willoughby area using TUFLOW, a two-dimensional hydraulic computer model. Together with site assessments of the affected properties, areas of local overland flooding were identified. Recommendations from this study are used in the Scotts Creek catchment. All flood studies are available at willoughby.nsw.gov.au.

2 Flooding mechanisms

This technical standard generally conforms with the principles for the development of floodprone land in the NSW Government's Floodplain Development Manual (2005) and the revised ministerial direction by Department of Planning on 31 January 2008 under Section 117 of *Environmental Planning and Assessment Act 1979*.

In line with the 2005 floodplain manual there are two main types flood producing mechanisms which result in flooding of properties in the City of Willoughby local government area: main stream flooding and local overland flooding.

2.1 Main stream flooding

This type of flooding can occur when the trunk drainage/creek system surcharge and flows inundate the surrounding floodplains. In the Willoughby area, the trunk drainage systems comprise sections of lined and unlined open channels of the main arms of the creeks and their major tributaries. They also include major pipelines along the routes of the natural channels, put in place as the catchments became urbanised.

Depths and velocities of flooding are sufficiently high as to cause provisional high hazard conditions to develop in at least part of the extent of land inundated. Depths of inundation of up to 2m and flow velocities up to 4m/s could typically occur in the lined sections of the channels during major storm events. The catchments are up to several square kilometres in area.

Depending on the extent and complexity of the floodwater, the location of structures in main stream flood paths can impact and alter the flow regime to the detriment of adjoining properties.

2.2 Local overland flooding

This type of flooding results from runoff that travels as sheet flow over grassed and paved surfaces in individual allotments, or along roads on the way to the trunk drainage system. It can also be from overland surcharges from the minor pipes in the catchment headwaters and the lateral sub-catchments bordering the trunk drainage systems. This type of flooding is further divided into two categories: major drainage and local drainage.

In line with the 2005 floodplain development manual, this technical standard distinguishes these two sub-categories by their characteristic depths of flooding and the potential danger to personal safety.

Usually, the depths and velocities of flow are not enough to create high hazard conditions. However, the discharge along overland flow paths increases with increasing catchment areas. Depending on local topographic conditions, high hazard areas may develop along the major overland flow paths before they reach the creek. Overland flooding also occurs with little warning time.

Structures located in the path of local overland flow could cause the water to be redirected or deflected to other adjoining properties. These impacts have to be mitigated in the design of proposed developments.

3 Objectives

The objectives of this standard are to:

a. provide consistent guidelines and criteria for developers and other land users of overland flow/floodprone properties in the City of Willoughby local government area when preparing development applications

b. ensure land identified by Willoughby City Council as subject to a flood related development control has a flood impact statement or flood risk study done before approval of new development

c. reduce the potential risks to property damage and loss of life arising from the development of overland flow and floodprone land, as well as minimise damage to private property during flooding events

d. prevent development intensification on land that is subject to a high risk of flood (H4 to H6)

e. ensure development on floodprone properties have to adopt measures to not exacerbate flooding on other properties

f. increase public awareness through education of the potential adverse impacts of development on properties adjoining overland flow/floodprone properties

4 Applications of this technical standard

This technical standard applies to all development that involves construction works on the site and located on private or public flood affected land in the Willoughby local government area. It

applies to development where approval is needed and development that is exempt from approval.

Flood affected lands are categorised into main stream flooding and local overland flooding (see Section 2).

If there is insufficient information within Willoughby City Council to determine the flood planning level or the minimum habitable floor level of a flood affected property, development on flood affected land must submit a flood study or overland flow assessment.

The study/assessment should determine the pre and post maximum water levels, depths at critical locations, velocity and extent of the flooding at the property. An experienced civil or hydrological engineer should be consulted.

The development application is also subject to:

- *Environmental Planning and Assessment Act 1979*, Section 79c
- NSW Government Floodplain Development Manual 2005
- Willoughby Local Environmental Plan
- other applicable planning instruments adopted by Willoughby City Council or NSW Government

The 2005 floodplain development manual defines floodprone lands as lands affected by the Probable Maximum Flood (PMF) – the largest flood that could physically occur in a location. From the perspective of an urban council, it is not feasible or economically desirable to alienate land in these areas from development. However, basement car parking must take this into account due to the risk present.

The flood planning levels adopted for this policy is generally the 1% annual exceedance probability (AEP) event, sometimes known as the 1 in 100 year average recurrence interval (ARI) flood plus a freeboard. This freeboard depends on the type of development, location, land usage and continuing risk.

5 Controls

5.1 General controls

For all development that is flood affected, a flood impact statement as a minimum must be provided to Willoughby City Council. This must be prepared by a suitably qualified engineer in line with the NSW Government's Floodplain Development Manual and address the various controls.

A detailed flood study is required for sites where Council has no detailed flood study, the footprint of the building is changing or overland flow paths are being altered. All major developments (works other than single dwellings and dual occupancy developments) will require a site specific flood study if the site is a flood affected lot or if the site is adjacent to a major drainage path or mainstream flooding.

To assist in preparing a flood impact statement or a flood study, a flood information certificate is available from Willoughby City Council for most flood affected properties. Details of the fees and application process are available at willoughby.nsw.gov.au.

Willoughby City Council has flood related planning controls for various categories of development within flood affected properties.

5.1.1 New development in areas subject to mainstream flooding or major drainage, including sites adjacent to flow paths

a. minimum floor level for buildings = 1% AEP water level + 500mm

b. minimum garage floor level including parking spaces within buildings = 1% AEP water level + 300mm.

c. minimum crest level for driveway to basement parking and all access points to basements – PMF water level or 1%AEP water level +500mm, whichever is higher

d. minimum floor level for carport = 1% AEP water level + 100mm

e. underside of any structure to be 300mm above 1% AEP flood level

f. flood evacuation route at the PMF level to be provided for sensitive developments, including hospitals, aged care, childcare and seniors living

g. flood evacuation route at 1% AEP +500mm level for all development other than sensitive developments

h. flood study required

5.1.2 New development in areas subject to local drainage or overland flow

a. minimum floor level for buildings = 1% AEP water level + 500mm

b. minimum garage floor level including parking spaces within buildings = 1% AEP water level + 300mm

c. minimum crest level for driveway to basement parking and all access points to basements – PMF water level or 1%AEP water level +500mm, whichever is higher

d. minimum floor level for carport = 1% AEP water level + 100mm

e. underside of any structure to be 300mm above 1% AEP flood level

f. construct on high side of property

g. flood evacuation route at 1% AEP +500mm level

h. flood impact assessment required; a flood study may be required where works potentially impact flood levels.

5.1.3 New pools and minor structures (such as a shed or pergola)

a. no pool or shed to be located in H4 to H6 hazard rating zones

b. structure is not to impede flood flows

c. minimum floor level for minor structures = 1% AEP water level + 100mm

d. flood impact assessment required for local drainage areas and flood study for mainstream flooding or major drainage areas

5.1.4 Alterations and additions in areas subject to mainstream flooding or major drainage, including sites adjacent to flow paths

a. minimum floor level for buildings = 1% AEP water level + 300mm

b. minimum floor level for buildings with sensitive use, including hospitals, aged care, childcare and seniors living = 1% AEP water level + 500mm

c. minimum garage floor level including parking spaces within buildings = 1% AEP water level + 100mm

d. minimum crest level for driveway to basement parking and all access points to basements – PMF water level or 1%AEP water level +500mm, whichever is higher

e. minimum floor level for carport = 1% AEP water level + 100mm

f. flood proof existing habitable areas where practical

g. underside of any structure to be 300mm above 1% AEP flood level

h. flood evacuation route for new works at 1% AEP +500mm level

i. flood study required

5.1.5 Alterations and additions in areas subject to local drainage or overland flow

a. minimum floor level for buildings = 1% AEP water level + 300mm or 500mm above existing natural ground

b. minimum floor level for buildings with sensitive use, including hospitals, aged care, childcare and seniors living = 1% AEP water level + 500mm

c. minimum garage floor level including parking spaces within buildings = 1% AEP water level + 100mm

d. construct works on high side of property

e. flood proof existing habitable areas where practical

f. flood evacuation route for new works at 1% AEP +300mm level

g. flood impact assessment required

5.2 Further requirements

For all developments on main stream or local overland flooding property, these further requirements should be considered:

a. compliance with requirements of other existing environmental planning instruments and relevant statutory authorities

b. obstruction to main stream or local overland flow path

c. cumulative effect of any encroachment and analysis on the impact to adjacent areas and the catchment as a whole

d. effect of proposed works on flood detention storage

e. future maintenance and upkeep of difficult to access locations or Stormwater systems

f. control of soil erosion of disturbed surfaces within the property

g. any other controls specific for the development recommended by a suitably qualified engineer or Willoughby City Council engineer

These developments will be treated case by case on their merits and the general principles of this technical standard.

5.3 Land subdivision, consolidation and rezoning

Proposals for subdivision/consolidation/rezoning of land in flood affected or high hazard properties must include a flood or overland flow study by an experienced civil or hydraulic engineer.

The intended land use of the allotments must be appropriate and in line with Willoughby City Council's flood related controls and other environmental planning instruments.

Willoughby City Council will not support any subdivision/rezoning unless in all subdivided/rezoned lots the proportion of flood free land is more than 50% of the proposed lot size and other planning requirements are complied with. Plans also need to show that there is sufficient flood free land, which will allow a building to be approved and constructed in line with the requirements of this section.

Generally, filling of the land is not supported by Willoughby City Council and filling of flood affected land or land in the flood path is strictly prohibited. Any minor filling/regrading of unaffected land is subject to Willoughby City Council's relevant environmental planning Instruments.

6 Redevelopment

When redevelopment of a flood affected property is proposed, the extent of the floodwater and the location of structures within flood paths which can alter the flow regime to the detriment of adjoining properties must be considered. Structures in the path of overland flow would cause the water to be redirected or deflected to other adjoining properties. The flood water could also cause 'afflux' (a rise in water level) upstream of the structure.

To reduce the impact of flooding on flood affected properties, all redevelopment must be located within the footprint of the existing structures, preferably clear of the 1% AEP event or the overland flow path. Any encroachment on these paths is not permissible unless a cumulative impact study of the floodwaters to both upstream and downstream properties is done by the applicant. Willoughby City Council must also be satisfied that there is no adverse impact to other land owners. A flood study, including an aflux analysis, will be required for any works beyond the existing building footprint.

Willoughby City Council will generally not support the filling of main stream or overland flow path on flood affected land unless it is shown that other upstream and downstream property owners are not affected by similar cumulative filling of adjoining and adjacent properties. Willoughby City Council must also be satisfied there are no adverse effects to other land owners.

7 Protection of properties and human lives

This technical standard also deals with the protection of human lives especially the safety of people who are young or frail. To achieve this, it is necessary that the velocity and depth of flow ratio within developments be within acceptable limits and that a safe flood evacuation route is provided for the occupants.

New development on flood affected properties must not only be sited away or above the standard flood level but also be analysed to include the cumulative effect of blockage if any of the flow and its effect to other adjoining and adjacent properties.

It may be necessary to flood proofed existing structures if these structures are significantly affected by overland flow, especially where people who are old and frail are to be accommodated in these buildings.

Residential development on land subject to flood risk categorised as H4 – H6 will not be permitted without a detailed flood study that clearly shows the development can be done without jeopardising public safety and access, property damage or adverse ramifications on the pre-developed flood regime.

8 Application of technical standard

This technical standard applies primarily to flood affected areas in the City of Willoughby area. Property affected by flooding is subject to flood related planning controls. Some properties within these areas will need to submit a flood study so that an accurate assessment regarding the level, depth, velocity and extent of the overland flow within the property can be determined. An experienced civil/hydrological engineer shall be consulted. In areas where Willoughby City Council has completed formal flood studies, a flood study (Appendix A) may not be required.

This standard also applies to properties subject to local overland flooding within the City of Willoughby area. Developments located on major drainage flooding areas must submit an overland flow study but developments on local drainage flooding areas do not in general require a flood study.

The assessment of a development application is subject to Section 79c of *Environmental Planning Assessment Act 1979*. Where relevant, compliance with this standard also will form part of the assessment.

9 Special requirements

Applicants of a development affected by main stream or major drainage flooding must provide satisfactory environmental, hydrological, hydraulics information including on mitigating adverse impacts to Willoughby City Council. It is highly unlikely we will consider structures in areas of high hazard, where depth, velocity and nature of the flow contribute risks to human lives and property damages without satisfactory mitigating measures.

Willoughby City Council will require supporting evidence to show that the development will not increase the hazards or risks to future owners or occupants and to adjoining properties.

For some developments, we may require additional flood planning control such as safe evacuation from the site and flood proofing of the existing structure. We will also consider the cumulative impact and effects of the development to adjoining properties.

Willoughby City Council has adopted the 1% AEP event as the standard flood in this policy. The largest flood that could conceivably occur is known as the probable maximum flood (PMF). The PMF is estimated from the probably maximum precipitation. Some developments may need to consider the effects of the PMF especially if the development may house young children, older people and people who are frail, and if the development is on or near the major drainage system.

Additionally, the effect of climate change has not been included in this standard.

Climate change relates to the generation of greenhouse gases due generally to the activities of humans. An accumulation of such gases in the atmosphere is understood to cause the planet to gradually warm up and bring about changes to the climate. These climatic changes are predicted to have significant impacts on temperature, air quality, sea levels, water temperature, evaporation rates and rainfall intensities.

In the Sydney metropolitan region, current modelling shows a trend towards an increase in both extreme rainfall frequencies and storm events, which increase the severity of the main stream and local overland flow and/or flood.

This means what is currently the 1% AEP (1 in 100 year ARI) rainfall event may become more frequent to say the 1.1% AEP (1 in 90.91 year ARI). This has serious implications for flooding frequencies.

Willoughby City Council may consider imposing a higher flood level than the current flood planning level if the nature and circumstances affecting the development warrants such consideration in the future.

10 The merit approach

When considering a development application, Willoughby City Council as consent authority uses the merit approach. It considers the environmental, technical, economic, sociological and flooding impacts of a development. This includes minimum flood planning levels, location of structures within the site, access, stability, safety, legal obligations and community responsibilities.

Environmental Planning and Assessment Act 1979 lists matters the consent authority must also consider.

11 Submission requirements

The applicant must show the proposed development is consistent with the objectives outlined in this standard.

The information below must be provided to support the development application on lands affected by overland flow and/or flooding:

- a detailed survey plan of the site and /or adjoining properties to Australian Height Datum (AHD) at 1:100 scale (or 1:200 for larger sites), prepared by a registered surveyor, showing

all existing buildings/structures that will affect the extent, depth and velocity of overland flow/flooding

- a flood or overland flow study (Appendix A) prepared by a suitably qualified engineer with suitable experience and eligible for Chartered Professional Engineer status with Engineers Australia

The flood or overland flow study must include:

a. a hydrological analysis (including plans and calculations) of the upstream catchment area for the 1% AEP and 20% AEP storm events[1]:

   the hydrological analysis is to include a catchment plan highlighting full upstream catchment area that generate the overland stormwater flows travelling through the site.

   The hydrological analysis model must be based on[2]:

   - for upstream catchment areas of less than 3 hectares, Rational Method is acceptable
   - for upstream catchment areas of 3 hectares and larger, a suitable runoff routing computer model, such as DRAINS or RAFTS shall be used
   - the estimation of time of concentration (Tc) must be justified – this may be done through the Kinematic Wave Equation
   - the fraction impervious (f) of the upstream catchment must be between 0.7 (for low density residential area) to 1.0 (for high density residential area, industrial and commercial area) unless it can be proven that other values are appropriate for a particular site

b. a hydraulic analysis (including plans and calculations) of overland flow path through the site for both pre-developed and post-developed scenarios.

   1. The overland flow path must be based on:
      - for open channel and natural creek: 1% AEP storm event of upstream catchment area
      - for enclosed drainage system (such as pipes and culverts), the larger of:
        - 1% AEP storm event of upstream catchment area with 50% blockage to the enclosed drainage system
        - 20% AEP storm event of upstream catchment area with 100% blockage to the enclosed drainage system
      - for basement car parking: PMF event with 50% blockage to the enclosed drainage system
   2. The hydraulic analysis model must be based on:
      - for local drainage or overland flow, Mannings Equation or HEC-RAS may be used
      - for mainstream flooding or major drainage, a Tuflow model or other model appropriate model as agreed by Willoughby City Council shall be used

[1] Willoughby City Council is in the process of modelling a number of catchments within the city and may be able to provide flow rates to consulting engineers where and when they become available. Where these flow rates are provided, Willoughby City Council will not accept lower flow rate values unless it can be clearly demonstrated that the nominal value is correct.

[2] Runoff coefficients tables and Intensity Frequency Duration (IFD) tables are provided in Appendix B of this standard.

3. 1:100 scaled site plans showing the extent and levels of the flood/overland flow path for both pre-developed and post-developed scenarios
4. longitudinal sections (at vertical scale 1:50, horizontal scale 1:100) showing the surface levels, flood/overland flow levels, flow profiles, hydraulic data, changes in grade and critical levels (such as finished floor levels) for both pre-developed and post-developed scenarios
5. cross-section details at 1:50 scale taken at the right angle to the flood/ overland flow path with a maximum spacing of every 5m for both pre-developed and post-developed scenarios and must include at least these locations:
    - at the immediate upstream of the property boundary
    - where the existing and proposed development/structure is closest to the flood/overland flow path
    - at the immediate downstream of the property boundary
    - other cross-sections as required where the flow path and/or drainage system being affected
    - note all levels shown on drawings and details must be to the Australian Height Datum (AHD).
6. Calculations of the velocity-depth product of the overland flow path. If the result exceeds $0.4m^2/s$, suitable open type fencing or other appropriate measures must be used to restrict access to such areas affected by hazardous overland flows.
7. Boundary fence details if they are within the extent of flood/overland flow path.

c. Finished floor levels for the proposed development must be nominated in the report, taking into account the freeboard requirements specified in this standard.

d. The flood/overland flow study must demonstrate that the proposed development will not impede the passage of floodwater to cause a rise (afflux) in the flood level upstream and/or increase the downstream velocities of flow for the flood standard. No structures and/or filling are permitted to be placed over the 1 in 100 year ARI overland flow path.

e. The study must include a hazard risk assessment for the site. The determination of the hazard and the assessment shall be in line with the current edition of Australian Rainfall and Runoff by Geoscience Australia.

f. The flood/overland flow study must be signed by the engineer declaring that the study has been done in line with Australian Rainfall and Runoff, the NSW Government's Floodplain Development Manual and Willoughby City Council's floodplain management technical standard.

12 Creation of restriction on the use of land

The applicant shall create a restriction over the land of the subject property for the affected overland flow path (Appendix C). This must be placed on the land title at NSW Land Registry.

The suggested wording for the restriction is:

> 'Do not place any unauthorised structure, wall, fence, fill or other item that may impede the 1% AEP flood event.'

The extent of the flood, in relation to the building footprint, must be shown on a scaled sketch attached as an annexure to the request forms.

13 Glossary

Afflux – the rise in water level (above normal) on the upstream side of an obstruction caused when the effective overland flow area at the obstruction is less than the natural width of the overland flow path immediately upstream of the obstruction.

Annual exceedance probability (AEP) – the chance of a flood of a given or larger size occurring in any one year, usually expressed as a percentage. For example, if a peak flood discharge of 50 m3/s has an AEP of 5%, it means that there is a 5% chance (that is one-in-20 chance) of a peak flood discharge of 50 m^3/s or larger occurring in any one year (see average recurrence interval).

Average recurrence interval (ARI) – the average period in years between the occurrences of a flood of a particular magnitude or greater. In a long period of say 1,000 years, a flood equivalent to or greater than a 100 year ARI event would occur 10 times. The 100 year ARI flood has a 1% chance (a one-in-100 chance) of occurrence in any one year.

In the economic life of structures, there is a 23% chance of the 100 year ARI event or greater occurring in a 30 year period, a 50% change of occurrence in a 70 year period and a 60% chance within a 100 year period.

Catchment – the land area draining through the main stream, as well as tributary streams, to a particular site. It always relates to an area above a specific location.

Consent authority – the council, government agency or person having the function to determine a development application for land use under the *Environmental Planning and Assessment Act 1979*. The consent authority is most often the council, however there are instances where legislation or an environmental planning instrument (EPI) specifies a minister or public authority (other than a council), or the Director General of the Department of Planning, as having the function to determine an application.

Flood – relatively high stream flow which overtops the natural or artificial banks in any part of a stream, river, estuary, lake or dam, and/or local overland flooding associated with major drainage before entering a watercourse, and/or coastal inundation resulting from super-elevated sea levels and/or waves overtopping coastline defences excluding tidal waves.

Flood affected properties – properties that are either encompassed or intersected by the flood.

Floodprone / liable land – land which is subject to inundation. Floodprone land is synonymous with flood liable land and floodplain.

Flood risk zone – the division of the floodplain into areas of varying flood risk according to the nature of flooding and the depth and velocity of floodwaters.

Flood study – detailed hydrologic and hydraulic assessment usually undertaken using modelling programs such as RAFTS, DRAINS, MIKE 11 or HECS-RAS to determine with more certainty the extent, velocity, volume and height of overland flows.

Freeboard – a factor of safety typically used when setting floor levels and levee crest levels. It is usually expressed as the difference in height between the adopted flood planning level and the flood used to determine the flood planning level. Freeboard provides a factor of safety to compensate for uncertainties in the estimation of flood levels across the floodplain, such as wave action, localised hydraulic behaviour and impacts that are specific event related, such as levee and embankment settlement, and other effects such as “greenhouse” and climate change. Freeboard is included in the flood planning level.

Garage – structure designed primarily for the storage of vehicles. Can be either a fully enclosed detached structure or a fully enclosed space separated from the main habitable floor area of a building.

Habitable – in a residential situation: a living or working area, such as a lounge room, dining room, rumpus room, kitchen, bedroom or workroom. In an industrial or commercial situation: an area used for offices or to store valuable possessions susceptible to flood damage in the event of a flood.

Hazard – a source of potential harm or a situation with a potential to cause loss. In relation to this technical standard, the hazard is flooding which has the potential to put life at risk or cause damage to property and local environment.

Hydraulics – term given to the study of water flow in waterways; in particular, the evaluation of flow parameters such as water level and velocity.

Hydrology – term given to the study of the rainfall and runoff process; in particular, the evaluation of peak flows, flow volumes and the derivation of hydrographs for a range of floods in a particular catchment or site.

Local drainage – local drainage problems in local overland flooding zones typically include direct surface runoff, surcharges and overflows from low points in kerbs, or overflows from the smaller pipes in the stormwater drainage system. Local drainage problems involve shallow depths of flooding up to 300 mm with generally little danger to personal safety. Problems generally arise because of deficiencies in building practice where floor levels are near finished ground levels.

Local overland flooding zone –area where flooding problems occur due to overland flow. Within those zones there may be two categories of flow based on increasing depth and velocity of inundation: local drainage and main drainage.

Main stream flooding – this type of flooding occurs when the trunk drainage systems surcharge and flows inundate the surrounding floodplains. In Willoughby City, the trunk drainage systems comprise sections of lined and unlined open channels of the main arms of the creeks and their major tributaries, as well as major pipelines which have been laid along the routes of the natural channels as the catchments became urbanised.

Major drainage flooding – major drainage problems in local overland flooding zones are categorised as the upper end of the scale of local overland flooding. Water depths are generally in excess of 300 mm (in the storm event used to derive FPLs). These conditions may result in provisional high hazard conditions.

Non-habitable – all other areas of a building not mentioned under the definition of ‘habitable’.

Overland flow – inundation by catchment runoff rather than overflow from a stream, river, estuary, lake or dam due to rising water level.

Provisional high hazard area – where land in the event of a 100-year ARI flood is subject to depths of inundation greater than 0.8 to 1 metre with little or no velocity associated with the flow. Wading would be unsafe for able bodied adults. This area may also be subject to local catchment flash flooding when the street system acts as floodways conveying relatively fast moving overland flow.

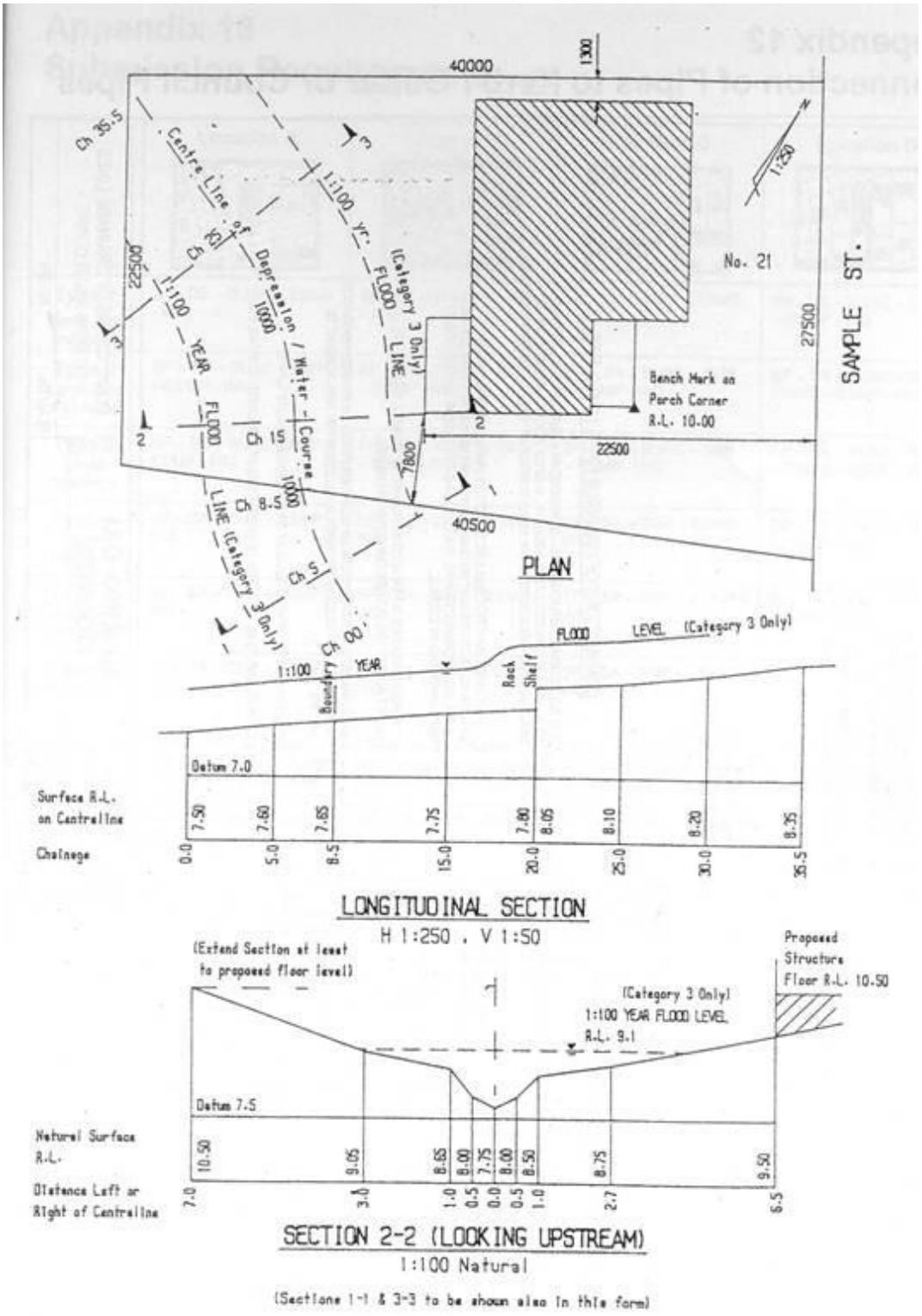
Probable maximum flood (PMF) – the largest flood that could conceivably occur at a particular location, usually estimated from probable maximum precipitation. Generally, it is not physically or economically possible to provide complete protection against this event. The probable maximum flood defines the extent of floodprone land, that is, the floodplain.

Probable maximum precipitation (PMP) – the greatest depth of precipitation for a given duration meteorologically possible over a given size storm area at a particular location at a particular time of the year, with no allowance made for long-term climatic trends (World Meteorological Organisation, 1986). It is the primary input to the estimation of the probable maximum flood.

Survey plan – plan prepared by a registered surveyor that shows the information required for the assessment of a development application.

Appendix A – Sample flood study

Figure 6 Typical flood study survey information



Typical Survey Information

Appendix B – Intensity–Frequency–Duration (IFD) tables

Table 18 Rainfall intensity in mm/hr for various durations and return periods

Duration (hours)	1 year	2 years	5 years	10 years	20 years	50 years	100 years
0.083	97.9	125.	159.	178	204.	237.	262.
0.100	91.8	118.	149.	167.	192.	223.	247.
0.167	75.2	96.7	124.	139.	159.	186.	206.
0.333	54.9	70.9	91.7	104.	120.	141.	156.
0.500	44.7	57.9	75.3	85.6	99.0	117.	130.
1.000	30.5	39.6	52.0	59.4	69.1	81.8	91.5
2.000	20.2	26.2	34.8	39.9	46.5	55.3	62.0
3.000	15.7	20.5	27.2	31.3	36.5	43.5	48.8
6.000	10.2	13.4	17.9	20.6	24.1	28.7	32.3
12.000	6.72	8.79	11.8	13.6	16.0	19.1	21.5
24.000	4.45	5.82	7.86	9.10	10.7	12.8	14.5
48.000	2.90	3.80	5.16	6.00	7.07	8.48	9.59
72.000	2.19	2.88	3.92	4.56	5.38	6.47	7.32

List of coefficients to equations for calculation of intensity frequency duration (IFD)

$\ln(I)=a+b(\ln(T))+c(\ln(T))^2+d(\ln(T)^3+e(\ln(T))^4+f(\ln(T))^5+g(\ln(T))^6$

I = Intensity in millimetres per hour

T = Time in hours

Table 19 Values of coefficients for IFD calculation

Years	a	b	c	d	e	f	g
1	3.4171	-0.5782	-0.0314	0.00714	0.000801	-0.0001313	-0.0000369
2	3.6781	-0.5738	-0.0320	0.00726	0.000784	-0.0001541	-0.0000311
5	3.9521	-0.5605	-0.0350	0.00702	0.001072	-0.0001620	-0.0000370
10	4.0849	-0.5537	-0.0365	0.00708	0.001194	-0.0001887	-0.0000353
20	4.2350	-0.5485	-0.0378	0.00720	0.001314	-0.0002139	-0.0000347
50	4.4041	-0.5427	-0.0392	0.00746	0.001396	-0.0002625	-0.0000286
100	4.5167	-0.5382	-0.0400	0.00733	0.001460	-0.0002589	-0.0000302

Values of $(t.I^{0.4})$

Where I = Intensity in millimetres per hour

Table 20 Rainfall duration intensity in minimum mm/hr for various return periods

DUR (hr)	DUR (min)	1 year	2 years	5 years	10 years	20 years	50 years	100 years
0.083	5	31.16	34.36	37.83	39.57	41.79	44.37	46.19
0.100	6	36.58	40.45	44.40	46.48	49.14	52.18	54.35
0.167	10	56.41	62.38	68.90	72.12	76.11	81.04	84.41
0.333	20	99.18	109.86	121.77	128.06	135.60	144.64	150.61
0.500	30	137.17	152.12	168.98	177.87	188.53	201.56	210.23
1.000	60	235.44	261.35	291.44	307.37	326.54	349.34	365.36
2.000	120	399.32	443.10	496.38	524.29	557.39	597.41	625.37
3.000	180	541.54	602.52	674.68	713.66	758.91	814.08	852.39
6.000	360	911.47	1016.59	1141.42	1207.40	1285.61	1378.65	1445.38
12.000	720	1542.70	1717.62	1932.35	2045.26	2182.63	2342.86	2456.45
24.000	1440	2616.41	2912.94	3284.97	3483.21	3716.34	3992.51	4196.72
48.000	2880	4409.12	4912.54	5552.04	5897.30	6297.39	6772.53	7114.10
72.000	4320	5911.01	6595.39	7461.02	7926.28	8468.31	9116.86	9578.29

Note:

- For use with kinematic wave equation (equation 14.2 in Australian Rainfall and Runoff, 1987 – see technical note 3).

Appendix C – Section 88(E)3 Instruments

Terms of restriction of use of land

To be used with Form 13RPA from Department of Lands.

The registered proprietors of the land enters into this covenant with the Willoughby City Council and agrees:

a. not to place any unauthorised structure such as a wall, fence, fill or other item that may impede the 1% AEP flood event on the land, with the exception of structures approved by Willoughby City Council.

Name of authority empowered to release or vary or modify restriction on the use of land referred to: **Willoughby City Council**

Attachment 3 – Technical standard 3 – sediment and erosion control

Definitions

Activity – the erection of a building; the carrying out of any work in, on, over or under land; or the subdivision of land.

Approval – a licence or permission or any authorisation under Part V of the *Environmental Planning and Assessment Act 1979*.

Authority – in relation to a development application, means: the Council having the function to determine the application; or the Minister or public authority or the Director where an environmental planning instrument specifies as having the function to determine the application.

Building worksincludes buildings or structures or part thereof.

Catchment – is that area within which rainfall will contribute to runoff at a particular point. The area included in catchment is determined by topographic features which on many buildings sites can range from a few square metres to several thousand square metres.

Construction site – is that portion of a site disturbed by the development and/or building and includes the areas where building materials are placed and access traversed by vehicles.

Development – in relation to land, means; the erection of a building on that land; the carrying out of a work in, on, over or under that land; the use of that land or of a building or work on that land, the subdivision of that land.

Diversion banks and catch drains – a bank is a ridge or embankment of compacted earth or blue metal aggregate. A catch drain is an excavated earth drainage ditch or path used to intercept and direct runoff to a desired location.

Erosion – is the process whereby gravity, rain, wind or wave action detaches soil particles and provides energy to move the particle. Types of erosion are:

Soil erosion – is the wearing away of the soil surface material by wind, water or gravitational effects. Natural rates of erosion are accelerated by some human activities such as removal of vegetation for building work exposing the underlying soil.

Stream bank erosion – occurs as a result of flooding or saturation of bank materials due to tidal movement or wave action.

Environmentally sensitive land – is land that is steeper than 18° of slope; liable to degradation due to erosion, sedimentation, inundation by sand/soil or water, salinity/acidity, invasion by exotic vegetation; or native vegetation and wetlands.

Flocculation – reverses the processes that cause dispersion of soil by causing finer particles to clump together into larger units or ‘flocs’ that can settle in a reasonable time or can be filtered. Flocculation is usually carried out in ponds or tanks with the most commonly used flocculation agents being gypsum or alum.

Geotextiles – are synthetic fabrics used to filter sediment or stabilise disturbed surfaces, there are several proprietary types of geotextile for a wide range of applications.

Landscape plan – is a plan showing the location, type and quantity of vegetation and structural elements to be placed on the site to gain visual amenity and screen sections of the site from public view or use.

Receiving waters – either natural water bodies, including rivers, streams (perennial or intermittent), flowing in natural channels with natural beds or in artificially modified channels or wetlands, either naturally formed or artificially modified, or tidal waters, including bays, estuaries or inlets, or constructed water bodies including waterways, ponds, or wetlands, bays or inlets, whether permanently or intermittently inundated with water.

Sediment – mineral or organic material that is being, or has been, moved from its site of origin by transporting agents such as water, wind and gravity to a lower position in the catchment, either above or below sea level. Types of sediment include:

Clay – are soil particles consisting of mineral particles less than 0.002 mm in diameter. Many of the properties of soil depend on the type and quantity of clay particles in the soil.

Sand – consists of particles consisting largely of quartz grains between 0.02 mm and 2.00 mm in diameter. Fine sand is defined as particles between 0.02 mm and 0.2 mm and coarse sand as those between 0.2 mm and 2.0 mm.

Silt – are particles between 0.002 mm and 0.02 mm in diameter.

Soil – is the loose aggregate of various sized small particles that covers the surface of the land. It consists of approximately 90% inorganic mineral material and 10% decayed plant and animal matter.

Sedimentation – the deposition of sediment, usually in locations such as a channel, along a fence, in an area of low slope or a sediment trap, dam or water body.

SEE – Statement of Environmental Effects as prescribed by the *Environmental Planning and Assessment Act 1997* which addresses matters such as any potential environmental impacts of the development, how they have been identified and what steps will be taken to protect the environment or to lessen the expected harm to the environment.

Site management plan – is a plan showing how potential erosion and sedimentation on a site resulting from approved building work, development or landscaping activity will be minimised or controlled.

Soil and water management plan – describes the planned measures to be undertaken at an activity site which will mitigate soil transport and control pollution to down slope lands and receiving waters.

Subdivision – ‘subdivision’, ‘subdivide’, and similar expressions see dividing land into parts.

Topsoil– is a part of the soil profile, typically the first soil layer (called the A1 horizon) below the ground surface that is darker in colour, more fertile and better structured than underlying layers.

Vegetation – means native and exotic trees, shrubs, understorey, ground cover and grasses.

1 Overview

1.1 Introduction

This technical standard is a code of practice for all building, construction and landscaping activity in the Willoughby City Council area. It covers all activity that involves the disturbance of the earth's surface, placement of fill or changes in the rate or volume of runoff entering a natural or built drainage system or flowing overland.

In pristine environments, soil erosion and sedimentation naturally occurs and our ecosystems have evolved over time to cope with this natural process. However, urban development and building activity causes significant land disturbance. This disturbance has resulted in tonnes of soil being eroded and deposited into nearby creeks and waterways at an accelerated rate causing severe environmental problems.

Building, construction and landscaping activities have a major impact on the downstream catchment receiving waters (Lane Cove River and Middle Harbour) when not properly managed.

Removal of vegetation as well as pollutants such as sand, silt, clay, soil and gravel (sediment) and other construction and landscape materials can cause these environmental impacts:

- increased stormwater flooding due to reduced capacity in waterways and blocked stormwater drainage systems
- increased degradation of terrestrial and aquatic ecosystems and habitats, especially aquatic habitat
- reduced air quality and traffic hazards due to increased dust problems
- reduced water quality in creeks, estuaries, river and harbour
- reduced aesthetics and recreational activities due to the infilling of creeks and estuaries

Due to the large number of construction sites within Willoughby City Council area at any given time, even small amounts of pollution from the individual sites can have significant cumulative effects.

Those involved in building and construction must be diligent in managing soil and water on construction sites to minimise soil erosion and sediment pollution and reduce the risk of environmental degradation.

1.2 Objectives

The aim of this technical standard is to ensure that developers/builders:

a. manage environmental and public safety risks during earth and construction works
b. control erosion and stabilise exposed soil surfaces
c. reduce the potential for the transportation of pollutants off site
d. provide a safe and effective framework for the installation and ongoing maintenance of sediment and erosion controls during earthworks, construction, building and landscaping activities

e. prevent the pollution of stormwater runoff draining into bushland, stormwater drainage systems and waterways.

1.3 Legal requirements

The *NSW Protection of the Environment Operations Act 1997* makes it an offence to pollute waters.

The term 'pollutes' means: to place in or on, or introduce into or onto, the waters (whether through act or omission) any material that changes the condition of the water. This includes soil, earth, mud, stones, sand, clay, or washing of such material.

The term 'waters' also has a broad definition, and includes not just harbour, river and creeks but also such things as artificial drains, channels and gutters designed to transport stormwater runoff, which are theoretically the headwaters of such water bodies.

A person may commit an offence if they place any matter (such as sand or soil) in a position where it is likely to fall, descend, wash or blow into any waters.

Severe penalties exist for offences under the Act, ranging from on-the-spot fines through to court imposed fines for each day the offence continues. The Act imposes liability on anyone who participated or contributed to the offence.

Willoughby City Council may also serve a cleanup or prevention notice to install or maintain erosion and sedimentation controls which carries an administration fee. Failure to comply with the cleanup or prevention notice may also result in a penalty.

1.4 Basic principles of erosion and sediment control

You must plan ahead when proposing to carry out any demolition, excavation, construction, building or landscaping works or installation of services.

Even if the proposed works are minor, a site plan showing where you intend to do carry out the works and store construction and building materials including spoil, sand and soil, should be drafted before starting any work. This is to decide where sediment and erosion control measures may be needed.

You should follow these principles when dealing with erosion and sediment control measures:

a. the most effective way of minimising soil erosion is to limit the area of soil disturbance and removal of existing site vegetation and controlling site activities throughout the project

b. establish sediment controls on the site before any earthworks commence or materials are stockpiled

c. control stormwater runoff on the site by reducing water volumes that need to be treated by site controls. This can be achieved by diverting surface water and rainwater away from the work and stockpile areas and reducing water velocities to minimise the potential for scour and sediment pollution

d. cover roof frames as soon as possible, install roof water drainage and connect to rainwater storage, onsite detention tank or the council drainage system to remove clean water from the site

e. conserve and cover topsoil on the site to reuse in revegetation works

f. cover disturbed soil with tarpaulin, straw or cut vegetation

g. stabilise the site through revegetation immediately construction work has been completed. Revegetation can be done in a staged way using temporary cover crops to provide short-term soil protection until permanent vegetation can be established

h. inspect and maintain erosion and sediment control measures throughout the project to reduce the potential for environmental degradation to occur

1.5 Other pollutants from construction sites

As well as pollutants such as sediment, there are several other potential sources of construction pollution. These include the following with suggested management techniques:

a. operations such as washing concreting tools and painting equipment, washing the cement residue from paved surfaces being prepared with an exposed aggregate finish and the cutting of bricks and tiles or other masonry should be done within the property boundary with suitable controls in place

b. potential pollutants such as building litter and waste, dust, fuels and oils and landscape inputs (like fertilisers and herbicides) need to be properly managed onsite

c. incorrect storage of chemicals used in the building process should also have suitable controls in place to minimise the risk of spillage and pollution incidents

2 Standards

Managing Urban Stormwater: Soils and Construction Volume 1, March 2004, by Landcom, NSW Government (The Blue Book) provides detailed information on sediment and erosion control management for construction sites. It includes information on the preparation of plans and techniques to be used to minimise sediment loss and control erosion.

Sediment and erosion control measures on construction sites within Willoughby local government area must be in line with the recommendations of The Blue Book.

3 Submission requirements

The total site area that will be disturbed during the course of the building and construction activity generally defines the minimum submission requirement for a development site.

This includes areas of cut and fill, removal of vegetation, unpaved driveways and access ways and stockpiling of building materials and soil.

Development applications for sites next to or near sensitive areas such as steep escarpments, public open space and waterways, may need to include more detailed management plans. This is at the discretion of Willoughby City Council.

The table below is a guide for submission requirements for site management of soil and water based on exposed surface area and the proposed development activity.

Table 21 Guide to submission requirements

Disturbed area	Typical activities	Submission requirements
<250m^2	Alterations and additions such as house extensions, small driveways, carports and garages which are not: Adjacent to public open space or a water course. Located on a slope greater than 18°. Involve cutting or filling the land which may alter the rate, volume or direction of overland flow.	Include a brief written statement on the proposed water management measures and expected performance levels in the SEE provided with the development application. Before disturbing any land or stockpiling any building materials appropriate sediment and erosion controls are to be installed and continually maintained to prevent pollution of downstream waterways.
250 to 2000 m^2	New house, commercial, dual occupancies, multi unit residential, small industrial complexes, subdivision and infrastructure construction. Alterations and additions that are: Adjacent to public open space or a water course. Located on a slope greater than 18°. Involve cutting or filling the land which may alter the rate, volume or direction of overland flow	A site management plan is to be provided to Willoughby City Council for approval before issue of any construction certificate. The plan should specify proposed sediment and erosion control measures for each stage of the construction activity such as demolition, excavation and construction. See Section 2.1 of this technical standard for submission details.
>2000 m^2	Large multi unit residential, commercial, institutional, industrial developments or development containing more than 10 car parking spaces	Include a soil and water management plan with the development application specifying proposed measures for the control of erosion and pollution of water both during and after construction. Calculations as to the need for a sediment basin should be included. See section 3.2 of this technical standard for submission details.

3.1 Site management plan

Willoughby City Council may require a site management plan to be lodged for consideration before issuing a construction certificate for a proposed development.

All development applications must include a site management plan if the proposed development will expose a soil surface area:

a. greater than 250m^2 up to 2000m^2

b. less than 250m^2 if the site:

- immediately adjoins public open space or a watercourse
- is on a slope exceeding 18 degrees
- involves cut or filling of the land which will alter the rate, volume or direction of overland flow
- where a new residential, commercial or industrial building is proposed to be demolished and/or constructed

The site management plan is to consist of a plan and written specification of proposed soil erosion minimisation and sediment control measures for the site during the demolition, excavation, construction and landscaping stages.

The plan must also consider these issues:

- truck routes to and from the site
- location and description of “all weather access and egress for vehicles” for the site
- location and type of diversionary drains and any other form of sediment and erosion control system
- sediment containment and/or dewatering methods including flocculation and monitoring program
- location and construction of temporary protective fencing/hoardings to the perimeter of the site
- location of site storage areas/sheds/equipment
- location of building materials/stock piles for construction
- dust suppression measures
- details of methods of temporary storage and disposal for demolition/waste materials
- protective measures for tree preservation
- schedule of proposed site stabilisation
- provisions for temporary sanitary facilities
- location and size of waste containers/bins
- frequency and nature of any maintenance program
- complaint management system including contact details of the person responsible for site management 24 hours a day, 7 days a week

If Willoughby City Council deems that the proposed building and construction activity poses a higher risk of erosion or polluting downstream waterways, the applicant may need to supply this information:

- soil characteristics such as depth, erodibility and dispersibility for all proposed exposed soil horizons
- topography such as ground slope and upstream catchment
- location relative to existing drains and waterways
- extent of cut and fill
- downstream vegetative buffer zones

3.2 Soil and water management plan

Development applications must include a soil and water management plan if:

a. the proposed area of soil surface exposure is 2000m^2 or more

b. the development contains 10 or more car parking spaces

c. new commercial, industrial or institutional development

The soil and water management plan must specify proposed measures to control erosion and pollution sources of water both during and after the construction phases.

Details on how to prepare a soil and water management plan are described in Chapter 2 of Managing Urban Stormwater: Soils and Construction Volume 1, March 2004, published by Landcom, NSW Government.

The soil and water management plan is to be provided to Willoughby City Council before issue of any consent for the proposed development.

4 Sediment and erosion control techniques

Each site will need to be individually assessed to ensure effective containment of sediment and erosion control during each stage of the development activity, particularly on large development sites.

These controls aim to ensure only clean water enters council stormwater drainage systems and natural waterways by:

a. diverting surface water around the site

b. minimising the volume of rainwater falling/flowing onto the site by installing the roof covering material and roof water drainage system as soon as possible

c. slowing down the surface water flows through the site

d. covering any exposed soil and stockpiles of soil, sand and spoil

e. placing erosion and sediment control systems at regular intervals on the site

f. preventing soil from leaving or entering the site on the wheels of vehicles or falling from the trays of vehicles

Erosion and sediment control measures on construction sites are to be in line with the requirements of The Blue Book. Techniques to be used include:

- sediment fences
- filter traps
- filter strips
- sediment basins
- diversion banks and catch drains
- stabilised site access
- site stabilisation

5 Maintenance

Proper maintenance of erosion and sediment controls is vital in protecting the environment from construction pollutants. Every rainfall event that generates runoff could reduce the effectiveness of the controls or expose problems that must be rectified before the next storm.

Consider these points in the maintenance of site controls:

a. the site should be actively managed to minimise potential pollution hazards at all times. A number of simple, cost effective methods are available to achieve this and delays due to wet weather or incurring fines can be expensive.

b. maintenance of controls should also account for changes to the site as construction proceeds. Drainage path locations and runoff velocities will change, site slopes, catchment size, type of runoff (sheet or concentrated flows) can be dramatically altered by site work. Additional controls or modifications to existing controls may be required to prevent soil loss off site

c. revegetation or other surface treatments like paving, heavy mulching or landscaping of the site must be done as soon as construction activities cease. It is vital that temporary sediment controls are not removed until the site has been effectively stabilised

d. current legislation requires that discharges from the site are of an acceptable standard and penalties will be imposed if pollution events occur. It is important to remember that the law does not recognise your inexperience or inability to meet acceptable water quality standards, the difficulties experienced on a specific site or how you feel on the day

5.1 Responsibilities of the site supervisor

A site supervisor must be nominated to regularly inspect site controls and initiate any repairs or maintenance activities. Inspections must be carried out daily when there is activity on the site and after any rainfall event that causes runoff.

The nominated site supervisor must supervise the installation of erosion and sediment controls, carry out maintenance inspections and initiate repairs if required.

This person will be Willoughby City Council’s contact for the maintenance of erosion and sediment controls at the site.

On large-scale development sites, the supervisor must keep a logbook to document site conditions and actions taken throughout the project. The book should be kept on the site and made available to any authorised person on request. Entries in the logbook should be made at least weekly, and immediately before forecast rain and after rainfall. Entries should include:

a. volume or intensity of rainfall

b. condition of soil and water management measures

c. condition of stabilisation material or vegetation

d. need for implementation of dust prevention measures

e. remedial works to be done

If the nominated supervisor is temporarily unable to perform inspections, another suitably experienced person should carry out and record the inspection. The nominated supervisor still has overall responsibility for the integrity of erosion and sediment controls on the site.

If the nominated supervisor is no longer involved in the project, a new supervisor is to be nominated by the applicant and Willoughby City Council notified in writing.

The site supervisor should be someone with a working knowledge of erosion and sediment control and preferably have overall responsibility for the site. The person could be the project

or area manager, a clerk of works, a builder, an owner/builder (depending on their experience) or specialist sub-contractor.

5.2 When to undertake maintenance inspections

The site supervisor will be required to inspect site controls every day before the start of any work and at the end of each day that activity occurs on site and after any rain event that causes runoff.

It is recommended that controls are inspected during rain events that threaten to exceed the capacity of the sediment controls so emergency measures can be put in place to reduce the potential for soil loss.

Appendix A – Guidelines for stormwater control in bushland areas

1 Introduction

Stormwater runoff from urban areas, particularly runoff from houses and roads, is a major cause of degradation in urban bushland areas.

Some of the problems created by uncontrolled runoff into the bushland reserves include:

- native vegetation is less able to cope with greater volumes of stormwater runoff and higher concentrations of nutrients, which causes dieback
- the extra nutrient and sediment pollution loads reduce water quality in the receiving waters
- stormwater transports seed and vegetative material from gardens and road edges which causes weed infestation
- stormwater outlets historically consist of a pipe outlet that allows stormwater discharge to spread across the bushland areas that do not have natural drainage lines. The stormwater flows to the receiving waters by spreading out across the slope and encouraging weed growth or by finding the quickest path and causing soil erosion problems

To protect the integrity of bushland reserves and their receiving waters it is essential to efficiently manage stormwater discharge outlets in bushland areas.

Approval is required from Willoughby City Council's infrastructure services division before any works is done in bushland reserves. Each site has individual characteristics that need to be addressed to ensure the effective capture and containment of stormwater runoff. We recommend having a preliminary meeting with a Willoughby City Council bushland manager to discuss options.

2 Procedure for managing stormwater in bushland areas

To reduce the impact of stormwater runoff on bushland areas and receiving waters, these guidelines apply:

a. construct an energy dissipation structure at the pipe outlet to reduce the velocity of the stormwater and reduce the incidence of scour

b. construct a pit or basin within private property for the retention of gross pollutants and sediment and to reduce the velocity of the stormwater flow

c. construct a well-defined, natural looking rock-lined channel. The channel should be incorporated into the existing landscape, drainage lines and rock outcrops

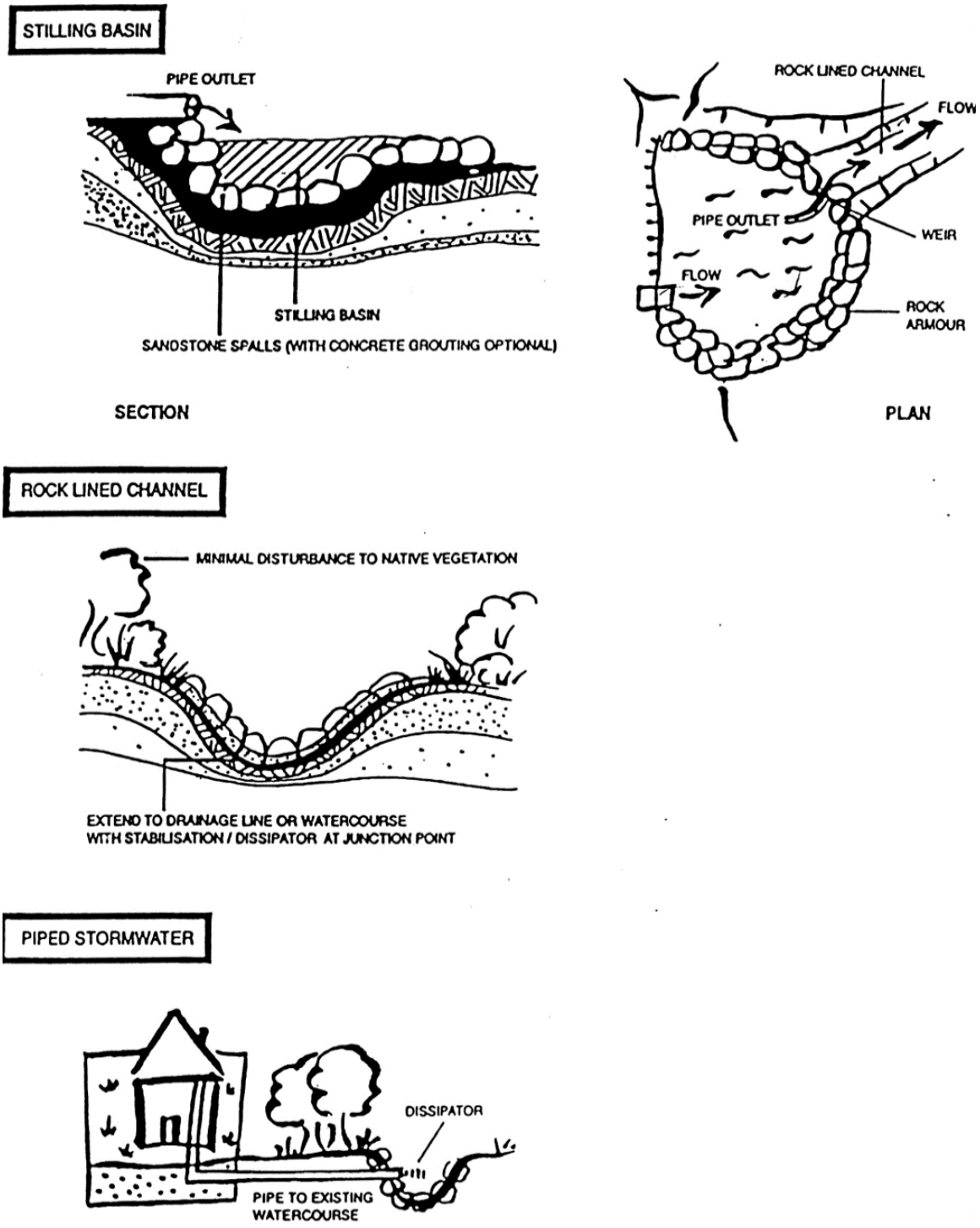
d. construct a sediment pond(s) along the channel to reduce the velocity of the flow, capture sediment and create habitat for local fauna

e. ensure the channel discharges into the nearest substantial drainage line or watercourse

f. rehabilitate any bushland area disturbed as a result of the stormwater channel works as soon as possible when works are complete

g. continually maintain the pit, channel and ponds

Figure 7 Methods for managing stormwater in bushland



2.1 Preliminary design work

When designing for stormwater discharge through bushland reserves, consider these issues:

a. determine the size of the catchment discharging through the channel and the scale of the channel required. The channels must be wide and deep enough to incorporate rock lining whilst maintaining the capacity of the peak flow. For example, an average household discharge pipeline is usually 90mm dia. so the internal sectional area of the rock-lined

channel should be around 0.5m^2 or 5 times the diameter of the pipeline. For large pipe diameters the channel size will need to be hydraulically and hydrologically assessed

b. determine the location of the channel which should include:

c. the location of the closest natural drainage line or watercourse that the proposed channel will discharge into and details on:

- the incorporation of the channel into the existing landscape, drainage lines and rock outcrops while protecting existing trees and vegetation
- the optimum route for the channel containing a minimal number of bends
- the most degraded area of bushland that will suffer least from construction disturbance

d. determine the most suitable location(s) for energy dissipation systems such as a pond(s). Ponds reduce the velocity of the water discharging from the stormwater pipeline. The size and capacity of the pond will depend on the volume and velocity of the peak flows

e. ponds must be located in areas where there is easy access for regular maintenance, such as the removal of any caught litter or sediment from the pond

Figure 8 A rock lined channel from a stormwater pipe outlet

2.2 Visual appearance of the rock-lined channel

There are a few techniques to keep in mind when designing a rock-lined channel that can improve the visual appearance of the construction. These include:

a. constructing channels and ponds below ground level as much as possible so that they blend into the natural landscape and allow surface runoff from the surrounding areas to also use the channel and ponds

b. using natural materials for the construction of the channel and ponds such as a combination of different sized rocks

c. using locally sourced materials (other than bush rock) where possible as they are more likely to blend in to the natural landscape

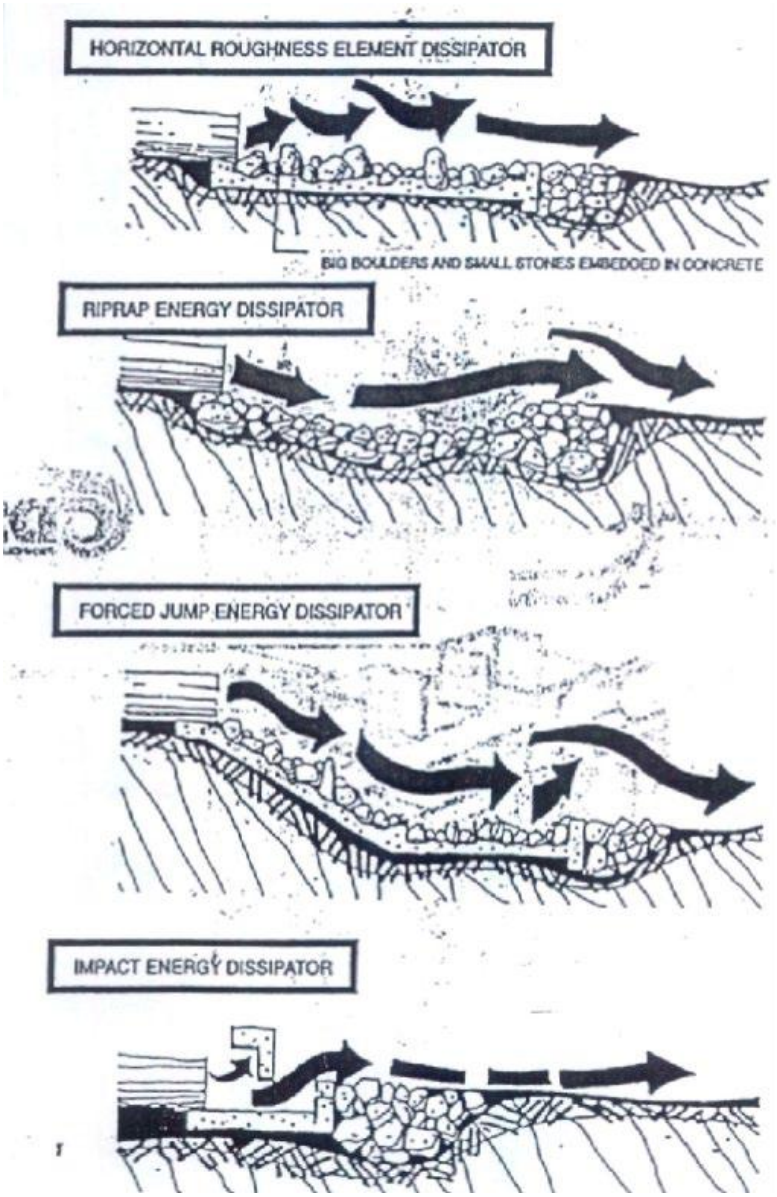
d. using the topography to direct the water. Channelling the water along the path of least resistance wherever possible gives it the appearance of being a natural drainage line and makes construction easier

2.3 Construction of energy dissipators

An energy dissipator must be constructed within the property boundary at the discharge point of the stormwater pipeline to reduce the velocity of the stormwater and for the retention of gross pollutants and sediment. Protection of the opposite bank of the system from scour may also be necessary dependant on the bank materials and the “jet” effect from the discharge pipe. This would also apply where the stormwater pipeline is discharging directly into a natural drainage line or watercourse.

Examples of systems that can be used for energy dissipation are shown below.

Figure 9 Four kinds of energy dissipators



2.4 Construction of rock-lined stormwater channels

To stop the spread of stormwater runoff across the bushland reserve, the water is to be directed or channelled to the nearest existing significant drainage line or watercourse. This may either involve digging a trench to the next layer of bedrock or constructing a rock-armoured channel.

These are the minimum requirements for any rock-lined channel works:

a. channels must be dug into the ground wherever possible so that the sides of the channel do not protrude above ground level. Where this is not possible, the sides of the channel may be mounded

b. when significantly changing the direction of the flow along the channel a pond must be incorporated at the bend

c. erosion control fabric should be laid along the base of the channel ensuring it overlaps the sides

d. the erosion control fabric should not be reinforced with plastic mesh and must not be exposed on completion of the channel construction

e. the base of the channel must be armoured with interlocking sandstone rock capable of withstanding high flows

f. spoil from the channel excavation should be used to form berms along the edges of the channel to divert overland flows from the surrounding area into the channel

Barriers such as large rocks should not be placed across the channel. While a barrier may slow the water to some degree it may also divert the water. This can place pressure on the sides of the channel causing the sides to erode and displace or the water may leave the channel entirely.

If it is necessary to slow the velocity of the water in the channel another pond should be incorporated into the structure. This will not only slow the velocity but will also create habitat for native fauna.

Figure 10 Rock-lined stormwater channel through bushland

2.5 Construction of sediment ponds

A sediment pond is used to slow the rate of flow of the stormwater and allow it to pool. As the water slows down, the energy of the water is dissipated causing the sediment carried in the water to settle to the bottom of the pond.

Requirements for building a sediment pond:

a. there must be a flat base lined with flat rocks free of obstructions such as protruding rocks, exposed erosion control fabric, roots etc to facilitate the removal of sediment

b. the pond walls may be constructed of mounded earth re-enforced with erosion control fabric and covered with sandstone boulders

c. the erosion or weed control fabric should not be exposed on completion of the pond construction

d. the pond outlet point should be located below the invert of the stormwater inlet pipeline but lower than the top of the pond walls. This ensures that the water is directed into the channel below. It also ensures that sediment doesn't cause the discharge pipeline to block and water to back-up the pipeline

2.6 Stabilisation of disturbed area

Any disturbed area must be stabilised on completion of the construction work. Berms should be covered with organic weed mat and planted out with a local native groundcover species to bind the soil.

3 Maintenance

Regular inspections and maintenance of the drainage systems long after the systems have been installed are essential for their continuing efficient operation. The property owner is responsible for regular monitoring and maintenance of any system in their property boundary.

Below is a general guideline for maintenance:

a. the pond will need regular inspections and removal of trapped sediment and gross pollutants. As the pond fills with sediment and gross pollutants it becomes less efficient during future storms

b. the channel should be regularly checked for signs of excessive erosion and blockages and cleared when necessary

Appendix B – Example method of site stormwater filtration before discharge

1 Introduction

This method may be useful for development sites that require substantial excavation works as part of the development and may need to be dewatered after rainfall events. This method is useful for shale and fine clay soils.

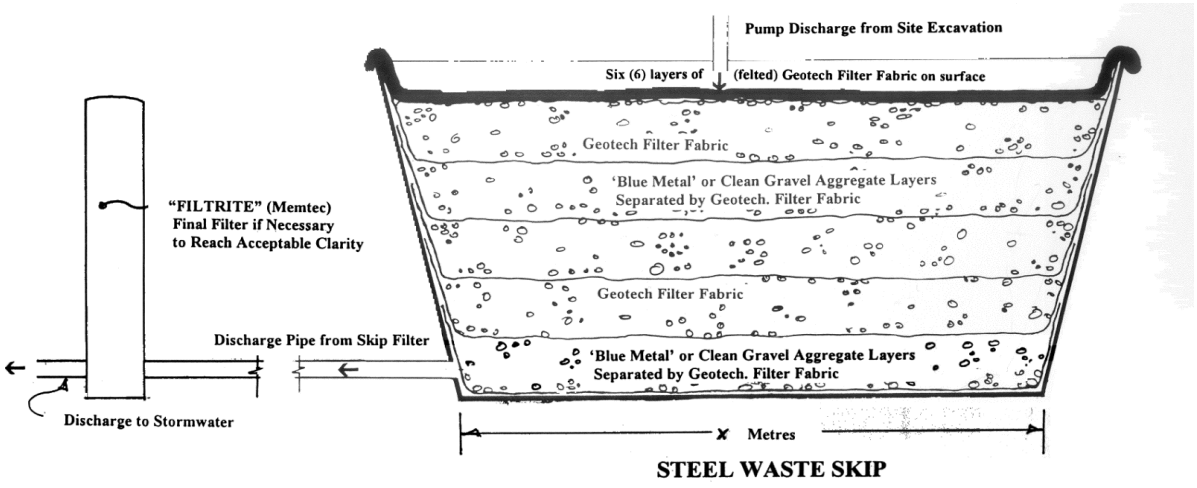
Figure 11 shows a site water filtration method based on a standard steel waste skip. This is provided with an outlet pipe fitted to one end near the base and filled with blue metal or clean gravel aggregate in five layers separated by a suitable geotech filter fabric so as to completely contain the contents.

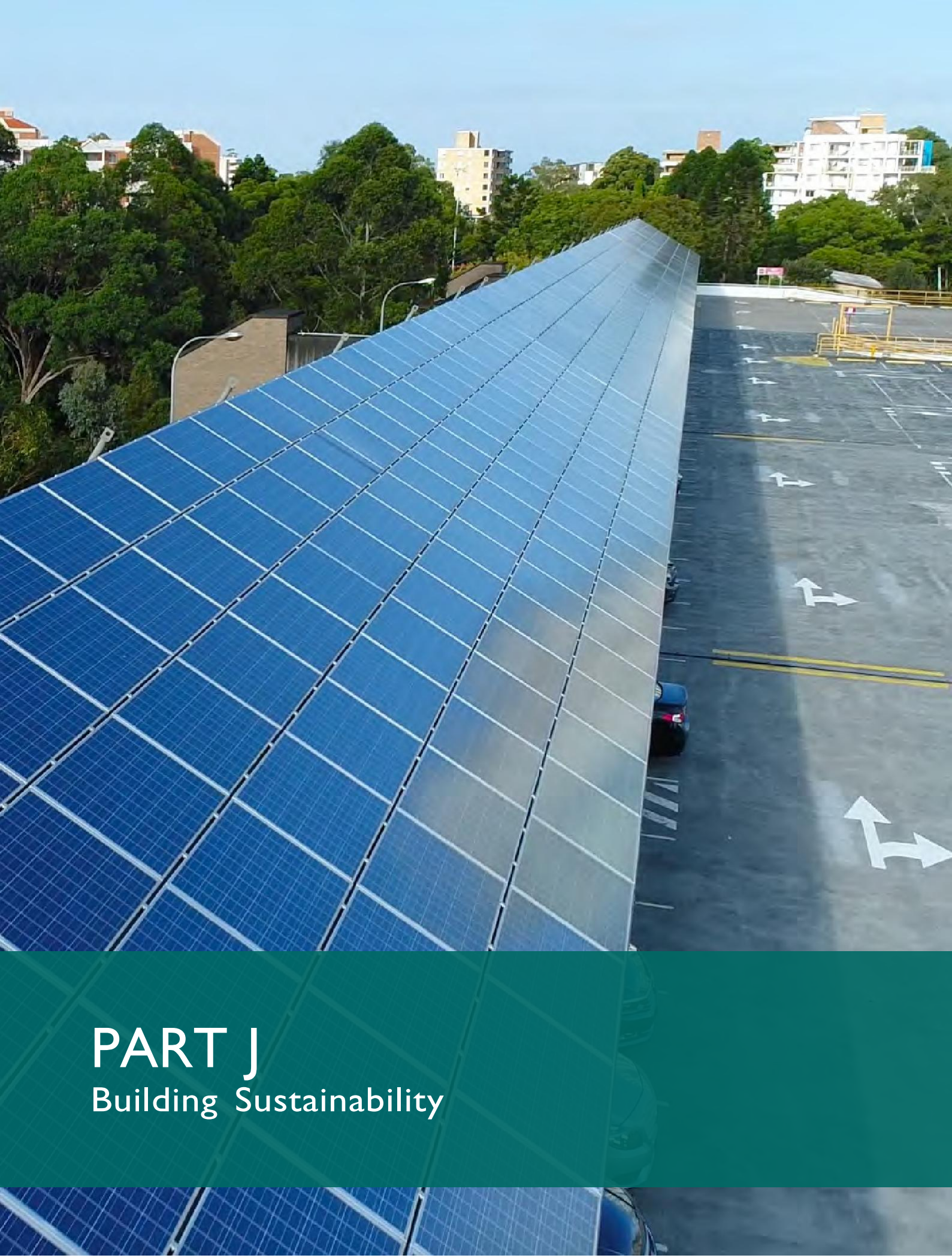
In operation, a settlement pit or tank within the site is used to settle the major sediment content of the site water by use of a proprietary flocculent over 24 hours before it is pumped into the top of the filter skip. The skip should be installed before starting excavation.

The water is filtered through the five layers of filter cloth before being discharged. In most cases the water should be of satisfactory clarity at that stage but it must be monitored regularly.

If the clarity is not satisfactory – it must be completely clear to the eye – or if the quality deteriorates over a period, a final filter such as a Filterite Memtec filter must be fitted between the skip filter and the final discharge point.

Figure 11 One method of site water filtration before discharge into the stormwater system





PART J
Building Sustainability

Part J: Building Sustainability

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1 Introduction

Willoughby City Council is committed to environmental sustainability and contributing to the conservation of natural resources and global environmental improvement.

This part of Willoughby Development Control Plan (Willoughby DCP) considers the objectives of *Environmental Planning and Assessment Act 1979* and Clause 1.2(2)(b) of Willoughby Local Environmental Plan (Willoughby LEP) for ecologically sustainable development.

1.1 Aim

The aim of this part is to apply best practice principles in the design and construction of developments to create energy efficient and environmentally sustainable buildings.

1.2 Objectives

The objectives of this part are to:

a. improve the design to achieve sustainable and energy efficient buildings with low greenhouse gas emissions

b. reduce waste and promote the adaptable reuse of existing buildings and encourage durable design and construction that is adaptable and low maintenance

c. encourage the use of renewable energy and alternative water supply

d. improve resident and employee comfort, health and wellbeing

e. reduce natural resource consumption and source materials responsibly

f. encourage sustainable water management

g. consider climate adaptation and resilience

h. promote sustainable transport management

1.3 Statutory controls

National and state controls provide mandatory requirements to meet energy efficiency targets in the design and construction of buildings. These controls are in the State Environmental Planning Policy (Building Sustainability Index: BASIX) 2004 (BASIX SEPP) and the National Construction Code.

The National Construction Code identifies the class of building for different types of development. Volume 1 has the requirements for Class 2 to Class 9 buildings. Volume 2 has the requirements for Class 1 and Class 10 buildings. Attachment 1 of this part has a guide to the building classifications.

1.4 Best practice

The objectives of this plan can be achieved by incorporating best practice design principles and initiatives into a development proposal. Attachment 2 includes criteria that should be considered to achieve the best possible outcome for energy efficient and sustainable developments.

The design principles in Attachment 2 will contribute to compliance with the statutory requirements and/or satisfy some of the components of a sustainability rating tool. Applying these principles is likely to reduce the need for active/mechanical heating and cooling and provide long-term cost savings by reducing energy consumption.

For more information and ideas to help integrate sustainability into a development project, see willoughby.nsw.gov.au/environment

2 Minor developments

2.1 Definition of minor developments

In this part, 'minor developments' include construction of, and alterations and additions to:

a. single dwellings

b. attached and detached dual occupancies

c. boarding houses not exceeding 300m^2 and 12 persons

d. attached dwellings of not more than two storeys

e. secondary dwellings

Alterations and additions include ancillary structures such as garages, carports, sheds, studios, swimming pools, gazebos and the like.

2.2 Building Sustainability Index (BASIX)

A BASIX Certificate must be included in a development application for all new residential construction. This includes alterations and additions to Class 1a, 1b, 2 and 4 buildings with an estimated cost exceeding $50,000. A BASIX Certificate is also required for the addition of a swimming pool or spa with a combined volume exceeding 40,000 litres.

This certifies that the residential development will meet the required environmental targets for energy, water and thermal comfort.

Notes:

- The BASIX SEPP states that a development control plan cannot include provisions that exceed the minimum BASIX standards. However, Willoughby City Council encourages proposals with sustainability initiatives that exceed minimum regulatory requirements.
- Willoughby City Council cannot approve a development application for residential development that exceeds $50,000 or a swimming pool/spa with a volume exceeding 40,000 litres if they do not achieve the minimum BASIX score.
- The BASIX certificate must be generated no earlier than three months before the development application date.

More details on BASIX are available from the Department of Planning and Environment at planningportal.nsw.gov.au/basix.

Applicants are encouraged to incorporate, wherever possible, the best practice design principles and initiatives detailed in Attachment 2.

3 Major developments

3.1 Definition of major developments

For this component of the plan, 'major developments' include construction of, and alterations and additions to:

a. residential accommodation comprising: attached dwellings with 3 or more storeys; hostels and boarding houses exceeding 300m^2 and 12 people; multi dwelling housing; residential flat buildings; seniors housing; and shop top housing

b. new buildings, alterations and additions, and refurbishment of commercial, industrial, indoor recreation facilities and institutional developments

Notes**:**

- Institutional development includes community facilities, educational establishments, childcare centres, places of public worship and public administration buildings.
- Indoor recreation facilities include gymnasiums, health studios, squash courts and indoor swimming pools.
- Depending on the size and scale, an authorised officer may determine that a proposed development is not considered ‘major development’.

3.2 General requirements

As a minimum, all major developments must consider the best practice design principles and initiatives outlined in Attachment 2.

Depending on the scale of development, the following requirements apply:

a. for developments with an estimated cost between $750,000 and $5 million, a sustainability performance statement must be included with the development application

b. for developments with an estimated cost between $5 million and $30 million, projects should seek to achieve a minimum 4-star rating using the most recent and relevant Green Star rating tool (or equivalent)

c. for developments with an estimated cost over $30 million, projects should seek to achieve a minimum 4-star rating and aspire to achieve a 5-star rating using the most recent and relevant Green Star rating tool (or equivalent)

Note**:**

- See section 3.3.5 for more information on Green Star requirements.

3.3 Sustainability performance statement

A sustainability performance statement must be prepared by a suitably qualified person and include details of the proposed method intended to be used to achieve an energy efficient and sustainable development. The statement must justify not including any of the design principles and initiatives in Attachment 2.

Willoughby City Council requires a compliance statement before a construction certificate is issued. The compliance statement must be prepared by a suitably qualified person. It must

verify the construction certificate plans include the provisions agreed to in the sustainability performance statement.

Note:

- Instead of a sustainability performance statement, applicants may choose to provide a Green Star rating report or similar rating tool.

3.4 National Construction Code

Section J of the National Construction Code requires the design of certain non-residential buildings to satisfy minimum standards to improve energy efficiency. These provisions relate to Class 3 and Class 5 to 9 buildings.

Compliance must be shown at the construction certificate stage. However, the design of an environmentally sustainable building needs to be resolved at the development application stage.

A compliance statement, prepared by a suitably qualified person, must be included with the development application for all Class 3 and Class 5 to 9 buildings. This is to confirm that the energy targets can be achieved in line with the 'deemed to satisfy' or 'performance solution' provisions under Section J of the code.

Notes:

- The Green Star rating tool or NABERS Commitment Agreement can be used to satisfy the 'performance solution' provisions under Section J of the code.
- Where BASIX is not applied to alterations and additions to Class 1 and 2 and Class 4 parts of buildings, the Section J provisions apply to ensure energy efficiency measures are incorporated.

3.5 Green Star

Green Star is a rating system for buildings and fitouts. The design and construction are assessed against nine categories, including energy, water, waste, and indoor environmental quality. A 4-star rating represents 'best practice', a 5-star rating is 'Australian excellence' and a 6-star rating is 'world leadership'.

For developments between $5 million and $30 million, Willoughby City Council encourages the use of Green Star or similar rating tool. If one of these tools is used for these types of development, a copy of the report must be included with the development application. We expect developments to achieve a minimum 4-star Green Star rating or a 'best practice' standard using a similar rating tool.

For developments that exceed an estimated cost of $30 million, applicants must show the design of the proposed development can achieve a minimum 4-star Green Star rating or a 'best practice' standard using a similar rating tool. We expect developments to achieve a minimum 5-star ('Australian excellence') rating under a Green Star or similar rating tool.

Note:

- Willoughby City Council requires a compliance statement to verify the Green Star rating can be achieved before a construction certificate is issued. The compliance statement must be prepared by a suitably qualified person.

3.6 National Australian Built Environment Rating Scheme (NABERS)

The commercial building disclosure program is a regulatory program that requires energy efficiency information to be provided in most cases when commercial office space of 1,000m^2 or more is offered for sale or lease. The program requires an up-to-date building energy efficiency certificate. This must be done by an accredited assessor in line with the National Australian Built Environment Rating System (NABERS).

The NABERS Energy for offices rating is a national rating system that measures building performance on a scale of zero to six stars. A zero-star rating means the building is performing well below average and has plenty of scope for improvement. A five-star rating is 'excellent'. A six-star rating is for market leading performance, with half the greenhouse gas emissions or water use of a five-star building.

For more information on the Australian Government program see cbd.gov.au

NABERS can also be used to rate developments such as apartments, hotels, data centres and office buildings, and to rate water, waste and indoor environment quality.

After construction and occupation, these types of development must show compliance with this plan by signing a NABERS Energy Commitment Agreement. This is an agreement to achieve a minimum five-star rating for the base building, whole building or tenancies as appropriate:

a. new residential developments comprising 10 or more apartments

b. new commercial office buildings with a net lettable floor area of 1,000m^2 or more

c. alterations and additions or refurbishment of existing commercial office buildings with a net lettable area of 1,000m^2 and estimated cost of work over $750,000

The NABERS Commitment Agreement must be submitted to the Department of Planning and Environment and a copy provided to Willoughby City Council before a construction certificate is issued.

The performance rating must be done once the building is fully operational and there is 12 months of energy data collection. A copy of the assessment report should be provided to Willoughby City Council for our records.

A NABERS Energy Commitment Agreement is not required if:

d. the upgrade works would negatively impact on the heritage significance of a heritage listed item under Schedule 5 of Willoughby LEP

e. the costs associated with the energy efficiency upgrade works are unreasonable when compared to the overall estimated cost of works for the alterations, additions and refurbishment

Notes**:**

- Any application that may impact on a heritage item must be supported by a heritage impact statement prepared by a suitably qualified heritage consultant.

- If it is asserted that the costs are unreasonable, the development application must be supported by a detailed cost report prepared by a registered quantity surveyor, itemising and verifying the cost of the required energy efficiency upgrade works.

For more information on NABERS see nabers.gov.au.

Attachment 1 – National Construction Code building classifications

This table details the class of building and the different types of development.

Table 1: National Construction Code building classifications

Class of building	Types of development
Class 1a	Single dwellings. This includes a detached house or semi-detached dwellings, semi-detached dual occupancies, terraces, townhouses and villa units with fire resisting walls and no basement carpark.
Class 1b	Boarding house, guest house, hostel, or the like, not exceeding 300m^2 and not more than 12 residents.
Class 2	Building containing two or more sole occupancy units. This includes dual occupancies/duplexes (where one unit is above the other), serviced apartments with a kitchen, apartments in shop top housing, and residential flat buildings.
Class 3	A residential building, other than a class 1 or 2 building, including; boarding house, hostel, backpackers accommodation, and retirement villages.
Class 4	A single dwelling within a class 5, 6, 7, 8 or 9 building.
Class 5	An office building used for professional or commercial purposes, excluding class 6, 7, 8 or 9 buildings.
Class 6	Shops and buildings used for supply of services directly to the public, including; café, restaurant, hairdresser's shop, public laundry, showroom, and service station.
Class 7a	A carpark.
Class 7b	Warehouses, storage/display of goods for sale by wholesale.
Class 8	A laboratory or building (factory) for the production, assembly, altering, repairing, packing, finishing or cleaning of goods for trade, sale or gain.
Class 9a	Healthcare building.
Class 9b	Assembly building including halls, libraries, schools, early childhood centres, church, theatre, nightclub/disco, cinema, and the like.
Class 9c	Aged care building.
Class 10a	Non-habitable building such as private garage, carport, shed or the like.
Class 10b	Non-habitable structures such as fences, retaining walls, swimming pools, or the like.

See more information at abcb.gov.au/resources.

Attachment 2 – Best practice design principles and initiatives

These design principles and initiatives should be incorporated into all new developments and where possible, into alterations and additions to existing buildings.

Objective a: Improve the design to achieve sustainable and energy efficient buildings

1. orientate buildings for passive solar control by maximising solar access in winter and minimising heat gain in summer
2. wherever possible, locate living and work areas within the optimum range of 20 degrees west and 30 degrees east of north
3. ensure solar access to existing solar panels or photovoltaic panels on adjoining properties is maintained for at least 3 hours
4. the area of north facing windows should be at least 10–15% of the floor area of the building
5. maximise natural ventilation through adequate window openings and flow paths
6. use appropriate thermal properties/mass for glazing and building materials
7. use effective external shading structures
8. landscaping: include deciduous trees adjacent to north facing windows and a minimum 70% local native plant species
9. buildings should include roof, wall and ceiling insulation that meets or exceeds minimum requirements of Section J in the National Construction Code
10. wherever possible, minimise east and west glazing area
11. maximise natural daylight availability
12. minimise external light pollution
13. provide energy metering to monitor annual energy consumption (MJ/m^2/year) and equivalent carbon emissions per annum (energy management system)
14. provide electricity sub-metering for developments that will consume more than 10,000kWh/annum
15. provide electricity sub-metering for lighting, air-conditioning and power within each tenancy and strata unit for multi-tenant commercial and industrial developments and strata subdivision of residential apartments
16. specify LED lighting
17. install a building management system as a minimum for all major developments
18. select energy efficient systems, appliances and equipment with efficiency controls to minimise use when not required

Objective b: reduce waste and promote the adaptable reuse of existing buildings

1. if a proposal involves demolition of an existing building, demonstrate that consideration has been given to the reuse of whole or part of the existing building

2. consider waste minimisation during design, construction and operation
3. demonstrate that at least 90% of construction and demolition waste (by mass) has been diverted from landfill
4. consider durable design and construction which is adaptable (futureproof) and low maintenance
5. provide for separation and storage of major waste streams onsite during operation including recyclable waste, general waste and composting
6. ensure waste storage rooms are easily accessible
7. set a target of 70% operational waste to be diverted from landfill including compostable organics and green waste

Objective c: encourage the use of renewable energy and alternative water systems

1. reduce the generation of greenhouse gas emissions by using renewable energy sources
2. consider the installation of photovoltaic and solar thermal hot water systems
3. consider rainwater harvesting and reuse to minimise potable water consumption
4. create more resilient, futureproofed buildings by using renewable energy systems and rainwater harvesting

Objective d: improve resident and employee comfort, health and wellbeing

1. maximise daylight while mitigating glare discomfort
2. design for good thermal comfort
3. allow for effective natural cross ventilation whenever possible
4. design mechanically ventilated areas to maintain $C0_2$ levels under 800ppm
5. consider biophilic design principles including provision of indoor plants
6. ensure pollutant generating activities such as high-volume printing equipment have a dedicated exhaust system that has a 100% return air exhausted directly to the outside or printers must have a low emission certificate
7. promote sustainable transport management

Objective e: reduce natural resource consumption and source materials responsibly

1. where possible, minimise natural resource depletion and prioritise responsible materials by specifying materials that:
    - have a high recycled/reused content
    - are responsibly sourced from non-invasive renewable sources free of modern slavery
    - are locally sourced materials
    - have transparent ingredients and supply chain
    - are free of toxic chemicals
    - are free of ozone depleting chemicals

2. aim to provide a use of lifecycle assessment to assist with design decisions and material selection.

Objective f. encourage sustainable water management

1. install water efficient appliances, fixtures and fittings
2. apply drought tolerant, low water use and native landscaping
3. use drip irrigation systems with soil moisture sensors
4. apply water sensitive urban design principles
5. use rainwater collection for reuse
6. collect and reuse greywater

Note:

- For more information on water management and sustainability, see Part I (Stormwater Management) of this plan.

Objective g: consider climate adaptation and resilience

1. provide a climate adaptation plan for major developments, identifying potential risks to people and to the project
2. ensure the climate adaptation plan addresses high and extreme risk factors during the design and operation of the development
3. consider these changes over the expected lifetime of the building:
    - increased average temperatures
    - increased maximum temperatures
    - increased severity of storm events
    - longer periods between rainfall
    - increased flooding risk and sea level rise

Objective h: promote sustainable transport management

1. major developments must submit a transport management plan (green travel plan)
2. car parking areas comprising 10 or more car parking spaces must install electric vehicle charging points at the rate of 1 charging point/10 car parking spaces
3. car parking areas comprising 10 or more car parking spaces must provide at least one car sharing space
4. if mechanical ventilation is required, the mechanically ventilated systems must install carbon monoxide monitoring and variable speed fans (refer to AS 1668.1 Car Park Ventilation)

Note:

- For more information on transport management, see Part F (Parking and Transport Management) of this plan.

PART K

Ventilation Stacks

Part K: Ventilation Stacks

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1 Introduction

The Lane Cove tunnel was completed and opened for traffic in March 2007.

Vehicle emissions from the tunnel are vented by two stacks: one at the western end, Sirius Road (Lane Cove Council local government area) and one at the eastern end, in the Artarmon industrial area, Marden Street (City of Willoughby local government area).

Both stacks have potential impacts on land in the Willoughby local government area.

Environmental assessments at design stage considered the potential impacts on buildings close to the stacks. The ventilation system was designed to ensure air impacts were acceptable to all existing buildings.

An air quality assessment was done to determine the likely characteristics of the plume and identify its zone of influence (a 3 dimensional zone also known as buffer volume). This is detailed the Lane Cove Tunnel Buffer Zone Analysis Report No: 21/06 Consulting Air Pollution Modelling and Meteorology, March 2008.

1.1 Land affected by this part

This part of the Willoughby Development Control Plan applies to land within an 800m radius of the stacks in Sirius Road, Lane Cove and Marden Street, Artarmon as shown in Maps A and B in Attachment 1.

1.2 Aim

This part aims to ensure new development does not reduce the effectiveness of the ventilation stacks.

1.3 Objectives

a. ensure a detailed assessment is carried out by a suitably qualified person to demonstrate that any new development is not exposed to excessive air pollution from the Lane Cove Tunnel ventilation stacks

b. ensure the assessment shows that any new development or alterations and additions to existing buildings do not alter the plume and wind conditions that may reduce the effectiveness of the ventilation stacks

2 Buffer zone analysis

The air quality assessment report defines a building overlay to ensure adequate separation between the vent stacks and any future proposed buildings. The derived overlay control has two main parts:

a. ensure sufficient separation of proposed buildings from the vent stack emissions so that occupants of those buildings would not be exposed to predicted pollutant concentrations that exceed air quality objectives

b. ensure sufficient separation of proposed buildings from the vent stack so that under any wind conditions the building wake would not interact with the vent stack plume and reduce the effectiveness of the vent stack

The height control/building overlay has been defined using conservative assumptions about the plume behaviour and meteorology. The overlay is used as a trigger to determine if more detailed site-specific investigation for a proposed building is needed.

The overlay for each vent stack is presented in the form of decision trees in Attachments 2 and 3 together with worked examples.

3 Submission requirements

The proponent is to assess the proposed building development against the objectives in Section 4 using the decision trees in Attachments 2 or 3 unless the development meets any of the exemption criteria below.

4 Exemptions

Exemption criteria:

a. the proposed development is outside the 800m radius from the stack (Attachment 1)

b. the proposal is for a change of use for commercial or industrial premises

c. the proposal is for alterations or additions to existing residential, industrial and commercial premises where the building floor area and building heights are not increased or altered

d. if the property is within the 800m radius of the Sirius Road stack:

    1. the building height of proposed new work is less than or equal to 13m (AHD 42)

    2. the building is located at a distance from the Sirius Road stack greater than five times the building height (when measured from the reference elevation level of 29m Australian Height Datum (AHD)

e. if the property is within the 800m radius of the Marden Street stack:

    1. the building height of the proposed new work is less than or equal to 24m (AHD 98)

    2. the building is located at a distance from the Marden Street stack greater than five times the building height (when measured from the reference elevation level of 74m AHD)

If the building is permitted under the decision tree assessment, no further action is needed unless Willoughby City Council considers a site specific investigation is necessary. The details showing that this process has been done are to be included with the development application.

A site specific investigation may be required due to the limitations of the decision tree process which applies more to isolated buildings and replacement of existing structures. For example, two or more new adjacent buildings that are larger than the existing buildings may together change the dispersion of the plume through building wake effects and need further investigation. Buildings of unusual shape or materials may also need further investigation.

5 Site specific investigation for non-exempt development based on the decision tree assessment

If a site specific investigation is required, this should be done by an appropriately qualified consultant. The consultant should first investigate that:

a. occupants of the proposed building would not be exposed to pollutant concentrations that exceed air quality objectives

b. the building wake from the proposed building under any wind conditions would not interact with the vent stack plume and reduce the effectiveness of the vent stack

Site specific investigations should then consider:

c. the detailed geometry of the proposed building

d. any changes to ambient air quality at the location of the proposed development

e. any changes to the vent stack pollutant emission rates

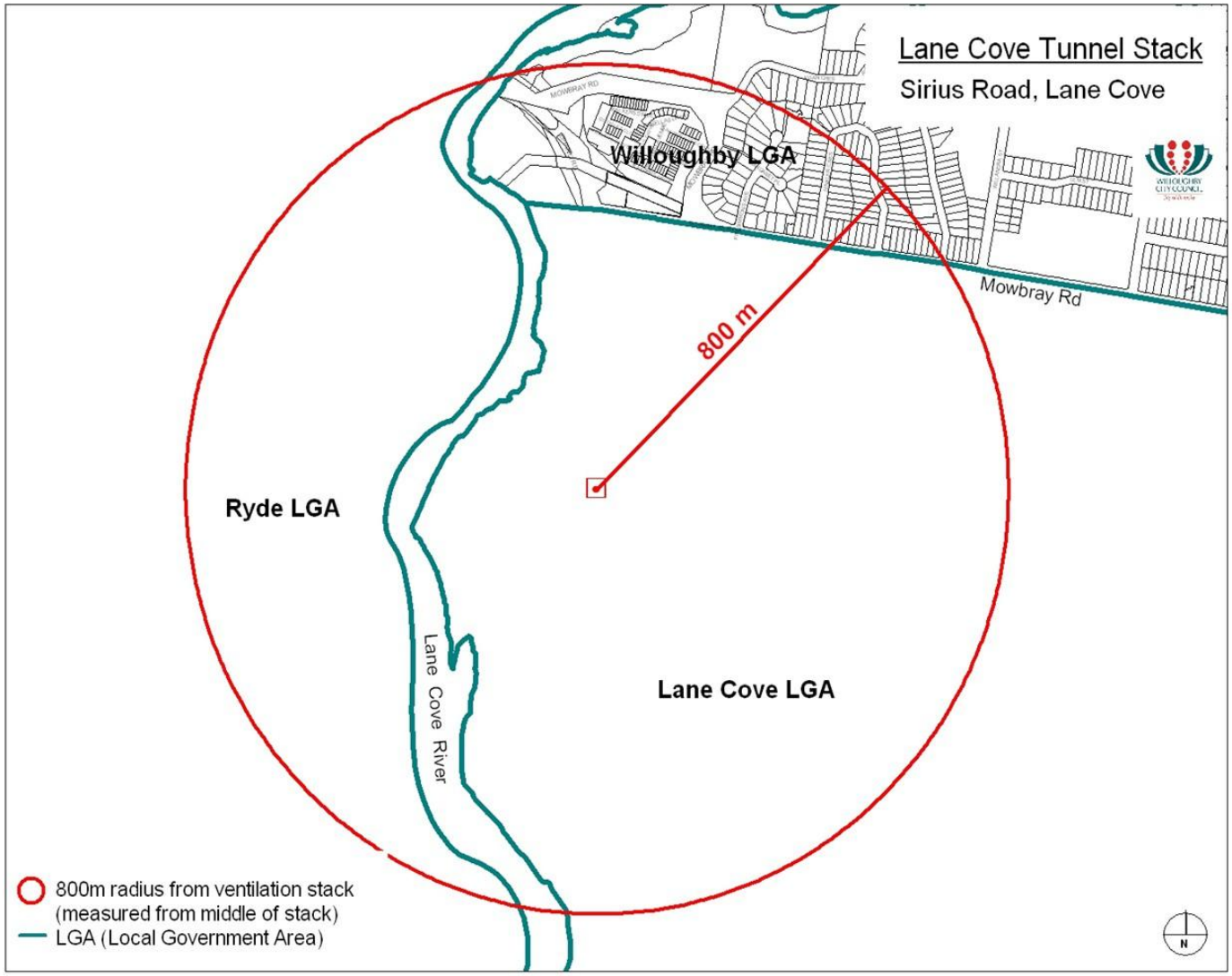
f. cumulative changes to the built form array around the ventilation stacks that may have altered the wind dispersion climate experienced by the Lane Cove Tunnel vent stacks

If the investigations show that the proposal is still marginal, physical scale modelling or computational fluid dynamic (CFD) modelling may be needed. This will more accurately determine if a new building development will satisfy the purpose of the control.

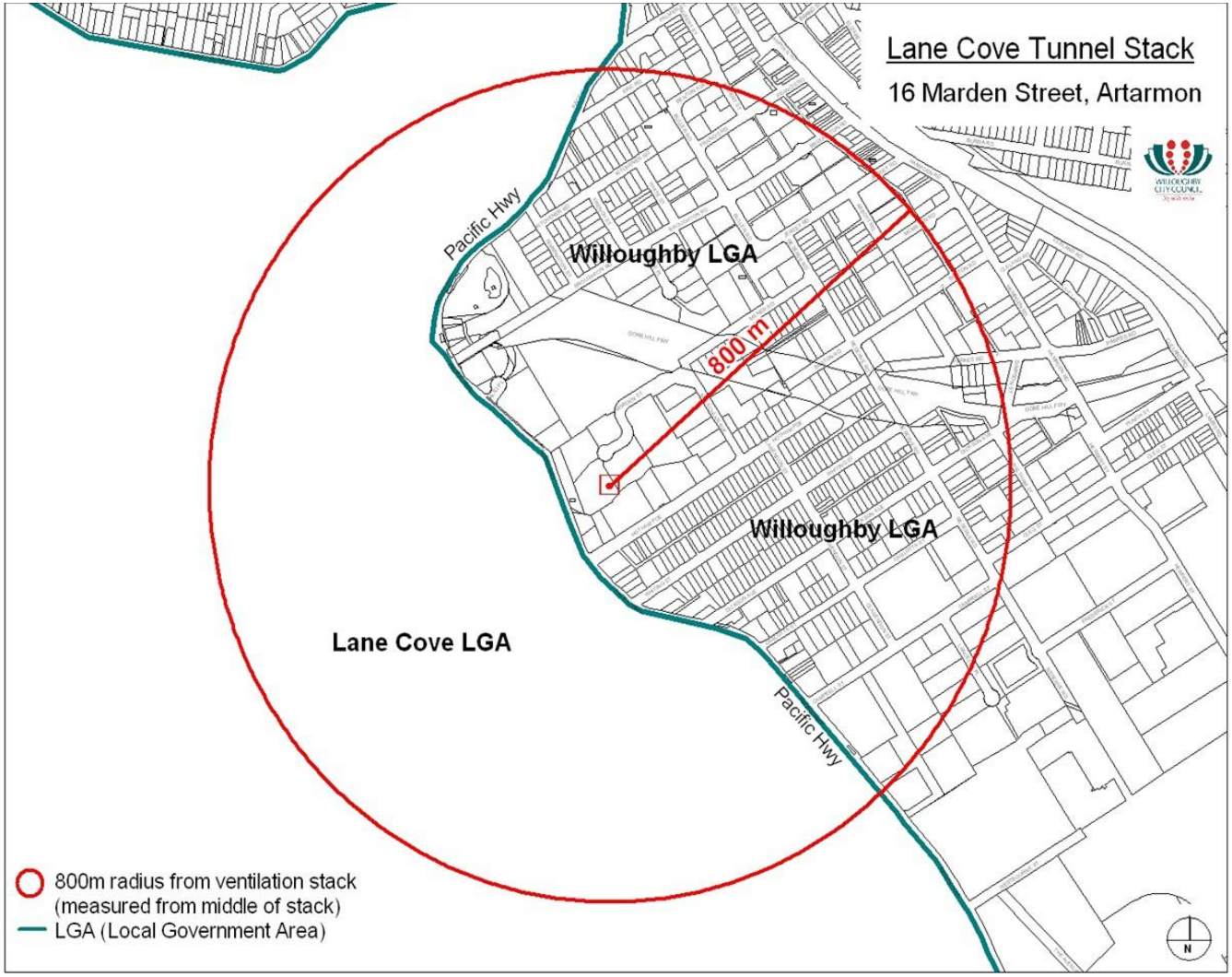
At the time of assessment the current up to date published methodologies should be used.

Attachment 1 – Maps of the Lane Cove tunnel ventilation stacks

Map A Lane Cove tunnel stack, Sirius Road, Lane Cove



Map B Lane Cove tunnel stack, 16 Marden Street, Artarmon



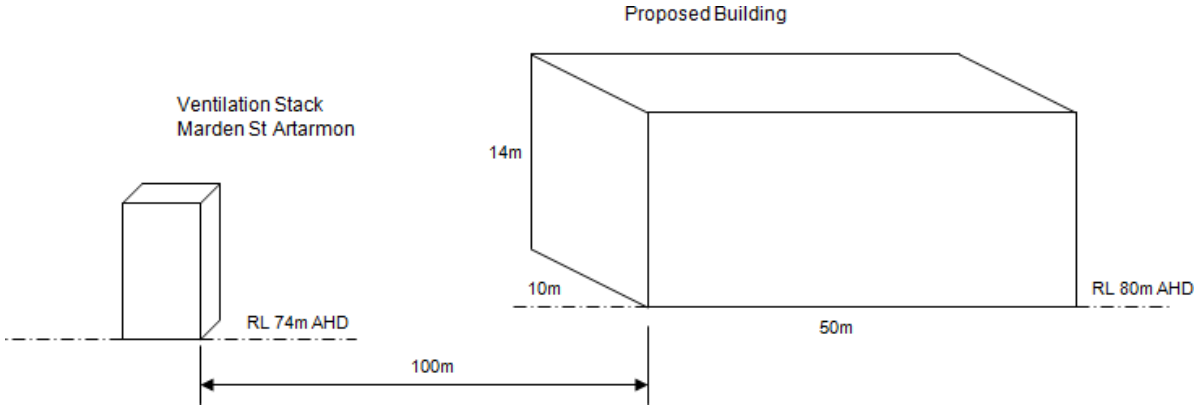
Attachment 2 – Assessment procedure for building height wake/plume interaction with Marden Street vent stack

Table 1 Procedure to assess the building height wake/plume interaction with Marden Street ventilation stack

Step	Task description	Action
Step 1	Measure the distance (Rb) from the Marden vent stack to the leading edge of the proposed building. If Rb > 800m	No height constraint
Step 1	If Rb ≤ 800m	Go to step 2
Step 2	Determine the proposed building height (Hb) measured from the ground level elevation at the base of the Marden stack (reference level of 74m AHD). If Hb ≤ 24m	Building permitted
Step 2	If Hb > 24m	Go to step 3
Step 3	On a site plan measure the projected building width (Wb) perpendicular to the line joining the building to the Marden vent stack.	Go to step 4
Step 4	Calculate the building aspect ratio A = Hb ÷ Wb Let A = 1 if Hb < Wb, that is if building width is greater than building height	Go to step 5
Step 5	Calculate the wake constraint height (Hwake) control using the trigger level curve and compare it to the proposed building height (Hb). Hwake = 0.2 × A × Rb If Hb ≤ Hwake	Go to step 6
Step 5	If Hb > Hwake	Reduce building dimensions (Wb or Hb) or go to step 7
Step 6	Calculate the plume constraint height (HT) at distance Rb from the limiting curve and compare it with the proposed building height (Hb). If Rb > 270m	Building permitted
Step 6	If Rb < 270m use plume constraint equation HT = (Rb + 90) ÷ 3 and compare Hb to HT. If Hb ≤ HT	Building permitted
Step 6	If Hb > HT	Reduce height (Hb) or go to step 7
Step 7	Detailed site-specific investigation required for the proposed building.	Contact an appropriately qualified consultant

Marden Street vent stack – example 1

Figure 1 Example 1 of proposed building interacting with Marden Street vent stack



A building that is 14m tall and 10m wide by 50m long (squat building) is proposed at a site located 100m to the east of the Marden Street stack on Hotham Parade. The reference level (RL) at the base of the proposed building is around 80m AHD.

Note:

- The RL ground elevation at the base of the Marden Street stack is 74m AHD. We need to determine RL of building at ground elevation (in this example we are using 80m AHD).

Using the table for building height wake/plume interaction with Marden Street vent stack the assessment steps are:

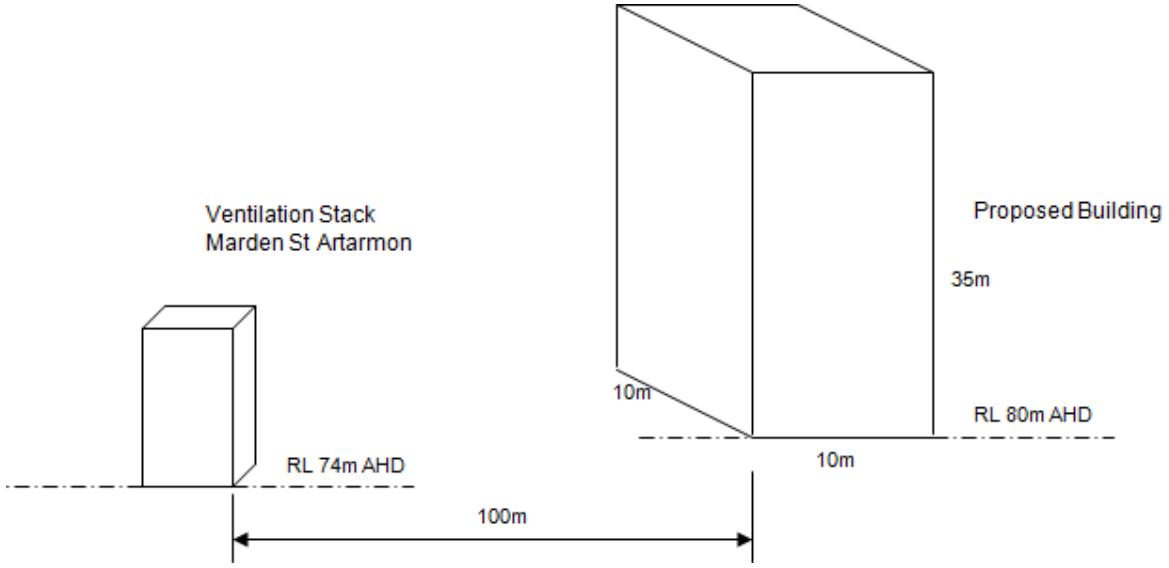
Table 2 Building height/wake plume assessment for Marden Street vent example 1

Step	Task description	Example 1	Action
Step 1	Measure the distance (Rb) from the Marden vent stack to the leading edge of the proposed building. Check if Rb ≤ 800m	Rb = 100m, therefore Rb < 800m	Go to step 2
Step 2	Determine the proposed building height (Hb) measured from the ground level elevation at the base of the Marden stack (reference level of 74m AHD)	Hb = height of building on plan - (Stack RL - Building RL) Hb = 14 – (74 - 80) Hb = 14 - (-6) Hb = 20m above elevation at stack base	As Hb ≤ 24m the building is permitted.

Therefore, in this example the building is permitted without further investigation. The building satisfies both the building wake constraint and the pollution plume constraint.

Marden Street vent stack – example 2

Figure 2 Example 2 of proposed building interacting with Marden Street vent stack



A building that is 35m tall and 10m wide by 10m long (tall building) is proposed at a site 100m to the east of the Marden Street stack on Hotham Parade. The reference level (RL) at the base of the proposed building is around 80m AHD.

Note:

- The RL ground elevation at the base of the Marden Street stack is 74m AHD. We need to determine RL of building at ground elevation (in this example we are using 80m AHD).

Using the table for building height wake/plume interaction with Marden Street vent stack the assessment steps are:

Table 3 Building height/wake plume assessment for Marden Street vent example 2

Step	Task description	Example 2	Action
Step 1	Measure the distance (Rb) from the Marden vent stack to the leading edge of the proposed building. Check if Rb ≤ 800m	Rb = 100m, therefore Rb < 800m	Go to step 2
Step 2	Determine the proposed building height (Hb) measured from the ground level elevation at the base of the Marden stack (reference level of 74m AHD)	Hb = height of building on plan - (Stack RL - Building RL) Hb = 35 - (74 - 80) Hb = 35 - (-6) Hb = 41m above elevation at stack base	Go to step 3 as Hb > 24m

Step	Task description	Example 2	Action
Step 3	On a site plan measure the projected building width (Wb) perpendicular to the line joining the building to the Marden vent stack.	In this example Wb = 10m	Go to step 4
Step 4	Calculate the building aspect ratio A = Hb ÷ Wb Let A = 1 if Hb < Wb	A = Hb ÷ Wb A = 41 ÷ 10 = 4.1	Go to step 5
Step 5	Calculate the wake constraint height (Hwake) control using the trigger level curve and compare it to the proposed building height (Hb) Hwake=0.2×A×Rb	Hwake=0.2×A×Rb Hwake=0.2×4.1×100=82m Hb=41m from step 2 Therefore Hb≤Hwake and the building satisfies the wake constraint	Go to step 6 as Hb ≤ Hwake
Step 6	Calculate the plume constraint height (HT) at distance Rb from the limiting curve and compare it with the proposed building height (Hb):	Given distance from stack Rb<270m use the plume constraint equation HT=(Rb+90)÷3 HT=(100+90)÷3= 63m Hb=41m Therefore Hb<HT and the building satisfies the plume constraint	Building permitted

If Hb was greater than Hwake or HT in the steps 5 or 6 above the wake constraint or plume constraint would not be satisfied and the dimensions of the proposed building would need to be reduced until compliance was achieved. Otherwise, further investigations must be conducted to determine if the proposed building can be permitted.

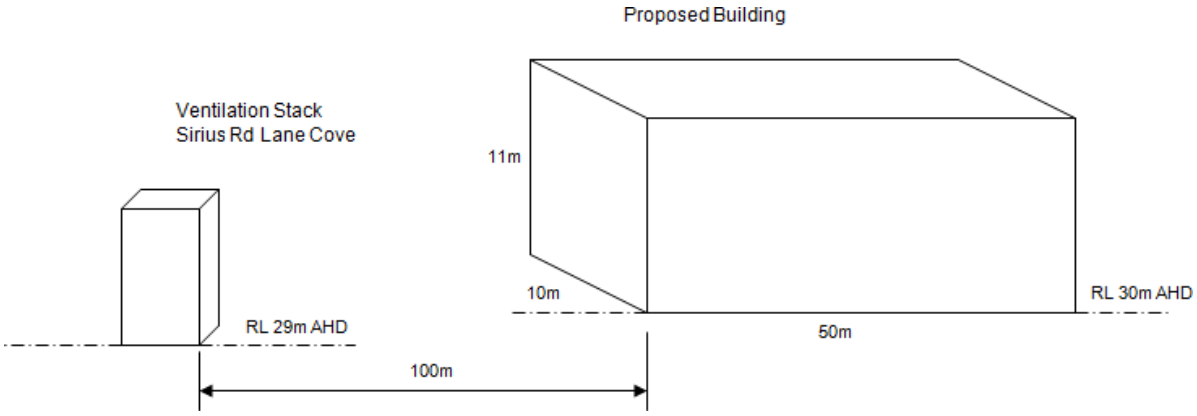
Attachment 3 – Assessment procedure for building height wake/plume interaction with Sirius Rod vent stack

Table 4 Procedure to assess the building height wake/plume interaction with Sirius Road ventilation stack

Step	Task description	Action
Step 1	Measure the distance (Rb) from the Sirius vent stack to the leading edge of the proposed building. If Rb>800m	No height constraint
Step 1	If Rb≤800m	Go to step 2
Step 2	Determine the proposed building height (Hb) measured from the ground level elevation at the base of the Sirius stack (reference level of 29m AHD). If Hb≤13m	Building permitted
Step 2	If Hb>13m	Go to step 3
Step 3	On a site plan measure the projected building width (Wb) perpendicular to the line joining the building to the Sirius vent stack.	Go to step 4
Step 4	Calculate the building aspect ratio A=Hb÷Wb Let A=1 if Hb<Wb	Go to step 5
Step 5	Calculate the wake constraint height (Hwake) control using the trigger level curve and compare it to the proposed building height (Hb). Hwake=0.2×A×Rb If Hb≤Hwake	Go to step 6
Step 5	If Hb>Hwake	Reduce building dimensions (Wb or Hb) or go to step 7
Step 6	Calculate the plume constraint height (HT) at distance Rb from the limiting curve and compare it with the proposed building height (Hb). If Rb>150m	Building permitted
Step 6	If Rb<40m, HT=20m If 40m≤Rb<120m, HT=0.5×Rb If 120m≤Rb<150m, HT=2×(Rb-90) Compare Hb to HT. If Hb≤HT	Building permitted
Step 6	If Hb>HT	Reduce height (Hb) or go to step 7
Step 7	Detailed site-specific investigation required for the proposed building.	Contact an appropriately qualified consultant

Sirius Road vent stack – example 1

Figure 3 Example 1 of proposed building interacting with Sirius Road vent stack



A building that is 11m tall and 10m wide by 50m long (squat building) is proposed at a site located 100m to the east of the Sirius Road stack. The reference level (RL) at the base of the proposed building is around 30m AHD.

Note:

- The RL ground elevation at the base of the Sirius Road stack is 29m AHD. We need to determine RL of building at ground elevation (in this example we are using 30m AHD).

Using the table for building height wake/plume interaction with Sirius Road vent stack the assessment steps are:

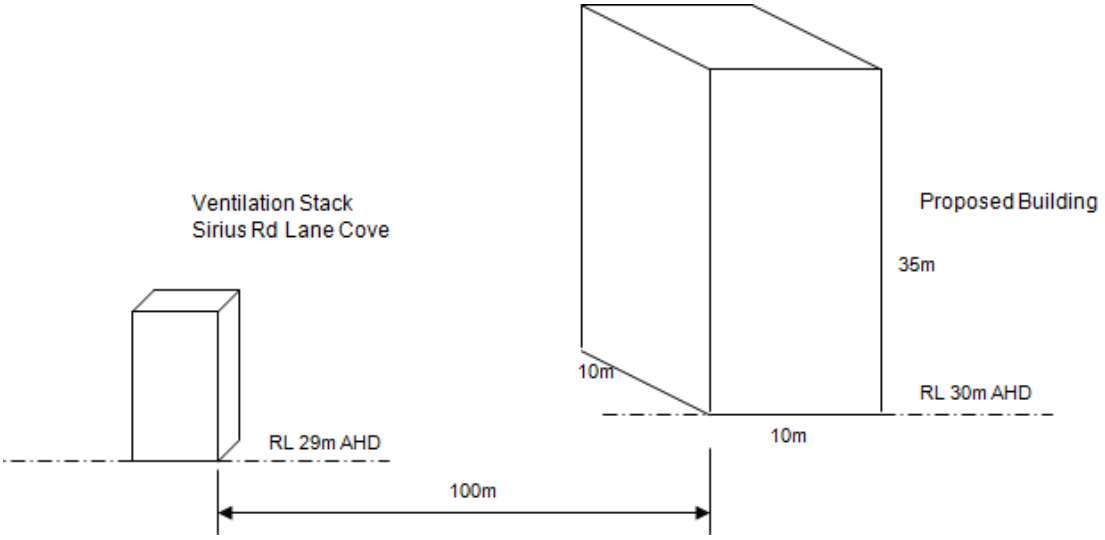
Table 5 Building height/wake plume assessment for Sirius Road vent example 1

Step	Task description	Example 1	Action or result
Step 1	Measure the distance (Rb) from the Sirius vent stack to the leading edge of the proposed building. Check if Rb≤800m	Rb=100m, therefore Rb<800m	Go to step 2
Step 2	Determine the proposed building height (Hb) measured from the ground level elevation at the base of the Sirius stack (reference level of 29m AHD)	Hb=height of building on plan-(Stack RL-Building RL) Hb=11-(29-30) Hb=11-(-1) Hb=12m above elevation at stack base	As Hb≤13m the building is permitted.

In this example the building is permitted without further investigation. The building satisfies both the building wake constraint and the pollution plume constraint.

Sirius Road vent stack – example 2

Figure 4 Example 2 of proposed building interacting with Sirius Road vent stack



A building that is 35m tall and 10m wide by 10m long (tall building) is proposed at a site located 100m to the east of the Sirius Road stack. The reference level (RL) at the base of the proposed building is around 30m AHD.

Note:

- The RL ground elevation at the base of the Sirius Road stack is 29m AHD. We need to determine RL of building at ground elevation (in this example we are using 30m AHD).

Using the table for building height wake/plume interaction with Sirius Road vent stack the assessment steps are:

Table 6 Building height/wake plume assessment for Sirius Road vent example 2

Step	Task description	Example 2	Action or result
Step 1	Measure the distance (Rb) from the Sirius vent stack to the leading edge of the proposed building. Check if Rb≤800m	Rb=100m, therefore Rb<800m	Go to step 2
Step 2	Determine the proposed building height (Hb) measured from the ground level elevation at the base of the Sirius stack (reference level of 29m AHD)	Hb=height of building on plan-(Stack RL-Building RL) Hb=35-(29-30) Hb=35-(-1) Hb=36m above elevation at stack base	Go to step 3 as Hb>13m

Step	Task description	Example 2	Action or result
Step 3	On a site plan, measure the projected building width (Wb) perpendicular to the line joining the building to the Sirius vent stack.	Measure the projected building width (Wb). In this example Wb=10m	Go to step 4
Step 4	Calculate the building aspect ratio A=Hb÷Wb Let A=1 if Hb<Wb	A=Hb÷Wb A=36÷10=3.6	Go to step 5
Step 5	Calculate the wake constraint height (Hwake) control using the trigger level curve and compare it to the proposed building height (Hb) Hwake=0.2×A×Rb	Hwake=0.2×A×Rb Hwake=0.2×3.6×100=72m Hb=36m from step 2 Therefore Hb≤Hwake and the building satisfies the wake constraint	Go to step 6 as Hb≤Hwake
Step 6	Calculate the plume constraint height (HT) at distance Rb from the limiting curve and compare it with the proposed building height (Hb)	Given distance from stack Rb<120m we need to use the plume constraint equation HT=0.5×Rb HT=0.5×100=50m Hb=36m Therefore Hb<HT and the building satisfies the plume constraint	Building permitted

If Hb was greater than Hwake or HT in the steps 5 or 6 above the wake constraint or plume constraint would not be satisfied and the dimensions of the proposed building would need to be reduced until compliance was achieved. Otherwise, further investigations need to be done to determine if the proposed building can be permitted.

PART L

Place Based Plans

Part L: Place Based Plans

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1 Introduction

This part of Willoughby Development Control Plan (Willoughby DCP*)* includes guidelines and further controls for developments in Chatswood Central Business District (CBD) and the local centres in the City of Willoughby local government area.

The performance criteria and controls in Part B (Residential Development) and Part D (Commercial Development) of Willoughby DCP apply to Chatswood CBD and the local commercial/retail centres. If there are inconsistencies between those parts and this part, the controls under this part prevail.

Section 13 of this Part includes controls for any future development of specific sites and areas. There have been a number of planning proposals approved by Council which are supported by site specific DCPs. These are also included in Section 13. ***If there is any inconsistency between Part L and another Part of the DCP, the controls under Part L prevail.***

1.1 Aim

The aim of this part is to ensure commercial, non-residential and residential development takes place in line with strategic planning for these localities. Development is to take place in a way that sustains and enhances the economic and environmental qualities of the Willoughby area, and the health and wellbeing of those local communities.

1.2 Objectives

The objectives of this part are to:

- establish a strong framework to guide development in Chatswood CBD and the local retail/commercial centres
- provide capacity for growth by increasing residential densities and create job opportunities by making provision for more non-residential floor space
- achieve exceptional design, and distinctive, resilient and vibrant centres
- create attractive and thriving local retail/commercial centres
- protect the heritage values of heritage listed items and ensure any new development integrates with the character of heritage conservation areas
- provide greening on and around buildings, and improve pedestrian and cycle links

2 Strategies and plans

A number of studies, strategies and plans have been adopted to set the direction for non-residential and residential development in key locations in the area.

2.1 Chatswood CBD Planning and Urban Design Strategy 2036

This strategy guides private and public development over the next 20 years. It includes expansions to the north and south of the existing CBD and provides growth prospects for

non-residential development in the E2 Commercial Core zone. It also creates further residential opportunities as part of the MU1 Mixed Use zone.

Chatswood CBD Planning and Urban Design Strategy 2036 is at willoughby.nsw.gov.au.

2.2 Willoughby Local Centres Strategy 2036

This strategy aims to revitalise the local economy and provides the framework for future planning controls. Those centres are:

a. Artarmon: the east and west side of Artarmon railway station.

b. Castlecrag: on Edinburgh Road between Eastern Valley Way and Rutland Avenue/The Postern.

c. North Willoughby: around the intersection of Penshurst Street and Victoria Avenue, extending south to Patrick Street.

d. High Street: bounded by McClelland Street and Glover Street.

e. Naremburn: on the western side of Willoughby Road and bounded by Quiamong Street, Bongalong Street and Glenmore Street.

f. Northbridge: along the major arterial roads of Sailors Bay Road, Eastern Valley Way and Strathallen Avenue.

g. Penshurst Street: along and north of Mowbray Road.

h. Willoughby South: along Willoughby Road and branching off Frenchs Road.

The strategy provides an indicative master plan for each local centre. It identifies key features to promote growth and shows how development can be achieved alongside opportunities to improve the public domain.

It also includes controls with a diagram to show how the desired scale of development can be achieved for each local centre.

Any variations from the adopted master plans and/or the layout and building envelopes identified in the scale of development diagrams must be justified. These must consider the performance criteria and controls under this and other relevant parts of Willoughby DCP.

Willoughby Local Centres Strategy 2036 is at willoughby.nsw.gov.au.

2.3 Willoughby Local Environmental Plan

Willoughby Local Environmental Plan (Willoughby LEP) ensures we meet the objectives, vision and guidelines in Willoughby City Council's adopted strategies. It includes controls for active street frontages and increased height limits and floor space for non-residential buildings and mixed use developments.

It provides minimum lot sizes for non-residential and mixed use developments and minimum non-residential floor space for mixed use developments in the outer parts of Chatswood CBD.

The plan supports design excellence. It has clauses to provide sun access to public spaces, maintain solar access to important areas in Chatswood, and address urban heat impacts.

3 Public art and prominent corner sites

To enhance the visual appearance of buildings in gateway or highly visible locations such as prominent street corner sites, major developments such as retail centres or shop top housing should make provision to integrate public art and/or unique façade treatment.

A major development proposal may need to include an art plan that identifies opportunities to integrate public art in line with Willoughby City Council's public art policy.

Prominent corner sites often define the main entry points into a commercial/business centre. Buildings on prominent corner sites should incorporate subtle changes in height or façade treatment by including measures such as:

- partial additional storey or parapet extension
- splayed setbacks
- public art
- landscape treatment
- other elements such as clock towers or spires

A change in colours or materials can also help to differentiate or accentuate a building on a prominent corner site.

4 Chatswood CBD

4.1 Character statement

Chatswood is a strategic centre in the Sydney metropolitan area. The CBD includes a vibrant mix of offices, major retail facilities, health facilities, educational establishments, cultural facilities, and high density residential accommodation.

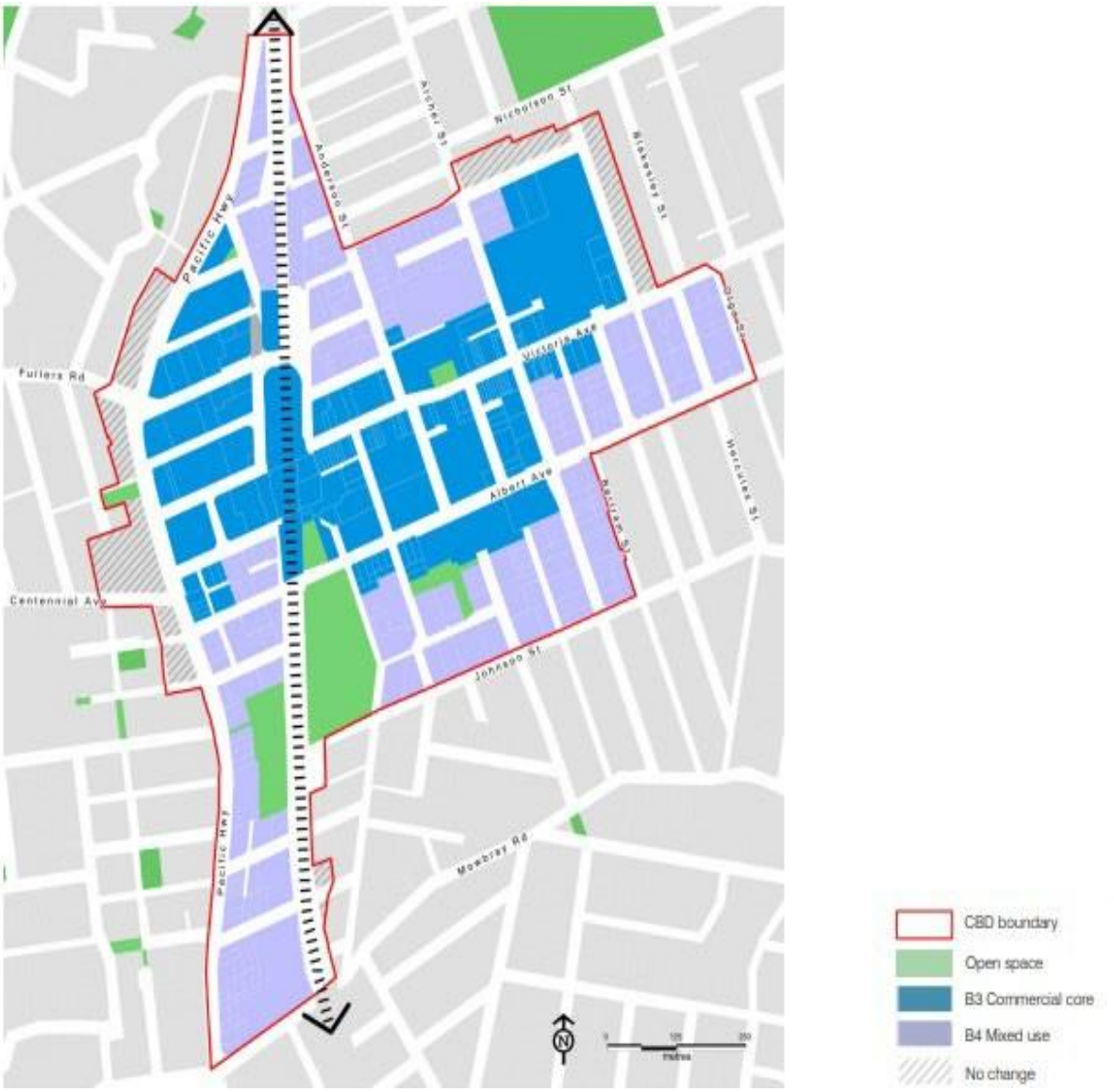
At its centre is a rail and bus interchange that provides direct connections to the Sydney CBD, other strategic centres, and residential areas across Sydney. Transport connections have been heightened by the opening of the metro northwest in 2019 with a new platform at Chatswood Station. Connections will be further improved by the opening of the metro city and southwest lines in 2024.

The controls in this plan relating to the E2 Commercial Core zone are designed to increase investment confidence in office development and protect these employment hubs from residential incursions.

The MU1 Mixed Use zone provides a mix of non-residential and residential around the E2 Commercial Core in line with Map 1 (Chatswood CBD Strategy Land Use Map). This is to help maximise returns on existing and planned investment in public infrastructure and ensure Chatswood remains a major employment centre in metropolitan Sydney.

The controls aim to maintain a compact, walkable city centre, and create exceptional urban design. They support easy pedestrian and bicycle linkages, a quality public domain that embraces the local character and heritage, and the greening of the centre.

Map 1: Land use map from Chatswood CBD Planning and Urban Design Strategy 2036



4.2 Performance criteria

Proposed development in Chatswood CBD should:

a. be in line with permitted development within the E2 Commercial Core zone, which prioritises non-residential land uses

b. allow residential growth as part of mixed use development surrounding the E2 Commercial Core within the CBD

c. maintain a diverse mix of uses, including retail, medical, educational, health, cultural and recreational activities

d. involve amalgamated sites for optimum development outcomes

e. create new publicly accessible spaces with links to existing public places

f. embellish and/or add to existing public spaces

g. adopt travel demand management to support active and sustainable transport

h. provide excellence in urban design

i. provide greening of the streetscape as well as green areas on and around new buildings to improve the visual quality, amenity for workers and visitors, and reduce the impacts of urban heat island effects

j. ensure sun access is provided to public places

k. reinforce Victoria Avenue as the primary pedestrian spine

l. increase activation of side streets from Victoria Avenue

m. create a pattern of mid-block links

n. ensure podium and roof top communal open spaces are designed to address issues of quality, safety and usability

4.3 Controls for Chatswood CBD

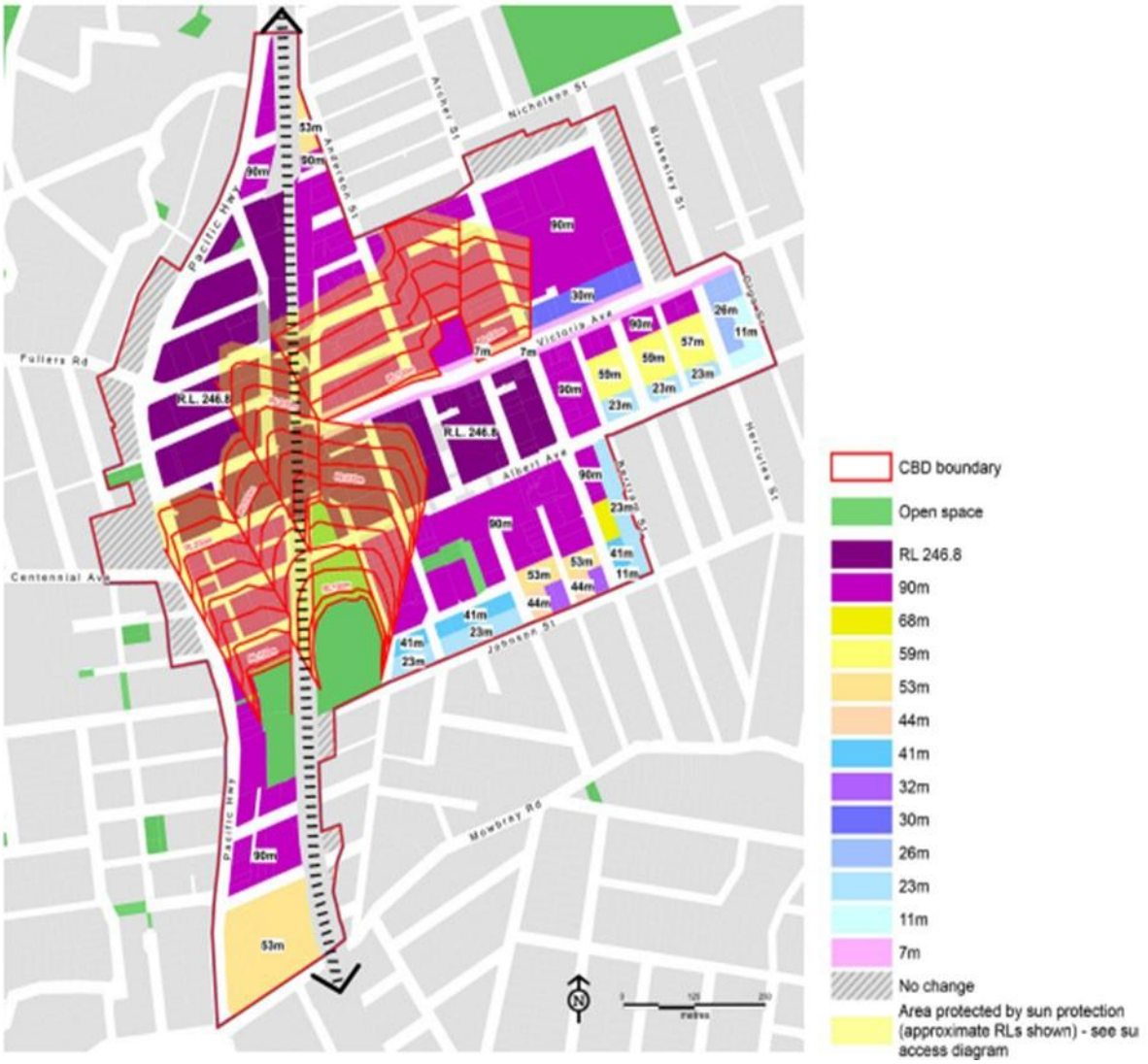
4.3.1 Built form

a. Height of buildings

The Height of Buildings Map in Willoughby LEP shows the maximum building heights.

Map 2 below shows the height of buildings and maximum reduced levels to ensure certain public spaces in Chatswood CBD are not impacted by excessive overshadowing.

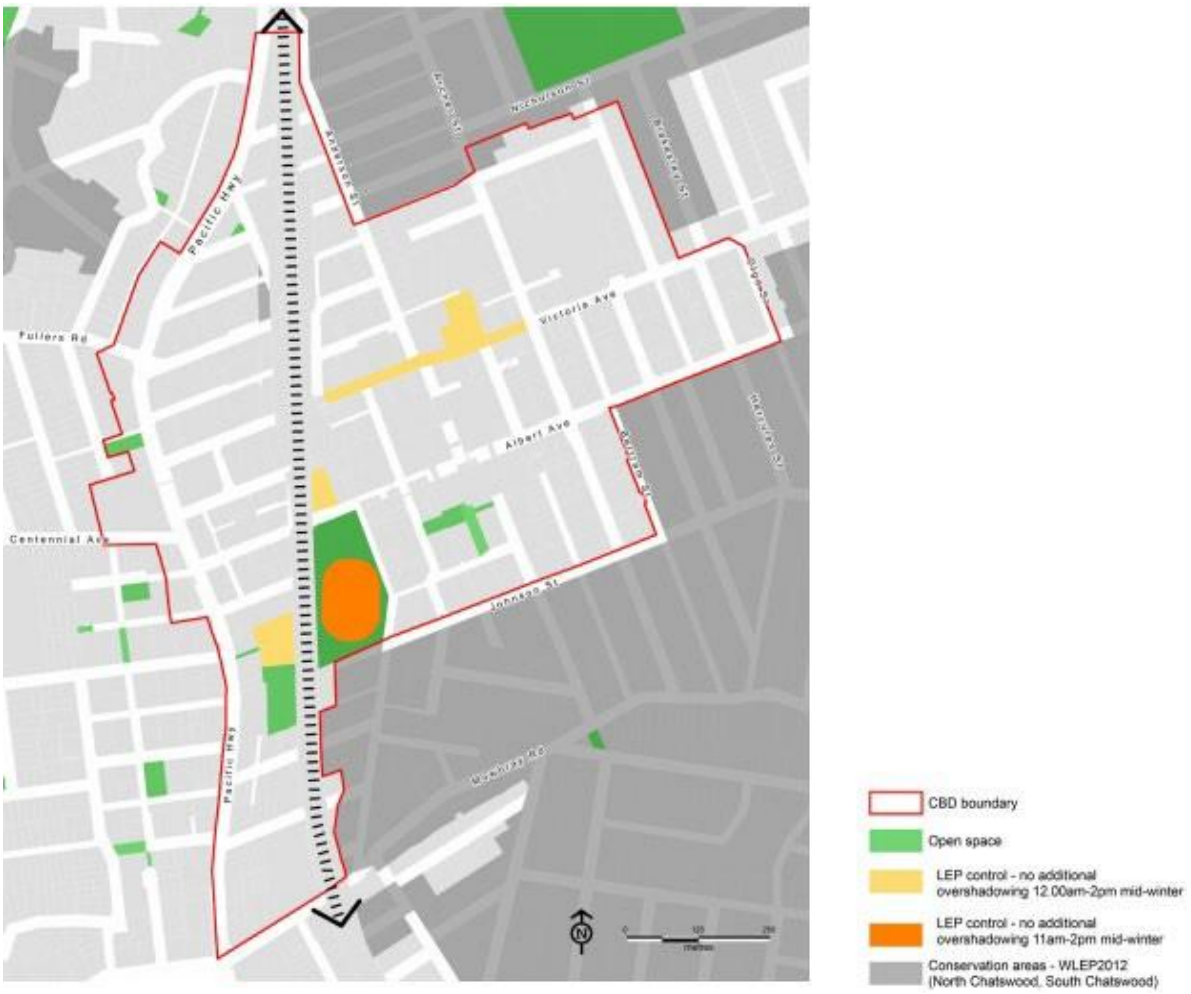
Map 2: Height map from Chatswood CBD Planning and Urban Design Strategy 2036



b. Sun access protection to key public spaces and the South Chatswood Conservation Area

Willoughby LEP includes provisions to protect certain public spaces in Chatswood CBD and South Chatswood Heritage Conservation Area from excessive overshadowing. The diagram below shows the areas that need to be protected from excessive overshadowing.

Map 3: Sun access protection map from Chatswood CBD Planning and Urban Design Strategy 2036



c. Minimum site area

The controls for minimum site areas for the E2 Commercial Core and MU1 Mixed Use zones are in Willoughby LEP.

This plan encourages site amalgamation to meet the minimum site requirements and ensure sites are not left isolated.

The objective of the minimum site area is to enable a development to achieve best outcomes by:

- providing the required setbacks to achieve slender towers and building separation whether onsite or on neighbouring sites. Objectives for setbacks are provided below in 4.3.4 Setbacks and street frontage heights.
- providing ground level public realm or areas accessible by the public on private land
- limiting and rationalising the number of vehicle entry/exit points
- providing parking and loading facilities in the basement with adequate area for onsite manoeuvrability to enter and leave the site in a forward direction
- maximising non-residential floor space and street activation at ground level

- maximising landscaping and deep soil planting

d. Slender towers

To achieve slender towers and adequate building separation, the maximum gross floor area at each level of a development should be no more than:

- 2,000m^2 for office developments
- 700m^2 for residential towers above the podium level in the mixed use zone

Note:

- The maximum gross floor areas is to be read in conjunction with other controls for Chatswood CBD (such as maximum floor space, height and minimum setbacks), as well as the site size and shape.
- These maximum floor plates, together with maximum floor space and height, and minimum setbacks, require an unusually large site to be achieved. Lesser floor plates than the maximum would be expected on standard size lots.
- If there is more than one tower on the same site, towers are not to be linked above the podium. Towers are to operate independently regarding lifts and services.
- The width is to be minimised on all sides to achieve a slender tower.
- Design elements are to be included that minimise the width, in particular on any wider tower side.

e. Lot pattern

Traditional building widths fronting the street of between 6m and 12m are to be retained along Victoria Avenue east.

f. Vehicular access

Only one entry area into and exiting a site is permitted. This is to minimise streetscape impacts, ensure vehicle entry points are rationalised and maximise ground level commercial floor space.

g. Loading/unloading facilities

All development is required to provide loading/unloading facilities on site.

All loading docks, including provision for garbage trucks and residential removal trucks are to be within basement areas with adequate onsite manoeuvrability. This is to ensure vehicles can enter and leave the site in a forward direction.

Note:

- Willoughby City Council does not generally support mechanical systems, such as turntables to facilitate vehicles entering and leaving the site in a forward direction. Non-mechanical solutions are sought.

h. Amalgamation and Site isolation

The objective of amalgamation is to enable a development to achieve best outcomes by:

- providing the required setbacks to achieve slender towers and building separation whether onsite or on neighbouring sites
- providing ground level public realm or areas accessible by the public on private land
- limiting and rationalising the number of vehicle entry/exit points
- providing parking and loading facilities in the basement with adequate area for onsite manoeuvrability to enter and leave the site in a forward direction
- maximising non-residential floor space and street activation at ground level
- maximising landscaping and deep soil planting
- If site isolation is unavoidable or inadequate area are available in the basement level, buildings are to provide for joined basement areas with 'break through' walls to provide vehicle access to adjoining sites.
- Also where site isolation is unavoidable, zero setback podiums are to provide 'break through' walls to encourage future efficient sharing of infrastructure.

i. Substations

Substations are to be integrated within buildings, not located in a street, open space, setback or area subject to a public right of way.

4.3.2 Greening Chatswood CBD

A range of approaches apply on a site-specific basis to provide permeability, publicly accessible open space and a 'green' ground plane.

These are designed to develop a comprehensive network for the centre of landscape and open space to create a green, well-connected CBD.

These controls apply to all high rise buildings:

a. all roofs up to 30m from the ground are to be green roofs (in particular podium roofs); these are to provide a green contribution visible to the street and a balance of passive and active green spaces that maximise solar access

b. a minimum of 20% of the site is to be soft landscaping, which may be on the ground, podium, and roof top levels of buildings; soft landscaping must be a minimum depth of 600mm

c. 'green' walls are encouraged

4.3.3 Links and public realm

a. All proposals must consider the potential for through links to public places. Pedestrian and cycling links are sought to improve existing access within and through the CBD (see Map 4). New linkages are sought where these are considered to be of public benefit. All such links must be:
    - a minimum of 3m wide

- provided with public rights of access and designed with adequate width, sympathetic landscaping and passive surveillance

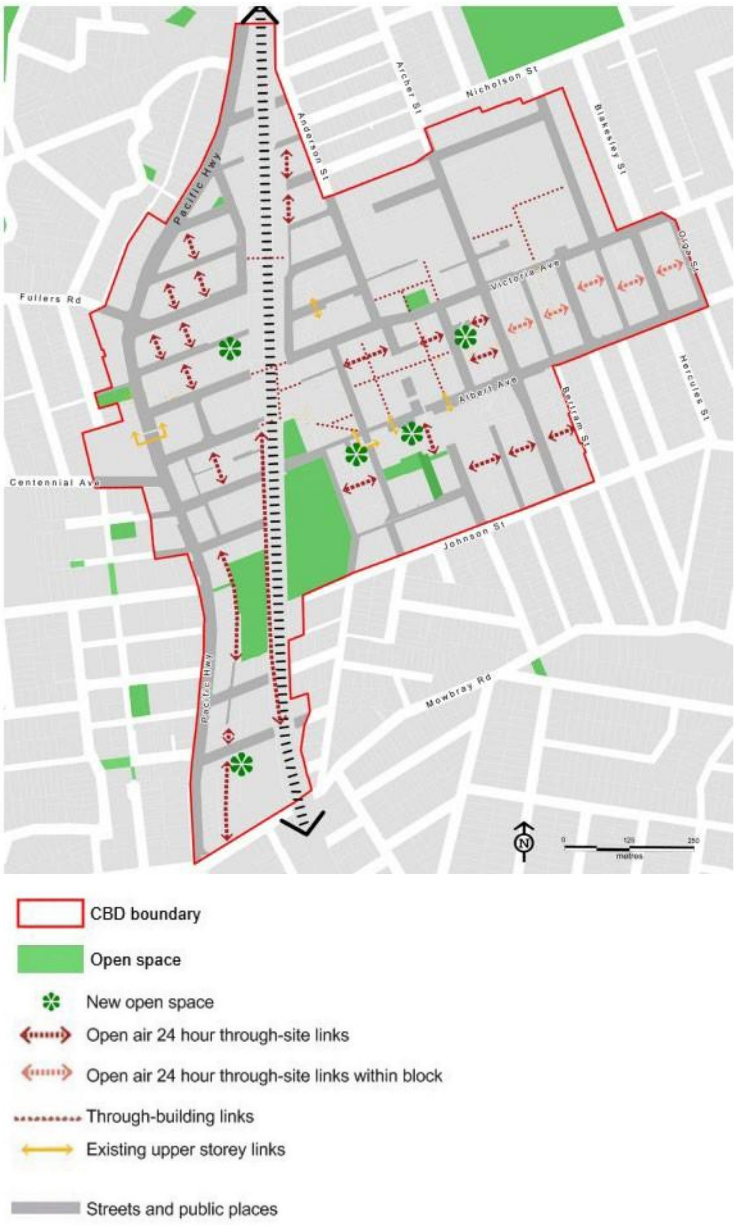
b. Public realm or areas accessible by the public on private land is required to be included in all E2 and MU1 redeveloped sites. These must be:

- designed to respond to context and nearby public domain
- visible from the street and easily accessible
- accompanied by public rights of way or similar to achieve a permanent public benefit

c. Through site pedestrian links and public realm or areas accessible by the public on private land are to:

- Be direct and accessible to all and have a clear line of sight between public places;
- Align with breaks between buildings so that views are extended;
- Be located and orientated to enhance the relationship between built form and open space.
- Be easily identified by users and include wayfinding signage;
- Be clearly distinguished from vehicle accessways and parking;
- Include materials and finishes such as paving materials and tree planting consistent with the proposed urban quality of the precinct;
- Be clear of both permanent and temporary obstructions or structures, including utility elements, advertising/promotional material, street furniture and outdoor dining.
- Include lighting to provide safety, amenity, character and night time quality appropriate to the link.
- Include potential for feature lighting within landscaped areas.
- Maximise soft landscaping and deep soil planting.
- Located to maximise solar access

Map 4: Through Site Links and Open Space Map Chatswood CBD Planning and Urban Design Strategy 2036



4.3.4 Setbacks and street frontage heights

The following performance criteria apply to setbacks and street wall heights in the Chatswood CBD: Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.

4. Protect all significant on-site trees and all street trees on Gordon Avenue, Hammond Lane and Nelson Street.

5. Contribute to slender tower forms.

6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

7. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

a. The minimum setbacks and maximum street frontage heights below apply to certain areas within Chatswood CBD. These areas are shown in Map 5.

   1. Victoria Avenue retail frontage:
      - maximum 7m street wall height at front boundary
      - minimum 6m setback above street wall to tower
   2. Urban core:
      - maximum 24m street wall height at front boundary
      - minimum 6m setback above street wall to tower
   3. Office core frontage:
      - 4m–12m maximum street wall height at front boundary
      - minimum 6m setback above street wall to tower
   4. Mixed use frontage with non-residential ground floor:
      - 6m–14m maximum street wall height at front boundary
      - minimum 3m setback above street wall to tower
   5. Pacific Highway frontage:
      - minimum 4m setback at ground level from front boundary (with exception of heritage sites)
      - maximum 7m street wall height
      - minimum 6m setback above street wall to tower
   6. Southern precinct:
      - minimum 6m setback at ground level from front boundary
      - no setback from podium to tower
   7. Albert Avenue south:
      - minimum 3m setback at ground level from front boundary, with intermittent wider open space
      - maximum 24m street wall height
      - 3m setback above street wall to tower
   8. Anderson Street interface:

- minimum 3m setback at ground level from front boundary
- 6m – 14m maximum street wall height
- minimum 1m setback above street wall to tower

9. Bertram Street interface:
   - minimum 6m setback at ground level from front boundary
   - maximum 7m street wall height
   - minimum 3m setback above street wall to tower
10. Albert Avenue north and Olga Street interface:
   - minimum 3m setback at ground level from front boundary
   - 6m – 14m maximum street wall height
   - minimum 1m setback above street wall to tower
11. Johnson Street interface:
   - minimum 12m setback at ground level from front boundary
   - no setback from podium to tower

Notes:

- On corner sites where different setbacks and street frontage height precincts meet, the greater setback and lower street wall height required on the primary frontage is to continue around the corner for a minimum distance of 6m. This is to provide a satisfactory transition between the setbacks and wall heights.
- With setbacks of 3m or more, including the Pacific Highway, deep soil planting for street trees is to be provided.
- Setbacks greater than the minimum are encouraged, particularly at the interface with low density residential conservation areas.
- Street wall heights lower than the maximum are encouraged at the interface with low density residential conservation areas.
- Where adjacent links or pathways involve public rights of way (not streets unless appropriate), stepped podiums are sought.
- Where a site adjoins a laneway and no footpath exists, a minimum ground level setback is to be provided to facilitate construction of a footpath.
- Ground level setbacks may be subject to public rights of carriageway to facilitate pedestrian and cycle paths through the Chatswood CBD, in appropriate locations, connecting with existing infrastructure and leading from CBD boundaries to the CBD centre. Consideration must be given to the anticipated future users of the setback, which may include pedestrians, cyclists and users of the proposed building. Setbacks, street wall treatments, and associated works are to be designed in a manner that minimises conflicts while accommodating the various users of the setback.

c. All towers above podiums in the E2 Commercial Core and MU1 Mixed Use zone are to be setback from all boundaries with a minimum 1:20 ratio of the setback to building height. This means if a building is:

   - a total height of 30m, a minimum setback from the side boundary of 1.5m is required for the entire tower on any side
   - a total height of 60m, a minimum setback from the side boundary of 3m is required for the entire tower on any side
   - a total height of 90m, a minimum setback from the side boundary of 4.5m is required for the entire tower on any side
   - a total height of 120m, a minimum setback from the side boundary of 6m is required for the entire tower on any side
   - a total height of 150m, a minimum setback from the side boundary of 7.5m is required for the entire tower on any side

Notes:

- The required setback of a tower will vary depending on the total height. It is not to be based on setback averages, or a stepped approach, but the full setback.
- Setbacks greater than the minimum are encouraged, particularly in locations at the interface with low density residential conservation area.
- Minimum setbacks and going beyond with additional setbacks where appropriate is considered an important way of achieving the desired outcome of slender towers.

d. Separation of buildings must be provided in line with the Apartment Design Guide (NSW Department of Planning and Environment, July 2015) for residential apartments within a mixed use development.

e. In the E2 Commercial Centre zone, all tower uses must have a minimum 6m setback from all boundaries above the street level wall height.

f. Balconies and any other building intrusions are not to encroach into setbacks.

4.3.5 Building Sustainability

a. MU1 Mixed Use Zone

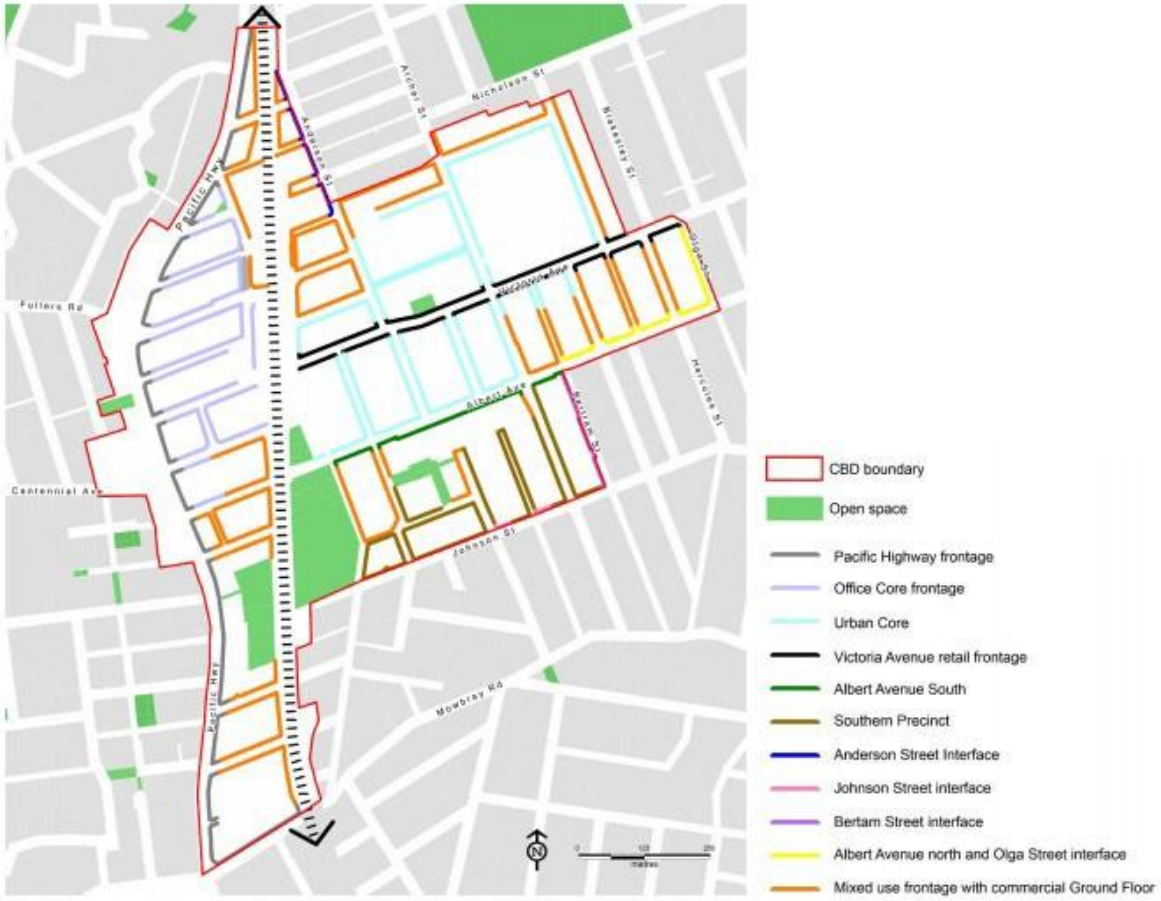
   All development is expected to achieve higher building sustainability standards.

   A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

b. E2 Commercial Centre Zone

   All development is expected to achieve of higher building sustainability standards. A minimum of 6 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

Map 5: Setbacks and street frontage heights map from Chatswood CBD Planning and Urban Design Strategy 2036



4.4 Additional controls for specific locations

4.4.1 Victoria Avenue east retail frontage

Victoria Avenue is the heart of Chatswood's retail activity and its shops contribute to the enduring quality of the centre. To maintain the great public spaces and the urban design quality of developments fronting Victoria Avenue, these principles apply:

a. clear and accessible vertical circulation

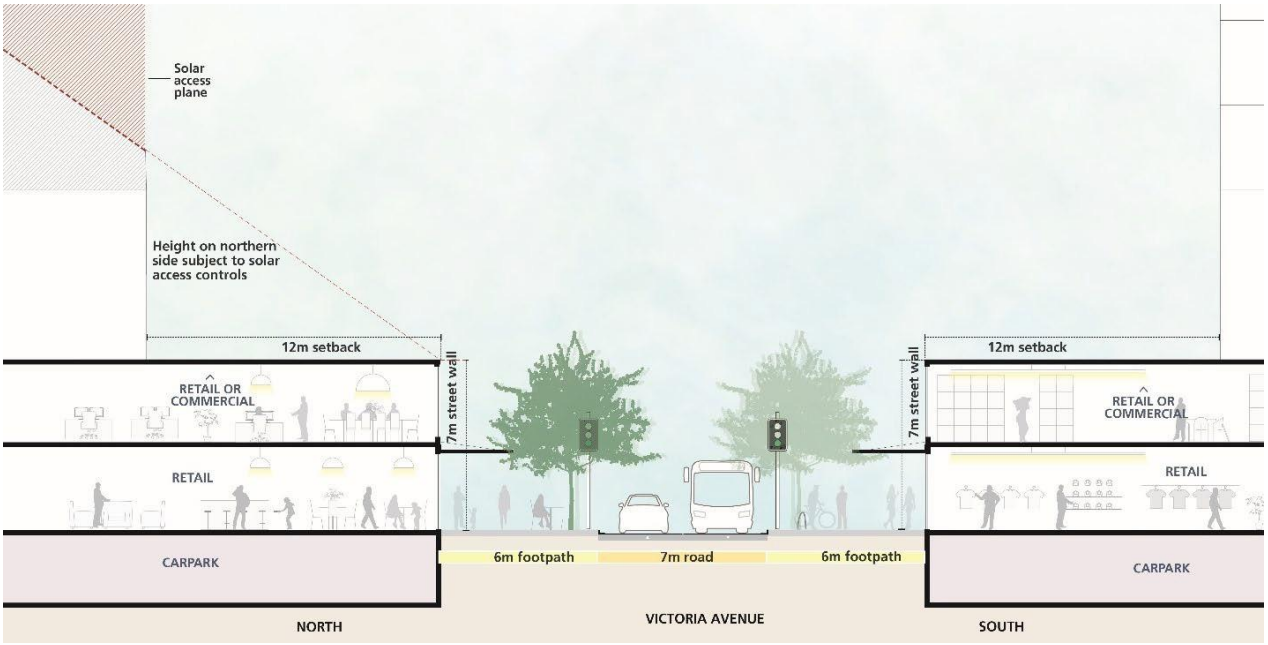
b. public use of rooftop spaces

c. provision of links to neighbouring sites

d. provision of car parking access from neighbouring sites

Figure 1 below shows an example of future development along the Victoria Avenue retail frontages.

Figure 1: Section of Victoria Avenue frontage



4.4.2 Pacific Highway – green setback

The eastern side of the Pacific Highway must provide a 4m wide deep soil zone setback and the western side must provide a 6m wide deep soil zone setback.

Willoughby City Council is seeking an unobstructed shared pedestrian and cycle path along the eastern side of the Pacific Highway (within Chatswood CBD). This must be factored into redevelopment of any site fronting the Pacific Highway.

Dimensions for the path are to be based on a 1.5m wide landscape verge and 3m wide shared path. The shared path may encroach on the 4m wide deep soil zone setback on a site, but if this is the case, a public right of way will be required.

4.4.3 Interface with low density residential conservation areas

Designs for developments at the interface with low density residential conservation areas must consider the surrounding low density heritage context. The objective is to minimise impacts by:

a. maximising setbacks and minimising street wall heights to provide separation and reduce bulk and scale (setbacks above minimum requirements may be sought in these locations)

b. providing deep soil planting at ground, landscaping on podium levels, and wherever possible, the upper levels

c. using design, façade treatment and high quality materials and finishes

5 Artarmon local centre

5.1 Character statement

The main part of Artarmon local centre is along the western side of Hampden Road, next to Artarmon railway station. A large portion is within Artarmon Heritage Conservation Area.

Shops on the western side of Hampden Road currently provide a range of local needs and services such as post office, banking, health services, takeaway food, cafes and restaurants. There are high rise residential units behind the main Artarmon shopping area to the west.

Wilkes Plaza, within Artarmon Heritage Conservation Area, is on the eastern side of the railway station. It provides a pleasant, popular open space area enclosed by cafes, restaurants and shops as well as being a well-used pedestrian thoroughfare.

The centre is in a valley, with the land climbing to the west of the railway line and the Wilkes Avenue plaza at a low point. Hampden Lane, to the rear of Hampden Road, is approximately level with the retail rooftops (two levels above Hampden Road). This slope results in around a 2-metre difference in some parts of the centre from the retail and the street.

Key attributes of the centre include the train station and a good level of pedestrian traffic. Artarmon has a strong village atmosphere and an active resident and business community concerned to retain and enhance its heritage character and sense of place. The built form generally presents two storey non-residential street frontages, with many of the retail facades demonstrating the heritage character of the precinct with architectural features and detailing. A narrow landscaped strip runs north–south between the railway line and Hampden Road, known as Artarmon Village Green.

5.2 Performance criteria

Proposed development in Artarmon local centre should:

a. maintain the two storey non-residential built form as shown in Figure 3 to provide a significant proportion of non-residential floor space for shop top housing developments

b. retain and refurbish the existing fine grain heritage frontages on Hampden Road, while allowing for site amalgamation to provide further shop top residential apartments

c. provide articulation to break up the length of walls to reduce the bulk and visual impacts of shop top housing above the second level

d. encourage green roofs, and roof top gardens in appropriate locations

e. maintain existing small-scale village character east of the railway line

f. maintain active street frontages

g. provide deep soil zones in appropriate locations to support existing and additional mature trees with wide tree canopies to improve the streetscape and reduce urban heat impacts

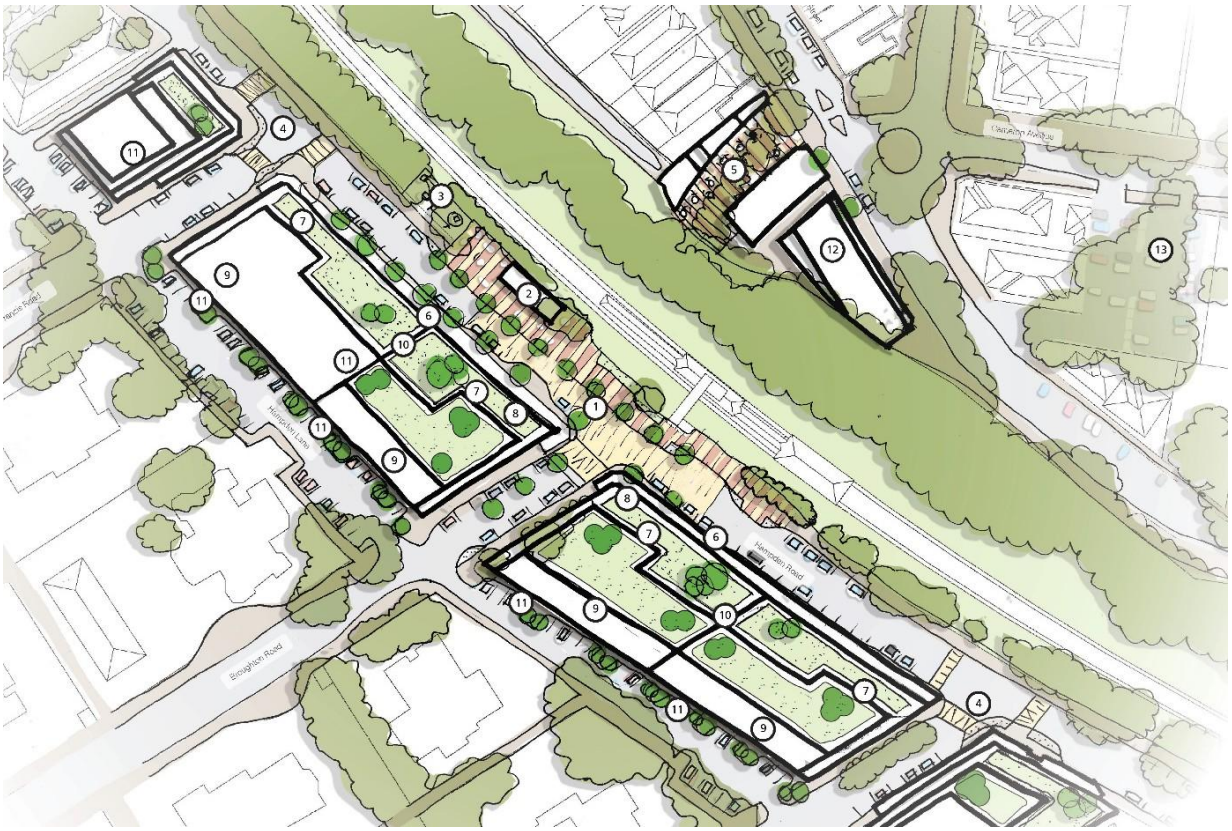
5.3 Master plan for Artarmon local centre

Any future development in Artarmon local centre must have regard to the following key features of the adopted master plan. The numbers below refer to actions to be carried out by Willoughby City Council and/or developers to achieve the outcomes of the adopted master plan.

1. Improvements to the public domain and pedestrian access around the station entry on Hampden Road and Broughton Road.
2. Construction of new toilets, small café pavilion and cycle storage.

3. Provision of new plaza and play space next to the pavilion.
4. Improvements to pedestrian crossings with new kerb blisters and thresholds to slow traffic.
5. Public domain improvements to Wilkes Avenue.
6. Retention of fine grain shop fronts.
7. New developments to provide upper level setbacks to minimise visual impact on Hampden Road.
8. Opportunity for lot amalgamation to provide a supermarket close to the station.
9. New developments to provide landscaping and good design elements along the Hampden Lane frontage.
10. New developments to provide roof gardens/green roofs.
11. New developments to make provision for at grade parallel car parking, interspersed with landscaped areas.
12. Any redevelopment of the library site should include a new community centre on the ground floor level.

Figure 2: Master plan for Artarmon from Willoughby Local Centres Strategy 2036



5.4 Controls for Artarmon local centre

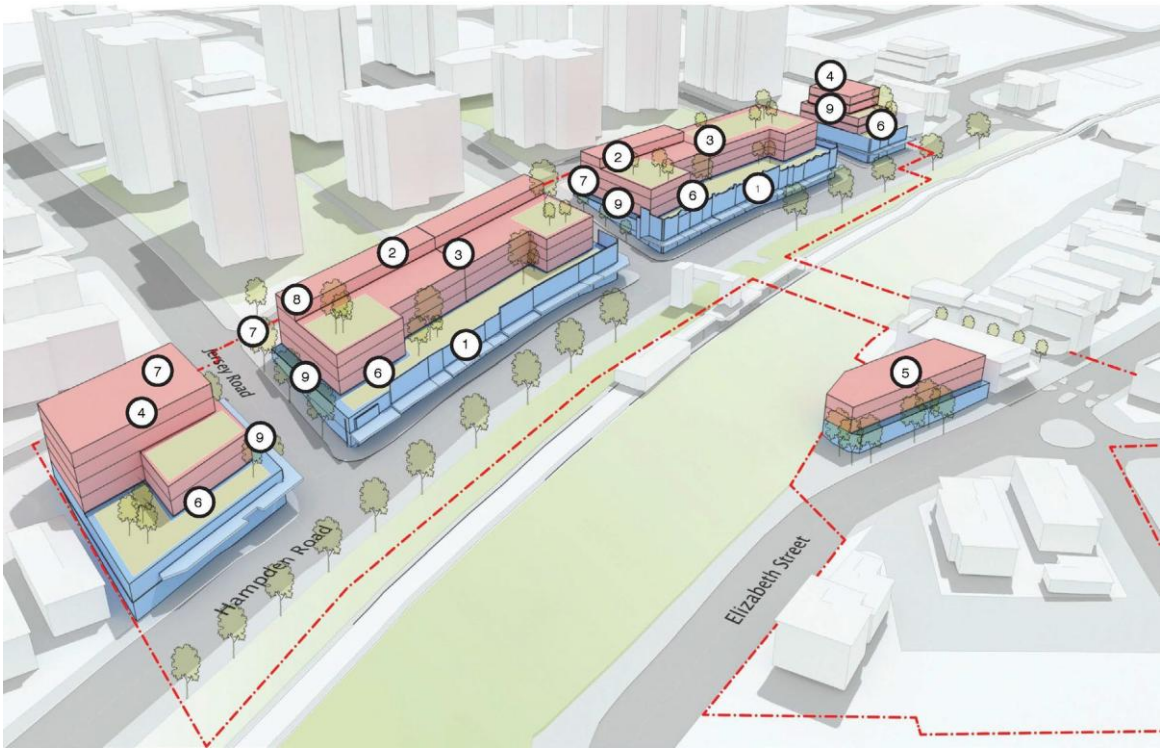
These controls are shown in Figure 3 and apply to any proposed new development in Artarmon local centre:

1. Retain the fine grain heritage frontage on Hampden Road.
2. Use the slope of the terrain to achieve 4 to 5 storey shop top housing developments between Francis Road and Jersey Road.
3. Amalgamate sites fronting Hampden Road, between Francis Road and Jersey Road to achieve a floor space ratio of up to 3:1.
4. Amalgamate sites fronting the eastern side of Jersey Road and the western side of Francis Road to achieve a floor space ratio of 3:1 and up to 6 storeys.
5. Maintain height of 3 storeys and floor space ratio of 1.3:1 on the library site.
6. Minimum 8m upper level setback (above the 2nd storey) to Hampden Road.
7. At grade vehicle access to car parking and loading/unloading area to be provided off Hampden Lane or side streets.
8. No upper level setback required to Hampden Lane.
9. Minimum 3m upper level setback (above the 2nd storey) from side streets.

Note:

- The roof top of shop top housing developments should provide communal (green) roof gardens.

Figure 3: Scale of development for Artarmon local centre



6 Castlecrag local centre

6.1 Character statement

The Castlecrag local centre is on Edinburgh Road between Eastern Valley Way and Rutland Avenue/The Postern.

The subdivision and layout of Castlecrag were designed by Walter Burley Griffin and the south eastern side of the centre is within the Griffin Conservation Area. The centre sits on a ridgeline with wide views across the district.

The centre has a mainly one to two storey built form character with a fine grain street frontage. The built form within the conservation area has a strong heritage character, referencing the original vision for Castlecrag.

The centre offers north-facing cafes and associated outdoor dining, complemented by retractable awnings on the southern side of the street and consistent fixed awnings on the north side.

The centre has been improved with new large unit UrbanStone paving, planting and street furniture. This includes more street trees and planting on the south side of Edinburgh Road. There is a strong desire within the community to retain the integrity of the design philosophy of Walter Burley Griffin for Castlecrag.

6.2 Performance criteria

Proposed development in Castlecrag local centre should:

a. reinforce the Griffin philosophy that the built form is subordinate to the surrounding natural environment

b. design buildings which are highly articulated with strong horizontal elements

c. encourage green roofs, and roof top gardens in appropriate locations

d. retain the fine grain built form of existing shopfronts on Edinburgh Road

e. ensure any redevelopment of the Quadrangle site provides new publicly accessible open space with good solar access

f. ensure any redevelopment of the Quadrangle provides a significant proportion of non-residential floor space as shown on Figure 11, with a pattern of fine grain shops along the Edinburgh Road frontage

g. maintain active street frontages

h. minimise vehicular access from Edinburgh Road

i. provide deep soil zones in appropriate locations to support existing and additional mature trees with wide tree canopies to improve the streetscape and reduce urban heat impacts

6.3 Master plan for Castlecrag local centre

Any future development in Castlecrag local centre must have regard to the following key features of the adopted master plan. The numbers below refer to actions to be carried out by

Willoughby City Council and/or developers to achieve the outcomes of the adopted master plan.

1. Retention of the Griffin Centre and ground floor façade, roof line and active street frontage.
2. Provide for a new park/green space.
3. Extend the curved façade of the Griffin Centre to reinforce The Postern.
4. Provide pedestrian links through to the Quadrangle site.
5. Retain mature boundary trees.
6. For any new development of the Quadrangle site, include a plaza with good solar access.
7. Provide roof top gardens and communal open space for any shop top housing.
8. Provide roof top gardens and communal open space for any shop top housing.
9. Retain at-grade car parking area.
10. Provide kerb blisters to improve mid-block crossing amenity.
11. Provide threshold and kerb blister treatments to improve pedestrian amenity.
12. For new developments, provide open space between shop top housing buildings.
13. Provide new vehicle access to the Quadrangle basement.
14. Provide a new left out vehicle access from the Quadrangle.
15. Create a green entry point at the intersection of Eastern Valley Way and Edinburgh Road.

Figure 4: Master plan for Castlecrag from Willoughby Local Centres Strategy 2036

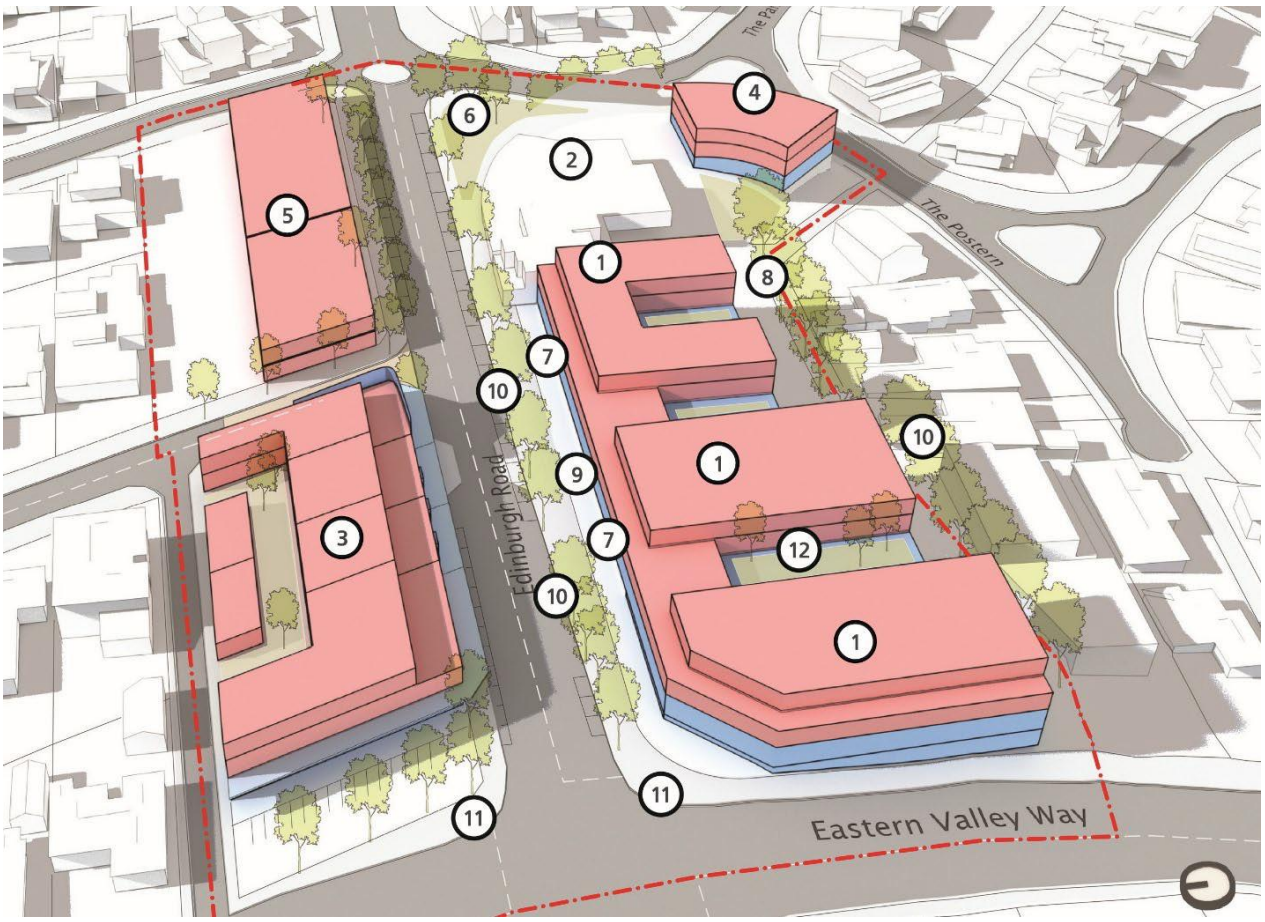
6.4 Controls for Castlecrag local centre

These controls are shown in Figure 5 and apply to any proposed new development in Castlecrag local centre:

1. Ensure a maximum of 4 storeys above the level at the Eastern Valley Way frontage and up to 3 storeys along Edinburgh Road (the additional storey below the Edinburgh Road frontage would use the topography of the site).
2. Retain the Griffin Centre which is identified as a heritage item.
3. Ensure a maximum of 3 storeys and 6m setback above the second level on the northern side of Edinburgh Road.
4. Ensure a maximum of 3 storeys for The Postern.
5. Require amalgamation for two development sites at 95–103 Edinburgh Road with vehicular access from the rear lane or side street.
6. Create public open space on the council-owned car park next to the Griffin Centre.
7. Provide a minimum 3m upper level setback (2nd storey) for shop top housing.
8. Maintain direct pedestrian links from the Quadrangle site to The Postern.
9. Ensure the built form does not restrict solar access along the footpath and adjoining public open space on the southern side of Edinburgh Road from 9am to 3pm on 21 June.
10. Retain the mature trees at the front and rear of the Quadrangle site.
11. Provide a ‘green’ landscaped entry point.

12. Provide a rooftop garden over the development with car parking area below.

Figure 5: Scale of development for Castlecrag local centre



7 North Willoughby local centre

7.1 Character statement

The North Willoughby local centre is at the intersection of Penshurst Street and Victoria Avenue. It is bounded by Patrick Street to the south, Terminus Lane to the north west and Power Lane to the north east.

North Willoughby, sometimes called East Chatswood, is on the fringe of Chatswood CBD, within a reasonable walking distance of around 1.5km from Chatswood railway station.

The existing built form character of the centre is a traditional streetscape defined by two-storey street frontages with a setback third storey. Buildings define key corners, such as the heritage listed Willoughby Hotel at the corner of Penshurst Street and MacMahon Street, and the mixed use building at the junction of Victoria Avenue and Penshurst Street. There is a range of businesses including medical suites, real estate agents, bakery, café and restaurants.

The centre offers excellent established street tree planting, particularly on Victoria Avenue and in the south of the centre. These trees provide shade on footpaths in the summer and a green leafy context. Generally, footpaths comprise a mix of tan herringbone brick pavers and

more recent asphalt infill treatment. There have been recent improvements with street trees and low-level planting along Victoria Avenue.

7.2 Performance criteria

Proposed development in North Willoughby local centre should:

a. retain the fine grain built form of existing shopfronts

b. ensure any new development respects the heritage value of the Willoughby Hotel on the corner of Penshurst Street and MacMahon Street

c. create high amenity retail connections to rear lanes

d. maintain active street frontages

e. encourage green roofs, and roof top gardens in appropriate locations

f. provide deep soil zones in appropriate locations to support existing and additional mature trees with wide tree canopies to improve the streetscape and reduce urban heat impacts

7.3 Master plan for North Willoughby local centre

Any future development in North Willoughby local centre must have regard to the following key features of the adopted master plan. The numbers below refer to actions to be carried out by Willoughby City Council and/or developers to achieve the outcomes of the adopted master plan.

1. Provide new high quality public open space (pedestrianised or traffic calmed) at the corner of Penshurst Street and Sydney Street.
2. Provide ground floor active street frontages to public open space.
3. Provide new through site pedestrian links to laneway networks.
4. Improve pedestrian amenity/crossings and public domain at the intersection of MacMahon Street and Penshurst Street.
5. Provide ‘gateway’ corner elements for any new developments at the intersection of Victoria Avenue and Penshurst Street.
6. Provide kerb blisters to improve pedestrian amenity.
7. Create new public open space at the corner of Royal Street and Victoria Avenue and relocate at grade parking into a basement car parking area.
8. New shop top housing developments to create open space connectivity and separation between buildings.
9. The increase in height and density of this site to facilitate the open space and basement parking identified in Item 7.
10. The height of any new development next to the Willoughby Hotel shall not exceed the eaves height of the heritage item.

Figure 6: Master plan for North Willoughby from Willoughby Local Centres Strategy 2036

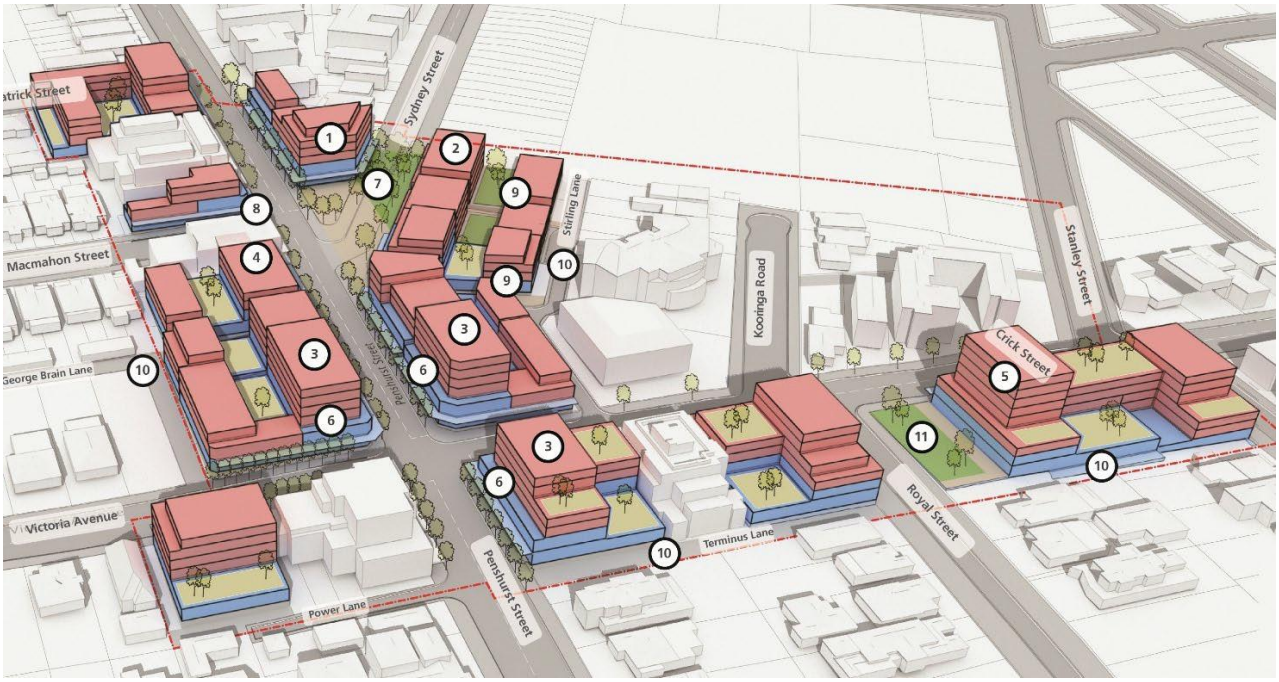
7.4 Controls for North Willoughby local centre

These controls are shown in Figure 7 and apply to any proposed new development in North Willoughby local centre:

1. Ensure a maximum of 5 storeys and floor space ratio of 2:1 for amalgamated lots directly fronting a new public open space.
2. Ensure a maximum of 5 storeys for amalgamated lots fronting Sydney Street within the local centre.
3. Ensure a maximum of 6 storeys and floor space ratio of 2.8:1 for amalgamated lots at the corners of the intersection of Victoria Avenue and Penshurst Streets.
4. Ensure development on sites immediately adjacent to the Willoughby Hotel does not exceed the current eaves height of the heritage item.
5. Ensure a maximum of 8 storeys and floor space ratio of 2.1:1 for amalgamated lots and provision of new public open space/plaza.
6. Ensure a minimum 3m upper level setback above two storeys along Penshurst Street and Victoria Avenue.
7. Provide public domain/open space improvements at the intersection of Sydney and Penshurst Streets.
8. Provide public domain and pedestrian improvements at the intersection of MacMahon and Penshurst Streets.

9. Provide minimum 4m wide through site pedestrian links to laneways.
10. Improve laneway access from Penshurst Street and Victoria Avenue.
11. For any new development identified in Item 5, provide for new public open space at the corner of Royal Street and Victoria Avenue with a minimum dimension of 15m by 30m.

Figure 7: Scale of development for North Willoughby local centre



8 High Street local centre

8.1 Character statement

High Street local centre is located along High Street and bounded by McClelland Street and Glover Street. The centre is a small strip of retail shops among one and two storey residential dwellings. It is a high amenity local centre, with a range of grocery and convenience offerings that serve the local neighbourhood.

The centre has convenient car parking provided at grade behind the shops on the northern side of Alexander Avenue. The church and associated school on the opposite side of the street have a key role in defining the character of the centre. The church adds civic and cultural elements and variety to the skyline.

8.2 Performance criteria

Proposed development in High Street local centre should:

a. retain and strengthen the good pedestrian access and through site connections from the retail shops

b. maintain and increase opportunities for active street frontages

c. increase fine grain shopfronts and maintain the active street frontages and outdoor dining areas

d. improve landscaped street frontages

e. encourage green roofs, and roof top gardens in appropriate locations

f. provide deep soil zones in appropriate locations to support existing and additional mature trees with wide tree canopies to improve the streetscape and reduce urban heat impacts

8.3 Master plan for High Street local centre

Any future development in High Street local centre must have regard to the following key features of the adopted master plan. The numbers below refer to actions to be carried out by Willoughby City Council and/or developers to achieve the outcomes of the adopted master plan.

1. Roof top gardens and communal open space to be provided for shop top housing.
2. Maintain and upgrade access and rear car parking area, north of Alexander Avenue; and maintain and upgrade rear lane car parking, south of Alexander Avenue.
3. Maintain through site/plaza links.
4. Relocate pedestrian crossings to better align with the through site/arcade links.
5. Improve pedestrian treatment and traffic calming measures at the intersection of High Street and Alexander Avenue.
6. Improve pedestrian treatment and traffic calming measures at the intersection of High Street and McClelland Street.

Figure 8: Master plan for High Street from Willoughby Local Centres Strategy 2036

8.4 Controls for High Street local centre

These controls are shown in Figure 9 and apply to any proposed new development in High Street local centre:

1. Maintain rear parking and laneway access.
2. Ensure a minimum 3m setback above 2 storeys along the High Street frontage.

Figure 9: Scale of development for High Street local centre

9 Naremburn local centre

9.1 Character statement

The Naremburn local centre is along Willoughby Road. It is bounded by Quiamong Street to the north, Willoughby Road to the east, Bongalong Street to the south and Glenmore Street to the west.

The centre is close to the Gore Hill Freeway. It is around 750m from St Leonards strategic centre and 3.3km from Chatswood CBD.

Naremburn has a series of two-storey, mid-century brick shop fronts with awnings and small unit paving that contributes to the village feel. Despite this, the centre is dominated by its relationship to Willoughby Road and the intersection with the Gore Hill Freeway.

Most of the built form along Willoughby Road has awnings that span the width of the footpath and provide pedestrians protection from the weather. The wide footpaths can accommodate multiple users, which encourages foot traffic as the predominant mode of transport in the area.

There are good trees around the northern end of the centre, providing a buffer between the street and the outdoor dining area. However, the landscape aesthetic is not continued to the southern portion of the centre. The centre includes two heritage items, at 284/284A Willoughby Road and 272–276 Willoughby Road. Retail in the centre is focused around food and beverage and some other services such as health and beauty and laundry. The centre lacks convenience stores such as grocery or banking.

9.2 Performance criteria

Proposed development in Naremburn local centre should:

a. retain the fine grain built form with heritage character

b. maintain active street frontages

c. encourage restaurants and cafes to provide outdoor dining areas

d. create through site activated arcade links

e. encourage green roofs, and roof top gardens in appropriate locations

f. provide deep soil zones in appropriate locations to support existing and additional mature trees with wide tree canopies to improve the streetscape and reduce urban heat impacts

9.3 Master plan for Naremburn local centre

Any future development in Naremburn local centre must have regard to the following key features of the adopted master plan. The numbers below refer to actions to be carried out by Willoughby City Council and/or developers to achieve the outcomes of the adopted master plan.

1. Retain the fine grain shop fronts/facades fronting Willoughby Road.
2. Provide through site/arcade links.
3. Retain heritage buildings/facades.
4. Provide public car parking and service access to the rear of lots fronting Willoughby Road.
5. Create additional high quality public space fronting Willoughby Road.
6. Provide pedestrian access for the public.
7. Provide roof top gardens and communal open space for shop top housing.
8. Provide an opportunity for a new public plaza.
9. Provide an opportunity for a small supermarket with basement car parking.

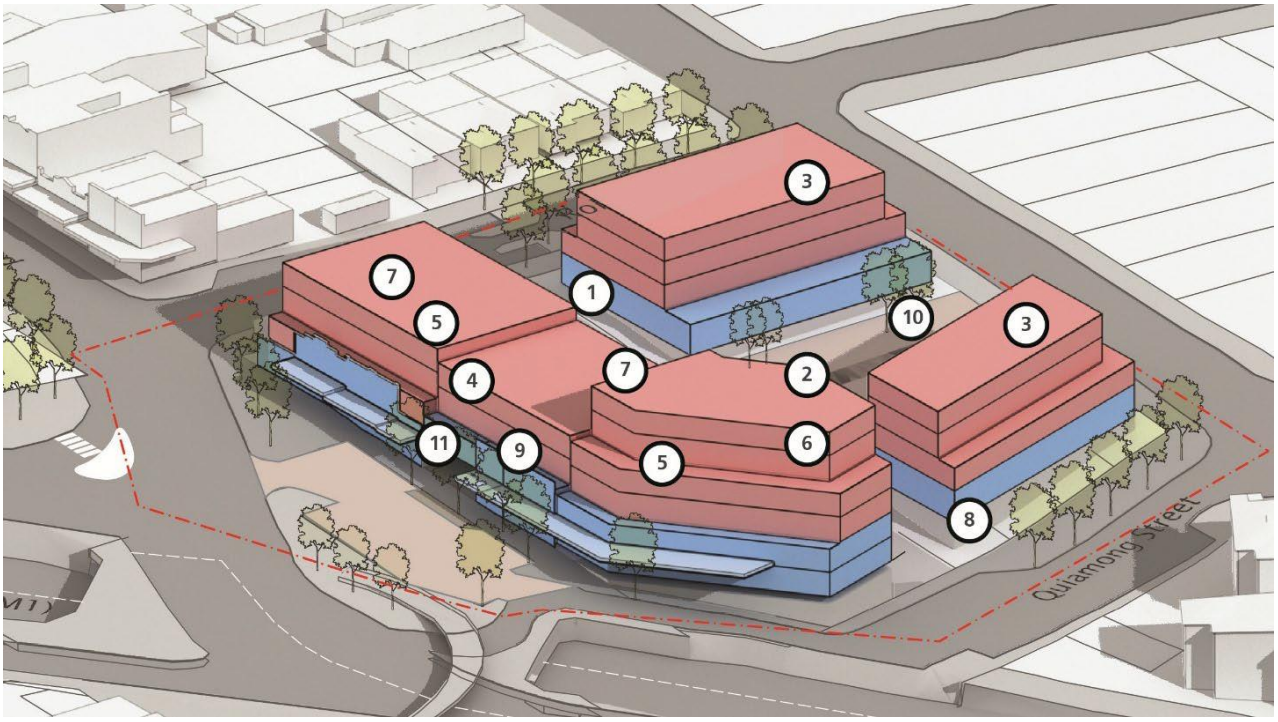
Figure 10: Master plan for Naremburn from Willoughby Local Centres Strategy 2036

9.4 Controls for Naremburn local centre

These controls are shown in Figure 11 and apply to any proposed new development in Naremburn local centre:

1. Ensure a maximum of 4 storeys.
2. Ensure a maximum of 6 storeys and floor space ratio of 1.9:1 for amalgamated lots on the north-east corner site.
3. Ensure a maximum of 4 storeys and floor space ratio of 1:1 for amalgamated lots with 3m setback above 2 storeys fronting Glenmore Street and Rohan Street.
4. Maintain fine grain shop fronts along Willoughby Road.
5. Provide a minimum 3m upper level setback above 2 storeys fronting Willoughby Road.
6. Additional 3m setback above 4 storeys fronting Willoughby Road.
7. Ensure the built form (including height and setbacks) of new development does not detract from the heritage values of adjacent heritage listed buildings.
8. Provide a minimum rear setback of 18m for lots fronting Willoughby Road to allow for future public parking and service areas.
9. Make provision for ground floor arcade/public through site link.
10. Make provision for new public space with active ground floor frontage.
11. Delete car parking spaces in front of shops and relocate car parking spaces within new developments to create improved public plaza on Willoughby Road.

Figure 11: Scale of development for Naremburn local centre



10 Northbridge local centre

10.1 Character Statement

Northbridge is at an entry point to the Willoughby local government area from the south. The business precinct is focused on a major arterial route along Sailors Bay Road, Eastern Valley Way and Strathallen Avenue.

Eastern Valley Way is a freight corridor and regular bus routes service the centre travelling north and south, and to and from the Sydney CBD.

Northbridge Plaza is on the northeast corner of Eastern Valley Way and Sailors Bay Road. It includes a major supermarket as well as 23 specialty shops. There is a large council car park at the rear of the plaza.

The Northbridge local centre lacks any significant outdoor open space area for community recreation and gatherings.

The topography of the centre is generally flat, with Sailors Bay Road running along the ridgeline. The built form character of the centre is generally a two to four storey street frontage. While the eastern side of the centre has a smaller lot pattern, the western side of the centre is characterised by larger lots and a bulkier built form. Some recent examples of shop top housing exist in the centre.

The eastern side of the centre benefits from increased street tree planting and landscaping, a pedestrian crossing at Bellambi Street, a generally consistent zero building setback to the street, and a higher level of fine grain street presentation. The majority of the centre is paved in a tan herringbone brick paver.

10.2 Performance criteria

Proposed development in Northbridge local centre should:

a. retain the fine grain built form along Sailors Bay Road, east of Strathallen Avenue

b. maintain and increase opportunities for active street frontages

c. encourage restaurants and cafes to provide outdoor dining areas east of Strathallen Avenue

d. create high quality design elements for buildings on prominent corner sites to identify the gateways entrances to the town centre

e. make provision for public open space and underground car parking at the rear of Northbridge plaza

f. encourage green roofs, and roof top gardens in appropriate locations

g. provide deep soil zones in appropriate locations to support existing and additional mature trees with wide tree canopies to improve the streetscape and reduce urban heat impacts

10.3 Master plan for Northbridge local centre

Any future development in Northbridge local centre must have regard to the following key features of the adopted master plan. The numbers below refer to actions to be carried out by Willoughby City Council and/or developers to achieve the outcomes of the adopted master plan.

1. Only non-residential floor space allowed for any new development on the northern side of Sailors Bay Road between Eastern Valley Way and Harden Avenue.
2. Provide new public open space above basement public car parking area.
3. Create a pedestrian laneway link.
4. Provide a range and mix of dwelling typologies, including opportunities for affordable housing and build-to-rent housing.
5. New residential development to provide basement car parking areas with vehicular access from Baringa Road only.
6. Provide streetscape improvements to Sailors Bay Road and Strathallen Avenue.
7. Provide a public domain with pedestrian access at the southern end of Bellambi Street.
8. Roof top gardens and communal open space to be provided for shop top housing.
9. Maintain and improve laneway network for servicing and deliveries.
10. Investigate opportunities for additional pedestrian crossing at existing intersections.
11. Provide blisters or central refuge for safer pedestrian crossing.

Figure 12: Master plan for Northbridge from Willoughby Local Centres Strategy 2036

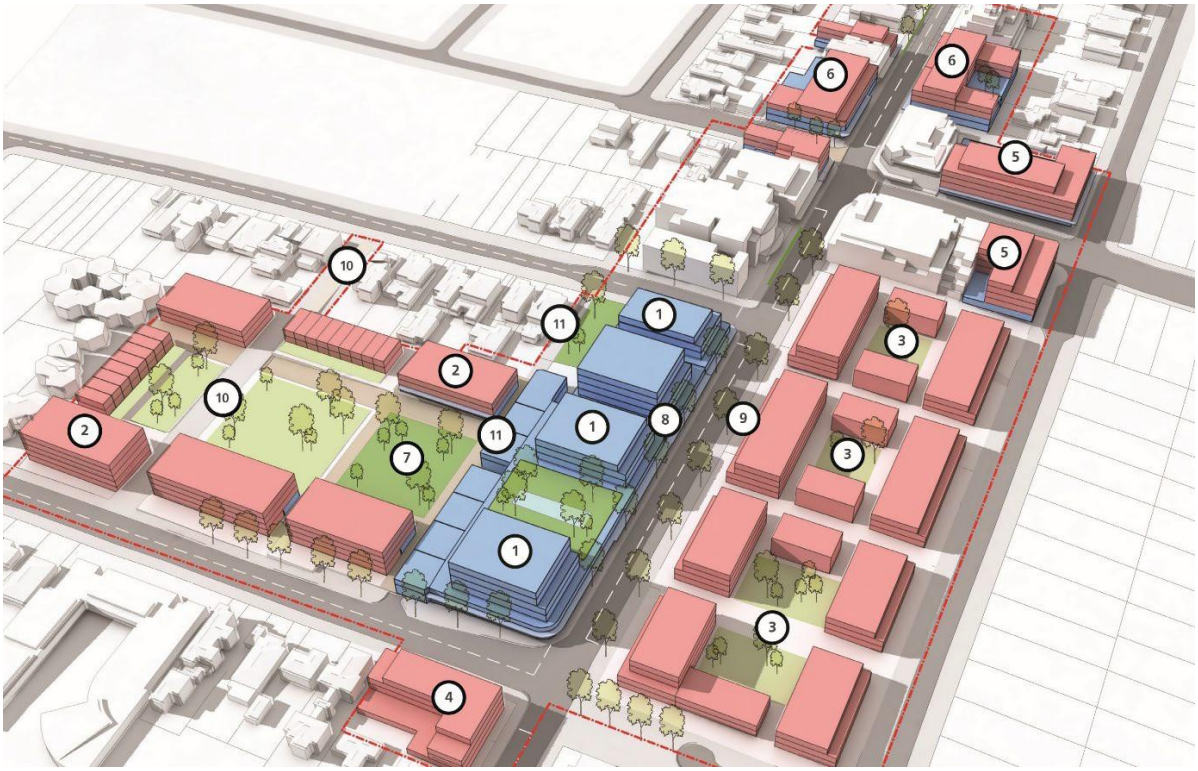
10.4 Controls for Northbridge local centre

These controls are shown in Figure 13 and apply to any proposed new development in Northbridge local centre.

1. Ensure a maximum of 6 storeys with provision for a roof top garden between two 6 storey building components for any new development on 83–113 Sailors Bay Road as shown on Figure 13.
2. Ensure a maximum of 3–5 storeys for a mix of residential apartment and townhouse developments with provision for large areas of open space as shown on Figure 13.
3. Ensure a maximum of 3 storeys along the northern side of Baringa Road and up to 4 storeys along Sailors Bay Road (new development sites must be amalgamated to achieve the minimum lot size of 1,100m^2 and a minimum frontage of 27m).
4. Ensure a maximum of 4 storeys for amalgamated lots on the north-west corner of Eastern Valley Way and Sailors Bay Road.
5. Ensure a maximum of 5 storeys for amalgamated lots in the E1 zone with a 1m setback on east and western side of Strathallen Avenue and a 3m setback on Baringa Road above the third level.
6. Ensure a maximum 4 storeys; or a maximum of 5 storeys if a second storey of non-residential floor space is provided.

7. Provide for new public open space with a minimum area of 2,000m^2 immediately behind any non-residential development along Sailors Bay Road.
8. Ensure a minimum upper level setback of 3m above 2nd storey and additional 6m setback above 4th storey for non-residential buildings along Sailors Bay Road.
9. Provide a minimum 6m deep soil landscaped setback area along the southern side of Sailors Bay Road, west of Strathallen Avenue.
10. Provide for a shared vehicle and pedestrian laneway from Eastern Valley Way to Harden Avenue.
11. Provide for a laneway and service access from Sailors Bay Road.

Figure 13: Scale of development for Northbridge local centre



11 Penshurst Street local centre

11.1 Character statement

The built form character of the centre is generally two-storey in the E1 Local Centre zone with a consistent build to boundary street address and awnings.

In the E3 zone running along the west of Penshurst Street, built form is generally two to three storeys, with inconsistent setbacks, landscaping, and awnings. Some buildings are substantially setback with at grade parking at the front. The lack of rear or side access has led to several access driveways interrupting the street footpath.

The E3 zone at the intersection of Mowbray Road and Penshurst Street is characterised by a taller built form of three to five storeys.

Despite the R3 medium density residential zoning south of Oakville Road, the residential character of built form facing the eastern side of Penshurst Street is low density with single storey detached dwellings and small garden setbacks to the street. This side of the street has ornamental pear trees and some landscaping along the footpath. There is minimal street tree planting elsewhere in the centre.

The area located south of Mowbray Road comprises shop top housing style premises including a recent development known as ‘The Mint’.

11.2 Performance criteria

Proposed development in Penshurst Street local centre should:

a. make provision to improve the pedestrian and cycling environment

b. improve service access to minimise impacts on the public domain along Penshurst Street

c. ensure active frontages are provided in the E1 and E3 zones

d. encourage green roofs, and roof top gardens in appropriate locations

e. provide deep soil zones in appropriate locations to support existing and additional mature trees with wide tree canopies to improve the streetscape and reduce urban heat impacts

11.3 Master plan for Penshurst Street local centre

Any future development in Penshurst Street local centre must have regard to the following key features of the adopted master plan. The numbers below refer to actions to be carried out by Willoughby City Council and/or developers to achieve the outcomes of the adopted master plan.

1. Roof top gardens and communal open space to be provided for shop top housing.
2. Provide shared vehicular access to reduce the number of driveways off Penshurst Street.
3. Investigate provision for safe pedestrian crossings at the intersection of Penshurst Street and Oakville Road.
4. Provide streetscape improvements and additional street trees along Penshurst Street.
5. Roof top gardens and communal open space to be provided for shop top housing.
6. Extend Medway Lane to Penshurst Street.
7. Retain at grade car parking area.
8. New development to incorporate enhanced visual features such as public art and/or unique façade treatment on the prominent corner site.
9. New development to incorporate enhanced visual features such as public art and/or unique façade treatment on the prominent corner site.

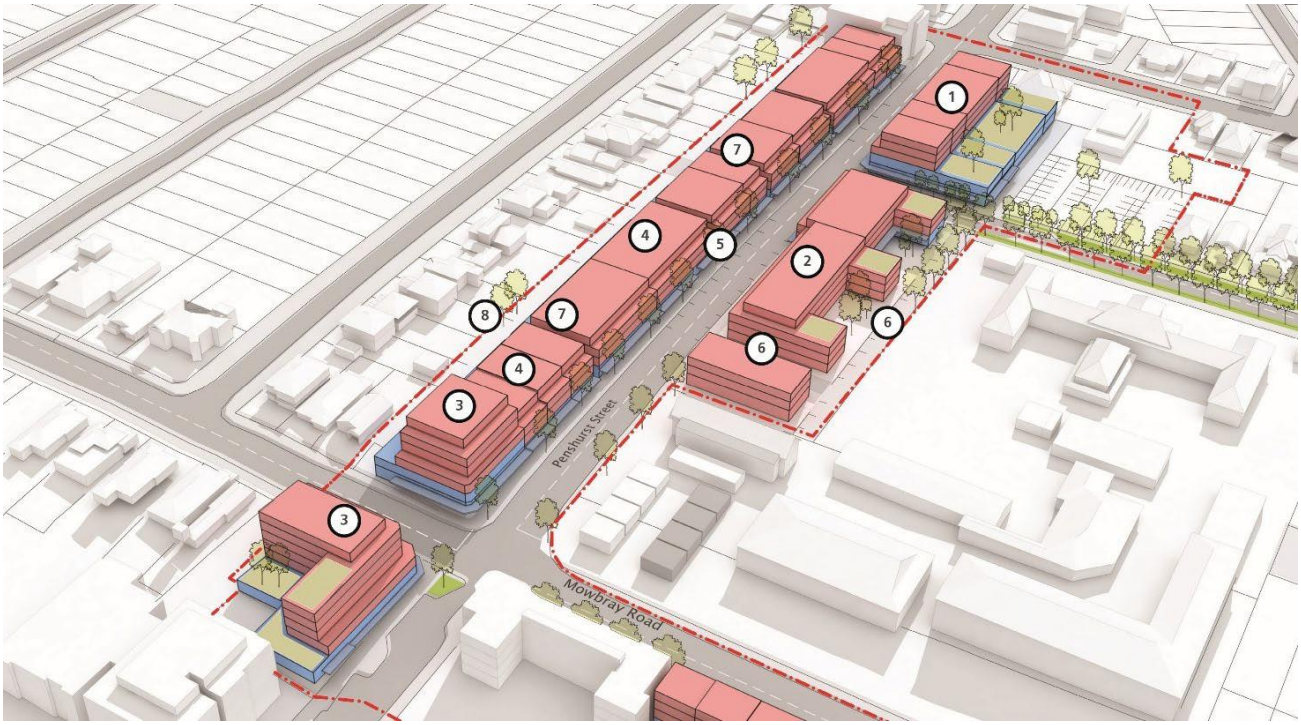
Figure 14: Master plan for Penshurst Street from Willoughby Local Centres Strategy 2036

11.4 Controls for Penshurst Street local centre

These controls are shown in Figure 15 and apply to any proposed new development in Penshurst Street local centre.

1. Ensure a maximum of 5 storeys and floor space ratio of 2.8:1 for amalgamated lots fronting Penshurst Street with a roof top garden above the second level comprising approximately 50% of the width of the site as shown on Figure 15.
2. Ensure a maximum of 4 storeys and floor space ratio of 2.1:1 on amalgamated lots in the R3 zone with roof top gardens at the rear above the third floor levels.
3. Ensure a maximum of 6 storeys and floor space ratio of 2.8:1 on amalgamated lots; with a minimum 3m setback from the rear boundary and a further minimum 3m setback above the second floor level for any new development on the north western corner of Penshurst Street and Mowbray Road; and roof top gardens for any new development on the south western corner as shown on Figure 15.
4. Ensure a maximum of 4 storeys and floor space ratio of 2:1 on amalgamated lots fronting Penshurst Street.
5. Ensure a minimum 3m setback above 3 storeys.
6. For new development, make provision to extend Medway Lane to join Penshurst Street.
7. Provide shared vehicle access between amalgamated sites to minimise the number of driveways on Penshurst Street.
8. Ensure a minimum 3m rear setbacks and a further minimum 3m setback above the first floor level for all new developments adjoining the rear yard area of properties along Ward Street.

Figure 15: Scale of development for Penshurst Street local centre



12 Willoughby South local centre

12.1 Character statement

The Willoughby South local centre is bounded by Penkivil Street and Harris Street to the north and Borlaise Street to the south. The commercial area mainly fronts Willoughby Road with some commercial areas branching off Willoughby Road in Frenchs Road. The centre has direct access to the Gore Hill freeway from Willoughby Road and is around 7.2km from the Sydney CBD.

The topography of the area slopes from north to south and there is a substantial east/west change in level between the area bounded by Frenchs Road, Tulloh Lane and Prentice Lane.

The existing buildings are generally one and two storeys with shop fronts to Willoughby Road. There have been some more recent mixed use developments which are three-storey in height. The Bridge View Hotel is a prominent building in this locality and is identified as a local heritage Item. The facades of the buildings at 549–553 Willoughby Road are also heritage listed.

There are two council car parks off Borlaise Street. However, they are not well sign-posted and have poor connection to the retail strip.

Sanders Park located to the west of the centre in Julian Street provides neighbourhood open space. There is also significant tree canopy creating a green network at the rear of properties on the western side of Willoughby Road.

12.2 Performance criteria

Proposed development in Willoughby South local centre should:

a. make provision for new and/or improved laneway connections

b. maintain and increase opportunities for active street frontages

c. encourage first floor non-residential uses

d. ensure active frontages are provided at the corners of local side streets off Willoughby Road and Frenchs Road, and to any internal plaza spaces

e. encourage green roofs, and roof top gardens in appropriate locations

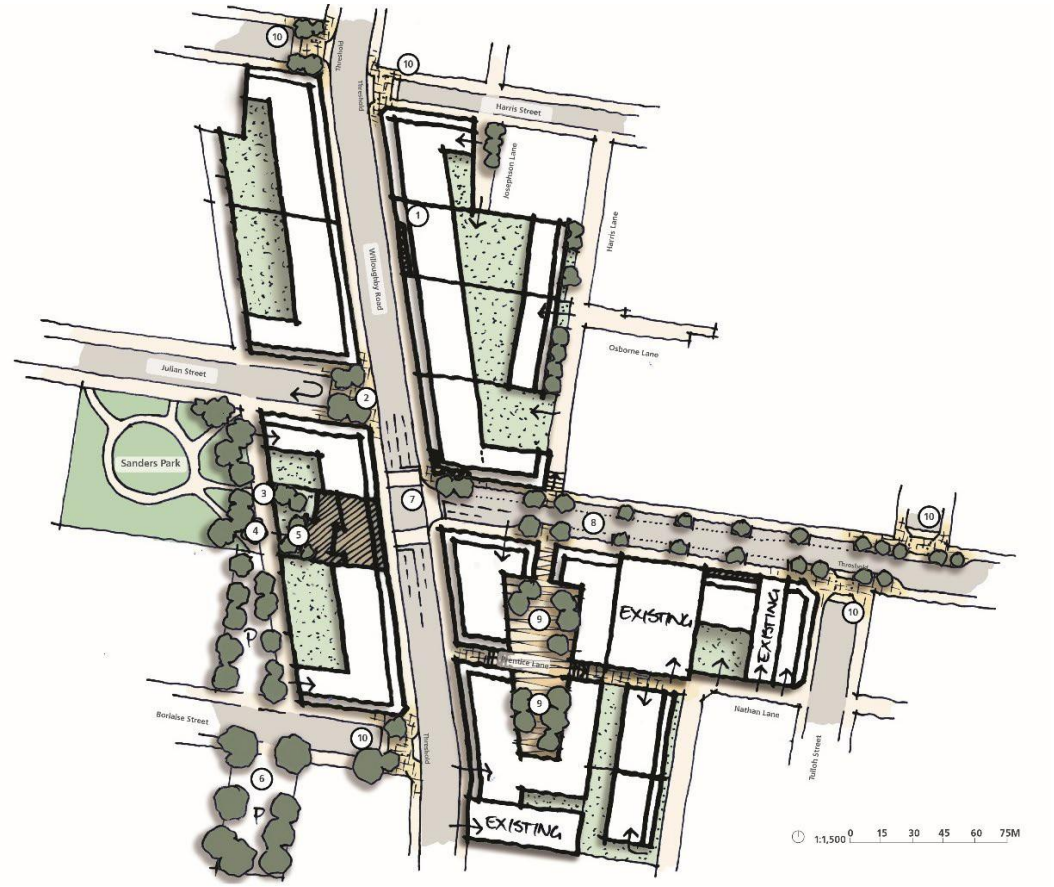
f. provide deep soil zones in appropriate locations to support existing and additional mature trees with wide tree canopies to improve the streetscape and reduce urban heat impacts

12.3 Master plan for Willoughby South local centre

Any future development in Willoughby South local centre must have regard to the following key features of the adopted master plan. The numbers below refer to actions to be carried out by Willoughby City Council and/or developers to achieve the outcomes of the adopted master plan.

1. Retain the fine grain shop fronts/facades fronting Willoughby Road.
2. Provide a public domain with pedestrian access at the eastern end of Julian Street.
3. Connect laneway between Julian Street and Borlaise Street.
4. Improve landscaping of Sanders Park along the frontage of the new laneway.
5. Upgrade outdoor space at the rear of the Bridgeview Hotel.
6. Retain car parking area off Borlaise Street.
7. Provide kerb extension on the northern side of Frenchs Road at the corner of Willoughby Road.
8. Provide streetscape improvements and additional street trees along Frenchs Road.
9. New developments to create a public plaza around Prentice Lane.
10. Improve pedestrian treatment and traffic calming measures at the intersections of Frenchs Road with Tulloh Street and Chiltern Road.

Figure 16: Master plan for Willoughby South from Willoughby Local Centres Strategy 2036



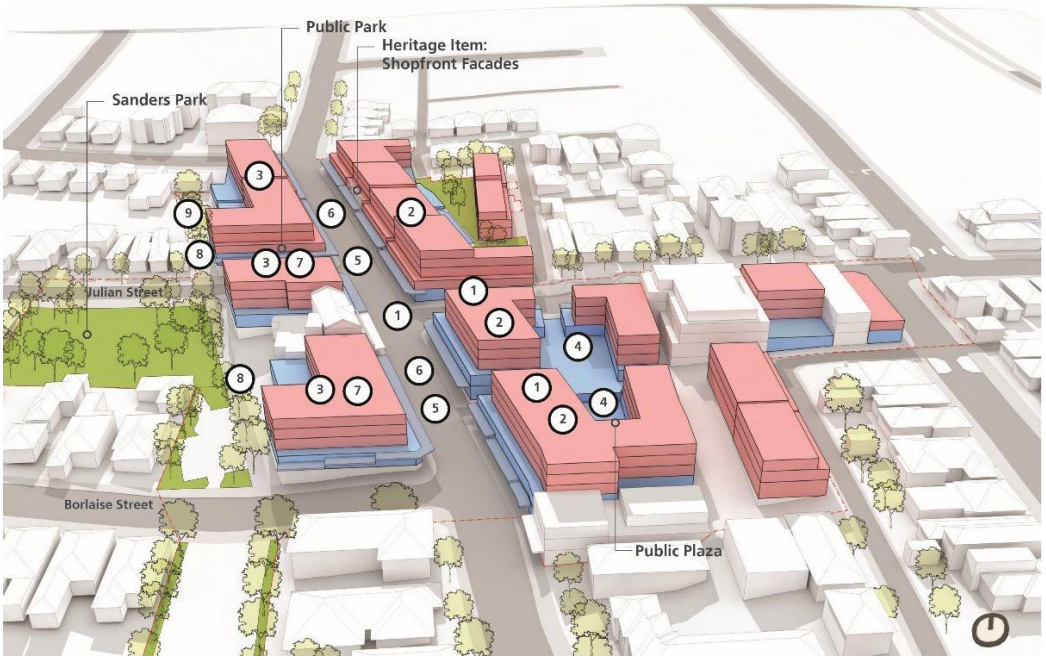
12.4 Controls for Willoughby South local centre

These controls are shown in Figure 17 and apply to any proposed new development in Willoughby South local centre.

1. Ensure a maximum of 5 storeys for amalgamated lots in the E1 zone on the eastern side of Willoughby Road.
2. Ensure a maximum of 2:1 floor space ratio for amalgamated lots east of Willoughby Road.
3. Ensure a maximum of 2:1 floor space ratio for amalgamated lots west of Willoughby Road.
4. Ensure a maximum of 3:1 floor space ratio for amalgamated lots with a frontage on the eastern side of Willoughby Road with a minimum non-residential floor space ratio of 1.5:1 (amalgamated sites must ensure there is an equitable distribution of public open space/plaza areas within the development).
5. Provide fine grain shop fronts that are consistent with existing development along Willoughby Road.
6. Ensure a minimum 3m upper level setback above 2 storeys fronting Willoughby Road.
7. For development on lots adjacent to the Bridgeview Hotel, between Julian Street and Borlaise Street, provide a consistent street setback to the heritage item above the first storey, and a 3m setback to the heritage item.

8. For developments to the west of Willoughby Road with access off Borlaise Street, consider vehicle/conflict and create activation for new developments next to Sanders Park.
9. For new developments west of Willoughby Road and north of Julian Street make provision for a 7m wide rear lane for vehicular access and servicing.

Figure 17: Scale of development for the Willoughby South local centre



13 Controls for specific sites

This section provides controls for any future development for specific sites and areas, and the site specific DCPs that support approved planning proposals in the Chatswood CBD and the Local Centres.

13.1 Chatswood CBD

The following specific sites and areas, and the site specific DCPs apply to the Chatswood CBD.

13.1.1 767 Pacific Highway, Chatswood

The existing façade of the heritage listed Chatswood Fire Station is to be incorporated within any future development on the site if it is required to be demolished for road widening.

13.1.2 61A Albert Avenue Chatswood (Mandarin Centre)

13.1.3.1 General

The controls contained in this site specific Development Control Plan apply to 65 Albert Avenue Chatswood. This land is bounded by Albert Avenue, Victor Street and Orchard Road as shown in Figure 18 below.

Figure 18: Site Aerial Map

1 Objectives of the Plan

The Objectives of the Plan are to:

1. Support the provision of non-residential development in the Chatswood CBD.
2. Enable the development of the site without impacting the viability of adjoining land.
3. Provide a building exhibiting design excellence in architectural form and materials.
4. Minimise traffic impacts on the surrounding road network.
5. Maximise street level activation.

2. Built Form

Performance Criteria

The built form of new development shall:

1. Achieve a slender tower/s on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties and streetscape.
3. Ensure visual and acoustic privacy and sun access.
4. Provide suitable areas for communal open spaces, deep soil zones and landscaping.
5. Minimise view loss for surrounding properties.

Controls

1. The width of each side of any tower, and design elements that contribute to building bulk, are to be minimised.
2. If there is more than one tower on a site, sufficient separation is to be provided in accordance with required setbacks to ensure slender tower forms are achieved.

3. Two towers must not visually read as one large tower. Towers are not to be linked above Podium and should operate independently regarding lifts and services.
4. The site layout, including any tower/s within the site, is to be in accordance with Figure 19.

Figure 19: Site Layout

3.0 Height of Building

Performance Criteria

The height of the new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties and the nearby key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable.

4.0 Setbacks and street frontage heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings are consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute to slender tower forms.
4. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

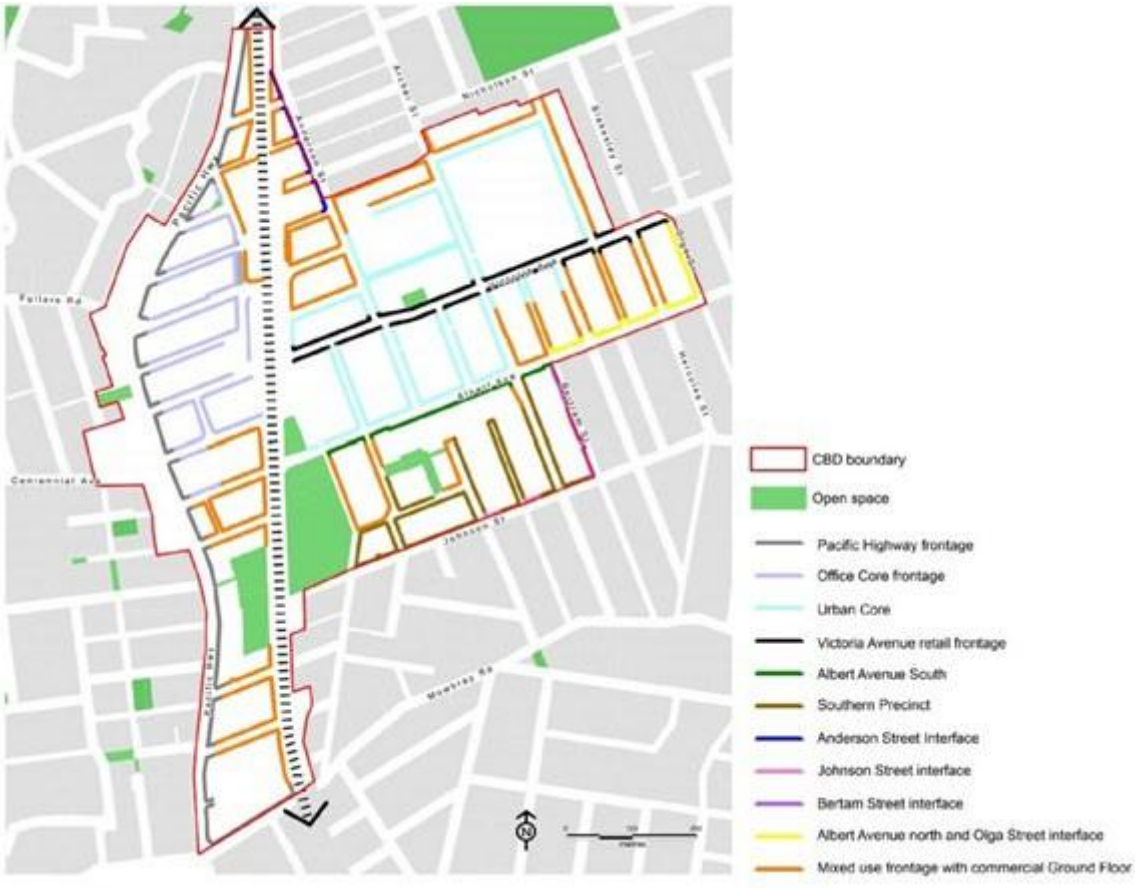
5. Ensure the street wall heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

The development is to have a maximum street wall height of 24m at any point.

1. Building setbacks and street wall heights are to be in accordance with Map 5 Setbacks and Street Frontage Heights. The setbacks are summarised as follows:
    a) Albert Avenue
        i. Maximum 24 metre street wall height at front boundary
        ii. Minimum 6 metre setback above street wall to tower.
    b) Victor Street
        i. Maximum 24 metre street wall height at front boundary.
        ii. Minimum 6 metre setback above street wall to tower.

Map 5: Setbacks and Street Frontage Heights



1. In addition to Controls 1 and 2:

   a) Setbacks may be greater and street wall heights may be lower.
   b) Additional ground level setbacks are sought that contribute to public realm.

5.0 Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements.
2. Extensive blank walls shall be avoided at street level.

6.0 Amenity

Performance Criteria

1. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application stage.
3. The development shall be designed to maximise solar access, cross ventilation, visual and acoustic privacy.

7.0 Open Space and Landscaping

Performance Criteria

1. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Albert Avenue and Victor Street.
2. Podium and roof tops are to be a combination of green and recreation spaces.
3. Street tree planting is to be provided.

Controls

1. Public domain improvements shall be provided to all street frontages to Council requirement
2. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
3. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
4. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
5. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8.0 Links

Performance Criteria

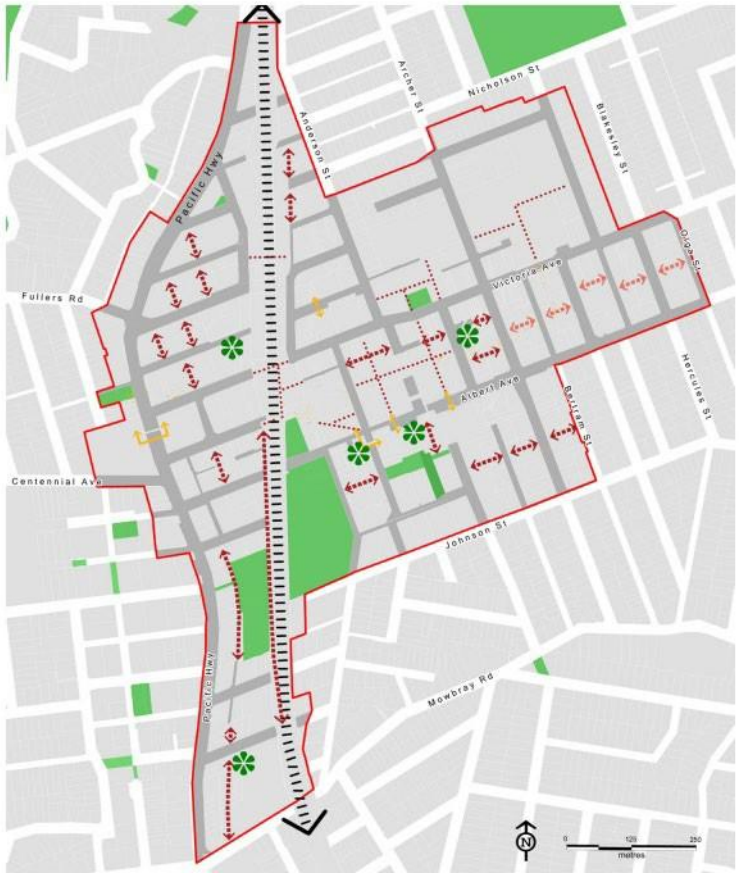
1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate public accessible through site links at:
    a) the Albert Avenue / Victor Street corner
    b) near the northern site boundary as it connects with Victor Street in accordance with Map 4.

2. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



CBD boundary

Open space

New open space

Open air 24 hour through-site links

Open air 24 hour through-site links within block

Through-building links

Existing upper storey links

Streets and public places

9.0 Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximise surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Albert Avenue and Victor Street.

2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10.0 Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site is not the cause of adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to the development is to be from one access point in Victor Street.
2. Vehicle egress from the development is to be from one access point in Orchard Road.
3. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
4. All car parking and loading facilities are to be located below ground level.
5. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.

11.0 Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages.

12.0 Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13.0 Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14.0 Building Sustainability

Performance Criteria

1. Achievement of design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage

15.0 Amusement Centres

General

1. These special controls apply to the development of amusement centres within the Mandarin Centre, Chatswood. All applications for amusement centres require development consent from the Council. When development applications for amusement centres are being assessed, local schools, Police and Council's Community Services Division will be consulted.

Objectives

1. Accommodate the establishment of a quality amusement centre within the Mandarin Centre;
2. Ensure that the operation of the amusement centre does not cause problems for schools within the area;
3. Ensure that the amusement centre is a functional and well supervised facility;
4. Achieve an open, pleasant and safe environment for all those wishing to play amusement devices; and
5. Ensure that the use does not cause any form of nuisance or social disturbance.

Location

1. Amusement centres must only be located within the managed centre.
2. Amusement centres shall not be located at natural ground level if the premises has direct access to a street, lane, mall or public place.

3. The entrance to the amusement centre shall not be located within 20m of the entrance of the managed centre.

Additional Considerations

1. In respect of any development application for an amusement centre, Council will take into consideration the following matters:
   a) any representations made by a statutory body or other interested persons in relation to the application;
   b) the necessity and desirability of making provision on site for the parking of cars and bicycles;
   c) the existing and future amenity of the neighbourhood;
   d) whether the proposed development is likely to cause a nuisance due to the emission of noise, vibration, congregations in public places, or otherwise adversely affect the environment;
   e) the social effect of the use in the locality and the public interest;
   f) access for people with disabilities; and
   g) the location of premises in relation to residential areas and schools.

Operational Requirements

1. Hours of operation
   a) Amusement centres shall not operate outside the approved hours of the Mandarin Centre.
2. Building Code of Australia
   a) Buildings will be required to comply with the requirements of the Building Code of Australia for a class 9(b) building and a “place of public entertainment”. The number of persons on the premises at any one time shall be limited to one (1) person per 1.2m^2 of gross floor area.
3. Management of Patrons
   a) No person who is in effective control of an amusement centre shall permit to enter or remain thereon any person under the age of eighteen (18) years except:
      i. at an hour or on a day when primary or secondary schools are not open; or
      ii. where the person submits satisfactory evidence to those in management that they are either not enrolled as a student at a primary or secondary school, or absent from school with the approval of the school.
   b) An amusement centre shall be attended during all hours of operation by at least one person, or if Council so decides depending on the size of the facility, two or more persons, and any such person(s) shall be recognisable as an attendant. The attendant(s) is to be responsible for the effective control of the centre and to ensure that the conditions of any consent applying to the centre are complied with so as to achieve the objectives of Part E3.3.3 of this plan.
   c) No alcohol or drugs are to be permitted on the premises at any time.

4. Amusement Devices
    a) An amusement centre shall not contain any amusement device which is or can be operated for profit or reward by the player(s) where such profit or reward is in the form of money.
5. Generation of Crowds
    a) The operation of the premises is not to adversely affect the amenity of the surrounding area, by allowing people to congregate in a public place.
    b) An applicant will need to submit details of appropriate management measures to satisfy this requirement.
6. Noise Transmission
    a) Materials of construction shall be such that the noise transmission rating shall achieve a sound level not exceeding 5dB/A above the ambient level of any adjoining building or public place. This does not include other occupancies within the managed centre.

13.1.3 58 Anderson Street Chatswood

General

These controls apply to land bounded by 58 Anderson Street, Wilson Street, and the North Shore Railway Line, as shown in Figure 20.

Figure 20: Site Aerial Map

- ---- The site

Objectives of the Plan

The aims and objectives of this plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of future development of surrounding properties.
3. Minimise traffic impacts on the surrounding road network.
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximises activation to Anderson and Wilson Streets.

2. Built Form

Performance Criteria

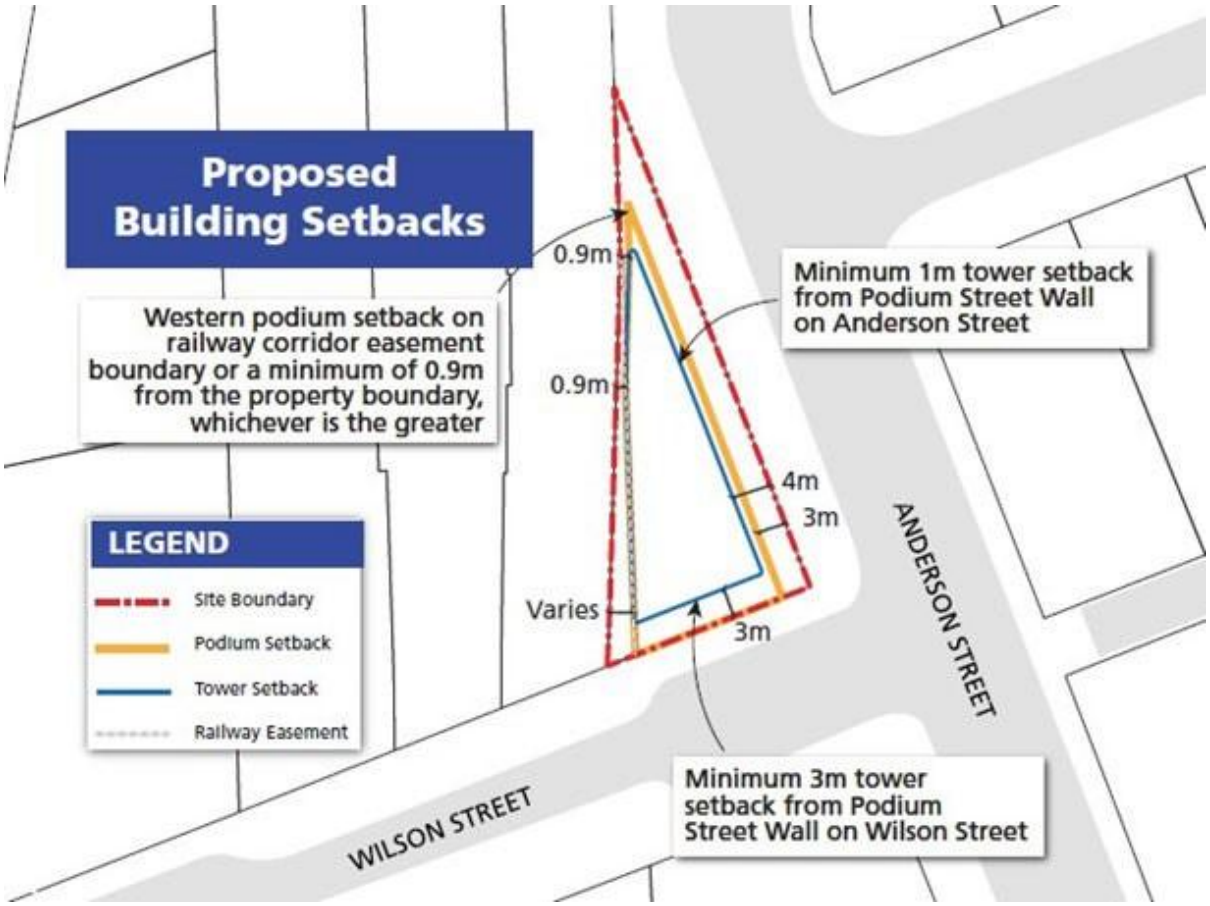
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access and views.
4. Provide suitable areas for communal open spaces, deep soil zones and landscaping.
5. Minimise view loss for surrounding properties.

Controls

1. The tower floor plate above podium is to be consistent with Figure 21.
2. The width of each side of any tower, and design elements that contribute to building bulk, are to be minimised.
3. The site layout is to be in accordance with Figure 21.

Figure 21: Site Layout



3 Height of Building

Performance Criteria

The height of the new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties and the nearby key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable.
4. Setbacks and street frontage heights.

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings are consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute to slender tower forms.
4. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

5. Ensure the street wall heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 “Street Frontage Heights and Building Setbacks”. The required setbacks are:
    a. Anderson Street frontage
        i. Minimum 3 metre setback at ground level from boundary.
        ii. 6-14 metre street wall height.
        iii. Minimum 1 metre setback above street wall.
    b. Wilson Street frontage
        i. 6-14m street wall height at boundary.
        ii. Minimum 3 metre setback above street wall.

Map 5: Street Frontage Heights and Building Setbacks

2. In addition to Control 1:
    a) Setbacks may be greater and street wall heights may be lower.
    b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. To maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units within the development and developments on adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. The development shall be designed to maximise solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include green landscaping.
5. Podium and roof tops are to be a combination of green and recreation spaces.
6. Street tree planting is to be provided.

Controls

1. All roofs up to 30 metres from ground level are to be green roofs. These are to provide a balance of passive and active green spaces that maximise solar access.
2. A minimum of 2 hours of direct solar access is to be provided to the public open space on the site.
3. Publicly accessible open space and green landscaping such as street trees will be required by all developments.
4. Communal open space for residents of the building is to be incorporated within/on the building, and include seating, recreational areas (e.g. barbeque area) and landscaping.
5. Any communal open space, with particular regard to roof top level on towers, shall be designed to address issues of quality, safety, and usability.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or as green walls of buildings.
7. Deep soil planting is to be provided within the setback to Anderson Street. Deep soil plantings include trees, shrubs, and grasses, and are to be unimpeded by buildings or structures below ground.
8. All publicly accessible open space is to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

8. Links

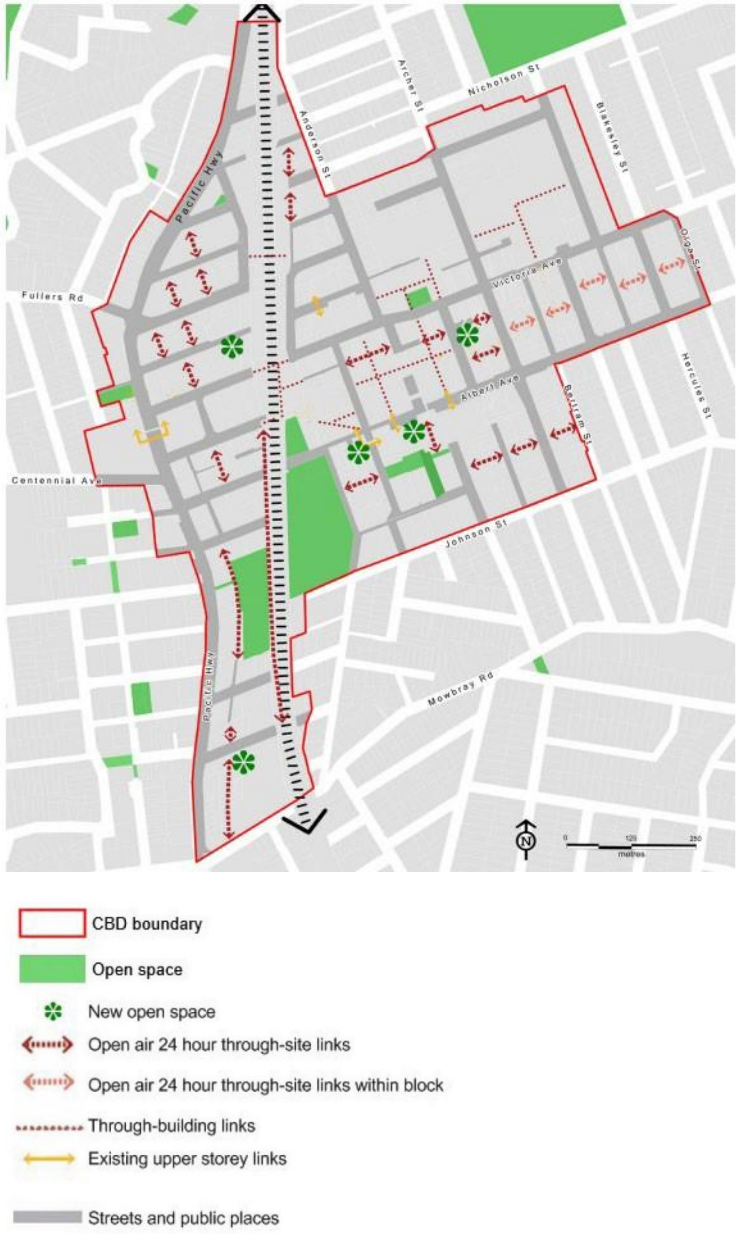
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate public accessible through site links in accordance with Map 4 below.
2. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximise surveillance of the public.

Controls

1. At ground level buildings are to achieve some active frontages to Anderson Street and Wilson Street.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site is not to cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Minimise car parking and encourage alternative transport options.

Controls

1. All vehicles are to enter and exit a site in a forward direction. Vehicle manoeuvring technologies such as turntables should only be provided where demonstrated to be necessary.
2. Vehicle access points are located and designed to achieve safety, minimise conflicts between pedestrians and vehicles, and create a high quality streetscape.
3. Traffic shall be restricted to left in/left out on the Anderson Street entrance, to be facilitated by the introduction of a median strip and constructed at the cost of the proponent and designed in consultation with Council.
4. Bicycle access/facilities and circulation along Anderson and Wilson Streets is required, including “filling the gaps” in the existing bicycle network across intersections.
5. Safe and secure on-site bicycle parking capacity including lockers and racks and end-of-trip facilities to meet the expected site demands are to be provided and designed to meet the relevant design standards.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised.

Controls

1. Any loading docks, including garbage, deliveries, and residential removal trucks are to be located in the basement or at the rear of ground floor areas. Loading docks may be permitted on the ground floor where it can be demonstrated that it is not practical to provide within basement levels.
2. Vehicular access to the site is to be via Wilson Street for non-residential deliveries and garbage collection and via Anderson Street for residential entries and exits.
3. A Waste Management Plan shall be submitted at Development Application Stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Achievement of design excellence shall include achievement of higher building sustainability standards.

Control

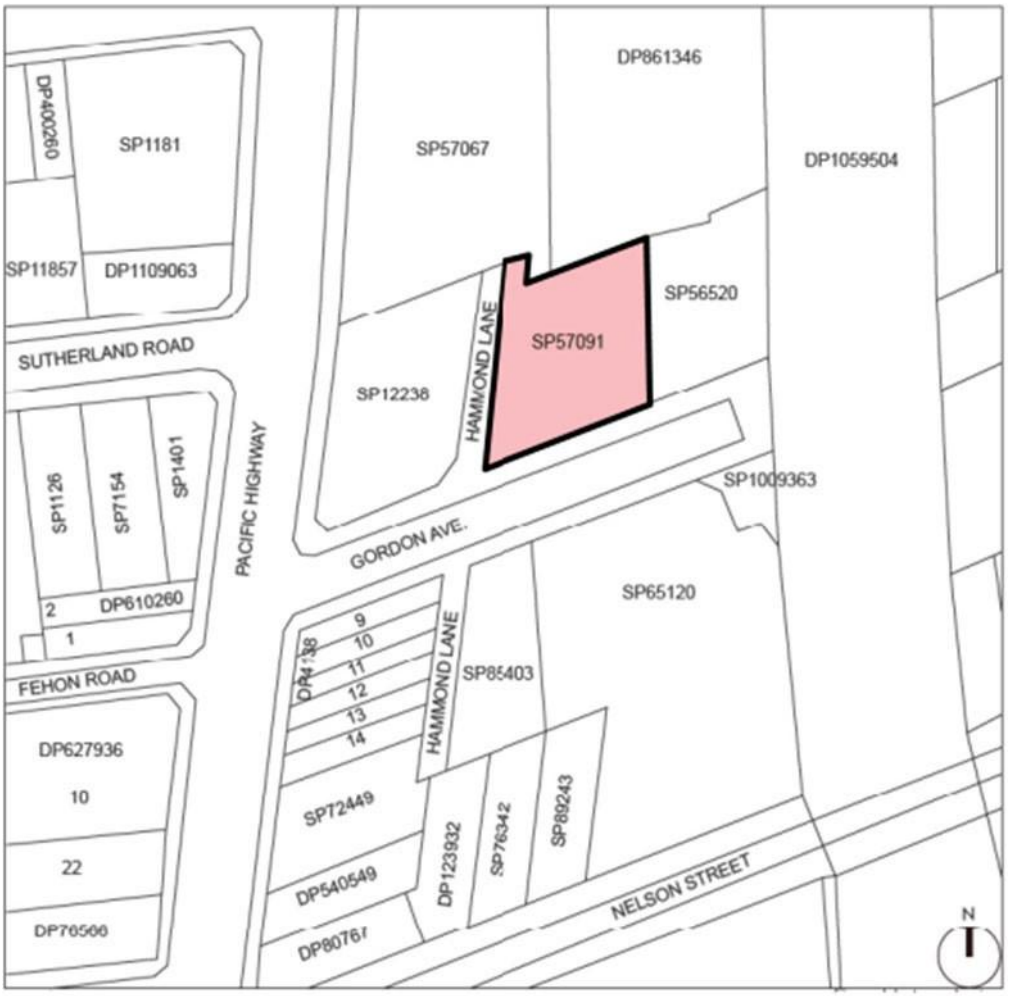
1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.4 5 - 9 Gordon Avenue Chatswood

1. General

The controls contained in this Site Specific Development Control Plan apply to land at 5-9 Gordon Avenue, Chatswood. The land is bounded by Gordon Avenue to the south, 655A and 655 Pacific Highway to the north, Hammond Lane to the west, and 1-3 Gordon Avenue to the east as shown on the Figure 22 below.

Figure 22: **Site Aerial Map**



Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to Gordon Avenue and Hammond Lane.

2. Built Form

Performance Criteria

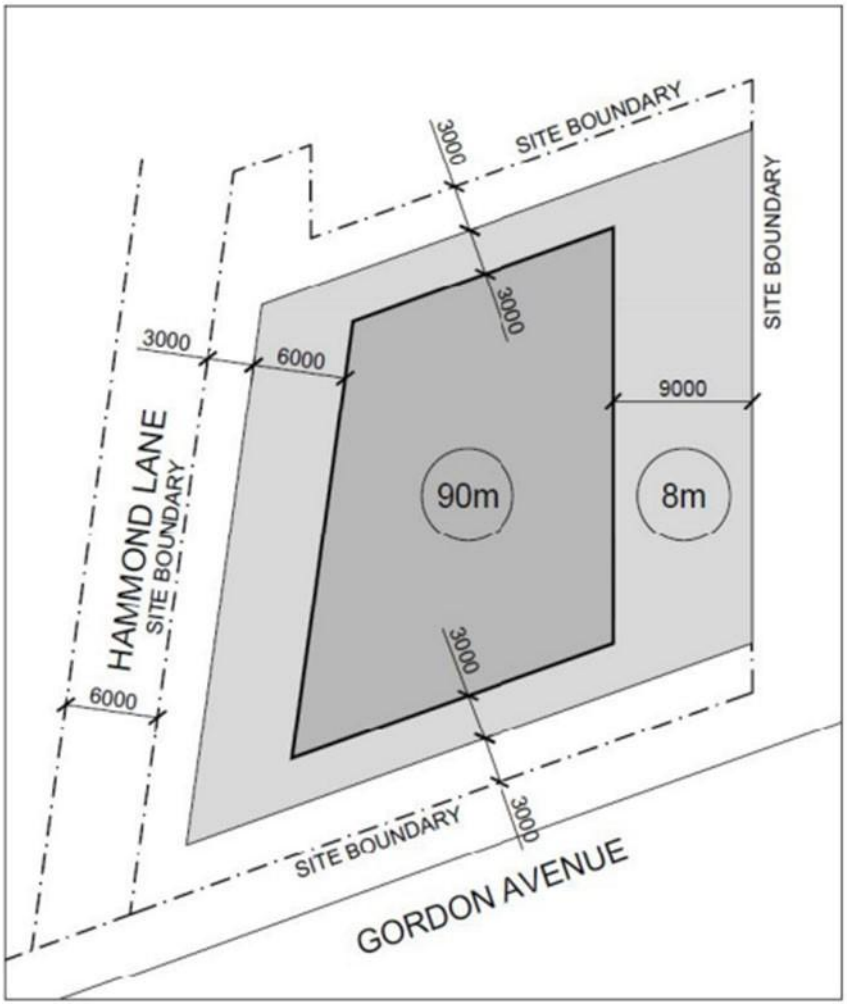
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 23.

Figure 23: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Contribute to slender tower forms.
5. Minimise the effects of adverse wind conditions at street level.

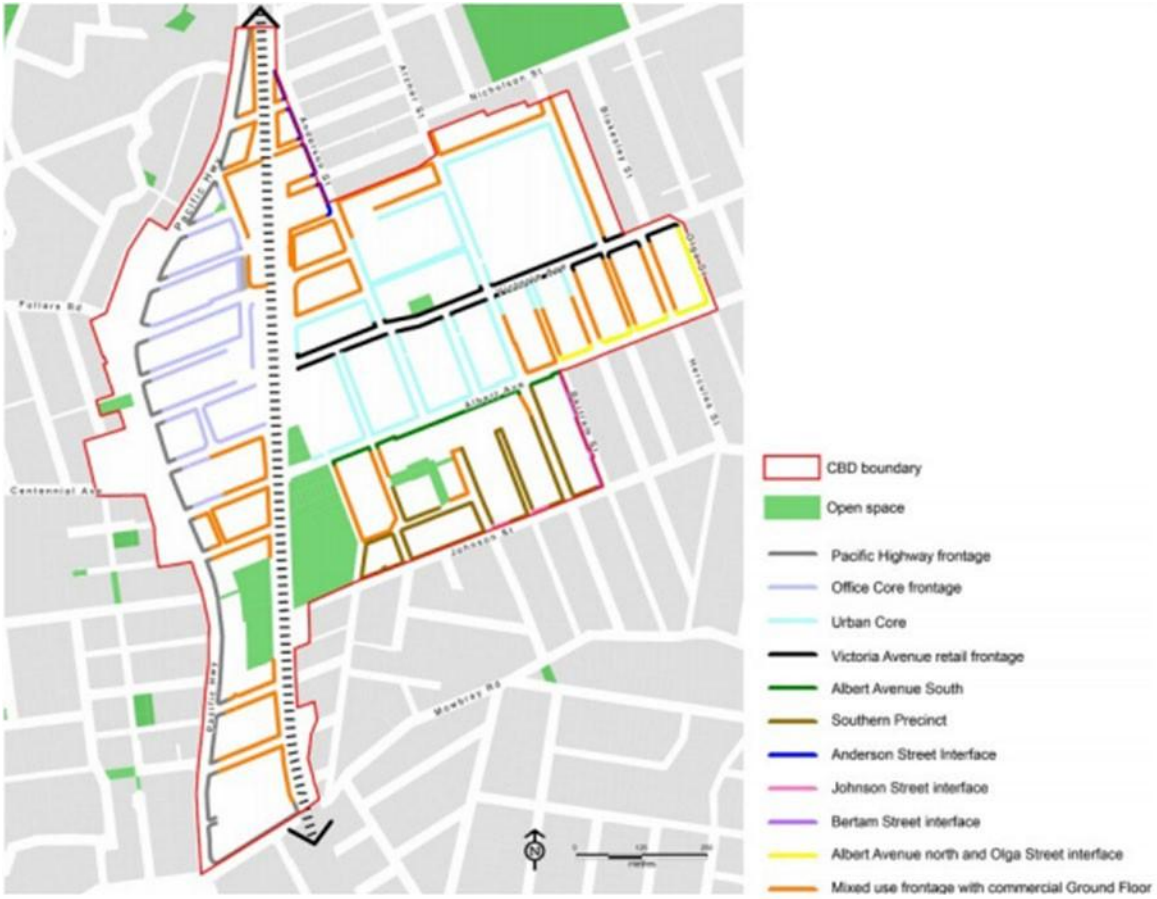
Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights). Setbacks are as follows:
    a) Gordon Avenue and Hammond Lane Frontage:
        i) 6-14 metre street wall height at front boundary (maximum two to four storeys).
        ii) Minimum 3 metre setback above street wall.
    b) In regards the tower, a minimum of 1:20 ratio of the setback to building height above the podium (eg. tower to be setback 3 metre above podium for a 60 metre building, 4.5 metre setback for a 90 metre building).

Map 5: Setbacks and Street Frontage Heights



2. In addition to Control 1:
    a) Setbacks may be greater and street wall heights may be lower.
    b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximise solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – in particular to Gordon Avenue, Hammond Lane and the neighbouring properties.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximise solar access.
3. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
4. Public domain improvements shall be provided to all street frontages to Council requirements.
5. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
6. Deep soil planting is to be provided within the setback to Gordon Avenue, Hammond Lane and setback to the northern (rear) boundary. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.

7. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
8. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
9. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

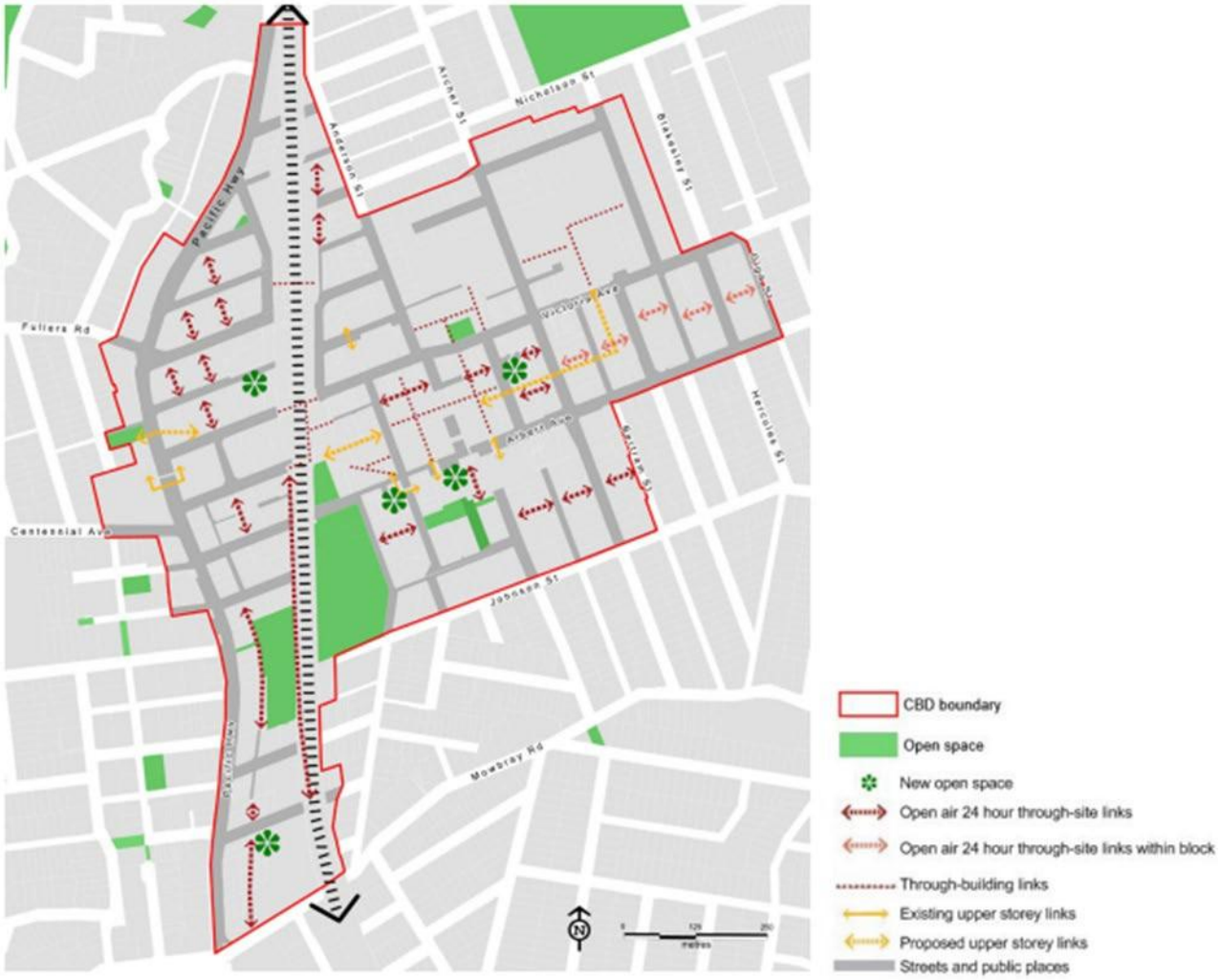
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximise surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Gordon Avenue and Hammond Lane.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.

2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be from one access point in Hammond Lane.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, are to be included in any future Development Application.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.5 753 Pacific Highway and 15 Ellis Street Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 753 Pacific Highway and 15 Ellis Street, Chatswood. The land is bounded by Pacific Highway to the west, Ellis Street to the south, Crispe Lane to the east and 755-759 Pacific Highway to the north as shown on Figure 24 below.

Figure 24: Site Aerial Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network.
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.

2. Built Form

Performance Criteria

The built form of the new development shall:

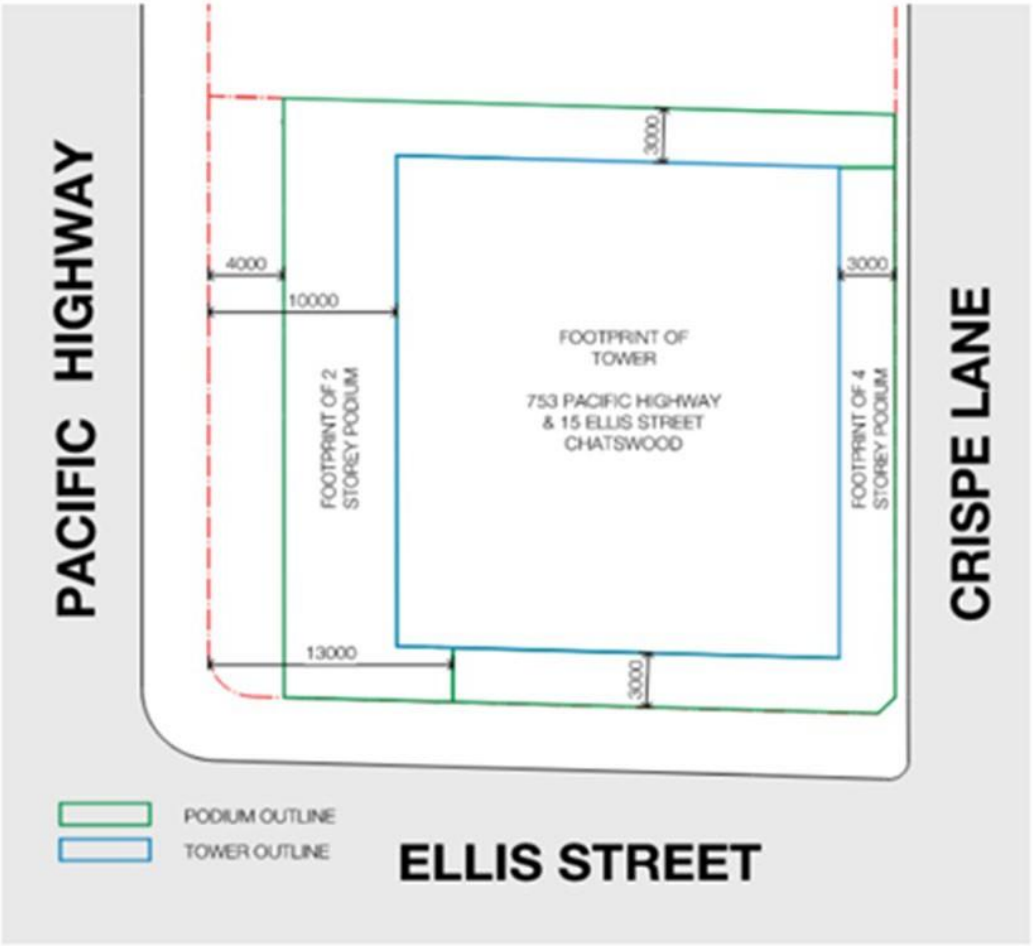
1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.

3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 25.

Figure 25: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Contribute to slender tower forms.
5. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

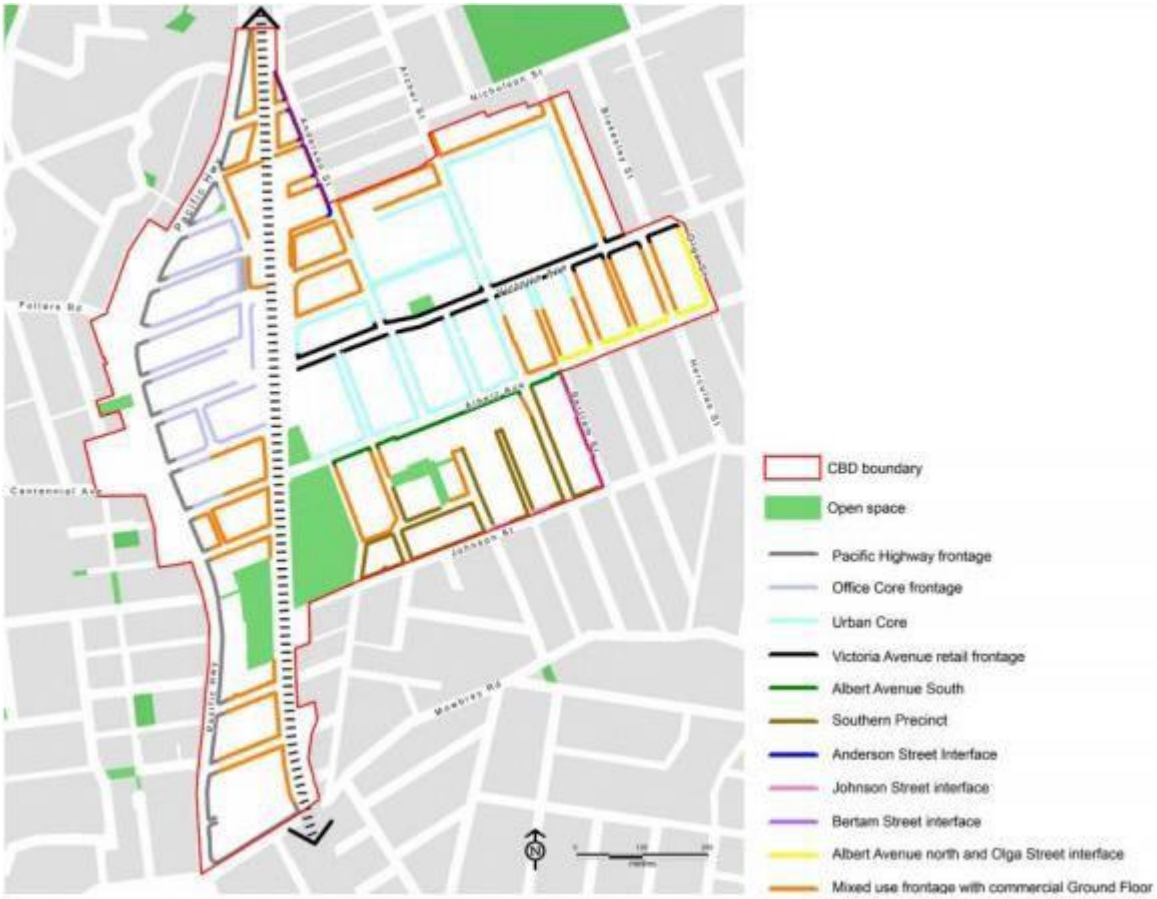
1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights). Setbacks are as follows:
    a) Pacific Highway (western) Frontage:
        i) Minimum 4 metre setback at ground level from front boundary.
        ii) Maximum 7 metre street wall height (maximum two storeys).
        iii) Minimum 6 metre setback above street wall.
    b) Ellis Street and Crispe Lane Frontage:
        i) 6-14 metre street wall height at front boundary (maximum two to four storeys).
        ii) Minimum 3 metre setback above street wall.

c) In regards the tower, a minimum of 1:20 ratio of the setback to building height above the podium (eg. tower to be setback 3 metre above podium for a 60 metre building, 4.5 metre setback for a 90 metre building).

Map 5: **Setbacks and street frontage heights**



2. In addition to Control 1:

a) Setbacks may be greater and street wall heights may be lower.

b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.

2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Pacific Highway, Ellis Street, Crispe Lane and neighbouring properties.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
3. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
4. Public domain improvements shall be provided to all street frontages to Council requirements.
5. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
6. Deep soil planting is to be provided within the 4m setback to Pacific Highway. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.

7. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
8. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
9. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

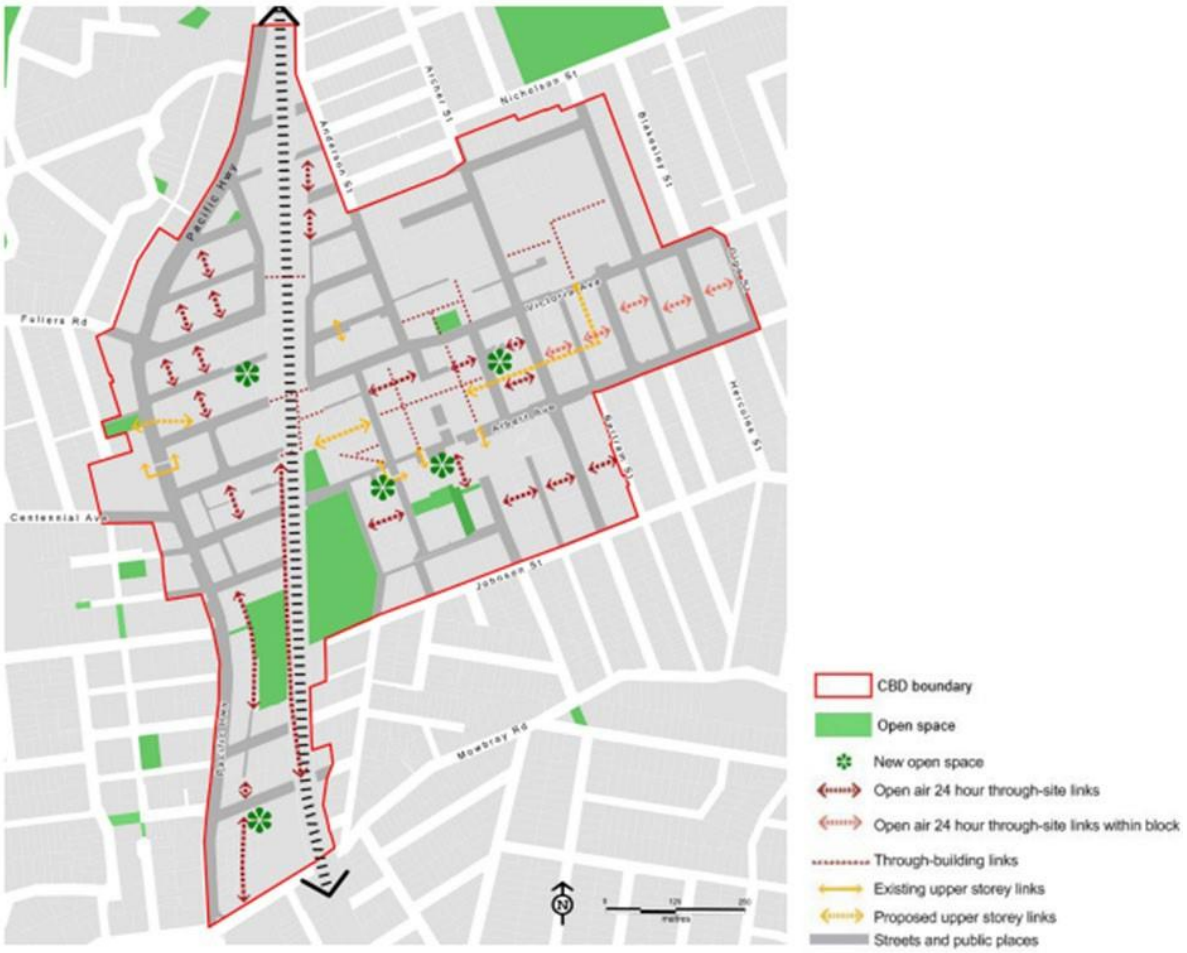
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Pacific Highway, Ellis Street and Crispe Lane.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.

3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be from one access point in Crispe Lane.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.6 871 – 877 Pacific Highway Chatswood

1. General

The controls contained in this Site Specific Development Control Plan apply to land at 871-877 Pacific Highway, Chatswood. The land is bounded by Pacific Highway to the west, 879 Pacific Highway to the north, the North Shore railway line to the east, and Wilson Street as shown on Figure 26 below.

Figure 26: Site Aerial Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to the Pacific Highway and Wilson Street.

2. Built Form

Performance Criteria

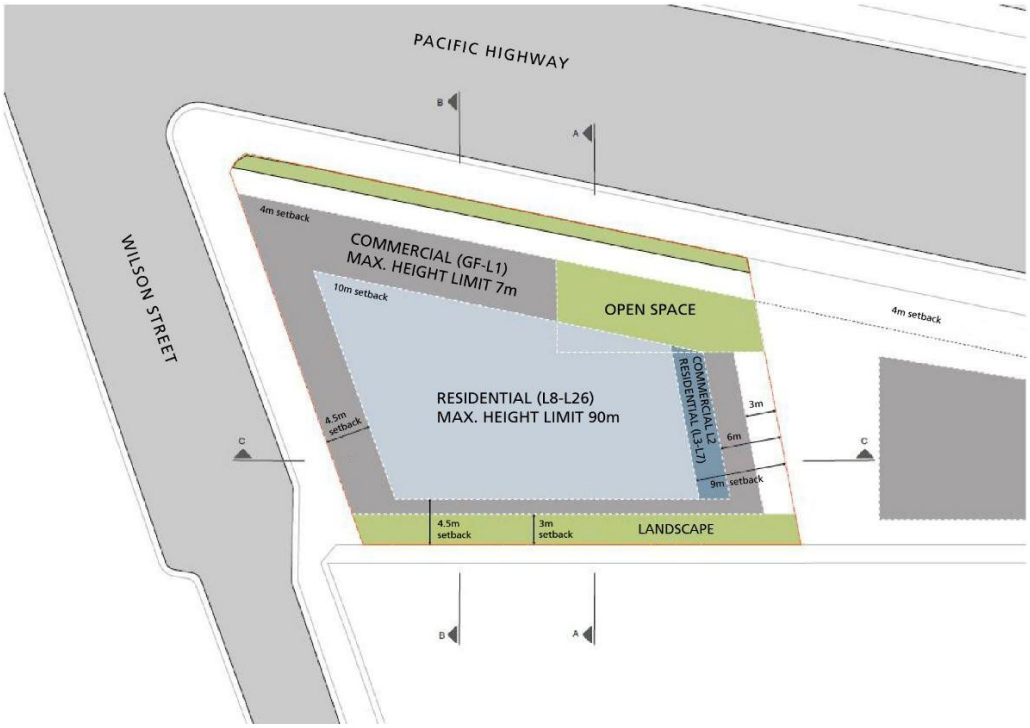
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 27.

Figure 27: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.

4. Contribute to slender tower forms.
5. Minimise the effects of adverse wind conditions at street level.

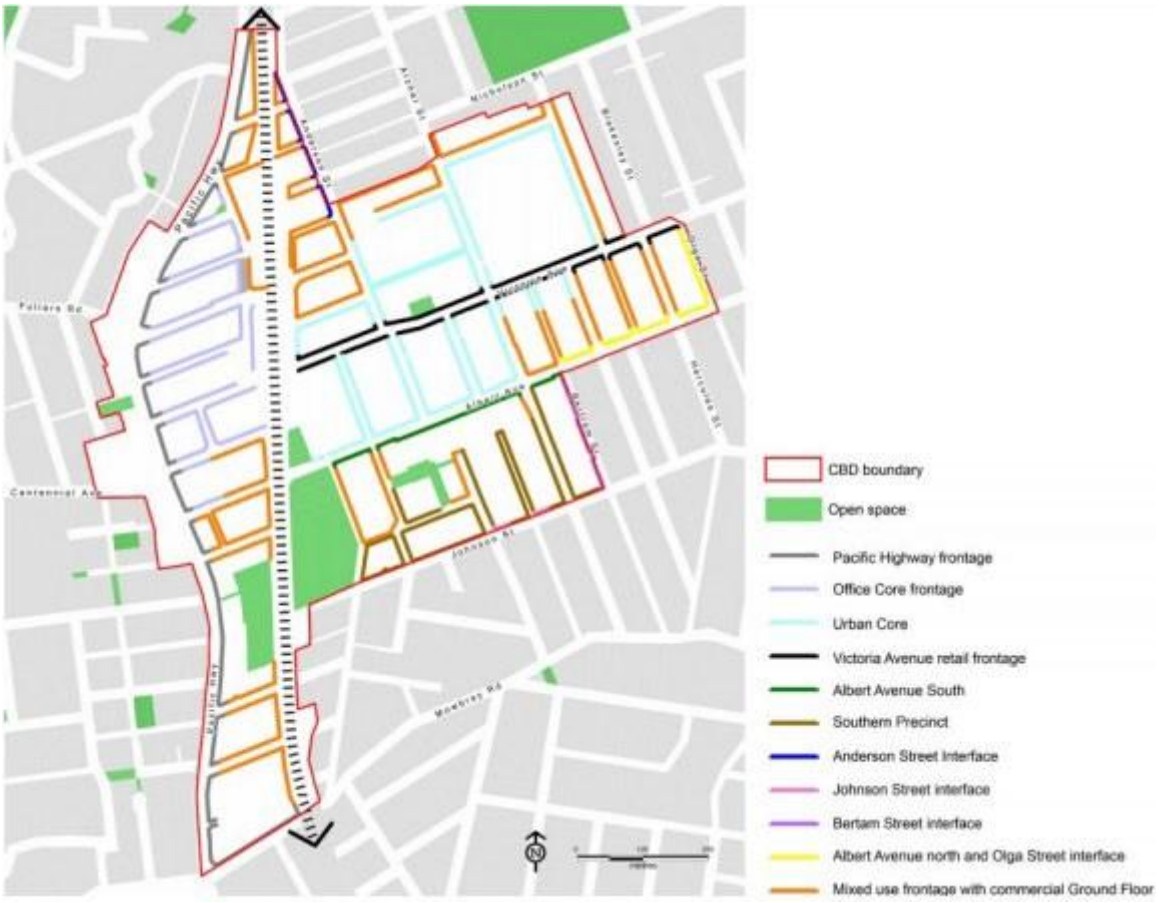
Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights). Setbacks are as follows:
    a) Pacific Highway (western) Frontage:
        i) Minimum 4 metre setback at ground level from front boundary.
        ii) Maximum 7 metre street wall height (maximum two storeys).
        iii) Minimum 6 metre setback above street wall.
    b) Wilson Street (southern) Frontage:
        i) 6-14 metre street wall height at front boundary (maximum two to four storeys).
        ii) Minimum 3 metre setback above street wall.
    c) Railway Line (eastern) Frontage
        i) 6-14 metre street wall height at front boundary (maximum two to four storeys).
        ii) Minimum 3m setback at podium levels.
    d) In regards the tower, a minimum of 1:20 ratio of the setback to building height above the podium (eg. tower to be setback 3 metre above podium for a 60 metre building, 4.5 metre setback for a 90 metre building).

Map 5: Setbacks and street frontage heights



2. In addition to Control 1:
   a) Setbacks may be greater and street wall heights may be lower.
   b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Pacific Highway, Wilson Street and the Railway Line.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. Large canopy tree planting must be provided along the Pacific Highway frontage within the 4m setback area.
3. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
4. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
5. Public domain improvements shall be provided to all street frontages to Council requirements.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.

7. Deep soil planting is to be provided within the setback to the Pacific Highway and setback to the eastern boundary (Railway Line frontage). Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
8. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

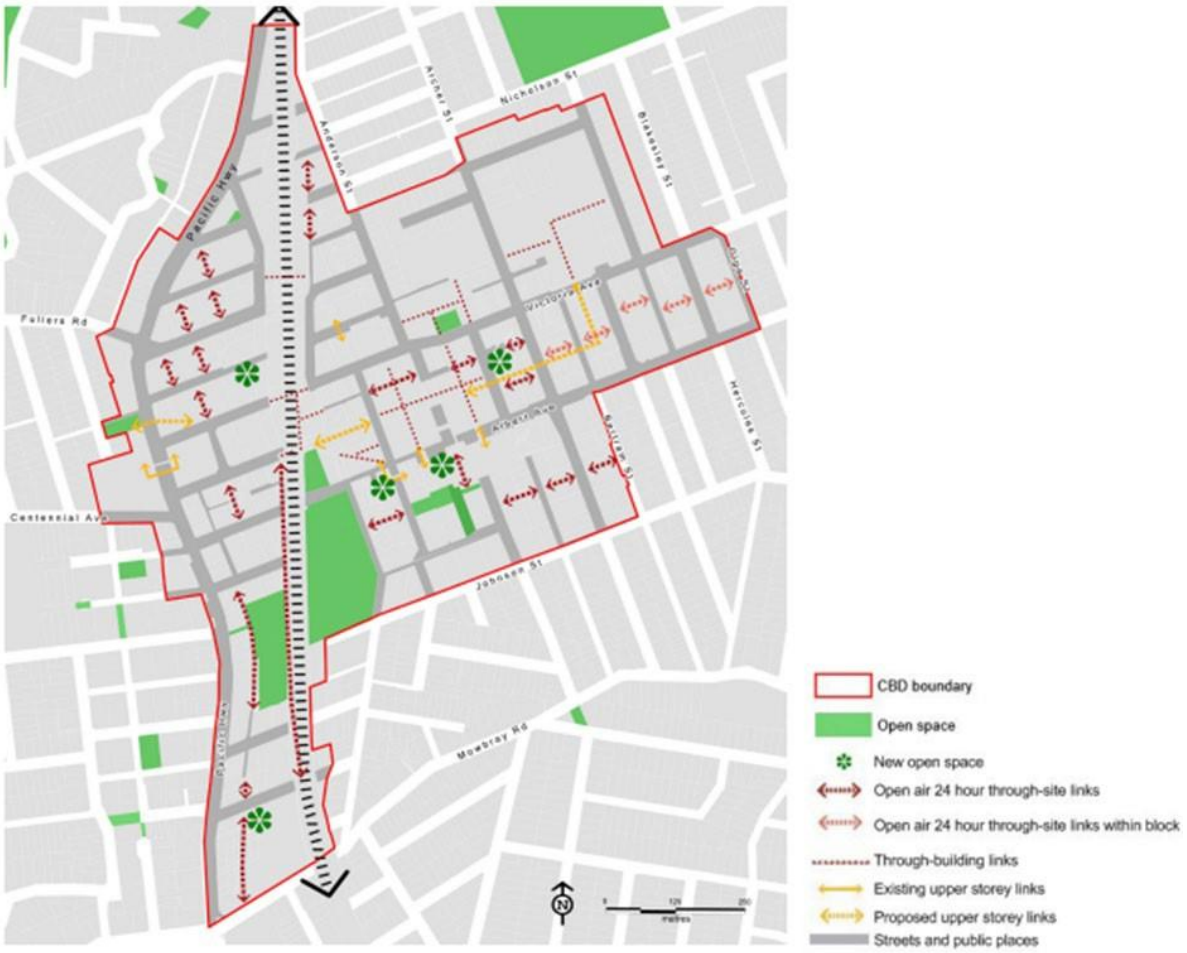
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. The rear landscape setback, along the railway (eastern) boundary of the site, should be unobstructed, along the entire length of the rear of the site and provide the opportunity for linking to a future public path to the north.
4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Pacific Highway and Wilson Street.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.

4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be from one access point in Wilson Street.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, are to be included in any future Development Application.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.7 3-5 Help Street Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 3-5 Help Street, Chatswood. The land is bounded by Help Street to the south, Cambridge Lane to the west, and McIntosh Street to the north as shown on the site map below.

Figure 28: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.

2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to Help Street, McIntosh Street and Cambridge Lane.

2. Built Form

Performance Criteria

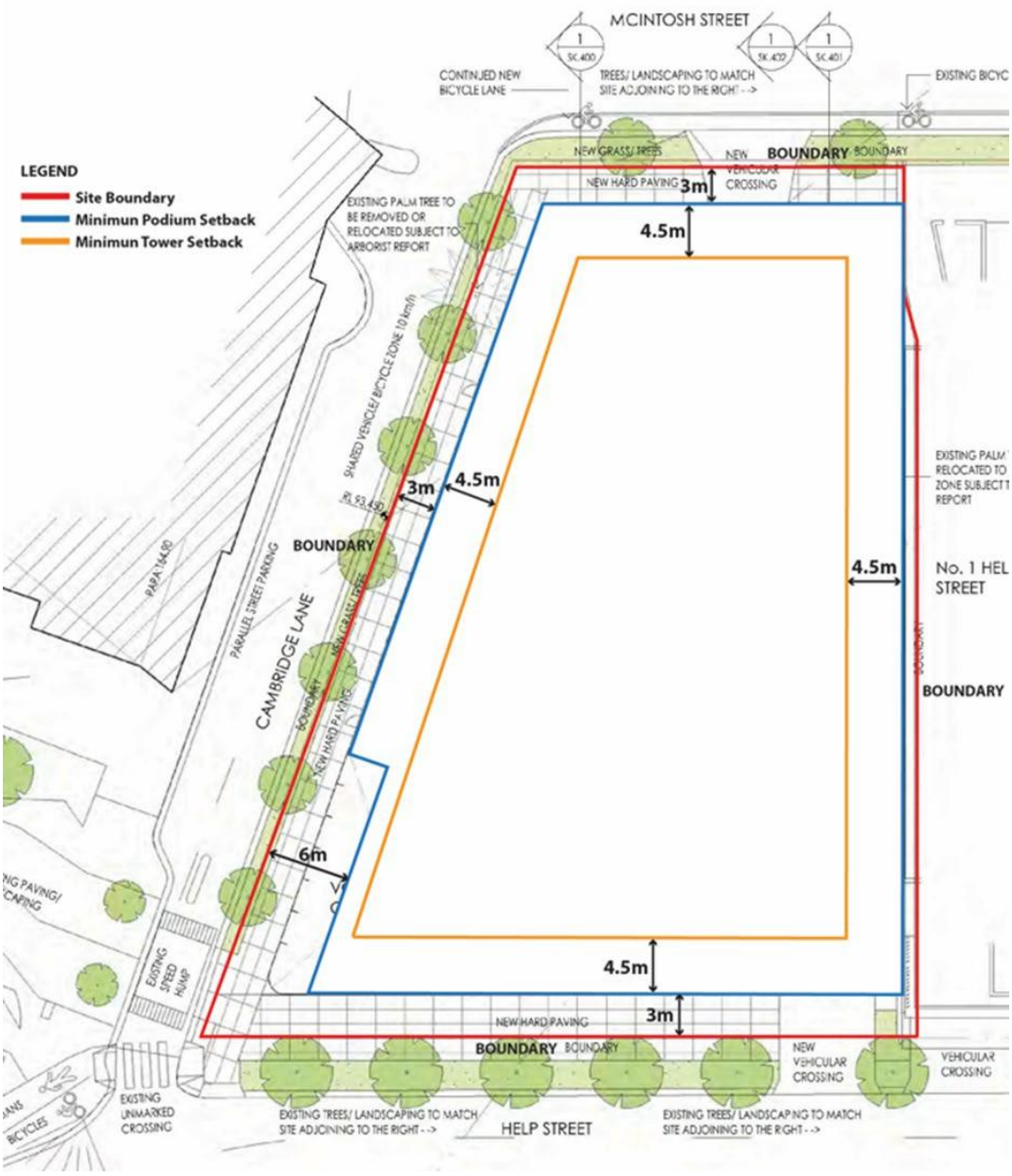
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides for a good level of amenity for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 29 subject to other requirements in this plan.

Figure 29: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.

2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level to deep soil areas, landscaping, and open space.
4. Contribute to slender tower forms.
5. Minimise the effects of adverse wind conditions at street level.

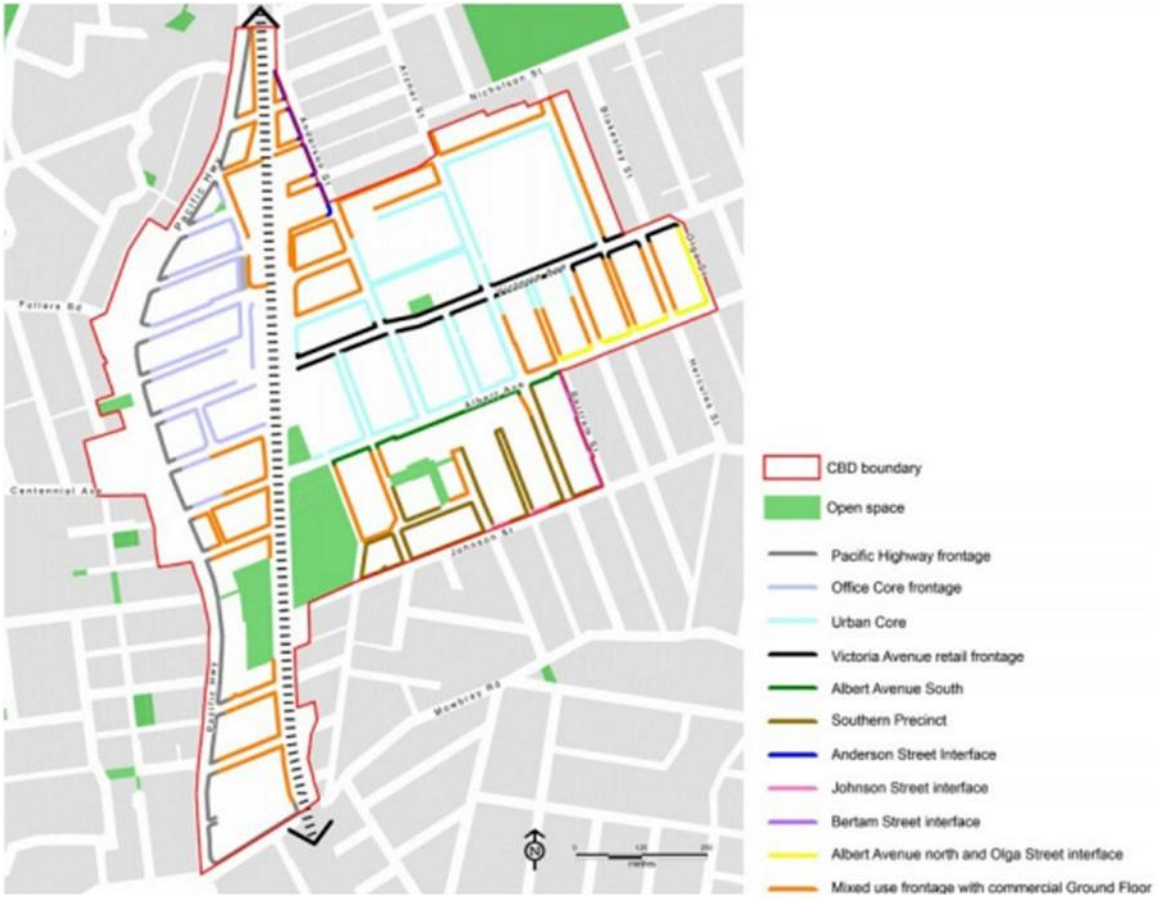
Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights) – except where additional setback is provided. Setbacks are as follows:
    a) Help Street, Macintosh Street, and Cambridge Lane frontages
        i) Minimum 3m setback at ground level from the street boundary to the podium street wall
        ii) 6-14m podium street wall height at the required setback
        iii) Minimum 7.5m setback from the street boundary to the tower
    b) Cambridge Lane
        i) Minimum 6m setback at ground level along the Cambridge boundary of the site, opposite the publicly accessibly open space at 1 Cambridge Street, for the purposes of publicly accessible open space. In this regard 3m is to comprise footpath and the remaining 3m is to comprise deep soil and landscaping planting.
    c) In regards the tower,
        i) Minimum 7.5m setback from the street boundary to the tower
        ii) Minimum 4.5m from any side (neighbouring) boundaries to the tower

Map 5: Setbacks and street frontage heights



2. In addition to Control 1:
   a) Setbacks may be greater and street wall heights may be lower.
   b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development, adjoining properties and properties on the other side of the street.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Help Street, McIntosh Street and Cambridge Lane.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be available as publicly accessible open space.
2. Tree planting must be provided along the Cambridge Lane frontage within the 6m setback area (3m being soft landscaping).
3. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
4. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
5. Public domain improvements shall be provided to all street frontages to Council requirements.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.

7. Deep soil planting is to be provided within the setback to Cambridge Lane. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
8. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

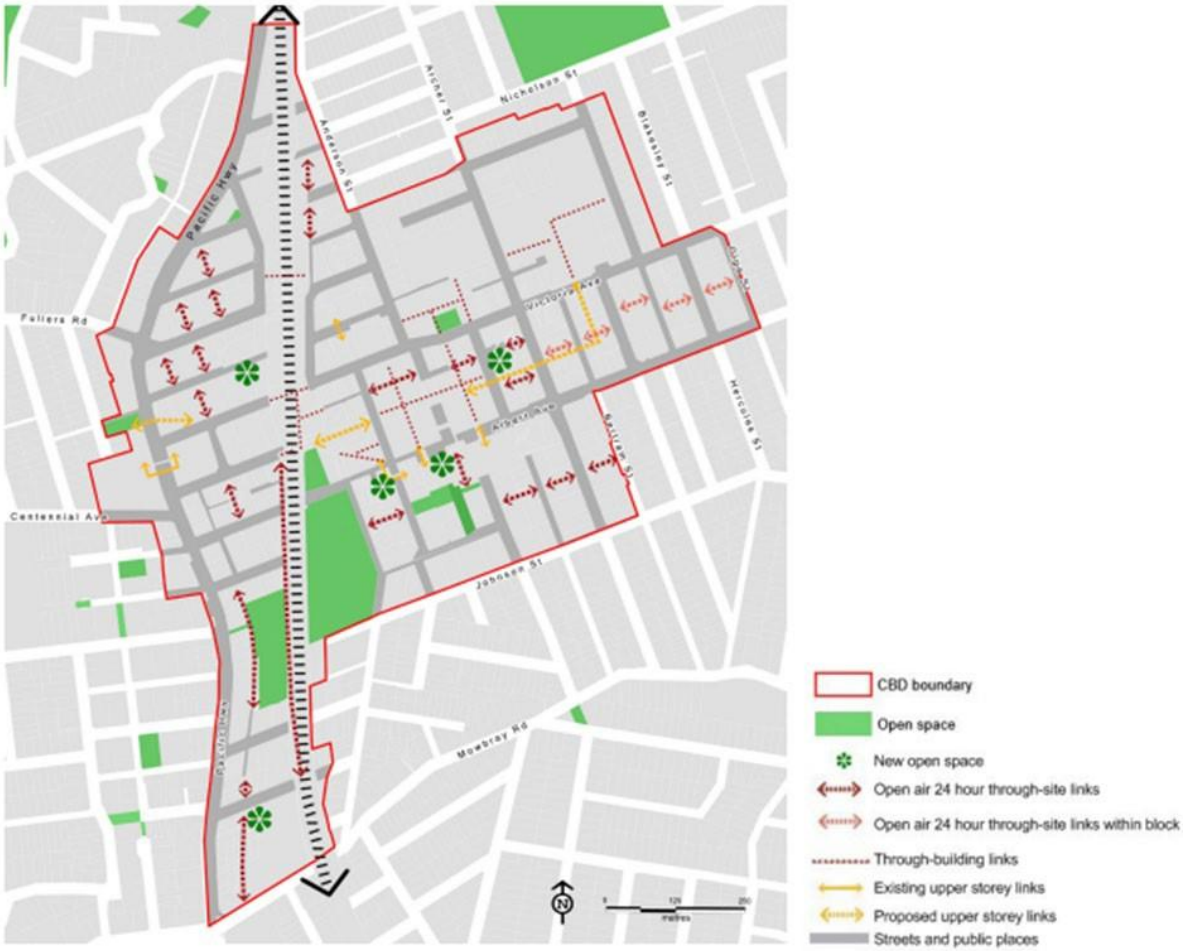
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. The ground level setback on Help Street, McIntosh Street and Cambridge Lane is to be unobstructed, with the only exception being ground level landscaping.
4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with an easement for public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Help Street, McIntosh Street and Cambridge Lane.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.

4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be minimised.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, are to be included in any future Development Application.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. Ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council’s Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.8 54-56 Anderson Street Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 54-56 Anderson Street, Chatswood.

Figure 30: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.

3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to Anderson Street, O'Brien Street and Wilson Street.

2. Built Form

Performance Criteria

The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 31.

Figure 31: Site Layout

3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.

3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Anderson Street, Wilson Street and O'Brien Street.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

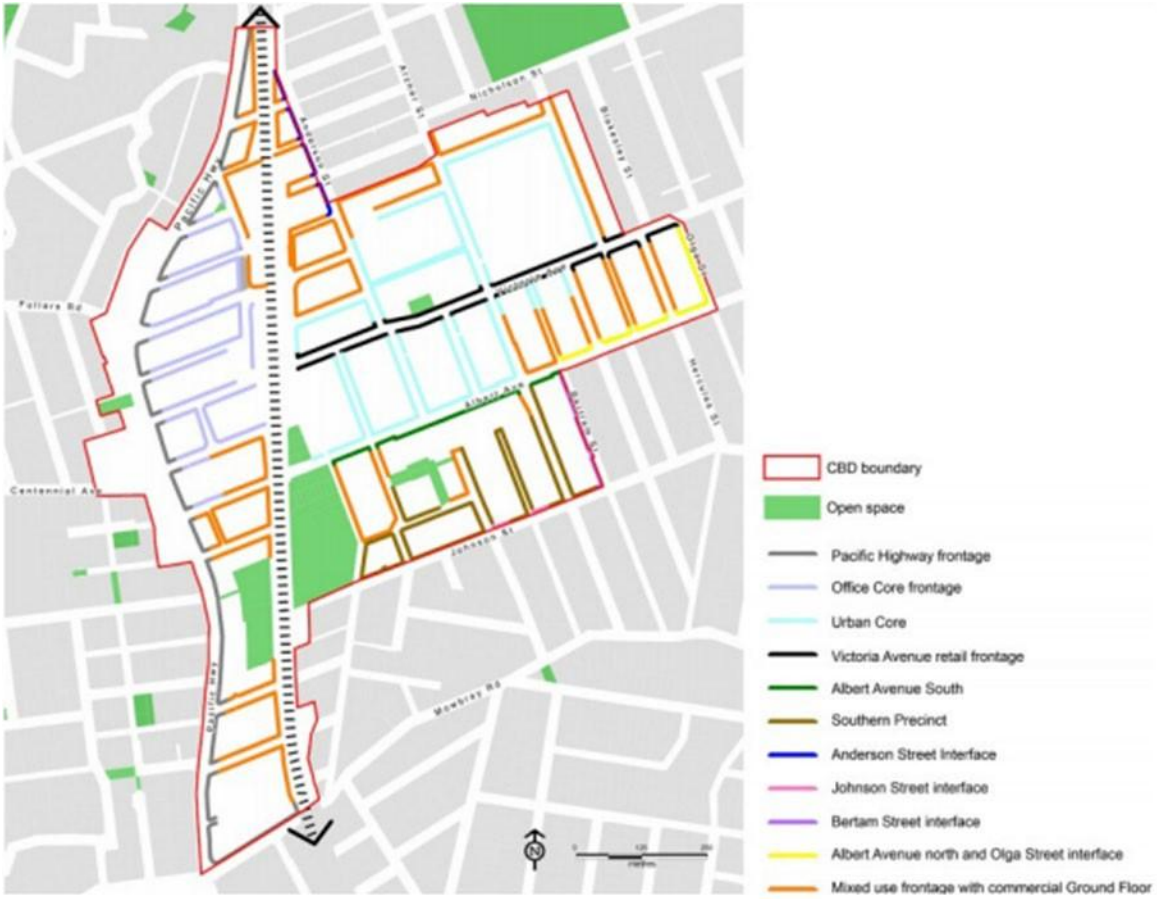
Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights). Setbacks are as follows:
    a) Anderson Street frontage
        i) Minimum 3m setback at ground level
        ii) Minimum 1m setback above street wall
        iii) 6-14 metre street wall height at front boundary (maximum two to four storeys).
        iv) In regards Point d) below, a minimum tower setback above street wall of 4.5m is required from the Anderson Street boundary where height is 90m.
    b) Wilson Street and O'Brien Street frontage:
        i) Minimum 3m setback at ground level
        ii) Minimum 3m setback above street wall
        iii) 6-14 metre street wall height at front boundary (maximum two to four storeys).
    c) North Shore Rail Line boundary
        i) Minimum 3m setback at ground level
        ii) Minimum 3m setback above street wall
        iii) 6-14 metre street wall height at front boundary (maximum two to four storeys).
    d) In regards the tower, a minimum of 1:20 ratio of the setback to building height above the podium from all boundaries (eg. tower to be setback a minimum of 2.65 metres above podium for a 53 metre high building, 4.5m for a 90m high building).

Map 5: Setbacks and street frontage heights



2. In addition to Control 1:
    a) Setbacks may be greater and street wall heights may be lower.
    b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including from the western side of the North Shore Rail Line.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Anderson Street, Wilson Street, O'Brien Street and the boundary with the North Shore Rail Line.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
3. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
4. Public domain improvements shall be provided to all street frontages to Council requirements.
5. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
6. Deep soil planting is to be provided within the 3m setbacks to Anderson Street, Wilson Street, O'Brien Street, and the North Shore Rail Line. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.

7. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
8. Large canopy tree planting must be provided along the Anderson Street, Wilson Street and O'Brien Street frontages within the 3m setback.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

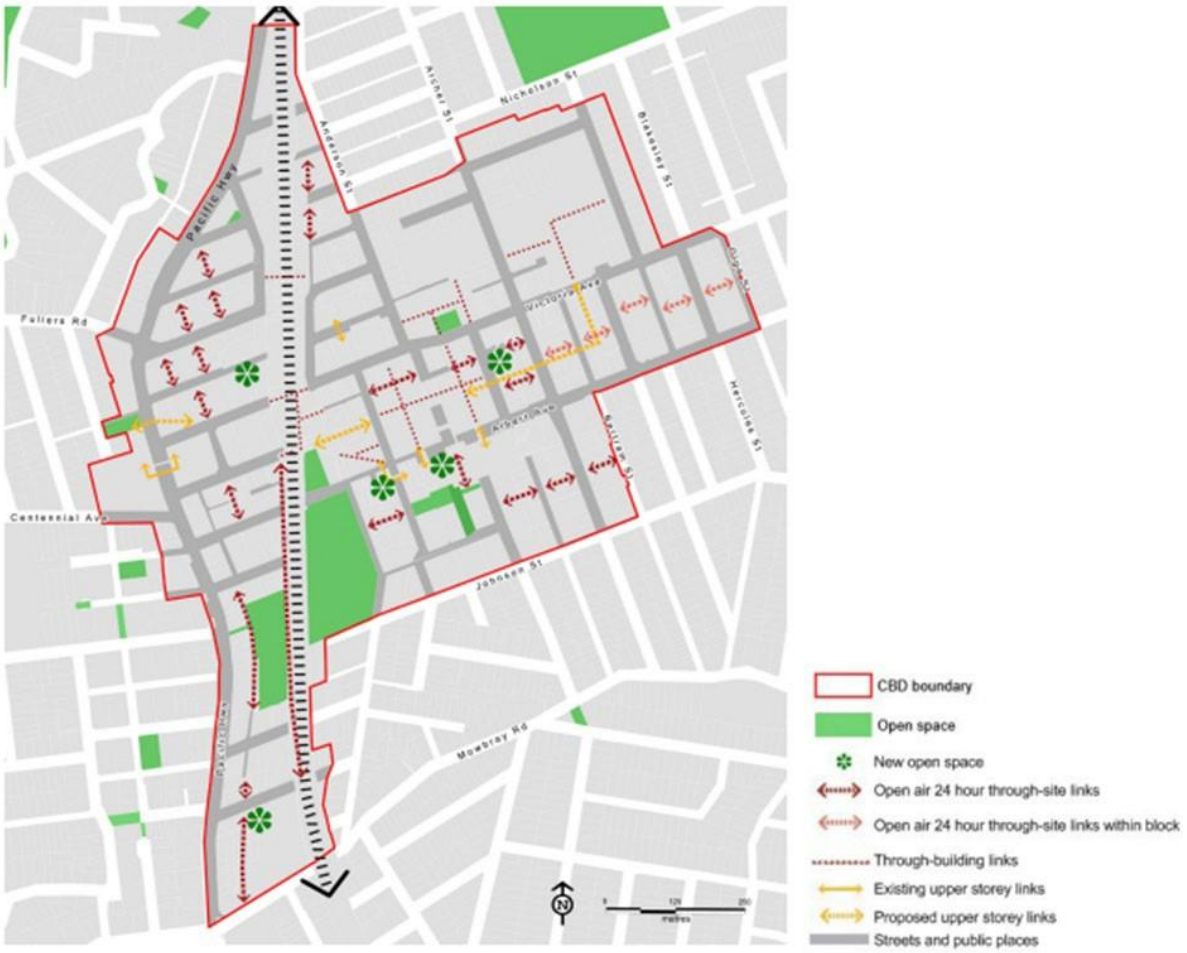
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below. In this regard a 3m wide publicly accessible through site link is required along the Boundary with the North Shore Rail Line, between Wilson Street and O'Brien Street.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. Public rights of way are to be provided on:
    - The 3m setback to Anderson Street.
    - The 3m setback to Wilson Street.
    - The 3m setback to O'Brien Street.
    - The 3m setback to the North Shore Rail Line.
4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Anderson Street, Wilson Street and O'Brien Street.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.

4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be from one access point in O'Brien Street.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    - Car parking provision is to be consistent with the requirements of SEPP 65 / constrained parking rates for new developments in the Chatswood CBD as supported by TfNSW.
    - A minimum of 1 secure bicycle parking space per apartment.
    - A minimum of 1 secure bicycle space per 100m² of non-residential /retail floor space.
    - Adequate end of trip facilities including lockers, showers, etc. for use by non-residential and retail tenants.
    - A bicycle rack within the site boundary for use by retail customers.
    - A minimum of three (3) freight and service vehicle spaces within the basement, in addition to the one (1) Medium Rigid Vehicle (MRV) space proposed within basement level 1 loading dock.
    - Swept path analysis for simultaneous MRV access and egress at the entry to the basement ramp to demonstrate that MRVs can safely pass each other.
    - Access arrangements and the operation of basement 1 to be amended to remove conflicts between vehicles accessing and egressing the site.
    - Evidence of consultation between the proponent and Sydney Trains, and any conclusions, to ensure that all relevant Sydney Trains matters of consideration are taken into account and are incorporated in the future design of the development.
    - Evidence of consultation with TfNSW and service providers in regards any impacts to the bus stop in front of 54 Anderson Street and the no stopping zone on Anderson Street, during construction or in end state.
    - A Traffic Management Plan for the construction phase, and future operation phase, to demonstrate that additional vehicular movements in and out of the site do not pose queuing issues along the Wilson Street overbridge, and potentially obstructing rail corridor access.
    - Updated traffic analysis and modelling.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

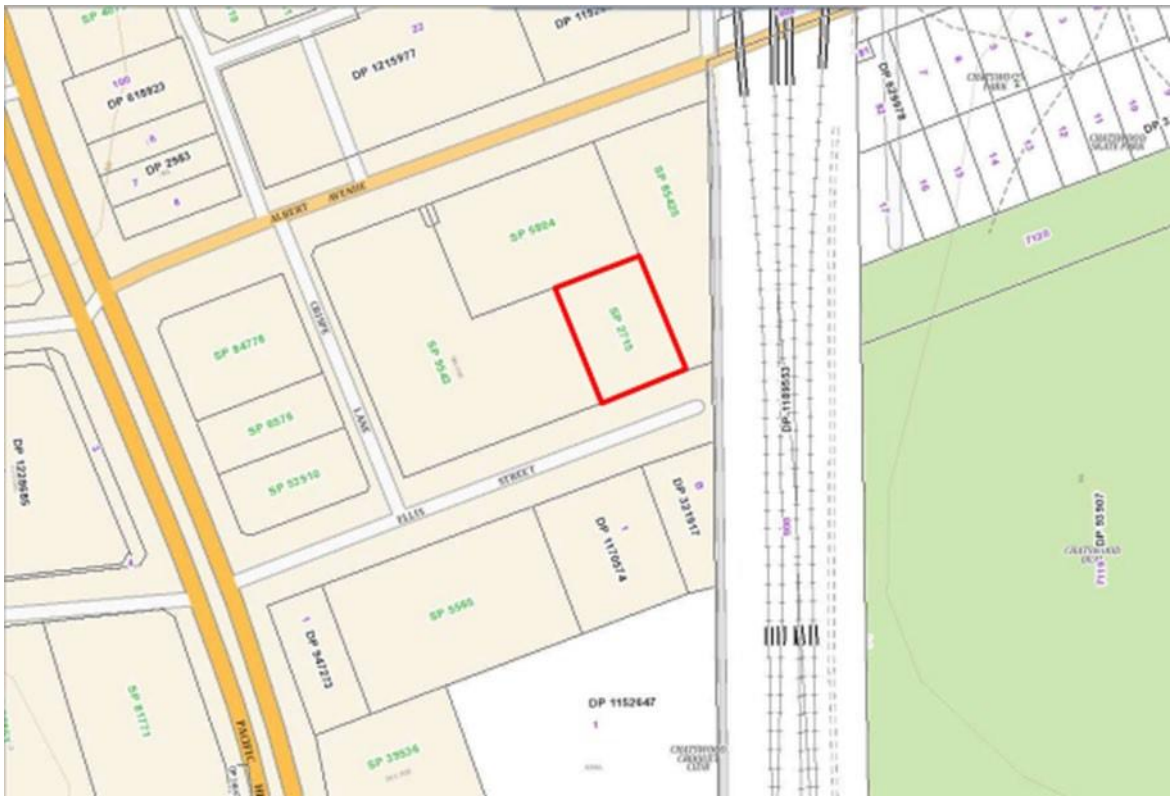
1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.9 3 Ellis Street Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 3 Ellis Street, Chatswood.

Figure 32: Site Map



Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maintains sun access to key public spaces.

2. Built Form

Performance Criteria

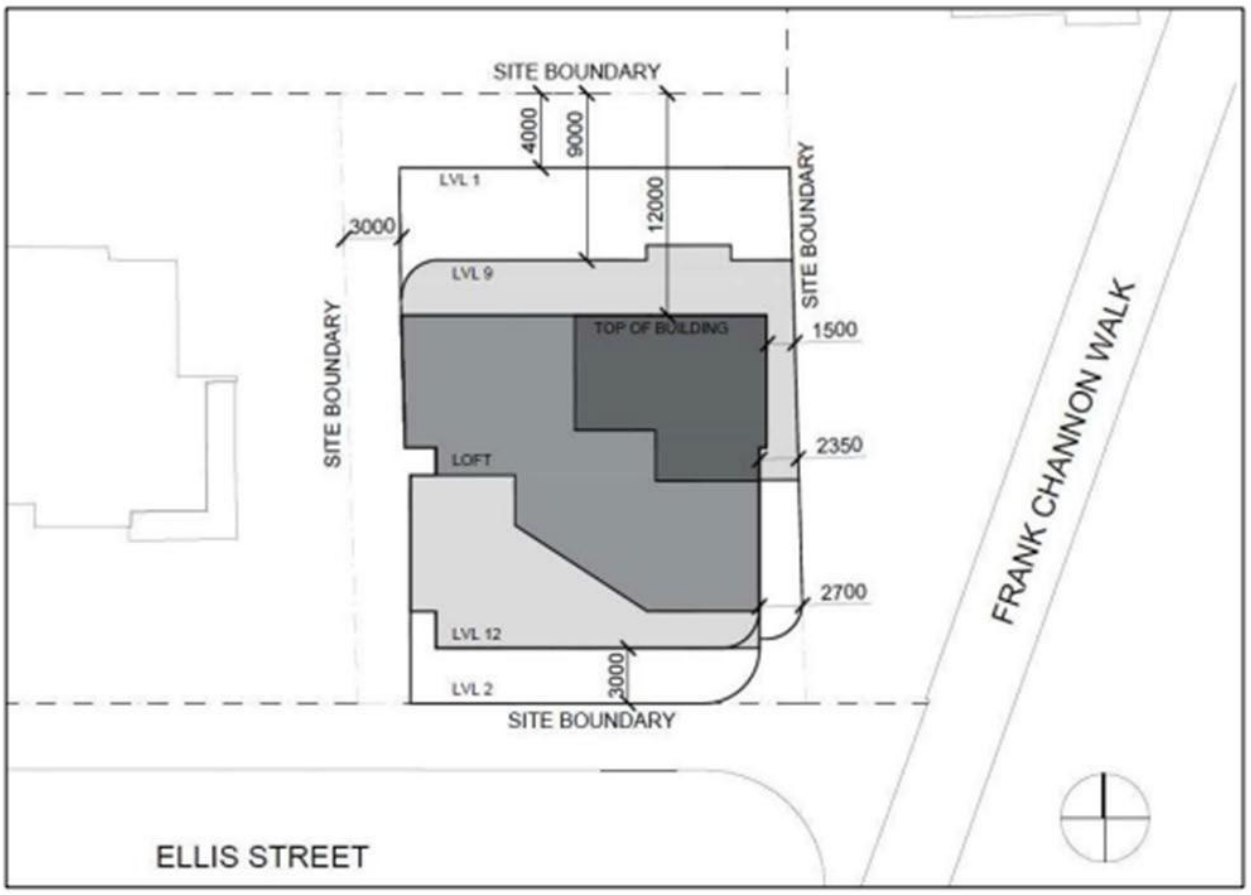
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 33.

Figure 33: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Contribute to slender tower forms.
5. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

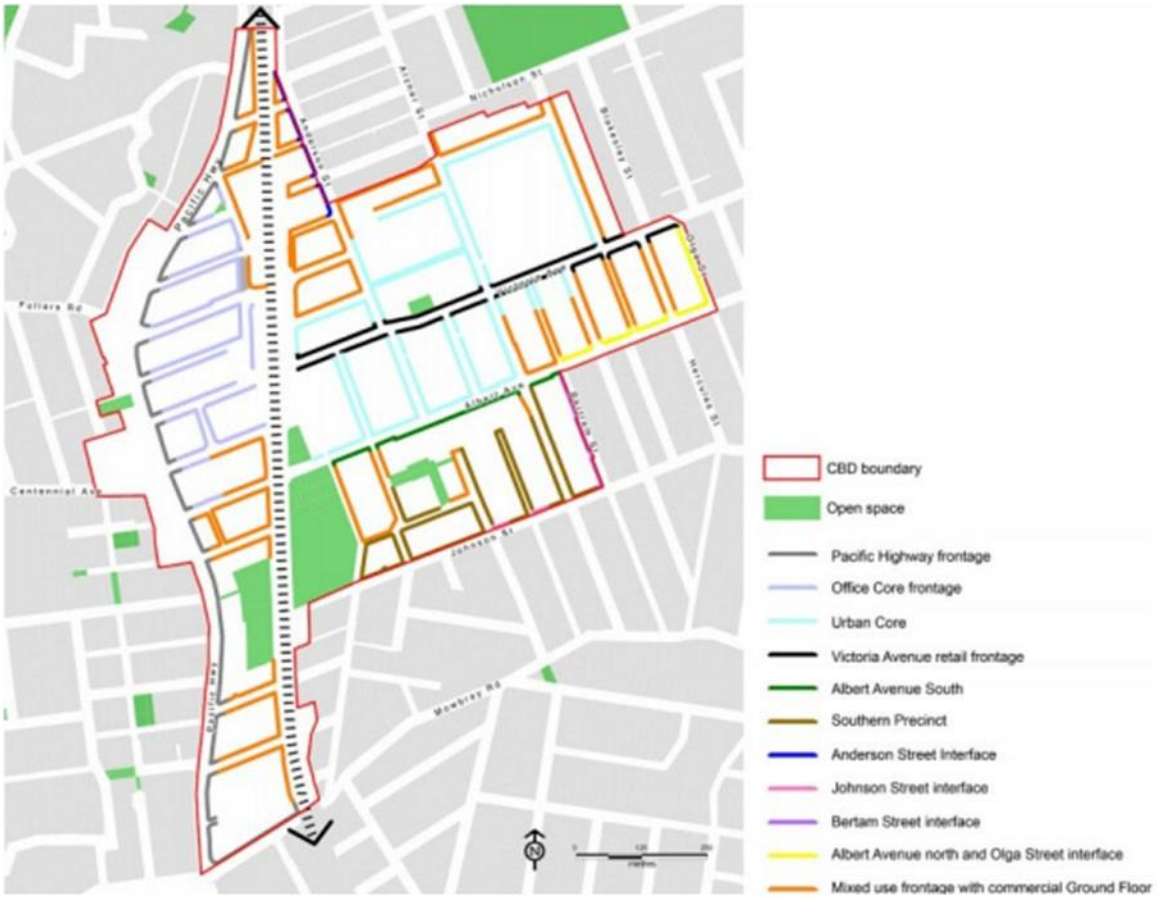
Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights). Setbacks are as follows:
    a) Ellis Street Frontage:
        i) Minimum 2 metre setback at ground level to front boundary
        ii) 6-14 metre street wall height (maximum two to four storeys).
        iii) Minimum 3 metre setback above street wall.
    b) Western boundary:
        i) Minimum 3 metre setback at ground level to western boundary.

c) In regards the tower, a minimum of 1:20 ratio of the setback to building height above the podium.

(eg. tower to be setback 2.2 metres above podium for a 44 metre high building)

Map 5: Setbacks and street frontage height



2. In addition to Control 1:
   a) Setbacks may be greater and street wall heights may be lower.
   b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including the Frank Channon Walk.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.

2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Ellis Street, neighbouring properties and the Frank Channon Walk.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
3. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
4. Public domain improvements shall be provided to all street frontages to Council requirements.
5. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
6. Deep soil planting is to be provided within the 2m setback to Ellis Street and the 3m setback to the western boundary. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.

7. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
8. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
9. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

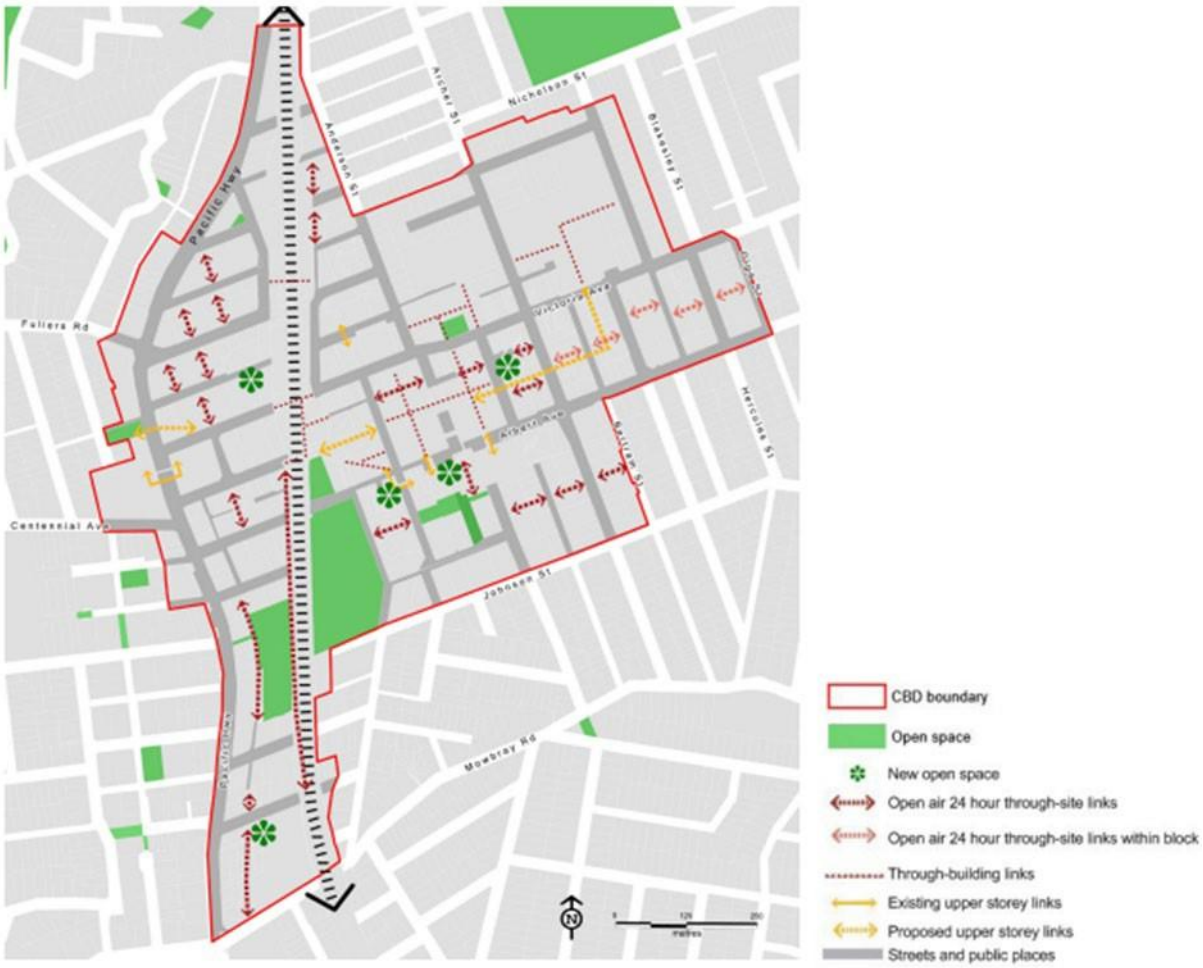
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. A public right of way is to be provided on the 2m setback to Ellis Street and the 3m setback to the western boundary.
4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Ellis Street.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.

4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be from one access point in Ellis Street.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. Car parking provision is to be consistent with the requirements of SEPP 65/constrained parking rates for new developments in the Chatswood CBD as supported by TfNSW.
6. A minimum of 1 secure bicycle parking space per apartment.
7. A minimum of 1 secure bicycle space per 100m² of non-residential /retail floor space.
8. Adequate end of trip facilities including lockers, showers, etc. for use by non-residential and retail tenants.
9. A bicycle rack within the site boundary for use by retail customers.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35mor more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.10 44-52 Anderson Street Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 44-52 Anderson Street, Chatswood.

Figure 34: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to Anderson Street, O'Brien Street and Day Street.

2. Built Form

Performance Criteria

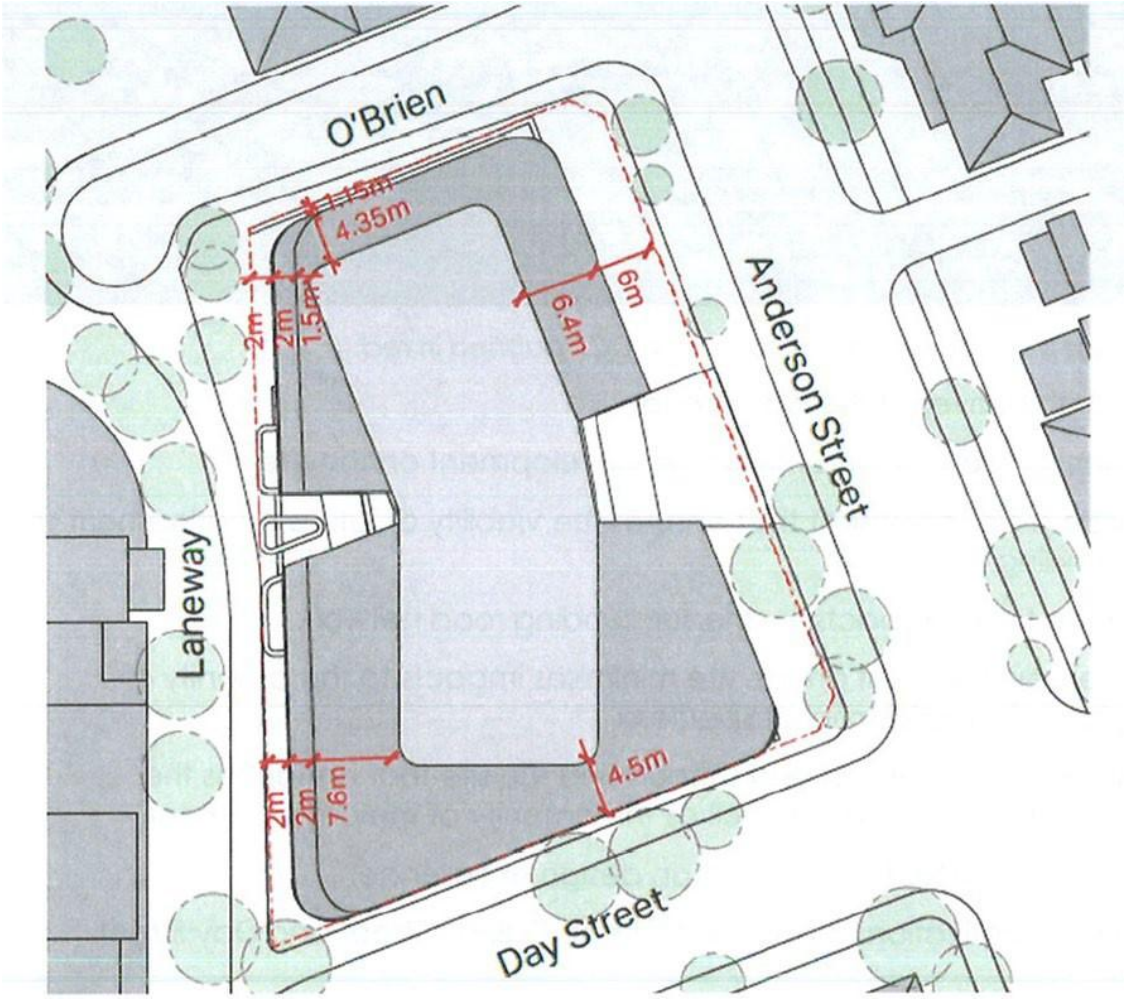
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 35.

Figure 35: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Anderson Street, O'Brien Street and Day Street.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

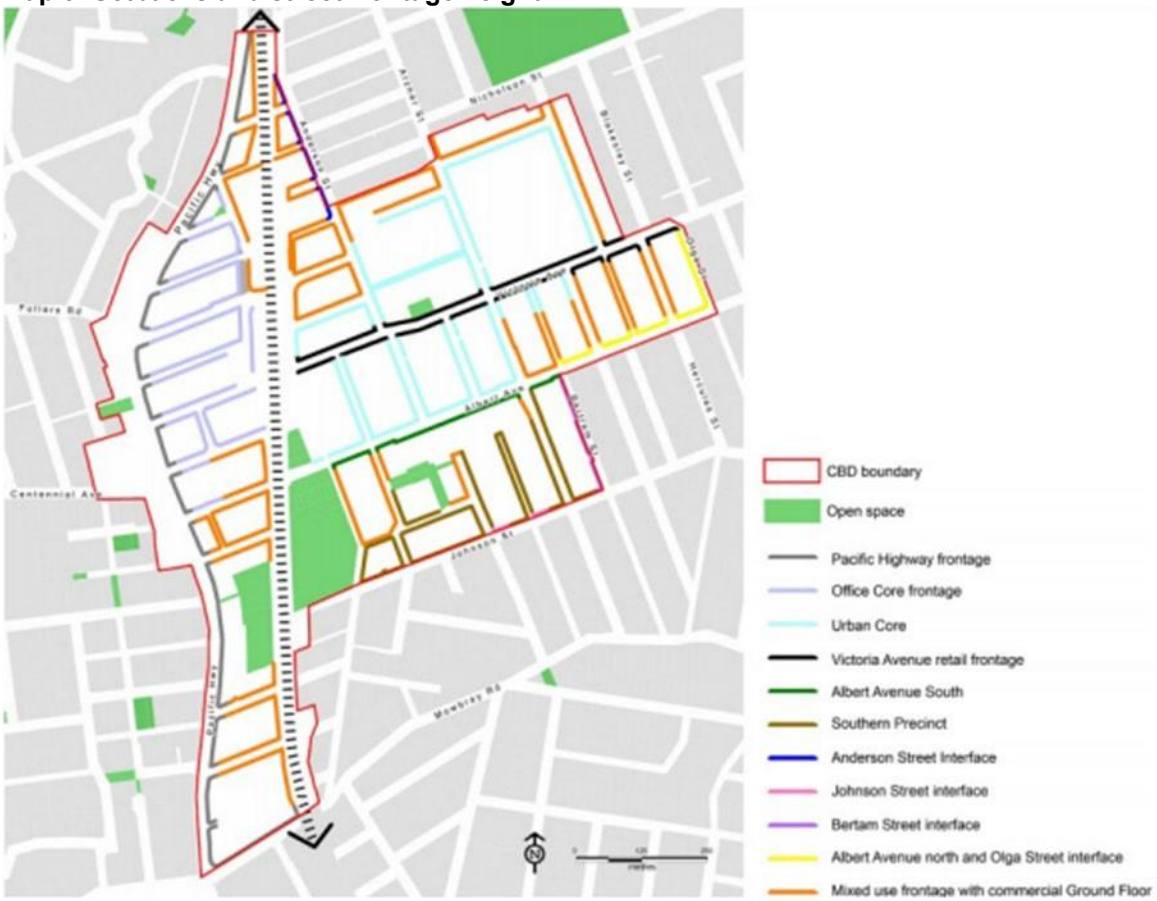
Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights) and the concept plans. Setbacks are as follows:
    1. Anderson Street frontage:
        1. Minimum 6m setback at ground Level
        2. Minimum 6.4m setback above street wall
        3. Maximum street wall height of 7m or two storeys.
    2. O'Brien Street frontage:

1. Minimum 1.15m setback at ground Level
2. Minimum 4.35m setback above street wall
3. Maximum street wall height of 7m or two storeys.

3. Day Street frontage:
    1. Nil setback at ground Level
    2. Minimum 4.5m above street wall
    3. Maximum street wall height of 7m or two storeys.
4. Western (rear publicly accessible pathway) boundary
    1. Minimum 2m setback at ground Level, with additional stepped 2m setback at Podium Level
    2. Minimum 7.6m setback above street wall
    3. Maximum wall height to rear pathway of 7m or two storeys.

Map 5: Setbacks and street frontage height



2. In addition to Control 1:
    a) Setbacks may be greater and street wall heights may be lower.
    b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including the eastern side of Anderson Street and the publicly accessible pathway to the western boundary.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Anderson Street, O’Brien Street, Day Street and the western boundary with the publicly accessible pathway.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
3. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
4. Public domain improvements shall be provided to all street frontages to Council requirements.
5. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
6. Deep soil planting is to be provided within the 6m setback to Anderson Street. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
7. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
8. Large canopy tree planting must be provided along the Anderson Street frontage within the 3m setback.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

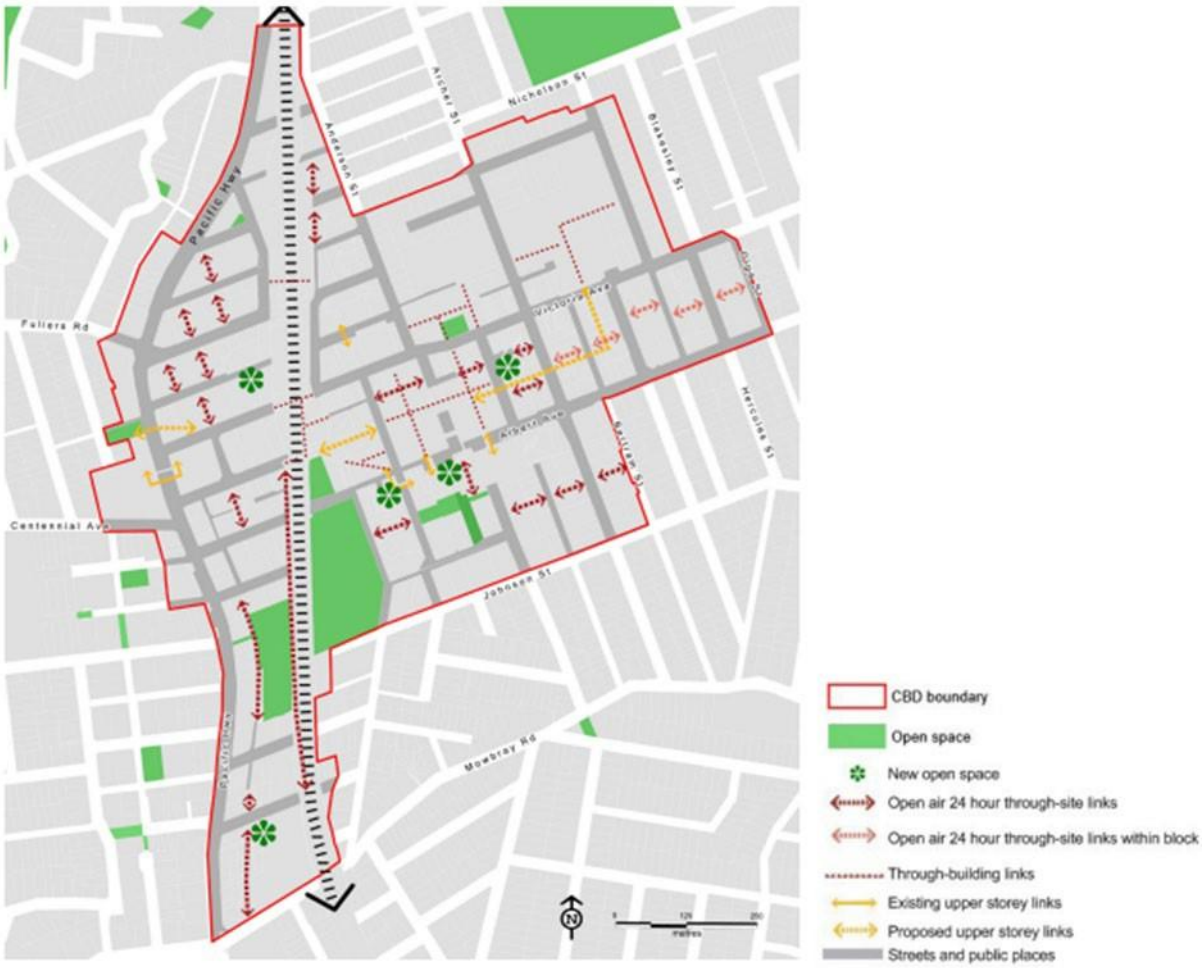
Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below. In this regard a 2m wide publicly accessible through site link is required along the western boundary, adjacent the existing publicly accessible pathway. This space is to function as a landscape embellishment to the existing publicly accessible pathway.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. Public rights of way are to be provided on:
    - The 6m setback to Anderson Street

- The 1.15m setback to O'Brien Street
- The 2m setback to the western boundary with the adjacent existing publicly accessible pathway.

4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Anderson Street, O'Brien Street and Day Street.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be from one access point in Day Street.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    1. Car parking provision based on a reduced car parking rate, consistent with the requirements of SEPP65 / constrained parking rates for new developments in the Chatswood CBD as supported by Transport for NSW (TfNSW).
    2. A minimum of 1 secure bicycle parking space per apartment.
    3. A minimum of 1 secure bicycle space per 100m² of non-residential /retail floor space.
    4. Adequate end of trip facilities including lockers, showers, etc. for use by non-residential and retail tenants.
    5. A bicycle rack within the site boundary for use by retail customers.
    6. A minimum of three (3) freight and service vehicle spaces within the basement, in addition to the one (1) Medium Rigid Vehicle (MRV) space proposed within basement level 1 loading dock.
    7. Evidence of consultation between the proponent and Sydney Trains, and any conclusions, to ensure that all relevant Sydney Trains matters of consideration are taken into account and are incorporated in the future design of the development.
    8. A Green Travel Plan.
    9. Updated traffic analysis and modelling.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.11 613-627 Pacific Highway Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 613, and 621-627 Pacific Highway, Chatswood.

Figure 36: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to Pacific Highway, Nelson Street and Hammond Lane.

2. Built Form

Performance Criteria

The built form of the new development shall:

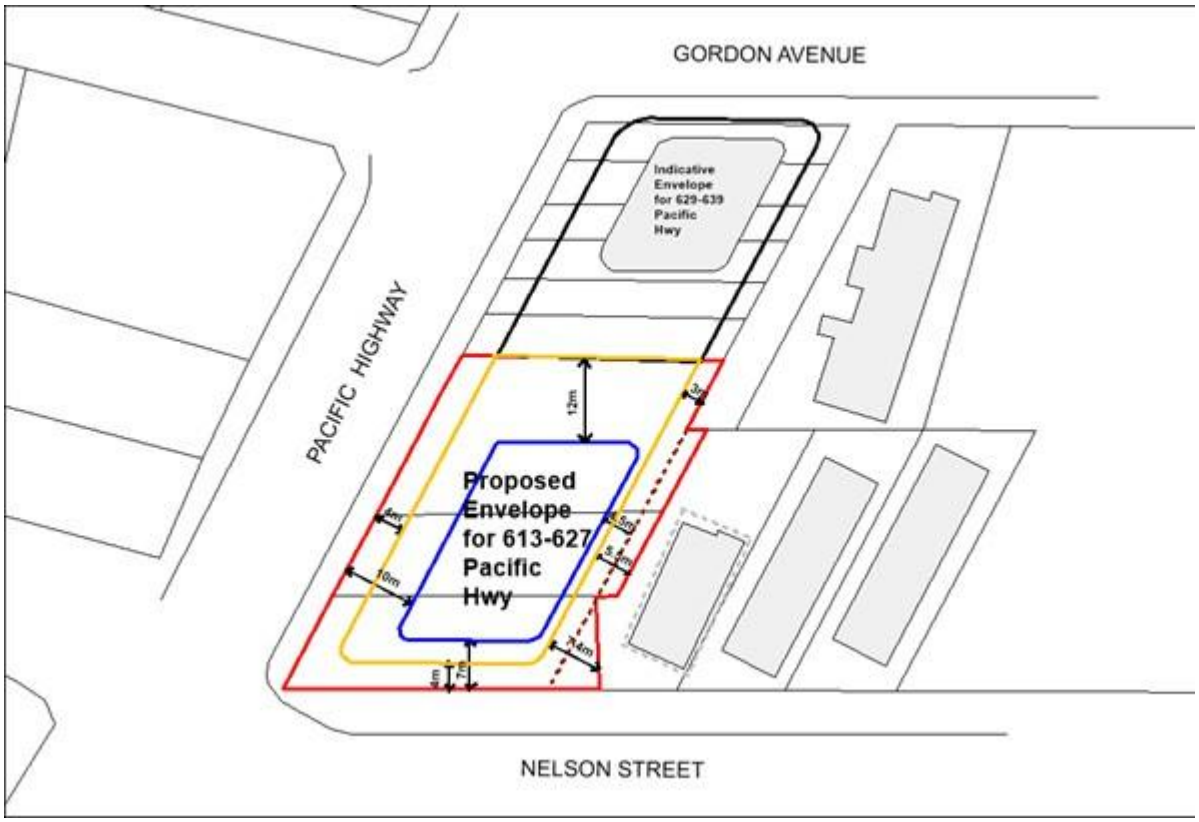
1. Achieve a slender tower form on the site.

2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 37.

Figure 37: Site Layout



____	Red Line	Site boundary
____	Yellow Line	Podium
____	Blue Line	Tower

3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Pacific Highway and Nelson Street.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights) – except where additional setback is provided. Setbacks are as follows:
    a) Pacific Highway frontage
        i) Minimum 4m setback at ground Level
        ii) Minimum 6m setback above street wall
        iii) Maximum street wall height of 7m or two storeys.

b) Nelson Street frontage:

i) Minimum 4m setback at ground Level

ii) Minimum 3m setback above street wall

iii) Maximum street wall height of 6-14m (two to four storeys).

c) Eastern boundary (from Hammond Lane to Nelson Street

i) Minimum 3m to 7.4m setback at ground Level

ii) Minimum 1.5m setback above eastern podium wall

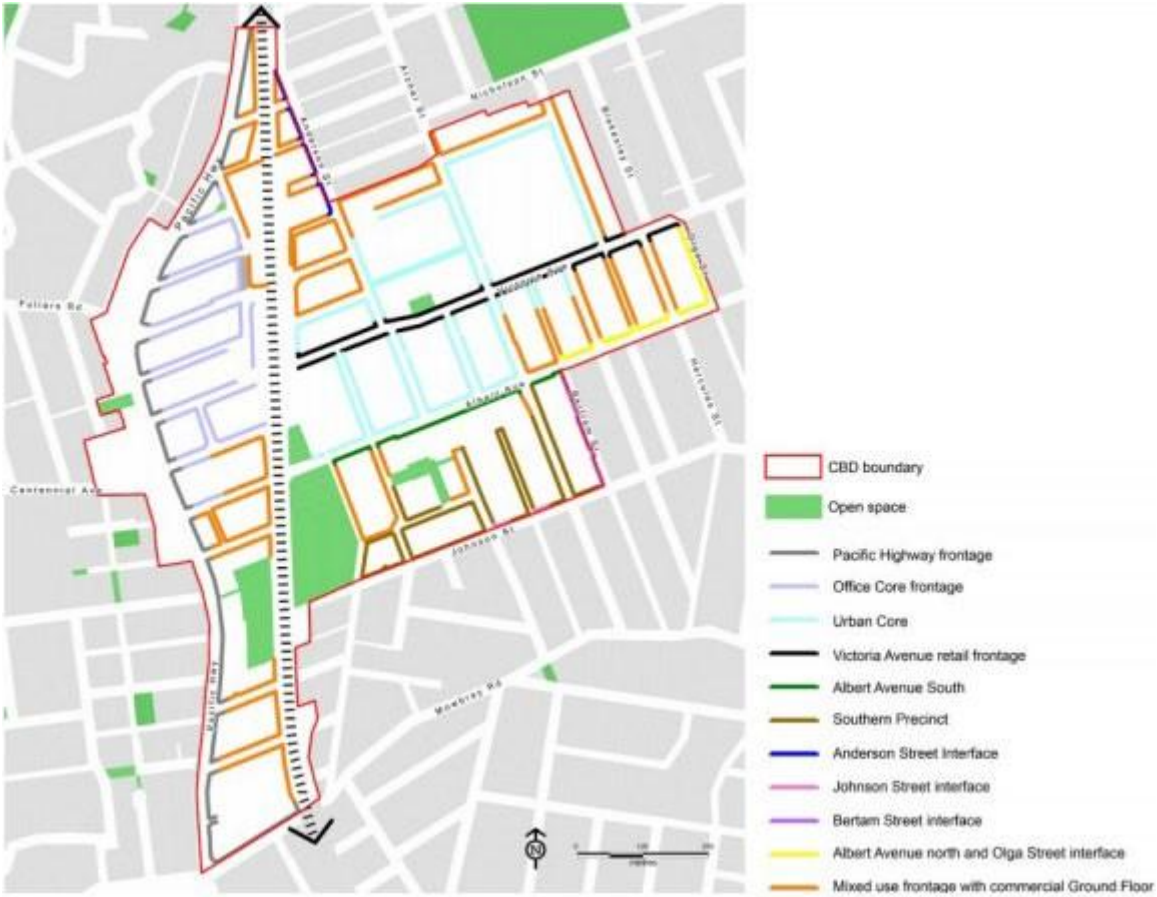
iii) Maximum street wall height of 6-14m (two to four storeys).

d) Northern boundary with adjoining property

i) Nil setback at ground Level

ii) Minimum 12m setback to tower

Map 5: **Setbacks and street frontage heights**



2. In addition to Control 1:

a) Setbacks may be greater and street wall heights may be lower.

b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including Pacific Highway, Nelson Street and Hammond Lane.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Pacific Highway, Nelson Street and Hammond Lane.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. Large canopy tree planting must be provided along the Pacific Highway frontage and Nelson Street frontage within the 4m setback area.
3. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
4. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
5. Public domain improvements shall be provided to all street frontages to Council requirements.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
7. Deep soil planting is to be provided within the 4m setback to Pacific Highway. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
8. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

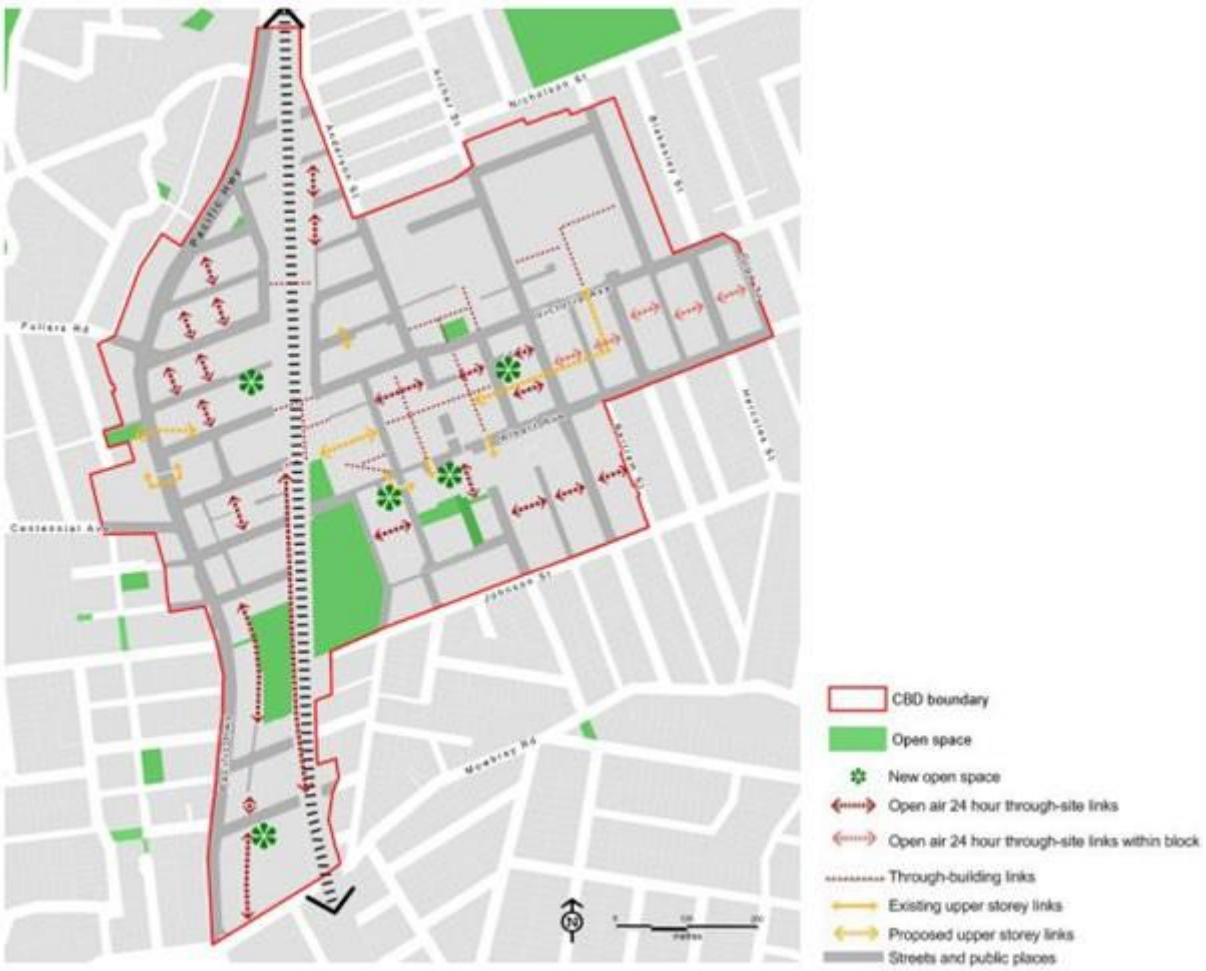
Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. Public rights of way are to be provided on:
    - The 4m setback to Pacific Highway.
    - The 4m setback to Nelson Street.

- The 3m to 7.4m setback along the eastern boundary (from Hammond Lane to Nelson Street) – being the through site link.

4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.
5. The through site link is to be integrated with the Hammond Lane setback at 629-637 Pacific Highway, Chatswood.

Map 4: **Through Site Links and Open Space**



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Pacific Highway, Nelson Street and Hammond Lane.

2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle and loading access is to be reviewed and master planned in conjunction with the adjoining site at 629-637 Pacific Highway, Chatswood. One vehicle and loading point in Hammond Lane is preferred for vehicle accessing and servicing of 613-637 Pacific Highway and 629-637 Pacific Highway, via a consolidated basement.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    a) Car parking provision based on reduced car parking rates, consistent with the requirements for new developments in the Chatswood CBD as supported by Transport for NSW.
    b) A minimum of 1 secure bicycle parking space per apartment.
    c) A minimum of 1 secure bicycle space per 100m² of non-residential /retail floor space.
    d) Adequate end of trip facilities including lockers, showers, etc. for use by non-residential and retail tenants.
    e) A bicycle rack within the site boundary for use by retail customers.
    f) A minimum of three (3) freight and service vehicle spaces within the basement, in addition to the one (1) Medium Rigid Vehicle (MRV) space proposed within the loading dock.
    g) A Green Travel Plan.
    h) Updated traffic analysis and modelling.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.12 629-637 Pacific Highway Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 629-637 Pacific Highway, Chatswood.

Figure 38: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to Pacific Highway, Gordon Avenue and Hammond Lane.

2. Built Form

Performance Criteria

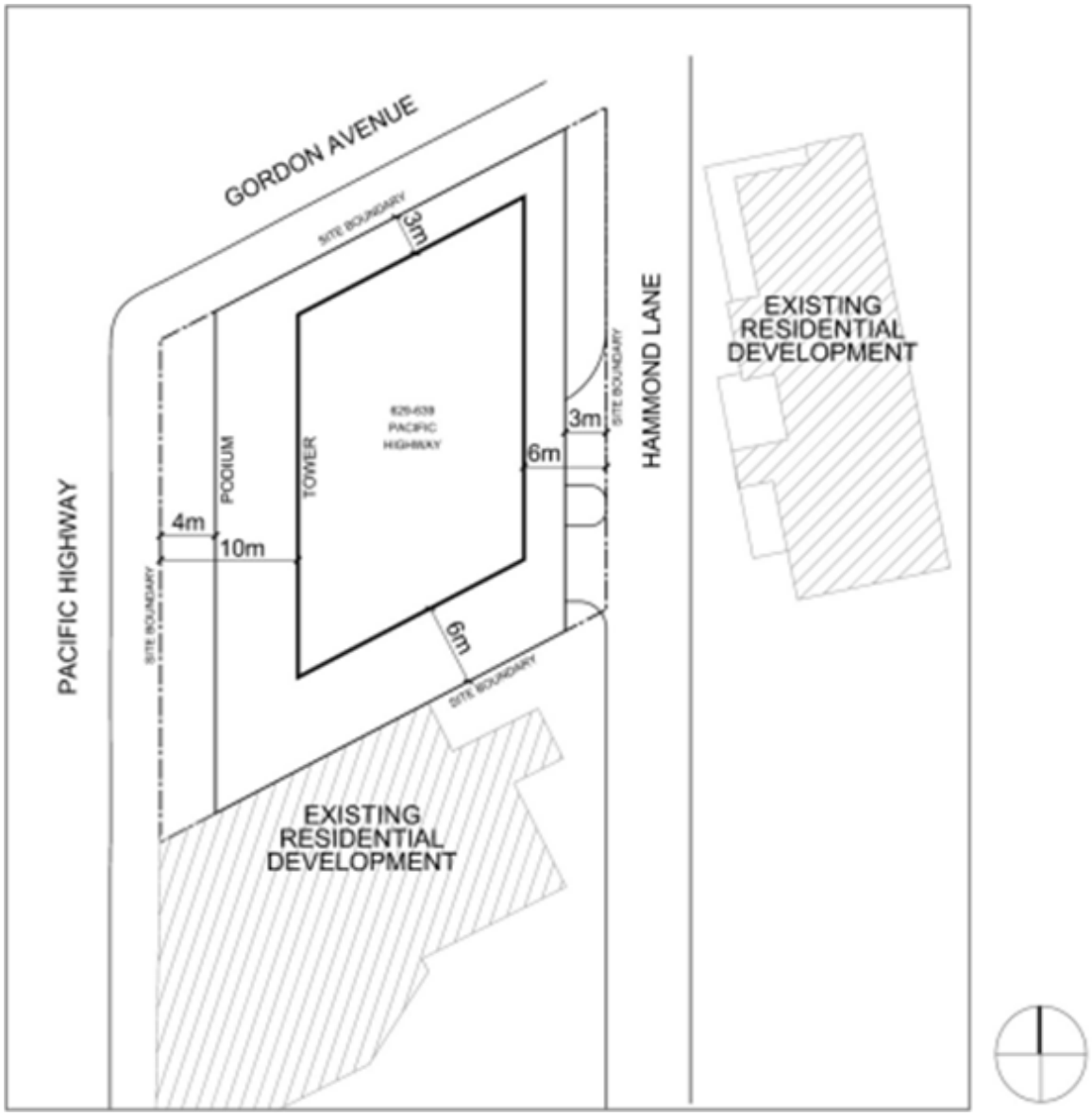
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 39.

Figure 39: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Pacific Highway and Gordon Avenue.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights) – except where additional setback is provided. Setbacks are as follows:
    a) Pacific Highway frontage
        i) Minimum 4m setback at ground Level
        ii) Minimum 6m setback above street wall
        iii) Maximum street wall height of 7m or two storeys.
    b) Gordon Avenue frontage:

i) Nil setback at ground Level

ii) Minimum 3m setback above street wall

iii) Maximum street wall height of 6-14m (two to four storeys).

c) Hammond Lane frontage

i) Minimum 3m setback at ground Level

ii) Minimum 6m setback above street wall

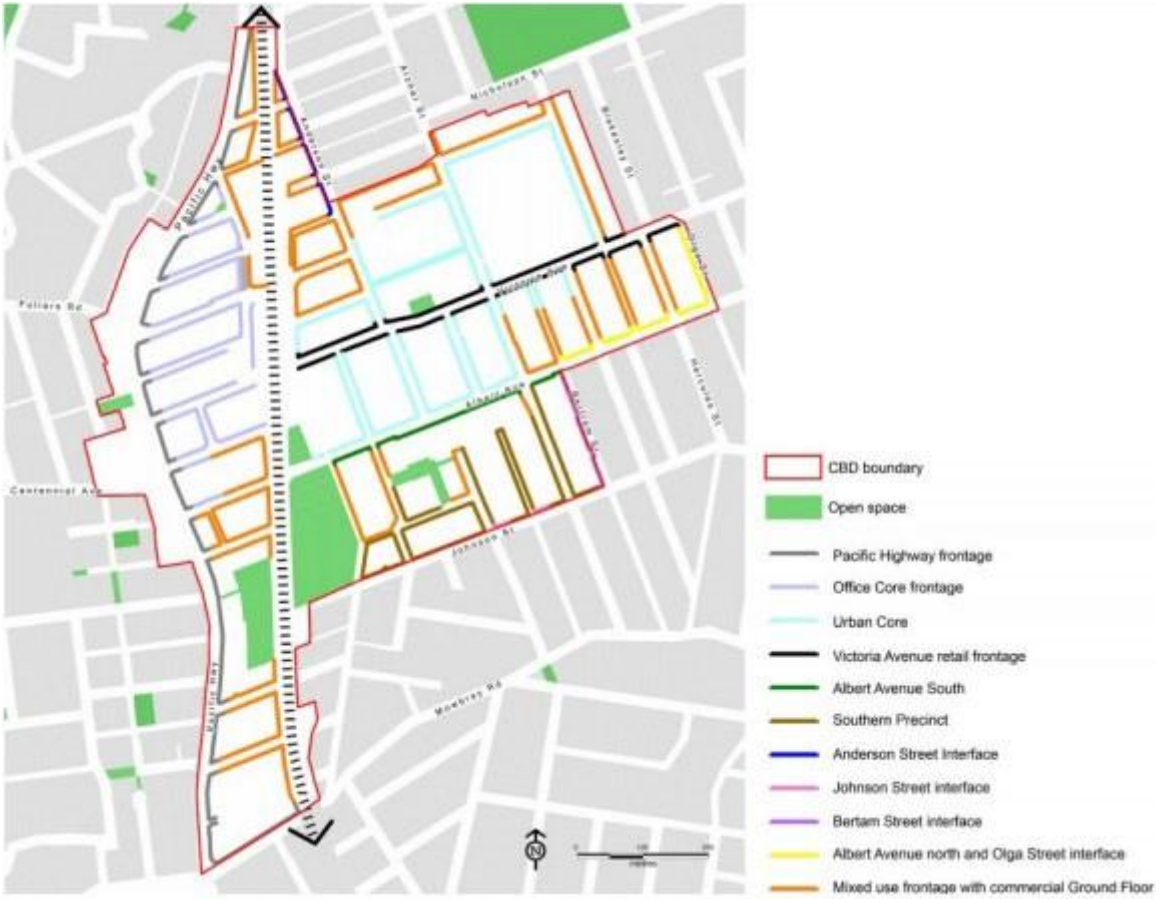
iii) Maximum street wall height of 6-14m (two to four storeys).

d) Southern boundary with adjoining property

i) Nil setback at ground Level

ii) Minimum 6m setback above street wall

Map 5: Setbacks and street frontage heights



2. In addition to Control 1:

a) Setbacks may be greater and street wall heights may be lower.

b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including Pacific Highway, Gordon Avenue and Hammond Lane.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Pacific Highway, Gordon Avenue and Hammond Lane.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. Large canopy tree planting must be provided along the Pacific Highway frontage within the 4m setback area.
3. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
4. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
5. Public domain improvements shall be provided to all street frontages to Council requirements.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
7. Deep soil planting is to be provided within the 4m setback to Pacific Highway. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
8. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

Performance Criteria

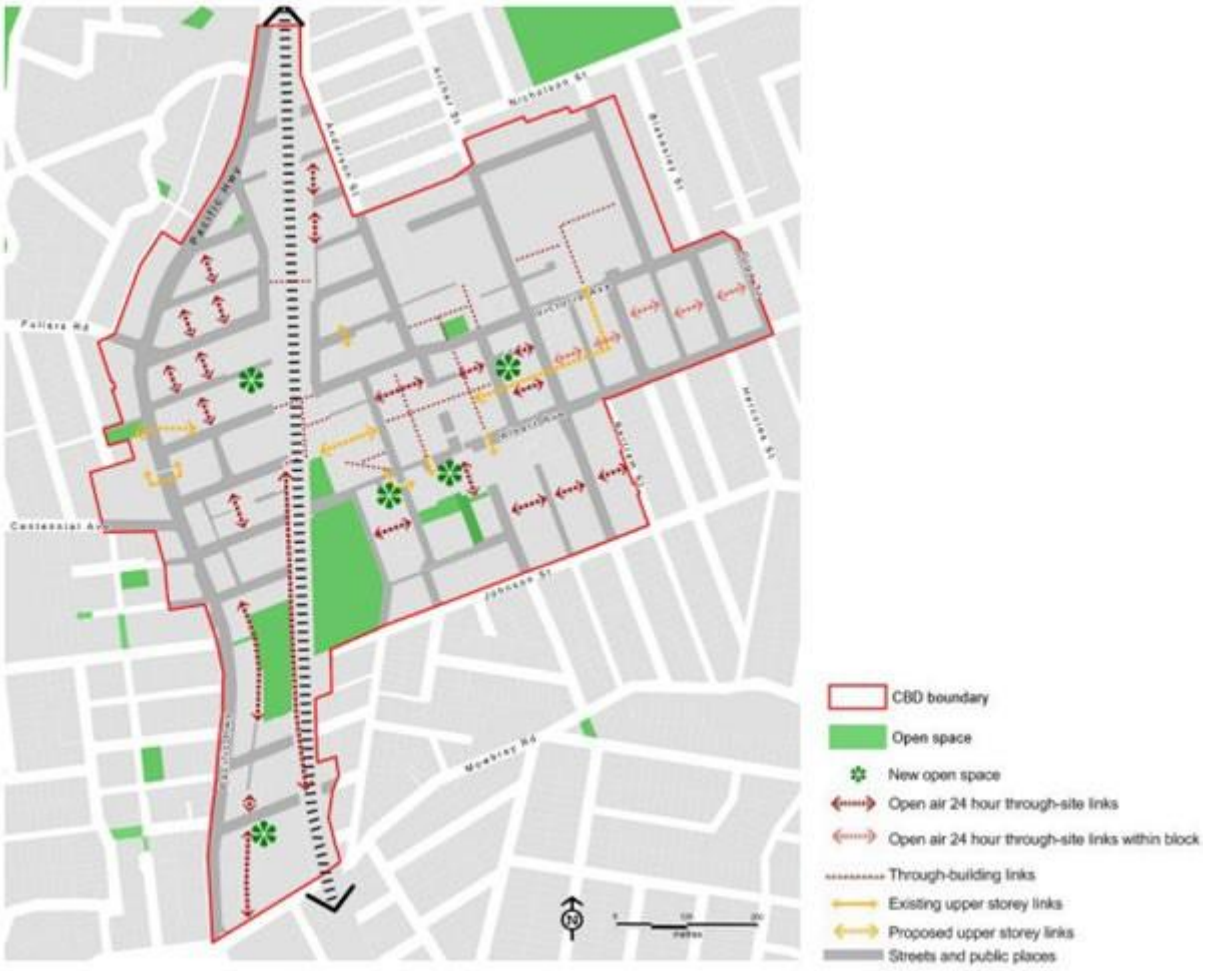
1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. Public rights of way are to be provided on:
    - The 4m setback to Pacific Highway.
    - The 3m setback to Hammond Lane.

4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.
5. The Hammond Lane setback is to be integrated with the through site link at 613-627 Pacific Highway, Chatswood.

Map 4: **Through Site Links and Open Space**



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Pacific Highway, Gordon Avenue and Hammond Lane.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle and loading access is to be reviewed and master planned in conjunction with the adjoining site at 613-627 Pacific Highway, Chatswood. One vehicle and loading point in Hammond Lane is preferred for the accessing and servicing of 629-637 Pacific Highway and 613-627 Pacific Highway, via a consolidated basement.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    a) Car parking provision based on reduced car parking rates, consistent with the requirements for new developments in the Chatswood CBD as supported by Transport for NSW.
    b) A minimum of 1 secure bicycle parking space per apartment.
    c) A minimum of 1 secure bicycle space per 100m² of non-residential /retail floor space.
    d) Adequate end of trip facilities including lockers, showers, etc. for use by non-residential and retail tenants.
    e) A bicycle rack within the site boundary for use by retail customers.
    f) A minimum of three (3) freight and service vehicle spaces within the basement, in addition to the one (1) Medium Rigid Vehicle (MRV) space proposed within the loading dock.
    g) A Green Travel Plan.
    h) Updated traffic analysis and modelling.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.

2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.13 815 Pacific Highway and 15 Help Street Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 815 Pacific Highway, and 15 Help Street, Chatswood.

Figure 40: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a commercial and non-residential development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to the Pacific Highway, Help Street and McIntosh Street

2. Site Amalgamation

Performance Criteria

1. Site amalgamation shall be provided to achieve the optimum development outcomes envisioned for the Chatswood CBD under the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. 815 Pacific Highway and 15 Help Street are required to be amalgamated.

3. Built Form

Performance Criteria

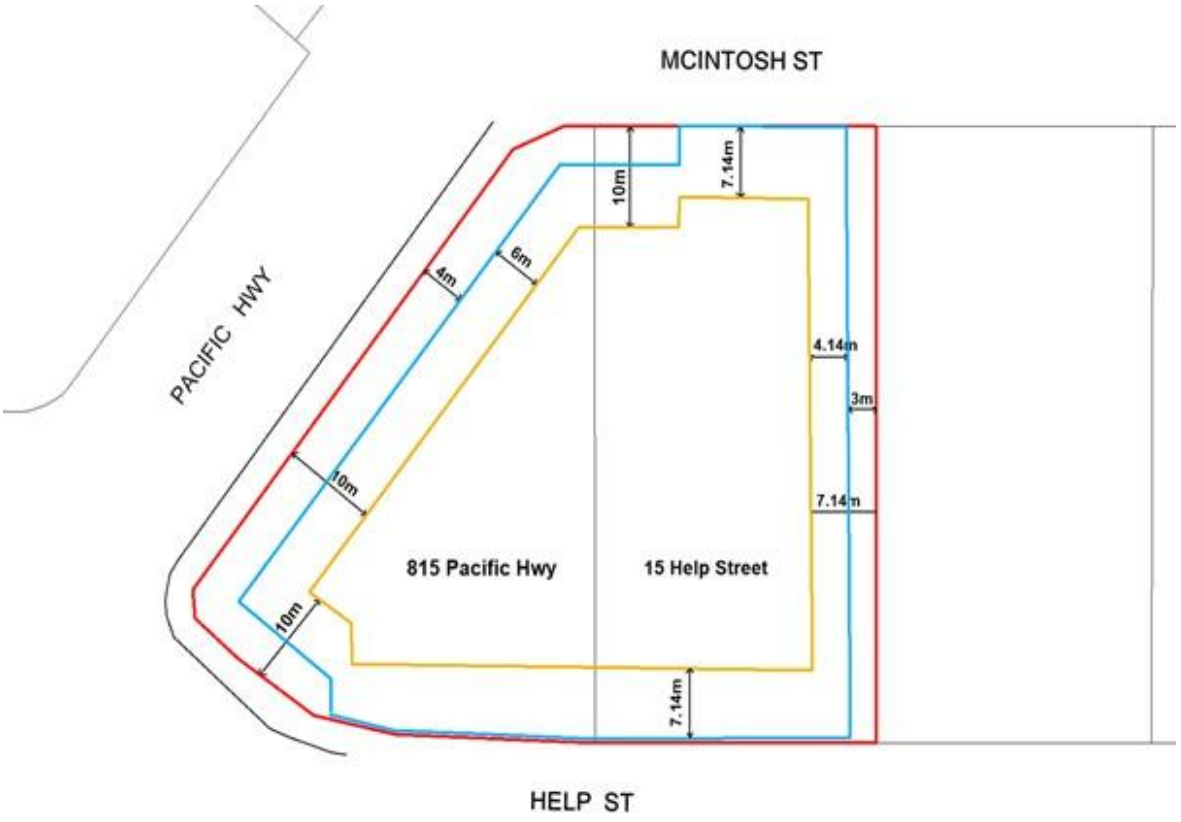
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
2. The building layout is to be in accordance with Figure 41.

Figure 41: Site Layout



Red line - boundary
Blue line - Podium
Yellow line - Tower

4. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

5. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Pacific Highway, Help Street and McIntosh Street.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights) – except where additional setback is provided. Setbacks and street wall heights are to be as follows:
    a) Pacific Highway frontage
        i) Minimum 4m setback at ground level
        ii) Minimum 6m setback above street wall
        iii) Maximum street wall height of 7m (two storeys).
    b) McIntosh Street frontage:

i) Minimum between 0 and 4m setback at ground level

ii) Minimum between 7.14m and 10m setback to tower

iii) Maximum street wall height of 4-12m (one to three storeys)

c) Help Street frontage

i) Minimum between 0 and 4m setback at ground level

ii) Minimum between 7.14m and 10m setback to tower

iii) Maximum street wall height of 4-12m (one to three storeys)

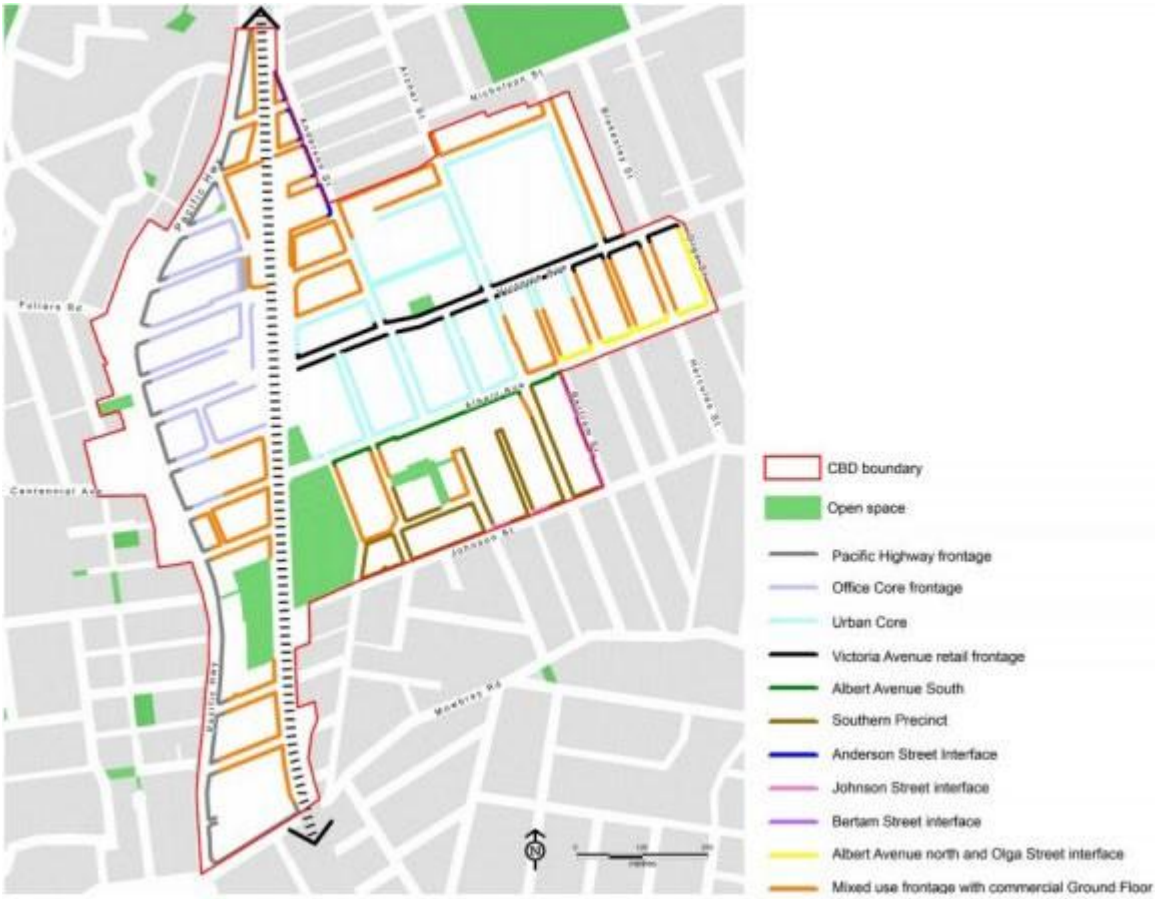
d) Eastern boundary with adjoining property

i) Minimum 3m setback at ground level

ii) Minimum 7.14m setback to tower

iii) Maximum street wall height of 4-12m (one to three storeys)

Map 5: Setbacks and street frontage heights



2. In addition to Control 1:

a) Setbacks may be greater and street wall heights may be lower.

b) Additional ground level setbacks are sought that contribute to public realm.

6. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including Pacific Highway, Help Street, McIntosh Street and the through site link on the eastern boundary.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

7. Amenity

Performance Criteria

1. Maximise solar access and ventilation.
2. Ensure visual and acoustic privacy in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. The building shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

8. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Pacific Highway, Help Street, McIntosh Street and the through site link on the eastern boundary.
6. Podium and roof tops are to be a combination of green and paved spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. Large canopy tree planting must be provided along the Pacific Highway frontage within the 4m setback area.
3. All roofs up to 30 metres from ground are to be green roofs.
4. Public domain improvements shall be provided to all street frontages to Council requirements.
5. A minimum of 2 hours sun access is to be provided to public open space on the site including the through site link.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
7. Deep soil planting is to be provided within the 4m setback to Pacific Highway. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
8. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

9. Links

Performance Criteria

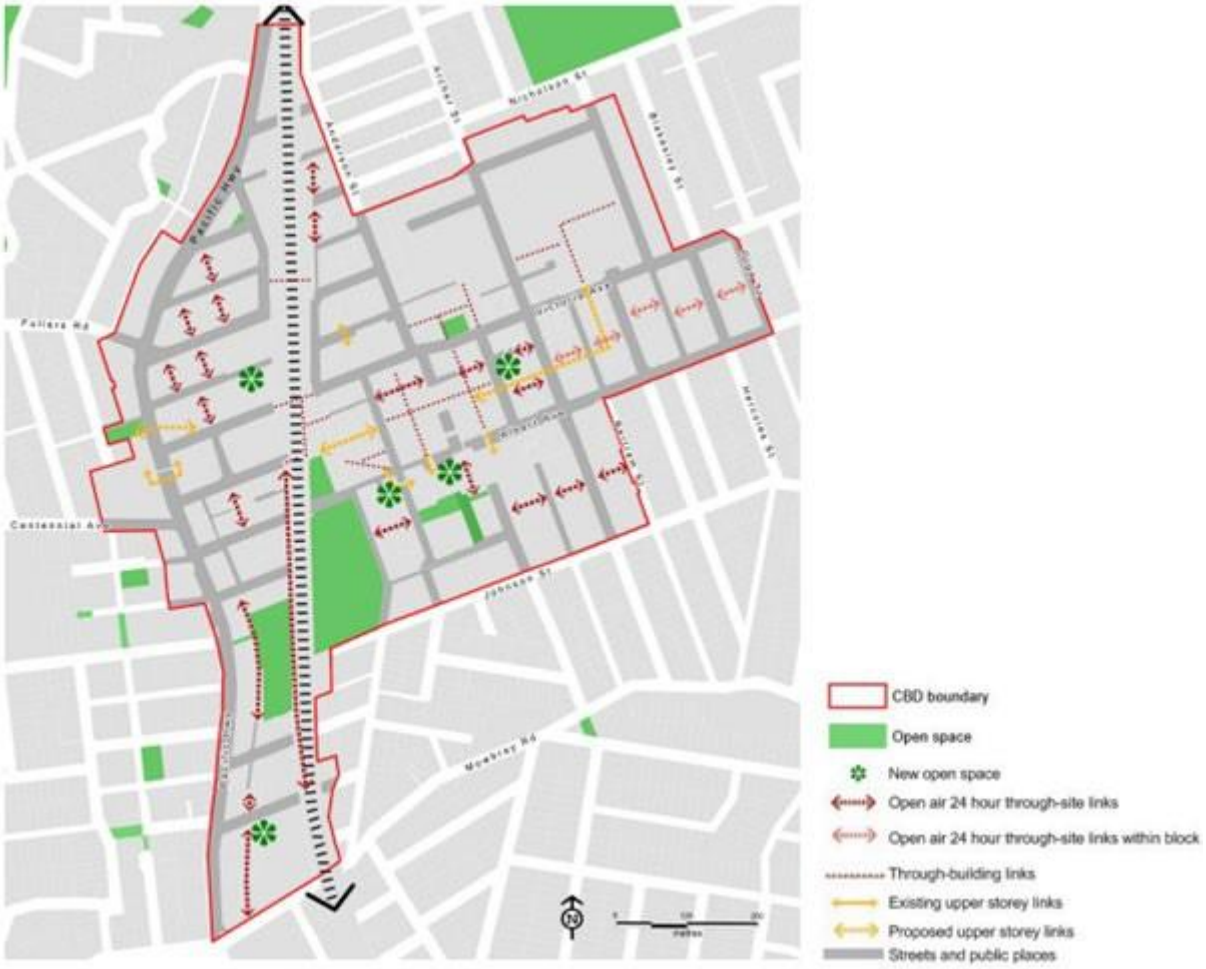
1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. Public rights of way are to be provided on:
    a) The 4m setback to Pacific Highway.
    b) The 3m setback along the eastern boundary (from Help Street to McIntosh Street) to provide a through site link.

4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



10. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Pacific Highway, Help Street, McIntosh Street and the through site link on the eastern boundary.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

11. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.

2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be from one access point in McIntosh Street.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level, in one consolidated basement over the site.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    a) Car parking provision based on reduced car parking rates, consistent with the requirements for new developments in the Chatswood CBD as supported by Transport for NSW.
    b) A minimum of 1 secure bicycle space per 100m² of non-residential /retail floor space for the first 10,000m^2 of floor space, and then 1 secure bicycle space per 200 m^2 of non-residential /retail floor space above 10,000m^2.
    c) Adequate end of trip facilities including lockers, showers, etc. for use by non-residential and retail tenants.
    d) A minimum of fifteen (15) freight and service vehicle spaces within the basement in addition to three (3) Medium Rigid Vehicle spaces.
    e) A Green Travel Plan.
    f) Updated traffic analysis and modelling.

12. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.

3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of employees from Electro Magnetic Radiation (EMR) emissions.

13. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

14. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

15. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 6 stars GBCA building rating is required. An assessment report is to be submitted at Development Application stage.

16. Heritage

Performance Criteria

1. Respect the built history of the Chatswood CBD.

Controls

1. A historical record must be made of the existing building at 815 Pacific Highway recognising its contribution to the development of Chatswood CBD prior to any demolition.

13.1.14 9-11 Nelson Street Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 9-11 Nelson Street, Chatswood.

Figure 42: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieves architectural and urban design excellence.
7. Maximise activation to Gordon Avenue and Nelson Street.

2. Built Form

Performance Criteria

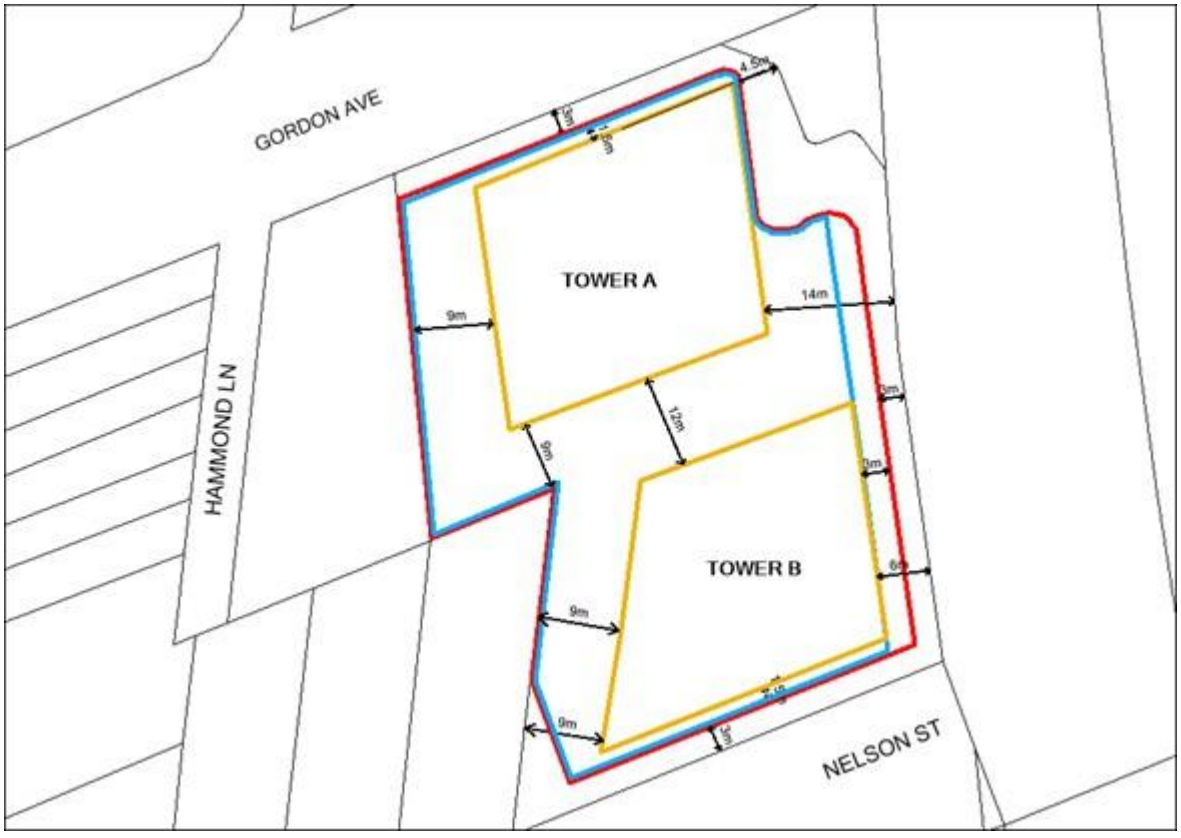
The built form of the new development shall:

1. Achieve a slender tower/s form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 43.

Figure 43: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Gordon Avenue and Nelson Street, and trees in the pocket park at the end of Gordon Avenue.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights) – except where additional setback is provided. Setbacks are as follows:
    a) Nelson Street frontage:
        i) Minimum 3m setback at ground level
        ii) Minimum 1.5m setback above street wall
        iii) Maximum street wall height of 8m (two storeys).
    b) Gordon Avenue frontage:

i) Minimum 3m setback at ground level

ii) Minimum 1.5m setback above street wall

iii) Maximum street wall height of 10m (two storeys).

c) Frank Channon Walk (eastern) boundary (from Nelson Street to Gordon Avenue)

i) Minimum 3m setback at ground level, with additional stepped 3m setback at Podium Level 1

ii) Nil setback above eastern podium wall

iii) Maximum street wall height of 10m (two storeys).

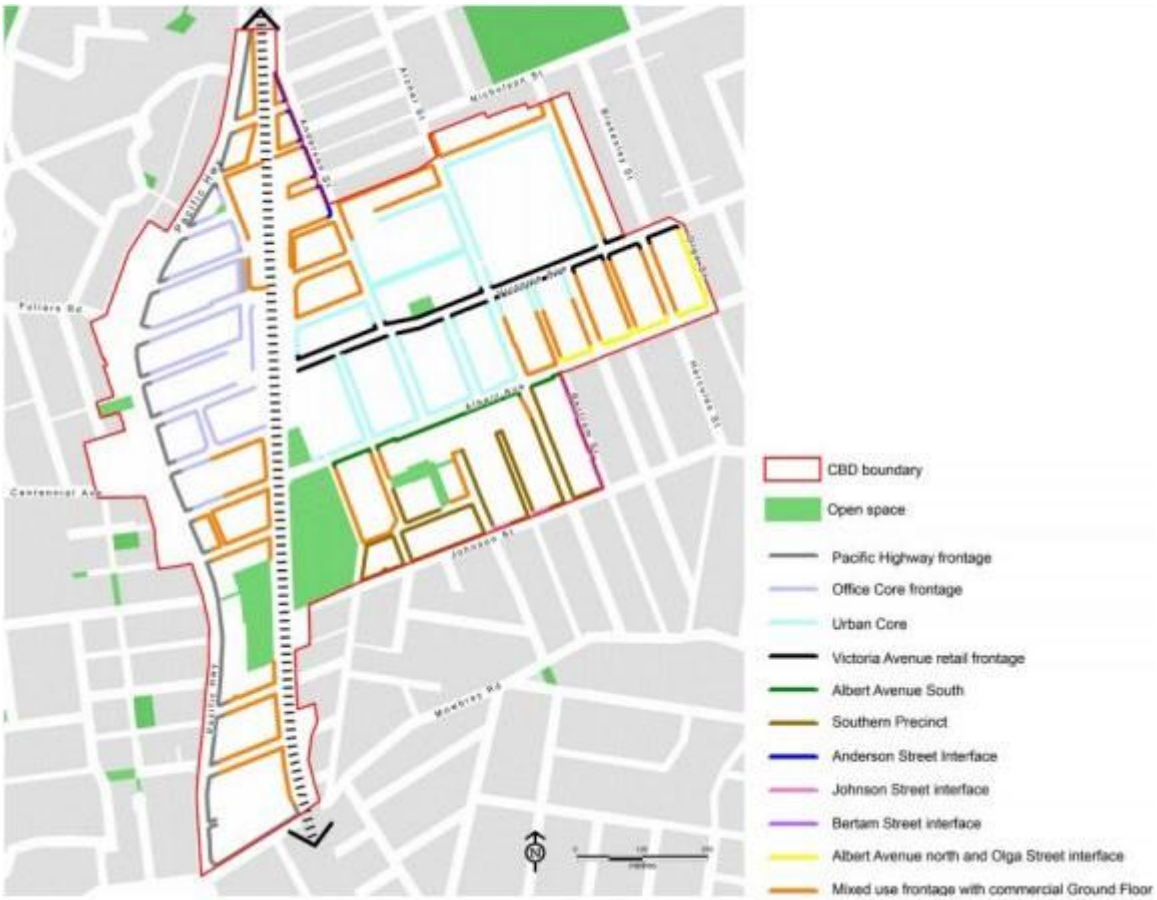
d) Western boundary with adjoining property

i) Nil setback at ground level

ii) Minimum 9m setback to tower

2. Balconies are not to encroach into setbacks.

Map 5: Setbacks and street frontage heights



3. In addition to Control 1:

a) Setbacks may be greater and street wall heights may be lower.

b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including Gordon Avenue, Nelson Street, the Frank Channon Walk and the pocket park at the end of Gordon Avenue.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Gordon Avenue, Nelson Street, the Frank Channon Walk and the pocket park at the end of Gordon Avenue.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. Large canopy tree planting must be provided along the Gordon Avenue and Nelson Street frontages within the 3m setback area.
3. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
4. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
5. Public domain improvements shall be provided to all street frontages, and the Frank Channon Walk and the pocket park at the end of Gordon Avenue frontages, to Council requirements.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
7. Deep soil planting is to be provided within the 3m setbacks to Gordon Avenue, Nelson Street and the Frank Channon Walk. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
8. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

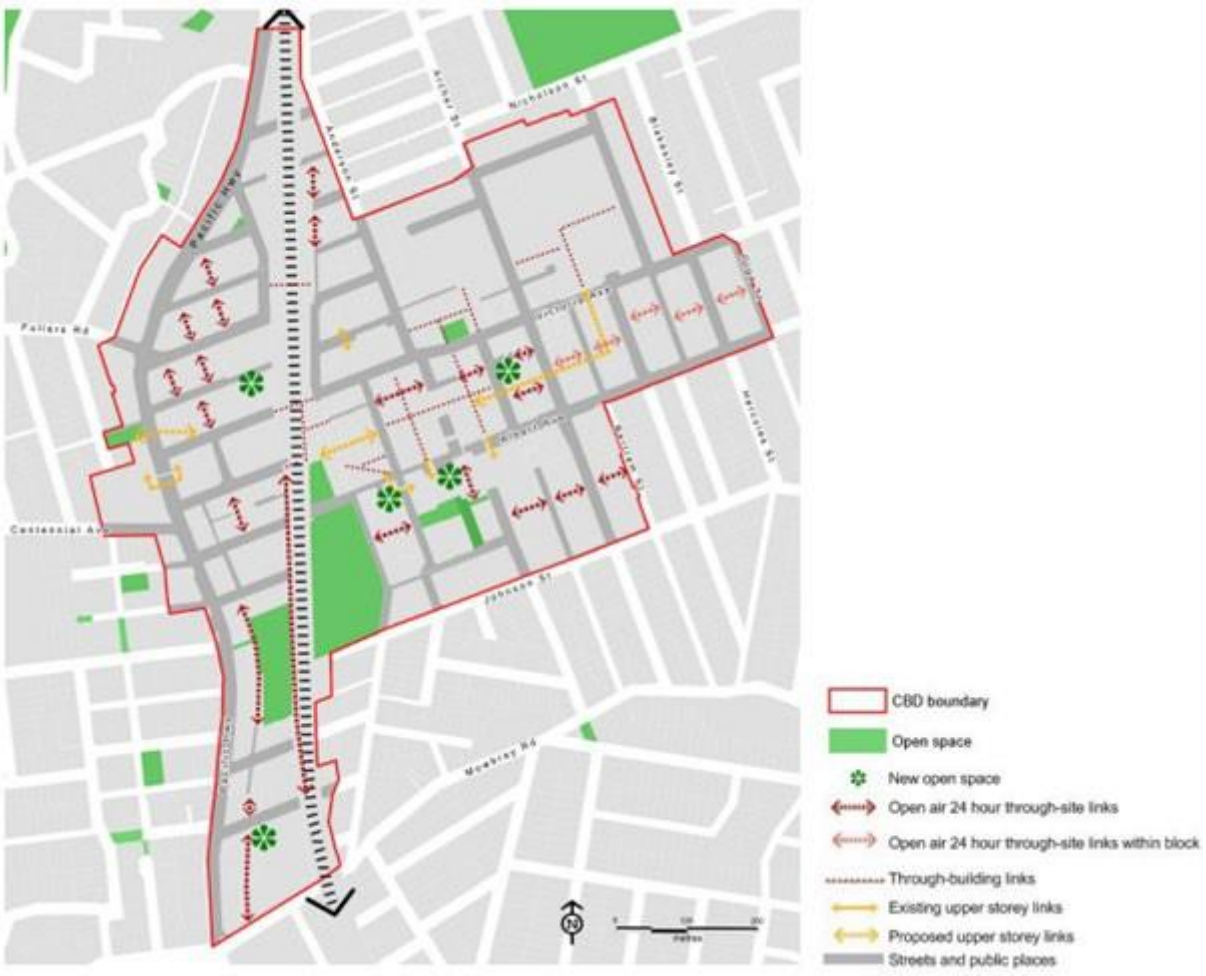
Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. Public rights of way are to be provided on:
    - The 3m setback to Gordon Avenue.
    - The 3m setback to Nelson Street.

- The 3m setback along the eastern boundary (from Gordon Avenue to Nelson Street) – adjacent the Frank Channon Walk and the pocket park at the end of Gordon Avenue.

4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Gordon Avenue and Nelson Street.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle and loading access is to be reviewed and master planned in conjunction with the adjoining land at 10 Gordon Avenue and 15-19 Nelson Street. One vehicle and loading point in Gordon Avenue is preferred for the block bounded by Nelson Street, Gordon Avenue, Hammond Lane and the Frank Channon Walk, via a consolidated basement.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level, utilizing physical solutions to ensure all vehicles (including loading vehicles) enter and leave the site in a forward direction.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    a) Car parking provision based on reduced car parking rates, consistent with the requirements for new developments in the Chatswood CBD as supported by Transport for NSW.
    b) A minimum of 1 secure bicycle parking space per apartment.
    c) A minimum of 1 secure bicycle space per 100m² of non-residential /retail floor space.
    d) Adequate end of trip facilities including lockers, showers, etc. for use by non-residential and retail tenants.
    e) A bicycle rack within the site boundary for use by retail customers.
    f) A minimum of three (3) freight and service vehicle spaces within the basement, in addition to the one (1) Medium Rigid Vehicle (MRV) space proposed within the loading dock.
    g) A Green Travel Plan.
    h) Updated traffic analysis and modelling.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council’s Public Art Policy.

Controls

2. Public Art is to be provided in accordance with Council’s Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.15 849, 853, 859 Pacific Highway, 2 Wilson Street and Lot 1 DP 1189541 Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 849, 853, and 859 Pacific Highway, 2 Wilson Street and Lot 1 DP 1189541 Chatswood.

Figure 44: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieve architectural and urban design excellence.

7. Maximise activation to Pacific Highway, O'Brien Street and Wilson Street.

2. Built Form

Performance Criteria

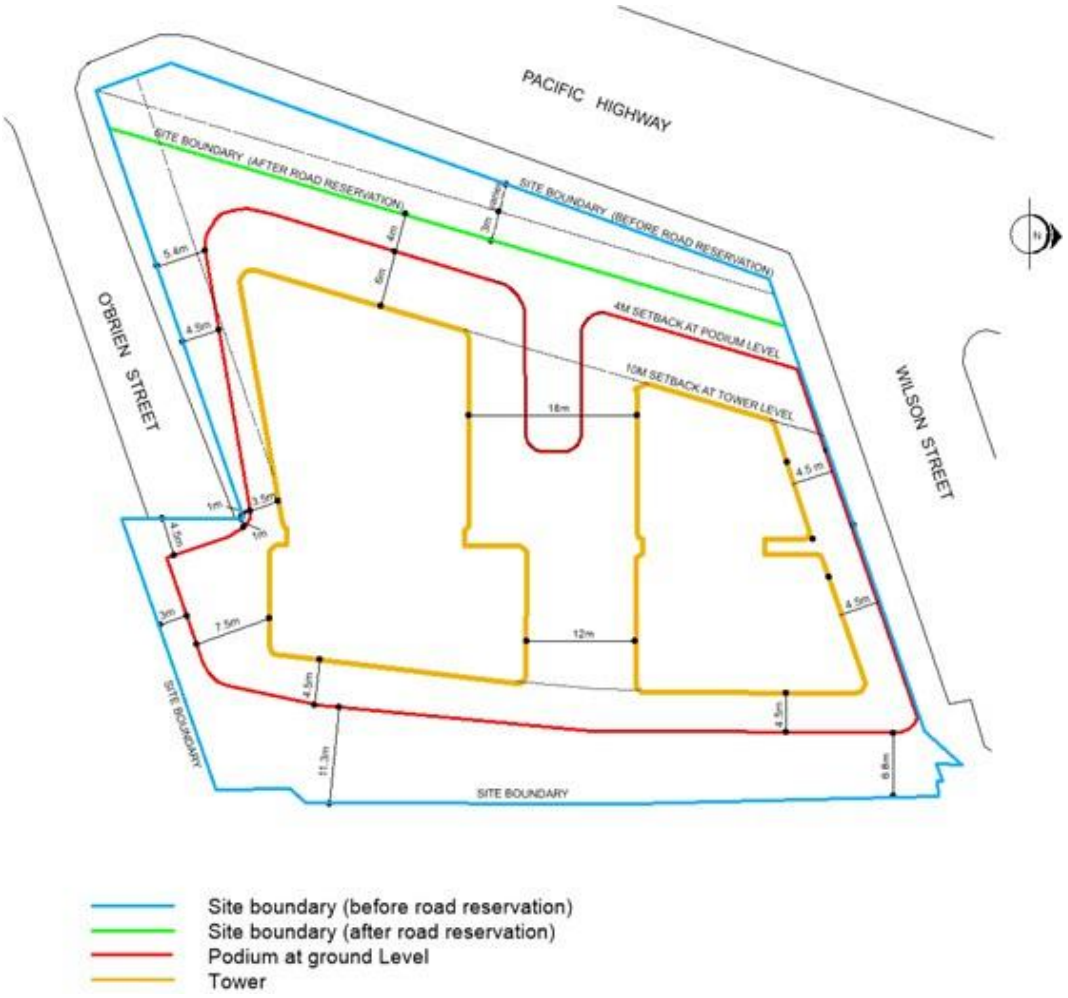
The built form of the new development shall:

1. Achieve a slender tower/s form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 45.

Figure 45: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Pacific Highway, O'Brien Street and Wilson Street, and trees in the setback to the North Shore Rail Line.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

1. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

1. The building setbacks are to be in accordance with Map 5 (Setbacks and street frontage heights) – except where additional setback is provided. Setbacks are as follows:
    a) Pacific Highway frontage:
        i) Minimum 4m setback at ground level
        ii) Minimum 6m setback above street wall
        iii) Maximum street wall height of 7m (two storeys).
    b) O'Brien Street and remaining southern boundary frontage:
        i) Minimum setback at ground level between 1m (where O'Brien Street meets corner of 849 Pacific Highway and Lot 1 DP 1189541) and 5.4m (where O'Brien Street meets SP2 Infrastructure (Classified Road) land adjacent Pacific Highway Note: Ground level setback at end of O'Brien Street addressed in iii) below.
        ii) Minimum 3m setback at ground level along remaining southern boundary
        Note: Ground level setback at end of O'Brien Street addressed in iii) below.
        iii) Minimum setback at ground level at end of O'Brien Street of between 1m and 4.5m
        iv) Minimum setback above street wall 3.5m (where O'Brien Street meets corner of 849 Pacific Highway and Lot 1 DP 1189541) and 3m (where O'Brien Street meets SP2 Infrastructure (Classified Road) land adjacent Pacific Highway
        Note: Tower setback at end of O'Brien Street addressed in v) below.
        v) Minimum setback above street wall 7.5m along end of O'Brien Street and remaining southern boundary

vi) Maximum street wall height of 7m (two storeys)

c) Wilson Street frontage

i) Minimum nil setback at ground level

ii) Minimum 4.5m setback above street wall

iii) Maximum street wall height of 7m (two storeys).

d) Eastern boundary frontage (facing North Shore Rail Line)

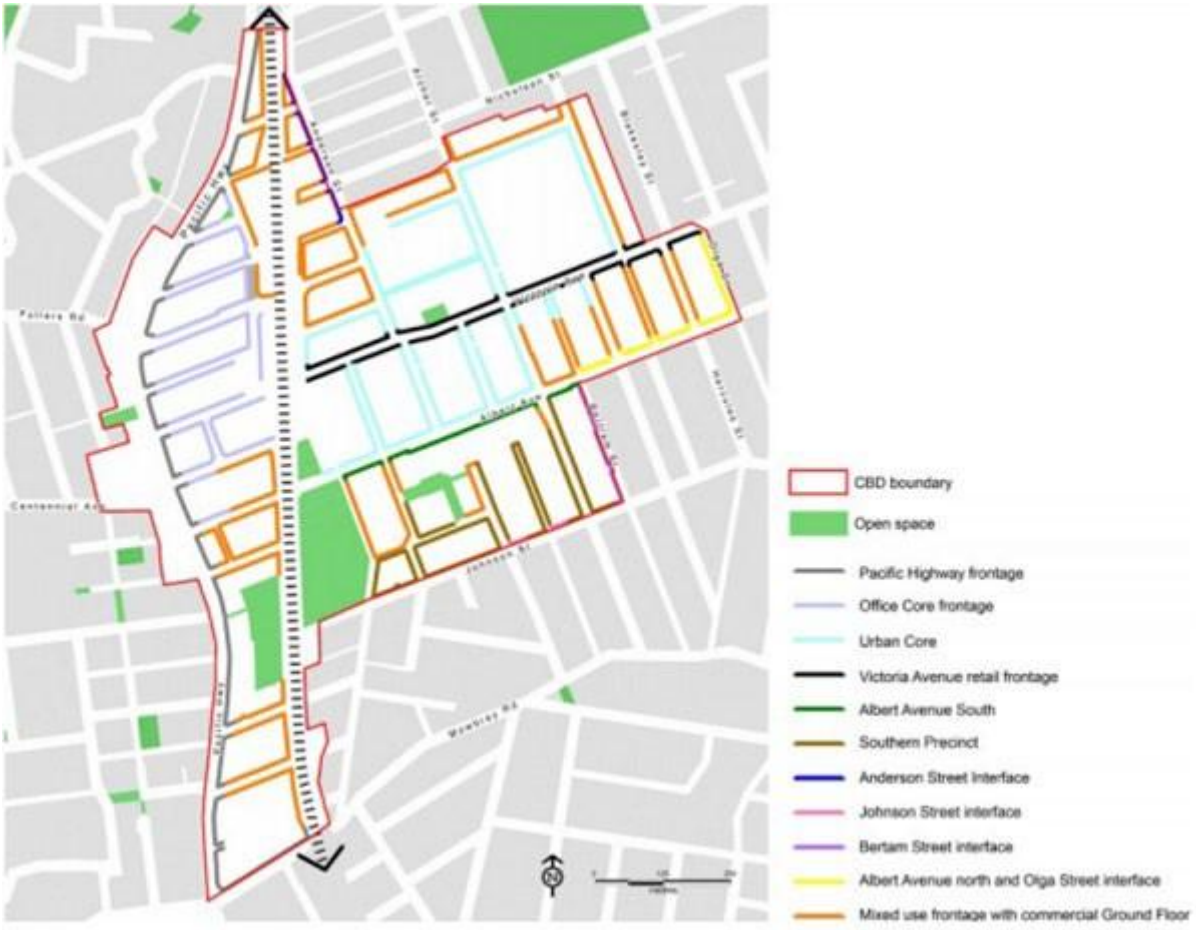
i) Minimum setback between 6.8m (Wilson Street end) and 11.3m (O'Brien Street end) at ground level

ii) Minimum 4.5m setback to tower

iii) Maximum podium wall height of 7m (two storeys).

2. Balconies are not to encroach into setbacks.

Map 5: Setbacks and street frontage heights



3. In addition to Control 1:

a) Setbacks may be greater and street wall heights may be lower.

b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including Pacific Highway, O'Brien Street, Wilson Street and the pocket park and through site link between O'Brien and Wilson Street, adjacent the North Shore Rail Line.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage addressing noise and vibrations from both the Pacific Highway and the North Shore Rail Line.
3. Residential units shall be designed to maximize solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Pacific Highway, O'Brien Street, Wilson Street and the pocket park and through site link between O'Brien and Wilson Streets, adjacent the North Shore Rail Line.

6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. Large canopy tree planting must be provided along the Pacific Highway, O'Brien Street frontages, and the pocket park and through site link between O'Brien and Wilson Street, adjacent the North Shore Rail Line, where any setback area is greater than 3m.
3. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximize solar access.
4. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
5. Public domain improvements shall be provided to all street frontages, and the pocket park and through site link between O'Brien and Wilson Streets, adjacent the North Shore Rail Line, to Council requirements.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
7. Deep soil planting is to be provided along the Pacific Highway, O'Brien Street frontages, and the pocket park and through site link between O'Brien and Wilson Streets, adjacent the North Shore Rail Line, where any setback area is greater than 3m. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
8. A landscape plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

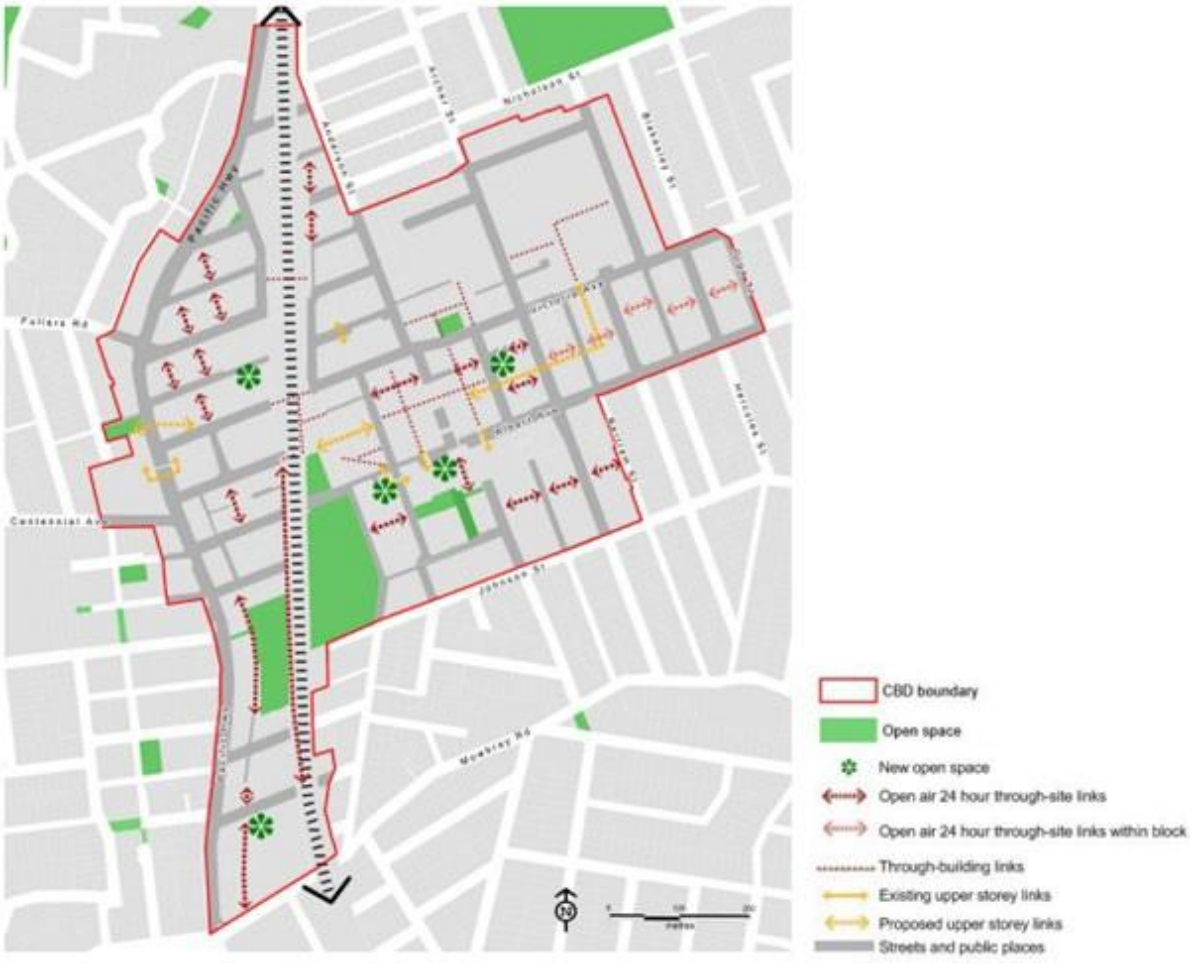
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. Public rights of way are to be provided on:
    - The 4m setback to Pacific Highway.
    - The setback between 1m (where O'Brien Street meets corner of 849 Pacific Highway and Lot 1 DP 1189541) and 5.4m (where O'Brien Street meets SP2 Infrastructure (Classified Road) land adjacent Pacific Highway.
    - The setback between 1m and 4.5m at the end of O'Brien Street.
    - The 3m setback to the remaining southern boundary.
    - The setback between 6.8m (Wilson Street end) and 11.3m (O'Brien Street end), adjacent the North Shore Rail Line.
4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximize surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Pacific Highway, O'Brien Street and Wilson Street.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle access to / egress from the development is to be from one access point in Wilson Street.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level, utilizing physical solutions to ensure all vehicles (including loading vehicles) enter and leave the site in a forward direction.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    a) Car parking provision based on reduced car parking rates, consistent with the requirements for new developments in the Chatswood CBD as supported by Transport for NSW
    b) All servicing, including drainage, to be outside the Sydney Train corridor.

c) Measures are to be included, to the satisfaction of Sydney Trains which prevent the throwing of objects onto the rail corridor

d) A Traffic Management Plan for the construction phase, and future operation phase, to demonstrate that additional vehicular movements in and out of the site:

i. Do not potentially obstruct access onto Transport Asset Holding Entity (TAHE land and rail corridor.

ii. Do not pose queuing issues along the Wilson Street overbridge, and potentially obstruct rail corridor access. The rail corridor access point/s must not be blocked at any stage during the construction and operation phases of future developments on the subject site. Rail bridge load restrictions must also be considered and adhered to, as related to construction-related vehicles

e) Documentation demonstrating compliance with the Sydney Metro Underground Corridor Protection Guidelines and/or Sydney Metro At Grade and Elevated Sections Corridor Protection Guidelines as applicable.

f) Infrastructure integral to redevelopment of the site not located in the existing TfNSW reservations and easements.

g) An amended Transport Impact Assessment addressing:

i. The cumulative impact on the surrounding active transport network, the identification of active transport links to existing school travel paths, and investigation regarding how the site can connect to the Principal Bike Network to encourage active transport uptake into the future.

ii. Consideration of the NSW Governments Movement and Place Framework (MAPF) and its Built Environment Performance Indicators

h) A Green Travel Plan.

i) Updated traffic analysis and modelling.

j) Assessment of sight lines for the vehicle ingress / egress point in Wilson Street, and amendments to design as necessary, with regard to traffic from the Pacific Highway and Wilson Street overbridge.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.

4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.16 10 Gordon Avenue, 15, 17 and 19 Nelson Street Chatswood

1. General

The controls contained in this Site Specific Development Control Plan applies to 10 Gordon Avenue, 15, 17 and 19 Nelson Street Chatswood.

Figure 46: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as well as the amenity of the development.
6. Achieve architectural and urban design excellence.
7. Maximise activation to Gordon Avenue, Hammond Lane Street and Nelson Street.

2. Built Form

Performance Criteria

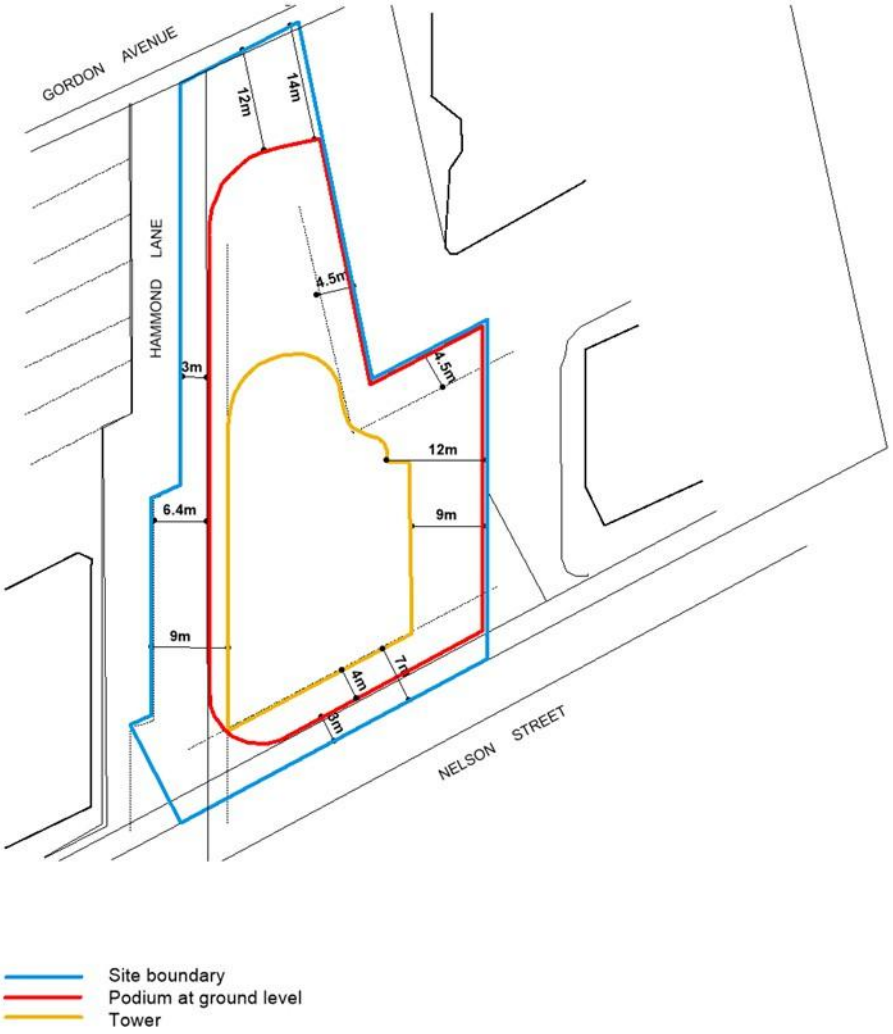
The built form of the new development shall:

1. Achieve a slender tower form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m^2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised.
3. The building layout is to be in accordance with Figure 47.

Figure 47: Site Layout



3. Height of Building

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the maximum building height.

4. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at ground level to contribute to public realm.
3. Contribute at ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Gordon Avenue, Hammond Lane and Nelson Street.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

7. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.

Controls

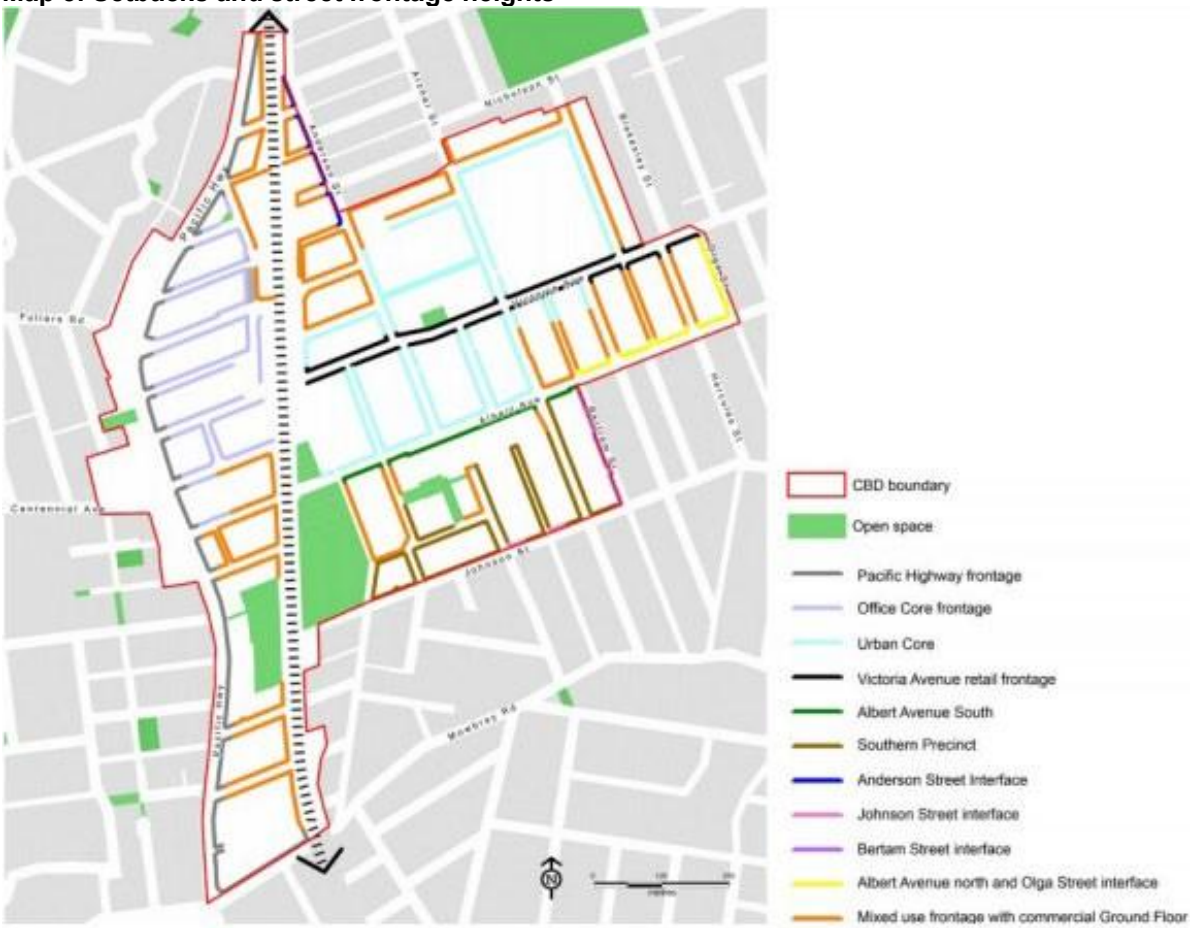
1. The building setbacks are to be in accordance with Figure 3 (Setbacks and street frontage heights) – except where additional setback is provided. Setbacks are as follows:
   a) Gordon Avenue frontage:
      i) Minimum 12 – 14m setback at ground level
      ii) Minimum 26m setback above street wall
      iii) Maximum street wall height of 8m (two storeys).
   b) Nelson Street frontage:
      i) Minimum 3m setback at ground level
      ii) Minimum 4m setback above street wall
      iii) Maximum street wall height of 8m (two storeys).

c) Western boundary frontage (to Hammond Lane and 613-627 Pacific Highway)
   i) Minimum 3m setback at ground Level to Hammond Lane
   ii) Minimum 6.4m setback at ground Level to 613-627 Pacific Highway
   iii) Minimum 2.6m setback above street wall to Hammond Lane and 613-627 Pacific Highway
   iv) Maximum street wall height of 8m (two storeys).

d) Eastern boundary frontage (to 9-11 Nelson Street)
   i) Nil setback at ground level
   ii) Minimum 9m setback to tower
   iii) Maximum podium wall height of 8m (two storeys).

2. Balconies are not to encroach into setbacks.

Map 5: Setbacks and street frontage heights



3. In addition to Control 1:
   a) Setbacks may be greater and street wall heights may be lower.
   b) Additional ground level setbacks are sought that contribute to public realm.

5. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including Gordon Avenue, Hammond Lane and Nelson Street.

2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

6. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage addressing noise and vibration from both the Pacific Highway and the North Shore Rail Line.
3. Residential units shall be designed to maximise solar access, cross ventilation, visual and acoustic privacy.

7. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.
2. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping.
5. Greening at the podium roof level is to be provided, with planting visible to the surrounding area – with particular regard to Gordon Avenue, Hammond Lane and Nelson Street.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. Large canopy tree planting must be provided along the Gordon Avenue and Nelson Street frontages, and the pocket park and through site link between Gordon Avenue and Nelson Street, where any setback area is greater than 3m.

3. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximise solar access.
4. A minimum of 2 hours of sun access is to be provided to the public open space on the site.
5. Public domain improvements shall be provided to all street frontages, and the pocket park and through site link between Gordon Avenue and Nelson Street, to Council requirements.
6. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on ground, podium and roof top levels or green walls of buildings.
7. Deep soil planting is to be provided along the Gordon Avenue and Nelson Street frontages, including the pocket park, and the through site link between Hammond Lane and Nelson Street, adjacent the boundary with 613-627 Pacific Highway, where any setback area is greater than 3m. Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
8. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

8. Links

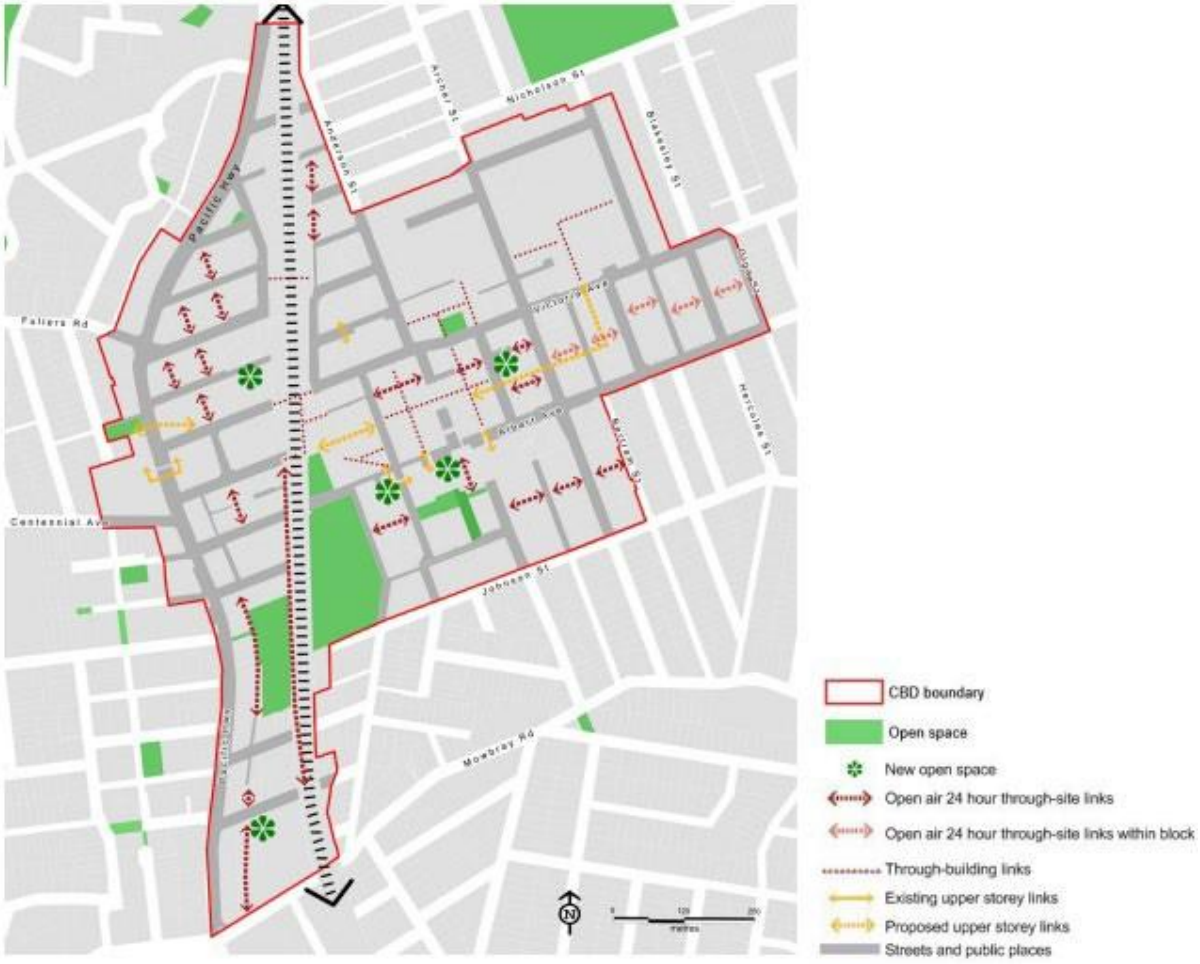
Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible open space is to include green landscaping.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Map 4 below.
2. Through site links and open space in addition to Map 4 is required on a site by site basis.
3. Public rights of way are to be provided on:
    - The 12 – 14m setback at ground level to Gordon Avenue.
    - The 3m setback at ground level to Nelson Street.
    - The 3m setback at ground level to Hammond Lane.
    - The 6.4m setback at ground level to 613-627 Pacific Highway.
4. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

Map 4: Through Site Links and Open Space



9. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximise surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Gordon Avenue, Hammond Lane Street and Nelson Street.
2. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

10. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site should not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular access points to the development.

4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.

Controls

1. Vehicle and loading access is to be reviewed and master planned in conjunction with the adjoining land at 9-11 Nelson Street. One vehicle and loading point in Gordon Avenue is preferred for the block bounded by Nelson Street, Gordon Avenue, Hammond Lane and the Frank Channon Walk, via a consolidated basement.
2. Vehicle access and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level, utilizing physical solutions to ensure all vehicles (including loading vehicles) enter and leave the site in a forward direction.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    a) Reduced car parking rates aligned with Council's Development Control Plan.
    b) An amended Transport report containing:
        i) Updated traffic analysis and modelling.
        ii) Analysis of the proposals cumulative impact on the surrounding transport network and impacts on active transport links to existing school travel paths.
        iii) Analysis of NSW Governments Movement and Place Framework (MAPF) and its Built Environment Performance Indicators.
    c) A report demonstrating compliance with the Sydney Metro
        i) Underground Corridor Protection Guidelines and/or Sydney Metro At Grade and Elevated Sections Corridor Protection Guidelines.
    d) A Green Travel Plan.

11. Waste Management, Loading and Services

Performance Criteria

1. All loading, unloading and servicing is required to occur on-site.
2. To ensure that adequate provision is made for waste storage and disposal.
3. Floor space at ground level is to be maximised, with services located in the basement.

Controls

1. All loading and unloading services are required to occur at basement level on-site.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in basement levels.
3. A Waste Management Plan shall be submitted at the Development Application stage.

4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

12. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

13. Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.

14. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

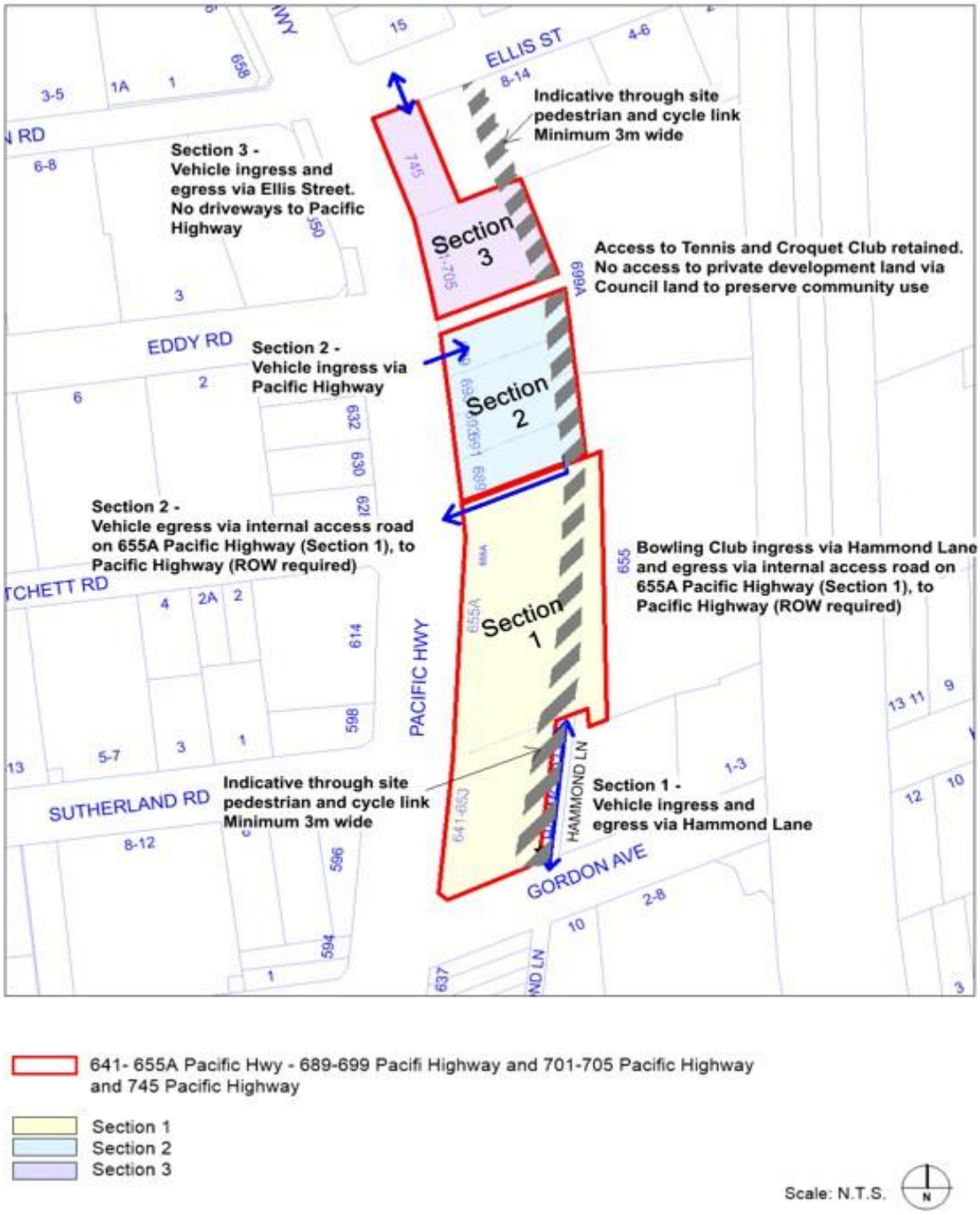
1. A minimum of 5 stars GBCA building rating is required. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

13.1.17 Precinct applying to the eastern side of Pacific Highway, between Gordon Avenue and Ellis Street, Chatswood (being 641-653, 655A, 689, 691-693, 695, 699, 701-705 and 745 Pacific Highway)

1. General

Inclusion of Precinct DCP applying to the eastern side of Pacific Highway, between Gordon Avenue and Ellis Street, Chatswood (being 641-653, 655A, 689, 691-693, 695, 699, 701-705 and 745 Pacific Highway)

Figure 48: Precinct Map



Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide Precinct guidelines for 641-653, 655A, 689, 691-693, 695, 699, 701-705 and 745 Pacific Highway Chatswood.
2. Facilitate provision of pedestrian and cycle through site links connecting with adjacent or envisioned future pedestrian and cycle links within the Chatswood CBD.
3. Minimise traffic impacts on the surrounding road network.

2. Precinct Amalgamation

Performance Criteria

1. Amalgamation shall be achieved to:
   a) Provide the optimum development outcomes envisioned for the Chatswood CBD under the Chatswood CBD Planning and Urban Design Strategy 2036; and
   b) Satisfy the requirements of Transport for NSW with particular regard to the Pacific Highway.

Controls

1. The Precinct is divided into three Sections, with the following sites required to be amalgamated:
   a) Section 1

      641-653 Pacific Highway and 655A Pacific Highway Chatswood.

   b) Section 2

      689 Pacific Highway. 691-693 Pacific Highway, 695 Pacific Highway and 699 Pacific Highway Chatswood.

   c) Section 3

      701-705 Pacific Highway and 745 Pacific Highway Chatswood.

Refer to Figure 48: Precinct Map

3. Links

Performance Criteria

1. Publicly accessible through site pedestrian and cycle links and open space are to contribute to the liveability of the Chatswood CBD.
2. The Precinct shall provide publicly accessible north / south and east / west through site links and open space.
3. Maximise the amenity of pedestrian and cycle through site links.
4. Encourage via design public use of through site links and public rights of way.
5. The development shall be consistent with Movement and Place principles.
6. Links are to encourage comfortable and safe movement.

Controls

1. The development is to incorporate publicly accessible through site links and open space in accordance with Figure 1 above (being north / south).
2. In Section 1, at ground level, a minimum 3m wide east / west through site link, connecting the north / south through site link with the Pacific Highway, along the northern boundary (with Section 2), separate of any vehicle egress.
3. In Section 1, at ground level, a minimum 3m wide east / west through site link, connecting the north / south through site link with the Pacific Highway, between the two towers.
4. Publicly accessible through site links are to be a minimum width of 3m (with greater width encouraged).
5. Through site links are to be the subject of public rights of way.
6. Through site pedestrian links are to:
   a) Be direct and accessible to all and have a clear line of sight between public places;
   b) Align with breaks between buildings so that views are extended;
   c) Be located and orientated to enhance the relationship between built form and open space.
   d) Be easily identified by users and include wayfinding signage;
   e) Be clearly distinguished from vehicle accessways and parking;
   f) Include materials and finishes such as paving materials and tree planting consistent with the proposed urban quality of the precinct;
   g) Be clear of both permanent and temporary obstructions or structures, including utility elements, advertising/promotional material, street furniture and outdoor dining.
   h) Include lighting to provide safety, amenity, character and night time quality appropriate to the link.
   i) Include potential for feature lighting within landscaped areas.
   j) Maximise soft landscaping and deep soil planting.
   k) Located to maximise solar access

4. Traffic and Transport

Performance Criteria

1. Development within the Precinct must be designed to provide adequate and safe access.
2. Development within the Precinct must not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular ingress and egress points within the Precinct, with particular regard to the Pacific Highway.
4. All vehicle ingress and egress to any site is to be in a forward direction.
5. Traffic and transport solutions are to be physical (not mechanical).
6. Minimise car parking and encourage alternative active transport options.

7. Encourage a street hierarchy that supports sustainable travel behaviour.

Controls

1. For Section 1 (641-653 Pacific Highway and 655A Pacific Highway Chatswood), vehicle ingress and egress is to be via Hammond Lane.
2. Within Section 1, Chatswood Bowling Club vehicle ingress and egress is to be via Hammond Lane. To this end rights of way are to be established.
3. Within Section 1, the six croquet greens car spaces access and egress is to be via Hammond Lane. To this end rights of way are to be established.
4. For Section 2 (689 Pacific Highway. 691-693 Pacific Highway, 695 Pacific Highway and 699 Pacific Highway Chatswood), vehicle ingress is to be left in via 699 Pacific Highway and egress left out via the existing internal access road located on 655A Pacific Highway. To this end rights of way are to be established.
5. The existing internal egress from 655A Pacific Highway to the Pacific Highway is to be blocked via physical measures for the exclusive use of properties to the north (identified as Section 2 in the Precinct Plan). This is not to be achieved by gates or barriers but by cul-de-sac or a similar solution whereby pedestrians are encouraged onto the site and are able to access through site links. This egress is not in any way to serve 641-653, and 655A Pacific Highway, the Chatswood Bowling Club at 655 Pacific Highway, the croquet greens or any other vehicles from Gordon Avenue.
6. For Section 3 (701-705 Pacific Highway and 745 Pacific Highway Chatswood), vehicle ingress and egress is to be via Ellis Street.

13.1.18 641-653; and 655A Pacific Highway Chatswood

1. General

The controls contained in this site specific Development Control Plan applies to 641-653, and 655A Pacific Highway Chatswood.

Figure 1: Site Map

Objectives of the Plan

The aims and objectives of this Plan are to:

1. Provide guidelines for a mixed use development on the site.
2. Provide a development that ensures the viability of adjoining and surrounding sites for future development.
3. Minimise traffic impacts on the surrounding road network
4. Ensure development on the site minimises impacts to the amenity of neighbouring residential properties.
5. Provide landscaping in and surrounding the site that enhances the presentation of the site as

well as the amenity of the development.

6. Embellish public domain on the site, to connect with existing and envisioned public domain outside the site.
7. Achieve architectural and urban design excellence.
8. Maximise activation to Pacific Highway, Gordon Avenue and Hammond Lane.

2. Site Amalgamation

Performance Criteria

1. Site amalgamation shall be provided to achieve the optimum development outcomes envisioned for the Chatswood CBD under:
   a) the Chatswood CBD Planning and Urban Design Strategy 2036 and
   b) Willoughby Development Control Plan Part L: Place Based Plans 2023, Section 13.1.17 Precinct applying to the eastern side of Pacific Highway, between Gordon Avenue and Ellis Street, Chatswood (being 641-653, 655A, 689, 691-693, 695, 699, 701-705 and 745 Pacific Highway).

Controls

1. 641 and 653 Pacific Highway and 655A Pacific Highway are required to be amalgamated.

3. Built Form

Performance Criteria

The built form of the new development shall:

1. Achieve a slender tower/s form on the site.
2. Achieve a site layout that provides a pleasant environment for the occupants and minimises impact on surrounding properties.
3. Ensure visual and acoustic privacy, natural ventilation, sun access, and views.
4. Provide suitable areas for communal open spaces, deep soil zones, and landscaping.

Controls

1. The maximum tower floor plate that applies to this site for residential towers above a podium is 700m2.
2. The width of each side of any tower should be minimised and design elements that contribute to building bulk should be minimised. Particular attention is drawn to the east / west elevation on this site.

4. Height of Buildings

Performance Criteria

The built form of new development shall:

1. Be consistent with the permitted Height of Buildings development standard applicable to the site.
2. Minimise overshadowing of surrounding properties, key public spaces and public domain.

Controls

1. The maximum building height is to include all structures located at roof level, including lift over runs and any other architectural features.
2. All roof top lift over runs or exposed structures are to be integrated with the building.
3. Flat roof areas shall incorporate useable outdoor recreation space where suitable, within the

maximum building height.

5. Setbacks and Street Frontage Heights

Performance Criteria

Setbacks shall:

1. Ensure the positioning of new buildings is consistent with the proposed streetscape envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
2. Be provided at Ground level to contribute to public domain.
3. Contribute at Ground level deep soil areas, landscaping, and open space.
4. Protect all significant on-site trees and all street trees on Pacific Highway, Gordon Avenue and Hammond Lane.
5. Contribute to slender tower forms.
6. Minimise the effects of adverse wind conditions at street level.

Street wall heights shall:

7. Ensure such heights are consistent with the street wall heights envisioned for Chatswood CBD and contained in the Chatswood CBD Planning and Urban Design Strategy 2036.
8. Be at a lower scale reflecting the location of the site towards the south western edge of the Chatswood CBD.
9. Be sympathetic, and respond positively, to what is located on neighbouring sites.

Controls

1. The building setbacks are to be as follows:
   a) Pacific Highway (western) frontage:
      i) Minimum 4m setback at Ground Level
      ii) Minimum 6m setback above street wall to tower
      iii) Maximum street wall height of 7m (two storeys).
   b) Gordon Avenue (southern) frontage:
      i) Minimum 1.5m setback at Ground Level
      ii) Minimum 3m setback above street wall to tower
      iii) Maximum street wall height of 7m (one to two storeys).
   c) Northern frontage adjacent 689 Pacific Highway
      i) Minimum 8m setback at Ground Level
      ii) Minimum 4.5m setback above Podium to tower
      iii) Maximum street wall height of 7m (one to two storeys).
   d) Eastern boundary frontage (facing Chatswood Bowling Club)

      Podium for Tower 1 (north tower, facing 689 Pacific Highway)
      i) Minimum setback between 12m (at northern end) and 16m (at southern end)
      ii) Minimum 4m setback above street wall to tower
      iii) Podium wall height of between 7m (one to storeys) and a maximum 14m (three storeys).

Podium for Tower 2 (south tower, facing Gordon Avenue)

i) Minimum 3m setback at Ground Level

ii) Minimum 4.5m setback to tower

iii) Podium wall height of between 7m (one to storeys) and a maximum 14m (three storeys).

2. Balconies are not to encroach into setbacks.
3. In addition to Control 1:
   a) Setbacks may be greater and street wall heights may be lower.
   b) Additional ground level setbacks are sought that contribute to public domain.
   c) Stepped podium setbacks are supported and encouraged to provide:
      - Elevated landscaping to neighbouring properties and the towers located on-site
      - Elevated landscaping to ground level public domain embellishment located on-site such as public rights of way to encourage sympathetic relationships with neighbouring sites and the greening of the Chatswood CBD.

6. Building Exterior

Performance Criteria

1. Buildings are to demonstrate a high design quality when viewed from the public domain and the surrounding area, including Pacific Highway, Gordon Avenue, Hammond Lane, the Chatswood Bowling Club and Tennis and Croquet Key Public Space.
2. Facade treatment and design is to be used to break down the mass and bulk of buildings.
3. High quality façade materials and finishes are to be used which contribute positively to the built environment and mitigate urban heat.

Controls

1. Facades are to be articulated and should incorporate recesses and projecting elements that do not encroach into required setbacks.
2. Extensive blank walls shall be avoided at street level.

7. Amenity

Performance Criteria

1. Maximise solar access and ventilation to residential units.
2. Ensure visual and acoustic privacy of residential units in the development and adjoining properties.
3. Improve pedestrian amenity surrounding the site.

Controls

1. A Wind Assessment shall be submitted at Development Application Stage.
2. A detailed Acoustic Assessment shall be submitted at Development Application Stage addressing noise and vibrations from the Pacific Highway, Gordon Avenue, Hammond Lane and the North Shore Rail Line.
3. Residential units shall be designed to maximise solar access, cross ventilation, visual and acoustic privacy.

8. Open Space and Landscaping

Performance Criteria

1. Landscaping is to soften and complement the development.

2. Landscaping at street level shall improve the amenity and attractiveness of the pedestrian environment.
3. The development shall provide publicly accessible links and open space.
4. Publicly accessible open space is to include meaningful green landscaping, providing corridors of locally indigenous vegetation that link open spaces to enhance environmental quality and optimise opportunities for habitat for native flora and fauna species.
5. Greening at the podium levels is to be provided, with planting visible to the surrounding area – with particular regard to Pacific Highway, Gordon Avenue, Hammond Lane, the Chatswood Bowling Club and Tennis and Croquet Key Public Space.
6. Podium and roof tops are to be a combination of green and recreation spaces.
7. Street tree planting is to be provided.

Controls

1. Open space at ground level shall be utilised as publicly accessible open space.
2. Large canopy tree planting must be provided:
   a) within the Pacific Highway setback,
   b) along the northern boundary of the site, and
   c) along the eastern boundary of the site to the Chatswood Bowling Club, where any setback area is greater than 3m.
3. To facilitate large canopy tree planting, deep soil planting is to be provided:
   a) within the Pacific Highway setback,
   b) along the northern boundary of the site, and
   c) along the eastern boundary of the site to the Chatswood Bowling Club, where any setback area is greater than 3m.
   Deep soil plantings include trees and shrubs, and are to be unimpeded by buildings or structures below ground.
4. All roofs up to 30 metres from ground are to be green roofs. These are to provide a balance of passive and active green spaces that maximise solar access.
5. A minimum of 2 hours of solar access is to be provided to the public open space on the site.
6. Public domain improvements shall be provided to all street frontages, and through site links, to Council requirements. This is to include road treatment and boundary to kerb treatment.
7. A minimum of 20% of the site is to be provided as soft landscaping, which may be located on Ground, Podium and roof top levels or green walls of buildings.
8. A Landscape Plan is to be provided at Development Application stage detailing all public domain at ground level, street tree planting, planting and space allocation at podium and roof top levels. This is to include species, container size at planting, spacing and approximate size at maturity.
9. Street tree planting is at the cost of the proponent, with location and species to be determined in consultation with Council at Development Application stage.
10. All existing aerial cables which may include for electricity, communications and other cables connecting to street poles and buildings around the site shall be removed and installed underground in accordance with the requirements of the relevant service authorities. Ausgrid lighting poles are to be provided to the requirements of Ausgrid for street lighting and shall be positioned compatible to the landscaping design around the site.

9. Links

Performance Criteria

1. The development shall provide publicly accessible through site links and open space.
2. Publicly accessible through site links and open space is to include green landscaping.

Controls

1. Public rights of way are to be provided on the subject site as follows:
   a) A minimum 4m wide for the full Pacific Highway frontage, being the setback to the Pacific Highway.
   b) A minimum 3m wide through site link, is to be provided for the frontage with Hammond Lane.
   c) A minimum 6m wide through site link is to be provided from Hammond Lane for the full length of the site north / south, to the north eastern boundary – separated from any vehicle access (except where unavoidable).
   d) A minimum 8m wide east / west through site link, connecting the north / south through site link with the Pacific Highway, along the northern boundary, separate of any vehicle egress.
   e) A minimum 6m wide east / west through site link, connecting the north / south through site link with the Pacific Highway, between the two towers.
   f) A minimum 1.5m setback to Gordon Avenue.
2. All publicly accessible open space and links are to be the responsibility of the relevant ownership entity, with formal public access to be created over these areas.

10. Active Street Frontages

Performance Criteria

1. To ensure that uses on the ground level contribute to the activation of the public domain.
2. To ensure that design and location of ground floor uses maximise surveillance of the public domain.

Controls

1. At ground level buildings are to maximise active frontages to Pacific Highway, Gordon Avenue and Hammond Lane.
2. Active frontages are to be maximised adjacent through site links.
3. A building has an active street frontage if all premises on the ground floor of the building facing the street(s) are used for the purpose of commercial premises or non-residential purposes and provide elements of visual interest when viewed from the street.

11. Traffic and Transport

Performance Criteria

1. Development must be designed to provide adequate and safe access to the site.
2. Development on the site must not cause adverse traffic impacts on the surrounding road system.
3. Minimise the number of vehicular ingress and egress points to the development.
4. All vehicles are to enter and exit the site in a forward direction.
5. Traffic and transport solutions are to be physical (rather than mechanical) on this site.
6. Minimise car parking and encourage alternative transport options.
7. Minimise impacts on, and be sympathetic to, publicly accessible through site links

Controls

641-653 and 655A Pacific Highway

1. Vehicle ingress and egress for the development is to be from one access point in Hammond Lane.
2. Vehicle ingress and egress is to be designed and located to achieve safety, minimise conflicts between pedestrians and vehicles and create a high quality streetscape.
3. All car parking and loading facilities are to be located below ground level, utilizing physical solutions to ensure all vehicles (including loading and garbage vehicles) enter and leave the site in a forward direction.
4. Other strategies for car parking reduction, such as reciprocal arrangements for sharing parking and car share, is to be included in any future Development Application.
5. The following is to be provided in any future development application:
    a) Reduced car parking rates based on Council's Development Control Plan.
    b) An amended Transport report containing updated traffic analysis and modelling having regard to other nearby made planning proposals, uplift under Willoughby Local Environmental Plan 2012 (Amendment 34), the Precinct Plan prepared by Willoughby Council for the Eastern side of Pacific Highway, between Gordon Avenue and Ellis Street, Chatswood, and any other applications including state significant development applications.
6. The existing egress from 655A Pacific Highway to the Pacific Highway is to be blocked via physical measures (not gates or barriers but by cul-de-sac or a similar solution whereby pedestrians are encouraged onto the site and able to access through site links) for the exclusive use of properties to the north and associated vehicles (identified as Section 2 in the Precinct Plan for the Eastern side of Pacific Highway, between Gordon Avenue and Ellis Street, Chatswood).
    a) In this regard appropriate rights of way are to be established (4m wide out of 8m wide setback). This egress is not in any way to serve the subject site, the Chatswood Bowling Club or any other vehicles from Gordon Avenue.

Chatswood Bowling Club

7. Chatswood Bowling Club vehicle ingress and egress is to be via Hammond Lane
8. Car parking for the Chatswood Bowling Club is to be on-site at 641-653 and 655A Pacific Highway and as follows:
    a) 6 car spaces at Ground Level.
    b) 35 car spaces at Basement level.

12. Waste Management, Loading and Services

Performance Criteria

1. Ensure all loading, unloading and servicing occurs on-site.
2. Ensure adequate provision is made for waste storage and disposal.
3. Ensure floor space at Ground level is maximised.
4. Ensure loading, unloading and servicing does not interfere with public rights of way and through site links.
5. Ensure undergrounding of aerial cables and provision of lighting poles to the requirements of Ausgrid and compatible with landscaping.

Controls

1. All loading and unloading services are required to occur at basement level on-site, and are to be designed to accommodate heavy rigid vehicles.
2. Other supporting functions such as garbage rooms, plant and other services are to be located in Basement levels.

3. A Waste Management Plan shall be submitted at the Development Application stage.
4. Substations are to be provided within buildings, not within the streets, open spaces or setbacks and not facing key active street frontages.
5. Substations are to be designed to ensure protection of residents from Electro Magnetic Radiation (EMR) emissions.

13. Design Excellence

Performance Criteria

1. Ensure high quality and varied design through the use of competitive design processes.
2. Implement a rigorous process to support good design outcomes.

Controls

1. All developments that have a height of 35m or more are subject to a competitive design process.
2. The competitive design process must be undertaken in accordance with the Willoughby Design Excellence Policy and Willoughby Design Excellence Guidelines.

14 Public Art

Performance Criteria

1. All redevelopments in the Chatswood CBD should contribute to public art in accordance with Council's Public Art Policy.
2. Public art is to promote place, sustainability and resilience.

Controls

1. Public Art is to be provided in accordance with Council's Public Art Policy.
2. Creative hoardings are to be provided during the construction process.

15. Building Sustainability

Performance Criteria

1. Design excellence shall include achievement of higher building sustainability standards.

Controls

1. A minimum of 5 stars GBCA building rating, or the equivalent, is expected. A higher rating is encouraged. An assessment report is to be submitted at Development Application stage.

16. Heritage

Performance Criteria

1. Respect the built history of the Chatswood CBD.

Controls

1. The front fence of 655A Pacific Highway is to be retained and incorporated into the design of the site.

13.2 Local Centres

The following specific sites and areas, and the site specific DCPs apply to the Local Centres.

13.2.1 100 Edinburgh Road, Castlecrag

Land to which this part applies

These special provisions apply to the land edged in yellow in Figure 49 with the legal description of lot 11 in DP 611594 and lot 1 in DP 43691.

Figure 49: Site area indicating boundaries, in yellow, of the site.

13.2.1.1 Application of this Part

This part is to be read in conjunction with:

- State Environmental Planning Policy (Housing) 2021 -
- Apartment Design Guide (ADG, 2015)
- Willoughby Local Environmental Plan 2012

Relationship to the Apartment Design Guide

The Apartment Design Guide (ADG, 2015) is the primary document for guiding the design of the residential development at the site. Where there is an inconsistency between the ADG and controls within this part, the provisions of the ADG prevail.

Relationship to other parts of the Willoughby DCP 2023

This part is to be read in addition to the parts listed above. Where there is an inconsistency between this Part and any other Part of the DCP in force, the provisions of this part of the WDCP prevail the extent necessary to achieve a design in accordance with that lodged as a Planning Proposal for the subject site.

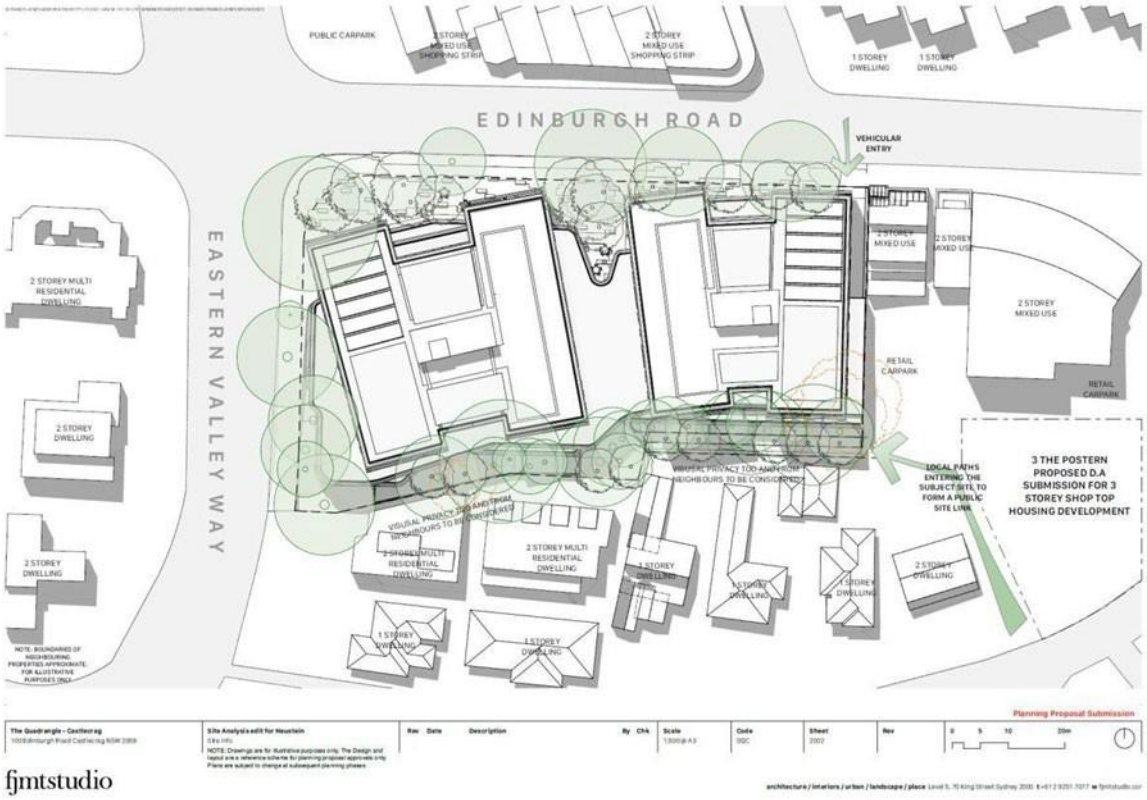
13.2.1.2 Site planning

Site master plan

Development of the site is to be carried out in accordance with the master plan scheme shown in Figure 50 and including the following siting and orientation of the buildings

- provision of site facilities
- vehicular and pedestrian access

Figure 50: Site master plan, November 2021, revision 06.



Building height

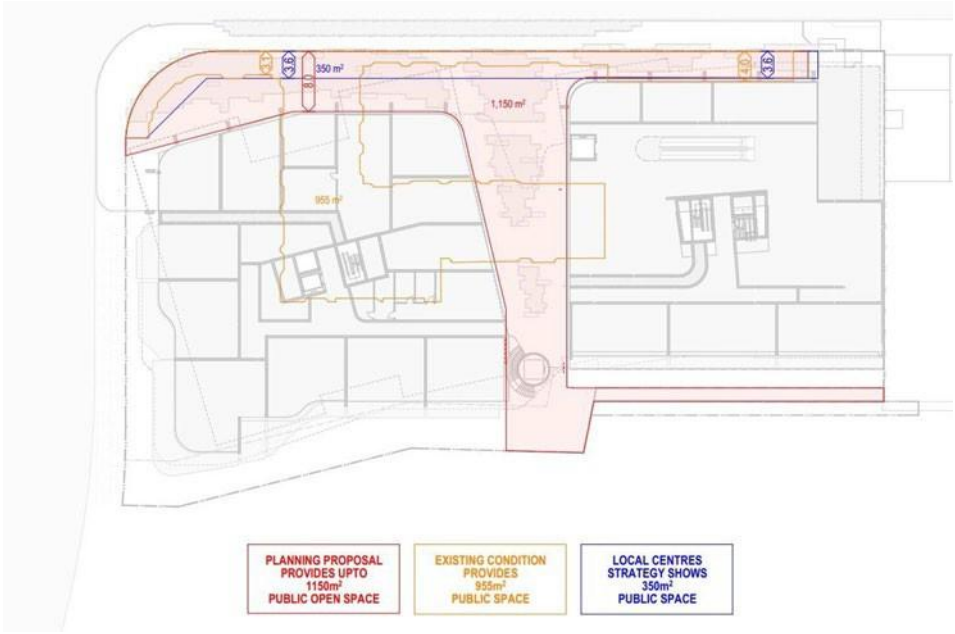
Controls

1. Development of the site is to be carried out in accordance with clause 4.3A of the Willoughby LEP.
2. Ensure a maximum of four storeys above the ground level at the Eastern Valley Way frontage and up to three storeys along Edinburgh Road.

Publicly accessible open space

Publicly accessible open space on site is to be provided as shown below in Figure 51.

Figure 51: Publicly accessible open space on site.



Setbacks

Setbacks are to comply with the building envelope plan identified by the red box outlines in Figure 52 and Figure 53 below.

Fig 52: Setback plan at ground level

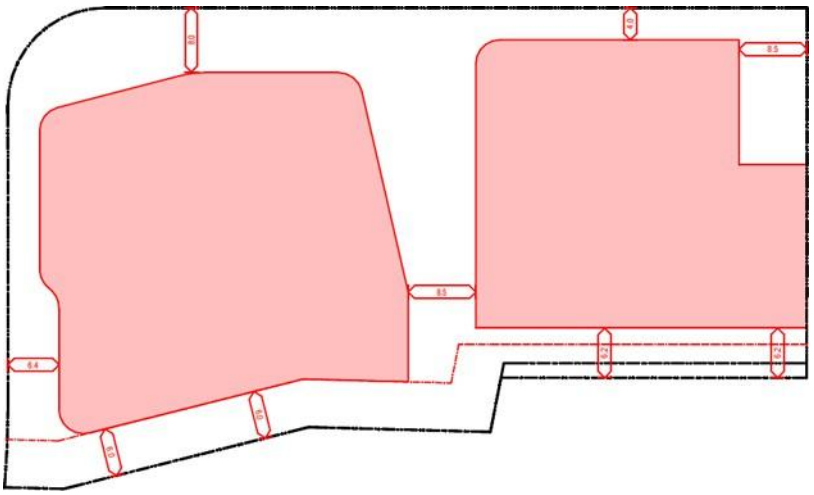
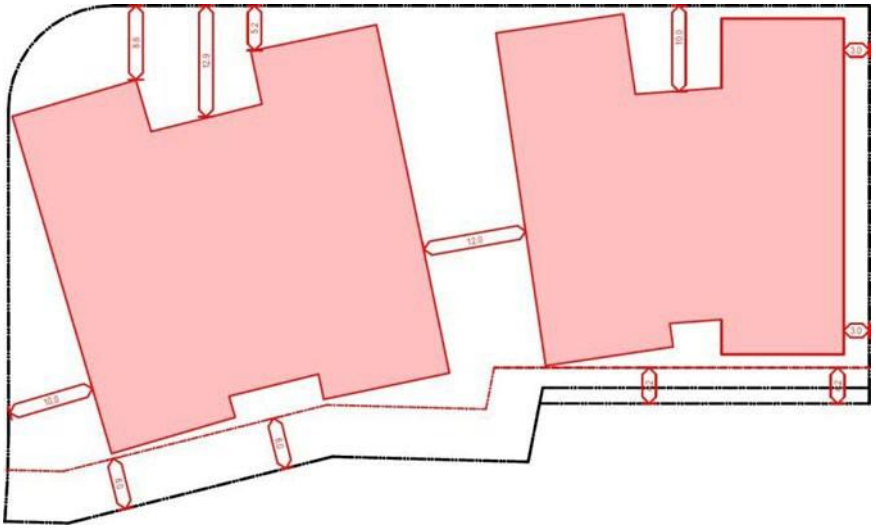


Fig 53: Setback plan at Level 1 and 2. Note setbacks are to include balconies.



Landscaping and tree preservation

Objectives

a. To maintain the existing healthy trees on site and those adjoining the site on Edinburgh Road and Eastern Valley Way.

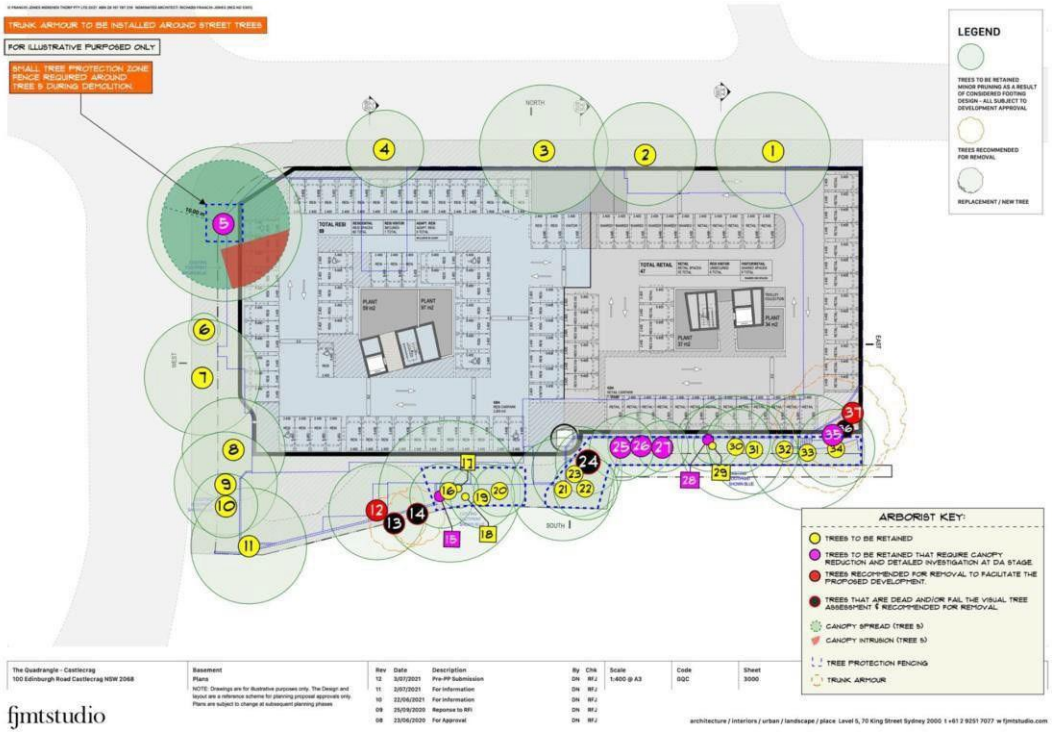
b. To maximise landscaped area on site commensurate with Height of Building and FSR controls.

c. To enhance the amenity of the site by planting additional trees where possible

Controls

1. Landscaping is to be provided as per the “Landscaping Concept” specified in “Planning Proposal Architecture and Landscape Report” by FJMT Studios (Aug. 2021).
2. Street trees adjoining the site on Eastern Valley Way and Edinburgh Road are to be retained as shown in Tree Preservation Plan (Figure 54).
3. Trees on the southern boundary, other than trees specified in the report of the arborist for removal, are to be retained as shown in Tree Preservation Plan (Figure 54). Trees subject to pruning are to be re-assessed at DA stage.
4. Additional trees are to be planted where shown on the landscape plan lodged with the Planning Proposal.

Figure 54: Tree preservation plan



Development controls Public open space

Objectives

a. Provide an open landscaped plaza, as shown on the public open space plan

b. Provide opportunities for café and public seating within the plaza

c. Allow for access to shops and cafes fronting Edinburgh Road

d. Provide a cross site through link to the walkway to The Postern

Control

1. Provide public access to the approximately 1,150 sqm of designated area defined in Figure 53.
2. Any public right of way over private land should be for the public to use but is to be maintained by the private land holder. This public right of way should be compliant for disabled access and give consideration to Safer By Design principles and guidelines.
3. A management plan for public open space areas is to be prepared by the applicant and approved by Council prior to approval of a development application on the site. The management plan is to address future management arrangements of areas in private ownership to be accessed by the public, including outdoor dining, landscaping, public access and ongoing maintenance responsibilities.
4. A footpath width on Edinburgh Road and Eastern Valley Way adjacent to the development that is 2.4m or wider as determined by the pedestrian load of the area and that generated by the development will be required. Any potential outdoor eating areas should be clear of this footway area.

Communal open space

Objectives

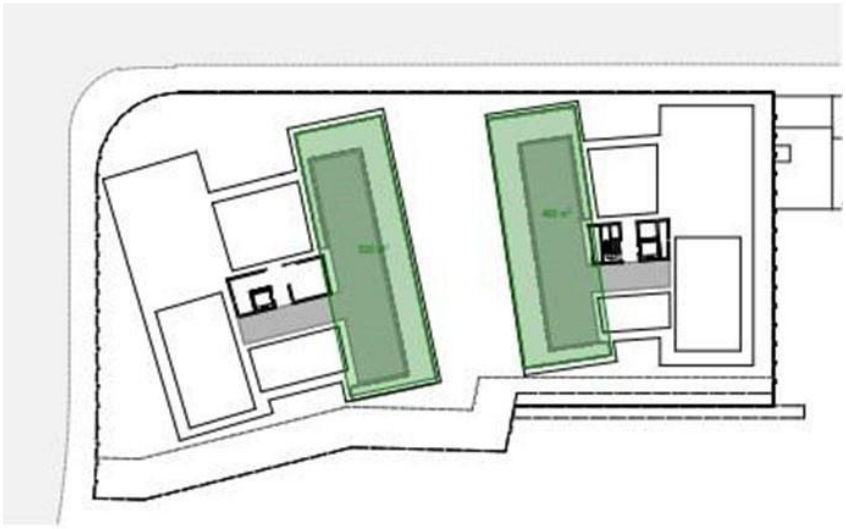
a. Provide outdoor recreation and relaxation opportunities for residents.

b. Maximise solar access to communal open space, both for the amenity of the residents and viability of landscaping.

c. Provide opportunities for residents to meet informally.

d. Provide landscaped communal open space on the roof of the pavilions generally as shown in Figure 55:

Figure 55: Indicative communal open space on the pavilion roofs.



Control

1. The area, extent and location of communal open space is generally as per Figure 55.
2. Application detailing how adverse wind impacts through and around the building are to be mitigated.

Colours and materials

Objectives

a. To achieve a materials, colours and architectural design sympathetic to the Griffin Legacy.

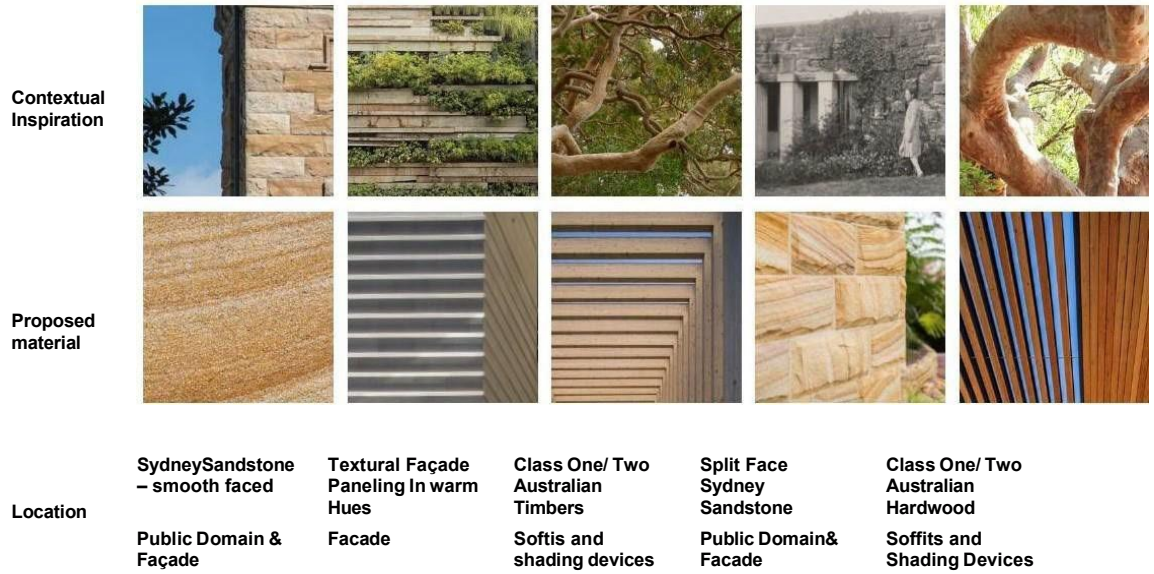
b. To use materials and colours that complement the adjoining Heritage Conservation Area.

Controls

1. The use of the materials and colours as identified in Indicative Material Palette is required to meet the stated objectives.
2. The use of coloured panels or cladding to achieve visual interest is not permitted.

Note: The following indicative but not exhaustive range of materials can be used to achieve the objectives above:

Figure 56: Indicative Material Palette



Access, Parking, and Transport

Objectives

a. The development shall meet Council's car parking requirements;

b. The number of vehicle access points to the development are minimised;

c. Traffic generated from the proposed development should be mitigated;

d. Vehicular access points are designed to minimise their impact on pedestrians and the flow of traffic; and

e. Vehicular access points should be unobtrusive in the streetscape but ensure visibility for motorists and approaching pedestrians.

Controls

1. Proposed vehicular access points are to be limited to no more than one ingress/egress from Edinburgh Road. Vehicular access is not permitted from Eastern Valley Way.
2. Future development of the site must ensure that measures are put in place so that development will have no significant impact on the efficiency and operation of the existing surrounding road network.
3. Active transport facilities including resident and visitor bicycle parking are to be provided.
4. Electric car charge points are to be provided within any basement car park.
5. The parking rates as specified in Figure 57 – Applicable Parking Rates are to be applied and any variation to these rates are to be agreed with Council.
6. Justification for an amended car parking rate must be accompanied by a Car Parking Demand Assessment. The Car Parking Demand Assessment must address the following matters, to the satisfaction of Council:
   i. The likelihood of multi-purpose trips within the locality which are likely to be combined with a trip to the land in connection with the proposed use
   ii. The variation of car parking demand likely to be generated by the proposed

use over time

iii. The short-stay and long-stay car parking demand likely to be generated by the proposed use

iv. The availability of public transport in the locality of the land

v. The convenience of pedestrian and cyclist access to the land

vi. The provision of bicycle parking and end of trip facilities for cyclists in the locality of the land

vii. The anticipated car ownership rates of likely or proposed visitors to or occupants (residents or employees) of the land

viii. Any empirical assessment or case study.

7. An additional 8 public car spaces are to be provided in addition to applicable DCP parking requirements.
8. A minimum of 4 accessible parking spaces are to be provided for use by the public on site.

Note: The space for an additional 8 public car spaces should not be deducted from the GFA of the building.

Figure 57: Applicable Parking Rates – 100 Edinburgh Road Castlecrag

Component	Parking rate (targets)
Residential	
1-bedroom unit	1 space per unit
2-bedroom unit	1 space per unit
3-bedroom unit	1.25 space per unit
Visitor spaces	1 space per 4 units
Supermarket	6 spaces per 100sqm
Retail	1 space per 25sqm
Restaurant	1 space per 75sqm

13.2.2 Site Specific DCP – 92-96 Victoria Avenue North Willoughby

To further guide the design of new development within the site, and to provide greater certainty regarding the future design outcome across the site, a site-specific development control plan (DCP) has been prepared to support the draft LEP amendment.

The DCP Controls are outlined in the following sections. These special provisions apply to the land edged in red in Figure 58 being 92-96 Victoria Avenue North Willoughby.

Figure 58: Site Plan

13.2.2.1 General Provisions

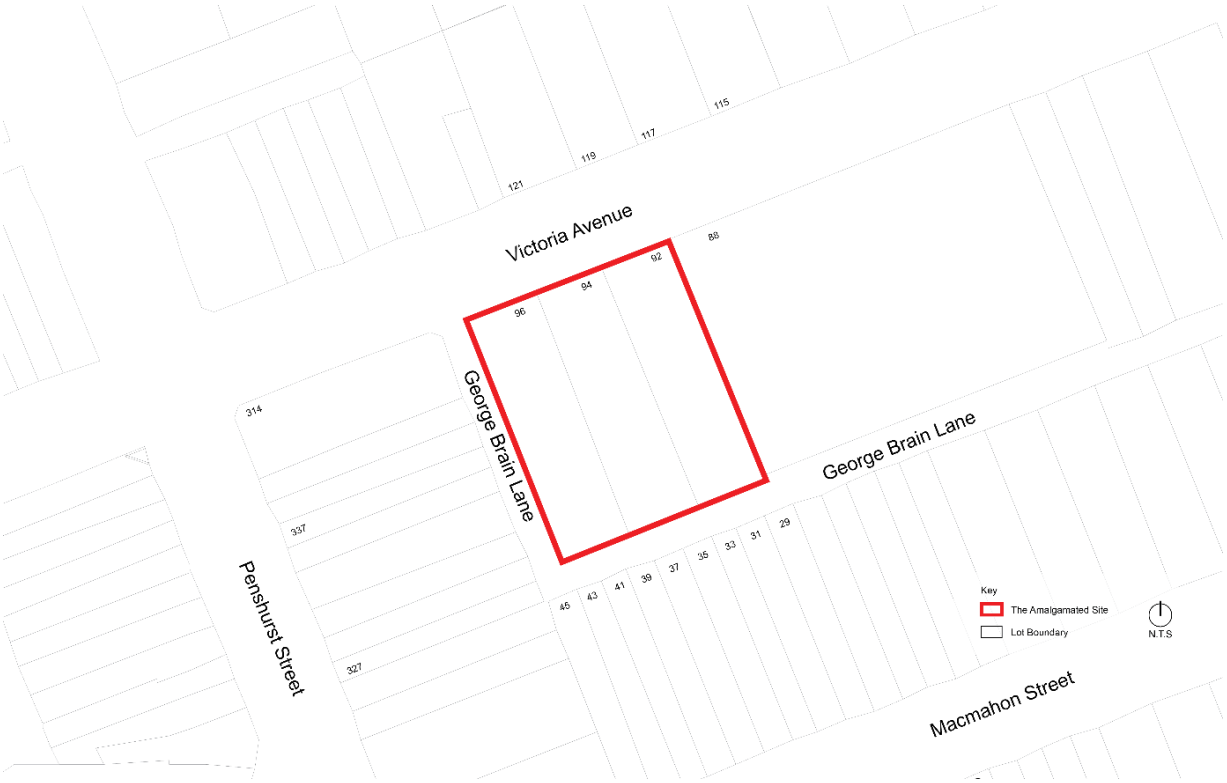
The aims and objectives of the DCP site specific provisions are:

1. To ensure that the site is amalgamated, so as to enable the substantial redevelopment of the site for medium density residential development;
2. To ensure that the development is complementary to the scale and character of development in the area;
3. To ensure that the development appropriately transitions in height across the site;
4. To ensure that new buildings on Victoria Avenue and George Brain Lane contribute to a quality streetscape;
5. To encourage energy efficient design and development;
6. To encourage high quality, built form outcomes and achieve design excellence;
7. To indicate the location of vehicle access points to the site enhance and improve surrounding streetscapes; and
8. To control adverse environmental impacts such as shadows from any development of the site.
9. Where there is an inconsistency between this Part and any other Part of the Willoughby DCP 2023 in force, the other provisions of WDCP will apply.

13.2.2.2 Specific Provisions

A – Allotment Size and Consolidation

Figure 59: Site Amalgamation Plan



Objectives

a. Allotments should be amalgamated to ensure that sufficient site area is available to achieve the stated development potential and adequate provision for landscaping and recreational open space;

b. Ensure that the site is amalgamated, so as to enable the substantial redevelopment of the site for medium density residential development; and

c. Ensure that the development is complementary to the scale and character of development in the area.

Controls

1. New development on the site will be required to consolidate all affected allotments.
2. The sites should be amalgamated in accordance with the Site Amalgamation Plan.
3. A development application for the substantial redevelopment of fewer than all of the allotments, the subject of this section of the DCP, must be accompanied by evidence that consolidation of all allotments has been reasonably attempted in accordance with the Land and Environment Court's Planning Principles.

B – Design Quality

Objectives

a. To encourage innovative, high quality architectural design of dwellings and their associated open space;

b. To achieve long term durability of design and finish;

c. To achieve a high standard of public and private amenity; and

d. To encourage high quality, built form outcomes and achieve design excellence.

Controls

1. In considering the quality of the architectural design, and notwithstanding a proposed development's compliance with the other provisions of any other part of this DCP, Council must be satisfied that:
   (a) the proposed development includes a high level of connectivity to the public domain and achieves building siting that provides street frontage for dwellings to Victoria Avenue and George Brain Lane; and
   (b) a high standard of architectural design, external materials and detailing will be achieved appropriate to the building type and location; and
   (c) visual interest is achieved when viewed from the public domain including but not limited to the façade design, use of landscaping, and building profile; and
   (d) the form and external appearance of the building will improve the quality and amenity of the public domain; and
   (e) any amelioration measures for environmental impacts such as architectural elements and landscaping are to be integrated into the design to achieve an overall high standard of design quality.

C – Height

Figure 60: Building Envelope Plan

Objectives

a. To achieve an appropriate height transition across the site responding to the existing and future desired context; and

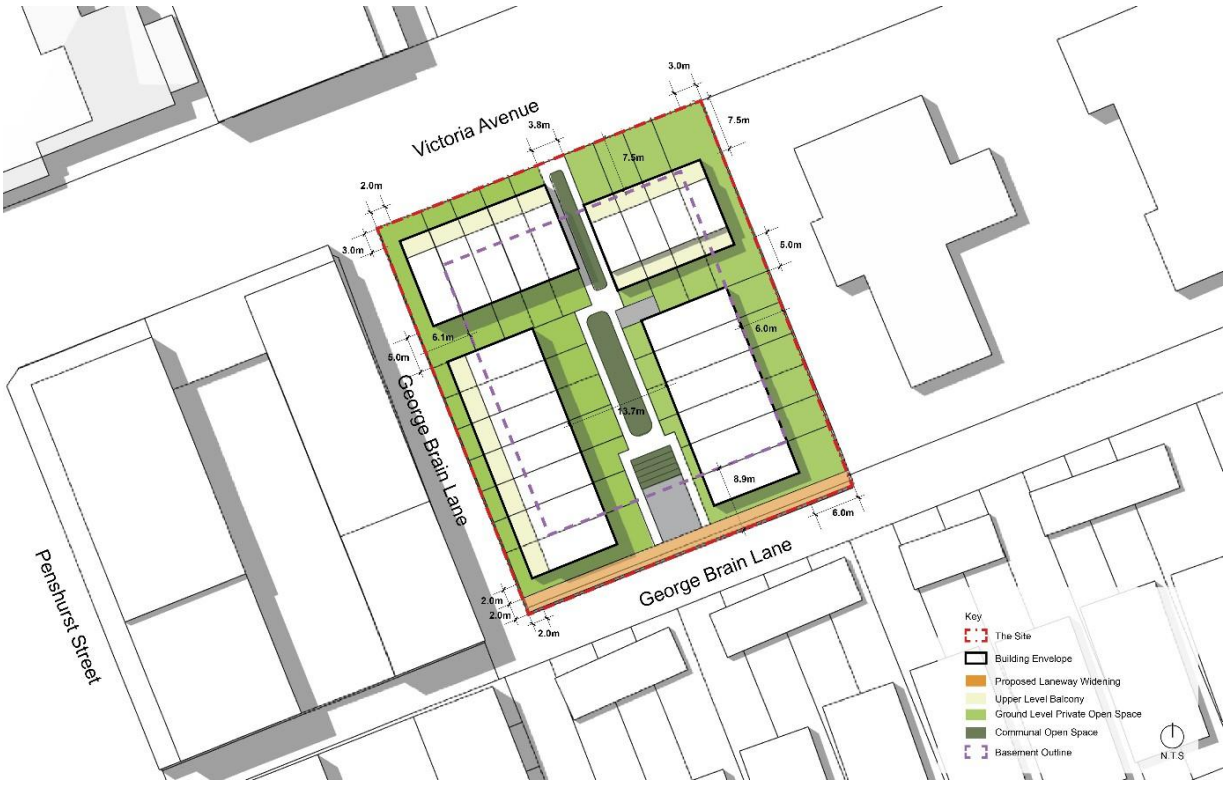
b. To provide various height controls that maintain reasonable solar access through the site and to nearby residential development.

Controls

1. The height of buildings in storeys are to be in accordance with the Building Envelope Plan.
2. The development is to provide a three-storey scale to the western edge of the site along George Brain Lane Street, transitioning to two storeys at the south east corner of the site.
3. The maximum wall height of buildings across the site is 9.5m.
4. Detailed design of roofs is to present as a terrace housing form. A variation of flat and pitched rooftops is encouraged to assist in transitioning height across the site and responding to the character of surrounding buildings.

D – Setbacks and Streetscape

Figure 61: Building Envelope and Setbacks Plan



Objectives

a. To ensure future development appropriately recognises the streetscape and contributes to the public domain.

b. To ensure residential development proposed on the site is responsive to the specific characteristics of the site and surrounding sites.

c. To ensure that the development proposed on the site is scaled to support the desired future character with appropriate massing and spaces between buildings.

d. To provide suitable areas for communal open spaces, deep soil zones and landscaping.

e. To provide for adequate privacy and amenity between dwellings.

f. To establish the desired spatial proportions of the street and define the street edge.

g. To manage a transition between sites to the east and west with different development controls such as height and land use.

h. To ensure the site can accommodate quality entries to communal areas and individual dwellings.

Controls

1. Setbacks are to be in accordance with the Building Envelope and Setbacks Plan.
2. The design of the development is not to be read as a single mass from Victoria Avenue or George Brain Lane but should be broken into separate buildings of varying scale.

3. The design of the building facades is to create a visually interesting form incorporating recessed and projecting elements.
4. Modulation of the street and courtyard elevations is required to reduce the scale and bulk of buildings and to give identity to different dwellings.
5. Setbacks must ensure sufficient separation to provide privacy, minimise overlooking, and ensure adequate solar access across the site and to adjoining properties.
6. Walls alongside boundaries and George Brain Lane should be modulated to avoid the appearance of large blank facades by articulating walls with projecting or recessed elements such as windows and doors.
7. Dwellings with frontages to Victoria Avenue and George Brain Lane are to provide dwelling entries at ground level.

E – Solar and Daylight Access

Objectives

a. To optimise the number of dwellings receiving sunlight to habitable rooms, primary windows and private open space.

b. To optimise solar access to areas of communal open space.

c. To ensure impacts on solar access to adjoining dwellings are minimised.

Controls

1. Living rooms and private open spaces of at least 70% of dwellings are to receive a minimum of 2 hours direct sunlight between 9 am and 3 pm on 21 June.
2. A maximum of 15% of dwellings are to receive no direct sunlight between 9 am and 3 pm on 21 June.
3. Development is to achieve a minimum of 50% direct sunlight to the principal part of the communal open space for a minimum of 2 hours between 9 am and 3 pm on 21 June.
4. Solar access modelling for the site is to have regard to the future height and massing of buildings within the North Willoughby Town Centre, as detailed in Council's Local Centres Strategy 2036.
5. Where an adjoining property does not currently receive the required hours of solar access, the proposed building ensures solar access to neighbouring properties is not reduced by more than 20%.
6. Note: The principal part of communal open space may include recreation and landscaping areas such as the communal gardens that afford residents privacy and amenity, as well as allowing them to relax and connect to the natural environment.

F – Open Space and Landscaping

Figure 62: Significant Tree Diagram

Table 1: Tree Inventory*

Tree No.	Botanical Name	Common Name	Spread (m)	Diameter at Breast Height (DBH) (mm)	Diameter at Base (DAB) (mm)	Tree Protection Zone (m)	Structural Root Zone (m)
1	Quercus robur	English Oak	25	1200	1500	14.4	3.9
2	Platanus x acerifolia	London Plane Tree	2	50	80	2.0	1.5
3	Liquidambar styraciflua	Sweet Gum	20	1100	1200	13.2	3.6
4	Platanus x acerifolia	London Plane Tree	12	400	500	4.8	2.5

** Tree Inventory (Table 1, P 7, Report: Preliminary Tree Assessment; 92-96 Victoria Ave, Chatswood 2020)*

Objectives

a. Landscaping is to soften and complement the development.

b. Landscaping is to retain and complement the existing mature tree plantings along Victoria Avenue.

c. Landscaping at street level shall improve the amenity and appearance of the pedestrian environment and public domain, as well as identify dwelling entrances.

d. Communal open space, where provided, is to be responsive to the pattern of development and surrounding street network.

e. Communal open space is designed to maximise safety.

Controls

1. The trees nominated on the Significant Tree Diagram and Tree Inventory are to be retained.
2. Planting within setback areas is to ensure clear sight lines between building entrances and the street, as well as throughout common property areas.
3. Significant tree and shrub planting is required in the front setback along Victoria Avenue, along the eastern side boundary and within the within the centrally located communal open space. Trees must reach a minimum height of 4m to provide adequate screening and privacy to dwellings and their private open space.
4. The redevelopment of the site is to provide the inclusion of soft landscaping. The applicant must be able to:
   (a) Provide high quality and attractive landscaping which enhances the setting of the buildings in the streetscape;
   (b) Provide landscaping which enhances the amenity of a development by allowing adequate open space, sunlight and shade; and
   (c) Maximise opportunities for deep soil and absorptive landscaped areas.
5. The maximum site coverage across the amalgamated site is 45%.
6. A minimum communal open space of 6% of the site area, and a minimum deep soil of 30% of the site area is to be provided across the amalgamated site.
7. Private open space should be located adjacent to the living room, dining room or kitchen to extend the living space.
8. Visual impacts of services should be minimised, including location of ventilation duct outlets from basement car parks, electrical substations and detention tanks
9. Communal open space should be readily visible from habitable rooms and private open space areas while maintaining visual privacy.
10. Communal open space should be well lit.
11. Boundaries should be clearly defined between communal open space, public open space, and private areas.
12. The minimum recreational open space is 55% of the site area.

G – Access and Parking

Objectives

a. The number of vehicle access points are to be minimised;

b. Traffic generated from the proposed development should be mitigated;

c. Vehicular access points are designed to minimise their impact on pedestrians and the flow of traffic; and

d. Vehicular access points should be unobtrusive in the streetscape but ensure visibility for motorists and approaching pedestrians.

Controls

1. Proposed vehicular access is to be limited to a single access point along George Brain Lane in accordance with the Building Envelope Plan. No vehicular access is permitted from Victoria Avenue.
2. The vehicle entrance should be designed to minimise impact on footpaths and pedestrians.
3. Vehicular access driveways and crossings shall be designed for simultaneous 2-way movements. All access driveway, circulation areas and car parking shall comply with AS 2890.1, 2890.2 and 2890.6. The grade of the access driveway shall be a maximum of 5% for the first 6 metres within the property and comply with AS 2890.
4. Fence / building splays in accordance with the requirements in AS 2890.1 are required on both sides of the vehicular access driveway.
5. All vehicles shall be able to enter and leave the site in a forward direction.
6. Electric car charge points are to be provided within any basement car park.